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Table Of Contents

Page 6: How Political Activity and Marketing by Meat Corporations Shape Food Culture By DoGyoung Bae

Page 12: Unequal Burdens of Climate Change: Occupational and Homelessness-Related Mortality, Injuries, and Labor Supply Losses By Jake Capampangan

Page 21: The Influence of Cultural Factors on the Role of Air Pollution in Accelerating Alzheimer's Disease Progression By Ava Singer

Page 32: A Bug, A Burden, A Son: Franz Kafka and the Literary Processing of Intergenerational Trauma-By Evelyn Yang

Page 47: Constructing Legitimacy After Usurpation: Statecraft, Culture, and Power under the Yongle Emperor By Chenzhe Huang

Page 61: How Does Nightly Sleep and Strategic Napping Affect Cognitive and Physical Performance Of Elite Soccer Players? By Ajay Kosaraju

Page 71: Personalizing Immunotherapy in NSCLC: Traditional Radiomics and the Emerging Potential of Habitat Radiomics By Eunchan Jung

Page 89: Through A Quantitative Correlational Study, How Does Proximity To Traditional Burial Sites Affect The Adherence To Groundwater Properties To Water Quality Standards As Well As Affect The Quantity Of Contaminants Compared To A Control in the Same Region? By Morgan Wagner

Page 103: Correlation Between Post-Meal Glucose Dynamics and Meal Components By Hyelin Chae

Page 115: The Dynamics of Globalization-Insights from the Trajectories of UAE Free Zones By Malika Uzun

Page 122: Musical Engagement in Preschool Classrooms for Children with Developmental Delays: An Observational Study By Ryan Qi

Page 131: Design and Development of a Household Black Soldier Fly Larvae Waste Conversion System By Elle Frazier

Page 143: Youth-Led Advocacy and Barriers to Education By Izabel Karrer

Page 161: From Disruption to Distrust: Conflict-Based Supply Chain Management By Harrison Lai

Page 171: Trauma, Attachment Relationships & Adolescence By Tanvi Polasi

Page 181: The Pursuit of Happiness Through Broken Promises and Sacred Lands By Amarpreet Kaur

Page 184: Biomechanics of Fast Bowling and Injury Risk in Cricket By Om Vyas

Page 203: The Impact of Social Media Health Misinformation on Adolescent Health Decisions: Implications for Public Health Policy By Anaya Kabir

Page 209: Not Quite My Tempo: An Analysis of Tempo Alterations and Preferential Ratings Across Differentiated Age Ranges By Hritik Patel

Page 224: Eliminating the “Improper Women”: Effects of the Indian Feminist Movement on the Devadasi System By Bonnie Shao

Page 234: "Trust the Country, Trust the Party": How Chinese Feminist Discourse Reproduces the Boundaries It Contests By Ada Hu

Page 248: From Rocket Science to Front Lines: Educated Unemployment and War Mobilization in Post-Sanctions Russia By Kangdi Zhao

Page 270: How School Culture and Parental Expectations Differently Affect the Academic Motivation and Mental Health of High School Students Across Gender Identities By Anshika Gupta

Page 281: The Exhaustion Divide: Gender Based Discrepancies in Sleep By Sofia Norton

Page 286: Playing for Passion or Pressure: Intrinsic vs. Extrinsic Motivation in Chinese Sports and Education By Cengan Zhang

Page 297: Feeling Invisible: AAPI Identity, Media Influence, and the Psychological Impact of Curriculum Absence By Evelyn Leung

Page 306: Ain't Worried: Selective Calm as a Systems Outcome in China's AI Moment By Liren Cao

Page 329: Water Quality Impacts on Reproductive, Birth, and Health Outcomes: What's in Your Water? By Hyeon-Joo Cha

Page 342: The Hidden Cost of Fashion: How Globalisation Shapes Sweatshops Worldwide By Natasha Bhatia

Page 350: Humanized vs. Dehumanized Opponents in First Person Shooter Games: How Character Design Impacts Adolescent Aggression By Sophia Hijjawi

Page 363: From Bully to Offender: A Mixed-Methods Study of How Bullying Can Escalate, the Role of Opportunity and Money, and How Media Distorts Our Understanding By Abhishek Raj

Page 379: Cardiac Arrest Prediction Using AI: A Literature Review By Devan Sharma

Page 384: Academic Accommodations: A Flawed System By Amber Olson

Page 390: Congressional Stock Trading and Market Performance: An Analysis of Government Officials' Investment Returns By Lucas Ahern and Eliot Meese

Page 399: The Interactions Between Gut-Microbiome Metabolites and T-cells in the Pathogenesis of Pediatric Celiac Disease By Reya Pandya

Page 415: Purkinje Cell Dysfunction in Autism Spectrum Disorders By Aastha Pendyala

Page 419: Which Machine Learning Classifier Model Most Accurately Detects Exoplanets and How Does Fast Fourier Transform Enhance This Process? By Carolina Camara

Page 429: Targeting Immune Evasion in Hodgkin Lymphoma: A Review of Modern Immunotherapy By Bhavin Patlola

Page 435: The Effects of Indian Classical Music on Mood By Ira Shellikeri & Sarah L. Hennessy

Page 446: The Effects of Commuter Rail Transit on Density: Evidence from American Cities By Yuncong Evan Liu

Page 474: The Transformation of Language's Metaphorical System in Algorithmic Media By Yaxuan He

Page 478: Happiness In a Box: A Quantitative Correlational Approach: How Blind Boxes Impact Hedonic and Eudaimonic Happiness in Singapore By Brooklynn Chow

Page 493: Is Cryptocurrency the Future of the World's Reserve? By Hayoon Jung

Page 499: The Science of Wormholes: Exotic Matter and the Plausibility of Human or Object Space Travel By Noelle Villani

Page 505: Analyzing How Noticing Hostile Architecture Affects Midtown Manhattan Residents' Empathy Toward Homeless Individuals By Alexa Zohar

Page 525: Bio-Inspired Aerodynamics in Modern Aircraft Design By Q. Khanh Hoang

Page 536: Effect of GLP-1RAs for Teenage Obesity on Anxiety By Rhea Srinivas

How Political Activity and Marketing by Meat Corporations Shape Food Culture

By DoGyoung Bae

Abstract

With global per capita meat consumption increasing substantially in recent decades, the human diet has undergone a process described as “meatification.” Although this trend is often framed as a natural outcome of economic development, this paper argues that today’s patterns of meat consumption are socially and politically constructed. Drawing on anthropological food studies, studies about corporate political activity, and product marketing scholarship, this study examines how corporations shape food culture by reinforcing and perpetuating ideas about meat consumption. Results indicate that there are two primary mechanisms through which meat industry actors shape food culture. First, through corporate political activities such as lobbying, corporations influence dietary guidelines to sustain high levels of meat production and consumption. Second, through marketing, they construct and perpetuate cultural meanings of meat by associating it with masculinity and status and embedding meat within national identity. This paper has implications for anthropology, politics, economics, and business because it highlights how corporations transform food culture and human–animal relations globally.

With global per capita meat consumption increasing by approximately one-third since 1980 and meat production output tripling since 1970, the human diet has undergone a phenomenon referred to as “meatification.” That is, people around the world are consuming more meat than ever before, including in countries that have not historically been high-consumption societies. For example, in China, per capita meat consumption has increased to approximately 60 kg annually—a 50% increase since 1995 (Sage 3). Some scholars argue that this trend is a natural consequence as countries become more developed because meat consumption is “a sign of affluence and modernity” (Carolan). However, the association of meat with affluence, as well as the rapid increase in consumption, is neither accidental nor necessarily natural. As Nungesser and Winter assert, meat consumption is “socially and culturally induced, regulated, and restricted,” and is embedded within broader political and economic frameworks. More specifically, dominant ideas about modern meat production and consumption were developed in a specific cultural framework and have been disseminated to global markets through mass media, advertising, foreign retailers, food service brands, and other food industry actors, including corporations seeking to expand production into developing regions (Sage 8).

Given that this shift is both intentional and human-driven, the purpose of this paper is to examine the methods by which corporations reinforce and perpetuate cultural narratives surrounding meat consumption. The literature review draws on anthropological food studies, research on corporate political activity, and scholarship on product marketing. The findings indicate that global meat corporations employ two primary mechanisms to influence food

culture: lobbying and product marketing. These strategies shape public perceptions of meat on a large scale, thereby transforming food cultures not only within individual nations but globally.

Lobbying and Nutritional Guidelines

Corporations influence government through a range of corporate political activities, including “campaign contributions, direct lobbying, government membership on company boards, political action committees, and even bribery,” in order to ensure that their products remain viable on the market (Lawton et al. 2). Meat corporations and other industry actors are no exception; they invest substantial financial resources to shape policies related to food regulation, waste management, and food labeling, thereby ensuring the continued production of meat at low cost. To maximize profits derived from low-cost production, these actors also seek to influence national discourse surrounding meat consumption, increasing the likelihood that consumers will purchase meat frequently and in large quantities. One strategy involves lobbying to influence dietary recommendations that would impact meat consumption.

A notable example of this is the Nordic Nutrition Recommendations (NNR), which serve as the foundation for food-based dietary guidelines across Nordic countries. The most recently updated NNR advises limiting red and processed meat consumption to 350 g per week for health reasons, and even lower amounts for environmental reasons. According to Wood et al., red meat industry actors in Nordic countries not only opposed these recommendations but also attempted to dissuade governments from adopting them by reframing high levels of red meat consumption as beneficial for both human health and the environment (2). Their claims included claims such as: “We should eat some meat”; “Livestock farming benefits the environment”; and “Reducing livestock numbers could lead to negative impacts.” A closer examination of these claims provides insight into how industry actors seek to shape policymakers’ perceptions of meat and, ultimately, influence national nutrition policies.

First, red meat industry actors asserted that Nordic populations require increased meat consumption for nutritional reasons. They argued that meat is superior to legumes, grains, fruits, and vegetables because plant-based foods do not “contain essential nutrients of sufficiently high bioavailability to replace, for example, the iron or zinc provided by meat,” and that meat consumption is necessary to prevent morbidity. They also challenged scientific evidence linking meat consumption to heart disease, claiming that “moderate” consumption does not increase disease risk. Furthermore, they contended that, “given the global demand for food,” reducing recommended meat intake would be “unwise” (Wood et al. 8-9) However, these claims lack scientific evidence and selectively omit information in order to portray red meat in a more favorable light. For instance, they emphasize nutritional benefits without addressing the health risks associated with high levels of meat consumption. Additionally, their arguments imply that meat-free diets cannot be healthy or nutritious, a claim that is not supported by existing research. The assertion regarding global food demand is also logically flawed, as domestic consumption of Nordic-produced meat has little bearing on broader global food supply and demand.

Second, industry actors argued that reducing recommended meat intake would lead to a decline in livestock numbers and, consequently, lead to reduced biodiversity in the region. This claim is misleading because the relationship between livestock production and biodiversity in Nordic pasturelands is weak. Wood et al. demonstrate that meat consumption is not a primary driver of biodiversity loss or the decline of semi-natural pastures; rather, direct economic support for farmers maintaining these landscapes is a key factor in preserving biodiversity (12). Notably, although all of these arguments are presented as scientifically grounded and evidence-based, they are frequently misleading or false. This suggests that industry actors are primarily motivated by economic and political interests rather than by concerns for public health or environmental sustainability.

This pattern extends beyond the Nordic context to other Western countries. Clare et al. found that meat industry actors in the United Kingdom similarly emphasized the environmental “benefits” of livestock, such as their role in maintaining open landscapes, when engaging with policymakers about nutritional policy. In the United States, pressure from meat corporations contributed to the removal of environmental sustainability considerations from the 2015 goals established by the Secretaries of Agriculture and Health (Wood et al. 12). Moreover, the United States Department of Agriculture’s food pyramid has long been subject to intensive lobbying by the meat industry (Ammann). Given that such dietary guidelines influence school meal programs and medical recommendations, lobbying efforts in this area represent a broader attempt to control the official representation of meat at the institutional level.

Shaping our Plate through Marketing

In addition to exerting political influence, global meat corporations also seek to shape public perceptions through the direct marketing of meat products to consumers. Marketing both establishes and reaffirms ideas around meat consumption, encouraging consumers to take actions that become part of an accepted norm or culture (Bogueva and Phau 264). This section outlines three primary ways in which marketing shapes cultural ideas about meat consumption.

First, as Evans and Miele argue, the meat industry presents animal products in ways that obscure their origins (7). In English, for example, the names of animal products are often different from the names of the animals themselves—pig meat becomes pork, sheep meat becomes mutton, and cow meat is called beef. This distinction encourages a cognitive separation between “food” and “animal,” reducing the likelihood that consumers will reflect on the implications of meat consumption. In addition, corporations construct appealing visuals that divert attention from the processes through which animals are transformed into food. Most cuts of meat by meat corporations do not include recognizable features like eyes or lips. As Bogueva and Phau note, advertisements, grocery store displays, and packaging frequently feature “happy pastoral scenes or mouth-watering images of ready-to-eat foods,” rather than realistic depictions of animal agriculture (271). These strategies are designed to make consumers more comfortable with what they are eating. In this way, marketing establishes the cultural idea (myth) that food is simply food and has nothing to do with animals.

Second, marketing portrays meat as a status symbol to make it more appealing to consumers. Historically, meat has been associated with masculinity, strength, and aggression. This association can be traced to early acts of gathering meat, or hunting, as a male-dominated arena; to fight wild animals required strength, as such, conquering animals became a sign of dominance and virility (Sumpter 106). Because women were often excluded from hunting practices, meat consumption—particularly activities such as grilling—became culturally associated with masculinity. In modern times, meat is often framed as a “vehicle for resurrecting manhood” in response to shifting gender roles (Bogueva and Phau 267).

Restaurants, food brands, and meat corporations create advertisements that reinforce the notion that meat consumption enhances masculinity and that “compromised masculinity can be regained through meat consumption” (Bogueva and Phau 267). For instance, in a Burger King advertisement, a man is visibly dissatisfied when served a small portion of meat at a fine dining restaurant; he subsequently leaves, singing “I am a man...,” and seeks out a larger portion at Burger King (Rogers). This narrative implies that “real men” consume large quantities of meat. Similarly, Hillshire Farm commercials from the early 2000s depict men chanting “Go meat!” while grilling in their manicured backyards in American suburbia. The contrast between almost tribal chanting and the quiet pastel landscape positions harkens back to Bogueva and Phau’s assertion about meat as a way to reclaim masculinity in contemporary contexts. Such advertisements make meat more appealing to consumers, not because of nutritional value, but because of what it symbolizes; meat corporations capitalize on the cultural association between meat and masculinity and continue to perpetuate the idea to increase firm profits.

Third, corporations also promote meat as an important product for national image and culture, making customers desire meat to maintain a sense of tradition. In Australia, for example, Meat & Livestock Australia (MLA) produced Australia Day advertisements featuring a “lambassador” and the slogan “put another steak on the barbie.” Bogueva and Phau explain that lamb has been heavily marketed as a central component of Australian identity. Particularly among younger adults, shared meat consumption is framed as a cornerstone of social life. They also note that chicken, though not as central to the construction of Australian identity as lamb, is marketed as a locally produced product, encouraging consumers to support the domestic economy (268-269). Similar strategies are evident in the European Union, where the Research Executive Agency has funded several campaigns promoting animal products to both domestic and international audiences in 2016 (Fernandez and Almiron 69). Campaigns such as “Pork Lovers Europe,” “Trusted Pork from Europe,” and “European Authentic Pleasure” present meat as culturally significant, high-quality, and emblematic of luxury. Through these efforts, meat is positioned not merely as a food item, but as a symbol of identity, tradition, and lifestyle.

Conclusion

The “meatification” of the human diet globally is the result of carefully constructed narratives and reinforced by corporate influence. Through corporate political activities such as lobbying, meat industry actors shape country-level nutritional guidelines and public policy to

protect and expand their market presence. At the same time, marketing enables them to shape public narratives about meat consumption more directly. By obscuring the origins of meat, associating it with masculinity and status, and embedding it within national identity, corporations actively shape the amount, frequency, and occasion that consumers consume meat. Mass media enables the export of these ideas to countries that have not been high-consumption societies historically. Taken together, it becomes clear that what humans eat is being shaped by more than taste; economic and political factors are significant contributors to a system that perpetuates high levels of meat consumption.

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Unequal Burdens of Climate Change: Occupational and Homelessness-Related Mortality, Injuries, and Labor Supply Losses By Jake Capampangan

Abstract

This paper synthesizes peer-reviewed research from 2009 to 2025 on climate-induced temperature extremes and their health and productivity impacts, focusing on workers and people experiencing homelessness and also disparities across communities. Both extreme heat ($>90^{\circ}\text{F}$) and extreme cold ($<20^{\circ}\text{F}$) generate a U-shaped rise in mortality and residential energy use with respect to temperature, while high ambient temperatures combined with metabolic heat increase direct heat illnesses and indirect injuries (falls, machinery mishandling) through fatigue and impaired cognition. Climate change has already cut global labor capacity by about 10 % and may erode it by 30 to 40 % by century-end, especially in agriculture and construction (De Sario et al., 2023). Demographic disparities due to climate change are also evident across gender, race, and age. Women have lower physiological heat tolerance, but men sustain higher injury spikes due to occupational segregation. Furthermore, Hispanic workers face the highest heat-related fatality rates because of socioeconomic barriers and limited insurance (Gubernot et al., 2015). Lastly, infants, the elderly, and younger workers experience the greatest mortality and injury increases. To mitigate these inequities, this paper recommends an overarching policy program including national heat-stress standards that cover “lone workers” (workers who perform tasks without direct supervision and are at increased risks for occupational injuries), green-infrastructure investments to correct historic heat-inequity, and monetary incentives (tax credits and subsidies) for firms and vulnerable populations to adopt cooling technologies and improve housing situations.

Introduction

Climate change has significantly reshaped not only the environment but also the conditions under which people live and work. Rising global temperatures, attributed to climate change, have intensified the frequency and severity of both extreme heat and cold events, creating higher risks for public health and economic productivity (Page and Sheppard, 2019; Drescher and Janzen, 2024; Deschênes and Greenstone, 2011). Mortality rates have risen among workers in outdoor occupations such as agriculture, construction, and mining during periods of extreme heat and cold, while productivity in these sectors also declines as temperatures pass safe working conditions (Gubernot et al., 2015; Tigchelaar et al., 2020; Spector et al., 2016; De Sario et al., 2023; Turner et al., 2025). Because Black and Latino workers are disproportionately represented in these heat-exposed industries, the health and economic consequences of climate change also reinforce existing social and economic inequalities (Berberian et al., 2022; Amoadu et al., 2023; Tigchelaar et al., 2020; Turner et al., 2025). Beyond the labor force, marginalized populations face increased vulnerability to extreme temperatures due to prolonged exposure and limited access to protective resources, further deepening existing social inequities.

In this paper, I review and synthesize available literature on climate change-related heat risks to health and productivity with a focus on workers and other vulnerable populations, particularly people experiencing homelessness. I find that exposure to temperature extremes creates a non-linear, often U-shaped distribution for both mortality and workplace accidents. Specifically, days with temperatures exceeding 90°F and those below 20°F significantly increase annual mortality rates and residential energy consumption. In outdoor workplaces, the combination of high ambient temperatures and metabolic heat generated from physically demanding tasks elevates risks for heat illness and traumatic accidents, such as falls and machinery mishandling, which are often driven by heat-induced fatigue and impaired cognitive performance. These risks are not limited to outdoor sectors but can also extend to indoor manufacturing and warehousing, where cooling systems may be insufficient.

For people experiencing homelessness, mortality risk rises at temperature extremes, with Lin et al. (2024) finding the strongest effect at extreme heat above the 90th percentile, a J-shaped pattern in Clark County, and a more U-shaped pattern in Los Angeles County. Heat waves further intensify morbidity among PEH by triggering underlying cardiovascular and respiratory conditions, resulting in increased emergency department visits that are frequently underreported or misclassified. Beyond direct health impacts, climate change has substantially been detrimental to labor productivity, with heat exposure already reducing global work capacity substantially. In climate-exposed industries, extreme heat leads to the loss of up to one hour of work per day as labor is reallocated to indoor leisure (Zivin and Neidell, 2014).

I further identify persistent demographic disparities in these outcomes. While women generally exhibit lower physiological heat tolerance, men suffer higher heat-related injury surges due to their overrepresentation in high-risk outdoor sectors like agriculture and construction. Gubernot et al. (2015) found that Hispanic workers had a 3.2 times higher risk of occupational heat-related death, pointing to several plausible contributing factors, including poverty, language barriers, housing, healthcare access, and training. Additionally, while infants and the elderly face the highest mortality risks, younger workers in their 20s and 30s exhibit the most significant relative increases in workplace injuries during extreme heat events.

Finally, I argue that a policy response should include the development of worldwide heat-stress management plans that cover “lone workers” (workers who perform tasks without direct supervision and are at increased risk for occupational injuries), funding of public cooling infrastructure in neighborhoods burdened by heat inequities, and the integration of green urban planning to reduce ambient temperatures. To support these projects, governments should provide monetary incentives, such as tax credits and subsidies, to help small and medium-sized firms acquire cooling technologies and improve long-term shelter for vulnerable populations.

Methodology

In this paper, I conduct a qualitative literature review, analyzing peer-reviewed studies from Google Scholar and JSTOR published from November 2009 to June 2025 that focus on various effects of climate change. I required studies to focus on examining how climate

change-driven temperature extremes affect occupational health and productivity in both outdoor and indoor sectors; disparities in racial and ethnic backgrounds in outdoor workplaces; and other vulnerable populations exposed to the outdoors, such as people experiencing homelessness. I searched for terms that combined three different categories: climate-related terms (“heat-related mortality,” “heat stress,” “climate change,” “climate risk,” “climate change and productivity,” and “climate change and health”); health impact terms (“heat-related injury,” “occupational mortality,” and “temperature-related workplace injuries”); and productivity impact terms (“labor supply,” “labor productivity,” and “productivity loss”). I applied the following exclusion criteria: studies lacking a quantitative heat-exposure metric and studies with insufficient sample sizes.

Literature Review

Climate-related injury and mortality risks in outdoor workplaces

Rising temperatures linked to climate change have led to an increase in injury and mortality risks for those in outdoor occupations. Workers performing demanding tasks in hot environments produce substantial metabolic heat, which in combination with high ambient temperature and humidity, can elevate the risk of heat strain, heat illness, and accidents (El Khayat et al., 2022). Evidence from the United States shows that agricultural and construction workers are among the most at-risk groups. Using the Census of Fatal Occupational Injuries (CFOI), Gubernot et al. (2015) identified 359 occupational heat-related deaths between 2000 and 2010 (average annual rate 0.22 per million workers), with the highest risks in agriculture and construction and among men and Hispanic workers. Conditions for Hispanic workers also might be exacerbated due to less than half of U.S. crop workers having access to health insurance, with the majority being Hispanic. Ultimately, these inequities can be attributed to the fact that a third of farmworkers have incomes below the federal poverty line (Tigchelaar et al., 2020). Using the California Worker’s Compensation System, Park et al. (2021) found that days with highs between about 85–90°F increase same-day injury risk by roughly 5–7 percent, and days above 90–100 °F increase risk by 10–15 percent, relative to mild days. They also document elevated injuries in both outdoor industries (agriculture, construction, and utilities) and indoor industries (manufacturing, wholesale trade, and warehousing), suggesting that inadequate cooling systems and the cognitive effects of heat are important factors in occupational risks. However, Park et al. found that heat-related injuries are not limited to “heat illness” diagnoses but that heat indirectly causes injuries such as falls, mishandling of machinery, or being struck by vehicles, which is consistent with heat-induced fatigue and reduced reaction times. Similarly, Deschênes and Greenstone (2011) found that there are similar increases in mortality rates above 90°F (0.11% increase) and below 20°F (0.07-0.08% increase). Annual residential energy consumption was raised at both ends of the distribution, where days above 90°F and below 20°F had a 0.4% and a 0.2-0.3% increase, respectively.

Extreme cold temperatures can also lead to an increase in risks of workplace accidents. Deschênes and Greenstone (2011) show that mortality and energy use both increase non-linearly

at temperature extremes. Relative to moderate days, very hot ($>90^{\circ}\text{F}$) and very cold ($<20^{\circ}\text{F}$) days are each associated with higher annual mortality and increased residential energy consumption, creating significant health and adaptation costs at both tails of the temperature distribution. Drescher and Janzen (2024) show a similar U-shaped relationship for accidents: days with maximum temperatures below 0°C (“ice days”) increase occupational accidents by about 6.3 percent, and days $\geq 30^{\circ}\text{C}$ increase accidents by about 7.4 percent, compared with mild days. Cold stress constricts blood vessels, impairs motor skills and dexterity, and increases the risk of slipping, frostbite, and other severe injuries, making accidents more frequent and serious on extreme cold days (Drescher and Janzen, 2024).

Climate-related hazards to other vulnerable populations

Another category of people who are exposed to climate-related risks are people experiencing homelessness (PEH) because they spend prolonged periods outdoors and often lack access to sufficient shelter that keeps them protected from extreme temperatures. A study of two U.S. counties, Clark County and Los Angeles, shows that daily temperature is strongly associated with mortality among PEH, with mortality risks increasing at both hot and cold extremes (Lin et al., 2024). Results in Los Angeles corroborate this finding by showing a U-shaped correlated graph between daily temperature and mortality among PEH with higher mortality rates at both extremes of the temperature distribution. However, in Clark County, they found that the association was J-shaped, with most temperature-attributable deaths occurring at the highest temperatures and lowest at moderate temperatures.

Heat waves and high-temperature days can also intensify the morbidity of PEH. Schwarz et al. (2022) report that heat waves in San Diego significantly increase emergency department visits among PEH, particularly for younger and older adults, and those with psychiatric diagnoses. They also found how heat waves can trigger underlying cardiovascular, respiratory, and mental health conditions, leading to higher emergency department visits even when the visits were not coded as “heat-related.”

Impacts of climate change on labor productivity

Climate change has substantially reduced labor productivity in heat-exposed and outdoor occupations. Wet Bulb Globe Temperatures (WBGT) measure the risk of heat-related illness in direct sunlight by combining temperature, humidity, cloud coverage, wind speed, and sun angle. Kjellstrom et al. (2009) show that as WBGT rises, physiological limits force workers to take more frequent breaks or reduce effort to avoid dangerous heat strain. Once WBGT exceeds $26\text{--}30^{\circ}\text{C}$, work capacity will decline sharply in heat-exposed industries, limiting economic advancement in tropical countries.

De Sario et al. (2023) estimate that current heat exposure already reduces labor productivity by nearly 10%, with future projections of 30–40% losses under worst-case climate change conditions by the end of the century if no additional adaptation occurs, with the most affected sectors being agriculture and construction. The International Labour Organization

similarly projects that by 2030, 2.2 percent of global labor productivity will be lost due to heat stress, with working-hour losses between 4–8 percent in Southern and Eastern Asia and Western Africa (Turner et al., 2025).

Using micro-data from manufacturing firms in India, Somanathan et al. (2015) find that worker productivity in tasks that are not especially heavy or fully outdoors still declines by 4 to 9 percent per additional degree on already-hot days and that sustained heat increases absenteeism. These daily-level effects will accumulate into an estimated reduction of at least 3 percent in Indian manufacturing output over recent decades relative to a no-warming state. Similarly, Zivin and Neidell (2014) show that, for workers in climate-exposed industries such as agriculture, construction, and outdoor services, daily maximum temperatures above about 85 °F lead to reductions of up to an hour of work per day, with most of the decline occurring late in the day as heat-induced fatigue accumulates. Therefore, work time is reallocated primarily from labor to indoor leisure, suggesting that heat reduces the marginal productivity of labor and creates tendencies towards cooler environments when possible.

As days of extreme heat continue, work at night or in the morning in outdoor sectors such as construction will become more prevalent, leading to further losses of wages for workers and decreasing productivity. Rode et al. (2024) found that thermal comfort was an important part of workers' desires, contributing to 2.9% of annual income on average. They also found that rising temperatures did not have a substantial effect on indoor workers. Casey et al. (2024) studied the effects of rising temperatures on the construction sector, which is a big part of investment. They showed that losses in productivity today from rising temperatures will slow aggregation of capital for future use. They also found that by 2200, extreme heat will reduce the U.S. capital stock by 5.4% and annual consumption by 1.8%.

Evidence from Europe and other high-income settings also points to how both extreme cold and hot temperatures can degrade performance. Studies found an overall non-linear relationship representing a U-shape graphically between economic productivity and temperature (Burke et al., 2015; Deschênes and Greenstone, 2011). Burke et al. (2015) found that productivity peaked at an annual temperature of 13 degrees Celsius, or 55.4 degrees Fahrenheit, and declined strongly at extreme temperatures across all countries. Exposure to extreme cold temperatures can impair tactile sensitivity, muscle function, finger mobility, and grip strength, reducing overall manual performance in labor industries (Ray et al., 2019).

Climate-related risks to residential living conditions

Climate change also heightens risks in residential environments, especially for those in substandard housing, buildings without air conditioning, and older structures with poor insulation against both heat and cold. For low-income households and communities of color, energy poverty further restricts financial growth. Even when cooling or heating equipment is present, high energy costs may limit use, forcing residents to endure unsafe indoor temperatures. Meanwhile, Deschênes and Greenstone (2011) show that hotter and colder days both raise

residential energy consumption, implying that climate change will drive up energy demand for cooling in summer and also for heating in winter in some regions.

Racial, ethnic, gender, and age-based disparities in exposure to climate-related risks

Climate-related hazards do not affect all populations uniformly. Many studies identify systematic differences in exposure and outcomes across race, ethnicity, gender, and age.

Gender

Women are at greater risks of heat-related performance losses and heat-related health risks than men. Amoadu et al. (2023) find that women have lower heat-dissipation capacity than men, leading to higher core-body temperatures under similar work conditions. However, the effect of extreme cold or ice days is significantly larger for females than males, but vice versa for hot days (Drescher and Janzen, 2024). U.S. workplace-safety data confirms this finding that men experience substantially larger injury rates on hotter days above 90°F: 8.4 % for men vs. 3.6 % for women (Park et al., 2021). Indeed, the gender gap is partly mediated by occupational segregation and gender norms: men dominate physically demanding outdoor occupations (construction, agriculture, mining) where heat exposure is the highest, whereas women are more represented in indoor or less heat-intensive occupations, lessening overall heat-related mortality for women despite a lower physiological tolerance. Furthermore, Hispanic women in agricultural occupations experience higher odds of heat-related illness, with 80% of Latinx farmworkers in Florida experiencing at least one heat-related illness and 40% experiencing 3 or more (Berberian et al., 2022).

Race/Ethnicity

Racial minorities such as Blacks, Hispanics, Latinos, and Native Americans are disproportionately affected by climate change and repeatedly show higher heat-related mortality and morbidity across occupational and community settings (Gubernot et al., 2015; Berberian et al., 2022; Hoffman et al., 2020; Kontor et al., 2025). Hispanic workers contribute to a majority of outdoor sectors such as agriculture and construction and therefore experience the highest heat-related fatality rates in the U.S. (Gubernot et al., 2015; Tigchelaar et al., 2020). In an analysis of Washington State farmworkers, Spector et al. (2016) show that each 5°C increase in Humidex raises the odds of traumatic injury by 1.45, with Hispanic workers bearing a disproportionate share of these injuries. Moreover, limited healthcare access, language barriers, and restraints on legal status can further restrain adaptations for these demographic groups (Tigchelaar et al., 2020). It is important to note that more than three quarters of agricultural workers are foreign-born, with most coming from Mexico, and only about half are legally authorized to work in the U.S. Furthermore, the average education level among crop workers is eighth grade, with about 71% who are also illiterate (Tigchelaar et al., 2020).

Age

Older adults (65 or older) and young adults (18 to 44) are most susceptible to climate-change-related risks. Deschênes and Greenstone (2011) found that mortality rises at both hot and cold extremes, with the effect especially pronounced for infants and older adults, and residential energy consumption also rises at both tails of the temperature distribution. Similarly, Park et al. (2021) find that a day above 90 °F increases injuries by roughly 7–8 % for workers in their 20s and 30s but by only 2–5 % for those in their 50s and 60s, even though older workers have higher baseline mortality rates. Schwarz et al. (2022) found that homeless populations in San Diego reinforce the age disparity: the younger (18-44) and older (65 or older) homeless adults have higher emergency-department visit rates during heat waves than middle-aged PEH.

Discussion

In the literature review, I observed that heat exposure leads to a variety of undesirable outcomes. In this section, I propose several ways in which governments may assuage these heat-related risks.

Heat protection technologies in worksites

Given the future adverse effects of climate change, policymakers should prioritize investment in advanced heat-protection technologies for high-exposure work sites such as construction, agriculture, manufacturing, and other physically demanding occupations to mitigate the rise in workplace accidents, injuries, and productivity losses. Providing cold towels, electric fans, adequate liquids, and on-site shade is cost-effective and would provide great relief to many workers in these outdoor sectors. Workers should also be provided breathable clothing, as it would allow them to dissipate their body heat more rapidly and cool down faster. However, many agricultural workers wear burdensome personal protective equipment (PPE) and protective clothing to shield against harmful pesticides, dust, and other chemicals. Future research should focus on adaptive PPE that would protect workers from both the heat and hazardous chemical exposure.

Public cooling infrastructure

City governments should fund and maintain accessible public cooling stations, water-refill points, and permanent shade structures, especially in neighborhoods burdened by historic heat inequities and among people experiencing homelessness, to curb the elevated mortality observed even during moderate heat days. The absence of these amenities is linked to higher levels of heat-related deaths among PEH, even on moderately warm days.

Heat-stress management plans

To protect workers from the dangers of exposure to extreme temperatures, employers should develop heat-stress management plans that include regular breaks in shade, access to water at all times, and acclimatization protocols. Existing heat safety mandates such as California's Heat Illness Prevention Standard provide a template for future preventative measures: mandatory water, shade, paid rest breaks, and training for temperatures above 95 °F.

However, it excludes “lone workers.” Extending the standard nationally and explicitly covering lone and remote workers would close this inequity.

Federal heat-illness legislation

The pending OSHA’s federal Heat Illness Prevention Standard is still being developed and will be the first federal mandate that will protect both indoor and outdoor workers (Alahmad et al., 2024). To maximize its protective range, the standard should require employers to conduct site-specific heat-risk assessments annually, incorporate safety measures for lone workers and for occupations where personal protective equipment (PPE) can restrict cooling of body temperatures, and adopt Wet Bulb Globe Temperature (WBGT) thresholds.

Addressing demographic heat-related inequities

Heat exposure is compounded by past land-use decisions that concentrate low-income and minority residents in extremely vulnerable communities. Green-infrastructure investments such as tree planting, green roofs, cool pavement, and urban canopy expansion have been shown to reduce surface temperatures, improve air quality, and lessen heat-related health disparities across demographics. Integrating these measures into city planning, with dedicated funding for marginalized communities, can diminish climate inequities.

Government financial support

Because many of these systems aiming to mitigate heat-related risks can be cumbersome financially, governments should provide subsidies, tax credits, or low-interest loans to small and medium-size firms for acquiring cooling equipment. Public investment should also focus on cooling centers, shaded public space, and housing or shelter improvements for people experiencing homelessness and residents of heat-burdened neighborhoods, since PEH face sharply elevated heat-related mortality and emergency department visits during hot periods.

Conclusion

In conclusion, I found that occupational heat stress, amplified by climate change, constitutes an escalating threat to worker health, safety, and economic productivity. High ambient temperatures increase mortality, occupational injury, and heat-related illnesses while also limiting total productivity and labor supply. Outdoor laborers, migrants, children, the elderly, women, racial minorities, and people experiencing homelessness disproportionately bear these burdens, reflecting historic inequities across these communities. Furthermore, heat-induced performance risks are task-specific, with perceptual-motor tasks declining above 30-33 °C WBGT and both physiological effects and behavioral pathways (fatigue, impaired decision-making) contributing to accidents. Adaptation measures such as hydration, work-rest cycles, shading, climate-controlled recovery areas, and engineering controls can mitigate but not fully eliminate impacts, and current policies (e.g., OSHA standards and California heat-illness prevention) have shown limited effectiveness. Moreover, new projections for future climate

scenarios find that heat-related workplace injuries, losses, and social costs of carbon will significantly increase. To avoid the worst of these harms, it is critical that countries implement fair and equitable policies and investments in green infrastructure and cool workplaces, including strong heat-stress standards for all workers. Additional research and data would be needed in low and middle-income countries and among populations that have been overlooked to protect the health and productivity of their workers in a warming world.

The Influence of Cultural Factors on the Role of Air Pollution in Accelerating Alzheimer's Disease Progression By Ava Singer

Abstract

Existing literature has found that air pollution has led to an increase in the incidence of Alzheimer's disease and an increase in progression in those who were previously affected. However, few studies have explored the role of cultural factors that influence one's exposure to air pollution and how this variation in exposure may impact the progression of disease. This article, therefore, focuses on the question of whether cultural factors influence the ability of air pollution to accelerate the progression of Alzheimer's disease. I propose that the relationship between air pollution and the progression of Alzheimer's disease is shaped by cultural practices that influence everyday exposure. For instance, the use of facial coverings to avoid skin tanning or mask-wearing norms that reduce exposure to air pollution. This article explores four cities with similar levels of air pollution using a novel exposure-reduction index measure, including cultural norms, to compare the PM_{2.5} levels across time. By connecting cultural practices with environmental differences in exposure, my findings suggest that wearing masks or face coverings could be a useful practice for those most at risk of developing Alzheimer's.

Introduction

Alzheimer's disease is one of the most urgent health challenges in the world today, affecting over 55 million people. As populations age, scientists and policymakers have tried to understand not just the biological roots of Alzheimer's. But the environmental and social conditions might make it worse. In the past decade, air pollution, especially fine particulate matter known as PM_{2.5}, has become a major focus. These tiny particles are small enough to enter the bloodstream and even reach the brain, where they can trigger inflammation and oxidative stress. Yet, the connection between pollution and brain health isn't the same everywhere. In some countries, high pollution doesn't seem to match up with faster cognitive decline. In others, small increases in exposure appear to have large effects. That gap suggests something else is taking place, something shaped by culture, healthcare systems, or the way that disease is measured or responded to.

This study asks a simple but important question: How do cultural factors influence the ability of air pollution to accelerate the progression of Alzheimer's disease? Here, cultural factors refer to the social, behavioural, and institutional conditions that affect how people are exposed to pollution or how easily they can access healthcare. The focus is on PM_{2.5}, fine particulate matter less than 2.5 micrometers in diameter, a standard measure of air quality used by the World Health Organization. Alzheimer's progression refers to how the share of the population with Alzheimer's changes over time. By bringing together environmental and cultural indicators, this project aims to explain why countries with similar pollution levels can experience such different Alzheimer's outcomes.

To explore this, I use data from seven countries: Japan, the United States, Mexico, China, Sweden, India, and South Africa, using datasets spanning from 2010-2019. Two Ordinary Least Squares (OLS) regression models form the basis of the analysis. The first model tests the relationship between air pollution (PM2.5) and Alzheimer's progression. The second adds controls for average temperature, healthcare coverage, and cultural trends, such as mask usage or beauty norms tied to sun exposure. Comparing the two models helps reveal how cultural and structural factors may shape, or even soften, the link between the environment and neurological decline.

The results show that PM2.5 alone does not significantly predict Alzheimer's progression once cultural and healthcare factors are accounted for. In other words, where and how people live, what care systems are in place, and what social habits are possessed appear to matter just as much as the air they breathe. Countries with stronger healthcare access and consistent diagnostic systems tend to report higher rates of progression. On the other hand, those with limited medical infrastructure report lower rates, regardless of pollution exposure.

This study aims to add something new to the conversation about Alzheimer's disease and environmental conditions. It connects environmental science to culture, showing that pollution's impact on the brain is not just a biological process. It is also a social one. These findings point toward the need for public health policies that consider not only cleaner air, but better accessibility to diagnosis and care.

Literature review

Recent research increasingly supports a strong link between environmental exposures, particularly air pollution, and the incidence of Alzheimer's disease. Long-term exposure to pollutants such as PM2.5 has been linked to inflammation, neuronal damage, and the buildup of amyloid plaques in the brain. All of these factors are the hallmarks of Alzheimer's disease. As a result, researchers have begun to frame air pollution as a neurological risk factor, not just a respiratory or cardiovascular concern (Calderón-Garcidueñas et al). However, much of the literature focuses narrowly on physical exposure and its biological outcomes. The social and cultural contexts that shape how individuals experience and respond to environmental conditions have often remained unexamined.

Several large-scale studies reinforce the link between pollution and cognitive decline. One study involving over 300,000 participants found that long-term exposure to ambient air pollution significantly increased the likelihood of developing dementia (Zhang et al.). The same research also showed that maintaining a healthy lifestyle may reduce that risk, even in polluted environments. These findings suggest that environmental impacts do not inevitably cause cognitive decline, especially when paired with preventive measures. Still, discussions of behaviour in these studies tend to generalize lifestyle choices without considering the cultural norms that may influence them. Factors such as daily routines, family roles, or beliefs about aging can vary significantly between populations. However, these variations are rarely addressed.

Other researchers have extended this inquiry to include the effects of climate factors such as heat waves and precipitation extremes. One study linked heat-related illness to higher dementia risk later in life (Kuo et al.). Another study found faster rates of cognitive decline in individuals exposed to unusually high levels of rainfall. (Zhang et al.). Exposure to rainfall is understood as a contributing factor to mental health issues, which are then understood as a contributing factor to dementia incidence. However, exposure is typically framed as something that happens to individuals rather than something that is influenced by their behaviours and choices. Very few studies ask how communities prepare for or respond to these conditions in ways that reflect their culture or their access to adequate resources.

Some research indirectly points toward the importance of culture, though it does not address dementia directly. A study on mask-wearing norms in East Asian countries highlights how shared values and historical experiences have shaped preventive habits in response to environmental threats (Ahmad). Wearing masks in response to pollution, illness, or dust storms is considered the norm in many of these East Asian countries, which may reduce harmful exposures over time. Similarly, research on colorism and beauty standards in Chinese communities notes that many people avoid sun exposure due to the cultural association between lighter skin tone and higher social status (Chen et al.). This has led to the widespread use of umbrellas, long-sleeved shirts, and face coverings. While these practices are not intended as preventive measures regarding health, they reveal how cultural values can influence exposure levels in ways that biomedical-focused studies often ignore.

Overall, the literature on air pollution and Alzheimer's reveals a consistent limitation. Most studies treat exposure as a universal and biologically determined variable. This particular framing overlooks the social and cultural values that influence one's behaviour and shape an individual's risk. Even studies that may include demographic breakdowns rarely go beyond listing ethnicity or region. Ultimately, the variation in culture, as it relates to memory, aging, and health preventive measures, is not fully explored.

This paper aims to respond to the gap in previous literature. It asks about the role culture plays in the relationship between environmental exposure and the incidence of Alzheimer's disease. By reviewing both health research and conducting a cultural analysis, the goal is to expand the current understanding of how air pollution affects cognitive health. This particular approach emphasizes the need to move beyond generalizations and to consider how culturally specific practices may protect or expose certain populations in ways that are not yet fully recognized.

Methods

This research focuses on seven countries that represent a range of economic development levels, geographic regions, and cultural contexts: Japan, the United States, Mexico, China, Sweden, India, and South Africa. These countries were selected to ensure both diversity and global relevance in examining the relationship between air pollution and the incidence of dementia. Each country differs in its weather, its geographical locations, and its levels of

pollution. The timeframe for data collection spans from 2010 to 2019, allowing for a longitudinal analysis of changes and patterns over the past decade. This timeframe was chosen because it looks at data across nine years, allowing for enough time to see changes in health outcomes given cultural practices and preventive measures against pollution. This is important because it would not be expected that health outcomes would show up immediately in the data once cultural practices change. The time series data also allows me to account for fixed effects by country-year to account for anomalies across the observations. To assess how cultural and environmental factors interplay to influence dementia rates, I constructed an additive measure that incorporates both pollution levels and proxies for cultural attitudes toward physical ideals and environmental responses. By integrating the cultural aspect with air quality indicators such as the PM2.5 levels, the study captures not just the environmental exposure but also how cultural contexts might mediate, reduce, or amplify its effects.

I run two models with two different outcome variables, which both capture the relationship between environmental and cultural factors and their combined influence on Alzheimer's progression rates. The first is Ambient Air Pollution, or the PM2.5 Concentration (WHO). This variable measures the annual concentration of fine particulate matter (PM2.5) in the air, and is expressed in micrograms per cubic meter. PM2.5 consists of particles with a diameter of 2.5 microns or less, which are small enough to be inhaled deeply into the lungs and are linked to long-term health risks. Some of those health risks include cardiovascular disease, respiratory illness, and neurological disorders such as dementia. Annual levels of PM2.5 are drawn from the World Health Organization (WHO) Ambient Air Quality Database in this study. This dataset provides standardized, internationally comparable air pollution data reported by governments and environmental agencies. PM2.5 levels are used as a proxy for environmental health risk and air quality trends across countries. These values are measured in micrograms per cubic meter and represent a key indicator of population exposure to harmful airborne pollutants. For this study, the levels were extracted for each country from 2010 to 2019, using the most reliable and publicly available data submitted to the WHO during this period.

The second outcome variable is the Rate of Alzheimer's progression, which measures the rate of change in Alzheimer's disease and other dementias over time. This measure is based on the age-standardized prevalence share (percent of population) from Our World in Data's "Prevalence of Alzheimer's Disease and other Dementias". The rate of Alzheimer's progression is an important variable in this study because it allows for a comparison alongside the other variables.

For explanatory variables, I capture cultural and structural factors that may mediate or amplify the relationship between air pollution and Alzheimer's progression. These factors include indicators of cultural behaviour, healthcare access, and environmental exposure.

The first is Cultural Trends (Google Search Interest). This variable captures public interest in culturally relevant behaviours that may indirectly influence individual exposure to air pollution. The purpose of including this measure is to examine how cultural practices may act as a protective or risk-enhancing factor regarding dementia incidence. To understand how culture

may shape the response to environmental risk, I used Google Trends to track public search interest in culturally relevant terms such as “hijab”, “mask”, and “skin whitening”. These terms were chosen because they reflect local values and behaviour related to sun and pollution exposure. For instance, interest in “hijab” may indirectly reflect cultural practices that would cover the mouth and nose, therefore reducing exposure to pollutants, while “skin whitening” can reflect cultural ideals that shape daily routines and sun avoidance. Search interest data was extracted from each country annually from 2010 to 2019. Each keyword’s popularity is reported as a relative score (from 0 to 100), where 100 indicates peak search interest during the time window and the opposite for lower scores. These values were used to bring in an additive index of culturally influenced behaviour related to environmental exposure. The index was created by averaging the normalized scores across the three search terms for each country year. Higher index values suggest a greater prevalence of behaviours or values that could shape exposure to air pollution.

I also capture country-specific fixed-effects, with country-year data, and healthcare accessibility to control for broader structural differences that may affect the relationship between air pollution and Alzheimer's progression. Including fixed effects helps account for country-level characteristics, such as governance, infrastructure and cultural norms. The second is Healthcare access. This variable measures the extent of free or universal health care coverage across each country in the dataset. It was constructed using data from the OECD Health Statistics: Government and Compulsory Health Insurance dataset. To scale this measure, I coded countries on a three-point scale, where a 0 means the country has no universal compulsory coverage and access depends mainly on private insurance or out-of-pocket payment, a 1 represents partial government or compulsory coverage, and 2 represents full or near-universal coverage.

2 = Yes (countries with universal or near-universal coverage, greater than or equal to 98% of the population)

1 = Partial (countries with government or compulsory coverage available to a majority of the population, but not universal or with restricted access)

0 = No (countries with no universal compulsory coverage, where access depends mainly on private insurance or out-of-pocket payment)

For this research, seven countries were coded: Japan, the United States, Mexico, China, Sweden, India, and South Africa. Based on the OECD documentation, Japan, China, and Sweden were coded as 2. The United States, Mexico, and South Africa were coded as 1, and India was coded as 0. This classification remained stable across the period 2010-2019, even though some policy reforms (e.g., the Affordable Care Act in the United States) altered coverage rates within categories, but without changing their overall coding.

Finally, the last is the annual average temperature. This variable measures the average annual surface temperature of each country. Data was extracted from the Global Data Lab Yearly Average Surface Temperature dataset, which reports national-level average temperatures in degrees Celsius. For each country, I recorded the annual average from 2010 to 2019. Using annual means provides a consistent indicator of long-term climate exposure while smoothing over daily and seasonal variation.

Results

Data from the seven countries, Japan, the United States, Mexico, China, Sweden, India, and South Africa, were collected between 2010 and 2019. The descriptive statistics for each variable are shown in Table 1. The average temperature across all countries ranged from roughly 3 °C to 23 °C, with a mean of approximately 14.6 °C. This range in the dataset reflects a balanced mix of temperate and tropical climates. PM2.5 concentrations varied substantially, averaging about 26 $\mu\text{m}/\text{m}^3$ across countries, with higher long-term exposure in rapidly industrializing nations. The Average rate of Alzheimer's progression varied by country, while the sample was approximately 1.1 percent of the population, rising over time in most countries. Overall, the descriptive data reveal considerable variation in climate, pollution exposure, and disease prevalence.

Table 1: Average Values of Control Variables by Country (2019-2019)

Country	Average Temperature	Average Mask Trends	Average Skin Whitening Trends	Average Hijab Trends	Average Free Health Care	Average Rate of Alzheimer's Progression
China	12.89	43.63	0.94	6.99	2.00	0.86
India	22.74	51.61	53.85	30.76	0.00	0.24
Japan	13.24	69.34	28.72	37.53	1.90	2.29
Mexico	20.94	67.88	11.30	26.86	1.00	0.50
South Africa	18.02	33.28	41.35	42.97	1.00	0.38
Sweden	5.74	55.85	16.00	44.08	2.00	1.70
United States	12.07	53.30	62.35	32.12	1.00	1.34

Determinants of Alzheimer's Disease Progression Rate		
	Dependent variable:	
	Rate of Alzheimer Progression	
	Reduced Model (1)	Full Model (2)
PM2.5 Rate	-0.004 (0.004)	-0.008 (0.004)
Average Temperature		-0.001 (0.022)
Free Healthcare		-0.211* (0.086)
Mask Google Trends		-0.002 (0.001)
Skin Whitening Google Trends		-0.001 (0.001)
Hijab Google Trends		0.0004 (0.001)
India	-0.599*** (0.041)	-0.947** (0.274)
Japan	1.269*** (0.142)	1.180*** (0.149)
Mexico	-0.484*** (0.112)	-0.742** (0.226)
South Africa	-0.613*** (0.119)	-0.927*** (0.197)
Sweden	0.652*** (0.164)	0.520* (0.228)
United States	0.301 (0.157)	0.007 (0.194)
Year	0.020*** (0.004)	0.028** (0.009)
Constant	-39.786*** (7.821)	-55.349** (17.368)
Observations	70	70
R ²	0.988	0.990
Adjusted R ²	0.987	0.988
Residual Std. Error	0.083 (df = 61)	0.078 (df = 56)
F Statistic	636.238*** (df = 8; 61)	442.057*** (df = 13; 56)
Note:	Standard errors in parentheses. *p < 0.05, **p < 0.01, ***p < 0.001	

Figure 1: Determinants of Alzheimer's Disease Progression Rate (OLS Models with and without Controls)

I then ran two OLS models, one with PM2.5 as the only explanatory variable to show the direct relationship between air pollution and the rate of Alzheimer's progression across countries. The second OLS model includes additional controls to account for cultural, structural, and environmental differences. These additional controls include average temperature, healthcare coverage, and cultural trend indicators to show how these factors may mediate or reduce the observed effect of pollution. Both models included country-year fixed effects.

The results indicate that while PM2.5 concentration shows a positive association with Alzheimer's progression in Model 1, the relationship weakens once the additional controls are

introduced in Model 2. This finding suggests that cultural and structural factors play a substantial role in moderating the effects of pollution on cognitive decline.

Discussion

The purpose of this study was to investigate the relationship between ambient air pollution, specifically PM_{2.5} levels, and the progression rate of Alzheimer's disease across seven countries between 2010 and 2019. The regression analysis revealed that PM_{2.5} concentrations were not statistically significant predictors of Alzheimer's progression once cultural and structural controls were introduced. This finding accounted for approximately 99% of the variation ($R^2 = 0.99$) in progression rates across the sample, suggesting that country-specific and institutional factors play a larger role in explaining the differences in disease outcomes than air pollution alone.

These findings are important because they challenge the prevailing assumption that pollution directly accelerates Alzheimer's progression across all contexts. These findings contrast existing literature, which often identifies air pollution as a contributing factor to faster cognitive decline (Peters et al, 2019; Park et al, 2024). This study's findings highlight that the strength of that relationship may depend heavily on cultural, healthcare, and diagnostic contexts. One likely explanation for this discrepancy is international variability in reporting and the diagnosis of Alzheimer's progression percentages. For instance, Japan reported the highest percentages of progression, rising from 1.9% in 2010 to 2.6% in 2019, even though its PM_{2.5} levels were relatively modest (10-14). In contrast, India and South Africa consistently reported progression percentages below 0.5%, despite much higher PM_{2.5} averages (50-60). This discrepancy suggested that the observed statistical trend may be less about pollution itself and more about the differences in health care systems and diagnosis techniques.

Other contextual factors also appear relevant. Countries with stronger healthcare access, such as Sweden and Japan, tend to report higher progression percentages compared to nations where diagnostic resources may be limited. Cultural practices may further influence whether Alzheimer's progression is formally recorded. In regions where family caregiving is the norm and institutionalized care is limited, official statistics may underestimate true progression outcomes. This finding implies that policymakers should be cautious when using national dementia data to conclude environmental health relationships. Instead, efforts to standardize diagnostic criteria and improve healthcare access could enhance the accuracy of cross-country comparisons.

Several limitations must be acknowledged. First, the study relies on secondary, country-level data, which may mask important within-country variation in both pollution exposure and dementia outcomes. Second, cultural proxies (such as Google Trends data) are indirect measures of social behaviour and may not capture the full complexity of cultural practices related to pollution avoidance or healthcare engagement. Third, Alzheimer's progression rates are dependent on reporting quality and diagnostic standards, which differ widely across countries and may introduce measurement bias.

Future studies should integrate environmental data with clinical research that directly measures Alzheimer's progression in patient populations. Complementary qualitative research, such as interviews with nurses and other frontline healthcare providers, could also clarify how progression is recognized and documented across different cultural and healthcare settings.

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A Bug, A Burden, A Son: Franz Kafka and the Literary Processing of Intergenerational Trauma-By Evelyn Yang

Abstract

A wide range of authors have written about trauma in literature, yet despite Franz Kafka's significance as a modernist writer—well known for works such as *The Metamorphosis* and *The Trial*—the connections between his works and his personal traumas remain under-examined. This paper addresses that literary gap by exploring how Kafka's use of characterization, genre, and form offers profound insight into his lived experiences with psychological and emotional trauma. Throughout his childhood and adult life, Kafka experienced an onslaught of emotional violence from his father, often in the form of manipulation and guilt. As a result, this paper argues that Kafka's portrayal of complex human relationships through innovative forms of creative expression serves as a literary method for processing his internalized feelings of shame and inferiority. The analysis of his oeuvre ranges from letters to short stories, particularly *Letter to His Father*, *The Judgement*, and *The Metamorphosis*. By examining how Kafka blends autobiographical detail with experimental literary techniques, it is evident that the therapeutic and redemptive possibilities of art and writing shape Kafka's writing. Additionally, a variety of genres offers insight into new ways of processing trauma, where types of literature such as fantasy allow Kafka to represent the often catastrophic effects of trauma. In conversation with scholars such as Cathy Caruth, Andrew Barnaby, James Pennebaker, and Aurther Scherr, this paper situates key elements of Kafka's works in broader discussions of trauma theory and literary production. Ultimately, the following question is raised: What is lost or obscured in literary analysis when the interior lives and emotional histories of writers are ignored?

Introduction

Franz Kafka is undeniably best known for his fictional works, such as *The Metamorphosis* (1915) and *The Trial* (1925), which showcase the excellence of his creative literary craft. Born to a Jewish family in Prague, Kafka worked as a lawyer while writing his legacy of works on the side. For the majority of his life, Kafka struggled with anxiety and familial issues. Together, these issues manifested in the expression of turmoil and other elements within his works. As Harry Steinhauer notes, "Kafka has captured the anxiety, suffering, turmoil, the frustrated hope and longing that constitute the human condition" (Steinhauer 391). It is from these themes of his writing that the style "kafkaesque" was coined, which continues to be prevalent in modern works today. However, despite the acclaim for many of his works, Kafka's literature is typically not studied through the lens of trauma. When looking deeper into his works, it is clear that Kafka often portrays familial relationships in parallel to the nature of his own relationships. While this may not be a new concept, as Andrew Barnaby writes, "literary trauma might be understood as just one way (or maybe a series of related ways) of grappling with the crisis of subjectivity in which the experience of finding oneself at a loss points to a

consciousness breached [...]” (Barnaby 22), Kafka, in particular, uses his writing to process, confront, and heal traumas from his personal relationships.

In particular, Kafka’s depiction of trauma-inflicted relationships acted as an approach to process his psychological and emotional trauma. The characterization of familial relationships in his works reflects Kafka’s complex perspectives on his own troubled relationships. The works of *Letter to His Father*, *The Judgement*, and *The Metamorphosis* reveal how different genres and forms of characterization provided Kafka a space to understand his experiences, as well as the inseparable nature of literature and artists with trauma. By examining Kafka’s writing through the framework of intergenerational trauma, a stronger understanding of the impact of his family on his literary legacy can be gained.

Although Kafka’s illustration of trauma is most evident in his fictional works, his contributions to literature extend beyond the form of fiction. Collections of his letters—addressed to various lovers, friends, and notably his father—have also been published, therefore, contributing to the complexity of his oeuvre. Nevertheless, Kafka’s collection of letters raises ethical concerns about whether such personal correspondence should be included as part of his legacy, challenging traditional boundaries between private writing and literary art. In *Letter to His Father*, Kafka confronts his father and recalls traumatic events from his childhood that shaped their relationship. However, the letter is not just a personal correspondence, but rather another form of art for Kafka to express the resentment and fear he held for his father. By doing so, the letter also allows Kafka to gain closure towards his unresolved childhood traumas—an example of how his writing provided a space to understand his experiences.

Although he never sends the letter to his father, Kafka’s decision to preserve the letter suggests its significance for reconciling their relationship. There is no way to know what sort of reconciliation would have occurred—perhaps Kafka wished for his father to apologize for his actions throughout Kafka’s childhood to improve their relationship. Alternatively, it is possible Kafka sought validation for his pain. Regardless, both the intention of Kafka to process his trauma through epistolary form and his decision not to send the letter reveal that his letters serve as a unique form of emotionally charged literary art. Furthermore, it is an example of how artists often use their craft as a way to express and process their experiences. In Kafka’s case, he centers *Letter to His Father* around his unresolved childhood traumas. His letter reveals how the impact of trauma can cause artists to express themselves in different ways.

While the meaning behind Kafka’s letter has the explicit intention of confronting his father, it also shares a commonality with many of his other artworks. This suggests that there is an underlying intention of processing complex relationships and their power dynamics. According to Steinhauer, “In his fiction he set up various models, imagined situations that suggest different ways of reading the riddle of life. His models are described in such general terms that they admit of the most diverse, antithetical interpretation” (Steinhauer 395). Kafka, across different forms and genres, ultimately returned to the varied and recursive effects of his trauma, including in his letters. This indicates that Kafka’s epistolary form of art found in his personal letters should be studied in accordance with his fictional works. To gain a deeper

understanding of Kafka and his oeuvre, it is essential to look past what is explicitly written and consider the complex implications of the epistolary form.

Kafka's trauma regarding his father and family must be considered in relation to the relationships he constructs in his literary works. For the majority of his life, Kafka's relationship with his father, Hermann Kafka, was marked by a commanding and imbalanced power dynamic. Hermann's actions as a father and his "approach made his son feel that he had little to offer others and that even when he experienced success, he did not deserve it" (Hallaç 190). The shame Hermann enforced is evident in *Letter to His Father*, after Kafka describes his father's efforts to support their family: "I could enjoy what you gave, but only in humiliation, weariness, weakness, and a sense of guilt" (Kafka Letter 135). Hermann's insistence that his struggles and work were above his son's implies Kafka's inferiority to and obligation to him. In this way, Kafka's individual experience of psychological and emotional trauma was formed. According to Giulio Perrotta, psychological trauma is events which "[generate] impotence and vulnerability, capable of causing such severe stress to threaten the integrity of the person's psychophysical balance [...]" (Perrotta 1). Kafka's environment, filled with harsh influences and a lack of care, caused stress and impacted his understanding and interaction with the world around him. His father's approach and cause of psychological trauma continued well into Kafka's adult life, where he experienced "indecision and guilt, [and a] chronic tendency to blame himself [...]" (Hallaç 190). These deeply internalized patterns of shame and guilt become the emotional and psychological framework through which Kafka constructed his fictional relationships.

Kafka's choice of profession further strained this unstable family dynamic; as Douglas Youvan notes, "the role of the intellectual or artist often stood in stark contrast to the role of a dutiful family member, and this tension alienated many writers from their familial circles" (Youvan 5). Hermann, a businessman, could neither understand nor approve of Kafka's choice to be a writer, which only intensified his disappointment and disapproval of Kafka. His resentment likely continued to be a part of his traumatic behavior toward his son into adulthood, where Kafka repeatedly felt judgment from his father. As a result, Kafka "fled everything that even remotely reminded [him] of [his father]," seeking a separation from the cause of his humiliation and psychological trauma (135). As Kafka describes in the *Letter to His Father*, he also partly broke away from his father's business because of the shameful way Hermann treated his employees. He writes, "you were so greatly superior to all those who ever came to learn the business from you that nothing they ever did could satisfy you, and you must, [...] in the same way be forever with me too" (Kafka Letter 137). The commanding nature of his father was inescapable even in a completely different setting, and Kafka evidently continuously experienced psychological and emotional trauma in different aspects of his life as a result of his father's actions. The traumatic experiences of Kafka's upbringing impacted him so much that circumstances similar to his own continuously appear in a plethora of his works.

Kafka's tenuous family relationships significantly shape his construction of certain connections, such as those between a father and son. In his depiction of familial relationships, Kafka seemingly connects the characters in his narratives to his own family through their role in

the family structure: the commanding and vindictive father, the obedient mother, and the disappointing and cowardly son. This is evident across many of his works, from *The Judgement* to *The Metamorphosis*, where his male protagonists often find themselves in distressing situations that circle back to their fathers. In *The Judgement*, Georg Bendemann's father accuses him of not caring for and appreciating him as a father, eventually ordering him to commit suicide. Similarly, in *The Metamorphosis*, when Gregor Samsa experiences metamorphosis, his father does not treat him with sympathy; in fact, the two times he interacts directly with Gregor, he inflicts violence upon him. These father-son relationships seem to be characterized by his own experiences. For example, in *Letter to His Father*, Kafka writes, "You had (unconsciously) always suppressed my power of decision and now believe (unconsciously) that you knew what it was worth" (160).

Similarly, in *The Judgement*, it is the protagonist's decision to inform his friend of his engagement that causes his father to shame the protagonist. Kafka himself identifies the shame many of his characters experience due to their fathers; he writes that the results of his father's actions were "[...] the weakness, the lack of self-confidence, the sense of guilt [...]" in himself (155). Guilt, in particular, is a manifestation of Kafka's trauma. Although it is an emotion that everyone can feel, Kafka seems to be tormented by it, often in regard to his father or his life. In this sense, Kafka's trauma and guilt are so severe that they affect his actions. Kafka voices guilt as a reason behind his inability to marry: "for my sense of guilt actually originates in you, and is filled with such conviction of its uniqueness—indeed, this feeling of uniqueness is an essential part of its tormenting nature [...]" (163). As Kafka writes, guilt impedes his ability to live fully or pursue marriage, in this instance. When looking closer, his guilt reveals the significance of considering the specific aspects of trauma that marked his life. In turn, it is important to closely inspect the multitude of traumas present in his writing, often through relationships and their impact on characters. Although not all of his works parallel the circumstances of his familial trauma, Kafka makes specific connections from his trauma to his fictional relationships.

Kafka's connections to his works prompt the larger question of whether his literary choices concerning his characters' familial trauma are a larger pattern in literature at the time of his writing. Early literary scholars have argued that an artist's personal experiences cannot be connected to their artwork (Wimsatt 468). Nevertheless, many of Kafka's works can be better understood when considering his personal traumas. Without experiencing familial trauma, Kafka may not have written a plethora of works exploring emotionally traumatic father-son relationships. In addition, during this period of his life, the traumatic nature of familial relationships may have been a standard. Kafka unquestionably was not the only person who experienced trauma from his family; many of the ideologies his parents and other families held concerning occupation, marriage, etc., were significant sources of his internal conflicts, and possibly for other children.

Kafka's writing pushes a narrative that breaks the normalization of abusive and traumatic familial relationships, drawing attention to the harmful actions that stemmed from the beliefs parents held. This is seen when Kafka's own father normalizes his harmful actions, as he writes,

"you charge me with it in such a way as to make it seem my fault [...] while you aren't in the slightest to blame, unless it be for having been too good to me" (Kafka Letter 116). Although it is clear that Kafka is resisting psychological manipulation from his father, he is also resisting the standard which enabled his father to be so psychologically abusive in the first place. It is revealed through Kafka's writing that his experiences are authentic, and the argument that artists must be detached from an expression of their trauma may be an inadvertent form of victim-blaming.

No matter what type, trauma leaves a significant impact on people, and for writers in particular, as shown by Kafka, their work can be a form of psychological processing. Cathy Caruth points out that "Trauma, as an experience, as a missed experience, causes you to ask, precisely, why it is that all the frameworks you have previously used, all the models, aren't adequate to describe this experience" (qtd. in Brochard and Tam 51). Coined as "literary trauma," his unconscious or conscious decision to write about his trauma "might be understood as just one way (or maybe a series of related ways) of grappling with the crisis of subjectivity in which the experience of finding oneself at a loss points to a consciousness breached [...]" (Barnaby 22). All of this helps explain why Kafka chose to grapple with his traumatic experiences by turning to writing. However, this concept can be applied to artists beyond writing, who may only be able to process and find closure through creative expression truly. Therefore, disconnection between an artist and their art, especially concerning the long-term consequences of trauma, is not possible. Through writing, Kafka confronted his emotional and psychological intergenerational trauma, using distinct characterizations of familial relationships to represent his own.

Kafka's Life and Personal Traumas

Kafka's mother, Julie Kafka, seemed to embody the stereotype of an obedient wife and mother mostly, never objecting to his father's actions toward him (Hallaç 192). In essence, Kafka's mother's quiet compliance with his father's authoritarian behavior contributed to his childhood traumas. In *Letter to His Father*, Kafka states that "Mother unconsciously played the part of a beater during a hunt," underscoring her complicity in his father's authority and the trauma Kafka endured while growing up (Kafka Letter 132). Through his mother's turning a blind eye to his father's actions, Kafka experienced psychological harm from her implied justification. As Arthur Scherr wrote, "Unable to rely on his mother for *support*, [Kafka] *felt threatened and isolated, perceiving that unconsciously* she did not care enough about his well-being to protect him against his father's outbursts [...]" (Scherr 262). When Kafka's own mother seemed to agree with his father's actions, her complicity suggested that she did not care for him either. Kafka consequently suffered from a lack of care from both of his parents. Although Kafka's father may have been the primary source of the trauma Kafka experienced, his mother's inaction undeniably compounded its effects.

Likely mimicking his own mother, the role of mothers within Kafka's works is similar—they do not offer care or support to their children, even in distressing situations. In *The*

Metamorphosis, the protagonist's mother and father are unable to face the protagonist after he turns into a bug, providing no refuge for him after his unexpected transformation. However, the protagonist's mother attempts to care for him throughout the novel, and notably saves him from death, "with her hands clasped around his father's neck as she begged for Gregor's life" (Kafka *Metamorphosis* 91). However, ultimately, at the end, his mother takes part in the emotional rejection which drives the protagonist to his death. In *The Judgement*, the protagonist's deceased mother mirrors Kafka's mother's passiveness; her absence in the story suggests that even had she been alive, she similarly would have remained complicit in the father's harmful actions. Even when Kafka's mother was present in his life, she did not protect him from his father and even inadvertently encouraged his behavior. For Kafka, who constantly experienced distressing conditions, his mother's emotional vacancy equated to her absence in his life. Both mother figures in *The Metamorphosis* and *The Judgement* represent the different aspects of Kafka's mother: her lack of care and contradicting maternal instinct, but also complicity, which seemingly is the most harmful for Kafka and his characters.

Kafka's experience and trauma from familial relationships deeply affected his ability to form and maintain strong personal relationships outside of the family, creating a pattern of emotional distance and self-doubt. Throughout his life, Kafka was engaged multiple times, but they were always eventually broken off. Kafka's mother "had subtly convinced him that he was too unstable to have a mature physical and emotional relationship with a woman" (266). His unhealthy relationship with his family affected his perception of relationships, leading to his inability to sustain healthy relationships. In particular, he was engaged twice to a woman named Felice Bauer, and they interacted primarily through letters for five years, from 1912 to 1917. According to Keith Cushman, the time Kafka spent corresponding with Felice "was also a time of great creative breakthrough: '*The Trial*, '*The Metamorphosis*,' and '*The Judgment*' all originated during these years, and '*The Judgment*' is directly related to [their] relationship" (Cushman 187). However, many of those works centered around familial relationships, except for *The Judgement* involving an engagement. Kafka, although an adult and in a growing romantic relationship, was still writing works that reflected intergenerational trauma. The centering of familial relationships reflects how Kafka was inhibited by his trauma and was not able to grow in other relationships as a result. This was likely due to the combination of trauma he experienced as a child and the continued presence of his parents throughout adulthood, continuing their harmful actions toward him (Scherr 262). Kafka's lifelong exposure to psychological trauma, through his familial relationships, undoubtedly contributed to his inability to sustain stable relationships across his personal life.

Representing Trauma Through Characterization

Generally, throughout many novels, characterization often holds a significant influence on the dynamics and complexity of interpersonal relationships. For Kafka's works in particular, his characterization dictates how characters interact with each other and the orientation of power structures within relationships. As discussed earlier, Kafka tends to characterize father figures

and sons with contrasting characteristics, such as domination and obedience, or superiority and shame. In *Letter to His Father*, Kafka writes to his father, "I had lost my self-confidence where you were concerned, and in its place had developed a boundless sense of guilt" (Kafka Letter 144-5). To reiterate a previous point, guilt is a significant aspect of Kafka's trauma. In particular, the familial structures and dynamics Kafka incorporates into his work further emphasize the lingering impact of trauma. Thus, the reason the sons in his works become polar opposites of their fathers is due to their inability to rebel without experiencing further psychological or emotional harm. In this way, whether intentional or not, Kafka processes his traumatic experiences with his father through works involving oppressive fathers and shameful sons.

In *The Judgement*, for instance, the protagonist, Georg Bendemann, is characterized as uncertain of his choices and fearful of his father, mirroring Kafka's own unhealthy relationship with his father. When Georg's father accuses him of failing to take care of their family,

Georg shrank into a corner, as far away from his father as possible. A long time ago, he had firmly made up his mind to watch everything with the greatest attention so that he would not be surprised by any indirect attack, a pounce from behind or above (Kafka Judgement 13).

Gregor's instinctive reaction to flee from his father suggests previous experiences with his father's anger and unpredictability. This passage also implies that he had grown accustomed to "indirect attack[s]," likely from recurring events during his youth. Georg's father exhibits harsh and irrational behavior toward his son, particularly when he quickly accuses Georg of trying to suppress him, even in what would typically be the caring act of helping him into bed. Convinced in himself, Georg's father says, "But fortunately a father doesn't need to be taught how to see through his own son" (12). Georg's father uses his authoritative familial position to rationalize his behavior toward Georg, wielding their father-son relationship to his advantage.

Georg's father adopts the role of a god, dictating to himself to be the only one who can see through Georg, and ultimately the only one who has power over him. His later statement, "'Do you think I didn't love you, I, from whose loins you sprang?,'" is an example of the emotional manipulation that seems to haunt Georg, as his father attempts to guilt him into believing that he scorns him out of fatherly love (14). Moreover, despite his father's uncaring and manipulative actions toward him, Georg does not seem to resent, but rather fear his father. Perhaps it is out of fear and obedience that he follows his father's command to take his own life through drowning.

In this emotionally charged instance, Kafka displays the extremities of an unhealthy father-son relationship. Kafka's characterization of literary characters and their relationship reveals the normalization of cruel treatment due to emotionally traumatic experiences. Not only does it give a voice to the perspectives of sons who experienced trauma under their fathers, but it also suggests how Kafka viewed his own relationship with his father long after. In *Letter to His Father*, Kafka describes his father's reaction to his engagement with Felice Bauer; he accuses Kafka of being seduced by Felice Bauer and shames him. Kafka writes, "You have hardly ever

humiliated me more deeply with words and shown me your contempt more clearly,” just as Georg’s own father shames him in *The Judgement* (Kafka Letter 160). *The Judgement* is dedicated to Felice Bauer, perhaps due to the inspiration Kafka drew from his own father’s reaction to their engagement. Although it is unknown if Kafka would’ve driven himself to such lengths as Georg, the culmination of the manipulation, guilt tripping, and cruelty Kafka experienced with his father suggests this alternative reality.

In *The Metamorphosis*, the protagonist, Gregor Samsa, experiences a transformation from human to insect—symbolizing the alienation Kafka experienced from his family and his feelings of inferiority. Although Gregor’s relationship to his family is unclear prior to his change, it is evident that his transformation causes his familial relationships to change drastically. Gregor becomes ostracized and mistreated which ultimately drives him to his death. A clear parallel emerges between Gregor and Kafka in their shared sense of inferiority, shaped by oppressive familial dynamics and an inability to meet their parents’ expectations.

Kafka’s decision for the novel to take on a fantastical and absurdist style, unlike his other works, also reflects his openness to exploring his trauma through the lens of different genres. Although it is not certain why Kafka chose to do so, it may have been because of what the genre of fantasy offers for exploring familial relationships, alienation, and neglect in the novel. As Canavan and Wald write, “[Speculative Fiction] conspicuously takes such estrangement as its central change: world making, after all, necessarily implies a form of world breaking—or at least introspection” (Canavan and Wald 241). The “central change” in this scenario is evidently Gregor’s physiological metamorphosis, which then sets off a series of events that lead to his downfall.

Gregor’s becoming an insect creates a clear distinction between himself and his family. He is an outsider, shut inside his room and unable to act as he once did as a human. A possibility of what Kafka wanted to imply with *The Metamorphosis* is his own alienation from his family, where even if he did not turn into a non-human form, perhaps his family viewed him as or on the same level as one. The emotional neglect that Gregor experiences throughout the novel likely reflects what Kafka experienced from his family, and particularly his father. Undeniably, his family plays a significant role in their inability to accept Gregor as a bug and all that follows, which causes his death.

Gregor’s father also inflicts physical violence in addition to the emotional trauma Gregor experiences—it is unclear if Kafka’s father was violent in addition to his psychologically traumatic actions—and not only does Kafka further explore mental trauma, he also points out the complexities of trauma and its different forms by including physical abuse. For Gregor in particular, he faces rejection and shortly after his transformation, violence from his father, forcing him into isolation:

Pitilessly, Gregor’s father drove him back, making hissing sounds like a savage [...] from behind his father gave him a strong push, which was literally a deliverance, and he flew far into the room, bleeding violently. The door was

slammed behind him with the stick, and then at last there was silence (Kafka *Metamorphosis* 71).

While suddenly becoming a bug is a jarring concept, instead of following a parental instinct, such as worrying about and caring for his son's well-being, Mr. Samsa hurts and alienates Gregor. Similar to *The Judgement*, the father figure lacks compassion for his son and instead humiliates him. When Gregor's father makes "hissing sounds like a savage," he strips Gregor of any remaining traces of humanity left in him, only able to see him as a bug. Even when Gregor is hurt—"bleeding violently"—he only cares about getting rid of his presence, and even hurts him in the process. The violence continues when Gregor escapes his room and causes his mother to faint in horror; his father responds by throwing apples at him. Gregor becomes injured when an apple "landed right on his back and got stuck in it; Gregor wanted to drag himself forward, as if this startling, incredible pain would disappear if he moved to a different spot [...]" (91). Although Gregor undeniably experiences physical pain from his injury, we can speculate that his emotional pain is greater.

Not only has his own mother fainted at the sight of him, but his own father has taken to violence in anger once again. By wanting to "drag himself forward," Gregor wishes to distance himself from the cause of his physical and emotional pain. The violent reactions of Mr. Samsa to Gregor on separate occasions reveal a pattern about the turbulent state of their relationship prior to his transformation. If Mr. Samsa's immediate instinct is to harm his child, it implies a lack of care and humanity in his perception of Gregor. It may also reflect Mr. Samsa's view of Gregor as only a bug, justifying his continuous violence.

However, Mr. Samsa is not the only one who contributes to the eventual downfall of Gregor; similarly to his own life, Kafka connects the trauma he experienced aside from his father to the novel by depicting Gregor's sister, Grete, alienating him and further reinforcing Gregor's inferiority. Although Gregor suffers primarily from physical violence, his sister's actions of severance emotionally harm him. This is clear at the end of the novel when Grete alienates Gregor to convince their father to get rid of him:

'You must try to get rid of the idea that this is Gregor. The fact that we've believed it for so long is the root of misfortune. How can it be Gregor? If this were Gregor, he would have realized long ago that human beings can't live with such a creature [...]' (104).

Although Grete is initially tasked with caring for Gregor, it is evident that she eventually comes to resent him, no longer seeing him as her brother. She names him as the "root of misfortune," further dehumanizing him to be their family's enemy, no longer a member. Once again, it is Gregor's transformation that allows her to do this; if he were a human, she would continue to consider him her brother. Grete's hostility suggests that the familial love she appeared to have does not extend past species or her expectations for her brother. In this way, it is clear

that Grete's love is conditional, and she is similar, if not the same, to Gregor's father. By excluding Gregor from his humanity, Grete justifies their father's violent actions and seemingly encourages them.

Grete's separation of Gregor as her brother is similar to how Kafka's relationship with his parents ended. Kafka is likely expressing the mindset of his parents after expressing their disappointment in him, ultimately disconnecting him from the role of being their son. The concept is central to Kafka's trauma: his upbringing supposedly had the intention for him to be an ideal son, yet it was his upbringing that caused him to become the opposite. Once Kafka lost his importance—just as Gregor lost his own by no longer being the sole provider for his family—his parents could no longer see him as their son. As a result, both Kafka and Gregor become burdens. Kafka was a failure, while Gregor not only loses his significance, but also needs a new level of care and attention. In this way, Gregor's transformation and what occurs as a result can also be considered Kafka's metaphorical interpretation of his relationship with his parents. The metamorphosis represents their reaction to Kafka becoming a writer; by defying their expectations, he is turned into a bug, and his parents can no longer see the son they thought to be their son.

The inability of Kafka's parents to accept his writing, or Gregor's nonhuman transformation, alienates him from his parents. Although Gregor dies from the isolation and neglect, Kafka does not—physically, at least. Nevertheless, a lack of support for his profession, in addition to his preconceived psychological and emotional trauma, likely worsened Kafka's mental state regarding his trauma. In this way, he emotionally died like Gregor, unable to handle his guilt of disappointing his family. He felt like a bug, just as Gregor became one. It is undeniable that *The Metamorphosis* reveals Kafka's feelings of guilt and insignificance, specifically regarding his family, through the metaphor of an insect, which compares Kafka's and Gregor's lives to those of an insect.

Despite Gregor's care for his family, Grete undermines it to justify their harmful actions of alienating and physically abusing him. She claims that he would understand that "human beings can't live with such a creature," a parallel to Kafka's life, where his relationships, marked by trauma, caused tension and self-isolation. However, even though Gregor metamorphoses into an insect, he proves Grete's claim incorrect as he continues to attempt to support them. One of Gregor's first concerns after his family learns of his transformation is supporting them, thinking "the whole future of Gregor and his family depended" on his job (68). Amid all the confusion of his transformation, Gregor can only think of his family, yet he is not met with the same care. Kafka, although aware of the trauma inflicted by his father, tried to repair their relationship. These actions were an effort to prove that he was not an inferior being. In this way, Kafka's persona and literary characters are about themselves above their hardships to care for their families, a clear sign of humanity. Even when his own father reacted negatively to his decisions and actions, Kafka persisted in doing acts of care for their relationship that mirrors his choice to write *Letter to His Father*. The acts of Kafka and Gregor, the protagonist of *The Metamorphosis*,

are selfless, yet neither is met with a positive outcome. Gregor eventually dies from neglect and isolation, and Kafka continuously experiences and relives his trauma from his family.

Kafka's portrayal of Gregor's perseverance when confronting familial trauma not only underscores his emotional resilience but also highlights how shifting literary genres—as represented by his turn to fantasy—allowed him to explore trauma in innovative ways. Kafka's turn to fantasy reveals that when it comes to works involving trauma, genre is irrelevant. Trauma is not a uniform experience, and those who choose to process their trauma through their art, like Kafka, may have varying ways of doing so. Restricting trauma to one genre may have hurt Kafka, as he would have been unable to understand his experiences in different contexts. Kafka's choice of genre in fantasy speaks to his use of writing to process his trauma, and it is an effective method for him to move past his experiences truly.

Literary Implications of Depicting Intergenerational Trauma

Although Kafka's own experiences are clear within the emotional and psychological trauma of his characters, some scholars may argue that art should be understood independently of the creator's personal context and background. This claim lacks scrutiny when pitted against the strong relationship between Kafka's trauma and literary works. It is possible for art to be understood in a different context—as everyone can have different interpretations. At the same time, considering the intention of artists when creating is vital. Not only do people's intentions help guide a general understanding of art, but for some artists, the art they create can also help them process their experiences, specifically traumatic ones. When explaining the effects of writing about traumatic experiences, James Pennebaker writes, “somehow in the process of writing down their deepest thoughts and feelings about the [traumatic] experience, people get to organize the experience in a very emotional way” (qtd. in King & Holden 359). Rather than ignoring or struggling with the trauma he endured, Kafka was able to ultimately understand and process his experiences through writing in both the epistolary and fictional form.

Considering the repressiveness of Kafka's environment from his father and his inability to confront him, it is likely that Kafka was not given much opportunity to communicate to others what he experienced directly. Consequently, it is possible that his traumatic experiences were repressed until he first started writing about them. Writing gave Kafka a space where he was free from the judgment and scorn of his father, where he could safely process his experiences and tumultuous relationships. In specific regard to works centered around familial relationships, Kafka's history of intergenerational trauma within his own family cannot be detached. It is because of his experiences and his trauma that Kafka was able to write so many emotionally riveting stories, taking inspiration from his own life. By acknowledging Kafka's history of intergenerational trauma, specifically regarding his father's oppressiveness, his works can be better interpreted. For instance, if Kafka's writing in epistolary forms is considered to be art, then it is essential for his relationships with those he is writing to be clarified. Without it, the reader may be confused about how or why Kafka writes to them in different tones. Factors for writing to a specific audience can include resentment or affection, due to the nature of their relationship

with them. When addressing his parents, it would be understandable for him to be more respectful toward his parents than toward his fiancée, due to the nature of the respective relationships.

However, in *Letter to His Father*, Kafka's resentment toward his father may initially seem out of place. Unless one considers the troubled histories between Kafka and his father, there is no good reason for him to be so bitter toward his father from the outset. In the first line of the letter, Kafka writes to his father, "You asked me recently why I maintain that I am afraid of you. As usual, I was unable to think of any answer to your question [...]" (Kafka Letter 115). Although his feeling of inferiority may be inferred without prior knowledge of his relationship with his father, Kafka's emotion is more visceral when his turbulent connection to his father is understood. Further into the letter, Kafka writes about his father's reaction to disapproving his business, "You then tried [...] to extract, nevertheless, some little sweetness for yourself from my dislike of the business, of your work— a dislike that was after all very distressing to you—by asserting that I had no business sense [...]" (138). His father's reaction may appear normal, but if his relationship with Kafka and the rest of his family is known, it can instead be read as manipulative. To have "some little sweetness," his father convinces himself and Kafka that he is not suited for the work, rather than accepting that Kafka dislikes it. This instance gives insight into their relationship as well: as soon as his father loses his control over Kafka, he attempts to take it back by putting him down.

However, this is not the only moment of manipulation Kafka mentions in the letter. When Kafka expresses his desire to distance himself from their family, he says that his father "believe[s] I do everything for other people out of affection and loyalty, and for [him] and the family nothing, out of coldness and betrayal" (144). Once again, as Kafka realizes that his familial experiences are harmful, his father attempts to manipulate his son through guilt. His father continues to rid himself of blame and instead displaces it on Kafka, accusing him of betraying and abandoning his family, when they have done the same to him. Understanding his father and the role his actions played in Kafka's development is essential; without it, both Kafka himself and his work can be misinterpreted. The unhealthy relationships of characters and the trauma they experience can be normalized due to the nature of relationships at the time. Not only does this completely disregard the blatant trauma characters experience, but it also disregards Kafka's trauma by relation. He processed and wrote about his personal experiences regarding his trauma in his works, and it is Kafka who is ultimately being misunderstood.

Fundamentally, Kafka's only means of confronting his father is through a form of art, his writing. Arthur Scherr notes that "writing had been Kafka's way of 'coming to terms' with despair," which likely included the trauma he experienced as a child (Scherr 273). Although he never gives the letter to his father, he is still able to process parts of his trauma and finally address his father in a way he could not in life. As James Pennebaker says, "just being able to put together a coherent and meaningful story about the trauma is therapeutic if there is a caring or interested person to read it" (qtd. in King and Holden 359). Although Kafka's father may not have been a "caring or interested person" to read Kafka's letter, it was still done with the intent to

address his father directly concerning the traumatic events he experienced. The letter itself gives context to the rest of Kafka's writing and the true motivation behind creating works like *The Metamorphosis* and *The Judgement*.

Many of Kafka's other works are centered around familial relationships; the trauma he experiences holds such an impact on him that he not only writes a letter to his father, but also centers his trauma in his other works. It is not always about closure, either; in *The Judgement*, the protagonist dies at the command of his father. Kafka may have also been reconciling with the possibility that his relationship with his father would never be repaired. Those who experience trauma due to another's actions may not want to reconcile with them or confront them directly. Perhaps Kafka does both through works like *Letter to His Father*, where he expresses the desire to reconcile with his father, but seems to understand that it would be nearly impossible to, due to his father's nature. He indirectly confronts his father and possibly accepts that their relationship is irreparable. In addition, it is feasible that Kafka's art was able to provide a different type of closure: that Kafka's relationship with his father and his trauma was not caused by himself. His works showcase different characters with similar father-son relationships, yet in each one, a pattern of unhealthy family dynamics persists. Ultimately, the way the sons are raised under their fathers, to be docile and shameful, is what impacts their lives and causes further trauma as adults. Kafka realized, eventually, as written in his works, that it was the actions of fathers that caused the turmoil in the relationships they had with their sons.

Conclusion

Although Kafka may not often be considered as an author who depicts intergenerational trauma in his works, it is crucial to understand the role it played in both his personal and professional writing. The trauma Kafka experienced as a child, and throughout the majority of his adult life, manifested in the characterization of his works, attributing similar qualities and patterns to himself and his family. Kafka's choice to include psychological abuse in his literary works not only reflects his inability to move past his trauma but also the desire to work through it by writing. His art was the one place his parents could not or did not want to understand, leaving Kafka a space where he could express himself without experiencing belittlement or shame from his parents. As a result, many of his traumatic experiences followed, causing him to turn to both epistolary and creative writing as ways to process his experiences.

In many of the works explored in this essay, namely, works centered around familial relationships, the protagonists, the sons, often do not have stereotypical happy endings. Ideal happy endings could vary: perhaps the family reconciles with the son, or perhaps the son can completely detach himself from his family and trauma. However, neither possibilities reflect Kafka's reality. His epistolary and fiction writing suggests that he never truly processed his trauma, even if writing provided an outlet. It was the actions of Kafka's family to try to compel him to become what they wanted as an ideal son, one who behaved as they wanted and could control, that caused Kafka to become so psychologically damaged. Their methods of upbringing made it impossible for their ideal son to become a reality.

Ultimately, a lack of happy endings reveals a world in which characters are unable to grow and thrive because of the trauma they experience, which is perhaps a reflection of how Kafka understood his own life. Kafka saw himself as someone trapped by not only his trauma, but also the continuous presence of the ones who caused it. However, this does not mean he did not make an effort to move past his parents' actions and build a life for himself. Kafka wrote a letter to his father, confronting him about the actions that shaped his childhood. He continued writing despite his family's disapproval. Furthermore, a less pronounced effect of Kafka's trauma is that many of those attempts to make amends with his family did not succeed. For example, his letter was never delivered to his father, and many of his engagements failed. The fear and obedience he learned from his father's actions often restrained him, and his inability to have healthy and stable relationships stemmed from his trauma. However, in spite of these 'failures,' Kafka's choice to keep writing is an act of resistance against his trauma and his family. Instead of looking for explicit reconciliation, Kafka's decision to turn inward through writing speaks to the importance of processing and healing from within. There is no correct way to work through trauma, but different methods may be more powerful than others. Kafka, who understood the ones who caused his trauma, knew the best way was to gain closure himself, rather than confronting those who continually failed him.

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Constructing Legitimacy After Usurpation: Statecraft, Culture, and Power under the Yongle Emperor By Chenzhe Huang

Introduction

Few rulers in Chinese history have been judged as ambivalently as the Yongle Emperor (r. 1402–1424). To later generations, he stands as one of the most formidable sovereigns of the Ming dynasty—a military strategist, institutional architect, and patron of ambitious cultural and political projects. Yet many of his contemporaries saw his accession to the throne as troubling and controversial. Yongle seized power from his nephew, the Jianwen Emperor, in a brutal three-year civil war, directly contravening the dynastic and moral injunctions laid down by his father, the Hongwu Emperor, founder of the Ming dynasty. Confucian political thought valued filial obedience, hierarchical order within the imperial clan, and loyalty to the reigning sovereign. By rebelling against his nephew, Yongle appeared to violate not only political norms but also ancestral authority.

The legitimacy crisis he faced was therefore profound. Although Yongle justified his rebellion as a necessary intervention against corrupt officials who had misled the young Jianwen Emperor, such reasoning could not fully erase the perception that he had usurped power from a divinely sanctioned ruler. The moral weight of Confucian expectations was a real political force in Chinese political culture at the time.¹ Yongle's challenge was not simply to rule effectively, but to transform a contested seizure of power into rightful imperial authority.

This paper contends that the Yongle Emperor's political philosophy relied on the strategic use of law, scholarship, and symbolic state-building projects to convert military victory into moral legitimacy. Through bureaucratic reorganization, monumental acts of imperial construction, and the politicization of cultural mastery, Yongle reshaped the political imagination of his dynasty. His reign demonstrates how political philosophy in practice could fuse strength and moral order, transforming personal ambition into a durable model of imperial rule that shaped Ming governance for generations.

Historical Background

The world that shaped the Yongle Emperor was forged by his father, the Hongwu Emperor (r. 1368–1398), founder of the Ming dynasty. Hongwu's rise to power came out of the chaos of the collapsing Mongol Yuan dynasty. Born into poverty and having lost his parents amid social unrest and famine under the chaos of late Yuan rule, he joined one of several rebel armies seeking to overthrow Mongol authority. He eventually defeated rival rebel leaders, entered Nanjing, and established the Ming dynasty in 1368.

Hongwu's volatile early career profoundly influenced the political culture he created. Having witnessed corruption and administrative collapse under Yuan, he developed an intense suspicion of officials and a determination to prevent bureaucratic abuse. His anti-corruption

¹ Hok-lam Chan, "Legitimizing Usurpation: Historical Revisions under the Ming Yongle Emperor (r. 1402-1424)." In Philip Yuen-Sang Leung (ed.), *The Legitimation of New Orders: Case Studies in World History* (Hong Kong: Chinese University Press, 2007): 75-130.

campaigns were famously severe. Large numbers of officials were investigated, imprisoned, or executed. The Mingshi records episodes in which accused administrators continued to perform their duties in chains before ultimately being put to death.² Hongwu established secret investigative organs to root out graft and pursued wide-ranging inquiries that often implicated associates and networks of officials.³ The severity of these purges at times left the state struggling to staff its own bureaucracy.

Yet Hongwu did not implement a uniform policy mandating execution for all those connected to corrupt figures. He instead combined harsh enforcement with attempts to maintain agrarian stability. Official salaries were so modest that they were often paid in grain and silver. The emperor sought to prevent exploitation of peasants by regulating quotas and attempting to control excessive local extraction. At the same time, Hongwu's economic vision prioritized agriculture over commerce. Merchants were viewed with suspicion, and commercial expansion was tightly constrained. Hongwu's agrarian and anti-bureaucratic focus was pronounced enough that his most famous communist biographer even labeled him a "peasant revolutionary."⁴ Nevertheless, military strength, rather than mercantile dynamism, was seen as the primary guarantor of state security and prosperity.

Zhu Di, the future Yongle Emperor, was born into this militarized and highly disciplined political environment. As the fourth son of Hongwu, he did not grow up as the presumptive heir. His childhood was shaped not by courtly luxury but by campaigns and frontier administration. He gained direct military experience under prominent Ming generals such as Xu Da, participating in northern expeditions against Mongol forces. By 1390 he had earned a reputation as a capable commander, and he was entrusted with guarding Beiping (modern Beijing), a strategic northern stronghold.⁵

Yongle was not the intended successor. Hongwu designated his eldest legitimate son, Zhu Biao, as crown prince. Zhu Biao held significant administrative authority and was widely respected. But his death in 1392 destabilized the succession order. According to dynastic rules, the throne should pass through the legitimate line, yet Hongwu ultimately chose Zhu Biao's son, Zhu Yunwen, as heir. When Hongwu died in 1398, the young Zhu Yunwen ascended as the Jianwen Emperor. Almost immediately, the new emperor moved to curtail the power of his princely uncles, whose regional military forces posed potential threats. Several princes were stripped of authority, banished, or demoted. Zhu Di, governing from Beiping, became a primary target of suspicion. Facing the gradual erosion of his military command, he reportedly feigned madness to deflect scrutiny while preparing resistance.

In 1399 Zhu Di launched what he framed as a righteous campaign to remove "treacherous officials" misleading the emperor—a justification grounded in a clause of the Huang-Ming

² *History of the Ming Dynasty*. <https://ctext.org/wiki.pl?if=gb&res=410835&remap=gb>.

³ Charles O. Hucker, *The Ming Dynasty: its Origins and Evolving Institutions* (Ann Arbor, Mich.: Michigan Monographs in Chinese Studies, 2020): 46-47.

⁴ Han Wu, *Zhu Yuanzhang: Founder of the Ming Dynasty*. Translated by Frederick W. Mote (New York: Columbia University Press, 1976).

⁵ Shih-shan Henry Tsai, *Perpetual Happiness: the Ming Emperor Yongle* (Seattle: University of Washington Press, 2002).

Zuxun (Ancestral Instructions of the Ming) permitting princely intervention under certain circumstances. The ensuing three-year civil war, known as the Jingnan Campaign, culminated in Zhu Di's victory in 1402. The Jianwen Emperor disappeared from the historical record, and Zhu Di assumed the throne, adopting the reign title Yongle ("Perpetual Happiness").⁶

Thus, when Yongle came to power at age forty-two, he inherited not only a vast empire but also the structural tensions bequeathed to him by his father: suspicion of officials and princes, strict legal discipline, a focus on agriculture, and ambivalence concerning commerce. His reign would be shaped by the need to stabilize these forces, resolve the contradictions of the princely system, and convert a violent seizure of power into a durable political order.

Centralized Rule and Subjugation of the Scholarly Class

In the aftermath of the Jingnan Campaign, the Yongle Emperor faced an immediate and pressing problem: how to transform a violently seized throne into a morally legitimate one. His solution was not to dismantle existing institutions, but to strategically reshape them. By consolidating bureaucratic authority while simultaneously subordinating the scholarly class, Yongle constructed a centralized political order that fused administrative efficiency with ideological control.

Yongle initially justified his rebellion by framing it as a moral intervention against "traitors" who had corrupted the court of his nephew, the Jianwen Emperor. This narrative became foundational to his political philosophy. Upon taking power, he institutionalized this claim through historical revisionism and legal measures designed to erase the legitimacy of his predecessor. The Jianwen reign was retroactively abolished, its years reclassified under the Hongwu era, thereby symbolically denying its independent existence. At the same time, Yongle translated rhetorical accusations of treachery into systematic purges. Officials associated with the Jianwen court were punished, exiled, or executed, sending a clear message that loyalty to the previous regime would not be tolerated.

The most dramatic example of this policy was the execution of the scholar-official Fang Xiaoru, whose refusal to legitimize Yongle's accession became a defining moment in the emperor's consolidation. When ordered to draft an imperial proclamation affirming the new regime, Fang instead denounced Yongle as a usurper. His execution—along with the unprecedented extermination of "ten generations" of his associates and forcing some of his female relatives into brothels—was highly unusual, as the official account of the incident in the *Book of Mingshi* demonstrates:

When the Emperor Chengzu [Yongle] departed from Beijing, Yao Guangxiao had entrusted him with a request: "On the day the city falls, he [Fang] will certainly not surrender. Please do not kill him. If you kill Xiaoru, the seed of scholarship in all of China will be extinguished." Emperor Chengzu nodded. Now, he wanted to have Xiaoru

⁶ Edward L. Dreyer. "The Reign of the Yung-lo Emperor." In *The Cambridge History of China*, Vol. 7: *The Ming Dynasty, 1368–1644, Part I*, edited by Frederick W. Mote and Denis Twitchett (Cambridge: Cambridge University Press, 1988): 221–271.

draft an imperial edict. When summoned, Xiaoru wept bitterly, his sobs echoing through the palace halls. Emperor Chengzu descended from his throne and consoled him, saying: “master, do not trouble yourself.” . . . Emperor Chengzu then turned to his attendants and handed Xiaoru writing materials, saying: “Issue a proclamation to the empire; it can only be drafted by you.” Xiaoru threw the brush on the ground and wept while cursing. “If I must die, let me die. But I will not draft your decree!” Emperor Chengzu became furious and ordered him to be executed by being torn apart in the marketplace. Xiaoru accepted his fate with composure and composed a final poem: “Heaven has brought chaos and separation—none knows its cause. Treacherous ministers have succeeded in their schemes, plotting the state's affairs as if they were games.” [Fang said:] “Loyal ministers, filled with anger, shed blood and tears. With this I die for my sovereign—what else could I possibly seek? Alas! May the people not blame me!”⁷

Fang’s resistance and tragic fate demonstrate that scholarly prestige and moral authority, long central to Confucianian political philosophy, would no longer stand above imperial authority. He was later remembered as a model scholar and official in Chinese history. In fact, Fang’s sacrifice was so memorable and politically potent, that future Ming emperors attempted to restore his reputation: Yongle’s son, the Rensong Emperor, pardoned many of the officials who had been punished by Yongle for their association with Fang, and the Wanli Emperor even opened a temple in his memory, which is still open today. Through such acts, Yongle decisively subordinated the literati, transforming them from a moral liability into pillars of state power.

Yet Yongle did not reject the scholar-official class outright. Instead, he restructured its function within a more tightly controlled bureaucratic framework. The civil service examination system, inherited from earlier reigns, was refined and expanded to ensure a steady supply of talented administrators. By introducing regional quotas that balanced representation between the North and South, Yongle addressed concerns of fairness while reinforcing the legitimacy of the system. Success in the examinations became the primary pathway to political influence, allowing the emperor to reward merit while simultaneously binding scholars’ advancement to imperial favor.

At the institutional level, Yongle further strengthened centralization through the development of the Grand Secretariat. Although the office functioned in practice as a coordinating body akin to a prime ministerial institution, its members were deliberately ranked lower within the official hierarchy. This structural ambiguity was crucial. It enabled the Grand Secretariat to manage the increasing administrative burden of a growing empire—drafting edicts, reviewing memorials, and coordinating policy—while ensuring that ultimate authority remained firmly in the emperor’s hands. In effect, Yongle created a system in which bureaucratic expertise was indispensable but politically subordinate.

⁷ *Book of Mingshi*, <https://ctext.org/wiki.pl?if=gb&chapter=861002&remap=gb>.

This reconfiguration of governance reflected Yongle's vision. He recognized both the necessity and the independence of the scholarly elite. As Yongle himself stated at a gathering with officials in summer 1403:

The purpose of establishing various agencies and appointing graded officials at the court was to govern the people. And the way to achieve the governance of the people was to protect and to feed them; that's all. It is based on this belief that, after I assumed the emperorship, I dutifully followed the instructions of my father and the established laws.⁸

By integrating scholars into a centralized system that he controlled through examinations, appointments, and institutional design, the wily emperor neutralized their independence while harnessing their capabilities. The result was a bureaucratic system that projected stability, competence, and Confucian legitimacy, even as it rested on the foundations of military conquest.

Through these measures, Yongle transformed the Ming state into a more centralized and disciplined structure. His policies did not simply mark a new level of consolidation; they redefined the relationship between emperor and scholar. In doing so, he laid the groundwork for a model of governance in which intellectual life, administration, and authority were tightly interwoven..

Reconfiguring Imperial Space: Capital Relocation, Monumental Power, and Strategic Defense

If Yongle's restructuring of the bureaucracy consolidated authority within the state, his approach to the geography of this state projected that authority outward. Through the relocation of the capital, the Yongle Emperor transformed the Ming state into a spatial expression of his political philosophy. While there was certainly a military and economic rationale beyond shifting the capital Beijing, this move was made with deliberate symbolism.

The transfer of the capital from Nanjing to Beijing was formally completed in 1420. This decision marked a milestone in the history of the Ming dynasty. Under the Hongwu Emperor, the dynasty had been rooted in the agrarian and economically rich south. Yongle's move shifted the political center northward, closer to the steppe frontier and the persistent threat of Mongol incursions. He transformed the Ming into a northern-facing empire, one that defined its identity by projecting authority outward.

The strategic logic of this relocation was clear. Even early in his reign, northern defenses had begun to weaken. In 1403, Mongol forces raided Liaobei, exposing the vulnerability created when Yongle vacated his former power base in the north to seize the throne in Nanjing. By relocating the capital to Beijing—his longtime princely seat—he restored direct imperial oversight over the frontier.⁹ This allowed for quicker military mobilization and reinforced the symbolic message that the emperor himself was responsible for the empire's security against

⁸ Tsai, *Perpetual Happiness*, 80.

⁹ Frederick W. Mote, *Imperial China 900–1800* (Cambridge, MA: Harvard University Press, 1999): 225.

incursions from the nomadic steppe. Yongle further reinforced this posture through his personal leadership of multiple campaigns against Mongol groups.

At the same time, the move carried significant political implications. Nanjing remained a center of resistance, where many officials and elites had supported the Jianwen Emperor and viewed Yongle as a usurper. Beijing, by contrast, was a region where Yongle had ruled for decades and commanded stronger loyalty. Relocating the capital therefore weakened the influence of hostile southern elites while embedding the regime in a space more favorable to his authority. In this sense, geography allowed Yongle to reshape to recenter the empire and with it the core loyalties underpinning his rule.

The construction of the imperial complex in Beijing was an enormous undertaking, involving more than 200,000 laborers, craftsmen, and engineers. Spanning roughly 101 hectares, the complex occupied a central position in the city and was enclosed by substantial defensive features, including Taiyi Lake, a broad moat known as the Jade River, and high outer walls. At its core stood the Forbidden City, a vast rectangular enclosure measuring approximately 961 by 753 meters. Within this space, Yongle commissioned a series of major palace halls aligned along a central north–south axis, accompanied by numerous additional buildings with yellow-tiled roofs arranged around courtyards of varying size. Security was further reinforced by an inner wall, supported by an extensive timber framework, painted in vermilion, and punctuated at each corner by elaborate tower structures.¹⁰

The construction of the Forbidden City and consequent transfer of personnel to the new capital imposed an enormous burden on the Chinese economy. In the aftermath of severe lightning storms that damaged several palaces, a scribe named Li Shimian mustered the courage to criticize the emperor directly:

The capital is the foundation of the world, but the people are the foundation of the capital. If the people feel secure, then the capital will be secure, and if the capital is secure, then the foundation of the country is solid, and the world will be at peace. . . . But since the beginning, even the carpenters and masons have used your name to force people through years of rehabilitation to move out of their homes, thereby creating a new army of homeless people. At present, a starving multitude in Shandong, Henan, Shanxi, and Shaanxi eat nothing but tree bark, grass, and whatever crumbs they can find. Others, in desperation, are forced to sell wives and children for their own survival. . . . In contrast, tens of thousands of Buddhist monks and Daoist priests, who were brought here to pray in various temples, daily consume hundreds of piculs of rice.¹¹

This passage highlights both the human cost of Yongle's move to Beijing and the emperor's determination to carry it through. While it begins with the principle that the people are the foundation of the state, it quickly reveals the suffering caused by forced relocations, famine,

¹⁰ Tsai, *Perpetual Happiness*, 125.

¹¹ Tsai, *Perpetual Happiness*, 126.

and exploitation. At the same time, the continuation of large-scale construction and resource allocation shows that the project proceeded despite these hardships. Importantly, the critique in the passage assumes that the emperor's justice is the central standard: the problem is not imperial authority itself, but the failure to govern in a way that protects the people. Unfortunately, Li's appeal was unsuccessful. Yongle had him put in prison.

By 1418, officials were required to conduct state business in Beijing, and by 1420 the new capital was fully operational. This transition demanded extraordinary resources. The construction of the new imperial city and especially the Forbidden City within it required over a decade of labor and the mobilization of hundreds of thousands of workers. Vast quantities of timber, stone, and precious materials were transported across long distances, while large-scale hydraulic engineering projects, including canal expansions, were undertaken to ensure the steady flow of grain and supplies from the fertile south to the northern capital.

These monumental efforts were not simply displays of wealth or administrative capacity; they were central to Yongle's strategy of legitimization. The very act of building a new capital on such a scale signaled that his regime possessed the mandate, resources, and stability of a true dynastic center. The Forbidden City, in particular, embodied this vision. Its imposing architecture, strict spatial design replete with symbolic meaning, and alignment with cosmic order projected an image of imperial authority that transcended the contested origins of Yongle's rule. Through construction, Yongle inscribed his legitimacy onto the empire's new capital.

The relocation of the capital can also be understood as an attempt to distract from the memory of usurpation. By abandoning Nanjing—the seat of his nephew's former power—and establishing a new imperial center, Yongle distanced his reign from the circumstances of its violent beginning. It seems possible the rancor of Fang's rebellion and Li's criticism continued to trouble him, as is evident in the following proclamation:

Since ascending the throne, I have not forgotten, whether eating or sleeping, the people of Beijing. The prefectures and counties under its jurisdiction have repeatedly suffered from floods and droughts; in many years the harvest has failed. There has been a lack of food, and people have been forced to flee. I am deeply moved with compassion for them. Therefore, all outstanding taxes and grain levies owed up to the fifth month of the fourth year of the Yongle reign are to be completely remitted. As for the various categories of taxes and levies, beginning in the sixth year of Yongle, they shall continue to be exempted for three years, in order to somewhat express my intentions.¹²

Concern for the population was meant to demonstrate that the emperor maintained the profile of a wise Confucian emperor. The new capital allowed Yongle to associate his authority in a space associated with strength, order, and continuity rather than rebellion.

¹² "Imperial Edict on the Completion of the Beijing Palaces."
<https://zh.wikisource.org/zh-hans/%E6%AC%BD%E5%AE%9A%E5%8F%A4%E4%BB%8A%E5%9C%96%E6%9B%B8%E9%9B%86%E6%88%90/%E6%98%8E%E5%80%AB%E5%BD%99%E7%B7%A8/%E7%9A%87%E6%A5%B5%E5%85%B8/%E7%AC%AC145%E5%8D%B7>.

Taken together, these policies reveal how Yongle used geography. The relocation of the capital, the construction of monumental architecture, and the strategic orientation toward the northern frontier were all interconnected elements of a vision. They demonstrate how physical space could be reshaped to delineate the contours of a new ideology. Yongle extended his project of state-building beyond institutions and into the very landscape of the Ming empire.

Cultural Mastery: the Yongle Encyclopedia and Zheng He's Explorations

While the move to Beijing was a monumental endeavor, Yongle's most ambitious efforts at legitimization emerged through projects of cultural mastery and diplomatic skill. Two of these epoch-changing efforts were the compilation of the massive Yongle Encyclopedia and the emperor's sponsorship of the maritime expeditions of Zheng He. They represented not only displays of imperial power, but a deliberate attempt to claim intellectual and global authority. Paradoxically, it is precisely these cultural projects, often viewed as secondary to military or administrative reforms, that constituted the most important element of Yongle's political philosophy. As the Record of Emperor Taizong of the Ming states:

Although the existing works have a system, they are not broad in their collection and only record general outlines. You [the scholars] should follow my will: collect all books written since the beginning of writing—including the Classics, Histories, Philosophies, and Literary Collections, as well as the teachings of the Hundred Schools, astronomy, geography, yin-yang, medicine, divination, Buddhism, Taoism, and the various arts—and compile them into a single work. Do not be deterred by its vast scale.¹³

The comprehensiveness and breathtaking scope of the project was made clear to the scholars compiling the encyclopedia. Through them, he transformed cultural mastery into a visible instrument of asserting legitimacy.

Confucian political culture, with its emphasis on filial obedience and respect for ancestral authority, figured prominently in the motivations behind the encyclopedia. To rebel against the succession order established by one's father was a violation of the principles that had underpinned imperial rule for more than a millenium. Much of what later generations understood about Yongle's reign was shaped by Qing dynasty scholars writing retrospectively, often through a Confucian lens that emphasized this breach. This was the reason that Yongle appropriated cultural mastery as a medium to overwhelm all critiques of his rule and usurpation of power.

¹³ *Record of Emperor Taizong of the Ming* Volume 21 (July 19, 1403).

[https://zh.wikisource.org/wiki/%E6%88%90%E7%A5%96%E6%96%87%E7%9A%87%E5%B8%9D%E5%AF%A6%E9%8C%84/%E5%8D%B7011#%E5%8F%B2%E5%BC%8F%E7%B7%A8%E5%B9%B4%E5%BC%8F%E6%98%8E%E5%AF%A6%E9%8C%84%E5%BC%8F%E5%A4%AA%E5%AE%97%E5%BC%8F%E5%8D%B7%E4%BA%8C%E5%8D%81%E4%B8%80_%E6%B0%B8%E6%A8%82%E5%85%83%E5%B9%B4%E5%85%AD%E6%9C%88%E8%87%B3%E4%B8%83%E6%9C%88\(P.377\)](https://zh.wikisource.org/wiki/%E6%88%90%E7%A5%96%E6%96%87%E7%9A%87%E5%B8%9D%E5%AF%A6%E9%8C%84/%E5%8D%B7011#%E5%8F%B2%E5%BC%8F%E7%B7%A8%E5%B9%B4%E5%BC%8F%E6%98%8E%E5%AF%A6%E9%8C%84%E5%BC%8F%E5%A4%AA%E5%AE%97%E5%BC%8F%E5%8D%B7%E4%BA%8C%E5%8D%81%E4%B8%80_%E6%B0%B8%E6%A8%82%E5%85%83%E5%B9%B4%E5%85%AD%E6%9C%88%E8%87%B3%E4%B8%83%E6%9C%88(P.377)).

The process of creating such a book was clearly demanding and could not be entrusted to just any scholar, particularly one whose reliability might be in question. In 1403, after Yongle issued the order for the compilation of this work, he appointed Xie Jin to oversee this significant undertaking. Xie Jin, a jinshi (a “Presented Scholar” who passed the highest level of the imperial civil service examination) of 1381, was widely recognized for his brilliance. He had been also highly valued by the Hongwu Emperor for his honesty and his willingness to express independent views. The degree of trust placed in him by Hongwu during his service is illustrated in *Mingshi*, Biography 35:

Xie Jin demonstrated exceptional intelligence from a young age. In the twenty-first year of the Hongwu Emperor’s reign, he passed the imperial examination and attained the jinshi degree. He was subsequently appointed as a Zhongshu shiji shi (a junior official in the Secretariat), where he earned considerable respect and was frequently summoned to attend to the emperor. On one occasion, while the emperor was in the Dapo West Room, he addressed Xie Jin as follows: “In formal terms, ours is a relationship of ruler and subject; yet in sentiment, it is like that of father and son. You should therefore speak without reservation on all matters.” That same day, Xie Jin submitted a sealed memorial of ten thousand words.¹⁴

Following the turmoil of the Jianwen Emperor’s reign, Xie Jin entered the service of the Yongle Emperor after the latter’s victory in Nanjing. He was promoted to Hanlin Academician Reader-in-Waiting and was tasked with serving as a principal editor of the Record of Emperor Taizong of the Ming. Owing to the trust placed in him, he was also appointed chief editor of the new encyclopedic project.

Upon receiving imperial instructions to compile the work, Xie Jin convened 147 scholars and, in 1404, presented an initial version titled the Great Collection of Literature. This one-year effort, however, failed to meet Yongle’s expectations and was rejected, placing considerable pressure on the scholars to undertake substantial revisions. Given that Xie Jin remained occupied with the compilation of the Record of Emperor Taizong, the Yongle Emperor subsequently appointed additional co-editors, including Yao Guangxiao and Zheng Ci. In 1407, the revised work—later known as the Yongle Encyclopedia (Yongle Dadian)—was presented to the emperor and received high praise. As Yongle remarked in the imperial preface to the Yongle Dadian:

Beginning with the distant past and extending to the present, the work draws on a wide range of sources and gathers material extensively, organizing and synthesizing it into a comprehensive and profound compendium.¹⁵

¹⁴ *Mingshi Biographies* 35.

<https://ctext.org/wiki.pl?if=gb&chapter=391284&remap=gb>.

¹⁵ Imperial Preface to the Yongle Dadian.

<https://zh.wikisource.org/zh-hans/%E5%BE%A1%E8%A3%BD%E6%B0%B8%E6%A8%82%E5%A4%A7%E5%85%B8%E5%BA%8F>.

After approving the content, the Yongle Emperor ordered that copies of the work be produced, and the entire project was completed in 1409. The Yongle Encyclopedia comprised 22,877 juan (volumes), including 60 volumes of tables of contents and references, and was ultimately collated into 11,095 booklets. Each volume measured approximately 30 centimeters in width and 50 centimeters in height. The Yongle Encyclopedia is without doubt a monumental achievement in Chinese history. Despite the tragic loss of much of the work over centuries of upheaval and warfare, its sheer scale and ambition represent a significant milestone in the development of Chinese literary culture.

The compilation of the Yongle Encyclopedia was the clearest expression of this ambition. Announced in 1403, the project aimed to gather all recorded knowledge into a single, unified work under imperial authority. As Yongle declared, scholars were to collect writings “since the beginning of writing,” encompassing the Classics, histories, philosophy, religion, medicine, astronomy, and the arts. The process itself reinforced this dynamic. Its sheer scale defied practical use. Rather, its significance lay in what it represented: a monumental claim that the Ming state, under Yongle, possessed both the capacity and the authority to encompass all human knowledge.¹⁶

In early modern Chinese political culture, such undertakings were historically associated with periods of exceptional prosperity, as they required immense financial resources, a highly developed scholarly class, and an effective examination system capable of producing qualified officials. Yongle’s reign met these conditions, but his motivations extended beyond the conventional desire for cultural achievement. By bringing together the totality of Chinese knowledge, he symbolically placed himself at the apex of civilization. The encyclopedia thus functioned as a form of intellectual centralization, paralleling his political centralization of the bureaucracy. It was not for nothing that the volumes have been estimated to contain the equivalent of 8,000 years of material.¹⁷ Thus, the Yongle Encyclopedia was not commissioned primarily for any practical use in governance, or even as a public display, but for purely symbolic effect. It proclaimed: the state is here; the state is powerful; the state can command the intellectual universe itself. This was cultural mastery not as passive preservation, but as a politicized demonstration of power.

Equally significant, though directed outward, were the maritime expeditions led by Zheng He. Born Ma Sanbao into a Muslim family, Zheng He was raised within an Islamic cultural environment and is said to have harbored an early desire to travel to Mecca, the holiest site in Islam. In 1381, at around the age of ten, he was captured during the Ming conquest of Yunnan and forced to become a eunuch. He was subsequently brought to the Ming court and entered the service of Zhu Di. When Zhu Di launched his rebellion, Zheng He was among his early supporters and distinguished himself during the conflict. Following the success of the

¹⁶ Duncan M. Campbell. “The Huntington Library’s Volume of the *Yongle Encyclopaedia* (*Yongle Dadian* 永樂大典): A Bibliographical and Historical Note.” *East Asian History*, no. 42 (2018): 1–24.

¹⁷ Lauren Christos, “The Yongle Dadian: the origin, destruction, dispersal and reclamation of a Chinese cultural treasure.” *Journal of Library and Information Science*, Vol 36: 1 (2010): 83.

Jingnan Campaign, Zhu Di ascended the throne as the Yongle Emperor and, in recognition of his service, bestowed upon Ma Sanbao the name Zheng He.

The motivations behind Yongle's sponsorship of maritime expeditions have been widely discussed. One major explanation, recorded in the *Mingshi* (Biography 192), is that the Yongle Emperor suspected the defeated Jianwen Emperor had fled overseas and sought to trace his whereabouts. At the same time, the voyages served to project imperial power abroad and to display the wealth and prestige of the Ming state. Under these motivations, a series of expeditions was launched. During Yongle's reign, Zheng He led six major voyages to the "Western Oceans," which resulted in the establishment of tributary relations with numerous polities and helped resolve diplomatic tensions that would shape regional interactions for generations. His fleets reached over thirty locations, including Java, Sumatra, and Calicut, and extended as far as the east coast of Africa.

One of the most consequential outcomes of these expeditions was the consolidation of trade networks, particularly in the Strait of Malacca. Zheng He's intervention in local succession disputes helped install a ruler favorable to Ming interests, who in turn supported tributary relations and facilitated commercial exchange. This contributed to the emergence of Malacca as a major entrepôt and a key node in regional trade.

At a time when maritime activity had been heavily restricted under earlier Ming policies, these voyages represented a dramatic shift. Between 1405 and 1433, Zheng He commanded a series of massive expeditions across the Indian Ocean, reaching Southeast Asia, South Asia, the Middle East, and the eastern coast of Africa. These voyages were unprecedented in scale, involving tens of thousands of personnel and hundreds of ships equipped with advanced technology and weaponry.¹⁸ They sought to establish diplomatic relations, secure tribute, and integrate foreign polities into a Sino-centric world order. More recent scholarship has challenged the notion of the voyages as exclusively cultural and economic, noting that they entailed military aggression in places like Vietnam.¹⁹ Yet regardless of these specific motivations, the broader significance of the voyages lay in their symbolic content: projecting cultural mastery on a global scale.

Zheng He's expeditions did not aim at territorial conquest in the conventional sense. Instead, they enacted a form of symbolic domination. By arriving with overwhelming force, distributing gifts, and receiving tribute, they staged encounters in which foreign rulers acknowledged the supremacy of the Ming emperor. These interactions extended Yongle's legitimacy beyond China's borders, transforming him into a ruler recognized at least ceremonially across much of the known world.

Taken together, the Yongle Encyclopedia and Zheng He's expeditions reveal a coherent strategy: the transformation of cultural and intellectual achievement into a form of political legitimacy backed by military power. These projects extended Yongle's authority across two dimensions, knowledge and geography. Ironically, it is this dimension of Yongle's that may be

¹⁸ Edward L. Dreyer, *China and the Oceans in the Early Ming Dynasty, 1405-1433* (New York: Pearson/Longman, 2007).

¹⁹ Geoff Wade, "The Zheng He Voyages: a reassessment." *JMBRAS*, Vol. 78 Part 1 (2005): 37-58.

the most important. His other efforts could not resolve the moral ambiguity of his accession. It was through cultural mastery, the ability to command knowledge, to assert the authority to promulgate memory, and to project power across the world, that Yongle most effectively redefined his image. These projects did not erase the origins of his rule, but they reframed them within a narrative of grandeur and supreme cultural authority.

Conclusion

In conclusion, the reign of the Yongle Emperor and his governing philosophy were not solely directed toward improving the welfare of the realm, but more fundamentally toward securing and sustaining the long-term legitimacy of his rule. By invoking the legal and ideological precedents established by his father, the Hongwu Emperor, Yongle exploited ambiguities within the political order to justify his rebellion. Although he carefully framed his actions in official histories and public proclamations, his accession remained contested among both officials and segments of the broader population. As a result, many of his policies can be interpreted as efforts to win over those who had previously been loyal to the Jianwen Emperor. His expansion of the role of scholarly elites in government, for instance, may partly reflect broader trends of the period, but it also served the clear purpose of cultivating support and enhancing his reputation.

At the same time, it would be reductive to view Yongle's actions as purely self-serving. Regardless of his underlying motivations, his policies contributed significantly to what is often regarded as one of the most prosperous periods of the Ming dynasty. His decision to relocate the capital to Beijing stands out as a particularly consequential measure. While it initially served to consolidate his authority and project imperial power, it also reshaped the political and strategic geography of the empire in ways that proved enduring.

Yongle's patronage of major cultural and diplomatic initiatives further reinforced his authority. The compilation of the Yongle Dadian and the sponsorship of the maritime expeditions led by Zheng He demonstrated his capacity to mobilize vast resources in ways that enhanced both imperial prestige and practical influence. These projects not only projected the power of the Ming state outward but also contributed to the construction of an image of the emperor as a ruler of unparalleled vision and capability.

Taken as a whole, the evidence suggests that Yongle's reign exemplifies the active construction of political legitimacy through a combination of governance, ideology, and symbolic action. His ability to transform a contested accession into a widely accepted and enduring model of imperial authority highlights the dynamic nature of legitimacy in the Ming dynasty. More broadly, it offers insight into how rulers can consolidate power in the face of initial instability, turning vulnerability into a foundation for long-term strength.

It is also worth considering whether the Yongle Emperor made critical mistakes. This question is not straightforward, since imperial historiography often avoids framing an emperor's actions as "mistakes" in explicit terms. Nonetheless, some of Yongle's policies had significant consequences. For example, his strong support for Zheng He's voyages, combined with his

extensive military campaigns (many of which he personally led) placed enormous strain on the state's finances. These undertakings required vast resources and contributed to fiscal pressure on the imperial treasury. By the end of his reign, the empire was left in a less stable economic condition, and it fell to his son and grandson to restore financial balance and prosperity with more limited means. From one perspective, one could argue that these policies brought the empire close to financial exhaustion.

Another area open to criticism is Yongle's harsh treatment of officials who had been loyal to the Jianwen Emperor. Although the Jianwen reign had its shortcomings, many of these bureaucrats were competent and principled. Yongle's reprisals against them were often severe, targeting individuals who had demonstrated loyalty in accordance with Confucian ideals of service. While their reputations were rehabilitated in later generations, the brutality of these purges remains a troubling aspect of his rule.

At the same time, from a broader historical perspective, these decisions may be interpreted as strategic investments that enhanced Ming prestige and influence, making it difficult to judge them purely as errors.

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How Does Nightly Sleep and Strategic Napping Affect Cognitive and Physical Performance Of Elite Soccer Players? By Ajay Kosaraju

Abstract

This is a systematic review that examines the impact of nightly sleep quality and duration and the use of strategic naps on cognitive and physical performance. The review will compile and analyze existing studies to find the relationship between sleep related factors and performance measures. The metrics include physical ability, cognitive function, recovery, injury risk, and fatigue. By including research that objective sleep measurements and performance metrics (sprint times, scanning frequency, and fitness scores), the review aims to identify consistent patterns. The end goal of this review is to offer insights and recommendations on how optimizing sleep and daytime naps can be used to enhance performance for soccer players.

Introduction

In the United States, approximately 50% of the population have participated in competitive sports throughout high school, highlighting the relevance of optimizing athletic performance. As high performing athletes find ways to excel, one of the most frequently asked questions is: "What can I do off the field to improve my performance?" Physical training, nutrition, and mental preparation are the more well known components of athletic success, however recent research has shown an increasing emphasis on sleep duration and quality to achieve peak performance. High-quality sleep has been shown to contribute significantly to physical recovery, motor skills, decision-making, and emotional regulation, factors that are particularly crucial in soccer. Soccer performance also relies heavily on reaction time, sprint speed, agility, coordination, and sustained focus, all of which can be enhanced through adequate sleep and strategic napping. Furthermore, the benefits of sleep extend further than soccer performance. Numerous studies have linked sleep quality with academic success and emotional resilience, especially among teens who participate in competitive sports. This is relevant considering the growing pressure on young athletes to perform well both on the field and in academic settings. This paper will further explore and summarize the current scientific understanding of how sleep, specifically nighttime sleep and napping, affects soccer performance. Through an in-depth analysis of recent studies, this review aims to highlight the important role of sleep in enhancing athletic ability, recovery, and long-term athlete health and development.

Methodology

To create this review, information was compiled from a set of credible sources, including fifteen peer-reviewed studies, academic research papers, and journals relevant to the topic. These sources were selected using databases such as PubMed, Kaggle, and Google Scholar ensuring that each source directly addressed the research question. The data collection followed a consistent criteria to ensure reliability across datasets. The information that was gathered was

compiled into a spreadsheet for analysis. The spreadsheet was used to make a series of graphs and visual representations to illustrate key trends, patterns, and correlations. The sources were split into 4 groups: Experimental Nap Intervention Studies, Observational & Monitoring Studies, Systematic & Narrative Reviews and Survey-Based Behavioral Studies to fully encompass the key trends and data.

Experimental Nap Intervention Studies: These studies manipulated nap duration, timing, and conditions to determine their effects on athletic performance, cognitive function, and physiological responses. Most used randomized crossover designs, which reduce individual performance variability by having each participant experience all conditions. (Controlled Nap vs. No Nap Designs)

“Daytime Napping Benefits Passing Performance and Scanning Activity in Elite Soccer Players” by Masaki Nishida: National-level male soccer players (excluding goalkeepers) followed a standardized 3 day baseline sleep protocol (6–8 hours a night) and were denied caffeine/alcohol pre-experiment. Participants either napped for 40 minutes (14:00–14:40) in a dark, air-conditioned room (~77°F) or remained quietly awake. Post-nap, cognitive flexibility and attention were tested via Trail Making Test A/B (a test that assesses visual scanning and attention span), soccer passing ability was assessed using a modified Loughborough Soccer Passing Test (10 timed passes), and scanning activity (head turns) was visually counted. Nap verification used a smartphone sleep-tracking app called Sleep Meister.

“Impact of 30-minute and 90-minute Naps on Aerobic and Anaerobic Intermittent Performance in Collegiate Soccer Players” by Xu A: Twenty five male university soccer players (national-level, with 10 hrs a week of training) completed a randomized crossover of three nap conditions: no nap, 30-min nap (14:00–14:30), and 90-min nap (13:00–14:30). Naps were taken in a dark, quiet, climate-controlled room (68–74°F) in a semi reclined position, with a 1-hour post-nap recovery to minimize sleep inertia. Performance tests included the Running-Based Anaerobic Sprint Test (6 × 35 m sprints, 10s recovery) measuring peak/minimum/average power and fatigue index, and the 30–15 Intermittent Fitness Test to determine maximal running speed, (VIFT), heart rate, and perceived exertion.

“The Impact of Strategic Napping on Peak Expiratory Flow and Respiratory Function in Young Elite Athletes” by Ahmet Kurtoğlu: Twelve elite youth athletes (ages 12.5 years), competed in the high jump and sprint events with no nap, 25-min nap, and 45-min nap conditions. (14:00 start) The nap took place in a dark, quiet room, preceded by a 10-min acclimatization period and followed by a 5-min warm-up. Pulmonary function (lung health) was measured using a MIR spirometer (FVC, FEV, FEF25–75%, PEF). Heart rate monitoring was conducted using Fitbit Charge 3 devices.

“Physiological Response and Physical Performance After 40 min and 90 min Daytime Nap Opportunities” by Omar Boukhris: Sixteen male amateur athletes (soccer, rugby, handball) completed three conditions, no nap, 40-min nap, and 90-min nap, in a controlled environment. Performance was assessed through 5-meter shuttle runs for total and maximum distance. Body

measures included resting and exercising heart rate, blood pressure, heart rate variability, perceived exertion (RPE), and body temperature.

“The Acute Effects of 25-Versus 60-Minute Naps on Agility and Vertical Jump Performance in Elite Youth Soccer Players: The Role of Individual Chronotype” by Ozgur Eken: Sixteen elite U19 male soccer players underwent no nap, 25-min nap, and 60-min nap conditions, randomized with 48-hour rest between sessions. Tests included the Illinois Agility Test, a change-of-direction drill with a soccer ball, and countermovement jump (CMJ) for anaerobic power. Sleep quality and mood were assessed pre-and post-nap using the Morningness-Eveningness Questionnaire, Pittsburgh Sleep Quality Index, and Profile of Mood States Questionnaire.

Observational & Monitoring Studies: These studies did not alter nap behavior experimentally but tracked sleep and performance patterns using wearables, self-reports, or data collection during training and competition periods.

“Sleep Architecture of Elite Soccer Players Surrounding Match Days As Measured By WHOOP Straps” by Nicole Sanders: Six elite English Premier League players (average age 28) were monitored for two seasons using WHOOP wearable devices across 13 afternoon matches. Sleep metrics (total sleep time, efficiency, light/deep/REM stages, disturbances, onset/offset times) were recorded for the night before the match, the night after, and the subsequent night. External physical load was quantified using GPS tracking during matches.

“Elite Soccer Athlete's Sleep: A literature review” by Aldo C. Silva: Multiple observational studies assessed sleep duration, efficiency, and onset latency in soccer players, correlating these with anaerobic performance, muscle strength, sprinting ability, and injury incidence. Studies reported matchday sleep disruptions (especially after evening games and losses), reduced sleep during congested fixture periods, and significant associations between poor sleep efficiency and increased injury count, absence days, and severity.

“Relationships between Sleep, Athletic and Match Performance, Training Load, and Injuries: A Systematic Review of Soccer Players” by Filipe Manuel Clemente: A narrative synthesis identified common sleep disruptors in soccer, including evening training/matches, travel, and heavy match loads. It reviewed literature linking sleep deprivation to declines in anaerobic capacity, reaction time, and motor learning. Systematic & Narrative Reviews: These works summarized existing experimental and observational soccer studies to draw broader conclusions and provide practical recommendations.

“Is Daytime Napping an Effective Strategy To Improve Sport-Related Cognitive and Physical Performance and Reduce Fatigue? A Systematic Review and Meta-Analysis of Randomised Controlled Trials” by Arthur Eumann Mesas: Systematic review of 37 athlete nap studies (ages ≥ 12) examining nap duration (20–90 min, occasionally 120 min), timing (13:00–16:00), and outcomes. Reported consistent benefits to sprinting, endurance, strength, jump ability, reaction time, attention, vigilance, and reduced perceived exertion, particularly under sleep deprivation.

“To Nap or Not to Nap? A Systematic Review Evaluating Napping Behavior in Athletes and the Impact on Various Measures of Athletic Performance” by Lastella M: Narrative review summarizing practical guidance for athletes: optimal nap length 20–90 min, best timing 13:00–16:00, and 30-min wake period before competition to dissipate sleep inertia.

“Sleep and Performance in Professional Athletes” by Jesse D Cook: Narrative review comparing sleep patterns of elite athletes with non-athletes, reporting shorter sleep durations, lower efficiency, longer sleep latency, and increased disturbances in athletes. It summarized intervention studies on sleep extension, jet lag effects, and Olympic-level performance improvements following targeted sleep programs.

Survey-Based Behavioral Studies: These studies used questionnaires to evaluate sleep quality, behaviors, and barriers to optimal rest among different athlete groups.

“Characterising and Comparing the Sleep Characteristics and Behaviours of Female and Male Soccer Players: A Cross-Sectional Survey of an Elite Soccer Club” by Nicole Sanders: Senior male, senior female, and U21 male soccer players completed the Athlete Sleep Screening Questionnaire (ASSQ) and Athlete Sleep Behaviour Questionnaire (ASBQ). Metrics included a Sleep Difficulty Score (SDS), global behavior scores, and individual behavior items. Comparisons revealed that senior female players reported the highest sleep difficulty, while U21 male players reported the lowest.

Results

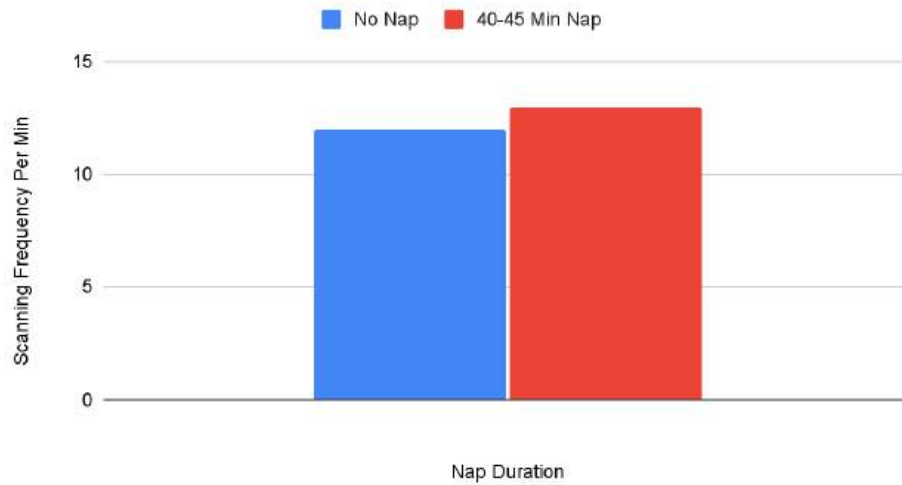
Sleep Data: Total Sleep Time (TST) Average Soccer Player: Night Before Match: 440 ± 37 min, Night of Match: 393 ± 41 min, Night After Match: 459 ± 36 min

General Sleep Statistics in Athletes: Sleep latency (Time in bed until asleep) : 18 minutes (vs. 5 min in controls), Sleep efficiency: $\sim 80.6\%$ in athletes (vs. 88.7%), Resting HR during sleep: Pros: 52 bpm; Amateurs: 59 bpm, Recovery scores: Pros > Amateurs by 10%

Cognitive Performance: Napping athletes outperformed non-nappers by 6% in cognitive tasks in the Trail Making Test A/B (a test that assesses visual scanning and attention span) and scanning behavior during soccer passing drills. Although the p-value approached significance ($p < 0.06$), it still shows a meaningful trend that favors napping.

The 30-minute nap condition did not show significant cognitive improvements, while the 90-minute nap improved anaerobic metrics, suggesting that deeper sleep stages may be required for cognitive enhancement.

Scanning Frequency (Cognitive Performance)

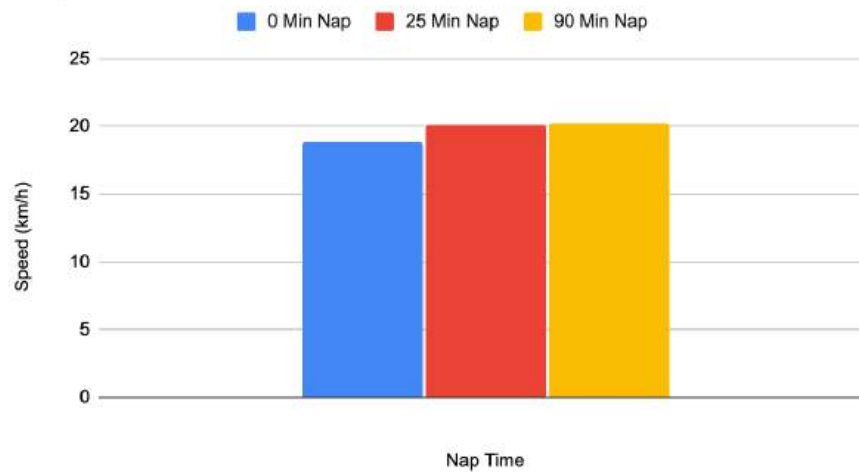


Anaerobic Power and Speed: Peak Power (Watts per kilogram, W/kg): No Nap (N0): 5.32, 30-min Nap (N30): 5.56, 90-min Nap (N90): 5.71, Significant improvement ($p = 0.007$)

Total Distance Ran in the Fitness Test (TD): N0: 846.3m, N40: 871.7 m, N90: 894.2 m
N90 significantly better than both N0 and N40 ($p < 0.001$)

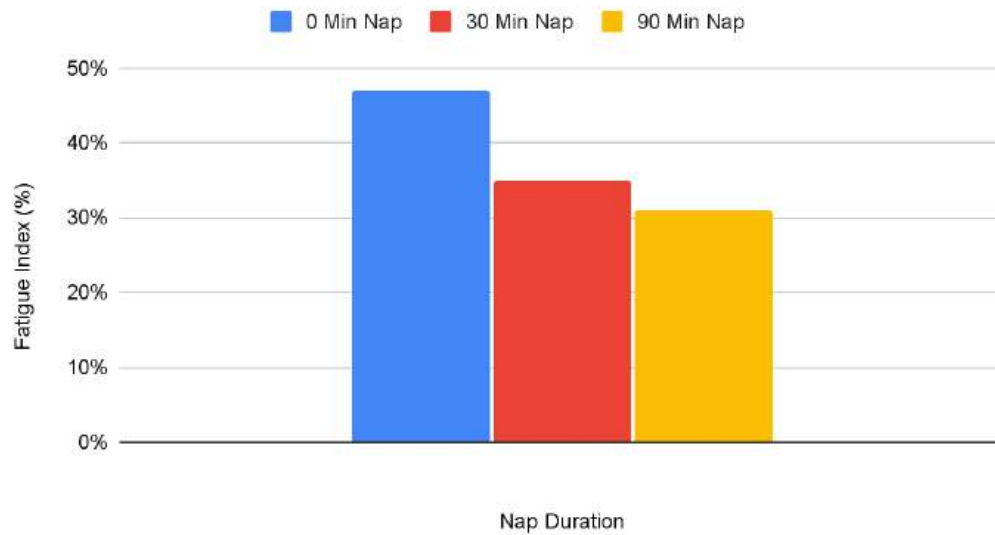
Player Top Speed (km/h), N0: 18.9, N25: 20.1, N90: 20.2, N25 shows significant improvement over N0

Players Top Speed (km/h)



Fatigue Index: Fatigue Index (Lower is better): N0: 30%, N30: 26%, N90: 25%

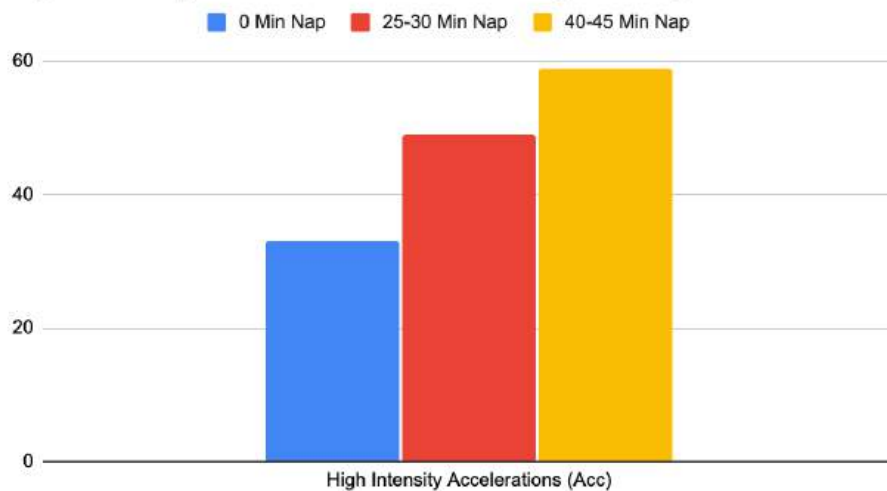
Fatigue Index (Lower is better)



Agility Tests: Agility Test Times (seconds), No Nap: 16.3, 25-min Nap: 16.1, 60-min Nap: 15.7, Significant difference between N60 and other groups ($p < 0.05$)

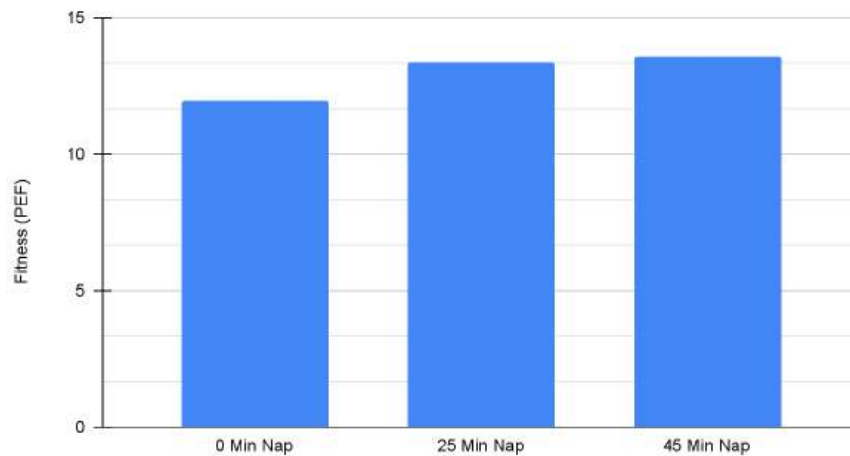
Jump Duration (milliseconds), No Nap: 55, 25-min Nap: 57, 60-min Nap: 56, No significant improvement in jump height or power across nap conditions ($p > 0.05$)

High Intensity Accelerations Per Game (90 mins)



Respiratory Metrics Peak Expiratory Flow: (PEF):No Nap: 6.25 L/s, 25-min Nap: 6.31 L/s, 45-min Nap: 6.68 L/s

Fitness (Peak Expiratory Flow Score)



Subjective Sleep Measures: Perceived Exertion (Method for rating the intensity of physical activity) (RPE Scale 6–20): RPE Scores (Post-Exercise), N0: 15.4 ± 1.2 , N40: 13.8 ± 1.0 , N90: 12.9 ± 0.8

Mood & Sleep Quality, Post-nap mood increased in N60 group by ~12%, Sleepiness (Karolinska Sleepiness Scale): decreased by 2.3 points post-nap

Napping Trends: Optimal nap duration: 20–60 minutes, Best timing: 1:00 PM – 4:00 PM, 90-minute naps yielded the greatest improvements in: Reaction time (~8–11% faster), Shooting accuracy (~9–15% better in elite athletes), Mental vigilance (fewer errors in concentration tasks)

Percent Increase Over No Nap

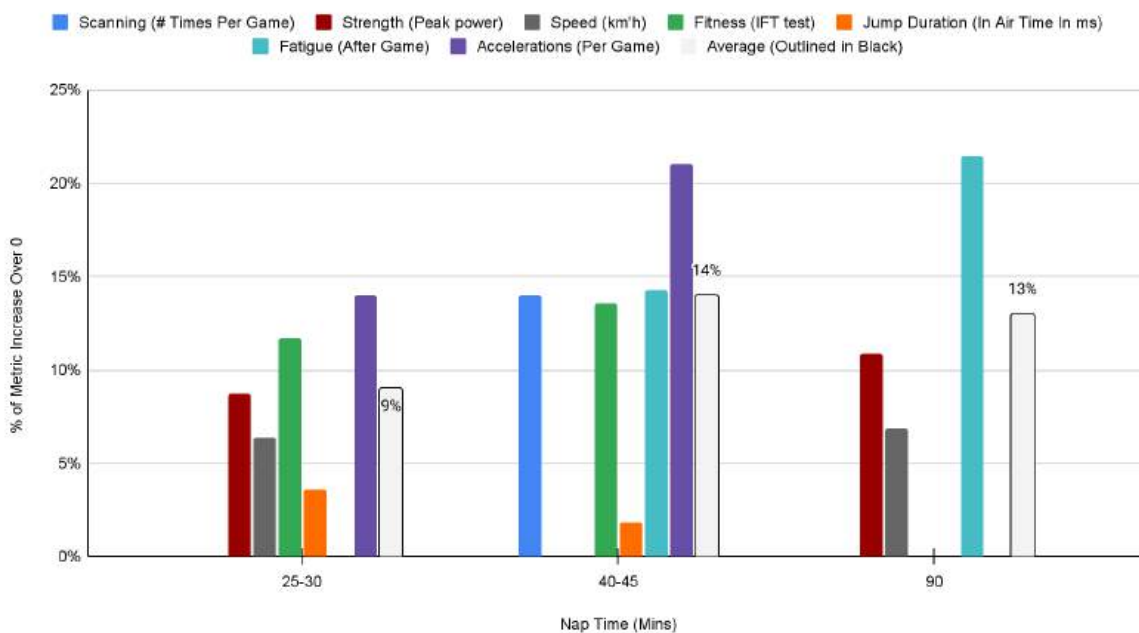


Figure 6. Averaged results of percent increase over no nap. Figure shows the metric percent increase over 0 nap in the key physical and cognitive pillars of the game. For example, speed (in grey) increases by 6% after a 25-30 minute nap.

Discussion

This section analyzes results across the cognitive, physical, and recovery areas to determine how nightly sleep and strategic naps influence soccer players' abilities. The data was put together by compiling the results from all studies into a spreadsheet and creating graphs to help analyze. These graphs displayed performance improvements in nap duration (0 min, 25–30 min, 40–45 min, 60 min, and 90 min). By averaging the percentage gains across physical and cognitive metrics, the graphs made it clear that performance was consistently the highest in the 40–45 minute nap range, leading to the conclusion that a 45 min nap provides the best balance between restfulness and readiness before a game.

Cognitive Performance: The cognitive analysis focused on visual scanning, attention span, and decision-making speed. The graph shows cognitive test scores in nap durations (0 min, 30 min, 90 min) and showed an upward trend, with nappers outperforming non-nappers by around 6%. While the p-value ($p < 0.06$) was small, the direction of improvement across studies indicates a substantial benefit.

Longer 90 minute naps provided slightly better cognitive performance because of deeper sleep stages such as slow-wave and REM sleep, which support memory function and problem-solving. However, these extended naps also increase post-nap grogginess or “sleep inertia,” which lowers short-term alertness. The graph revealed that 40–45 minute naps produced similar cognitive benefits to the 90-minute naps but with reduced inertia effects. This supported the conclusion that moderate-length naps taken 1–2 hours before competition are the most practical for improving mental readiness, reaction time, and focus during matches.

Physical Performance: To evaluate physical effects, data on peak power, sprint speed, total distance, agility, and fatigue index were compiled. The graphs highlighted clear progressions with increasing nap length up to 90 minutes. Average peak power improved from 5.32 W/kg (no nap) to 5.71 W/kg (90-min nap), while top speed in fitness tests increased from 18.9km/h to 20.1 km/h. Similarly, the fatigue index dropped by 7%, showing greater energy efficiency and endurance.

When graphing these metrics, the biggest improvement happened between the no-nap and 40–45-minute nap, suggesting that the performance gain is present within this midrange nap window. Agility tests mildly supported this pattern, with jump times improving from 55ms (no nap) to 57ms (60-min nap). These results, when averaged across all physical tests, showed that short to moderate naps enhanced explosive movements, speed, and coordination,

Recovery and Physiological Readiness: Recovery metrics, including perceived exertion (RPE), resting heart rate, mood, and respiratory performance, were important in determining overall readiness. The RPE scores and mood charts showed downward and upward trends, showing that athletes felt less fatigued and more alert following naps. For example, perceived

exertion dropped from 15.4 (no nap) to 12.9 (90-min nap), while mood scores increased by approximately 12% in the 60-minute nap group.

Peak expiratory flow (respiratory strength and recovery) rose from 11.94 L/s (no nap) to 13.34 L/s (45-min nap), which supports the idea that short to moderate naps restore the body. When combining all of the recovery metrics, graphs revealed an average of 14% increase over the no-nap group which is consistent with the same optimal nap window in the cognitive and physical areas.

Determining the Optimal Nap Strategy: By summarizing all graph data onto a single chart, a clear trend emerged. While 90-minute naps occasionally provided the highest individual scores, the most balanced improvements across all metrics occurred within the 40–45-minute range. This duration offers sufficient time to enter restorative sleep stages without entering deep slow-wave sleep, which can cause post-nap grogginess.

Additionally, the timing of these naps (between 1:00 and 4:00 PM) also aligns with the circadian dip in alertness, making it the best period for recovery before late afternoon or evening matches. Based on these consistent trends, the data support the conclusion that a 40–45-minute nap taken two hours before kickoff maximizes cognitive sharpness, physical performance, and recovery without negative effects.

Conclusion

This systematic review examined how nightly sleep and strategic napping affect the cognitive and physical performance of elite soccer players. Across fifteen studies, the evidence showed that optimal sleep quality and properly timed naps at 40-45 minutes enhance reaction time, agility, endurance, and mental focus. These benefits occurred without disrupting nighttime sleep, indicating that daytime napping can be an effective and sustainable performance strategy.

However, several biases were present in the data. Most studies used small, male-only samples from elite or collegiate players, limiting generalization to other groups. Many experiments took place in controlled lab settings rather than real match environments and focused on short-term effects instead of long-term outcomes. Additionally, the reliance on self-reported sleep measures and wearable data could present some inaccuracies.

The findings of this research extend beyond soccer. Optimizing sleep and rest can improve focus, recovery, and resilience in other fields such as medicine, education, and military. Future studies should include more diverse participants, examine long-term effects, and use advanced monitoring tools for greater accuracy. In conclusion, sleep and strategic napping are not just forms of rest but important tools for enhancing performance, recovery, and well-being in athletes and the broader population.

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Personalizing Immunotherapy in NSCLC: Traditional Radiomics and the Emerging Potential of Habitat Radiomics By Eunchan Jung

Abstract

Radiomics is an artificial intelligence (AI)-based image analysis technology that extracts diverse feature information from amounts of medical images, helping diagnose and predict the prognosis of various cancers. Monitoring the progress of immunotherapy in patients with non-small cell lung cancer (NSCLC) is challenging due to atypical responses, and invasive testing for programmed cell death protein ligand 1 (PD-L1) expression in tissue is considered the only effective method for predicting prognosis. Radiomics has demonstrated remarkable performance in treatment planning and predicting prognosis in many studies. Unlike traditional radiomics, habitat radiomics, a recently emerging approach, reflects intratumoral heterogeneity and the tumor microenvironment. Since habitat radiomics identifies different subregions or cell subpopulations by clustering similar features, it provides more detailed and dynamic information about tumors, improving predictive accuracy, and enabling personalized treatment plans. Furthermore, habitat radiomics correlated with tumor-infiltrating lymphocytes (TILs) and PD-L1, suggesting its potential as a noninvasive method for predicting prognosis. A few studies show that habitat radiomics are also correlated with tumor invasiveness and immune-related adverse events (irAEs). However, further studies on reproducibility and standardization, as well as large-scale prospective studies, are needed. This paper comprehensively reviews existing research papers on traditional radiomics and habitat radiomics, verifies the current state of research, and suggests research directions for developing habitat radiomics, still in its infancy, into a scalable tool for personalized immunotherapy in non-small cell lung cancer patients.

1. Introduction

Lung cancer is the most common malignant tumor worldwide and a leading cause of cancer-related deaths (Bray *et al.*). Adenocarcinoma is the most common histologic subtype of lung cancer, and non-small cell lung cancer (NSCLC) accounts for 85% of all lung cancers (Luo *et al.*).

Immunotherapy using immune checkpoint inhibitors (ICIs) has recently been a gamechanger in the field of cancer therapy, including lung cancer (Tang *et al.*). ICIs reorganize the tumor microenvironment, strengthen anti-tumor immune responses, and initiate indirect cytotoxicity by generating inflammation (Bagchi *et al.*). Although ICIs are known to yield promising results, not all patients benefit equally from ICIs treatment. Response rates to ICIs can vary depending on a patient's unique molecular, histological, or genetic characteristics. Chemotherapy can directly inhibit cancer cell growth and rapidly reduce tumor burden, but while ICIs may take time to show effects and the rate of tumor burden reduction may be slower, their effects can persist even after treatment is stopped (Milanese *et al.*). Therefore, there may be different interpretations when reading medical images to evaluate the effectiveness of immunotherapy (Bagchi *et al.*). Programmed cell death protein ligand 1 (PD-L1) expression is

the only validated predictive biomarker in NSCLC, and response rates vary depending on different thresholds of PD-L1 expression (Davis and Patel)(Reck *et al.*). However, tumor PD-L1 expression for predicting immune responses is primarily obtained through invasive tissue biopsies, and the limited biopsy samples make it difficult to understand the heterogeneity within the tumor and its microenvironment comprehensively. Tumor heterogeneity remains a persistent challenge in cancer treatment, and intratumoral heterogeneity encompasses phenotypic diversity, including cell surface markers, genetic abnormalities, growth rate, apoptosis-inducing characteristics, disease progression, and treatment resistance (Prasetyanti and Medema). This heterogeneity leads to varying responses to ICIs in NSCLC, and consequently, tumor biomarkers that overlook intratumoral heterogeneity and only reflect the characteristics of the entire lung tumor have shown limited predictive power in treatment response (Song *et al.*). Immune-related adverse events (irAEs) are common in clinical trials and can impede the effective use of ICIs. With the widespread use of ICIs in cancer treatment, the annual incidence and number of irAEs have increased significantly (Kennedy and Salama). A comprehensive analysis of 13,051 cases using big data revealed that the pooled incidence of irAEs ranged from 54% to 76%. IrAEs can cause significant discomfort to patients and, in severe cases, can be life-threatening. Therefore, early detection of irAEs is crucial for optimal patient care. Differentiating patient responses to ICIs and assessing the risk of various irAEs is essential for developing individualized treatment plans (Xu *et al.*).

Radiomics, an artificial intelligence (AI)-based image analysis technique, efficiently extracts diverse feature information, such as shape, texture, and signal intensity, from massive amounts of medical images. Radiomics has been widely applied to the diagnosis, prognosis, and monitoring of treatment response in lung cancer, demonstrating its clinical value and potential for further advancement (K. Liu *et al.*). Furthermore, radiomics are being actively researched as a complementary tool to traditional invasive biopsies for PD-L1 in ICIs treated patients. But traditional radiomics analysis has overlooked intratumoral complexity, making it difficult to evaluate the impact of intratumoral heterogeneity on treatment efficacy comprehensively. While traditional radiomics studies have focused on the entire tumor, the recently emerging habitat imaging approach is characterized by identifying different regions or cell subpopulations within the tumor, analogous to the biological microenvironment surrounding it (X. Zhang *et al.*)(Prior *et al.*).

This review examines traditional radiomics and habitat radiomics studies to explore challenges and solutions in applying habitat radiomics to immunotherapy decision-making in patients with NSCLC. We summarize and validate current research progress and suggest research directions for developing early-stage habitat radiomics into a scalable tool for personalized immunotherapy in patients with NSCLC.

2. Radiomics

2.1 Overview of Radiomics

Based on the hypothesis that images are data, data obtained from diagnostic medical images have demonstrated clinical applicability in improving treatment decision-making and outcome prediction, along with clinical, pathological, laboratory, and therapeutic characteristics, and have spawned a new research field called "radiomics" (Aerts *et al.*)(Gillies *et al.*). Several studies have identified associations between radiological features, molecular markers, immune cell infiltration patterns, and clinical outcomes in various malignancies (Trebeschi *et al.*; She *et al.*; Gong *et al.*; J. Wu *et al.*). Radiomics extracts quantitative features related to tumor pathophysiology from medical images, digitally encodes them, and ultimately transforms them into high-dimensional data suitable for analysis (Lambin, *et al.*). The process of radiomics consists of several basic steps, including image acquisition (i.e., CT, MR, and PET/CT), identifying tumors and their surroundings as volumes of interest (VOIs), extracting and qualifying high-dimensional features from the VOIs, and mining extracted features with artificial intelligence to develop classifier models (Z. Liu *et al.*).

2.2 Radiomics for the prediction of immune related biomarker

Immunotherapy is an emerging cancer treatment, but it is not effective for all patients, making patient selection crucial (Bagchi *et al.*). Immunohistochemistry (IHC)-based PD-L1 expression analysis is widely used in clinical practice due to its correlation with immunotherapy efficacy (Reck *et al.*). However, it requires invasive specimen collection, and due to intratumoral heterogeneity, IHC results obtained from a small number of biopsy samples may not be representative of the entire tumor (Kinker *et al.*). With the advancement of AI, interest in predicting PD-L1 levels using radiomics analysis has recently increased. Radiomics allows for the analysis of quantitative biomarkers across the entire tumor, thereby aiding in identifying patients who will benefit from ICIs therapy, predicting prognosis, and monitoring treatment effectiveness (Lambin, *et al.*)(Gillies *et al.*).

2.3 Atypical imaging presentations of ICIs-treated patients

According to World Health Organization (WHO) criteria and the Response Evaluation Criteria in Solid Tumors (RECIST), a reduction in lesion size or the absence of new lesions indicates a positive response to cancer treatment. Conversely, an increase in lesion size or the appearance of new lesions indicates disease progression (PD), which signals treatment failure and leads to treatment discontinuation. Complete tumor disappearance (complete remission), reduction in tumor size and numerical stability (stable disease), partial remission, and disease progression are established patterns of response to treatment (Beer *et al.*).

In patients receiving ICIs treatment, an initial increase in tumor size followed by a delayed response, or the appearance of new lesions before a decrease in tumor size, may occur, but these response patterns do not necessarily indicate disease progression (Beer *et al.*) This phenomenon is referred to as pseudoprogression and occurs in less than 10% of cancer patients receiving ICI treatment, and specifically in approximately 5% of patients with NSCLC (Park, *et al.*). Pseudoprogression is defined as an increase of more than 20% (at least 5 mm) in

the sum of the longest diameters relative to the lowest point, progression of a nontarget lesion, or the appearance of a new lesion not identified in follow-up studies (Sheikhbahaei *et al.*; Tazdait *et al.*). This is based on biological factors such as transient immune cell infiltration, necrosis, or edema, or early tumor growth before the immune response is fully activated (Chiou and Burotto)(L. Zhou *et al.*). Therefore, radiologically, tumor cell size may appear to increase and may be mistaken for tumor cell proliferation (Chiou and Burotto)(Gopalakrishnan *et al.*). Furthermore, new lesions observed in pseudolesions are typically caused by lymphocytic infiltration into the tumor area, which is not detected on baseline imaging (L. Zhou *et al.*)(Ma *et al.*). Pseudoprogression cannot be predicted by PD-L1 expression, and the onset of pseudoprogression can vary from day to month after initiating ICIs treatment (Park *et al.*). Therefore, the development of radiologically relevant criteria for immune response to determine pseudoprogression. Additionally, these new criteria recommend performing additional imaging evaluations 4 to 8 weeks later, rather than immediately declaring disease progression (Fujimoto *et al.*).

Hyperprogression reflects severe tumor growth that occurs early after ICIs treatment and is associated with poor overall survival, high metastatic burden, significantly shortened survival, and poor performance status (Kas *et al.*). Translational and *in vivo* studies have been conducted to elucidate the biological mechanisms of hyperprogression. These include T regulatory cell proliferation, T effector cell depletion, modulation of tumor-promoting immune cell subpopulations, aberrant inflammation, and activation of oncogenic pathways. However, no definitive answers have been found (Ding *et al.*). The incidence of hyperprogression has been reported to range widely, from 5.9% to 43.1%, and this is likely due to differences in hyperprogression assessment methods, necessitating the establishment of unified criteria (Park, *et al.*)(Stéphane Champiat *et al.*)(S Champiat *et al.*).

Simultaneous shrinkage and progression of multiple lesions detected on the same imaging study, or a decrease or less than 20% increase in the size of a target lesion from baseline after the appearance of a new lesion, is called a dissociated response and may be related to tumor heterogeneity or differences in drug distribution between tissues (Humbert and Chardin). Figure 1 illustrates the adverse reactions that may occur in NSCLC patients receiving ICI treatment.

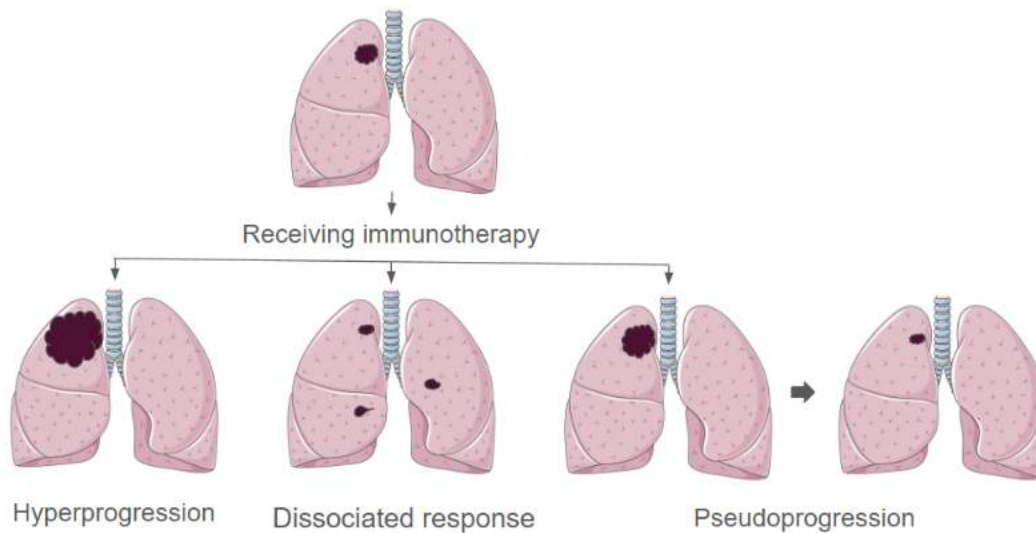


Figure1. Atypical presentations in patients with NSCLC treated with ICIs. Hyperprogression refers to severe tumor growth that occurs after ICI treatment and is associated with poor prognosis. Dissociation response is the simultaneous reduction and progression of multiple lesions observed in the same imaging study. Pseudoprogression is a phenomenon in which the tumor initially increases in size, then shows a delayed response, or new lesions appear before the tumor's size decreases.

2.4 Radiomic and atypical response to immunotherapy

To avoid the unfortunate misdiagnosis of pseudoprogression as disease progression and subsequent treatment interruption, timely differentiation between pseudoprogression and true disease progression is essential for clinical decision-making. Features extracted from radiomics have been shown to be useful in distinguishing inflammatory infiltrates from tumor cell proliferation (H. Zhou *et al.*)(Basler *et al.*). evaluated the ability of radiomics features, lesion volume, and blood markers to distinguish pseudoprogression from true disease progression at 3 months. Among them, the blood-radiomics model showed the best predictive performance, suggesting that it is a useful biomarker for early detection of pseudoprogression (Basler *et al.*). Ji *et al.* have used radiomics to distinguish between pseudoprogression and true disease progression, demonstrating significant area under the curve (AUC), 0.806 (Ji *et al.*). Identifying high-risk patients is crucial because hyperprogression indicates a poor prognosis after ICIs treatment. However, no predictive biomarker has yet been shown to reliably identify hyperprogression risk (S Champiat *et al.*)(Petrova *et al.*). To assess the accuracy of predicting progression in patients with NSCLC treated with ICIs, Vaidya *et al.* conducted a study using a noninvasive pretreatment CT-based radiomics model. They found that the texture and vascular pattern of the peritumoral tissue on baseline CT scans were positively correlated with progression (Vaidya *et al.*). Several studies have shown that composite models utilizing radiomics model can model can accurately predict hyperprogression, with outcomes comparing standard progression disease (AUC 0.933) (He *et al.*) or non-hyperprogression (AUC 0.865) (Valentinuzzi *et al.*).

3. Habitat Radiomics

3.1 Habitat Radiomics as an Extension of Traditional Radiomics

Although radiomics features have demonstrated excellent performance in predicting the efficacy of ICIs treatment in lung cancer, these models fail to capture the spatial heterogeneity of tumors (Ye *et al.*). Because intratumoral heterogeneity is complex and challenging to comprehensively study immunotherapy, understanding intratumoral heterogeneity is crucial for evaluating clinical efficacy and discovering new targets for cancer immunotherapy (Hinohara and Polyak)(Dagogo-Jack and Shaw). In addition, tumor cells can promote proliferation and survival by changing the surrounding microenvironment, and the microenvironment can also affect tumor cells. Therefore, diagnosis methods that reflects the habitat are needed for accurate diagnosis and prognosis prediction (Tabassum *et al.*).

NSCLC shows diverse histological subtypes, and this complexity complicates treatment plans planning and prognosis prediction (Ettinger *et al.*). While traditional radiomics focuses only on the tumor itself without considering histological diversity, habitat radiomics focuses on reflecting spatial heterogeneity and biological features within the tumor (Tabassum *et al.*). In other words, while traditional radiomics analyzes the overall characteristics of tumors, such as shape, texture, and signal intensity, habitat radiomics divides tumors into several areas that reflect biological changes in tissues, such as necrosis, hypoxia, and high cell density. Based on traditional radiomics, habitat radiomics clusters similar features to identify different subregions or cell subpopulations clearly (X. Zhang *et al.*)(Prior *et al.*). Researchers validated the histological relevance of habitat radiomics by matching pixel-level spatial imaging data with histological findings (Park, *et al.*). Reflecting that tumors are not homogeneous and are surrounded by a diverse microenvironment that influences treatment response and outcome, habitat radiomics considers the distribution of features across the entire tumor periphery. This provides more detailed and dynamic information about the tumor, improves predictive accuracy, and enables the development of personalized treatment plans (Cai *et al.*).

3.2 Habitat Radiomics Model Construction

In a typical workflow for habitat radiomics, images are first acquired from CT, MRI, and PET/CT. By extracting radiomics features from each tumor VOIs pixel using specialized software, quantitative radiomics features of each pixel can be visualized for each pixel. The K-means clustering algorithm is used to cluster the radiomics features of each pixel within the tumor VOIs. The Calinsky-Harrabasz index (CH) is used to determine the optimal number of subregions (Ye *et al.*). These features can be broadly categorized into three-dimensional geometric shape, image intensity, and texture features (Lambin, Rios-Velazquez, *et al.*). Independent samples t-tests can be used to identify features that significantly differ between the subjects being compared. Pearson correlation coefficients can be calculated to eliminate the potential influence of habitat volume on radiomics features and identify highly correlated features, allowing for correlation analysis with clinically meaningful outcomes. The selected

factors are then combined with radiomics features. Using the the least absolute shrinkage and selection operator (LASSO) algorithm, 10-fold internal cross-validation based on a minimum criterion is used to construct a model with improved generalization performance. This data is analyzed and correlated with clinically meaningful outcomes (Dong *et al.*)(Shang *et al.*). After statistical or machine learning techniques are used, predictions or classifications for specific oncological outcomes can be generated. In machine learning, classification is a supervised learning task that infers functions from labeled training data. Various classification algorithms, such as logistic regression (LR), random forests (RF), and support vector machines (SVMs), can be used to build predictive models (Kothari *et al.*). Model performance can be assessed using receiver operating characteristics (ROC), area under the curve (AUC), accuracy, sensitivity, specificity, positive predictive value, and negative predictive value. Decision curve analysis (DCA) can be used to evaluate a model's clinical benefit across different risk thresholds (Ye *et al.*). These tools can be used to assess whether a model has clinical significance. Figure 2 shows the process of habitat radiomics through a schematic diagram.

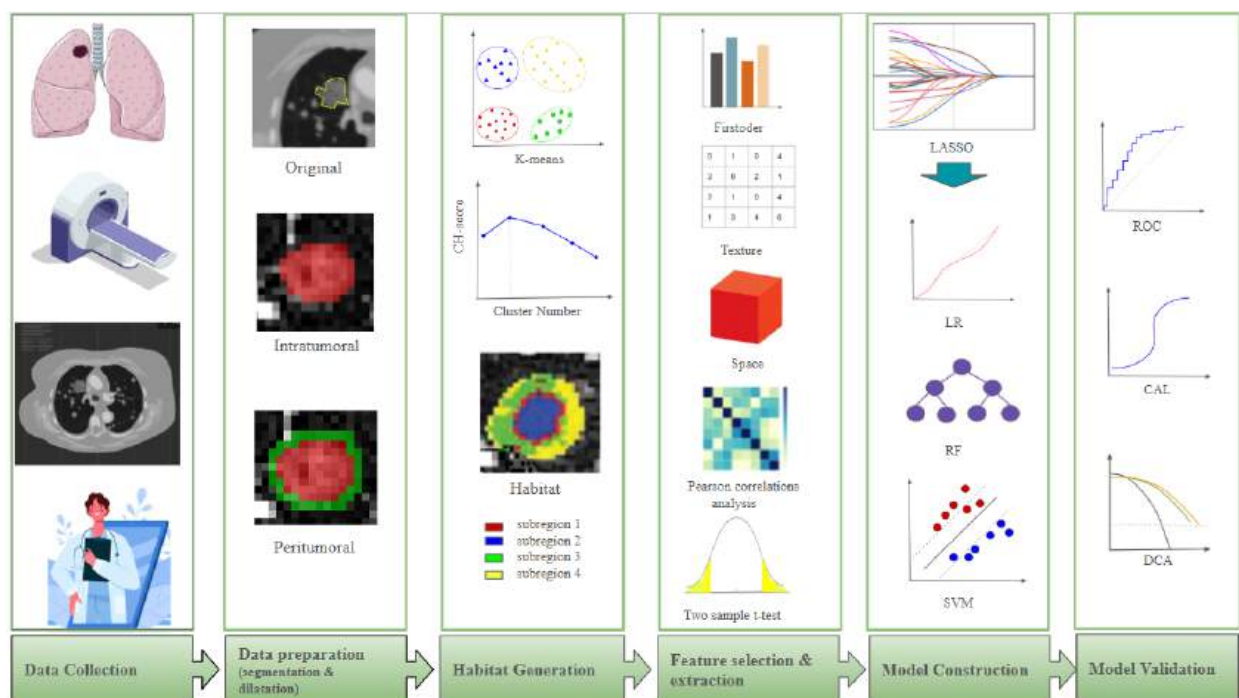


Figure 2: The overall workflow of habitat radiomics. CAL, calibration; CH, Calinski-Harabasz; DCA, decision curve analysis; LASSO, least absolute shrinkage and selection operator; LR, Logistic Regression; RF, Random Forest; ROC, receiver operating characteristic; SVM, Support Vector Machine.

3.3 Habitat Radiomics in NSCLC Immunotherapy: Current Studies

In a study by Zhang *et al.* to predict the efficacy of first-line ICIs-based therapy in patients with NSCLC, habitat radiomics more accurately depicted the relationship between tumor internal structures and immunotherapy response than traditional radiomics. This approach, which

integrates the tumor proportion score, clinical characteristics, and habitat radiomics, emerged as a new method that significantly improved predictive performance (Z. Zhang *et al.*). In a radiomics study conducted by Wu *et al.*, a habitat radiomics model focused on tumor subregion showed better predictive performance than a model derived from the entire tumor (Y. Wu *et al.*). Huang *et al.* reported that habitat radiomics (AUC 0.889) had strong predictive superiority compared to traditional radiomics (AUC 0.766), 3D deep learning (AUC 0.733), clinical (AUC 0.625), and PD-L1 expression models (AUC 0.714) (Huang *et al.*). Khorrami *et al.* focused on predicting cancer recurrence in early-stage NSCLC patients using a habitat model that integrates features based on stability and discriminant criteria. The model demonstrated high discriminatory power with an AUC of approximately 0.76 in the validation dataset and proved effective in predicting cancer recurrence after surgery (Khorrami, Bera, *et al.*). These studies provide a solid foundation for applying habitat models to personalized treatment planning, particularly in NSCLC patients. Table 1 shows studies that performed traditional or habitat radiomics in NSCLC patients who received immunotherapy.

Table 1. Summary of radiomics studies in patients with non-small cell lung cancer treated with immune checkpoint inhibitors.

Study	Pipeline (software / extraction / statistics)	Radiomics approach	Volume of interest	No. of patients	Endpoint	AUC	Model characteristics
Mu et al., 2020(Mu et al.)	PET/CT radiomics; resampling; LASSO; Cox regression	Multimodal PET/CT + KLD + clinical nomogram	Primary lung tumor	194	DCB, PFS, OS	0.83 (retrospectiv e test), 0.81 (prospective test); 0.86 with clinical integration	Multimodal PET/CT radiomics signature (mpRS) integrating PET, CT, and KLD features; combined with clinical variables in a nomogram
Valentinuzzi et al., 2020(Valent inuzzi et al.)	PET texture features; z-score normalization; logistic regression	Baseline PET radiomics signature	Primary lung tumor	30	Response prediction; survival-rel ated outcome	0.90	Baseline PET radiomics signature derived from primary tumors in pembrolizumab-treated patients
Zhu et al., 2023(Zhu et al.)	Multi-lesion segmentation; attention-based MIL; ridge regression	Weighted multi-lesion radiomics + clinical model	Multiple intrapulmo nary lesions	233	DCB	0.80 (radiomics alone), 0.86 (clinical + radiomics)	Multiple-lesion radiomics model using attention-based multiple-instance learning; combined with clinical features
Farina et al., 2023(Farina et al.)	syngo.via/3D Slicer; PyRadiomics; longitudinal analysis	Deep-radiomi cs + clinical ensemble	Primary tumor or largest lesion	264	DCB, OS	0.74 (PFS6), 0.70 (PFS9)	Longitudinal deep-radiomics model integrating baseline and early follow-up CT with clinical variables
Cousin et al., 2023(Cousi n et al.)	Baseline + on-treatment CT; RF/linear models; Cox regression	Delta-radiomi cs model	Index lesion or all RECIST target lesions	188	6-month response; OS	0.80 (RF), 0.77 (GLM)	Lesion-based and delta-radiomics models; delta-radiomics outperformed baseline-only approaches

Janzen et al., 2025(Janzen et al.)	In-house MATLAB; core/edge/ring features; logistic regression	Peritumoral radiomics + clinical model	Tumor-centered lesion with peritumoral ring	114	Disease progression vs. disease control	0.81	Peritumoral radiomics model for first-line pembrolizumab response prediction, combined with clinical variables
Cai et al., 2024(Cai et al.)	K-means habitats; 3D ResNet; LASSO; SVM	Hybrid habitat radiomics/deep learning	Intratumoral habitats	246	Immunotherapy response / durable benefit	0.865	Habitat-based radiomics with K-means-defined subregions combined with 3D deep learning features
Wu et al., 2025(Y. Wu et al.)	K-means + ReMap; PyRadiomics; LASSO; LR/XGBoost	Habitat nomogram + clinical features	Whole tumor subdivided into 3 habitats	343	Prognostic outcome; irAE prediction	0.817 (internal), 0.899 (external)	Habitat radiomics and whole tumor radiomics nomogram integrating clinical features
Zhang et al., 2025(Z. Zhang et al.)	Arterial-phase CECT; habitat clustering; PD-L1/clinical integration	Habitat imaging integrated model	Intratumoral habitat subregions	220	PFS / outcome prediction for first-line immunotherapy	0.758; 0.869 with integration	Multi-institutional habitat imaging model using arterial-phase CECT; integrated model combined habitat imaging with PD-L1 TPS and clinical metrics
Huang et al., 2025(Huang et al.)	ITK-SNAP; K-means; PyRadiomics; multi-classifier comparison	Multicenter habitat radiomics model	Whole tumor with limited peritumoral context, subdivided into 3 habitats	590	Immunotherapy benefit / response prediction	0.814	Habitat-based machine learning model using K-means-defined subregions and multiparametric CT-derived features

Abbreviations: RF, Random Forest; PFS, Progression-Free Survival; GLM: Generalized Linear Model; OS, Overall Survival; DCB, Durable Clinical Benefit; VOI, Volume of Interest; irAE, Immune-Related Adverse Event; KLD, Kullback–Leibler divergence

Tumor-infiltrating lymphocytes (TILs) in the tumor microenvironment play an essential role in the immune response to cancer. As a component of anti-tumor immunity, tumor infiltration of CD8+ T cells is known to be associated with better outcomes of cancer immunotherapy (Tay *et al.*). It has been demonstrated that NSCLC patients with high TIL infiltration exhibited treatment response and survival benefits after immunochemotherapy (Park, Ock, Kim, *et al.*). The prognostic significance of TILs and the clinical need for noninvasive monitoring tools have been recognized. In a study using pre- and post-treatment CT images of 135 NSCLC patients treated with ICIs, Khorrami *et al.* confirmed the concordance of radiomics with TIL infiltration by comparing them with TIL infiltration in biopsy samples and showed that radiomics were associated with response and overall survival in NSCLC patients receiving ICIs (Khorrami, Prasanna, *et al.*). Giulia Mazzaschi *et al.* found that texture, effect, and margin of cancer on noninvasive CT scans are directly associated with PD-L1 expression and tumor-infiltrating lymphocytes (TILs). They developed a predictive model based on CT-extracted features, which were also correlated with the prognosis of NSCLC patients (Mazzaschi *et al.*). In a study by Geng *et al.*, a habitat radiomics analysis was used to predict TIL status after neoadjuvant immunochemotherapy in patients with NSCLC, demonstrating excellent

predictive performance (Geng *et al.*). This study demonstrated that habitat radiomics and machine learning can provide interpretable imaging biomarkers that can guide personalized immunotherapy.

Considering the diverse and overlapping CT findings of lung adenocarcinoma, it is challenging for radiologists to predict the invasiveness of cancers associated with poor prognosis accurately. Growing nodule size often reflects the proliferation and increased invasiveness of tumor cells (Wang *et al.*). In addition, larger nodules are more likely to contain solid components and exhibit invasive growth patterns (Kakinuma *et al.*). For the classification of invasiveness of lung adenocarcinoma over the past decade, traditional radiomics studies assuming a uniform distribution of features within the region of interest (ROI) have been conducted (Jiang *et al.*). Radiomics of tumor peripheries has been found to be useful in predicting the invasiveness of lung adenocarcinoma (L. Wu *et al.*). Different tumor habitats exhibit distinct growth and invasive patterns, with different responses to treatment (Lee *et al.*). Shang *et al.* created a habitat based on the index of similarity between voxels to maintain spatial heterogeneity and predicted the invasive nature of lung cancer by extracting radiological features. These habitats included both tumor and tumor peripheries, and compared with non-invasive tumors, invasive tumors showed a more regular distribution of tumor habitat, largely classified into central, peripheral, and peritumoral regions. They suggest that the lung cancer microenvironment differs between invasive and non-invasive tumors, and that these differences can be assessed using habitat radiomics (Shang *et al.*). These studies demonstrate that habitat radiomics can quantitatively assess the invasiveness of lung adenocarcinomas, potentially improving the accuracy of treatment planning and prognosis.

irAEs related to immunotherapy are caused by the activation of an immune response against healthy tissue and can affect any organ or system. Timely diagnosis and prompt management are crucial, depending on the severity of irAE. If left untreated, irAEs can develop into life-threatening complications. Therefore, early diagnosis and monitoring of irAEs are important (Reynolds *et al.*). Blood test markers, clinical symptoms, and medical imaging, including CT, ultrasound, magnetic resonance imaging (MRI), X-ray, and PET/CT, are used to detect irAEs, but early detection of irAE remains challenging. Radiomics can help identify early, invisible signs of irAEs (H. Zhou *et al.*). According to Wu *et al.*, a predictive model using CT-based habitat radiomics and machine learning techniques demonstrated excellent performance in predicting treatment responses and irAEs to immunotherapy in NSCLC, with internal and external AUC validation. Habitat radiomics not only provides independent prognostic information beyond clinical factors but also predicts the occurrence of irAEs in NSCLC. Immunotherapy may be recommended for patients at low risk for irAEs and with a favorable response to ICIs. In contrast, patients at high risk for irAEs and with an inadequate response to immunotherapy may benefit from cautious administration of ICIs and closer monitoring for irAEs. Habitat radiomics is expected to improve the screening of patients with NSCLC and to help distinguish those most likely to benefit from immunotherapy from those at high risk for irAEs (Y. Wu *et al.*). This will accelerate the development of precision

immunotherapy for NSCLC and provide valuable information for the development and management of follow-up treatment plans for personalized treatment.

3.3 Limitations of habitat radiomics and future direction

Since most habitat radiology studies rely on retrospective analysis of patient data, they can be susceptible to data bias and potentially influence outcomes. Furthermore, retrospective analysis often results in a significant reduction in the number of cases, thereby reducing the sample size required to predict long-term prognosis. Multicenter retrospective studies of habitat radiology have typically shown significant differences in image acquisition parameters using multiple models. Furthermore, the use of multiple models in retrospective studies can lead to poor reproducibility, resulting in suboptimal study design, high technical complexity, data overfitting, and incomplete reporting. Habitat radiomics are still in their infancy, and further studies must be conducted to identify the optimal model. Patient positioning, section thickness, pixel or matrix size, and other factors vary across studies due to the different software used. Furthermore, institutions use variable image extraction methods, resulting in differences in reconstruction algorithms. Statistical methods used vary across studies (Table1). These make comparisons difficult and reduce reproducibility. Therefore, large-scale studies using AI models with unified image acquisition methods and consistent reconstruction algorithms are essential. Manual tumor delineation is highly variable, both reader-to-reader and intra-reader, and labor-intensive, making it unsuitable for large-scale dataset analysis required in radiology. Future studies should consider introducing automatic tumor segmentation methods to improve feature extraction efficiency and ensure stability and consistency of the process. Additionally, habitat radiomics have shown differences in the definition, size, and number of subregions. Therefore, research is needed to develop a standardized method for defining subregions. To minimize these issues, large-scale, multicenter, prospective studies that use standardized data collection, preprocessing, and expert-annotated validation are needed. Habitat radiology can continuously learn, and advance based on data acquired through this standardized infrastructure and calibration.

To enhance the biological significance of habitat images, methods for incorporating more features into habitat generation should be explored. Furthermore, it is necessary to study the association with less studied tissue biomarkers, such as tumor mutational burden, beyond PD-L1. Studies using tumor habitat radiomics to predict irAEs after ICI treatment have been limited due to the complex and unknown mechanisms of irAEs. Combining habitat radiomics with other clinical factors is necessary to predict life-threatening irAEs. Studies using habitat radiomics to quantify tumor mutations burden and predict irAEs in patients with NSCLC could contribute to personalized dosing and standardization of ICI treatment, thereby improving patient prognosis.

4. Conclusion

Habitat radiomics has the potential to predict prognosis and facilitate clinical

decision-making noninvasively, efficiently, economically, and rapidly for NSCLC patients receiving ICIs treatment. Identifying tumor heterogeneity through habitat radiomics and machine learning provides interpretable imaging biomarkers that can improve diagnostic accuracy and guide personalized immunotherapy plans. Large-scale, prospective, multi-center clinical trials are needed to move habitat radiomics beyond the trial phase and establish a practical habitat radiomics platform for personalized immunotherapy in NSCLC. This requires fully automated and reproducible tumor and habitat segmentation, standardized image acquisition and reconstruction algorithms across scanners and institutions, and validated AI models integrated into clinical decision-making workflows. With standardized infrastructure and calibration, habitat radiomics can continuously learn from data and evolve into a core decision-support tool that actively supports patient selection, risk stratification, and treatment plans in personalized NSCLC immunotherapy.

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Through A Quantitative Correlational Study, How Does Proximity To Traditional Burial Sites Affect The Adherence To Groundwater Properties To Water Quality Standards As Well As Affect The Quantity Of Contaminants Compared To A Control in the Same Region? By Morgan Wagner

Introduction

In the United States alone, each year “the amount of wood needed to produce enough coffins...equates to building 4.5 million homes” while “1.6 million tons of reinforced concrete and over 100,000 tons of steel” are utilized to meet the demand of coffins and headstones (Forstmann et al. 2023). These figures help to display the prominent use of traditional burials currently, in one country alone. As a result of how heavily resources are allocated to these industries, environmental considerations have risen as a priority to global powers, with people around the world asking how to create more sustainable industries, such as the essential and inevitable funeral industry. Across various case studies, connections between traditional burial methods and environmental harm have arisen. In many cases, areas surrounding traditional burials have shown signs of contamination from the resources used in this burial method such as the materials for coffins and embalming fluids. As a result, scientists continue to investigate how proximity to cemeteries affects the quality of groundwater. While there are many individual case studies examining groundwater properties in close proximity to traditional burial sites, there is a need to examine groundwater properties out of proximity from cemeteries in similar regions to establish a better control and disprove a null hypothesis that there is no correlation between proximity and contamination. This would help to confirm that traditional cemeteries are sources of groundwater contamination globally, and contamination results are not caused by additional unrelated factors. The study will also further define how contaminants from traditional burials such as heavy metals and embalming fluids affect the environment. Most importantly it will help to establish if there is a need for alternative burial methods in order to preserve the natural environment.

Traditional Burial Sites Definition

Before understanding the environmental issues that traditional burials cause, it is important to give clear boundaries for what they are. Traditional burials sites overall are cemeteries that cause significant disturbances to the local ecosystem. This results from manmade burial markers, embalming of the bodies, and the use of nonbiodegradable coffins (Marshall 2011). Only those that reject all these criteria would be considered eco-friendly green burials, avoiding the negative impacts of traditional burial practices such as embalming.

Necessity of Groundwater

Before delving into the effects that practices like embalming leave on groundwater, it is also important to establish the importance of groundwater and the importance of it remaining pure. Around the world people depend upon resources such as groundwater to survive as it is the

leading source of accessible freshwater, accounting “for 99% of liquid freshwater on Earth and is the source of one quarter of all the water used by humans” (UNESCO 2022). This water is accessible to rural populations and provides about half of the water used globally for domestic purposes. In addition, water contamination is one of the highest global sources of health risks as “around 1.5 billion...rely on groundwater resources to fulfill their water needs” (Razzaq et al. 2020). Overall, a reported 75% of diseases “are caused precisely by the consumption of poor-quality water” (Mikhaylichenko et al. 2023). This displays how critical groundwater contamination may be to the communities that depend upon it and potentially lethal, depending on the scale of the contamination.

Environmental Impact of Embalming Fluids

Along with the potentially wasteful allocation of resources concerning traditional burials, an additional concern is the persistence of commonly used embalming fluids in the environment. Long after a corpse has decomposed, these chemicals, used to preserve the body from decomposition after death, remain in the ground contaminating it. The depth at which these caskets are lowered into the ground allows a significantly higher chance for contamination as these chemicals permeate through the soil and eventually find their way into the surrounding groundwater. Usually, these fluids contain harsh chemicals such as “a carcinogenic formaldehyde-based solution” that can even be unsafe for the morticians exposed to it (Forstmann et al. 2023). Because of these toxic components of these embalming fluids, there is a significant risk of disease in communities surrounding these traditional burial sites that depend upon the groundwater for their survival. Even now banned chemicals used in the embalming process still hold a great impact on communities in the present.

This issue has been very prevalent with Civil War era cemeteries, for example, in which many embalming fluids often contained an even more potent and lethal substance than formaldehyde-arsenic before it was banned in the 1900s (Chiappelli 2008). Substances such as this serve a severe risk to those who use groundwater, especially if unfiltered. The fact that arsenic is no longer allowed in embalming fluids does not negate its current environmental impact. Those living in communities surrounding these types of cemeteries still face the risk of having their groundwater contaminated by arsenic.

Current Groundwater Investigations

Currently, groundwater contamination from cemeteries has been found around the world and tested in a variety of ways. For example, one study based in Benin City, Nigeria investigated the sources of contamination for groundwater in the area, collecting thirty samples (Enzenwa et al 2023). Heavy metal contamination was predominantly found across the different wells and was linked to contamination leaking from a nearby burial site. The contaminated water was measured by the World Health Organization, Environmental Protection Agency (EPA), and national standards as poor quality and not suitable for consumption. The contaminants displayed a high carcinogenic risk to citizens of those areas and residents needed an alternative source of

water immediately. Another study conducted in Tennessee analyzed the prevalence of arsenic and formaldehyde in cemetery plots of different ages examining both soil and groundwater. In the end across eight samples including both soil and groundwater, it was discovered that only one sample presented a significant formaldehyde prevalence (Richardson et al 2024). This study implied that there may be a lower risk of contamination from formaldehyde and arsenic from burial sites than previously assumed. This contrast highlights the variability in contamination across different sites and underscores the importance of including a geographically similar control to isolate proximity as the true cause of contamination rather than site-specific factors.

Standards of Water Quality

There are various standards that indicate the quality of water. The Canadian Model of Water Quality or CCME QWI model is “widely used worldwide by researchers to evaluate water sources and determine the degree of contamination” as well as the guidelines of the United States EPA (Al-Saffawi 2019). Additionally, it is common for countries to set and maintain their own individual standards. For the CCME QWI model it detects contamination through comparing the number of individual tests that exceed the standards over the total number of variables and the total number of tests and several other calculations to assess water quality on a 100-point scale. It individually tests the pH of the water, total dissolved solids, total alkalinity, calcium, magnesium, total hardness, sodium, potassium, sulfate, and nitrite. These measurements are particularly relevant to this study as they provide a standardized framework for comparing groundwater quality across different geographic conditions. While the CCME model provides a broad assessment of water quality, cemetery-specific contamination studies also focus on heavy metals particularly associated with burial practices. While the CCME is a widely utilized model for comparing groundwater quality it is also important to note that many other studies conduct chi-square tests after comparing these models to the EPA/national standards in order to get a more detailed comparison based on individual heavy metals or contaminants. Specifically in the case study of Benin City, international guidelines such as EPA standards and national standards are both regarded as important for finding accurate reflections of the data presented (Enzenwa et al 2023).

Gap In Research

While individual case studies of these traditional cemeteries potentially causing water contamination have been examined, the connection between traditional cemeteries and groundwater needs further verification. There is a need for direct comparison between a proper control and one of these sites. Many studies individually observe the properties of groundwater in proximity to traditional burials but lack the use of a control in the same region to compare results to properly solidify that proximity causes groundwater contamination and are not regional factors. For example, one of the previous studies I referenced in Nigeria examines data from locations only within proximity to burial sites and uses the EPA and Nigerian standards for water quality as a type of control to compare rather than other groundwater sources distant from any

burial sites. The addition of a control of a groundwater sample distant from any traditional burial sites would help to establish that the proximity to one of these sites is the cause of contamination. This gap exists in many studies due to the scope of further resources; it would require studying and analyzing an additional sample location through primary research. This study seeks to address this gap by answering the following question: Through a quantitative correlational study, how does proximity to traditional burial sites affect the adherence of groundwater properties to water quality standards as well as affect the quantity of contaminants compared to a control in the same region? Through addressing this gap, it can be better analyzed whether or not the close proximity to traditional burial sites is the true cause of this contamination or potential unrelated outside factors.

Study Selection Criteria

Based on preliminary research, the most efficient design is to study the differences in contamination of groundwater based on proximity to traditional burials through a correlational study. I am unable to conduct a primary study, considering the limitations on access to wells for sampling and the proper instruments for examination. Therefore, I will examine different primary studies and will collect data on the properties of contaminated groundwater in different areas around the world. I will search through different academic databases such as EBSCOhost and Research Rabbit to find different studies examining groundwater properties. I will search for two studies per country: one within proximity of the traditional burial site and one outside of proximity of the site. The EPA defines the distance at which contaminants affect groundwater as approximately one mile (EPA 2015). I broadened these criteria to five-mile radius as multiple studies I investigated had a larger affected radius around the site and to ensure that I had access to viable data for my experiment. I also believed that a five-mile radius would serve better to ensure the control is not affected by any burial sites. The criteria that the studies must be within the same country aims to best account for regional differences as to ensure groundwater property differences could be attributed to proximity to traditional burial sites and not the region. The five-mile radius was chosen to ensure proximity was a factor in one study and not the other.

Data Collection and Adjustment

After finding as many viable countries as possible, I will mark down in a chart all the overlapping properties examined in each study. The country that has the most overlap between the two studies within varying proximities will be the one I chose to study. After this process I will add the overlapping properties into one chart for the ability to easily compare quantities. I will then adjust values by averaging samples or converting units to make each study more accurately comparable such as in converting microsiemens to millisiemens for example.

Statistical Analysis

For my statistical analysis, I will conduct multiple comparisons to determine whether there is a significant difference in the number of contaminants found between the two sites. I will

determine how contaminated each site is through a comparison to the United States EPA standards, as they are more internationally recognized as well as the country's national standards secondarily. Additionally, I will use a chi-square test to determine if there is a significant difference in the number of contaminants found between the sites in-proximity and out-of-proximity. This type of test was found as best fit for comparison as one of the core studies examined in this comparison justified its use to determine a significant difference between the samples and standards (Enzenwa et al 2023). Chi-square is appropriate here as the data is categorical, comparing whether each property meets or fails established water quality standards. After accepting or rejecting the null hypothesis, I will also compare the individual quantities of contaminated materials to these standards to determine the degree of contamination.

Ethical Considerations

In order to ensure adherence to ethical standards, this research design was looked over and approved by the Institutional Review Board at my high school, which is registered with the National Institute of Health. See Appendix.

Findings

The following findings detail the process of selecting a case study location, adjusting raw data for comparison, and statistically analyzing the relationship between proximity to traditional burial sites and groundwater quality. After searching through research papers, looking for studies in and out of proximity of traditional burials in the same country, three countries appeared to be viable locations to base a case study on examining the role that proximity plays in groundwater quality: Tanzania, Brazil, Nigeria. Ultimately, I concluded that Nigeria would offer the best results to analyze as the papers shared 15 total common properties of groundwater available to analyze. In addition, the study of the in-proximity groundwater focused its effort on determining samples from two different locations, providing additional sampling data to interpret whether their proximity is a major factor in groundwater contamination. This case study provided the properties of electrical conductivity, pH, BOD₅, Cl, NO₃, SO₄, PO₄, Cd, Pb, Ni, Zn, Fe, Na, K, and Ca concentrations to examine in the study. Figure 1 below displays all of the different groundwater properties examined throughout all of the studies and indicates the presence of each within the different papers. The proximity of the locations to traditional burial sites also fit the criteria I was examining as those outside of proximity were over 20 miles away from the closest traditional burial site and that in proximity was within 1.5 miles of a burial site.

Figure 1

Ground Water Properties Studied in Each Paper

	Tanzania (In Proximity)	Tanzania (Out of Proximity)	Nigeria (In Proximity)	Nigeria (Out of Proximity)	Brazil (In Proximity)	Brazil (Out of Proximity)
Turbidity				X	X	X
EC	X	X	X	X	X	X

pH		X	X	X		X
TS				X		
TDS	X	X		X	X	
TH				X		
DO		X		X	X	
BOD5			X	X		
Cl		X	X	X		X
NO3	X	X	X	X		X
SO4	X	X	X	X		X
PO4	X		X	X		
Cd			X	X	X	
Pb			X	X	X	
Cr				X	X	
Ni			X	X	X	
Zn			X	X	X	
Fe			X	X		X
Na		X	X	X		
K		X	X	X		
Ca		X	X	X		X
Mg		X		X		X
Alkalinity			X			
NH4			X			
COD			X		X	
Mn			X			
Cu			X		X	
ORP					X	
NH3	X				X	
Total phenols					X	
Total phosphorous					X	
Salinity %	X					
NO3-N	X					
NO2	X					X
CO3		X				
F		X				X
HCO3		X				
Temp C		X				
RSC		X				
SAR		X				
MAR %		X				
KR		X				
Depth		X				X
Static Level						X
Dynamic Level						X
Flow rate						X

Color						X
Alkali-HCO						X
Alkali-CO						X
Alkali-OH						X
Silica						X
Hardness						X
Total Residue						X

After determining Nigeria as the best location of study, adjustments were made to the raw statistical data in order to provide a more accurate comparison between locations. For the in-proximity study, a total of 6 samples were taken between 2 separate locations each beside 2 different cemeteries. As the out of proximity was an average of data across different samples, I decided to take an average of the samples of each location at the in-proximity locations to best reflect this. In addition, only borehole data from out of proximity study was used, as the in-proximity study exclusively recorded borehole samples. Data found and to be used in a chi-square test comparison is listed in Figure 2.

Different properties were also taken out of the chi-square comparison between the different locations as there was an insufficient amount of data or standards for determining water quality based upon them did not exist. These properties include PO_4 , K, Cd, BOD_5 , Zn, and Ni as data regarded in the data table was marked ND (No Data) or information regarding water quality standards was not present in EPA or national guidelines.

Figure 2

Average Quantity of Different Ground Water Properties Based on Location and Proximity to Traditional Burial Site

Ground Water Properties	Location Out of Proximity	Location In Proximity (Cemetery 1)	Location In Proximity (Cemetery 2)
Electrical Conductivity (ms/cm)	530	.24	.04
pH	5.36	3.98	6.91
Cl mg/L	4.5	49.41	16.79
NO_3 mg/L	96.25	3.99	2.06
SO_4 mg/L	5.5	3.27	2.66
Pb mg/L	.04	.18	.13
Fe mg/L	.01	.52	1.51
Na mg/L	12.8	21.82	.47
Ca mg/L	21	15.83	3.69

In order to statistically analyze whether proximity contributed to the groundwater quality, values from both the United States Environmental Protection Agency and Nigerian Standard for Drinking Water Quality (NSDWQ) were used. The EPA's limits and values were used as the

main basis for determining groundwater quality and the NSDWQ as a supplement if EPA data was not provided for, represented by N/A. EPA guidelines were used primarily because, as referenced in the Standards for Water Quality section, they are applied not only nationally in America but globally as well. Figure 3 displays these guidelines.

Figure 3

NSDWQ and EPA Standards for Contaminant Levels

Property	NSDWQ	EPA
EC micro siemens/cm	1000	N/A
pH	6.5-8.5	6.5-8.5
Cl mg/L	250	250
NO ₃ mg/L	50	10
SO ₄ mg/L	100	250
Pb mg/L	.01	0
Fe mg/L	.3	.03
Na mg/L	200	30-60
Ca mg/L	200	N/A

Analysis

In order to account for the role proximity played in the number of different groundwater contaminants between the sites a chi-square test was conducted. It would help to determine if there was a significant difference in the numbers of groundwater contaminants that deviated from the standards for groundwater quality. Each value was assigned a YES/NO depending on whether or not it fell into the accepted ranges outlined by the EPA or NSDWQ for safe drinking water. NO was given if it fell outside of the range, and YES was given if the value fell within the range. The presence of indicators exceeding EPA and NSDWQ standards can be viewed in Figure 4.

Figure 4

Adherence of Ground Water Properties to EPA/NSDWQ Standards

Ground Water Properties	Location Out of Proximity	Location In Proximity (Cemetery 1)	Location In Proximity (Cemetery 2)
Electrical Conductivity (ms/cm)	No	No	No
pH	Yes	Yes	No
Cl mg/L	No	No	No
NO ₃ mg/L	Yes	No	No
SO ₄ mg/L	No	No	No
Pb mg/L	Yes	Yes	Yes
Fe mg/L	No	Yes	Yes
Na mg/L	No	No	No
Ca mg/L	No	No	No

For the first location out of proximity to a traditional burial site, three values fell outside of the accepted range: the lead, pH, and NO₃ values. In the second location within proximity to a traditional burial site, three values fell outside of the accepted range by the NSDWQ and EPA: the pH, Pb, and Fe values. In the third location within proximity to a traditional burial site, two values fell outside of the accepted range: the Pb and Fe values. The number of properties abiding by and exceeding EPA and national standards can be viewed in Figure 5.

Figure 5

Number of Groundwater Properties Abiding By and Exceeding EPA/National Water Quality Standards

	Location 1 (Out of Proximity)	Location 2 (In Proximity Cemetery #1)	Location 3 (In Proximity Cemetery #2)
Properties abiding by standards	6	6	7
Properties exceeding standards	3	3	2

Chi-Square Test

In order to determine whether or not a correlation exists between proximity to traditional burial sites and the number of contaminants present in groundwater a chi-square test was conducted. In Figure 5 a chart was created to track the number of properties that each location abided by and failed to adhere to EPA and NSDWQ standards. A p-value of .05 was used for testing.

After conducting the chi-square test, a value of .355 was calculated falling under the critical value of 5.991. These results conclude that the null hypothesis must be accepted that there is no correlation between the proximity to traditional burial sites and the number of contaminants exceeding EPA and NSDWQ standards.

While this test accounts for correlation between proximity and overall adherence to water quality standards, there is a need to more closely examine which specific variables exceeded those standards and the implications of such results.

Heavy Metal Contamination Indicators

It is of high importance to note the similarities of contaminants between both locations in proximity to traditional cemeteries. Both locations in proximity show excessive levels of iron and lead present. The presence of these specific contaminants shows consistency with the claim that traditional cemeteries may cause heavy metal contamination from resources used in coffin making. In the location in proximity to cemetery 1, the lead concentration was .18 mg/L about 18 times greater than the limits set by the NSDWQ. The location in proximity to cemetery 2 also remains 13 times greater than the limits set by the NSDWQ. While the location out of proximity

to traditional burial sites also exceeded the limits of .01 mg/L, it is important to note that the value was .04 mg/L only 4 times the limit.

These results suggest that lead contamination could be a common variable across Nigeria in their groundwater sources, as all samples show evidence of contamination, but, as evidenced by the comparisons of how much greater the contamination is in sites closer to traditional cemeteries, it could show that these sites contribute greatly to the lead concentration being that high. While lead at any level in potable water is a cause for concern, the levels found at sites in close proximity to traditional burial sites pose an even greater risk to human health. At such levels, cardiovascular effects, decreased kidney function, and reproductive problems are at high risk for adults, while pregnant women could face reduced growth of the fetus and premature birth. Children are the most susceptible to lead poisoning and risk behavioral and learning problems, lower IQ and hyperactivity, slowed growth, hearing problems, and anemia from the consumption of water such as this (EPA 2018).

Another heavy metal evident only in the locations in proximity to traditional burial sites was iron. The presence of iron only at burial sites indicates that there may be a correlation between proximity to traditional cemeteries and excessive iron concentration. While this is a possibility, there is, along with the previous indications of higher lead concentrations in groundwater in closer proximity to traditional cemeteries, more tests are needed to determine this. At the out of proximity site, the iron concentration fell well within the limits set by the EPA at a concentration of .01 mg/L compared to a limit of .03 mg/L. While in comparison both sites in proximity to a traditional burial site far exceeded this with a concentration of .52mg/L at the first location and 1.51 mg/L at the second location. The first location represents 17 times the limit, and the second location represents 50 times the limit. This shows the presence of significant iron contamination at both sites in proximity and a clear distinction between sites that could be attributed to differing proximities to cemeteries. Unlike high lead concentrations, high iron concentrations pose significantly less risk to human health (Minnesota Department of Health 2025).

Another notable factor in comparing the results of the groundwater studies is noting the pHs that fall out of range at the location out of proximity to a traditional burial site and at one of the locations in proximity to a traditional burial site. Both the EPA and the NSDWQ are specific that the pH range of groundwater should fall into about 6.5 - 8.5. Two locations fell out of this range as one of the in-proximity locations fell short at 3.98, and the out-of-proximity location fell short at 5.36. Both results show the water to be significantly acidic compared to the more neutral 6.5 -8.5 range. Acidic water can potentially cause significant risks to human health including in association with heavy metal leakage: diarrhea, nausea and vomiting, abdominal pain, chills, weakness, shortness of breath, suppression of the immune system, and organ damage (McGrane 2020). Most likely though, considering lack of evidence in previous studies of pH change, the pHs that fall out of the acceptable range are unrelated to the proximity to traditional burial sites.

Results

Throughout the results of the analysis, it is clear that in this case study, there is a quantifiable difference not just in the samples that come from a site in proximity or out of proximity to a traditional burial site, but even between the two locations in proximity. The results of the chi-square test confirm that the null hypothesis is accepted, meaning there is no correlation between the proximity to traditional burial sites and number of groundwater contaminants, and conflicts with my original hypothesis. Beyond the overall statistical results, examining individual contaminants reveals further patterns worth noting. While there was no significant difference found in the overall number of contaminants found at each site, I did note significant differences in individual contaminant quantities between the studies. One pertinent consistency found between both sample sets in proximity to a traditional burial site was significant values of heavy metal such as iron and lead found. In both cases, at both sites the quantities of the values far exceeded limits set by the EPA. This remained consistent with results from other studies indicating that heavy metal contamination is deeply associated with traditional burial site proximity. While excessive quantities of lead were found at the site out of proximity of a traditional site the quantity was far less than those found at sites in closer proximity and much closer to the limit. Across one of the studies in proximity to a traditional burial site and one study out of proximity pH was found outside of accepted range. This discrepancy is likely unrelated to the proximity to the traditional burial site and a product of other factors. Further research would be required to establish this as clearly associating to any proximity factors. Nitrate concentrations also exceeded limits set by the EPA. Again, like with pH, this is likely an unrelated factor and would need further evidence to prove a definite association.

Overall, my results appear to remain mostly consistent with previous findings developed by other researchers. Heavy metal contamination described in other studies appears to remain a consistent factor most potentially found in sites closer to traditional cemeteries rather than being a more established result of other factors affecting water quality in the region.

Limitations

The certainties of my results have limitations in terms of the primary data available for analysis, which is likewise limited in certainty to the accuracy of the methods of the experimenters conducting the experiments essential to the findings of my study. In order to formulate a more complete understanding of how proximity affects groundwater contamination many more sample points would be needed. An additional limitation is the possibility that observed contamination results from site-specific factors unrelated to burial site proximity rather than proximity itself. The seven-year gap between studies analyzed also presents another limitation in research as two studies conducted in the same year would likely provide the most accurate results as environmental conditions, land use, or burial practices may have changed over that period, potentially affecting the comparability of the data.

Future Research

This study opens several avenues for future research as results of this study serve implications for the field related to groundwater contamination. This study could serve as a reference point for those working on further mitigating contamination from essential groundwater resources and determining ways to protect human health from these hazards. Analyzing additional contamination indicators such as arsenic in future studies would also be essential in properly determining the connections between groundwater contamination and proximity to traditional burial sites. Additional primary research also is essential to determining correlations, specifically research conducted within the same year would be best for comparability. Multiple studies determining groundwater properties from different distances in proximity to the same traditional burial sites would also help indicate an even more concrete understanding into how traditional cemeteries may or may not contaminate groundwater and eliminate uncertainties that other factors, potentially regional, are affecting the supply instead.

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Correlation Between Post-Meal Glucose Dynamics and Meal Components By Hyelin Chae

Abstract

CGM studies often analyze meals as independent observations. We revisited the CGMacros dataset, which contains 45 adults monitored for 10 days with minute-level glucose data and detailed meal logs. This dataset allowed assessment of how meal macronutrients relate to post-meal glucose dynamics while accounting for within-participant correlation. We computed incremental area under the curve (AUC), relative peak rise, time to peak, and spike count for each meal from the provided aggregated metrics. Negative AUC values were retained. Metrics were also aggregated across meals for each participant. Associations between mean nutrient intake and mean glucose metrics were quantified using Spearman's rank correlation. A linear mixed-effects model with a random participant intercept was fitted to the meal-level data. Mean carbohydrate intake across participants was weakly correlated with mean AUC ($p = 0.23$) and relative peak rise ($p = 0.27$). Mean protein showed a modest correlation with relative peak rise ($p = 0.42$). The mixed-effects model estimated that each gram of carbohydrate was associated with a $28.6 \text{ mg} \cdot \text{dL}^{-1} \cdot \text{min}$ increase in AUC ($p < 0.001$). Protein showed a small negative association ($-13.2 \text{ mg} \cdot \text{dL}^{-1} \cdot \text{min}$ per gram, $p = 0.015$), whereas fat showed no significant effect. These modest associations highlight the variability of glycemic responses and emphasize the importance of appropriate statistical treatment for repeated measures.

Keywords

AUC, mixed-effects model, postprandial glycemia, continuous glucose monitoring, wearable sensors

1. Introduction

1.1 Background

Blood glucose regulation after eating, also known as postprandial glycemia, is critical for metabolic health. Elevated and prolonged post-meal glucose levels are associated with an increased risk of chronic diseases, including type 2 diabetes and cardiovascular disease. Prior work shows that individual factors such as age, body mass index (BMI), and gender modulate glucose responses. With aging, individuals often exhibit higher glucose excursions, partly due to increases in BMI. Specifically, each 10-year increase in age has been associated with a 2–3.5 mg/dL increase in peak postprandial glucose, and each 1-unit increase in BMI with a 0.7 mg/dL increase (Calvo-Malvar, et al.). Women tend to have higher late postprandial glucose levels than men (Calvo-Malvar, et al.).

Carbohydrates are a primary determinant of post-meal glucose increases. According to a study, proteins and fats can also influence response timing (Margaritis, et al.). High-fat intake may delay gastric emptying, which can alter glucose absorption kinetics. Increased protein intake raises the proportion of gluconeogenic amino acids, potentially increasing endogenous glucose

production 3–5 hours after feeding (Margaritis, et al.). Within approximately 3.5 hours of consuming high-protein meals, glucose levels may rise. An author reported that fatty meals can lead to a secondary rise in glucose levels within 2 hours after an initial decline (Abdou, et al.). However, relatively few studies have examined these effects under real-world conditions. Detailed data on interactions between macronutrients and personal factors, such as age and BMI, remain limited.

1.2 Gap

So far, few datasets combine continuous glucose monitoring with detailed dietary intake data in free-living adults. Studies describe the CGMacros dataset as a valuable resource, comprising 10-day records from participants with varying metabolic profiles consuming diverse macronutrient compositions (Gutierrez-Osuna, et al.). Wearable devices enable the collection of real-world physiological data.

Previous studies have rarely used multimodal data to examine correlations among these variables simultaneously. In particular, it remains unclear how carbohydrates, fats, and proteins each contribute to overall glucose exposure and peak responses under free-living conditions. Moreover, few studies have quantified multiple post-meal dynamics in relation to macronutrient composition in real-world settings. Many prior analyses examined one factor at a time and did not integrate meal-level and personal-level factors. Because variable interactions can influence glucose responses, consistent examination of these correlations is important. The CGMacros dataset helps address this gap by providing detailed meal logs and high-quality CGM data over ten days from 45 participants.

2. Methods

This study followed a multistep pipeline. We loaded the aggregated per-meal metrics file (GlucoseMealMetrics_Full.csv) and the combined CGM file (CGM_Combined_Rebuild_v3.csv). Negative AUC values were retained to avoid bias. We computed participant-level mean AUC, relative peak rise, time to peak, and spike count, as well as mean nutrient intakes, to account for within-subject correlation. Spearman’s rank correlation was used to quantify associations between mean nutrient intake and mean glucose metrics across participants.

We then fitted a linear mixed-effects model to the meal-level data with a random intercept for each participant. The model regressed AUC on carbohydrate, protein, and fat grams to estimate nutrient associations while accounting for repeated measures. Meals were divided into high- and low-nutrient groups using the median value for each nutrient. Welch’s t-tests and the nonparametric Wilcoxon test were used to compare mean AUC and peak rise between groups. We also compared AUC across meal types using one-way ANOVA.

2.1 Dataset

The study used the CGMacros (v1.0.0) dataset released on PhysioNet. It comprises 45 individuals (16 prediabetic, 15 normoglycemic, and 14 with type 2 diabetes). Data were collected between 2021 and 2024 (Gutierrez-Osuna, et al.). Participants had BMIs ranging from 21 to 46 kg/m² and were aged 18 to 69 years. They were instructed to wear continuous glucose monitors for ten days. Devices included the Dexcom G6 (5-minute intervals) and the Abbott FreeStyle Libre (15-minute intervals) (Gutierrez-Osuna, et al.).

Participants logged all meals using a smartphone app, including quantities of carbohydrates, proteins, fats, fiber, and sugar, and took before-and-after food photographs. Standardized breakfasts (protein shakes) and lunches (Chipotle meals covering a range of macronutrient compositions) were provided. Dinners were ad libitum. Participants were instructed to space meals by ≥ 3 hours (Gutierrez-Osuna, et al.). The dataset provides minute-level CGM glucose data and, at each meal timestamp, a full macronutrient breakdown. All participants provided informed consent under IRB-approved protocols.

2.2 Data Processing

Our analysis focused on the aggregated metrics file *GlucoseMealMetrics_Full.csv*, which contains one row per meal. Meals with missing nutrients or metric values were removed. Negative AUC values were retained, as they may occur when glucose falls below baseline, for example, after high-protein meals or during physical activity.

To account for repeated measures, metrics were aggregated at the participant level. For each individual, we computed mean values of AUC, RelPeak, TimeToPeak, SpikeCount, carbohydrates, protein, and fat intake.

2.3 Group Definitions

Demographic variables were described in the source data; however, the provided tables did not include individual subject IDs linked to these traits. Therefore, we did not form subject groups based on age, sex, or weight. Instead, for comparative analyses, meals were stratified by nutrient quantity. For each nutrient, the median value across all meals was calculated, and meals were labeled as high or low depending on whether their value was above or below the median.

Meals were also categorized by type: Breakfast, Lunch, Dinner, and Snack. Mean AUC and peak rise were compared between high- and low-nutrient groups and across meal types in cross-group analyses.

2.4 Data Analysis

All analyses were performed using Python 3.11 with the following libraries: pandas, numpy, scipy, and statsmodels. Spearman's rank correlation was used to assess monotonic relationships between participant-level mean nutrient intake and mean glucose metrics.

For the mixed-effects model, AUC was specified as the dependent variable, with carbohydrate, protein, and fat grams included as fixed effects. A random intercept for each participant accounted for between-participant variation. The model was fitted using restricted

maximum likelihood (REML), and coefficient estimates, standard errors, z-statistics, and p-values were extracted. To compare high- and low-nutrient groups, a two-sample t-test with Welch's correction for unequal variances was performed. When distributions were skewed, results were confirmed using the nonparametric Wilcoxon rank-sum test. A significance threshold of $p < 0.05$ was used. One-way ANOVA was conducted to test differences in mean AUC across breakfast, lunch, dinner, and snack categories. Pairwise Welch t-tests were performed if the ANOVA indicated significance.

3. Results

Our analysis yielded three main findings. First, at the participant level, mean carbohydrate intake showed only a modest positive association with mean AUC and peak rise ($p = 0.23$ – 0.27), as shown in Figure 1. Protein demonstrated a moderate positive association with peak rise, whereas no clear association with fat was observed. The mixed-effects model indicated that each gram of carbohydrate was associated with a $28.6 \text{ mg} \cdot \text{dL}^{-1} \cdot \text{min}$ increase in AUC ($p < 0.001$). Protein showed a small inverse association ($-13.2 \text{ mg} \cdot \text{dL}^{-1} \cdot \text{min}$ per gram, $p = 0.015$), while fat was not significantly associated with AUC. High-carbohydrate meals had significantly higher AUC than low-carbohydrate meals in unpaired t-tests, although the effect size was moderate.

Overall, these results suggest that carbohydrate intake contributes to glucose exposure; however, inter-individual variability remains substantial, and macronutrient effect sizes are modest. Recognizing such variability is important for personalized nutrition approaches and for applications that generate meal recommendations based on CGM data.

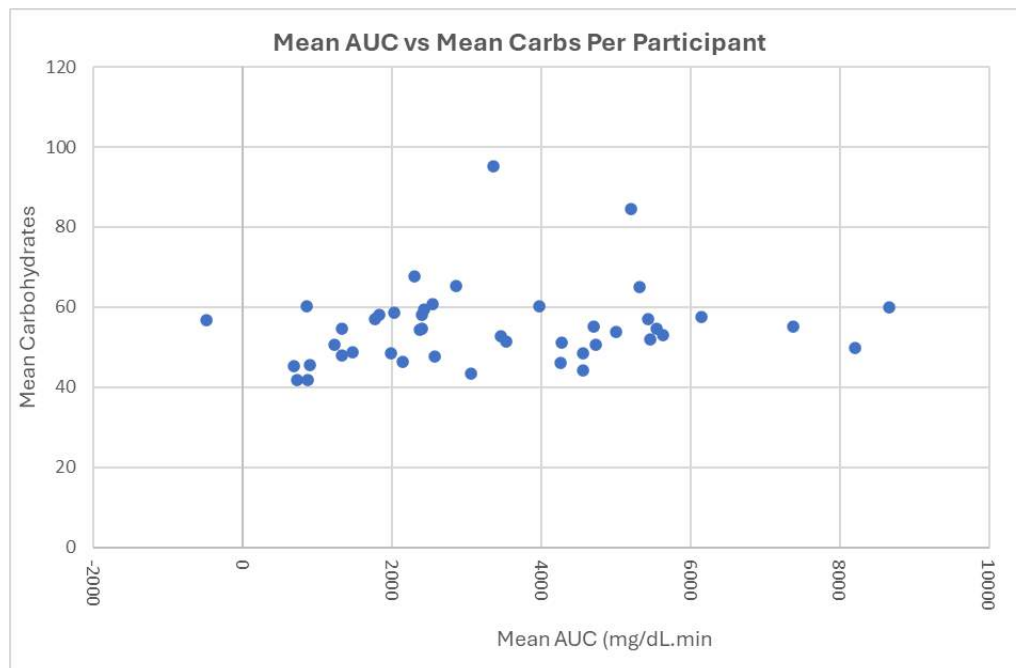


Figure 1. Scatterplot of Mean AUC vs Mean Carbohydrates per Participant.

The aggregated dataset contained 1,636 meals from 45 participants. Meals with missing values were removed, while meals with negative AUC values were retained. This resulted in 1,304 complete observations. The distribution of AUC values was right-skewed, as shown in Figure 2. The median AUC was 1,637 $\text{mg} \cdot \text{dL}^{-1} \cdot \text{min}$, although some meals had very high values (up to 25,856 $\text{mg} \cdot \text{dL}^{-1} \cdot \text{min}$). A few meals had negative AUC values as low as -28,986 $\text{mg} \cdot \text{dL}^{-1} \cdot \text{min}$. The mean relative peak rise across meals was 44.5 $\text{mg} \cdot \text{dL}^{-1}$ (SD 39.1), time to peak averaged 77.9 minutes (SD 50.8), and spike count averaged 1.18 spikes per meal. At the participant level, the mean AUC was 3,351 $\text{mg} \cdot \text{dL}^{-1} \cdot \text{min}$ (SD 4,673). Mean carbohydrate intake ranged from 15.3 g to 168.7 g per meal (mean 54.9 g; SD 34.4), mean protein intake ranged from 5.2 g to 129.1 g (mean 31.5 g; SD 25.2), and mean fat intake ranged from 0.9 g to 82.7 g (mean 21.0 g; SD 18.3). The table below summarizes these distributions.\

Table 1

Variable	Mean	SD	Min	Median	Max
AUC ($\text{mg} \cdot \text{dL}^{-1} \cdot \text{min}$)	3 351	4 673	-1 852	1 637	10 667
Relative peak ($\text{mg} \cdot \text{dL}^{-1}$)	46.6	32.9	-15.8	39.2	142.6
Time to peak (min)	79.8	48.0	17.2	72.5	180.0
Spike count (spikes/meal)	1.16	0.56	0.33	1.0	2.75
Mean carbohydrates (g)	54.9	34.4	15.3	43.1	168.7

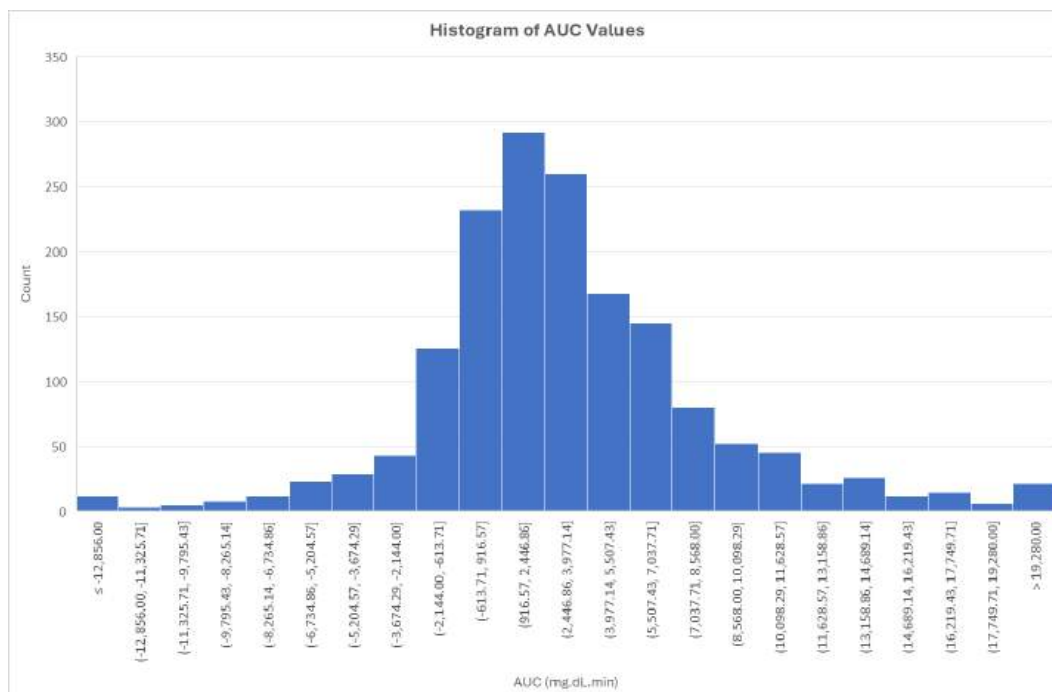


Figure 2. Histogram of AUC Values.

3.1 Spearman Correlations Between Participant-Level Means

Table 2 shows Spearman correlation coefficients (ρ) between mean nutrient intakes and mean glucose metrics across participants. Modest positive associations were observed for carbohydrates with mean AUC ($\rho = 0.226$, $p = 0.14$) and mean relative peak rise ($\rho = 0.270$, $p = 0.073$). Protein demonstrated a moderate positive association with mean relative peak rise ($\rho = 0.424$, $p = 0.004$) and a mild positive trend with mean AUC ($\rho = 0.269$, $p = 0.075$). Small, non-significant correlations were observed for fat. No nutrient showed a meaningful correlation with mean time to peak or mean spike count. These associations can be considered weak to moderate, as no absolute ρ values exceeded 0.5.

Table 2

Spearman Correlation Coefficient Between Mean Nutrient Intakes and Mean Glucose Metrics

Glucose metric	Nutrient	ρ	p-value	Interpretation
Mean AUC	Carbohydrates	0.226	0.14	modest positive association
Mean AUC	Protein	0.269	0.075	mild positive trend
Mean AUC	Fat	0.186	0.222	weak, not significant
Mean relative peak	Carbohydrates	0.270	0.073	modest positive association
Mean relative peak	Protein	0.424	0.004	moderate positive association
Mean relative peak	Fat	0.257	0.088	weak positive trend
Mean time to peak	Carbohydrates	-0.059	0.700	no association
Mean time to peak	Protein	0.206	0.174	small positive trend
Mean time to peak	Fat	0.085	0.579	no association
Mean spike count	Carbohydrates	0.029	0.850	none
Mean spike count	Protein	-0.278	0.065	weak negative trend
Mean spike count	Fat	-0.160	0.293	weak negative trend

3.2 Mixed-Effects Model Results

Table 3 summarizes the coefficients from the mixed-effects model for meal-level AUC. The intercept (baseline AUC) was 2,167 $\text{mg} \cdot \text{dL}^{-1} \cdot \text{min}$. Carbohydrate had a positive coefficient of 28.6 $\text{mg} \cdot \text{dL}^{-1} \cdot \text{min}$ per gram ($p < 0.001$). This indicates that, on average, each additional gram of carbohydrate was associated with an approximately 28.6 $\text{mg} \cdot \text{dL}^{-1} \cdot \text{min}$ increase in incremental AUC after adjusting for protein and fat. Protein had a negative coefficient of -13.2 $\text{mg} \cdot \text{dL}^{-1} \cdot \text{min}$ per gram ($p = 0.015$), suggesting a modest attenuating association. Fat had a small, non-significant coefficient. The random intercept variance was significant, reflecting substantial between-participant variability.

Table 3

A Summary of the Mixed-Effects Model Coefficients for Meal-Level AUC

Predictor	Coefficient (β)	SE	z-statistic	p-value
Intercept	2167	162	13.4	<0.001
Carbohydrates	28.6	3.6	8.0	<0.001
Protein	-13.2	5.4	-2.4	0.015

Fat	4.5	5.0	0.9	0.37
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3.3 Nutrient-Group Comparisons

We compared high- versus low-nutrient meals using the median nutrient value as the cutoff. For carbohydrates (median = 30 g), high-carbohydrate meals had a mean AUC of 4,262 $\text{mg}\cdot\text{dL}^{-1}\cdot\text{min}$ compared with 2,395 $\text{mg}\cdot\text{dL}^{-1}\cdot\text{min}$ for low-carbohydrate meals. The Welch t-test yielded $t = 12.1$, $p < 10^{-27}$, indicating a statistically significant difference. High-carbohydrate meals also had a slightly higher mean peak rise (44.2 $\text{mg}\cdot\text{dL}^{-1}$ vs 39.1 $\text{mg}\cdot\text{dL}^{-1}$, $p = 0.002$), although the effect size was smaller.

Protein and fat groups showed less consistent differences. High-protein meals had a modestly lower mean AUC than low-protein meals (3,075 vs 3,589 $\text{mg}\cdot\text{dL}^{-1}\cdot\text{min}$, $p = 0.035$), consistent with the negative coefficient observed in the mixed-effects model. A slightly higher mean AUC was observed for high-fat meals (3,468 vs 3,179 $\text{mg}\cdot\text{dL}^{-1}\cdot\text{min}$); however, this difference was not statistically significant ($p = 0.11$). Nonparametric Wilcoxon tests produced similar results.

3.4 Meal-Type Comparisons

One-way ANOVA comparing AUC across meal types yielded $F(3, 1300) = 33.3$, $p < 10^{-20}$, indicating that mean AUC differed by meal type. Post hoc pairwise t-tests showed that breakfast meals had significantly higher mean AUC (5,783 $\text{mg}\cdot\text{dL}^{-1}\cdot\text{min}$) than lunch (2,111), dinner (2,942), and snacks (1,976) (all $p < 10^{-12}$). These differences may partly reflect the standardized breakfast shakes and fast-casual lunches provided in the study design.

4. Discussion

4.1 Process Review and Key Results

We analyzed the CGMacros dataset using participant-level averages and mixed-effects models to account for repeated meals within individuals. Our pipeline preserved negative AUC values, applied rank-based correlations, and fitted a random-intercept regression model. The results revealed modest associations. Carbohydrates were positively associated with incremental AUC and peak rise. Protein showed a weak inverse association with AUC and a moderate positive association with peak rise. Fat had little effect within the 2-hour window. High-carbohydrate meals produced higher AUC than low-carbohydrate meals. Additionally, breakfast meals, particularly the standardized protein shakes, had the highest mean AUC among meal types.

4.2 Comparison with Previous Studies

Our findings both align with and differ from previous studies. The cross-sectional CGM study by another author reported that higher dietary intake, along with greater age, BMI, and HbA_{1c}, was associated with higher post-meal glucose levels (Calvo-Malvar, et al.). They found that each 10-year increase in age was associated with a 1.9–3.5 $\text{mg}\cdot\text{dL}^{-1}$ increase in postprandial

glucose (Calvo-Malvar, et al.). Our analysis did not include demographic covariates; however, the association between mean carbohydrate intake and AUC supports the importance of glycemic load. The mixed-effects coefficient was $28.6 \text{ mg} \cdot \text{dL}^{-1} \cdot \text{min}$ per gram of carbohydrate. Thus, carbohydrate quantity appears to be a primary contributor to glycemic exposure in this dataset.

Studies on macronutrient composition also provide partial agreement with our findings. The author reported that adding fat delays the glucose peak and reduces early postprandial glucose, while increasing later glycemic exposure (Li, et al.). Our analysis, limited to a two-hour window, found only a small and non-significant effect of fat on AUC and time to peak. This may reflect the shorter observation window and the mixed composition of free-living meals. The author reported that protein-rich meals produced lower iAUC and peak glucose compared with carbohydrate-rich meals (Ekberg, et al.). In our mixed-effects model, protein had a negative coefficient, which is consistent with a potential attenuating role. However, the participant-level correlation between mean protein intake and mean AUC was weakly positive. Individuals who consume more protein may also consume more total calories or have different meal patterns, which could partially explain this discrepancy.

4.3 Limitations and Future Directions

This study has several limitations. A major limitation was that demographic data linking individual meals to age, sex, and BMI were not available in the aggregated file. Therefore, we could not stratify analyses by these characteristics. Future releases of CGMacros and similar datasets could include anonymized demographic identifiers to enable stratified analyses. The sample size of 45 participants is modest, which limits statistical power and generalizability. In addition, we considered only a two-hour postprandial window. Fat and protein may exert effects beyond this period; therefore, including longer windows and modeling the full glycemic curve could reveal additional patterns.

Future studies should incorporate data on physical activity, sleep, and stress from wearable devices, as these factors may help explain residual variability. Comparisons across age, BMI, and sex groups will also be important to clarify how demographic factors may modify nutrient effects.

5. Conclusion

This study used the CGMacros dataset to assess associations between meal macronutrients and post-meal glucose dynamics. Carbohydrates were positively associated with glycemic exposure; however, they explained only a modest portion of the observed variability. Protein showed a weak inverse association with AUC, suggesting a potential attenuating effect. In contrast, fat had little influence within the 2-hour window. High-carbohydrate meals and breakfast meals were associated with higher incremental AUC, although overall correlations across participants were moderate. These findings highlight the complexity of postprandial glycemia and underscore the value of continuous wearable data in advancing personalized nutrition approaches.

6. Appendix

Data Availability

The code below has been used to compute the relationship between meal components and post-meal glucose dynamics using the CGMacros dataset. To execute the script, ensure the 'GlucoseMealMetrics_Full.csv' file is in the working directory, and install the required libraries. Below is the code:

```
import pandas as pd
import numpy as np
from scipy import stats
import statsmodels.formula.api as smf
import matplotlib.pyplot as plt

# Load per-meal metrics
df = pd.read_csv('GlucoseMealMetrics_Full.csv')

# Remove meals with missing values
df = df.dropna(subset=['AUC', 'RelPeak', 'TimeToPeak', 'SpikeCount',
                      'Carbs', 'Protein', 'Fat'])

# Participant-level aggregation
mean_df = df.groupby('SubjectID').agg(
    mean_auc=('AUC', 'mean'),
    mean_relpeak=('RelPeak', 'mean'),
    mean_timetp=('TimeToPeak', 'mean'),
    mean_spike=('SpikeCount', 'mean'),
    mean_carbs=('Carbs', 'mean'),
    mean_protein=('Protein', 'mean'),
    mean_fat=('Fat', 'mean')
).reset_index()

# Spearman correlations between mean nutrients and mean glucose metrics
metrics = ['mean_auc', 'mean_relpeak', 'mean_timetp', 'mean_spike']
nutrients = ['mean_carbs', 'mean_protein', 'mean_fat']
corr_results = []
for metric in metrics:
    for nut in nutrients:
        rho, pval = stats.spearmanr(mean_df[metric], mean_df[nut])
        corr_results.append({'metric': metric, 'nutrient': nut,
                             'rho': rho, 'pval': pval})
```

```

correlations = pd.DataFrame(corr_results)
print(correlations)

# Mixed-effects model for meal-level AUC
model = smf.mixedlm('AUC ~ Carbs + Protein + Fat', df,
                    groups=df['SubjectID'])
fit = model.fit()
print(fit.summary())

# High vs low nutrient group comparisons
results = []
for nut in ['Carbs', 'Protein', 'Fat']:
    median_val = df[nut].median()
    high = df[df[nut] > median_val]
    low = df[df[nut] <= median_val]
    for metric in ['AUC', 'RelPeak']:
        tstat, pval = stats.ttest_ind(high[metric], low[metric], equal_var=False)
        results.append({'nutrient': nut, 'metric': metric,
                        'tstat': tstat, 'pval': pval})
group_tests = pd.DataFrame(results)
print(group_tests)

# Meal type ANOVA
anova = stats.f_oneway(*(df[df['MealType'] == m]['AUC'] for m in
df['MealType'].unique()))
print('ANOVA F-statistic:', anova.statistic, 'p-value:', anova.pvalue)

# Scatter plot: AUC vs Carbs (all meals)
plt.figure()
plt.scatter(df['Carbs'], df['AUC'], alpha=0.3)
slope, intercept, *_ = stats.linregress(df['Carbs'], df['AUC'])
xs = np.linspace(df['Carbs'].min(), df['Carbs'].max(), 100)
plt.plot(xs, slope * xs + intercept)
plt.xlabel('Carbohydrates (g)')
plt.ylabel('AUC (mg/dL·min)')
plt.title('Per-meal AUC vs. Carbohydrate content')
plt.tight_layout()
plt.savefig('scatter_auc_vs_carbs.png')

# Histogram of AUC values

```



```

plt.figure()
plt.hist(df['AUC'], bins=40)
plt.xlabel('AUC (mg/dL·min)')
plt.ylabel('Count')
plt.title('Distribution of incremental AUC values')
plt.tight_layout()
plt.savefig('histogram_auc.png')

# Participant-level scatter: mean AUC vs mean Carbs
plt.figure()
plt.scatter(mean_df['mean_carbs'], mean_df['mean_auc'])
slope_m, intercept_m, *_ = stats.linregress(mean_df['mean_carbs'],
mean_df['mean_auc'])
xs_m = np.linspace(mean_df['mean_carbs'].min(), mean_df['mean_carbs'].max(), 100)
plt.plot(xs_m, slope_m * xs_m + intercept_m)
plt.xlabel('Mean carbohydrates (g)')
plt.ylabel('Mean AUC (mg/dL·min)')
plt.title('Mean AUC vs. Mean carbohydrates per participant')
plt.tight_layout()
plt.savefig('mean_auc_vs_carbs.png')

```

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The Dynamics of Globalization-Insights from the Trajectories of UAE Free Zones

By Malika Uzun

Research Question

How do the trajectories of UAE free zones expose the concrete mechanisms through which globalisation is planned, regulated, and lived, and what aspects of these mechanisms remain underexplored in the literature on free zones and globalisation?

Abstract

Over the last four decades, the Jebel Ali Free Zone (JAFZA) in Dubai has come to represent the promise of free zones and special economic zones in the transformation of a small Gulf state into a global trade hub. This paper asks: How do the trajectories of UAE free zones expose the concrete mechanisms through which globalization is planned, regulated, and lived, and what aspects of these mechanisms remain underexplored in the literature on free zones and globalization? This paper argues that JAFZA is more than a successful policy instrument, but a planned enclave with its own regulatory and institutional system, partially autonomous from the broader UAE system of law and economy, and that it has been used by Dubai to attract foreign direct investment, to pursue export-oriented industrialization, to create jobs, and to diversify its economy away from oil.

Empirically, the paper presents a qualitative case study of the emergence and evolution of JAFZA. It examines the founding legislation and regulations, policy and marketing materials published by the Government of Dubai and DP World, and existing literature on Dubai, the United Arab Emirates, and Gulf political economy, to read across these sources to examine how the legal, institutional, and infrastructural frameworks of JAFZA have been constructed, modified, and explained over time. This approach enables the paper to specify the mechanisms by which JAFZA shapes and is shaped by globalisation, such as unique ownership structures, taxation, and customs regimes, and to illustrate how these mechanisms shape the day-to-day activities of businesses and government agencies in the free zone. The significance of the study is that it goes beyond the “success story” narrative and refutes the notion that UAE free zones are a generic tool of liberalisation. The paper presents a fresh perspective by showing that free zones are tangible locations where the design, governance, and experience of globalisation are articulated. On a conceptual level, the article moves the frontiers of the notion of JAFZA as a “space of exception” that is constitutionally and juristically different, and as a “laboratory of globalization” where state power, capital, and globalization are recombined.

Keywords

Jebel Ali Free Zone; JAFZA; DP World; Dubai; United Arab Emirates; free zones; special economic zones; globalisation; trade hubs; foreign direct investment; global supply chains.

“Our flagship Jebel Ali Port and the adjacent Jebel Ali Free Zone together form the

largest seaport free zone combination in the Middle East, providing an integrated logistics and business hub for the region.” DP World, Annual Report and Accounts 2019, p. 20

Introduction

Containers piled up in Jebel Ali Port, trucks entering and leaving massive storage warehouses, and employees commuting through the gates of the Jebel Ali Free Zone (JAFZA) every morning may appear to be ordinary images of trade and logistics, but they are also instances where globalisation is actively produced. Globalisation is commonly defined as one of the key drivers of the contemporary world economy, often conceptualised as an increasingly rapid expansion of the flows of goods, finance, labour, and knowledge across national borders (Held et al., 1999, p. 2). A significant part of this literature focuses on the increasing international economic interdependence, which may create the impression that globalisation is an abstract, non-spatial process that occurs homogeneously across national space. This approach often neglects the spatially specific and uneven ways in which globalisation is actually produced, organized, and lived. In contrast, research in economic geography and political economy highlights the importance of globalisation being rooted in specifically located sites such as ports, cities, and special economic zones (SEZs), which are not merely conduits for global flows but actively constitute them (Hanieh, 2011, pp. 7-9; Easterling, 2014, pp. 13-16). Of these, JAFZA in Dubai has become an important free zone in the United Arab Emirates (UAE) that links local economic activities to the global economy of production, logistics, and trade.

This paper adopts a political economy approach to examine globalisation in general, and the JAFZA case in particular, to examine how Dubai has utilized a free zone that is institutionally and legally different from the rest of the UAE to manage its globalisation. This paper contends that JAFZA should not only be regarded as a successful policy instrument of globalization but also a particular instrument through which globalization is designed, governed, and lived. More specifically, this paper contends that JAFZA is a “space of exception” that carefully designs rules of ownership, taxation, labour, and regulation to attract and organize global capital, as well as reworking state sovereignty. In this respect, this paper will show that globalisation in the UAE is not an abstract process but a set of practices that are spatially engineered and politically contested.

In terms of analysis, the paper will be divided into the following sections. The first section will situate SEZs and free zones in the existing literature that are policy-oriented and critical, pointing out that most of the literature views zones as instruments of development or models of regulation rather than as spaces where globalisation is actively produced. The section will then introduce the literature on JAFZA and free zones in the Gulf to identify what has been said and what has not. The second section will discuss the historical and institutional development of JAFZA in the context of Dubai’s transformation from an oil economy to a trade and services economy. The third section will discuss the development of JAFZA’s legal, regulatory, and infrastructural framework with particular attention to the production of capital flows and labour governance. The conclusion will reflect on the implications of JAFZA for the

design of globalisation in the Gulf and its contribution to the debate on SEZs, exceptions, and infrastructure.

Literature Review: SEZs, Exception, and Gulf Political Economy

The use of SEZs and free zones as an approach to economic integration into the global economy has been extensively discussed in the literature of development economics and policy analysis. As an approach to national economic policy, SEZs are typically designed as an approach to foreign direct investment attraction, export-oriented industrialization, and job creation (Farole & Akinci, 2011, p. 3; Zeng, 2015, pp. 2-3). Much of this literature evaluates SEZs in terms of their economic effectiveness, often making a distinction between “static” gains (such as employment and export growth) and “dynamic” gains (such as technology diffusion, human capital development, and productivity enhancement) (Farole & Akinci, 2011, pp. 4-5). In this literature, zones are typically represented as purely technical, non-politicized policy instruments, with no attention paid to issues of state transformation, territorialization, or social relations.

However, a distinct interpretation of SEZs and free zones is offered by critical perspectives in political economy and economic geography. Neoliberalism as exception, as defined by Aihwa Ong, theorizes SEZs as a space of exception where some areas are partially exempt from national laws to allow for new types of capital accumulation (Ong, 2006, pp. 4-5). In this sense, globalization results from processes of differentiation, whereby particular ownership, labour, and taxation regimes are applied differently, rather than via standardized liberalization. Keller Easterling also points that the infrastructural and administrative perspectives of global authority, using the wording “extra statecraft” to emphasize the role of logistics, zoning, and spatial standards as the foundation of globalisation (Easterling, 2014, pp. 15-16). In the Gulf, Adam Hanieh locates ports, logistics hubs, and free zones within broader processes of capital accumulation and class formation, arguing that these spaces both integrate Gulf states into global capital and reinforce alliances between state actors and the leading business groups (Hanieh, 2011, pp. 140-142).

These advances notwithstanding, there are several areas that still need to be filled. First, SEZs are still largely studied either as instruments of development policy or as regulatory ideal types, and not as the locations where the practices of globalisation are actually produced and negotiated in particular institutional and spatial arrangements (Farole & Akinci, 2011, pp. 3-6; Ong, 2006, pp. 4-7). Second, there is a relatively small amount of literature that seeks to integrate policy-focused SEZ research with critical political economy and economic geography approaches, particularly through case studies that examine the actual design, governance, and experience of a specific zone over time (Easterling, 2014, pp. 13-20; Massey, 2005, pp. 68-75). Third, although there is a growing literature on free zones and ports in the Gulf, JAFZA itself has been all too often mentioned briefly as a success story or as part of a larger narrative about Dubai’s development, and not as a mechanism through which globalisation is actively managed (Soenen & Buncombe, 2000, pp. 97-99; Ulrichsen, 2016, pp. 115-118).

JAFZA and the Design of Globalisation

The emergence of JAFZA in the mid-1980s has to be understood in the context of Dubai's overall political and economic vision to shift its development paradigm from being oil-dependent to trade and service-led development (Ulrichsen, 2016, pp. 115-118). Instead of merely reacting to the emergence of global markets, Dubai had a proactive vision to locate itself in the newly emerging global networks of production and circulation. The vision to develop JAFZA and the Jebel Ali Port was one of the most significant development decisions made by Dubai, which sought to integrate production, storage, and distribution functions at one location and make it highly networked. This vision also represents the broader transformation of global capitalism towards logistics capitalism, where value creation increasingly comes from the efficiency of circulation and coordination rather than production (Cowen, 2014, pp. 1-5; Mezzadra & Neilson, 2013, pp. 130-135).

Right from the start, JAFZA was characterized by a space of exception. Companies operating in the zone enjoyed 100 percent foreign ownership, long-term lease agreements, exemption from corporate tax and customs duties, and a highly streamlined administrative process. These conditions were more than just a form of economic liberalization; they represented a transformation in the spatial organization of state sovereignty and regulatory power. In Ong's words, JAFZA can be understood as a "space of exception" where "normal" national regulatory standards are suspended or selectively varied in order to lure and regulate global capital (Ong, 2006, pp. 95-97). According to Soenen and Buncombe, a "market-oriented regulations regime was circumscribed within a territorially bounded area" (Soenen & Buncombe, 2000, p. 99). This enabled Dubai to fully integrate into the global economy while maintaining stricter control over regulatory and social conditions outside the zone. In other words, globalization in the UAE was not implemented in a uniform manner over the entire national territory but was "carefully territorialized" within designated zones such as JAFZA.

Over the years, JAFZA has grown from being a rather classical export-oriented manufacturing zone to a more intricate logistics, services, and governance platform. Today, the relevance of JAFZA does not reside in its financial incentives, which are now common to many other competing free zones, but rather in the complexity of its governance architecture. "One stop shop" services, the computerization of customs procedures, and the strong operational nexus with the port facilities managed by DP World have cumulatively enabled JAFZA to emerge as a highly efficient gateway through which international trade is facilitated and coordinated. This is not dissimilar to the SEZ literature, which has found that successful SEZs rely less on financial incentives and more on high-quality governance and infrastructure (Farole & Akinci, 2011, pp. 19-23). JAFZA would thus seem to be a case study that illustrates the emerging trend in globalization from "incentive-driven" competition to "governance-driven" competitiveness.

However, JAFZA also highlights the selective nature of globalisation. On the one hand, JAFZA provides companies with stability, efficiency, and predictability, but on the other hand, the labour regimes in the zone reflect the Gulf-wide phenomenon of having migrant-majority workforces with differentiated legal status and rights. As Hanieh argues, these regimes are

crucial to Gulf accumulation regimes, which facilitate high mobility of capital but make only very limited social and political claims (Hanieh, 2011, pp. 80-88). From the perspective of Ong, the spaces of exception are not only facilitators of capital mobility but also producers of membership with differential access to rights and protections based on economic utility rather than citizenship (Ong, 2006, pp. 78-82). From this perspective, JAFZA is a case that illustrates globalisation as a process that is deliberately planned and managed through spatial planning, regulatory differentiation, and infrastructural power. It is more than a simple conduit for global capital; it is also a place where global capital is carefully managed, and the benefits of globalisation are differentially allocated.

The development of JAFZA, therefore, provides a case in point that illustrates how modern globalisation functions in a very specific way through the design of nodes that are simultaneously open to capital and closed to labour and space. Through the examination of JAFZA as an institutional and spatial formation, it will be clear from this paper how a single free zone can be a device through which globalisation is actively governed.

Conclusion

This paper has explored how the UAE free zones and specifically JAFZA exemplify the ways in which globalisation is designed, governed, and lived. In line with the introduction's critique of abstract and "placeless" conceptualisations of globalisation (Held et al., 1999, pp. 1-3) and the main body's exploration of SEZs in development economics and critical political economy (Farole & Akinci, 2011, pp. 3-6; Ong, 2006, pp. 4-7; Hanieh, 2011, pp. 7-12), this paper contends that JAFZA cannot be seen simply as a successful policy tool of liberalisation. Rather, it has been shown that JAFZA is a concrete mechanism through which globalisation is actively designed, governed, and lived, and that it does so in ways that rework state sovereignty, labour, and territorialisation. First, it has been shown that globalisation in the UAE is engineered through the active production of 'spaces of exception' that are circumscribed with specific legal and institutional arrangements. Rather than liberalising the national economy in a uniform way, Dubai has instead adopted a policy of 'space of exception' production, whereby special arrangements regarding ownership, taxation, and regulation are selectively constructed to assemble global capital (Ong, 2006, pp. 95-97).

This paper argues that this exceptional state of affairs has enabled Dubai to achieve a strategic location in the global production and logistics chain while exercising a tighter political and social control in the remaining national territory. In this respect, JAFZA illustrates the "neoliberalism as exception" paradigm, whereby global integration is achieved not through the retreat of the state but through the re-territorialization of state power in enclaves.

Secondly, the paper has illustrated how modern globalization is being increasingly governed via infrastructure and administration, as opposed to fiscal incentives. In tracing the development of JAFZA from an export-oriented manufacturing zone to a more complex logistics and administrative platform, the main body of the paper illustrated how the "one stop shop" process, the digitalization of customs, and the very close integration of the zone with Jebel Ali

Port and DP World's infrastructure have enabled JAFZA to be developed into a highly efficient interface for the coordination and disciplining of global flows (DP World, 2019, pp. 18-21). This paper contends that such a development can be seen to confirm Easterling's (2014, pp. 13-20) concept of "extra statecraft," whereby infrastructural and administrative systems, as opposed to markets, become the primary instruments of globalization.

Third, the social disparity of zone-based globalization has been brought to light by the JAFZA example. According to the Gulf accumulation model, companies that operate in the zone benefit from efficiency, predictability, and legal security (Hanieh, 2011, pp. 80-88). Conversely, labour regimes that align with the Gulf accumulation model are typified by limited rights, stratified legal status, and migrant dependency. This essay argues that JAFZA encompasses the key topic of modern globalization, according to which a similar institutional structure that facilitates mobility and economic integration increases organized social inequalities, social hierarchies, and exclusions.

However, the implications of these results extend well beyond Dubai, and it is possible to discern a number of wider implications for the debate about globalisation and development. As free zones multiply across the Global South, the implications of the JAFZA case, it would seem, are first, that their success is less dependent on tax incentives and more on state capacity and infrastructural complexity (Farole & Akinci, 2011, pp. 19-23). Second, there are clearly questions of sustainability and replicability, and what happens to the spaces and populations that exist outside of these zones of globalisation?

Finally, this paper contends that the case of JAFZA proves that the globalisation of the present day is best conceptualised as "globalisation in the making," a determinate and contested process of a particular kind assembled in particular sites through political, legal, and infrastructural decisions, rather than a self-evident effect of the workings of impersonal markets. The UAE free zones, like JAFZA, actualise a particular form of zone-based globalisation that is carefully crafted, controlled, and promoted for its ability to regulate time, space, and capital, while the social and political implications of this process are often left unattended in the long run. In this way, this paper concludes that free zones must be conceptualised not merely as instruments of development policy, but as important laboratories of the future geographies of globalisation and its inequalities.

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Musical Engagement in Preschool Classrooms for Children with Developmental Delays: An Observational Study By Ryan Qi

Abstract

This exploratory observational study examined the responsiveness of preschool-aged children with developmental delays to live music and instrument play in classroom settings. Across fifty-five children observed over two years, more than ninety percent demonstrated positive engagement with musical activities. Instrument play elicited slightly higher responsiveness than passive music listening. Engagement levels appeared broadly comparable across gender, class size, number of teachers, and student-to-teacher ratio. These findings highlight music as a broadly accessible and inclusive medium for classroom-based engagement. The results further emphasize the value of incorporating both active and passive musical experiences into educational environments. By observing children's behavior during music sessions in their school setting, this study contributes to the existing literature by providing classroom-based evidence relevant to the integration of inclusive music practices in early childhood special education. Future research should investigate longer-term patterns of engagement and explore how music-based activities can be adapted to support a wider range of learners.

Introduction

Music has long been recognized as a powerful component of early childhood education (Dumont et al.). For children with developmental delays, particularly those on the autism spectrum, music-based activities have been widely used to support communication, social interaction, and emotional regulation. Unlike language, music often offers more flexible approaches for participation, allowing children to engage through listening, movement, or sound exploration (Suthers and Niland).

Research consistently demonstrates that music can support attention, emotional expression, and communication in children who may struggle with traditional verbal or social approaches (Kern and Aldridge). Studies have shown that passive music exposure (such as listening to familiar songs) can reduce anxiety and promote calmness (Katagiri). Active music-making, including drumming or improvisation, has been associated with improved joint attention, motor engagement, and social reciprocity (Geretsegger et al.; Wigram and Gold).

One of the unique benefits of music in educational settings is its capacity to accommodate multiple modes of participation, making it particularly supportive for children with autism who often present with diverse developmental profiles (Reschke-Hernández). Group-based music activities can also promote peer interaction and reduce social isolation in classroom contexts.

Despite these benefits, few studies have examined music engagement in naturalistic classroom settings using simple observational tools. Much of the existing literature, including large-scale studies (Bieleninik et al.), has focused on structured or clinical contexts rather than everyday classroom environments. This gap highlights the need for classroom-based data that

can inform scalable, inclusive music practices in early childhood education. Research has also emphasized the importance of balancing methodological rigor with the study of applied music in everyday settings (Dumont et al.).

In classrooms, structured musical experiences are not always consistently integrated into the daily routine for children with developmental challenges. This study explores how preschool-aged children with developmental delays respond to music-based activities within their own classrooms. Specifically, we investigate how children respond to two forms of music engagement: listening to live music and playing musical instruments.

This observational study was conducted over two years (2024 and 2025) in a preschool program serving children with developmental delays. The analysis includes data from 55 children and examines patterns of responsiveness across years, classroom contexts, and selected demographic characteristics. By observing children's behavior during music sessions in their school setting, this study provides classroom-based evidence relevant to the integration of inclusive music practices in early childhood special education. This study is intended as an exploratory, naturalistic observational investigation conducted in real-world classroom settings, aiming to provide practical insights rather than establish causal relationships.

Methods

This study was conducted at a preschool program serving children with developmental delays. The objective was to document behavioral responses to two types of musical engagement: (1) passive listening to live music (sing-along sessions) and (2) active interaction with musical instruments.

Participants and Setting

This study was conducted in a preschool program serving children aged approximately 3-5 years with developmental delays, including a significant proportion (~ 96%) diagnosed with autism spectrum disorder (ASD). Classrooms were part of a structured early childhood special education program. A total of 55 children were observed across two academic years (2024 and 2025). No individually identifiable data were collected, and all observations were conducted in routine classroom settings.

Session Design

Music sessions were conducted during regular school hours within classroom settings. Each session lasted approximately 40-45 minutes and was led by a small team of trained facilitators with prior experience in music-based engagement activities. Sessions followed a consistent structure across classrooms and years, including (1) live singing of familiar children's songs and (2) opportunities for active engagement with instruments such as drum pads, keyboards, and glockenspiels. The same general format and set of songs were used across sessions to maintain consistency in delivery.

Data Collection

Children participated in small classroom groups, and their responses were observed and recorded individually during each session. Observations were conducted by facilitators immediately following each session using consistent criteria. Each session was observed by two independent observers who recorded engagement responses separately and then compared their assessments to ensure consistency. High inter-observer agreement was observed, as engagement behaviors were generally distinct and consistently identified across observers. Observations were supplemented by video recordings, which were used for review and verification of engagement responses. To further reduce potential coding bias, one observer remained consistent across all classrooms and years, supporting continuity in observational standards.

Engagement was operationally defined as any observable positive behavioral response, including but not limited to smiling, clapping, vocalization, body movement (e.g., rocking or dancing), eye contact, or reaching toward the music source or instrument. Responses were coded as “Yes” if any such engagement was observed and “No” if no observable engagement or avoidance behavior was present.

Classroom-level variables such as gender, class size, number of teachers, and student-to-teacher ratio were included to explore whether common structural characteristics of early childhood classrooms influenced engagement. These variables were selected based on their practical relevance to educators implementing music-based activities in classroom settings. For each child, the following information was recorded:

- Classroom ID
- Gender (male/female)
- Response to live music (yes/no)
- Response to instrument play (yes/no)
- Class size (number of children present; range: 4-8)
- Number of teachers present (range: 2-5)
- Student-to-teacher ratio (range: 1.25-2)
- Observation year (2024 or 2025)

Statistical Analysis

A total of 55 observations were included in the analysis, representing 18 children from 2024 and 37 children from 2025. The data were entered and analyzed using Microsoft Excel and Python for descriptive summary statistics and exploratory comparisons by year, gender, and class characteristics. Chi-square tests of independence were used to examine whether children’s engagement responses differed across categorical variables such as gender (girls vs boys) and class size (small vs large). The chi-square test evaluates whether observed differences in response patterns across groups are greater than what would be expected by chance alone (Agresti). Statistical significance was defined as a p-value below 0.05, indicating that the observed association is unlikely to be due to random variation (Gravetter and Wallnau). Given the exploratory nature of this study and the relatively small sample size, statistical results are interpreted descriptively and should not be considered definitive evidence of causal relationships.

Results

A total of 55 preschool-aged children with developmental delays, primarily autism, were observed across music sessions held in 2024 and 2025. Figure 1 displays the proportion of preschool-aged children who demonstrated observable engagement during classroom music sessions. The figure shows four categories: children who responded to both live music and instrument play, children who responded only to instrument play, children who responded only to live music, and children who showed no observable response.

Overall, 90.9% (50 out of 55) of children demonstrated a positive response to either live music, instrument interaction, or both (Figure 1). 80.0% of children responded positively to live music, as observed through smiling, clapping, swaying, and dancing. 81.8% of children engaged with musical instruments, as observed through reaching for, tapping, exploring, and playing the instruments. Most children responded to both types of musical engagement (Figure 1), with smaller percentages responding to only one or neither: 10.9% responded only to instrument play, 9.1% responded only to live music, and 9.1% showed no observable response to either musical stimulus.

		Response to live music?		
		YES	NO	total
Response to instrument?	YES	70.9% (39)	10.9% (6)	81.8% (45)
	NO	9.1% (5)	9.1% (5)	18.2% (10)
total		80.0% (44)	20.0% (11)	100.0% (55)

Figure 1. Distribution of observed child engagement responses to live music and instrument play during classroom music sessions.

Trends Across Years

Children observed in 2025 showed slightly higher response rates than those in 2024:

- Response to Live Music: 78.4% (2025) vs. 72.2% (2024)
- Response to Instrument Play: 83.8% (2025) vs. 77.8% (2024)

These descriptive statistics are also presented for comparison in Figure 2. This figure compares response rates to live music and instrument play among preschool-aged children

during classroom music sessions conducted in 2024 and 2025. Response rates are shown as percentages for each year and engagement type.

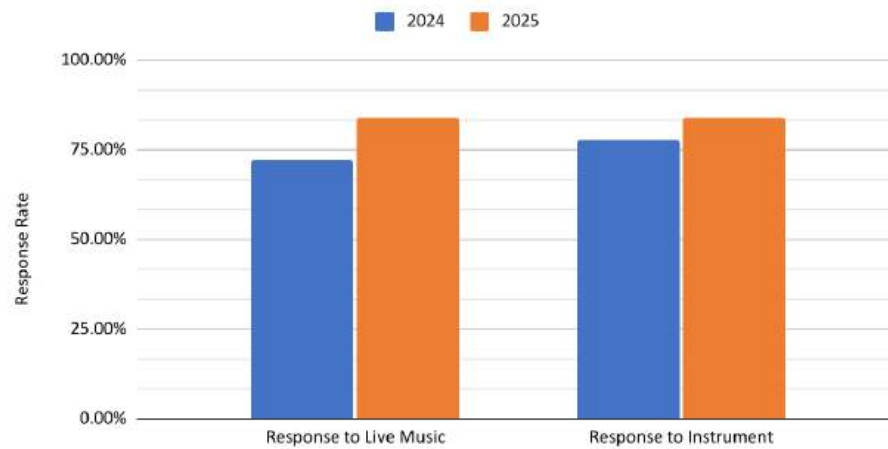


Figure 2. Observed response rates to live music and instrument interaction among preschool-aged children during classroom music sessions in 2024 and 2025.

Gender

No statistically significant differences in response rates were observed between genders. For live music, 82.4% of girls (14 of 17) responded positively, compared with 78.9% of boys (30 of 38). For instrument play, 76.5% of girls (13 of 17) and 84.2% of boys (32 of 38) demonstrated engagement. To confirm whether there were significant differences in response rates between genders, chi-square tests of independence were conducted. The results suggest that gender was not associated with differences in responsiveness to either musical stimulus ($p = 0.77$ for live music response, $p = 0.49$ for instrument response).

Class Size

No statistically significant differences in response rates were observed between small and large class sizes. The data were divided into two groups based on class size: small class size (5 or fewer) and large class size (6 or more). For live music, 75.8% of children (25 of 33) from small classes responded positively, compared with 81.8% (18 of 22) from larger classes. For instrument play, 81.8% of children (27 of 33) in small classes and 81.8% of children (18 of 22) in larger classes actively engaged. Chi-square tests did not indicate a statistically significant association within this sample between class size and responsiveness to either musical stimulus ($p = 0.59$ for live music response, $p = 1.00$ for instrument response).

Teacher Count

No statistically significant differences in response rates were observed based on the number of teachers in the classroom. The data were divided into two groups based on teacher count: lower teacher count (2-3 teachers) and higher teacher count (4-5 teachers). For live music, 77.8% (14 of 18) of children in classrooms with fewer teachers and 81.1% (30 of 37) in classrooms with more teachers demonstrated engagement. For instrument interaction, 88.9% (16 of 18) of children in classrooms with fewer teachers and 78.4% (29 of 37) in classrooms with more teachers responded positively. Chi-square tests suggested no statistically significant association within this sample between teacher count and responsiveness to either musical stimulus ($p = 0.77$ for live music, $p = 0.34$ for instrument play).

Student-to-Teacher Ratio

To better assess classroom staffing more holistically, the student-to-teacher ratio was evaluated. The data was divided into two groups: low student-to-teacher ratio (less than 1.5) and high student-to-teacher ratio (greater than or equal to 1.5). For live music, 76.9% (20 of 26) of children in low-ratio classrooms and 82.8% (24 of 29) in high-ratio classrooms responded positively. For instrument interaction, 80.8% (21 of 26) and 82.8% (24 of 29) demonstrated engagement in low-ratio and high-ratio classrooms respectively. Chi-square tests indicated no statistically significant association within this sample between student-to-teacher ratio and responsiveness ($p = 0.59$ for live music, $p = 0.85$ for instrument interaction).

Classroom-Level Variation

Engagement levels varied across classrooms. Responsiveness to live music and instrument play was 100% in some classes, while other classes showed variable but generally high response rates. This variability may reflect differences in classroom dynamics, sensory environment, or developmental profiles.

Overall, the results indicate that classroom structure and demographic characteristics were not associated with substantial differences in observed engagement during music sessions, highlighting the broad accessibility and adaptability of music across classroom environments.

Discussion

This study explored how preschool-aged children with developmental delays, primarily autism, responded to two forms of musical engagement - listening to live music and actively interacting with instruments - in natural classroom settings. The findings demonstrate that music appeared to function as an effective pathway for engagement, with over 90% of children responding positively to at least one form of music exposure. These findings may be particularly relevant for educators seeking low-resource, effective strategies to support engagement in inclusive classroom settings.

Music as an Inclusive Medium

The high overall engagement rate reinforces existing research that identifies music as an accessible and motivating medium for children with autism and related developmental challenges. Because music offers multiple modes of participation, it may reduce some of the social and linguistic barriers that these children often face. In this study, responsiveness did not vary meaningfully by gender, class size, teacher count, or student-to-teacher ratio, suggesting that music has an inherently broad and inclusive appeal across classroom contexts. From a practical perspective, these findings suggest that music-based activities do not need to be highly specialized or dependent on specific classroom structures to be effective. Educators may incorporate short, flexible music sessions, such as 15-minute musical activities, into daily routines to support engagement, even within time-constrained classroom routines.

Active Participation and Engagement

Although both passive music listening and instrument play were well received, slightly more children responded to instrument interaction. This pattern aligns with prior literature emphasizing the value of active music-making in educational and supportive contexts (Dumont et al.). Giving children opportunities for direct interaction - such as banging a drum, pressing a key, or experimenting with sound - may foster choice, agency, and sustained engagement (Suthers and Niland).

The fact that a small number of children responded only to instruments or only to music highlights the value of offering multiple modalities within music-based activities. Flexibility in delivery allows educators and teaching artists to accommodate individual sensory preferences, motor abilities, and comfort levels.

Access Across Classrooms

The variation in response rates did not appear to follow any pattern related to class size, number of teachers, student-teacher ratio, or gender. This supports the idea that music is not constrained by structural or demographic factors. Instead, musical engagement may relate more to session quality, adult facilitation, and openness to interaction than to static classroom metrics (Kern and Aldridge; Wigram and Gold).

Conclusion

This study provides preliminary evidence that music - both passive listening and active participation - serves as a widely accessible and engaging medium for preschool-aged children with developmental delays. Across multiple classrooms, genders, and years, the majority of children responded positively to musical stimuli, with slightly higher engagement observed during instrument interaction.

Importantly, engagement did not appear to be influenced by class size, teacher count, student-to-teacher ratio, or gender, underscoring music's flexibility across classroom contexts. These findings suggest that integrating music-based experiences into preschool classrooms may

offer consistent opportunities for inclusive engagement without the significant need for individualized accommodations.

The variability in individual and classroom response patterns also highlights the importance of offering multiple forms of musical engagement. What captivates one child may differ from another, and the dual approach of listening and instrument play supports broader participation and responsiveness.

Limitations

This study has several limitations. The sample size was relatively small and limited to preschool-aged children within a single program, which may reduce generalizability. In addition, limited variation in class size, number of teachers, and student-to-teacher ratio may have reduced the ability to detect differences across classroom conditions. The study relied on observational data collected in natural classroom settings; although consistent criteria were used and observations were supplemented by video recordings for review and verification, the absence of formal inter-rater reliability statistics and standardized behavioral assessment tools limits the precision of the findings. Additionally, the binary coding of engagement (yes/no) may not capture variations in intensity or duration of responses. As an exploratory observational study, the design does not support causal conclusions regarding the effects of music on engagement.

Despite these limitations, the study provides preliminary classroom-based insights into how music-based activities may support engagement among children with developmental delays. Future research should include larger and more diverse samples, incorporate standardized assessment tools, and examine longitudinal or experimental designs, which may help further clarify the specific developmental benefits of classroom-based music programs.

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Design and Development of a Household Black Soldier Fly Larvae Waste Conversion System By Elle Frazier

1. Introduction

Global food waste has become a growing problem as our population continues to grow, driving urbanization and generating more organic waste. The United Nations estimates that about one-third of all food production is wasted each year. (Jiang et al.) This food waste contributes to greenhouse gas emissions, overuse of landfills, and wasted resources. Therefore, managing organic food waste more sustainably has become a global goal. Black soldier fly larvae (BSFL) are a solution to this problem. They can convert organic waste into frass, a valuable resource for improving plant health, and create larval biomass. Because of this ability, the soldier fly has been widely studied for its role in circular economies and as a valuable resource. Despite BSFL's biological efficiency, there is a market gap for household use- or small-scale community-use systems. Many existing systems are for industrial-scale facilities and require extensive manual labor. They require tasks such as feeding larvae, combining substrate, draining leachate, and harvesting larvae with direct contact with the insect. These challenges can create both organizational barriers and unrealistic expectations for maintenance and operability.

This paper examines the barriers to the current limit on the adoption of small-scale BSFL systems. It identifies potential design requirements to improve the accessibility and usability of these systems by reviewing existing literature on system design, food safety, odor control, moisture control, and containment, and uncovering barriers to adoption. Based on the findings, this paper proposes a conceptual prototype for household BSFL systems that reduces user interaction while enabling effective processing of larvae. This work aims to support the development of a more easily adoptable system for insect-based waste management.

2. Background Information

Black soldier fly larvae (BSFL), the larval stage of *Hermetia illucens*, have attracted attention for their ability to serve as a sustainable method for converting organic waste. The BSFL life cycle includes eggs, larval, prepupal, pupal, and adult stages. The larval stage is the primary feeding phase during which the larvae rapidly break down organic waste and process it. BSFL can feed on a wide range of waste, including organic materials, food waste, agricultural waste, and even animal-based manure. (Kenchanna et al.) During the larval stage, they rapidly break down waste, converting it into larval biomass and frass. Larval biomass is valued for its high protein and lipid content. (Su et al.) Frass has a potential use as a soil amendment and fertilizer. Understanding the biology of BSFL is crucial for designing these processing systems. BSFL spend most of their lives actively feeding, whereas adult flies do not. (Sivanantharaja and Gnaneswaran) As larvae mature into their prepupal stage, they stop feeding and migrate away in search of a dry spot away from waste to pupate. (Sivanantharaja and Gnaneswaran) The natural biological behavior is important in the context of BSFL systems as it can be used to develop a self-harvesting system. That being said, BSFL performance depends strongly on their

environmental conditions. It is shown that substrate, moisture, temperature, and airflow can all affect BSFL mortality. (Tognocchi et al.)

Existing BSFL systems vary widely in their scale and complexity. At their smallest scale, there are DIY or household bins that require manual feeding and harvesting. (Food and Agriculture Organization) Whereas large-scale systems include controlled harvesting, containment, and environmental management. This gap is important because the requirements of a small-scale BSFL processing system differ from those of agricultural or industrial operations. Household users are less likely to be experts or comfortable with direct interaction with larvae; as a result, design questions are especially important when adapting BSFL systems into small-scale usable technology.

3. Methods

This paper uses a focused literature review to examine the design barriers affecting small-scale BSFL systems and to identify a small-scale, touchless household use case. Sources were selected to understand three main aspects: biological requirements, larval requirements, and operational challenges in waste-processing systems. The goal of the review is to synthesize findings most relevant to containment, system maintenance, odor control, and overall user experience. Literature was collected using keywords such as black soldier fly larvae, composting, food waste, bioconversion, insect protein feed, frass fertilizer, odor control, moisture management, and small-scale rearing systems. The review uses approximately 20 sources, most of which were published between 2018 and 2025. Sources included peer-reviewed journal articles, household rearing guides, extension publications, and technical reports. Priority in the use of sources was given to BSFL waste conversion efficiency, environmental requirements, and practical system operation, particularly in a small-scale system. This review used a human-centered design perspective to identify factors that influence adoption and usability. The review informs the identification of design requirements and guides the development of a conceptual prototype for a small-scale BSFL system.

4. Identified Barriers in Existing Small-Scale BSFL Systems

Black soldier fly larvae promise results in the bioconversion fields; however, in practice, it takes more to maintain a system for households or homesteads. While the biological use of BSFL is effective, the user comfort and efficiency remain unexamined. Many operations require constant maintenance and handling of the flies. Typical tasks when tending to BSFL include churning the bins, adding compost, draining leachate, mixing substrate, and manually removing BSFL from the frass when harvesting. (Purdue University) When an individual is interested in harvesting BSFL, their demand to be involved and comfort level with insects drive the components of the system. This section explores the specific handling requirements, psychological barriers, and adoption consequences of BSFL.

A prominent barrier to obtaining and maintaining a small-scale BSFL system is manual handling of the tasks associated with housing BSFL. This includes opening bins to add food for

the larvae, tray rotation, combining substrates, moisture correction, manual larval harvesting, and cleaning residue buildup. The Black Soldier Fly Composting Guide by Catherine Terrell and Laura Ingwell, Department of Entomology, provides a plan for building a bin to house the larvae. It requires using a gallon tote with PVC pipes and adding substrate to create an environment for the BSFL. This is a low-cost option, and the materials can be easily obtained. (Purdue University) This system requires manual observation of humidity levels, compost addition, bin churning, and moisture correction.

In a research study by Kevin Urrutia Avila et al., the authors developed a smart bin that automated BSFL production. Their research highlights the need to reduce the amount of tending to the system. The critical aspects include: 1) monitoring the overall growth in the larvae population using computer vision techniques, 2) aerating and mixing the larvae-containing substrate with an automated mechanism, and 3) logging multiple sensor readings to control the larvae environment. (Urrutia Avila et al.) They created a system that could be monitored via a real-time clock (PCF8523) and found that it needed to be aerated once a day by the spindle. They perform thermal imaging to evaluate the aeration. In the end, a multimodal visual perception was created, with a mechanism design, a sensor-based system, and automatic aeration. (Urrutia Avila et al.) They suggest that the current traditional bins require significant participant involvement and reinforce the barrier of maintaining a BSFL system.

A household rearing guide published by The Bug Picture outlines a system with four basins: one for food/waste and three for the larvae. (“Black Soldier Fly Larvae Rearing”) While the system is advertised as low-cost and accessible, it requires manual labor. It includes the daily feeding, manual harvesting, and manual drying of the grass. The guide assumes a high level of user comfort with larvae and routine maintenance, which may not apply to everyone. Research on user-friendly options for BSFL bin design found that food waste alone produced strong larval output but also generated high moisture levels, which make maintenance more difficult. (Ahmad et al.) Adding moisture control material would make maintenance and operation easier.

4.1 Psychological Barriers

Beyond manual handling, exposure to larvae and repeated contact with them can cause psychological stress. Although insects do ecological good on our planet, it does not stop most people from feeling negative towards them. Attitudes include emotions and cognitive beliefs that influence behavior. (Lorenz and Eisner) When someone has a negative attitude towards an insect, it can create fear and disgust. Insects provoke negative emotions in people. (Kellert; Bjerke and Østdahl; Schlegel and Rupf) It is found that more educated and experienced people spend less time feeling disgust toward insects. In the Shepardson (2002) study, they found that many children's opinions about insects were influenced by their interactions, including stings and bites. (Shepardson) It is found that humans view insects as they do pathogens, which creates avoidance, and that humans have evolved to avoid things associated with disease. (Lorenz and Eisner) This creates a barrier because BSFL systems require repeated encounters with the larvae, which can be difficult for users who experience psychological discomfort around insects. In

addition, creating decomposing waste can increase discomfort, which disconcerts non-expert users. Because of these perceptions, systems requiring direct user contact can face limited adoption. If adopted, users must be highly motivated or expert handlers.

4.2 Adoption Barriers

When adopting a system, users are less likely to adopt it when they perceive a chance of larvae escaping or leaking. When BSFL are in the prepupal stage, they exhibit wandering behavior to fulfill their need to pupate. Because of this, it can cause problems in the system. Each time larvae escape, this decreases the user's comfort level with the insect; therefore, this concept should be a key part of the design requirements and include sealed interfaces and controlled migrational paths. Odor and moisture instability are also major concerns for potential consumers considering adopting a BSFL system. Processed organic waste can produce strong odors, hindering its widespread application. (Michishita et al.) Digestion from BSFL alters the microbial flora, which can heighten odor. In most large-scale facilities that host BSFL, open containment releases strong odors; when BSFL are small-scale and hosted in closed containment, the odors produced are significantly less. (Michishita et al.) Information on VOC emissions and microbial 167 is severely limited, preventing us from knowing the full extent to which these affect toxin emissions in BSFL systems.

Controlling moisture levels in BSFL can be challenging. Many substrates added to the compost have a high moisture content, leading to low BSF biomass and high mortality. (Laksanawimol et al.) This can become an issue because if the user recognizes an increased level of moisture or a strong odor, they are more likely to abandon the system, regardless of the environmental benefits it offers. One potential solution to mitigate moisture content is to incorporate various dry substances into the mixture. (Laksanawimol et al.) It was found that rice bran and rice husk had the best results in mitigating the moisture. This was because rice bran has a high water-holding capacity and nutritional content, but its downsides include the potential to raise temperatures, potentially leading to larval mortality. (Laksanawimol et al.) On the other hand, rice husk improved larval lifespan and growth. (Laksanawimol et al.) Because of these barriers, a key design requirement for a system would be moisture-buffering and odor-management solutions, such as filtration, aeration, and substrates.

4.3 Food Safety and Pathogen Risk

Small-scale systems cannot assume that BSFL will fully eliminate pathogens in processed compost. There are conflicting sources offering views on how BSFL removes *Salmonella enterica*. Some have found that it partially removes, while others have found that it fully removes. De Smet et al. (2021) found that when larvae were on chicken feed, they did not significantly remove the pathogens in the substrate. However, the initial rate of bacterial growth slowed, but it did not remove the *Salmonella*. (Beskin et al.) This means that sanitation and substrate control are needed to achieve the cleanest outputs. In household use, the system design

must address safety constraints. The design implications are that it must prioritize larval containment and post-harvest treatment standards.

5. Design Requirements

The barriers that were outlined in section 4, manual labor concerns, psychological barriers, adoption barriers, and the safety risks, all make adopting a small-scale household BSFL system hard to commit to. Transforming these barriers into necessary engineering solutions creates a system that is safer and more user-friendly for non-expert users. Based on this literature review, these design concepts are for a household-scale BSFL bioconverter.

5.1 Touchless System Operation

Manual operation is a constraint for most systems, because it requires expert users and time that many people cannot afford. This includes: churning of the bins, extracting the larvae, adding compost, and draining the leachate. To lessen this barrier, the design requirements should include an automated BSFL self-harvesting mechanism to eliminate manual collection, removable frass trays that can be accessed without direct contact with the larvae or wet substrate, and tool-free points for cleaning and inspection. These improvements can make operations more passive and allow for more users to adopt the household waste conversion system.

5.2 Containment and Escape Prevention

The psychological stress that has been linked to handling insects is an issue to the point where people do not want to adopt a system that involves BSFL. This discomfort is heightened even when there is a minor escape or the owner has to touch the larvae. Because of this, it is necessary to have fully sealed and enclosed chambers so the larvae cannot escape. The household system must have a controlled migration channel for the larvae. This will not only prevent larvae escape, but also substrate leakage.

5.3 Odor Management

Odor has been a factor of discomfort for the owner of the BSFL system. Thus, it is heightened in a household setting because it's a person's place of comfort, not commercial use or a warehouse. Design components to prevent odor include enclosed bin designs that prevent odor from escaping into the surrounding area. Controlled airflow from the system, with aeration or filtration, also helps limit odor.

5.4 Moisture Management

Moisture is a major barrier to the use of a BSFL system because high moisture levels can reduce biomass production, increase larval mortality, and create more odor. These can all discourage future household use of the system. To prevent this discouragement, the design will include a drainage path (mentioned in detail in the next section) for excess liquid from the

compost. In addition, it will include drybulking materials as substrates, such as dry bran or rice husks, and provide clear guidance to users on handling high-moisture food waste.

5.5 Food Safety Controls

Food safety considerations are important in a BSFL system, especially when the larvae will be used as animal feed. Research has shown that BSFL can reduce bacterial growth, although they do not eliminate all pathogens, such as Salmonella or Listeria, which can pose a threat to livestock. For household systems, this can be less of a threat if clearly defined feedstock restrictions are imposed to prevent high-waste materials from entering the system. A fully enclosed chamber, so there is no contamination to the environment, a separation between areas for waste input and for harvesting larvae, and clear sanitation instructions for the system will all add to the food safety standard and sanitation of the household system..

6. Proposed Engineering Concept

Based on the design requirements identified in section 5, a household-scale bioconversion system prototype was developed. This system integrates a BSF bioconversion system with a drying process and a collection bin, all enclosed in a single space, so the user has minimal contact with the larvae and creates an efficient system that requires less maintenance. The overall construction includes a barrel for feeding and decomposing waste, connected to an enclosed structure that processes the larvae via a migration pipe. The system has a structured workflow. First, a barrel for the input of live larvae and compost that the larvae will process. Then, when the larvae reach the prepupal stage, they will look for a migration tunnel that leads to a PVC pipe, which in turn leads into the processing chamber. When the larvae reach the processing box, they will pass through staged drying and finally reach a heated plate. The dried larvae are collected in a bin. This workflow allows for food waste production, larval harvesting and drying, and feed collection with minimal user contact.

The biological processing barrel has a capacity of 25 gallons. It is an opaque high-density polyethylene barrel that serves as the larva growth section. This is where the compost and the larvae are placed in the chamber. The barrel is designed to be enclosed, preventing any larvae from escaping or releasing odor. The barrel includes an 8x8-inch feeding hatch at the top to make it easy to add compost and prevent larvae from escaping. At the top of the chamber are ventilation holes covered in fine mesh to allow airflow to relieve pressure. Inside the barrel, there is a removable perforated false floor that separates the active feeding substrate from the leachate, allowing leachate to collect on the bottom. There is a spigot at the bottom front of the barrel to allow periodic leachate removal. This will control moisture and prevent conditions that could lead to strong odors.

The household scale bioconversion BSFL system uses the larvae's natural migratory behavior to guide them to a location away from the substrate to find a dry pupation site. This enables passive harvesting. To allow this behavior, a PVC pipe in the composting barrel connects to the processing chambers. The pipe is angled at approximately 32.5° to encourage larval

movement while preventing substrate movement. The conduits inside have a textured surface that provides traction for migrating larvae.

After reaching the end of the migrational conduit, the larvae enter a processing chamber where they are dried and processed. These larvae are enclosed in a box made of aluminum framing and UV-resistant HDPE panels, and a sealed, lockable door. The internal drying system includes multistage funnel drying racks, designed to slow the larvae's movement and increase the time spent drying. The funnels are offset and arranged at 20-degree angles. As the larvae pass through each funnel, the dedicated fans speed up the process. This drying stage reduces the larvae's moisture content, making them storable. It also keeps the processing stages contained in an environment that will prevent insects from escaping. The funnels are mounted on removable frames, allowing for the mechanism to be serviced and cleaned easily.

After the drying stage, the larvae are dropped onto a hot plate, which serves as the final precaution to ensure their termination. The plate is made of stainless steel and heated by an electrical element. The plate is at 70 degrees Celsius to ensure rapid termination. Safety features include an overtemperature cut-off and a drip shield. After thermal termination, the larvae drop into a collection bin below, with a pull-out tray for easy collection. The collection bins include a ventilation fan to reduce moisture and overall temperature. The tray system for the collection allows the owner to collect the finished product without damaging internal systems. It is easy to access.

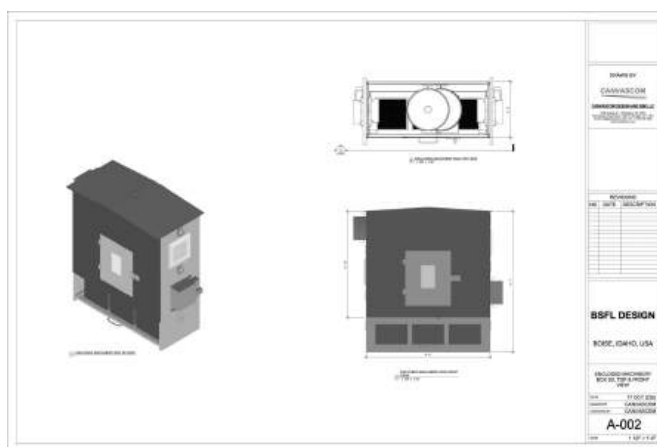


Figure 1. Isometric and orthographic views of the proposed household-scale BSFL bioconversion system, showing the enclosed processing unit, structural housing, and overall system configuration.

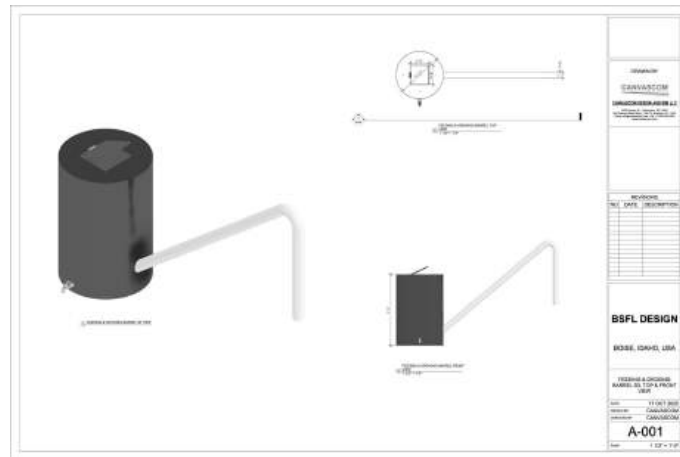


Figure 2. Biological processing barrel and migration conduit design, illustrating the 25-gallon composting chamber and angled PVC pipe used to guide prepupal larvae toward the processing unit.

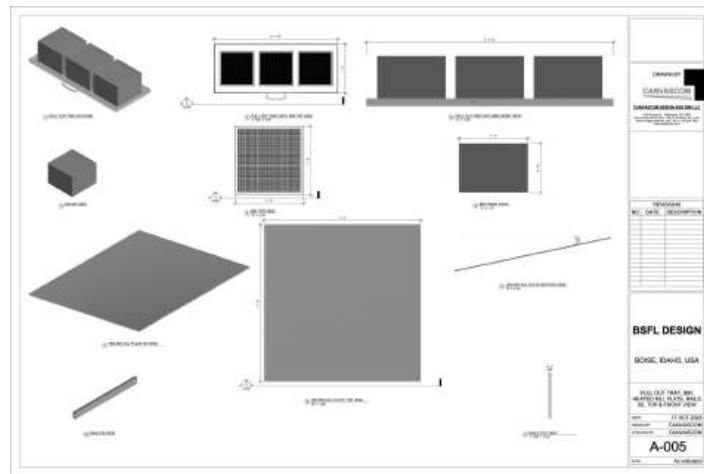


Figure 3. Exploded component layout of the BSFL processing system, including structural panels, internal supports, and modular elements for assembly and maintenance.

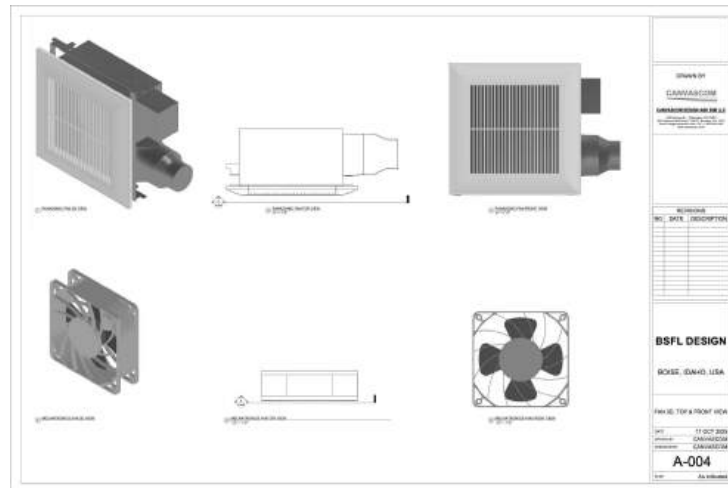


Figure 4. Ventilation and airflow components used in the processing chamber, including axial fans and filtration elements for moisture and odor control.

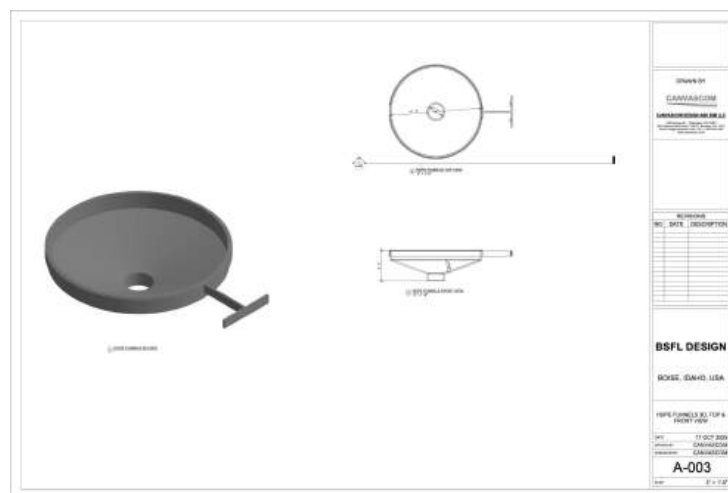


Figure 5. A multistage funnel drying system is used to reduce larval moisture content prior to thermal processing, designed to slow movement and increase drying efficiency.

7. Expected Performance and Use

BSFL are widely researched for their ability to process organic waste into larval biomass and frass. Previous studies have shown that BSFL can reduce food waste mass, though system performance depends on environmental factors such as moisture, aeration, and temperature. The proposed household-scale bioconversion system includes several elements to support stable processing conditions. The two primary outputs of the system are the larval biomass and frass. Larval biomass has been widely studied as a potential protein source for animals. Frass can serve

as a soil amendment. Proposed here could support several small-scale waste management scenarios. It is intended for use with household food waste, small farms, and educational sustainability programs. At the same time, it reduces the need for direct handling of larvae and improves containment, making it easier for non-expert users.

8. Limitations and Future Work

While the proposed system integrates design requirements identified in this review, the current study presents a conceptual prototype rather than an experimentally validated system. To validate performance, future work should test waste-processing efficiency, larval growth and survival rates, long-term reliability, and the extent of required upkeep. In addition to environmental performance, there is still a lack of information on testing and the specific environment in which the system would be located. For example, future studies should measure odor emission, moisture stability in the substrate, temperature variation within the enclosed system, and the effectiveness of the airflow system and cooling mechanisms. In addition, further work is needed to evaluate microbial safety in larval biomass and frass. Testing should evaluate pathogens' survival, contamination risks during harvesting, and the effectiveness of thermal termination, especially for larvae used as animal feed, and whether frass is used as a soil amendment. Future work for long-term deployment could include larger system sizes, integration with community waste programs, a fully solar-powered system, and a cost analysis for household adoption.

9. Conclusion

Black soldier fly larvae (BSFL) have demonstrated a promising method for converting food waste into valuable byproducts while themselves being a solid food source for animal feed. Despite their biological efficiency, the adoption of BSFL systems has remained limited. This review identifies several barriers that contribute to the industry's gap. In many existing systems, users are required to perform frequent manual tasks, such as turning the substrate and feeding system on and off, harvesting larvae, and monitoring conditions. These tasks often require direct interaction with the organisms, which is neither practical nor psychologically comfortable.

These design requirements were identified in the review to emphasize containment, odor control, less manual interaction, and simplified maintenance. By integrating these principles into design, systems can become more accessible and widespread worldwide. The conceptual design proposed in this review represents an initial step towards the broad adoption of insect-based waste-processing systems.

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Youth-Led Advocacy and Barriers to Education By Izabel Karrer

Abstract

This research paper examines barriers to education access and the role of youth-led advocacy in addressing these challenges within school communities. The objective of the study was to explore the obstacles students face in attending and succeeding in school, while also analyzing how student-led initiatives can contribute to meaningful solutions. To conduct this research, survey responses from 54 high school students and an interview with a student leader involved in advocacy work were analyzed. The findings suggest that while many students have access to basic educational resources, challenges such as mental health struggles, academic pressure, transportation issues, and lack of support still affect student success. The research also suggests that youth-led advocacy can help promote awareness, strengthen school communities, and encourage practical solutions to educational barriers.

Introduction

Education is widely considered one of the most important factors in creating opportunities for individuals and improving communities. However, access to education is not equal for all students. Around the world, many students face barriers that make it difficult to attend school consistently or succeed academically. These barriers can include financial limitations, lack of transportation, language differences, and limited access to school resources. Even in communities where schools are well funded, students may still struggle with challenges such as academic pressure, mental health concerns, and a lack of support systems.

Because of these challenges, improving access to education requires more than just building schools or providing materials. It also requires understanding the different obstacles that students experience and identifying solutions that address those challenges. In many cases, students themselves are able to recognize these issues because they experience them directly in their daily lives. As a result, youth-led advocacy and student initiatives have become an important way to raise awareness and develop solutions that improve educational access and support.

Student-led organizations and community initiatives allow young people to participate in identifying problems and creating meaningful change. Through activities like fundraising, awareness campaigns, peer support programs, and community outreach, students can contribute to improving education access both within their own schools and in other communities. These initiatives also help create stronger school environments by encouraging collaboration, leadership, and student engagement.

The purpose of this research paper is to examine the barriers that students face in accessing and succeeding in education and to explore how youth-led advocacy can help address these challenges. In order to better understand these issues, this research includes both survey data and interview findings. The survey, titled *“Barriers, Belonging, and Support in School Access,”* was distributed over a two-week period to high school students from several schools in

Broward County, including American Heritage School, NSU University School, Archbishop McCarthy High School, St. Thomas Aquinas High School, and others. A total of 54 students responded to the survey. In addition, this research includes an interview with a student leader involved in youth advocacy and community outreach through a school UNICEF organization. By analyzing these perspectives, this paper aims to identify common challenges students experience and evaluate the role that youth-led efforts can play in improving education access and support systems.

Understanding these barriers and potential solutions is important because students are not only affected by educational systems, but they can also contribute to improving them. By examining both student experiences and youth-led initiatives, this research highlights how student voices and advocacy can help create more supportive and accessible educational environments.

The first section of this paper examines global and national barriers to education access, including financial inequality, mental health challenges, discrimination, and academic preparedness. The second section explores international frameworks and youth-led advocacy models, focusing on organizations such as CAMFED, the Malala Fund, and Girl Up. The third section analyzes the survey and interview findings collected for this research and connects them to broader educational issues and the role of student advocacy. Finally, the conclusion summarizes the findings of the research, discusses the implications of youth-led involvement in education reform, and identifies areas for future research.

Rights & Global Targets

This section examines the international legal frameworks and global goals that shape access to education around the world. It explores how documents such as the Universal Declaration of Human Rights, the UN Convention on the Rights of the Child, and Sustainable Development Goal 4 define education as a human right and place responsibilities on governments to provide equal educational opportunities. In addition, this section analyzes how youth-led advocacy movements interact with these international frameworks. Although many of these legal instruments do not directly give authority to young people, they still provide the language and goals that student activists and organizations use to advocate for change.

1. International Legal Framework on Education

Education access has always been a crucial aspect of international human rights law. Various legal frameworks have given a pathway to social and youth-led movements in the promotion of access to education. In Article 26 of The Universal Declaration of Human Rights, everyone has the right to education and education should be free, at least at the elementary level (UDHR, 1948). This legal instrument establishes education as a universal right and sets higher education availability based on merit. The International Covenant on Economic, Social, and Cultural Rights (ICESCR, 1966), in Articles 13 and 14, also says that primary education should be required and free, and secondary and higher education should be available based on merit.

The UN Convention on the Rights of the Child (UNCRC, 1989) adds a youth-centered angle because it emphasizes children and young people as rights-holders. Specifically, in Articles 28 and 29, education should promote full development of the child and respected for human rights, and these articles expand earlier protections by naming children as indecent right-holders instead of treating them as people who only receive education without having a say.

The Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW, 1979) and the UNESCO Convention against Discrimination in Education (1960) also deal with fairness, saying governments have to prevent discrimination, especially against girls, in schools.

The Sustainable Development Goal (SDGs) come from the United Nations 2030 Agenda for Sustainable Development, and they outline global targets across areas like education, poverty, and health. Goal 4 (SDG 4) focuses on equity and measurable progress by 2030 to “Ensure inclusive and equitable quality education and promote lifelong learning for all” (United Nations, 2015). It focuses on giving both girls and boys equal chances and making sure education is affordable.

The table below outlines major international treaties and legal instruments that address education as a fundamental human right. It highlights how each document defines access to education, the responsibilities placed on governments, and the specific attention (if any) given to youth involvement in education policy.

Table 1
Key International Treaties and Frameworks Related to the Right of Education

Treaty/Framework	What It Says About Education	What Governments Have to Do	Is Education Mandatory?	Notes/Youth Focus
UDHR (1948)	Everyone has the right to education; elementary school should be free; higher education based on merit	Provide free primary school; make higher education accessible	Primary: yes; Secondary/Higher : not required	General human rights; doesn't specifically talk about youth-led movements
ICESCR (1966), Articles 13 & 14	Education should help people develop fully; primary school free and required; secondary and higher based on merit	Make primary school free and required; gradually expand secondary and higher education	Primary: yes; Secondary/Higher : merit-based	Builds on UDHR; legally binding for countries that sign it
UNCRC (1989), Articles 28 & 29	Education should help kids develop personality and respect human rights; all kids have right to school	Make primary school required and free; encourage secondary and higher; respect kids as rights-holders	Primary: yes; Secondary: encouraged	Focuses on kids and youth; indirectly supports youth participation
CEDAW (1979), Articles 10-11	No discrimination against girls/women in school; equal chances for all	Make sure girls have same opportunities in school; remove laws or rules that block	Varies by country: universal access	Focuses on gender equality; helps SDG 4 goals for girls

		them		
UNESCO Convention Against Discrimination in Education (1960)	No discrimination in education based on race, gender, language, religion	Remove barriers; give everyone same opportunities	Primary and secondary: yes in principle	Helps make education fair globally
SDG 4 (2015)	Make education inclusive and equal; promote lifelong learning	Creates policies to make school accessible, reduce gaps between boys and girls, make it affordable	Primary and secondary: goal is universal access	Has measurable targets by 2030; youth groups use this in their advocacy
UN Security Council Resolution 2250 (2015)	Recognizes youth role in peace and positive change	Include youth in decision-making	N/A	Not about education directly but shows international recognition of youth voices
UN Youth Strategy (2018)	Young people should help achieve SDGs, including education	Involve youth in planning and running education programs	N/A	Non-binding; supports youth-led advocacy indirectly

These global foundations focus on governmental responsibilities. States must remove barriers, have plenty of resources, and make sure there is fair opportunity for education. However, some governments do not follow these obligations due to economic inequality, political instability, or lack of initiative. This is where youth-led movements become critical. Even though the UDHR and UNCRC do not give these movements a role, they ensure that languages (meaning the rights and terms the treaties use) in these documents are held accountable and are fair. For example, the Malala Fund references SDG 4 and the UNCRC when advocating for girls' education globally (Malala Fund, 2021). Similarly, grassroots groups like Fridays for Future have invoked education rights, even though they are focused mostly on climate. The groups argue that young people cannot be effective leaders or innovators without the correct quality of education (UNICEF, 2021; Fridays for Future, n.d.). These examples show how international frameworks do not explicitly name youth-led movements, but they directly empower them by being full of shared vocabulary rights and obligations.

2. Recognition of Youth-Led Advocacy in International Law

It is important to ask if international treaties explicitly recognize youth-led advocacy in the field of education. When it comes to "hard law," it is usually no. Neither the UDHR nor the UNCR provides youth organizations with actual authority.

However, more recent developments somewhat recognize youth as partners in global policy making through documents like the UN Youth Strategy (2018) and UN Security Council Resolution 2250, which both acknowledge youth participation in policy work and social change. The UN Security Council Resolution 2250 (2015), for example, highlights the role of youth in promoting peace and security and recognizes them in making positive change (United Nations Security Council, 2015). While not education-specific, this resolution is important as it shows how international law is beginning to recognize youth as making changes in their own rights. Likewise, the UN Youth Strategy (2018) calls for young people to be involved in achieving the SDGs, including those connected to education (United Nations, 2018). Even though these documents are not legally binding like the UDHR or UNCRC, they still show a shift in international norms toward including youth voices when implementing global targets.

Ultimately, international legal instruments establish that the responsibility to provide education lies with governments, but they also provide legal support that youth-led movements rely on. Governments may be those officially responsible for these matters, but youth step up as leaders who push for change in their communities. Without their involvement, the commitments of SDG 4 and other treaties risk being aspirational and not actionable. The relationship between these frameworks and youth-led advocacy is both indirect and essential: the laws provide the foundation, and young people bring them to life through activism.

While international frameworks establish education as a universal right and encourage governments to improve access, significant barriers still remain in practice. The next section examines the current scope of these challenges, both globally and within the United States, in order to better understand the different forms educational inequality can take.

Barriers to Access and Quality in Education

This section examines the major barriers that continue to affect education both globally and within the United States. While educational challenges exist worldwide, they often appear in different forms depending on a country's level of economic development. In many developing regions, the main barriers involve access and infrastructure, including poverty, conflict, transportation, and lack of school resources. In contrast, developed countries more commonly struggle with issues connected to educational quality, such as academic burnout, declining proficiency rates, and mental health challenges. By examining both dimensions of the issue, this section highlights how educational inequality can exist even in very different environments.

In 2026, education worldwide faces significant challenges. According to UNESCO, approximately 259 million children and youth were out of school in 2018, with Sub-Saharan Africa and South Asia having the highest out-of-school rates (UNESCO, 2019). Factors contributing to these challenges include poverty, gender discrimination, and conflict. For example, in Gaza, over 660,000 children do not have education due to ongoing conflict and the destruction of schools (UNICEF, 2024).

Although barriers to education exist globally, they do not always appear in the same form. In many lower-income or conflict-affected regions, the primary challenge is educational

access itself. Students may be unable to attend school because of poverty, violence, transportation issues, lack of infrastructure, or discrimination. However, in many developed countries, students are more likely to have physical access to schools but still face challenges related to educational quality and student well-being. These issues can include declining academic performance, mental health struggles, burnout, and lack of engagement. Recognizing these two dimensions is important because it shows that improving education requires more than simply increasing enrollment. It also requires creating environments where students are supported and able to succeed academically and emotionally.

In the United States, education statistics have also shown concerning trends. The National Assessment of Educational Progress (NAEP) reported that in 2024, only 35% of high school seniors were proficient in reading and 22% were proficient in math, the lowest level in decades (NAEP, 2024). Many districts reported widening gaps between high and low income students during this period. Factors like increased screen time and reduced in-person learning are contributing to this decline.

The table below summarizes key barriers that continue to limit access to quality education globally and within the U.S.

Table 2
Primary Barriers to Accessing Quality Education in Global and U.S. Contexts

Barrier	Description	Impact on Education	Example/Statistic
Financial Restrictions	High tuition fees, limited scholarships, and rising education costs prevent students (especially from low-income backgrounds) from attending or completing school. (World Bank, 2023)	Reduces enrollments and increases dropout rates, especially in higher education.	Nearly half of U.S students report that cost is a major barrier to pursuing college (NASFAA, 2025).
Mental Health Issues	Stress, anxiety, and depression have become major challenges for students globally, especially after COVID-19 and academic pressure (UNICEF, 2024).	Decreasing focus, motivation, and attendance; leads to lower academic performance and higher dropout rates.	1 of 5 teens report that mental health struggles affect their school performance (UNICEF, 2024).
Lack of Family Support	Many students do not have enough guidance, resources, or encouragement from family to continue education (OECD, 2024).	Low parental engagement is linked to weaker academic success and leaving school early.	40% of students without family academic support are less likely to graduate high school (OECD, 2024).
Academic Preparedness	Students entering higher education sometimes lack foundational skills because of under-resourced schools or poor curriculum (NAEP, 2025).	Increases need for classes and lowers first-year college retention rates.	Only 37% of U.S. students are college-ready in math and reading (NAEP, 2025).
Discrimination and Inequality	Educational inequality based on race, gender, or economic status continues to limit opportunities worldwide (UNESCO, 2025).	Marginalized groups face restricted access to resources, qualified teachers, and safe environments.	In Sub-Saharan Africa, girls are still 20% less likely to complete secondary school than boys (UNESCO, 2025).

Over the past twenty-five years, global enrollment in education has shown steady improvement, though progress varies by region and economic development (UNESCO, 2023) (see Figure 1).

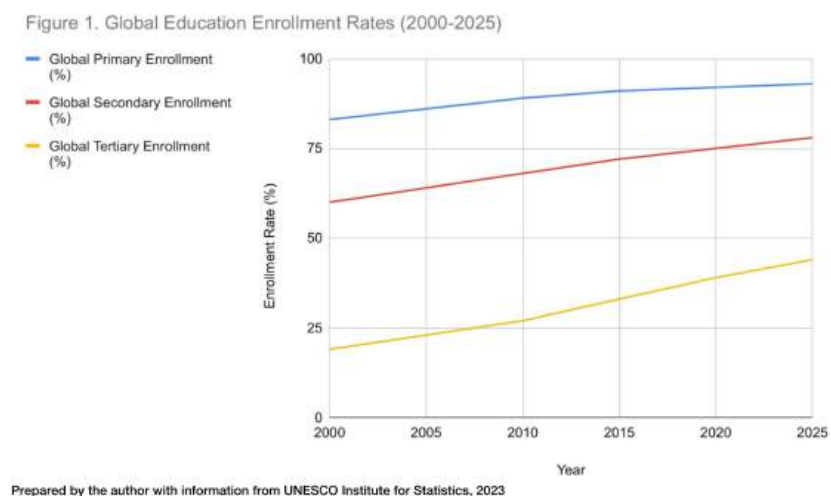


Figure 1. Global trends in educational enrollment over the past 25 years (UNESCO, 2023).

Despite progress in enrollment, millions of children remain out of school, especially in lower-income regions (see Figure 2). Enrollment has gone up steadily over the years, showing that some global efforts to expand education are working, but many kids still don't have access because of poverty, conflict, growing populations, and weak school systems (UNESCO 2025; UNESCO Institute for Statistics, 2025).

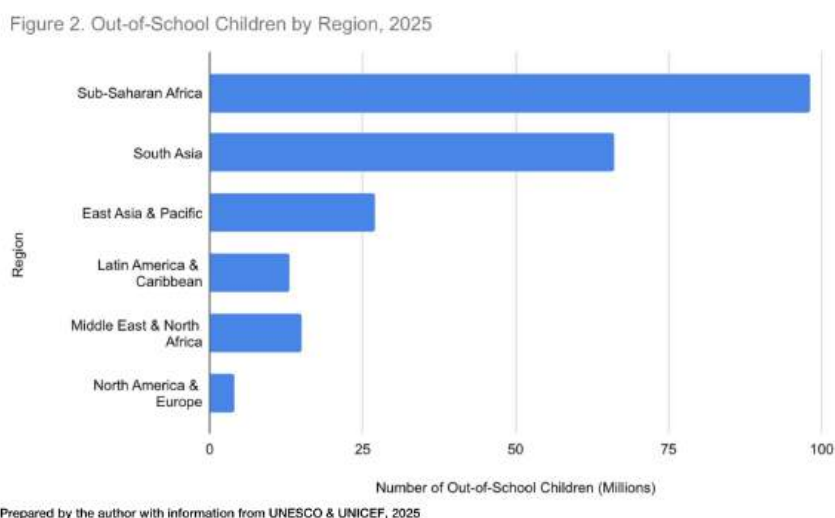
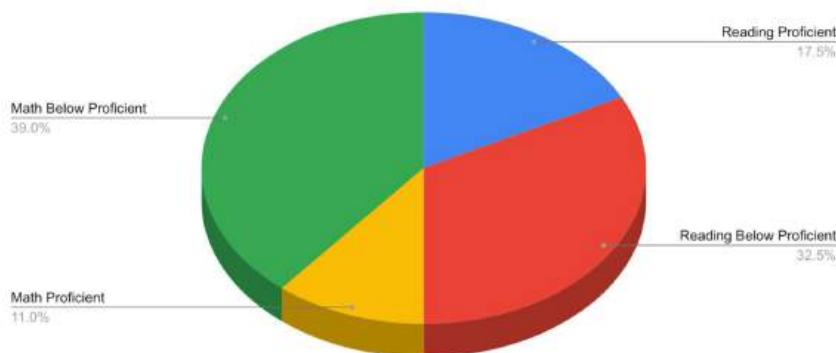


Figure 2. Number of children and youth out of school globally, highlighting persistent access gaps despite overall enrollment increases (UNESCO Institute for Statistics, 2025).

In the United States, academic proficiency among high school seniors has fallen to record lows, highlighting national challenges in educational quality (NAEP, 2024) (see Figure 3).

Figure 3. U.S. High School Seniors' Reading and Math Proficiency Levels, 2025

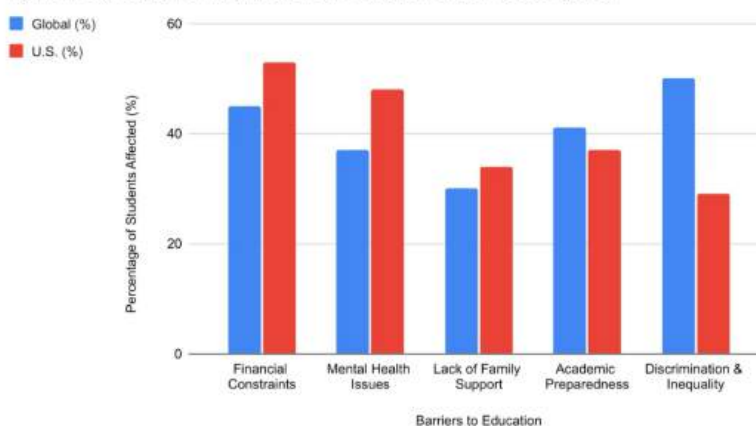


Prepared by the author with information from NAEP, 2024

Figure 3. Decline in academic proficiency among U.S. high school seniors in reading and mathematics, 2000-2024 (NAEP, 2024)

Figure 4 compares the main barriers to education across global and U.S. contexts, revealing that both face overlapping challenges, though their intensity differs (UNESCO and NAEP, 2024).

Figure 4. Comparison of Major Barriers to Education: Global vs. U.S. (2025)



Prepared by the author with data adapted from UNESCO (2025) and NAEP (2024)

Figure 4. Comparison of major barriers to education in global versus U.S. contexts showing differences in access-related and quality-related challenges (UNESCO; NAEP, 2024).

A key takeaway from the comparison in Figure 4 is that barriers to education can be understood through two main dimensions: access and infrastructure, and quality of education.

These two categories help explain why educational inequality exists in very different contexts, even between developing and developed countries.

Access and infrastructure barriers are more common in lower-income or conflict-affected regions. These include challenges such as poverty, lack of transportation, limited school buildings, unsafe travel conditions, and shortages of basic learning materials. In these cases, the main issue is not necessarily whether education exists in theory, but whether students are physically able to reach and participate in it. This is reflected in global data showing that millions of children remain out of school due to economic hardship and instability.

On the other hand, quality of education becomes a more dominant issue in developed countries, including the United States. In these contexts, students generally have access to schools and resources, but still face challenges related to academic performance, mental health, motivation, and engagement. The NAEP data showing declining proficiency levels highlights that being in school does not always guarantee strong educational outcomes. Instead, students may struggle with stress, burnout, or lack of support systems that affect how effectively they learn.

Understanding these two dimensions is important because it shows that improving education cannot focus only on enrollment or access. Even when access is achieved, students may still not be receiving a high-quality or supportive education. At the same time, improving quality alone does not address the millions of students who still face basic barriers to reaching school in the first place. This distinction helps frame education inequality as a multi-layered issue that requires different types of solutions depending on context.

Literature & Models

To better understand how youth-led advocacy operates in practice, it is important to examine existing organizations that translate educational theory into real-world action. The following section explores three key models: CAMFED, the Malala Fund, and Girl Up, that demonstrate different approaches to expanding educational access through youth engagement. These case studies provide evidence of how advocacy can function at the community, national, and global levels, and they help establish a framework for analyzing the impact of student-led initiatives. Together, they illustrate that youth involvement is not symbolic but structurally significant in shaping educational outcomes.

Youth-led advocacy has become one of the most effective ways to promote equal access to education. Government promises and international treaties may sound aspiring, but they fall short when there is no public pressure or community involvement. Youth-led movements and advocacy are efforts organized and run by young people to influence policy, raise awareness, and create tangible change in education and other areas. Youth organizations help fill this gap by connecting policy goals with real people and creating visible change. They do not wait for permission to lead. They identify problems within their own communities and act on them. Such groups as CAMFED, the Malala Fund, and Girl Up prove that youth leadership is not symbolic, it is imperative for progress.

1. CAMFED (Campaign for Female Education)

The Campaign for Female Education, known as CAMFED, demonstrates how education reform thrives when led by people who have lived through its challenges. Founded by Ann Cotton in 1993 in Zimbabwe, CAMFED started with a straightforward premise: if girls receive financial and emotional support to go to school, they will remain in school and eventually return to help others do the same. This is a cycle of support that is as strategic as it is charitable. By creating a network of over 250,000 young women graduates called the CAMA network, CAMFED turns former students into community leaders and advocates (CAMFED, 2025).

This works because the approach shifts from merely temporary aid to long-term leadership. The young women in the CAMA network act as mentors, business owners, and educators who influence local policies. Their success challenges a stubborn stereotype of students in low-income countries as passive learners without influence, instead, they are shown to have agency and the ability to shape education systems from within. According to CAMFED's 2024 report, over 6.7 million girls have benefited from its programs (CAMFED, 2025). That success comes not only from funding provided by private donors, foundations, and governments but from the sense of shared responsibility it builds among youth. The model proves that whenever young people are trusted to lead, they strengthen both communities and institutions. Expanding organizations like CAMFED would translate to investing in local leadership, rather than relying only on outside aid, which often fades away when funding does.

2. Malala Fund

Where the Global Partnership for Girls' and Women's Education focuses on grassroots development, the Malala Fund seeks many of the same outcomes but through advocacy, policy reform, and international pressure as the means of pushing governments to act. Founded by Malala Yousafzai a Pakistani activist for girls' education who survived an attack by the Taliban at age 15 and her father, Ziauddin, the organization brings global attention to the barriers that keep girls out of school: child marriage, poverty, and violence. But its influence goes beyond raising awareness. The Fund's support comes from private donations, corporate sponsors, and foundation grants, and its "Education Champion Network" directly supports young activists who lead change in their own countries (Malala Fund, 2025).

This model matters: It connects local activism to international accountability. Governments that have signed treaties like the UNCRC and ICESCR are promising education for all, but those promises often are not met without continued pressure from within. The Malala Fund keeps them in the public eye. For instance, efforts in Pakistan contributed to the passage of the Right to Education Bill in Khyber Pakhtunkhwa province, ensuring free education to millions of children (Malala Fund, 2025). This victory demonstrates just how youth-led advocacy can strengthen democracy by insisting on transparency and tangible action. Challenges include political resistance in some countries, limited resources for local champions, and safety risks for young activists.

The Fund also gives a global platform to young women who would otherwise be ignored in political discussions. When girls lead advocacy campaigns, they not only gain education themselves but also redefine what leadership looks like. And supporting more organizations like the Malala Fund would mean promoting policies with roots in the real lived experiences of our youth, rather than decisions made without them.

3. Girl Up

Girl Up, an initiative of the United Nations Foundation, was launched in 2010, with a focus on developing leadership among girls and young women. Rather than operating primarily through grants or the lobbying of government policy, Girl Up trains students to become activists in their own schools and communities. To date, the group has reached over 250,000 participants from 150 countries, training them on how to lead campaigns, meet with legislators, and demand equal access to education for girls (Girl Up, 2023).

The relevance of Girl Up lies in its structure: It operates through peer-to-peer networks that grant a sense of belonging and purpose to students. Teenagers are not only taught to speak up but also to mobilize public opinion and policy to effect change. This model builds confidence and civic awareness, while addressing global education goals. When students take the lead to campaign for menstrual equity or anti-violence policies, they help remove some of the barriers blocking girls from attending school. Challenges include maintaining engagement over time, funding for expansion, and ensuring cultural adaptation across different countries.

The success of this organization proves that activism does not have to wait until adulthood: high school students, if organized and informed, can move institutions in the direction of fairness and inclusion. Programs such as Girl Up also prepare future policymakers who understand education both personally and from a social perspective. Expanding these kinds of programs would result in a generation of young leaders who could sustain educational reform rather than rely on temporary government programs.

4. Why These Models Matter

These three organizations prove that youth-led advocacy isn't a minor, it's a necessity. International frameworks like SDG 4 and the UNCRC create moral and legal imperatives, but youth organizations give those commitments muscle. CAMFED turns education access into community leadership. The Malala Fund forces governments to act on their promises. Girl Up trains young people to advocate for themselves and others. Each approach reinforces the notion that education thrives when those affected by it are part of the solution. While all three emphasize youth-empowerment, CAMFED focuses on long-term community leadership, the Malala Fund emphasizes policy advocacy and international attention, and Girl Up emphasizes peer-to-peer activism and school-level campaigns. Each faces funding, scale, and political challenges that have to be managed to maximize their impact.

Also, disregarding youth involvement weakens the global movement of equal education. Policies, without groups like these, are in danger of being disconnected from real needs.

Supporting youth-led organizations means investing in lasting solutions, as young leaders remain in their communities and continue the work long after the campaigns are over. Their inclusion ensures that education is not treated as a charity but rather a shared responsibility.

Incentivizing more youth-driven models would help close the gap between what is promised in international law and what students experience in classrooms. When young people organize, they hold systems accountable, inspire peers, and turn global goals into local change. For education to be accessible and equitable, their leadership is not only helpful, it is essential.

Building on these models of advocacy, the next section turns to primary data collected through a student survey and interview. This empirical study connects the literature to real student experiences, offering insight into how barriers to education are perceived within school communities and how youth leadership is reflected in practice.

Empirical Study

This empirical study section presents the primary data collected to examine student experiences with educational access and the role of youth-led advocacy in addressing barriers within school environments. Using a mixed-methods approach that combined a student survey and an interview, this research connects the theoretical models discussed in the previous section to real-world perspectives from students. The following findings provide insight into patterns of access, support, and school climate as reported directly by participants, while also acknowledging the limitations of a small and non-representative sample.

Methodology and Research Design

In order to better understand barriers to education and the role of youth-led advocacy, this research used a mixed-method approach that included both a student survey and an interview. The survey, titled “Barriers, Belonging, and Support in School Access,” was distributed to high school students and received a total of fifty-four responses. The survey responses came from students attending several schools, including American Heritage School, NSU University School, Archbishop Edward A. McCarthy High School, and other schools within Broward County. Most respondents were eleventh grade students, which reflects the age group most represented in this study. It included sixteen questions covering topics like attendance, access to resources, school environment, and barriers to success. Most questions were multiple choice or Likert-scale, with some open-ended questions to allow students to explain their experiences in more detail. The full survey instrument is included in the Annex section of this paper.

In addition to the survey, an interview was conducted with a student leader who serves as co-president of a UNICEF club at her school. Her position is relevant because she is directly involved in youth-led advocacy efforts, including fundraising, awareness campaigns, and initiatives aimed at improving access to education. Her perspective provides insight into both local school environments and global education challenges through her involvement with UNICEF-related work.

While these methods provide useful insight, there are important limitations to consider. The survey sample is not representative of all students, as the majority of respondents attend

well-resourced private schools. Because of this, the findings cannot be generalized to all student populations. Instead, the data should be understood as exploratory. It highlights patterns and perspectives that may reflect broader issues, but does not definitively represent all students. However, these findings are still valuable because they provide a starting point for understanding student experiences and can serve as a baseline for future, more comprehensive studies.

Analysis of Survey and Interview Findings

The survey results and interview findings support the overall position of this research paper, which is that youth-led advocacy can help address barriers to education by identifying student challenges and encouraging solutions within school communities. While the sample is limited, the data suggests patterns that reflect both local experiences and broader educational issues.

One key finding from the survey is that most students reported having access to the basic resources needed to succeed in school. Specifically, 87.04% of students indicated that they almost always have what they need, including transportation, supplies, and technology. This suggests that many students in this sample attend relatively well-resourced schools. During the interview, the participant supported this idea by explaining that her school community faces fewer structural barriers due to its financial resources.

To better understand how access to basic resources affects student success, the survey asked students how often they have what they need to do well in school including transportation, supplies, and technology. The results of this question are shown in Figure 4.

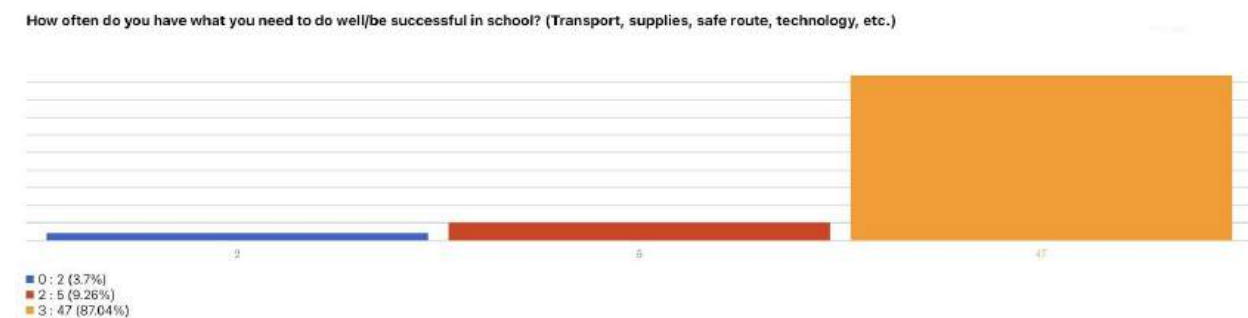


Figure 4. Student access to resources needed for school success, including transportation, supplies, and technology (author's survey data; survey included in Annex).

However, this does not mean that barriers to education are absent. Instead, the type of barriers appears to shift. While students in lower-income communities may struggle more with access to basic necessities, students in more privileged environments may face challenges related to mental health, stress, and academic pressure. This reflects broader research, which shows that educational challenges are not limited to access alone but also include well-being and engagement.

One of the most significant patterns in the survey data was the role of mental health and academic pressure. About 24.29% of students identified physical or mental health challenges as a barrier to performing well in school. In addition, many students used open-ended responses to

describe issues such as academic burnout, lack of motivation, and overwhelming workloads. Phrases like “too much homework,” “overwhelming pressure,” and “lack of time” appeared frequently. This suggests that even when material resources are available, students may still struggle to succeed due to internal and environmental pressures.

To identify the most common challenges students face, the survey asked participants to select the main obstacles that prevent them from performing well in school. The responses, summarized in figure 5, highlight patterns in both academic and personal barriers.

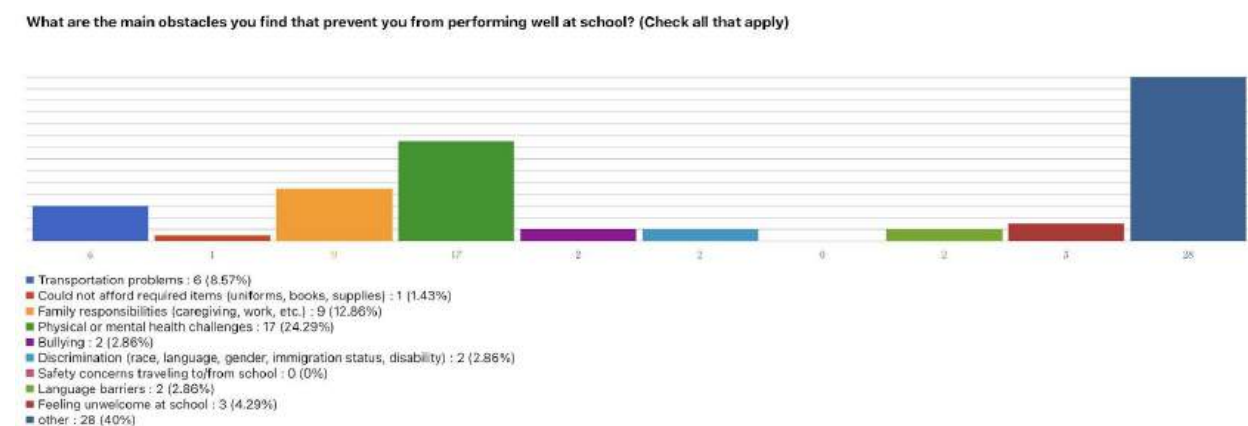


Figure 5. Reported obstacles that prevent students from performing well in school, including academic, personal, and environmental challenges (*author’s survey data; survey included in Annex*).

This idea is reinforced by the interview findings. The interviewee explained that mental health challenges, especially in competitive academic environments, can significantly impact attendance and performance. She noted that stress and anxiety can discourage students from attending school regularly. These findings align with broader trends in education, where increasing attention is being given to student mental health as a key factor in academic success.

Transportation also emerged as a recurring theme. Even though only a small percentage of students directly selected transportation as a barrier in the survey, it appeared frequently in open-ended responses. Many students suggested that more buses, improved transportation systems, or later start times would make it easier to attend school consistently. The interviewee also emphasized that transportation is a major barrier in many communities, especially in lower-income areas where students may not have reliable access to school.

Another important theme in the data is the role of school environment and belonging. A large majority of students reported feeling safe and comfortable at school, with over 87% agreeing that they feel welcomed. However, open-ended responses showed that students still see room for improvement. Many suggested creating stronger communities through clubs, reducing academic pressure, and improving teacher support. This indicates that while students may feel generally accepted, they still desire more supportive and balanced environments.

In addition to resources and academic challenges, the survey also explored how students feel within their school environments. Students were asked about their sense of belonging and support, and the results are presented in Figure 6.

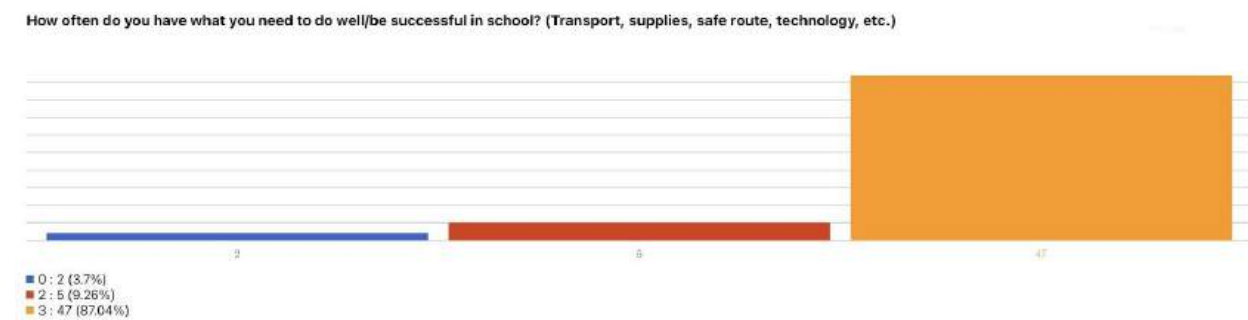


Figure 6. Student perceptions of school environment, including feelings of safety, belonging, and support (*author's survey data; survey included in Annex*).

The interview provided additional insight into this issue by highlighting how small actions can improve student experiences. The interviewee suggested that even simple initiatives, such as promoting positive messages or increasing mental health awareness, could have a meaningful impact on students. This connects directly to the role of youth-led advocacy.

Finally, both the survey and the interview emphasize the importance of student-led initiatives. The interviewee described how organizations like the UNICEF club organize fundraising events, donation drives, and awareness campaigns that support education access. She also provided a specific example of a student-led initiative called “Promote Her Period,” which placed menstrual health supplies in school bathrooms. This example demonstrates how students can identify specific needs and create practical solutions that directly impact their peers.

These findings suggest that youth-led advocacy can play a meaningful role in addressing educational challenges at the school level, particularly by providing targeted support within student communities. However, because this study is based on a limited and non-representative sample, the results should be interpreted as exploratory rather than definitive. While the data does not indicate that youth advocacy alone can resolve structural barriers, it does suggest that it can contribute to broader efforts to improve access to education when combined with larger policy initiatives.

Overall, this research indicates that barriers to education are complex and context-dependent, but also that students are not only affected by these challenges—they can actively participate in addressing them through advocacy and leadership. These findings will be further synthesized in the concluding section, along with supporting documentation including MLA and APA references and the annex containing the full survey instrument.

Conclusion

This research set out to better understand the barriers that affect students’ ability to access and succeed in education, as well as the role that youth-led advocacy can play in addressing

these challenges. Based on both the survey data and the interview findings, it is clear that while many students, especially those in well-resourced schools, have access to basic educational needs, this does not mean that barriers no longer exist. Instead, the type of challenges students face often shifts from material limitations to issues such as mental health, academic pressure, and lack of balance.

The survey results suggest that most students feel they have the resources needed to succeed, but many still struggle with stress, burnout, and motivation. Similarly, the interview highlighted that even in stronger school environments, students experience pressure that can affect attendance and performance. At the same time, both the survey and interview pointed to transportation, financial limitations, and lack of resources as major barriers in other communities, showing that educational inequality still exists on a broader level.

One of the most important takeaways from this research is the role of youth-led advocacy. The findings suggest that students are not only aware of the challenges they face, but are also capable of creating meaningful solutions. Examples such as student-led fundraising, awareness campaigns, and initiatives like providing menstrual health resources show that students can directly impact their communities. These efforts may not replace large-scale policy changes, but they can still play a significant role in improving access to education and supporting students in practical ways.

At the same time, it is important to recognize the limitations of this research. Because the survey sample was relatively small and not representative of all student populations, the findings cannot be generalized to all schools or communities. Instead, the data should be understood as a starting point that highlights possible patterns and areas for further exploration. Despite this limitation, the research still provides valuable insight into student experiences and the potential impact of youth involvement.

Future research should expand on these findings by including a larger and more diverse sample of students, especially those from underserved and low-income communities. It would also be beneficial to conduct multiple interviews with educators, policymakers, and students from different backgrounds to gain a more comprehensive understanding of educational barriers. Additionally, future studies could focus more specifically on measuring the long-term impact of youth-led initiatives to determine how effective they are in improving education access over time.

Overall, this research suggests that improving education access requires both structural changes and student involvement. While governments and institutions play a major role, students themselves are an important part of the solution. By listening to student perspectives and supporting youth-led advocacy, schools and communities can take meaningful steps toward creating more equitable and supportive educational environments.

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From Disruption to Distrust: Conflict-Based Supply Chain Management By Harrison Lai

Abstract

Global supply chains are increasingly exposed to geopolitical conflict, including tariffs, wars, and political instability. For multinational corporations, geopolitical conflicts have made supply chains more complex due to the operational, financial, and reputational risks associated. The central research question of this paper is: How do geopolitical conflicts influence supply chain diversification strategies of multinational corporations? This paper reviews existing research on supply chain vulnerability, risk management, supply chain diversification, geopolitical disruption, and corporate supply chain strategies. Existing literature shows that supply chains are vulnerable due to being optimized for efficiency and cost. The literature also demonstrates that supply chain diversification can reduce risk, improve reliability, and sometimes reduce inventory pressure. Most research addresses the central research question indirectly through conceptual frameworks, case studies, broad reviews, or models. This paper develops The Empty Shelf Theory. The theory argues that geopolitical conflict creates reputational damage for multinational corporations, causing corporations to diversify in order to protect timeliness, customer satisfaction, and reputation. Corporations are likely to prioritize diversification around bottleneck components, customer-facing products, and major logistic routes where failure would be most visible to the market. Because firm-level evidence remains limited, future work should test The Empty Shelf Theory using direct corporate data.

Introduction

Supply chains carry serious implications. Supply chain diversification typically means using a collection of strong, independent, and resilient suppliers. Diversification is crucial for multinational corporations to succeed overall, from simply minimizing and avoiding risk, to minimizing cost and more. In a time of increasingly complex and unprecedented geopolitical conflicts, supply chain diversification is now more important to multinational corporations than ever before. Geopolitical conflicts and political policy are heavily intertwined with the supply chains of multinational corporations. For example, during the Covid-19 pandemic, U.S. manufacturers struggled to keep up with mask demand as well as the overall short supply of masks in the U.S. caused by China withholding supplies intentionally (Chanis).

Taiwan's stronghold on semiconductors through TSMC, which supplies semiconductors to Apple, for instance, is not only crucial to the success and effectiveness of many products sold by multinational corporations, but also poses significant supply chain vulnerabilities posed by what experts and politicians worldwide call an essentially inevitable invasion of Taiwan by China. These dependencies also carry major U.S. national security implications. Strengthening supply chain diversification strategies can not only help multinational corporations minimize risk during major geopolitical conflicts like the invasion of Taiwan, but can also dramatically reduce the inventory investment of a multinational corporation, for example (Jain et al.). Strengthening

supply chain diversification strategies can look like using more suppliers than just one, strategically choosing manufacturing or sourcing locations based on limited geopolitical risk, or by using radio frequency identification tags, among many other ways.

This paper's central research question is: How do geopolitical conflicts influence supply chain diversification strategies of multinational corporations? Although existing literature widely agrees that diversification reduces supply chain vulnerability, most work approaches the issue indirectly through conceptual models, risk management frameworks, or case studies rather than examining firm-level strategic decisions directly. This literature review first examines how research frames the vulnerability of global supply chains and analyzes broader risk-management. Next, this paper addresses how literature frames diversification as a key risk-mitigation strategy, then evaluates research on the effects of geopolitical conflicts on supply chains, and then addresses the direct effects of geopolitical conflicts on companies. Finally, the paper evaluates the lack of direct firm-level data on supply chain diversification as a result of geopolitical conflicts, then offers a reputation-driven theory, and concludes with a summary of the paper.

Supply Chain Vulnerability

The existing literature agrees that global supply chains are highly vulnerable to risks across multiple stages of the supply chains and key structural components. In a case study on supply chain disruption by the Avian Flu pandemic, Kumar identifies vulnerabilities at nearly every stage of a retailer's global supply chain, especially where staff health hazards could spread disruption, which can lead to the flu's exposure at the raw materials process of the supply chain all the way to customers through the retailer's retail outlets. With a different model of a computer assembly company's supply chain, Kumar identifies exposure at multiple parts of the supply chain again, including at the supplier and at the retail outlets. The implications of the pandemic for the corporation are increasingly evident due to Kumar's closed loop diagrams, showing that the flu can overall lead to a decline in the retailer's sales, putting a strain on the corporation's financial resources (Kumar and Chandra). Overall, Kumar's case study demonstrates that supply chains have vulnerabilities at multiple stages. The effects are not just operational, but are also financial, since these disruptions reduce sales and strain overall corporate resources.

Like Kumar, Chanis highlights several risks and vulnerabilities, mainly the structural fragility of the supply chains. Chanis asserts that supply chains optimized for efficiency through the just-in-time inventory leads to those supply chains being more exposed. The complexity of these supply chains creates fragility, and firms often lack the knowledge of all lower-tier suppliers. In addition, corporate strategy focused on profit and efficiency leads to increased dependency on offshoring and outsourcing (Chanis). This is a key vulnerability because multinational corporations lack direct control over these suppliers. Also, the concentration of manufacturing in China creates a dependence on China across many sectors. Too much production is concentrated in one place, leading to significant vulnerabilities for global supply chains.

Kumar shows where vulnerability can occur, but Chanis demonstrates the severity of it.

Chanis explains that the structural reason for these vulnerabilities is that the firms have optimized for profit and efficiency, while Kumar finds the disruptions spreading across multiple stages of the supply chain. These supply chains have little redundancy, hidden suppliers, and single points of failure. Chanis shows that the operational and financial consequences demonstrated by Kumar are due to decades of offshoring and outsourcing. These two works explain that supply chains are vulnerable not only because these effects spread through interconnected systems, but also because the production is concentrated.

In addition to Kumar and Chanis' works, work by Sarathy reinforces, similar to Kumar and Chanis, that vulnerabilities occur at multiple parts of the supply chain, but focuses on security issues such as terrorism. Sarathy identifies containers as a major source of security concerns when it comes to global supply chains. Not only does smuggling often occur, but the use of a weapon of mass destruction at a container port could cost a trillion U.S. dollars due to the port closure. In addition to containers, Sarathy identifies vulnerabilities at the following factors of a global supply chain: Goods, factories, supply chain providers and partners, supply chain facilities, freight carriers, people, and confidential information. Among other concerns, Sarathy identifies firms that use smaller supply chain partners as being more vulnerable and less resilient because smaller partners are likely to be less able to withstand supply chain disruptions (Sarathy). Overall, the literature presents that vulnerability comes from weak supply chains, concentrated supply chains, and major sources of security risks at multiple points, all possibly leading to diversification as a necessary solution.

Risk Management Broadly

A large part of the literature explains how firms manage supply chain risk only indirectly, using conceptual frameworks, collaboration models, and managerial prioritization tools. First, work by Yu provides a conceptual model on supply chain risk management. Yu frames supply chain risk as an increasing problem due to globalization and outsourcing, like the other works indicate as well. Yu's article provides use to this paper because it helps explain the logic of risk mitigation by proposing relationships among variables. Specifically, Yu argues that supply chain integration is the key mediating mechanism between risk and firm performance. Their model suggests that supply chain integration is positively related to company performance. Thus, they show that firms manage risk better with stronger coordination and integration. Yu also presents the idea that organizational risk propensity moderates the relationship between risk and integration (Yu et al.).

Another work highlights collaboration as a risk-management mechanism. Ali and Shukran argue that supply chain risk can not be managed effectively at the firm level alone. They use a case study of the Australian citrus industry to show that firms are not only vulnerable to their own operational issues, but also to the actions and conditions of other supply chain partners. They treat collaboration as a central risk-management mechanism. It frames coordination and collaboration as central to maintaining profitability and continuity. They identify collaborative practices that reduce risk, such as information sharing, decision synchronization, resource

sharing, and goal congruence (Ali and Shukran). The work is ultimately another conceptual framework, not a direct study of diversification. It does contribute, however, by explaining how collaboration can moderate the relationship between risk management and firm performance.

Lee and Ulferts, on the other hand, provides a logical identification and prioritization process of risks as a mechanism to effectively manage supply chain risk. Lee and Ulferts frames supply chain risk management as a strategic planning process. They conclude that effective management requires firms to accurately identify, assess, and prioritize risks. They categorize risk into four major types: financial, operational, human resources, and information risk. Using an analytic hierarchy process, the study prioritizes risks according to strategy criteria like cost, quality, delivery, and flexibility. In the specific case study used, they end up ranking operational risk as the highest, followed by financial risk (Lee and Ulferts). They demonstrate the importance of systematic prioritization. They contribute to this paper by explaining how firms can structure risk assessment and mitigation.

These studies treat risk management as important, but at a conceptual or managerial level. Yu offers the closest to firm level decision making, but still lacks direct answers. Yu's point on organizational risk propensity helps argue that firms respond differently to risk. Ali and Shukran shift the focus to a network. They help highlight risk as interdependent. They explain that collaboration matters, but also shift the logic slightly as it is no longer simply a multinational corporations' decisions on diversification, but a network of supply chain partners. Lee and Ulferts focus on managerial processes, and they do demonstrate value in categorizing risks and prioritising those risks structurally. These sources together explain mechanisms of risk response without directly addressing diversification or geopolitical conflicts.

Supply Chain Diversification as a Risk-Mitigation Strategy

The existing literature broadly agrees that diversification is a major mitigation strategy. Jain writes on managing global inventory sourcing and key points related to diversification. Global sourcing alone tends to increase inventory investment. They find that a 10% shift from domestic to global sourcing increases inventory investment by 8.8% for the average firm. However, supplier diversification mitigates that large inventory burden. Increasing the number of suppliers can reduce the inventory strain created by global sourcing, and the work also finds that moving from single to dual sourcing reduces inventory investment by about 11%. Interestingly, Jain's work helps argue that diversification is not just about resilience, but can also improve efficiency while still maintaining a strong supply chain. Jain demonstrates that diversification is not just a measure against disruption, but a more dispersed supplier base can reduce effective lead times and lower safety-stock needs, meaning that diversification may improve inventory performance in addition to making the supply chain more resilient. Jain's work utilizes firm-level empirical evidence linking global sourcing practices to inventory investment (Jain et al.). Overall, Jain explains the tradeoffs involved in diversification. Concentrated sourcing may lower transactional frictions, but diversified sourcing can reduce lead times, improve reliability, and lower inventory pressure by reducing the need for safety-stock.

Rather than treating diversification as a universal practice, the existing literature also presents diversification as a conditional mitigation strategy depending on supplier reliability and inventory tradeoffs. Tomlin's work on supply chain risk mitigation strategies emphasizes diversification as an operational mitigation tactic. Tomlin uses a framework to show that sourcing sits alongside inventory as a major risk mitigation strategy. Tomlin's main contribution to this paper is that he argues that dual sourcing can be optimal when under significant disruption risk. He gives examples of Nokia's multiple-supplier sourcing and Chiquita's multiple location sourcing. He also utilizes a model to highlight that shifting to a more reliable supplier can be optimal (Tomlin).

Not only does Tomlin demonstrate that diversification can help reduce risk, but he also indicates that disruption characteristics affect the value of diversification as a mitigation strategy. He finds that sourcing improvements are favored over inventory adjustments especially when the disruptions are longer. Similar to Jain, Tomlin rejects single sourcing because it values efficiency over resilience. Tomlin shows this similarity with Jain by modeling tradeoffs between suppliers of different levels of reliability, leading to the conclusion of using multiple suppliers and more reliable suppliers. Also similar to Jain, Tomlin links inventory with sourcing strategies and generally finds that in certain cases, sourcing strategies are significantly preferable to inventory changes. Tomlin's results are generally more cautious than Jain's, as Tomlin suggests diversification as significant but conditional, while Jain emphasizes a more optimistic view of diversification, as Jain suggests that diversification can lead to both resilience and efficiency. Both works do acknowledge that diversification is a valuable risk mitigation strategy in global supply chains.

Geopolitical Conflicts and Global Supply Chains

The existing literature has a general consensus that geopolitical conflicts can be a distinct driver of supply chain risk. First, work by Xiong and Sun is the clearest source that shows geopolitical conflicts increase supply chain risk. The work identifies three concrete mechanisms of geopolitical conflicts increasing supply chain risk: lower supply chain resilience, greater customer concentration, and reduced inventory liquidity. This work is valuable because it uses empirical evidence from Chinese firms. It adds an element to the existing literature beyond conceptual models. Xiong and Sun, however, focus more on the effect of these conflicts rather than how firms diversify in response to these conflicts (Xiong and Sun).

Part of the literature demonstrates how patronage networks can complicate the already complex industry of conflict minerals, furthering the consensus that conflicts can lead to increased risk in supply chains. Vogel's work shows that conflict zones are shaped by informal political power rather than normal market logic that Western business is used to. His work demonstrates that geopolitical conflicts shift supply chains to be governed differently: by power, influence, and coercion rather than traditional corporate culture. Vogel demonstrates that patronage networks, smuggling systems, and political alliances distort the movement of goods, adding to significant supply chain risks. First, Vogel discusses a case of "boat busting," in which

a smuggling operation occurs where military officers and associated patronage networks become undeniably powerful, and as a result, this network then moves shipments and evades controls. These military officers use their military power in order to smuggle minerals, with minerals mysteriously disappearing and different actors of the patronage network constantly coming into play to influence movement of goods. Second, the article examines a succession dispute involving mining and armed groups. After a chief dies, legitimacy becomes a competition, political elites come into play to back different people claiming power, and factions aligned with local militias infiltrate mining committees. Vogel helps broaden the definition of geopolitical risk from being only tariffs, wars to include fragmented authority, post-conflict politics and informal control over supply chains (Vogel).

Xiong and Sun show measurable firm level effects of geopolitical conflicts, while Vogel establishes what these conflicts can look like first hand in the context of a supply chain. Xiong and Sun treat these conflicts as something that increases supply chain risk. On the other hand, Vogel demonstrates that conflicts can transform the supply chain environment itself. Together, these works demonstrate why conflict can push multinational corporations toward diversification, relocation, or outright avoidance of certain regions.

Geopolitical Conflicts and Corporate Adjustment of Global Supply Chains

The literature does identify common corporate adjustments to geopolitical conflicts, but it is limited and only discusses patterns broadly. The most direct work on this paper's central research question is a comprehensive literature review on the impact of geopolitical disruptions on supply chains overall, including the configuration, flow, and management of supply chains. While the work focuses on supply chains more broadly and not specifically supply chain diversification strategies, the work does offer significant insight into the impact of geopolitical conflicts on global supply chains. The work finds three key themes in the fifty articles: tensions between nation states, supply chain design, and technology. Using a very structured methodology to find these fifty articles, they highlight many important findings that are useful for this paper. First, focusing on literature about Brexit, they find that during geopolitical disruptions like Brexit, corporate leaders need to make decisions in "tight time frames," utilizing flawed information to deal with uncertainty. They found that companies planned to move manufacturing facilities to the European Union (EU) and planned to move distribution centers to the United Kingdom. The literature they analyzed found that these shifts were caused by advantages to do with the closeness of centers of demand and overcoming logistics like tariffs. In the section of the supply chain design theme, they find that actions are centered around moving manufacturing and storage facilities closer to end markets. They also find that the literature suggests businesses move suppliers closer to manufacturing facilities to avoid inconsistencies in materials (Bednarski et al.). However, even this seemingly direct work to this paper's central research question has its limitations. The Bednarski work gets close, but still tends to synthesize broad responses as it does not have a similar specific dependent variable of the supply chain diversification strategies of multinational corporations.

This work is distinct in that it does highlight specific actions taken by companies as a result of geopolitical conflicts through presenting other literature. While a work like Sarathy's gives significant recommendations to companies such as making their supply chains more resilient, using RFID tags, and using larger suppliers as they are likely more reliable and less susceptible to supply chain disruptions, Bednarski uniquely presents literature that highlights actions taken by European firms instead of offering broad, possible recommendations.

Gaps in the Literature

Overall, the literature is broad, useful, and methodologically diverse, but the existing literature tends to address the question indirectly and largely lacks direct firm level supply chain diversification decision making as a result of geopolitical conflicts. Not only is firm level data limited, but the literature does not clearly explain how geopolitical risk gets translated into corporate decisions. The literature stops before showing how executives respond in terms of suppliers, locations, and overall supply chain structure as a result of geopolitical conflicts. Conceptual frameworks, case studies, and broad reviews are the main pieces of literature, providing valuable insight for vulnerability, but does not show firms choosing strategies. The literature does not separate different types of diversification either. This causes the literature to miss important differences in how firms actually respond to geopolitical conflict. The literature also largely does not explain why two corporations could respond differently to the same geopolitical conflict. The nuances of decision making are missed. There are many potential reasons for these gaps. The main one is that firms largely do not publicly disclose strategy changes. Because sources are scarce, the literature tends to rely on frameworks and broad conclusions. Overlooking firm level evidence causes the literature to miss the sequence of decisions and tradeoffs executives make. Another important implication is that the literature may overstate diversification because of this gap. Diversification may be treated as a more effective solution to addressing risk caused by geopolitical conflicts than it actually is.

The Empty Shelf Theory

Under normal conditions, multinational corporations build supply chains mainly based on cost. But when geopolitical conflicts occur or the significant threat of geopolitical conflict, the corporation's reputations become at risk. The supply chain turns from a logistics problem to a reputation problem. When a geopolitical conflict occurs, customers' minds are not reshaped directly. Customers likely do not think about how their desired products will be affected by conflicts, but rather have the same expectations of fast deliveries and having favorite products in stock. While a supply chain might experience a geopolitical conflict directly through tariffs, damage to facilities or infrastructure, or full-blown war impacting the entire supply chain, customers do not typically experience these conflicts. However, they do experience late deliveries, stockouts, missed launches, and their favorite products not being available for them to purchase conveniently. Damage then moves from operations to the market. Retailers could lose trust because they don't have products in stock. Customers get frustrated that they can't buy their

desired products. Overall, because of geopolitical conflicts, a firm's reputation could significantly weaken. This reputation aspect leads to a possible theory on supply chain diversification during geopolitical conflicts: The Empty Shelf Theory.

1. Geopolitical conflict makes multinational corporations diversify supply chains not just to reduce disruption, but to protect timeliness, customer satisfaction, and reputation.

Reputation directly affects future business, so multinational corporations would likely strive to diversify supply chains in an attempt to maintain a solid reputation for being reliable and stable. Firms would strive to avoid the visible consequences of disruption on their customers and retail partners. Multinational corporations, in the event of geopolitical conflicts, would attempt to protect customer satisfaction and preserve trust in the brand. This changes how executives think about supply chains. They would likely begin thinking about what structure of supply chain makes their firm least likely to disappoint the market. Firms would more than likely diversify first in parts of the supply chain that are directly related to delivery speed and customer facing performance. Specifically, firms would likely diversify bottleneck components, components that are crucial to the entire product. These bottleneck components can stop the entire product from being sold, so firms would likely diversify to minimize risk for these bottleneck components. Geopolitical conflict makes firms think about which single supplier, country, or component could stop them from delivering the final product. For example, in the event of a geopolitical conflict, a car company may not urgently diversify suppliers and manufacturing for small interior parts of the car, but they may more likely diversify suppliers and manufacturing for semiconductors, batteries, or key engine parts. In addition, firms are more likely to diversify supply chain components for customer-facing products first. A multinational corporation's reputation isn't affected equally by every product. Corporations would diversify supply chain parts for favorite products and not focus much on smaller, less volume products. For example, a large company like Apple, when experiencing a geopolitical conflict, would likely take action to diversify supply chain parts for the Iphone, but not for an Iphone case. Firms would also diversify logistic routes and transportation nodes in addition to suppliers and manufacturing. During a geopolitical conflict, a firm will likely shift dependence to multiple ports, multiple shipping lanes, or different ports and shipping lanes. This shift creates redundancies and reduces the large-scale impact of a geopolitical conflict even if it costs more in the short term for a corporation. Over time, firms develop stability in revenue and profits by depending on repeat customers, and having repeat customers depends on having a reliable brand. Geopolitical conflicts make reputation-sensitive supply chain components more crucial to executives of multinational corporations.

Conclusion

To conclude, geopolitical conflict has become a major force shaping global supply chains, changing conditions, increasing overall uncertainty, and exposing concentrated dependencies. The literature consistently shows that supply chains are vulnerable when

prioritizing efficiency, offshoring, and region concentration as opposed to resilience. Diversification of these supply chains acts as one of the clearest risk mitigation strategies because it can reduce dependence on single suppliers and improve overall reliability. Diversification also reduces inventory pressure in some cases. The implications of this paper for business executives of multinational corporations are as follows. Supply chains must be treated as a leadership and geopolitical issue, not simply an operational one. Executives should actively evaluate concentration of suppliers, location risk, and emergency preparedness. For example, although Vogel's work is not directly about geopolitical conflicts impacting firm level decisions on supply chain diversification, the logic is that firms may avoid unstable regions altogether or treat them as unusually risky because the conventional business calculations do not work in these regions. In addition, executives should recognize that geopolitical risks require fast decisions under imperfect information. A major research implication is that future studies should move beyond frameworks to examine direct firm level decision making. Further research should examine how executives choose among supply chain locations and examine the direct relationship between these strategies and geopolitical conflicts. Future work can also strive to test The Empty Shelf Theory developed in this paper.

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Trauma, Attachment Relationships & Adolescence By Tanvi Polasi

Abstract

In recent years, adolescent mental health has emerged as a pressing public health crisis. Youth mental health is a defining issue of our time, with trauma playing a central role in shaping long-term psychological outcomes. Post-traumatic stress disorder (PTSD) in adolescents includes several clusters of symptoms: intrusive memories, avoidance of trauma-related stimuli, negative changes in mood and cognition, and heightened arousal responses, as outlined in the DSM-5 (American Psychiatric Association 5). While much research on PTSD in adolescents has focused on its biological, emotional, and behavioral effects, it is also important to consider how early relationships shape the development and persistence of PTSD symptoms. Additionally, more research is needed to explore the effectiveness of tailored interventions based on trauma type and attachment style, as current therapeutic approaches may not fully address the nuanced needs of adolescents experiencing diverse forms of trauma. Trauma-focused cognitive behavioral therapy (TF-CBT) is an evidence-based treatment designed for children and adolescents suffering from PTSD and other trauma-related symptoms. This paper highlights the critical role of attachment in adolescent PTSD, highlighting TF-CBT as an effective intervention that addresses both trauma symptoms and the caregiver-child relationship to promote lasting recovery.

Introduction

In recent years, adolescent mental health has emerged as a pressing public health crisis. Youth mental health as a defining issue of our time, with trauma playing a central role in shaping long-term psychological outcomes. Research suggests that between 3% and 15% of adolescent girls and 1% to 6% of boys will develop post-traumatic stress disorder (PTSD) following exposure to trauma (Hamblen and Barnett 366). The consequences of untreated PTSD are severe, potentially leading to emotional dysregulation, academic struggles, and impaired social relationships, significantly altering a young person's developmental trajectory. As the prevalence of trauma-related disorders continues to rise, understanding the relational factors that influence PTSD development and recovery is essential.

Caregiver-child relationships form the foundation of healthy emotional development, particularly during adolescence, a period marked by identity exploration, increasing autonomy, and heightened sensitivity to social experiences (Julian et al. 87). Adolescents who experience trauma often struggle with emotional regulation, interpersonal relationships, and academic performance. However, research suggests that attachment security may serve as a protective factor, buffering the long-term effects of trauma (Julian et al. 89). Conversely, insecure attachment can heighten vulnerability to PTSD, influencing both symptom severity and recovery outcomes. Given the critical role of attachment in adolescent well-being, it is essential to explore treatment approaches that not only address trauma symptoms but also strengthen the caregiver-child bond.

This paper examines the relationship between PTSD, attachment, and adolescent development, arguing that varying attachment styles significantly impact PTSD outcomes and

recovery trajectories. Specifically, we explore how attachment security influences resilience in trauma-exposed adolescents and how attachment-based interventions, such as trauma-focused cognitive behavioral therapy (TF-CBT), can promote healing. By analyzing existing research on PTSD symptomatology, the role of attachment in trauma response, and evidence-based treatment approaches, this paper contributes to the growing body of literature emphasizing the importance of relational interventions in adolescent trauma care. The following sections will first explore the nature of PTSD in adolescents, then discuss attachment theory as a framework for understanding trauma responses and finally evaluate TF-CBT as an effective treatment modality.

Post Traumatic Stress Disorder (PTSD)

PTSD in adolescents includes several clusters of symptoms: intrusive memories, avoidance of trauma-related stimuli, negative changes in mood and cognition, and heightened arousal responses, as outlined in the DSM-5 (American Psychiatric Association 5). The onset of PTSD follows exposure to a potentially life-threatening traumatic event, such as witnessing or experiencing abuse, or a natural disaster, which trigger intense emotional and physical reactions. According to the National Center of PTSD, the prevalence of PTSD in adolescents is estimated to be around 3% to 15% of girls and 1% to 6% of boys (Hamblen and Barnett 366). Symptoms must last for more than one month and cause significant distress or functional impairment (e.g., interfering with school performance or daily activities) to qualify for a PTSD diagnosis, with many individuals developing symptoms within three months of the traumatic event, although some may experience delayed onset. Key symptom domains of PTSD include re-experiencing traumatic events, avoidance behaviors, negative alterations in mood and cognition, and hyperarousal (i.e., exaggerated startle responses, increased paranoia and hypervigilance behaviors) (American Psychiatric Association 5). Adolescents with PTSD may experience intrusive memories and nightmares associated with the traumatic experience. Mood alterations can manifest as heightened anxiety, irritability, emotional numbness, sadness, and distress, which can also complicate their ability to form and maintain relationships (Lawson and Quinn 499). Avoidance can present as a reluctance to engage in activities or discussions related to the trauma, which can also isolate them from peers and hinder their social development (Nooner et al. 154).

Different types of traumas can lead to distinct symptom profiles in adolescents with PTSD, as highlighted in a study by Birkeland, Skar, and Jensen. For instance, exposure to interpersonal violence, including physical, emotional, or sexual abuse, is often linked to significant emotional instability and difficulties in building and sustaining relationships. Adolescents who endure this type of trauma may find it hard to trust others, have trouble understanding social cues, and exhibit attachment-related issues. The deep personal nature of such violence can result in overwhelming feelings of shame, guilt, and low self-worth, further disrupting their emotional well-being and social interactions. Meanwhile, trauma from accidents or natural disasters may result in hyperarousal symptoms, including an exaggerated startle response, irritability, and difficulty concentrating. They may also engage in increased avoidance behaviors, such as steering clear of places, situations, or conversations that remind them of the

traumatic event (Shenk et al. 118). Exposure to multiple instances of trauma can have a differential impact on someone's symptoms and trauma trajectory, possibly leading to more severe and persistent PTSD symptoms, increased risk of disorders such as depression and anxiety, and greater difficulty in emotional regulation and interpersonal relationships. This overall effect can result in a more complex clinical diagnosis, often requiring specialized treatment approaches to address the layered impact of repeated traumatic experiences (Herman 121). These distinct symptom profiles emphasize the importance of tailoring interventions to address the specific needs and challenges associated with the type of trauma experienced. Understanding the nuanced ways in which trauma affects adolescents can help clinicians provide more effective, individualized care that fosters resilience and recovery.

Adolescents with PTSD also often exhibit significant alterations in brain functioning, particularly in regions associated with executive functions, emotional regulation, and fear processing. The prefrontal cortex, a brain region that is responsible for rational thought and emotional regulation, typically shows decreased functioning in adolescents with PTSD (Lawson and Quinn 502). This impaired functioning can lead to challenges in coping and decision-making. In contrast, the amygdala, critical in processing emotions and fear responses, tends to be hyperactive. This heightened activity results in an exaggerated sensitivity to perceived threats, leading to excessive fear and anxiety even in relatively safe or non-threatening situations. Such overactivation reinforces a heightened state of arousal and vigilance, which are hallmark symptoms of PTSD. These neurobiological changes contribute to the persistent and debilitating symptoms associated with PTSD. These neurobiological changes like reduced prefrontal cortex activity and amygdala hyperactivity create a feedback loop that perpetuates the disorder's debilitating symptoms. The imbalance between these regions disrupts the adolescent's ability to regulate emotional responses and distinguish between real and perceived dangers. Understanding these mechanisms highlights the importance of therapeutic approaches focusing on "retraining" the brain. Techniques such as exposure therapy aim to gradually desensitize adolescents to trauma-related triggers by encouraging safe and controlled confrontation of distressing memories or situations, and mindfulness-based interventions to foster present-moment awareness and grounding in moments of overwhelm. This process helps to reduce hyperactivity in the amygdala and strengthen the regulatory role of the prefrontal cortex. Restoring balance within these brain systems can alleviate symptoms, improve emotional regulation, and support recovery, ultimately enabling adolescents to regain a sense of safety and control in their lives.

Research has highlighted the prevalence and impact of PTSD in adolescents, thereby providing a foundation for understanding the complexities of trauma in this population (Lawson and Quinn 500). However, significant areas need improvement, particularly in diversifying research samples to ensure findings are applicable across different demographics. Longitudinal studies are also necessary to track the evolution of PTSD symptoms over time, which could inform more effective, tailored interventions. Furthermore, integrating findings on co-occurring

mental health issues, such as anxiety and depression, would enhance treatment approaches, addressing the multifaceted nature of trauma in adolescents more comprehensively.

Attachment & the Relational World

While much research on PTSD in adolescents has focused on its biological, emotional, and behavioral effects, it is also important to consider how early relationships shape the development and persistence of PTSD symptoms. Attachment theory helps explain these connections by highlighting the role of internal working models, mental representations of the self, others, and relationships, in shaping how adolescents respond to trauma. These models, formed through early interactions with caregivers, influence how adolescents process traumatic experiences, manage emotions, and seek or avoid support. Exploring the link between PTSD and attachment offers valuable insights into how early relationships affect trauma outcomes, helping to create more effective and personalized treatments. These cognitive frameworks, formed in infancy, continue to shape an individual's expectations and interactions in relationships throughout life, influencing how a person perceives themselves and others (i.e., their sense of trust in relationships and self-worth). Internal working models (IWMs) consist of memories, beliefs, goals, and behavioral strategies related to attachment (Shaver and Mikulincer 17). IWMs operate hierarchically, with general representations at the top and relationship-specific ones below, allowing for variations across different relationships. IWMs are largely shaped in childhood by repeated interactions with caregivers. Secure attachments form when caregivers are consistently responsive, leading to positive self-image and trust in others, while insecure attachments generally result from inconsistent or neglectful caregiving, and can lead to anxious-preoccupied, dismissive-avoidant, or fearful-avoidant styles (Davis and Carnelley). Anxious-preoccupied individuals often desire closeness and reassurance but may fear abandonment, leading to heightened sensitivity in relationships. Dismissive-avoidant individuals tend to prioritize independence and may undermine the importance of emotional intimacy, sometimes avoiding deep connections. Fearful-avoidant individuals experience a mix of both, desiring closeness but also fearing rejection or betrayal, resulting in push-pull dynamics in relationships. These models significantly impact social cognition, emotion regulation, and psychological well-being, and often operate outside of conscious awareness. This unconscious influence shapes automatic responses to relationships, stabilizing interpersonal patterns but potentially reinforcing maladaptive beliefs and behaviors until they are consciously examined and addressed. Major life events such as losing a loved one, career transitions, marriage, divorce, or becoming a parent can challenge and reshape internal working models (Kendall-Tackett). For example, someone who has always believed that relationships are inherently unreliable may begin to change this belief after entering a stable, supportive relationship. On the other hand, a person who once felt secure in their career may develop self-doubt after unexpectedly losing their job. These experiences provide new emotional data that can reinforce or contradict previous expectations, leading to an evolution in how individuals perceive themselves and their relationships.

Attachment relationships can thus serve as both a source of trauma and a key factor in predicting a child's ability to recover from a traumatic event. Importantly, existing research shows that not all individuals develop PTSD following exposure to potentially traumatic events, and attachment can be a factor in prevention (Nooner et al. 158). For example, a securely attached individual may have internalized expectations that others will offer care and support, enabling them to seek connection and regulate distress effectively. In contrast, an individual with an insecure attachment may lack these expectations, potentially leading to avoidance of support or difficulty managing distress on their own.

Adolescence is a significant stage in which attachment relationships evolve, balancing the need for independence with continued reliance on caregivers for emotional security. Unlike childhood, where attachment is characterized by direct dependence on caregivers, adolescence involves a gradual shift in attachment needs toward peers while maintaining an internalized sense of security from parental figures (Vostanis 62). IWMs developed in early attachment relationships influence adolescents' ability to form close peer connections, regulate emotions, and establish a stable sense of self. Secure attachment during this period is linked to higher emotional well-being, greater resilience, and healthier social interactions, while insecure attachment can contribute to emotional distress, difficulties in managing relationships, and increased vulnerability to mental health challenges (Mónaco et al. 2554). Compared to adulthood, where romantic partners often fulfill primary attachment roles, adolescence represents a transitional phase in which caregivers remain crucial sources of support, even as teens push for independence. Caregivers who balance warmth, support, and autonomy enable adolescents to navigate this shift successfully, fostering emotional growth and long-term psychological well-being (Vostanis 93).

However, our understanding of these relationships remains limited, particularly regarding how specific trauma types interact with individual factors such as age, gender, and pre-existing mental health conditions. There are still open questions concerning the long-term effects of different symptom profiles, such as those characterized by emotional dysregulation, dissociative tendencies, etc. Understanding how these profiles evolve is crucial for developing more effective interventions. For instance, individuals with avoidant attachment tendencies may require different therapeutic approaches than those with anxious patterns, yet current models often take a one-size-fits-all approach. Additionally, trauma responses can vary widely, with some individuals exhibiting hyperarousal while others display emotional numbing. These differences require more personalized treatments. Cultural and developmental factors further complicate these trajectories, as attachment styles and trauma processing can be shaped by social norms, family structures, and early life experiences, highlighting the need for interventions that are sensitive to these influences. Additionally, more research is needed to explore the effectiveness of tailored interventions based on trauma type and attachment style, as current therapeutic approaches may not fully address the nuanced needs of adolescents experiencing diverse forms of trauma.

Trauma-Focused CBT

TF-CBT is an evidence-based treatment designed for children and adolescents who have PTSD and other trauma-related symptoms (Cohen and Mannarino 133; de Arellano et al. 591). What sets TF-CBT apart from other trauma treatments is its strong emphasis on the involvement of parents or caregivers, integrating an attachment component to promote healing and resilience (de Arellano et al. 592). This caregiver involvement is a crucial aspect of TF-CBT, helping to strengthen the child's support system and improve overall treatment outcomes (Cohen and Mannarino 135). Central to TF-CBT is the CBT triangle, which illustrates the interconnection between thoughts, feelings, and behaviors, helping clients recognize how their thoughts about traumatic events affect their emotional responses and actions (Cohen and Mannarino 136). This understanding is crucial for addressing maladaptive beliefs and fostering healthier coping strategies. Another key component of TF-CBT is gradual exposure, where children are gradually introduced to trauma reminders in a safe and controlled manner, allowing them to confront their fears and reduce avoidance behaviors (Cohen and Mannarino 137; de Arellano et al. 593). The trauma narrative technique further facilitates this process by encouraging children to articulate their traumatic experiences in detail, which aids in cognitive processing and emotional regulation, ultimately helping them reframe their understanding of the trauma (de Arellano et al. 593). Parent involvement is also integral to TF-CBT; caregivers participate in parallel sessions that equip them with skills to support their child's recovery while also addressing their emotional responses to the trauma (Cohen and Mannarino 139; de Arellano et al. 594). By integrating these components, cognitive restructuring, gradual exposure, trauma narratives, and caregiver involvement, TF-CBT provides a comprehensive framework that effectively addresses the complex needs of children affected by trauma (de Arellano et al. 594).

TF-CBT is a structured treatment designed for children and adolescents who have experienced trauma, organized into three phases represented by the acronym PRACTICE (de Arellano et al. 595). The first phase, "coping skills," encompasses several key components that lay the groundwork for effective therapy. It begins with "psychoeducation," where both children and caregivers learn about trauma and its effects, helping to normalize their experiences and prepare them for the therapeutic process (Cohen and Mannarino 140). Following this, the "relaxation" component teaches techniques such as deep breathing and progressive muscle relaxation, essential for managing anxiety and physiological arousal (de Arellano et al. 595). The next step, "affective modulation," focuses on helping children identify and regulate their emotions, enabling them to express feelings healthily (Cohen and Mannarino 141). This is complemented by "cognitive coping," where children learn to recognize and challenge negative thoughts related to their trauma, fostering healthier perspectives and reducing feelings of self-blame (de Arellano et al. 596; Cohen and Mannarino 142). The second phase, "processing," centers on creating a "trauma narrative." In this step, children are guided to recount their traumatic experiences in a safe environment, facilitating emotional processing and integration of fragmented memories (de Arellano et al. 596). This structured approach allows them to confront their trauma gradually, helping to reframe maladaptive beliefs and reduce associated distress (Cohen and Mannarino 143). Finally, the "treatment and consolidation" phase includes several

important components: In-vivo gradual exposure, where children are gradually exposed to real-life reminders of their trauma in a controlled manner; conjoint parent-child sessions, which foster open communication about the trauma and strengthen the bond between caregiver and child; and enhancing safety and future development, which ensures that the child feels safe while promoting healthy development through supportive parenting practices (Cohen and Mannarino 145; de Arellano et al. 597). Together, these phases provide a comprehensive framework that effectively addresses trauma symptoms while building resilience in children and their caregivers (Cohen and Mannarino 145).

TF-CBT is an effective treatment for children and adolescents experiencing PTSD and related symptoms. Research indicates that TF-CBT is particularly effective in reducing PTSD symptoms among children who have experienced various types of traumas, including sexual abuse, domestic violence, and natural disasters (Cohen and Mannarino 146). A review of ten randomized controlled trials demonstrated a high level of evidence supporting TF-CBT's efficacy, with significant improvements noted in trauma-reactive behaviors and emotional distress (Cohen and Mannarino 147; de Arellano et al. 598). Studies have found that TF-CBT not only helps reduce PTSD symptoms but also addresses associated issues such as depression and behavioral problems. However, the evidence for its effectiveness in these areas is less clear (Cohen and Mannarino 147). The treatment has been successfully implemented with diverse populations, including children exposed to war and exploitation, and has shown positive outcomes across different settings (de Arellano et al. 598). Importantly, TF-CBT's structured approach, which includes components like psychoeducation, relaxation techniques, cognitive coping strategies, gradual exposure to trauma reminders, and parental involvement, contributes to its effectiveness in fostering resilience and promoting recovery in young trauma survivors (Cohen and Mannarino 147; de Arellano et al. 599). Overall, the evidence supports the inclusion of TF-CBT as a covered service in health plans, as it provides a viable option for addressing the complex needs of children affected by trauma while enhancing their emotional well-being and coping skills.

Discussion

Adolescent PTSD is a complex condition with profound implications for emotional regulation, decision-making, and social functioning. Characterized by intrusive memories, avoidance behaviors, negative mood alterations, and heightened arousal, PTSD often stems from exposure to life-threatening events. Neurobiological changes, including decreased prefrontal cortex activity and hyperactive amygdala responses, further exacerbate symptoms, making effective intervention crucial. TF-CBT has emerged as a leading treatment, offering a structured approach that integrates psychoeducation, relaxation techniques, cognitive coping strategies, trauma narratives, and parental involvement. Research consistently supports its effectiveness across various trauma types and populations, highlighting its role in fostering recovery and resilience.

The significance of these findings underscores the importance of early intervention and evidence-based treatments like TF-CBT, which not only alleviate PTSD symptoms but also strengthen caregiver-child relationships, a critical factor in long-term recovery. By addressing both psychological and relational aspects of trauma, TF-CBT provides a comprehensive framework for healing, ensuring that adolescents regain emotional stability and social connectedness. Additionally, it is essential to consider complex PTSD (C-PTSD). This condition can arise from prolonged or repeated trauma, often leading to more severe emotional dysregulation, identity disturbances, and interpersonal difficulties (Van der Kolk 156). C-PTSD requires tailored interventions that go beyond standard PTSD treatments by addressing the pervasive impact of chronic trauma on attachment and self-perception (Van der Kolk 157).

Future research should prioritize expanding the diversity of study populations to ensure the broad applicability of treatment outcomes across different cultural and demographic groups. Studies tracking PTSD progression and TF-CBT's long-term effectiveness are necessary to refine therapeutic approaches further. Additionally, developing interventions tailored to specific trauma types, attachment styles, and co-occurring conditions such as anxiety and depression will enhance treatment precision. By addressing these gaps, the field can continue to improve outcomes for adolescents navigating the complex effects of trauma, ultimately promoting lasting psychological well-being.

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The Pursuit of Happiness Through Broken Promises and Sacred Lands

By Amarpreet Kaur

“The Pursuit of Happiness.” What does it mean? To our Founding Fathers, the pursuit of happiness meant a right through which one could achieve a meaningful life through self-improvement, and importantly, legal activities (Locke). Although, in 2026 the pursuit of happiness is often viewed as temporary personal fulfillment; the citizens of 1776 defined it as the right to own land and labor without interference from an outside force. Instead of only seeking happiness, they practiced it by making sure their rights were protected within the Declaration of Independence itself. A quote said by George Mason in the Virginia Declaration of Rights captures the essence of the pursuit of happiness, “All men are by nature equally free and independent and have certain inherent rights... namely, the enjoyment of life and liberty, with the means of acquiring and possessing property, and pursuing and obtaining happiness and safety” (Mason). By this, Mason guaranteed happiness rather than promising it; he assured citizens that the government should have no hand in taking away their property or freedom of speech. However, while George Mason and the founders fought to protect these rights of citizens, many Native Americans in 1776 and today feel that the guarantee never was a possibility for them. Specifically, the ‘force’ of the government has constantly interfered and limited the land of the Natives.

For citizens of 1776, King George III was the ultimate outside force at that time. By imposing the Quartering Acts and taking away due process for colonists, he prevented the pursuit of happiness from taking place, leading the Founding Fathers to write the Third and Fifth Amendments, contributing to the desire of revolution. The Declaration of Independence was the promise; the Constitution was the rulebook, which protected the rights of the citizens. To protect the pursuit of happiness, in 1791 the Bill of Rights included the Third and Fifth Amendments. This is because the Fifth Amendment protects citizens through due process, the ability to remain silent during interrogations, and the government must follow fair legal procedures before snatching away someone’s life, property, or liberty (U.S. Const. Amend. V). The Third Amendment directly protects the property and consent of citizens to refuse to house soldiers during wartime (U.S. Const. Amend. III). Before the American Revolution, Great Britain had passed Quartering Acts, which forced citizens to house soldiers in their homes without their consent. Following the Revolutionary War, it was considered illegal to ask this of citizens, and the amendment further guarded the authority of citizens. However, this was a double standard; when colonists fought for their own rights, they took away those same protections from Natives because colonists wanted Native land.

In 1763, King George III tried to protect Native land from expansion in the Proclamation of 1763; however, colonists saw this as a barrier to their happiness, and this fueled the fight for their own property rights (Great Britain). Natives were left out of the Declaration of Independence and the Constitution, being referred to as “merciless Indian savages, whose known rule of warfare, is undistinguished destruction of all ages, sexes and conditions” (Center for Native American Youth). Today although it has been an embarrassingly long time, the Natives

haven't "earned" the privilege to these rights. Through aggressive proposals by companies and governments to take over the lands of these Indigenous peoples, such as the Keystone XL Pipeline, Willow Project, and Trail of Tears controversy, Native lands have been fought over. During the Trail of Tears era, Natives filed several lawsuits, most notably *Worcester v. Georgia*. Natives won that case in the court, but were still kicked out of their homes by President Andrew Jackson. When tribes resisted the military was ordered to forcibly remove them regardless of whether it required violence (Cherokee Nation). "The Court held that Georgia had no power to impose its laws in the Cherokee Nation. Nevertheless, the determination of then-President Andrew Jackson to remove the Cherokees and other southeastern Indian nations from their aboriginal territories resulted in the 1835 Treaty of New Echota [and the Trail of Tears]" (National Park Service). The court's promise vanished on the Trail of Tears, where "happiness" was replaced by government sponsored death and displacement (National Park Service). We can only imagine what the Natives were being put through, just for living on their rightful lands, and losing the ability to pursue happiness as the law seemed to promise them.

Likewise, in the present day, Natives are faced with more brutal incidents. Industrial projects like the Keystone XL Pipeline and the Willow Project display modern instances where government interests have invaded upon Indigenous lands and water rights, sparking environmental concern (Baswan). In states across the U.S., Native Americans are being taken away from their homes by Immigration and Customs Enforcement, also known as ICE. Ordered by our current president, Donald Trump, Native people have been victims of racial profiling by ICE and have wrongfully been detained in Minnesota. "Native people have been here since time immemorial — there's no one that has been a citizen of this country longer than us," mentioned Peggy Flanagan, the highest-ranking Indigenous woman in statewide office (Schafer). As a student observing these contradictions, I believe what is happening today is truly a tragedy, because we don't respect the land and rights that Native people have been protecting and residing on longer than us. Promised by the Founding Fathers and the Constitution, those promises are being broken left and right. The government is interfering with the pursuit of happiness of Native Americans. The government has no right to be taking such measures to unjustly remove Natives off their own land. Improving this right requires more than ink on paper; it demands for a legal system that treats tribal sovereignty as equal to our own, ensuring that the 'pursuit of happiness' is a shared reality rather than a selective privilege. To solve these issues, we must vow that the next 250 years are defined not by the interference of government, but by the honoring of its original promises.

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Biomechanics of Fast Bowling and Injury Risk in Cricket By Om Vyas

Abstract

Fast bowling is a decisive component of cricket performance, capable of determining match outcomes through pace, movement, and sustained pressure on batters. However, it is also one of the most biomechanically demanding actions in sport, exposing bowlers to extreme ground reaction forces, rapid trunk rotation, and repeated asymmetric loading. As a result, fast bowlers experience the highest injury rates in cricket, particularly overuse injuries such as lumbar spine stress fractures, hamstring strains, and side strains, which can significantly affect player availability, development, and career longevity.

This review examines the biomechanics of fast bowling and their relationship to injury risk by synthesizing evidence from biomechanics, epidemiology, and sports medicine research. It outlines the key phases of the fast bowling action, identifies common injury patterns, and analyzes major biomechanical risk factors including mixed bowling actions, excessive trunk lateral flexion, high ground reaction forces, and poor load management. The review also explores the role of workload, fatigue, and asymmetrical loading in injury development, with particular emphasis on the vulnerability of youth fast bowlers.

Key findings highlight that injury risk is driven not by force generation alone, but by how forces are managed through technique, coordination, and recovery. Practical implications include early biomechanical education, technique modification, strength and conditioning interventions, workload monitoring, and the use of biomechanical screening and emerging technologies. Finally, the review identifies limitations in current research and outlines future directions, including greater focus on women's cricket, real-world match analysis, and fatigue-related injury mechanisms.

Introduction

Fast bowling plays a central role in cricket by providing teams with the ability to take wickets, apply sustained pressure, and control the tempo of a match. Unlike other bowling styles, fast bowling relies on delivering the ball at very high speeds—often exceeding 140 km/h—to reduce a batter's reaction time and exploit bounce, seam, and swing. At both professional and developmental levels, fast bowlers are therefore prized for their ability to influence match outcomes through speed and aggression.

Producing such pace, however, requires an exceptionally demanding biomechanical sequence. Fast bowling involves the rapid conversion of linear momentum generated during the run-up into rotational energy through the hips, trunk, and upper limbs. This kinetic chain operates within fractions of a second and must be precisely coordinated to maximize ball velocity while maintaining control. During this process, bowlers experience extreme ground reaction forces at front foot contact, rapid trunk rotation and lateral flexion, and high loads through the shoulders and lower limbs.

These physical demands explain why fast bowling is considered one of the most taxing roles in cricket. The action places repeated asymmetric stress on the musculoskeletal system, particularly the lumbar spine, due to the combination of rotation, extension, and side bending under high load. Because these forces are applied repeatedly across spells, matches, and seasons, fast bowlers are exposed to a substantially greater injury burden than other playing roles. These demands explain why fast bowlers experience a disproportionately high incidence of injury throughout their careers.

Because performance is generated through extreme loading, understanding how forces move through the body is essential for preventing injury.

Injury Prevalence and Why It Matters

Injury prevalence among fast bowlers is consistently higher than in other cricketing populations. Research indicates that annual injury rates for fast bowlers commonly range from approximately 12.5% to over 20%, with some studies reporting that more than half of all fast bowlers experience a significant injury during their careers. Lower back injuries are particularly prevalent, with a large proportion of adolescent and elite fast bowlers showing signs of lumbar stress reactions or stress fractures. Lower limb injuries, especially hamstring strains, also account for a substantial percentage of time-loss injuries in cricket.

These injuries have serious implications beyond short-term pain or missed matches. Time-loss injuries disrupt player development, reduce availability for selection, and can lead to chronic problems that shorten careers. For youth players, prolonged injury may limit skill acquisition and physical development, while for elite athletes, repeated injury episodes can undermine performance consistency and long-term participation at the highest level.

Fast bowlers experience higher injury rates than batsmen, spin bowlers, and wicketkeepers due to the unique demands of their role. While batsmen are more commonly affected by impact injuries to the hands or acute muscle strains, and spin bowlers experience lower-intensity rotational loads, fast bowlers are repeatedly exposed to high-impact landings and explosive trunk movements. The combination of high ground reaction forces, frequent sprinting, and heavy bowling workloads makes pace bowling the most physically stressful role in the game.

Injuries are common in both youth and elite fast bowlers, although the underlying reasons differ slightly. Youth fast bowlers are particularly vulnerable because their bones, growth plates, and connective tissues are still developing and less able to tolerate repeated high loads. Rapid increases in workload, poor technique, and inadequate recovery further increase injury risk during adolescence. Elite fast bowlers, despite superior strength and conditioning, face injury risk due to congested playing schedules, frequent transitions between formats such as T20 and Test cricket, and cumulative fatigue across long seasons. In both groups, the interaction between biomechanical stress and workload is central to injury development.

The purpose of this review is to synthesize current evidence on the biomechanics of fast bowling and examine how mechanical factors contribute to injury risk in cricket. By integrating

findings from biomechanics, epidemiology, and sports medicine, the review aims to explain why specific injuries occur so frequently in fast bowlers and how they can be prevented through informed intervention.

The scope of the review focuses on the biomechanical phases of the fast bowling action, including the run-up, pre-delivery stride, front foot contact, ball release, and follow-through. It examines common injury patterns affecting the lumbar spine, lower limbs, and shoulder complex, and identifies key biomechanical risk factors such as mixed bowling actions, excessive trunk lateral flexion, high ground reaction forces, asymmetrical loading, fatigue, and workload spikes.

Importantly, this review also considers practical applications for players, coaches, and academies. By translating biomechanical evidence into coaching strategies, strength and conditioning principles, workload management guidelines, and screening practices, the review aims to support safer technique development and long-term performance.

The review first outlines the phases of the bowling action, then links mechanical stressors to common injuries, before discussing workload, fatigue, and evidence-based injury prevention strategies.

Phases of the Fast Bowling Action (Biomechanical Breakdown)

Fast bowling is not a single isolated movement but a highly coordinated sequence of phases that together form an efficient kinetic chain. Each phase contributes to both ball velocity and injury risk, with errors or inefficiencies early in the action often amplifying mechanical stress later. Understanding these phases allows coaches and practitioners to identify where excessive loading occurs and how biomechanical faults translate into injury.

Run-Up Phase

The primary purpose of the run-up in fast bowling is to generate linear momentum that can be transferred into the delivery stride and ultimately into ball speed. A well-executed run-up builds rhythm, timing, and controlled acceleration, allowing the bowler to arrive at the crease in a balanced and repeatable position. Rather than simply sprinting at maximum speed, the run-up must align the body toward the target and prepare the bowler for efficient force transfer in the subsequent phases of the action.

Run-up speed has a direct influence on force generation and performance. As run-up velocity increases, linear momentum also increases, providing greater potential energy to be converted into rotational power during the delivery stride. However, this increase in momentum also elevates ground reaction forces at foot contact and places greater mechanical demands on the lower limbs, lumbar spine, and hamstrings. If a bowler lacks sufficient strength, coordination, or control, the additional forces generated by a faster run-up may exceed the body's capacity to absorb them safely.

An excessively fast or inconsistent run-up can therefore increase injury risk. Sudden accelerations, braking errors, or poor rhythm can create sharp workload spikes and disrupt

timing, forcing compensatory movements later in the action. These compensations commonly shift load toward vulnerable structures such as the lumbar spine and hamstrings, increasing the likelihood of strains and stress injuries. Consequently, the run-up plays a critical role not only in performance but also in setting the mechanical conditions that influence injury risk throughout the bowling action.

Pre-Delivery / Bound Phase

The pre-delivery or bound phase serves as the transition between the run-up and the delivery stride and is essential for energy storage and sequencing. During this phase, the bowler lands on the back foot and briefly enters a period of hip and knee flexion, pelvic tilt, and trunk positioning that allows elastic energy to be stored in the lower limbs and trunk. This stored energy is then released in a coordinated sequence as the bowler moves toward front foot contact, contributing significantly to ball speed.

Efficient coordination during the bound phase ensures that momentum generated in the run-up is directed forward and upward in a controlled manner. When timing and sequencing are optimal, force flows smoothly through the kinetic chain from the legs to the hips, trunk, and bowling arm. However, poor coordination during this phase disrupts the kinetic chain, resulting in energy loss through excessive vertical lift, sideways motion, or premature braking. When momentum is not transferred efficiently, the body compensates by increasing mechanical load on the lumbar spine, knees, and ankles.

Balance plays a crucial protective role during the pre-delivery phase. Because fast bowling is inherently asymmetrical, maintaining dynamic balance helps stabilize the trunk and pelvis, reducing uneven loading across the body. Good balance allows bowlers to control body alignment during high-speed transitions, minimizing excessive spinal movement and joint stress. Conversely, poor balance increases the likelihood of awkward landings and compensatory movements, elevating the risk of both acute and overuse injuries.

Front-Foot Contact (FFC)

Front-foot contact is widely regarded as the most critical phase of the fast bowling action in terms of injury risk. At this moment, the bowler must rapidly decelerate the body's forward momentum while simultaneously initiating powerful trunk rotation and arm acceleration. This rapid deceleration generates extremely high ground reaction forces, often several times the bowler's body weight, which are transmitted through the front leg and into the pelvis and spine. These ground reaction forces consist primarily of vertical and horizontal braking components that combine into a resultant force vector traveling up the kinetic chain. While these forces are essential for transferring momentum into ball speed, they also place substantial stress on the ankle, knee, hip, and lumbar spine. The ability of the body to absorb and redirect these forces safely depends heavily on technique, strength, and neuromuscular control.

Technical faults at front-foot contact significantly amplify injury risk. A stiff or hyperextended front knee reduces the leg's ability to act as a shock absorber, increasing the

transmission of force to the lumbar spine. Poor trunk alignment or excessive lateral flexion at this stage further concentrates stress on the vertebral structures, particularly the pars interarticularis. As a result, front-foot contact represents a critical point where performance demands and injury risk intersect most sharply.

Ball Release and Follow-Through

During ball release, trunk rotation plays a central role in generating bowling speed. Energy transferred from the lower body is amplified through hip–shoulder separation and rapid trunk rotation, creating a whip-like action that accelerates the bowling arm. Effective sequencing allows the trunk to rotate ahead of the arm, increasing angular velocity and maximizing ball release speed while maintaining efficiency.

However, this phase also places extreme mechanical stress on the lumbar spine. Ball release typically occurs while the spine is simultaneously extended, rotated, and laterally flexed under high load. This tri-planar movement pattern generates large compressive and shear forces on the vertebrae, particularly at the lower lumbar levels. Repeated exposure to these stresses, especially when combined with poor technique or fatigue, contributes to the development of lumbar stress reactions and fractures.

The follow-through phase is essential for safely dissipating the large forces generated during delivery. A smooth, well-controlled follow-through allows momentum to be spread across the body, reducing peak stress on individual joints and tissues. In contrast, a poor or restricted follow-through prevents effective deceleration, concentrating forces in the lumbar spine, knees, ankles, shoulders, and elbows. Over time, this concentrated loading increases the likelihood of both chronic overuse injuries and acute joint or muscle damage.

These phase-specific mechanical demands help explain why certain injuries repeatedly appear in fast bowlers, particularly in the lumbar spine, lower limbs, and shoulder complex.

Common Injuries in Fast Bowlers

Fast bowlers experience a distinct injury profile compared with other cricketing roles due to the extreme forces, repetition, and asymmetrical nature of the bowling action. Injuries most commonly affect the lumbar spine, lower limbs, and shoulder complex, reflecting the phases of bowling where mechanical stress is greatest. Understanding these injuries in relation to bowling biomechanics is essential for effective prevention and management.

Lumbar Spine Stress Injuries

Lumbar spine stress injuries are the most common and most serious injuries affecting fast bowlers. The repetitive nature of fast bowling exposes the lower back to extreme loading across multiple planes of motion, including extension, rotation, and lateral flexion. These movements occur under high ground reaction forces, particularly at front-foot contact and ball release, placing exceptional stress on the vertebral structures over time.

The vertebrae most frequently affected are L4 and L5, where mechanical demands are greatest during bowling. These segments experience high torsional, compressive, and shear forces as the trunk rotates rapidly while the pelvis is stabilized by the braced front leg. Injuries typically involve the pars interarticularis, a small bony region that connects the anterior and posterior elements of the vertebra and is particularly vulnerable to repetitive extension and rotation.

The primary injury mechanism involves repeated microtrauma rather than a single traumatic event. Each delivery creates small amounts of bone stress, and when workloads are high or recovery is insufficient, this micro-damage accumulates faster than the bone can repair. Mixed bowling actions, excessive lateral flexion, and poor front-leg mechanics further amplify spinal loading, increasing the likelihood of stress reactions and stress fractures. In severe cases, bilateral pars injuries may progress to spondylolisthesis, resulting in long-term pain and functional limitation.

Lower Limb Injuries

Lower limb injuries are also highly prevalent among fast bowlers due to the large braking forces and rapid deceleration that occur during the delivery stride. The knee and ankle of the front leg are particularly vulnerable, as they are responsible for absorbing and redirecting the high ground reaction forces generated at front-foot contact.

At the knee, fast bowlers commonly experience patellar tendinopathy, cartilage degeneration, ligament strains, meniscal injuries, and stress fractures of the tibia or femur. These injuries are associated with repeated high-impact loading, rotational forces, and inadequate shock absorption, particularly when the front knee is excessively stiff or hyperextended. While a braced front leg improves performance by facilitating momentum transfer, insufficient strength or control can lead to excessive joint stress.

Ankle injuries are similarly common and often result from forced plantarflexion and rapid braking during landing. Posterior ankle impingement, Achilles tendinopathy, ligament sprains, and stress fractures of the distal tibia or navicular bone frequently occur, especially in bowlers with poor landing mechanics or inadequate ankle mobility. Over time, repetitive exposure to these stresses leads to chronic pain and reduced performance.

Asymmetrical loading further contributes to lower limb injury development. Fast bowling places disproportionate stress on one side of the body, leading to strength and mobility imbalances between the dominant and non-dominant limbs. When these asymmetries exceed the body's capacity to compensate, they disrupt force absorption and increase the risk of overuse injuries, particularly at the knee and ankle.

Shoulder and Upper Limb Injuries

The shoulder complex is subjected to extreme mechanical demands during fast bowling, particularly during the arm acceleration and deceleration phases. The most stressful movements occur during late cocking, where the shoulder reaches maximal external rotation, followed by

rapid internal rotation, abduction, and horizontal adduction as the arm accelerates toward ball release.

These high-velocity movements place significant strain on the rotator cuff muscles, shoulder capsule, and ligaments. After ball release, the shoulder must rapidly decelerate the arm, generating large eccentric forces that further stress soft tissues. If strength, control, or kinetic chain efficiency is inadequate, these tissues are forced to absorb excessive loads, increasing injury risk.

Repetitive high-velocity arm actions lead to overuse injuries through the accumulation of microtrauma. Without adequate recovery, small tears in muscles, tendons, and ligaments fail to heal completely, resulting in chronic inflammation, tendinopathy, or structural damage. Common injuries include rotator cuff tendinopathy, shoulder impingement, elbow ligament injuries, and side strains of the oblique muscles, which are unique to the rotational demands of fast bowling. Muscle imbalances play a key role in upper limb injury development. Overdevelopment of the internal rotators and prime movers, combined with weakness in the external rotators and scapular stabilizers, compromises joint stability. This imbalance reduces control during deceleration and increases stress on passive structures, further elevating the risk of shoulder and elbow injuries.

Biomechanical Risk Factors for Injury

While fast bowlers are exposed to high forces by the nature of their role, injury risk is not determined by load alone. Instead, injuries arise from how these forces are generated, transferred, and absorbed through the body. Specific biomechanical patterns consistently emerge in the literature as key risk factors, particularly when combined with high workload and fatigue. This section synthesizes the major biomechanical contributors to injury in fast bowlers.

Bowling Action Classification: Front-On vs Side-On vs Mixed

Fast bowling actions are commonly classified as front-on, side-on, or mixed based on the alignment of the hips, shoulders, and feet during the delivery stride. In a side-on action, the shoulders and hips remain relatively aligned side-on to the batsman, while in a front-on action the chest faces more directly down the pitch. Both of these actions allow for smoother force transfer through the kinetic chain when executed correctly.

The mixed bowling action, however, is consistently associated with a higher risk of injury, particularly to the lumbar spine. In this action, the hips are often side-on while the shoulders rotate rapidly toward a front-on position at or just before front-foot contact. This misalignment forces the spine to compensate through rapid counter-rotation under high load, creating excessive torsional and shear forces.

The danger of the mixed action lies not in rotation itself, but in poor sequencing. When shoulder rotation occurs too early or too abruptly relative to pelvic motion, the trunk is exposed to combined extension, rotation, and lateral flexion simultaneously. This tri-plane loading is particularly damaging to the pars interarticularis and helps explain the strong association between mixed actions and lumbar stress fractures. Injury risk is further amplified when mixed

actions are combined with high workloads and fatigue, a pattern frequently observed in young and developing fast bowlers.

Trunk Rotation and Lateral Flexion

Trunk rotation is a fundamental component of fast bowling performance and acts as a primary driver of ball speed. By rotating the pelvis and trunk sequentially, bowlers generate angular momentum that transfers energy efficiently from the lower body to the bowling arm. Controlled hip–shoulder separation allows the arm to accelerate later in the sequence, producing a whip-like effect that enhances velocity.

However, trunk rotation becomes a significant injury risk when it is excessive, poorly controlled, or combined with large amounts of lateral flexion. Excessive side bending—particularly contralateral lateral flexion away from the bowling arm—redirects forces away from the hips and legs and into the lumbar spine. When this movement occurs under high ground reaction forces at front-foot contact, compressive and shearing loads concentrate on the posterior elements of the vertebrae.

Timing and coordination are critical moderating factors. Under fatigue or technical breakdown, bowlers often increase trunk rotation and lateral flexion to compensate for reduced lower-body force production. These compensatory strategies may preserve ball speed in the short term but significantly increase spinal stress. As a result, excessive trunk motion is not only a technical flaw but also a fatigue-sensitive injury mechanism.

Ground Reaction Forces and Front-Leg Bracing

Ground reaction forces (GRFs) are central to both performance and injury risk in fast bowling. At front-foot contact, bowlers experience extremely high vertical and horizontal braking forces as forward momentum from the run-up is rapidly decelerated. These forces are essential for transferring energy into trunk rotation and arm acceleration, but they also represent one of the greatest mechanical challenges to the musculoskeletal system.

Front-leg bracing plays a critical role in managing these forces. An optimally braced front leg provides a stable base that allows momentum to be redirected upward through the kinetic chain. This improves ball speed and reduces energy leakage through excessive trunk movement. However, if the front leg is too stiff or poorly controlled, shock absorption is reduced, causing forces to be transmitted directly to the knee, hip, and lumbar spine.

Importantly, greater force is not inherently better. Fast bowling actions occur within extremely short timeframes, making rate of force development and force direction more important than absolute force magnitude. Bowlers who can generate force quickly and manage it efficiently through coordinated movement patterns are more likely to perform well while avoiding injury. In contrast, poor force management—through stiff landings, misalignment, or delayed sequencing—elevates injury risk despite similar or even lower absolute loads.

Synthesis of Risk Factors:

Across techniques and phases of the bowling action, a consistent pattern emerges: injury risk increases when high forces interact with poor alignment, excessive trunk motion, or disrupted sequencing. Mixed bowling actions, excessive lateral flexion, high ground reaction forces, and suboptimal front-leg mechanics do not act in isolation. Instead, they compound one another, particularly under fatigue and high workload, creating a biomechanical environment where injury becomes increasingly likely.

Workload, Fatigue, and Overuse

Fast bowling injuries do not result from single movements alone, but from repeated exposure to high mechanical stress over time. Even biomechanically efficient bowling actions can become injurious when workloads exceed the body's capacity to adapt and recover. Understanding how workload and fatigue interact with biomechanics is therefore essential for explaining why injuries often develop gradually and why they frequently occur late in spells or matches.

Workload Measurement and Injury Correlation

Bowling workload in cricket is most commonly quantified by the number of balls bowled, which reflects total volume, along with overs per spell, total overs per day, and frequency of bowling across training and matches. These external load measures are increasingly complemented by internal load indicators such as rate of perceived exertion (RPE), heart rate, and ball release velocity to better capture the true physiological demand on the bowler. Strong evidence links high workloads—particularly sudden increases in bowling volume—to elevated injury risk. Research using the Acute:Chronic Workload Ratio (ACWR) demonstrates that when short-term bowling load greatly exceeds a bowler's recent training history, injury risk rises sharply. Acute workload spikes of more than 150–200% of a player's chronic load have been associated with a four- to five-fold increase in injury likelihood, particularly for lumbar stress fractures and hamstring strains.

Youth fast bowlers are especially vulnerable to workload-related injuries. Their developing bones, growth plates, and connective tissues are less able to tolerate repeated high forces, and rapid increases in bowling volume often outpace their ability to adapt. When high workloads are combined with immature technique or insufficient rest, micro-damage accumulates in the lumbar spine and lower limbs, frequently progressing to time-loss injuries.

Fatigue Effects on Biomechanics

Fatigue has a profound impact on bowling mechanics and plays a central role in injury development. As bowlers tire, their ability to generate force efficiently and maintain precise coordination declines. This leads to observable technical changes, including reduced run-up speed, softer front-leg bracing, slower arm acceleration, and decreased accuracy in line and length.

Under fatigue, bowlers often adopt compensatory movement patterns to maintain performance. These include increased trunk flexion and lateral flexion, altered pelvic control, reduced hip and knee flexion at front-foot contact, and greater reliance on upper-body rotation. While these adaptations may temporarily preserve ball speed, they shift mechanical load away from the legs and toward the lumbar spine and shoulder complex, increasing injury risk. Fatigue also disrupts proprioception and timing, making it harder for bowlers to consistently land in optimal positions or control arm path during delivery. As a result, forces that would normally be distributed across multiple joints become concentrated in specific tissues, accelerating overuse damage. These effects explain why injuries such as lumbar stress fractures, side strains, and hamstring tears are more likely to emerge as fatigue accumulates.

Why Injuries Occur Late in Spells or Matches

Injuries in fast bowlers frequently occur late in bowling spells or toward the end of matches, when cumulative fatigue and workload converge. By this stage, muscles have reduced capacity to absorb impact and stabilize joints, increasing reliance on passive structures such as ligaments, discs, and bone. High-intensity efforts performed under fatigue—such as pushing for extra pace or sprinting between deliveries—create particularly high-risk conditions. Late-stage fatigue also amplifies the effects of earlier workload spikes and minor technical inefficiencies. Small biomechanical flaws that are well tolerated early in a spell may become injurious as neuromuscular control declines. Without adequate recovery between spells or matches, microtrauma accumulates faster than tissue repair, resulting in both acute injuries and chronic overuse conditions.

Workload and fatigue are not separate from biomechanics; they directly influence how forces are produced and managed during fast bowling. Sudden workload spikes, insufficient recovery, and fatigue-induced technical breakdowns significantly increase injury risk, even in bowlers with otherwise sound technique. Effective injury prevention therefore requires not only biomechanical efficiency, but also intelligent workload monitoring and fatigue management.

Injury Prevention and Biomechanical Interventions

Given the high mechanical loads and repetitive nature of fast bowling, injury prevention requires a proactive, multifactorial approach that integrates biomechanical technique, physical preparation, and workload management. Research consistently shows that injuries are not inevitable; rather, they are strongly influenced by how forces are generated, transferred, and managed across the kinetic chain. When prevention strategies are grounded in biomechanics and supported by conditioning and monitoring, both injury risk and performance limitations can be substantially reduced.

Technique Modification

Biomechanically guided technique modification is one of the most effective methods for reducing injury risk in fast bowlers. Evidence indicates that changes such as improving shoulder

alignment, reducing excessive shoulder counter-rotation, and avoiding mixed bowling actions can significantly decrease stress on the lumbar spine. Similarly, promoting controlled front-knee flexion at front-foot contact—rather than a locked or hyperextended knee—enhances shock absorption and reduces spinal loading while preserving ball speed.

Effective technique modification focuses on improving efficiency rather than forcing a single “ideal” action. By minimizing redundant or poorly sequenced movements, bowlers are better able to manage ground reaction forces and maintain consistent mechanics under fatigue. Importantly, research suggests that these adjustments do not reduce performance; instead, they often improve ball speed, accuracy, and repeatability by enhancing force transfer through the kinetic chain.

The timing of technical correction is critical. Early intervention during youth development is most effective, as movement patterns are more adaptable and faulty mechanics are less ingrained. Practice environments—rather than competitive matches—provide the optimal setting for technique modification, allowing bowlers to integrate changes gradually through repetition and feedback without performance pressure.

Strength, Conditioning, Flexibility

While technique determines how forces are applied, physical capacity determines whether the body can tolerate those forces safely. Strength and conditioning programs for fast bowlers must therefore support both performance and injury prevention by targeting the demands of the bowling action.

Lower-body strength, particularly in the glutes, quadriceps, hamstrings, and calves, is essential for generating run-up speed, controlling front-leg bracing, and absorbing ground reaction forces. Equally important is core strength, which acts as a stabilizing brace for the spine during high-speed rotation, lateral flexion, and extension. Strong deep trunk muscles allow efficient force transfer from the lower to upper body while limiting excessive spinal motion that contributes to lumbar stress injuries.

The shoulder complex also plays a critical role in injury prevention. Strengthening the rotator cuff and scapular stabilizers improves joint stability during extreme ranges of motion and high-speed arm deceleration, reducing the risk of overuse injuries such as tendinopathy or ligament strain.

Flexibility and mobility—particularly at the hips—are essential for maintaining an efficient kinetic chain. Adequate hip mobility allows for greater stride length, smoother rotation, and improved load distribution, preventing compensatory stress on the lower back and knees. When combined, strength, stability, and mobility form the physical foundation that enables bowlers to sustain high workloads with reduced injury risk.

Workload Management Guidelines + Technology

Workload management is a cornerstone of injury prevention, especially for youth fast bowlers. Governing bodies such as the ECB have introduced age-based bowling limits that

restrict overs per spell, overs per day, and consecutive bowling days. These guidelines aim to control cumulative load and ensure adequate recovery during critical stages of physical development.

While many workload recommendations are increasingly supported by epidemiological evidence—particularly regarding sudden workload spikes—some coaching practices still rely heavily on tradition rather than data. This highlights the importance of continued research and education to ensure guidelines are consistently evidence-based and applied across all levels of the game.

Modern technology has significantly enhanced workload monitoring capabilities. Wearable GPS units, inertial measurement sensors, heart rate monitors, and smart apparel provide objective data on movement patterns, intensity, fatigue, and recovery. Artificial intelligence systems can analyze these inputs to identify early signs of overload or biomechanical deterioration, allowing coaches to intervene before injury occurs. By moving beyond simple ball counts, technology enables personalized workload management that balances performance demands with long-term player health.

Injury prevention in fast bowling is most effective when technique modification, physical preparation, and workload management are integrated into a cohesive framework. Biomechanically sound techniques reduce unnecessary stress, strength and mobility enhance tissue resilience, and intelligent workload monitoring prevents overload. Together, these interventions allow fast bowlers to perform at high intensity while minimizing injury risk and supporting long-term athletic development.

Implications for Youth and Elite Cricket

The biomechanical demands and injury risks associated with fast bowling have important implications for how players are developed, trained, and supported across different levels of cricket. While the fundamental mechanics of fast bowling remain consistent, the priorities and strategies for injury prevention differ between youth and elite environments due to differences in physical maturity, workload exposure, and competitive demands.

- ***Youth: skill foundation + safe loads + early technique education***

In youth cricket, the primary focus should be on long-term athletic development rather than immediate performance outcomes. Young fast bowlers are particularly vulnerable to injury because their bones, growth plates, and neuromuscular systems are still developing, making them less tolerant of the extreme forces generated during bowling. As a result, early biomechanical education is essential to help young players understand safe movement patterns, efficient alignment, and basic kinetic chain principles before high workloads are introduced.

Coaching at the youth level should prioritize sound technique, general athleticism, and controlled exposure to bowling loads. Strict adherence to age-based workload limits, adequate rest days, and careful monitoring of cumulative loads across school, club, and representative cricket are critical for preventing overuse injuries such as lumbar stress fractures. Strength and

conditioning programs should emphasize core stability, hip mobility, balance, and movement quality rather than maximal strength or speed.

Importantly, youth programs must also address systemic inequalities in access to coaching, facilities, and medical support. A holistic and inclusive approach—one that values late physical maturation and emphasizes skill development over early selection—can help reduce injury risk while broadening the talent pool and supporting sustainable participation.

- ***Elite: format switching + scheduling + data-driven recovery***

At the elite level, fast bowlers operate under significantly higher physical and psychological demands due to congested competition schedules, frequent travel, and rapid transitions between formats such as Test, ODI, and T20 cricket. These factors create substantial workload variability, increasing the risk of fatigue-related injuries despite high levels of conditioning and technical proficiency.

Elite injury prevention strategies must therefore focus on individualized load management, recovery optimization, and continuous biomechanical monitoring. Data from wearable technology and performance analytics allows teams to tailor training volumes, spell lengths, and rest periods to each bowler's physical profile and injury history. Managing cumulative fatigue—rather than just acute workload—is essential for maintaining performance and availability across long seasons.

Additionally, elite coaching environments must support psychological well-being by fostering open communication and reducing stigma around fatigue or injury reporting. Creating a “safe” environment where players can disclose discomfort early helps prevent minor issues from progressing into serious injuries and long-term absences.

- ***Coaches/academies: duty of care + screening + individualized plans***

Coaches and academies play a central role in translating biomechanical research into practical outcomes. Their responsibilities extend beyond technical instruction to include safeguarding player health, managing workloads, and supporting mental and physical well-being. Biomechanical screening, movement analysis, and collaboration with sports scientists and medical staff allow for early identification of injury risk factors and personalized intervention strategies.

Academies, in particular, are responsible for establishing structured development pathways that balance performance progression with injury prevention. By integrating technique education, strength and conditioning, workload monitoring, and athlete welfare into a unified framework, academies can better prepare fast bowlers for the demands of elite cricket while minimizing long-term injury risk.

The implications of fast-bowling biomechanics differ across youth and elite cricket but share a common goal: sustaining performance while protecting player health. Early technical education, controlled workloads, and holistic development are critical at the youth level, while elite cricket requires individualized, data-driven management of load, recovery, and fatigue.

Across all levels, coaches and academies serve as the key link between research and practice, ensuring biomechanical principles are applied effectively to support both performance and career longevity.

Limitations of Current Research

Despite significant advances in understanding the biomechanics and injury mechanisms of fast bowling, several limitations remain within the current body of cricket research. These constraints affect how confidently findings can be generalized to real-world match settings and applied across diverse playing populations.

One major limitation is the gap between laboratory-based biomechanical studies and actual match conditions. Laboratory analyses provide highly controlled and precise measurements of joint angles, forces, and movement sequencing, but they lack ecological validity. Real matches introduce variable factors such as fatigue, psychological pressure, pitch deterioration, weather conditions, ball wear, and fluctuating intensity, all of which influence bowling mechanics and injury risk. As a result, biomechanical patterns observed in controlled settings may not fully reflect how fast bowlers move and load their bodies during competitive play.

Methodological challenges also limit current research. Many studies rely on two-dimensional video analysis, which cannot accurately capture the complex three-dimensional rotations, lateral flexion, and multi-planar loading that characterize fast bowling. While three-dimensional motion capture systems provide greater accuracy, they are expensive, time-consuming, and often impractical for large-scale or in-field studies. Emerging marker-less and AI-based systems show promise but currently lack the precision needed to reliably assess joint loading and spinal mechanics.

Another significant limitation is the lack of representation across player populations. Much of the existing literature focuses on male, semi-elite, or youth players, with comparatively little research on elite professionals, female cricketers, or players from diverse ethnic and socioeconomic backgrounds. This creates gaps in understanding sex-specific injury mechanisms, long-term developmental trajectories, and the influence of cultural and environmental factors on injury risk and performance.

Finally, limited data sharing and longitudinal tracking restrict the ability to draw strong causal conclusions. Elite teams often withhold workload and injury data for competitive reasons, while many studies are cross-sectional rather than long-term. This makes it difficult to fully understand how fatigue, cumulative workload, and biomechanical adaptations interact over multiple seasons.

Future Research Directions

Future research in cricket biomechanics should aim to bridge the gap between controlled scientific analysis and the complex realities of match play, while expanding inclusivity and improving injury prediction. Advancements in technology, data integration, and research design

offer significant opportunities to improve understanding of fast bowling performance and injury risk.

One key direction is the increased use of emerging technologies such as AI-powered computer vision, wearable inertial measurement units (IMUs), smart equipment, and marker-less 3D motion capture. These tools allow biomechanical data to be collected in real match and training environments rather than exclusively in laboratories, improving ecological validity. Wearable sensors can continuously monitor joint angles, impact forces, workload, and fatigue, while smart bats and balls can provide insight into ball release characteristics, spin rate, and speed. Integrating these technologies enables a more complete, real-time understanding of how technique and load interact during competition.

Artificial intelligence and machine learning also present major opportunities for injury prediction and prevention. By combining biomechanical data with workload metrics, physiological markers (such as heart rate variability and sleep), and historical injury records, AI models can identify subtle deviations from an athlete's normal movement patterns that may signal increased injury risk. Future research should focus on validating these predictive models and determining which variables most accurately forecast overuse injuries such as lumbar stress fractures and hamstring strains.

Another critical research priority is expanding studies to under-represented populations. Greater emphasis is needed on women's cricket, including sex-specific biomechanics, injury patterns, hormonal influences, and workload tolerance as the women's game continues to professionalize. Longitudinal research tracking youth fast bowlers across growth and maturation phases is also essential to understand how early technique, workload exposure, and physical development influence injury risk later in a career. Additionally, broader inclusion of players from diverse ethnic and socioeconomic backgrounds would help identify systemic barriers and ensure findings are applicable across the global cricketing population.

Finally, future research should prioritize long-term, multi-season studies that integrate biomechanics, workload, fatigue, and performance outcomes. Standardized data collection and greater collaboration between researchers, governing bodies, and professional teams would improve data quality and allow stronger causal links to be established. This approach would move cricket science beyond isolated risk factors toward a comprehensive, predictive model of performance and injury.

Conclusion

Fast bowling is one of the most physically demanding skills in sport, requiring the rapid conversion of run-up momentum into high-velocity ball release through extreme forces, asymmetric loading, and complex multi-planar movements. This review demonstrates that injury risk in fast bowlers is primarily driven by a combination of faulty biomechanics, excessive and poorly managed workloads, and fatigue. Among the most significant biomechanical risk factors identified are mixed bowling actions, excessive trunk lateral flexion, poor lumbo-pelvic control, and the inability to safely manage high ground reaction forces at front foot contact. These factors

consistently place excessive stress on the lumbar spine, lower limbs, and shoulder complex, explaining the high prevalence of stress fractures, muscle strains, and overuse injuries in fast bowlers.

From a practical standpoint, injury risk can be substantially reduced through targeted biomechanical interventions and holistic athlete management. Technique modification that minimizes shoulder counter-rotation, avoids mixed actions, and promotes controlled front-leg bracing can lower spinal loading while maintaining performance. Strength and conditioning programs that prioritize lower-body power, core stability, shoulder resilience, and hip mobility are essential for safe force transfer along the kinetic chain. Equally important is workload management, particularly for youth fast bowlers, where age-appropriate bowling limits, adequate recovery, and gradual load progression are critical for protecting developing musculoskeletal structures.

Early biomechanical education emerges as a central theme for long-term injury prevention and performance sustainability. Teaching young bowlers efficient movement patterns before faulty techniques become ingrained reduces hidden tissue stress and builds resilience as workloads increase with age and competition level. At the elite level, integrating biomechanical screening, wearable technology, and data-driven monitoring enables individualized load management and early detection of fatigue-related breakdowns.

Ultimately, advancing fast bowling performance while reducing injury risk requires a shift from reactive treatment to proactive, evidence-based prevention. By combining sound biomechanics, intelligent workload management, technological innovation, and athlete-centered coaching, cricket can support safer development pathways and extend the longevity of fast bowlers at all levels of the game.

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The Impact of Social Media Health Misinformation on Adolescent Health Decisions: Implications for Public Health Policy By Anaya Kabir

Abstract

The current use of social media as one of the sources of health information has led to an alteration in the way adolescents perceive health information regarding nutrition, fitness, mental health issues, and treatments. While social media allows for greater dissemination of health education information, it also encourages the propagation of misinformation. The adolescents are prone to such misinformation due to their developmental vulnerability to peer pressure, emotionally charged behavior, and media illiteracy skills. This review presents existing research on the effects of health misinformation disseminated via social media on the health of adolescents. According to current literature, misinformation about dieting, body image, vaccines, alternative medicine, and self-diagnosis of mental conditions can have a detrimental effect on the adolescents' physical and psychological well-being. Additionally, social media algorithms, designed primarily to increase user engagement, can promote health misinformation and its negative effects even further. Available literature indicates that prolonged consumption of health misinformation can lead to disordered eating behavior, vaccine hesitancy, greater levels of anxiety, and excessive self-diagnoses among others. Apart from examining current research on the matter, this review will focus on structural reasons behind health misinformation proliferation and will explore ways of addressing this problem via public health interventions.

Keywords social media, misinformation, adolescents, mental health, TikTok, public health, digital health literacy

Introduction

Social media has revolutionized how adolescents obtain and understand health-related information. Sites like TikTok, Instagram, and YouTube are becoming the predominant sources for information about topics like dieting, physical activity, psychological well-being, prevention, and health care. Adolescents are spending considerable time on social media each day, and algorithmic influences shape what they see on these sites (Pew Research Center, 2025). Some recent research findings indicate that adolescents are coming across health-related information through social media channels more often than from health-care facilities or schools.

Even as social media broadens health education opportunities, it is also leading to rapid diffusion of health misinformation. In general, health misinformation consists of inaccurate or unsupported claims that could affect public perception and behavior regarding their health. The U.S. Department of Health and Human Services Advisory from the Surgeon General on Social Media and Youth Mental Health notes digital misinformation as a rising health threat due to its impact on the psychological and physical well-being of adolescents (HHS, 2023). Adolescents' vulnerabilities stem from their immaturity with regard to logical reasoning, emotional control,

and sense of self. Therefore, continuous exposure to deceptive information about weight loss, exercise routines, vaccines, nutritional supplements, and self-assessment of one's mental health state will have a profound effect on individuals' choices and well-being in the long term.

Methodology

This narrative literature review is done by conducting searches from Google Scholar and PubMed databases. The key terms used in searching for sources are "social media health misinformation adolescents," "TikTok mental health misinformation," "vaccine misinformation youth," "social media body image," and "adolescent self-diagnosis social media." Sources published between the years 2020 and 2026 have been prioritized since they would be relevant regarding current social media trends among adolescents. Peer-reviewed studies, systematic reviews, government health advisories, and public health reports have been considered. Sources that do not meet scientific credibility and relevance have been excluded.

Social Media as a Primary Source of Health Information

The role of social media in the lives of teenagers is significant due to several reasons, including convenience, accessibility, and advanced personalized mechanisms of information delivery. As opposed to conventional approaches to healthcare communication, social media platforms offer more entertaining and appealing ways of disseminating health information.

According to Freeman et al., it is common for adolescents to assess online health information not considering the credentials of the author but their own relatability and emotions instead. In such situations, influencers act as relatable peers who talk about various experiences related to diets, sports, skin care, and other health-related areas.

It should be noted that recommendation algorithms only amplify this effect by providing users with relevant content on a regular basis. Thus, once a teen starts reading particular pieces of health-related information on social media platforms, they get even more similar information.

Repeated exposure plays a critical role in creating an “echo chamber” effect, where misinformation becomes normalized and progressively harder to differentiate from factual advice.

One of the latest studies on the use of health information by Generation Z members found that adolescents often cite social media as one of their main channels for acquiring health-related knowledge (Healthline, 2024). This trend poses a significant problem since adolescents might not have enough media literacy to assess the credibility of their sources of information.

Diet and Fitness Misinformation

Misinformation about diet and physical fitness is one of the most common types of health-related misinformation aimed at teenagers. Social media sites often disseminate unattainable body ideals, extreme diets, and impractical exercise regimes that can adversely impact teenage perceptions of their bodies and health choices.

While "fitspiration" content often appears inspirational, it may lead to poor body image and distorted eating habits. According to Czubaj et al. (2025), consistent consumption of social media fitness content was linked to body image problems and unhealthy competition amongst teenagers. The popularity of viral "What I Eat in a Day" videos and quick weight loss fads encourages caloric deficit without stressing nutrition balance and medical guidance.

Moreover, online influencers market detoxification products, dietary supplements, and appetite suppressants by using pseudoscientific statements. Adolescents are particularly vulnerable to peer pressure and social comparison based on appearances, thus making them more prone to adopting dangerous eating practices and false expectations about body image.

Medical Misinformation and Vaccine Hesitancy

There has been an increase in misinformation in medical practice via social media networks, especially since the onset of the COVID-19 virus outbreak. Various forms of information that are unproven about vaccines, disease management techniques, and other alternative treatments have spread like wildfire in cyberspace, preceding the scientifically verified information to the audience, mainly adolescents.

According to Pierri et al., a few powerful social media networks contributed significantly to the generation of misinformation about the vaccine for the COVID-19 virus. The social media algorithm contributed significantly since the more emotionally charged false information often received more engagements than scientifically factual content.

Misinformation may cause adolescents to develop a mistrustful attitude towards healthcare workers and scientific institutions. Repeated exposure to false claims in this case may predispose adolescents to doubt healthcare professionals' advice and depend on online anecdotes.

Mental Health Misinformation and Self-Diagnosis

The mental health discourse in the age of social media has played a great role in reducing stigmatization of mental illnesses and promoting their understanding. Still, misrepresentation of information about psychological disorders through social media is rather prevalent. The short-form video content on anxiety, depression, attention-deficit/hyperactivity disorder, autism spectrum disorder, and dissociative disorders often portrays psychiatric disorders in a simplified manner.

Conte et al. (2024) found a number of concerns with mental illness information and representation on TikTok, such as the spreading of inaccuracies and glorification of psychological disorders. Similarly, Foster (2024) noted that social-media-facilitated self-diagnosis might lead adolescents to falsely identify themselves as suffering from psychiatric disorders without expert evaluation.

Excessive TikTok usage was associated with anxiety, emotional dysregulation, and compulsion towards social networking in adolescence, according to Jain et al. (2025). The constant exposure to idealized lifestyles and emotionally intense content might exacerbate social comparison, stress, and self-perception.

Factors Contributing to the Spread of Misinformation

There are several structural elements associated with social media that promote the fast propagation of health misinformation. First, social media relies on engagement-based algorithms, which favor posts that trigger considerable interaction rates. Misleading claims regarding health tend to draw more engagement than scientific facts do, increasing the likelihood of being shared.

According to Nguyen et al. (2023), the regulation of misleading videos on social media is challenging since such content can easily convey emotionally persuasive information without mentioning any scientific evidence. Hence, teenagers might find videos that contain misinformation more convincing than evidence-based explanations provided by medical professionals.

Misinformation dissemination can be promoted by confirmation bias. People usually tend to interact more with information supporting their ideas and perceptions, leading recommendation algorithms to continue presenting them with similar posts. As a result, adolescents can gradually become less exposed to credible medical data.

Public Health Implications and Potential Solutions

Tackling adolescent health misinformation requires coordinated efforts by educators, healthcare professionals, policymakers, researchers, and even social media companies. Improving digital health literacy education is one way of tackling this problem. Adolescents should be taught how to verify the credibility of sources, understand scientific information, identify misinformation tactics, and differentiate between personal anecdotes and scientifically verified medical information.

Healthcare professionals also have a key role to play in online health communication. According to Sharevski et al. (2024), more needs to be done to ensure healthcare professionals contribute to social media conversations to debunk health misinformation.

Finally, social media companies need to play their part. This includes improving algorithms that detect misinformation, increasing transparency in sponsored health information, and curbing the use of algorithms that increase visibility for demonstrable falsehoods. Heiss et al. (2024) have discovered that debunking evidence-based information in short-form videos may effectively minimize the spread of misinformation.

Conclusion

Social media has radically transformed how adolescents interact with health information. While it provides a platform for learning about health-related issues, it has also contributed to the widespread dissemination of misinformation related to diet culture, vaccine misinformation, alternative medicine, and mental health self-assessment.

Adolescents are prone to misinformation due to their developmental tendency to be influenced by peers and engage emotionally with content. Engaging with false health information on social media can impact adolescents' dietary habits, medical decisions, attitudes towards mental health care, and perception of the medical profession.

It is crucial to educate adolescents on digital health literacy to address misinformation. Social media companies need to regulate the information posted online, and there needs to be more involvement from health professionals on these platforms. Public health systems should be able to evolve alongside adolescents' social media usage trends to guarantee that they have access to valid and medically sound information.

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Not Quite My Tempo: An Analysis of Tempo Alterations and Preferential Ratings Across Differentiated Age Ranges By Hritik Patel

Abstract

The aspects that impact a listener's reaction to musical stimuli on a neurological basis are numerous, and many require richer analysis to ascertain how they do so. This study focuses on the variable of age, as aging has shown to impact the structures of the brain associated with auditory perception. Specifically, this study investigates the correlational relationship between the age of a listener and their preference for songs after isolated tempo alterations. This pilot methodological investigation benefits from separating tempo as a sole differentiable factor, as well as employing a wide age demographic, unlike many other studies who have attempted to analyze said relationship. The results demonstrate a slight preference for faster tempi at younger ages (14-18 years), with partiality decreasing as age increases. However, the overall intensity and significance of the results has limited statistical separation, contrasting the large degree of difference in liking concluded from other studies that did not utilize this method. This distinguishing factor suggests that in other studies, failure to isolate tempo as the only independent variable could contribute to a higher degree of variation in preference ratings, artificially strengthening their conclusions. If applied to a larger participant body, this method could reveal new conclusions about the presence of tempo as an important factor in neurological perception of music, potentially challenging previous notions about its importance, and identifying a flaw in numerous instances of prior research.

Introduction

"What is your favorite song?" is a common question often used to reveal aspects of an individual's identity. While many have various reasons for why their favorite song is favored, one aspect of interest is the technical elements of the song. Elements such as pitch, rhythm, timbre, and harmony greatly affect listener enjoyment when they are manipulated (Schäfer et al.). The human interpretation of music is a complex and multifaceted subject that is not yet fully understood. One element of music with a large body of research is tempo: the speed at which music is played or recorded.

The tempo of music is claimed to have many psychological interactions with a listener, with tempo being linked to many elements of human behavior, both externally (listening to music of a certain tempo) and internally (humans producing tempo, either by playing an instrument or performing behaviors like tapping their foot or hand, etc.). Although widely studied, these interactions have yet to be exhaustively researched. There is not yet a full understanding of the extent to which they are interconnected with the human mind. According to Zengyao Yang from Jiaotong University, "different music tempi significantly affected brain wave activity, particularly in the frontal, parietal, temporal, and occipital regions," (Yang et al.), showing how the tempo of music does not just affect a listener superficially, but also affects

internal functions of various regions of the brain. The tempo of music has tangible psychological effects, depending on whether it is uncharacteristically high or low.

Tempo has also been shown to have physiological effects. In areas such as athletics, these effects are especially prominent, with faster music being linked to higher athletic performance, higher exercise time, and endurance, and more (Thakare). What is even more interesting, arguably, is what physiological factors affect how humans perceive tempo. Of the factors that could impact performance, preference, and perception of tempo, what the current body of research has found to be particularly intriguing is age.

Literature review

Various scientific perspectives accompany the comparison of age and tempo in the brain. One consideration is how age affects elements of tempo perception in performers. In one study conducted by researchers Geoff Luck and Alessandro Ansani detail how, as performers age, a steady loss of motor function and ability to physically maintain a higher tempo leads to a decline in tempo of the music they make as they age. They state that, “Results revealed a slight tempo increase from early adulthood to age 30, followed by a marked, linear slowdown with age across the remainder of the lifespan.” (Luck and Ansani). Their motor function directly affects the dynamics of their tempo capabilities, showing that some physical aspect is also present in how age affects tempo (Ragot et al.).

Tempo changes related to age are not solely physical, however. Certain studies suggest that the “internal metronome” of people (expressed through actions like tapping feet) is subject to change as a person ages. A section of the volume *Experimental Aging Research* details a study in which subjects of varying age were asked to tap at their preferred speed, and the tempo was approximated. The study found that “tapping rates were consistently and significantly slower on average in the older subjects than the younger ones” (Vanneste et al.). Aging can affect how people perform temporal functions, and physical and psychological factors influence these functions. Tempo seems to be less of an external factor in sound for the brain to interpret and more of an internal function that responds to external stimuli and can be affected by age, just like other brain functions, such as memory (Suave). Yet, there is a notable difference between internal and external tempi. Internal tempi can be described as the tempi Suave’s study was evaluating: tempo that was created by the participant, whereas external tempi, reacting to external stimuli and judging preference as such, is evaluated separately in other studies.

The ability to interpret tempo is important to consider when comparing age with tempo preferences. If there is a link between age and the ability to draw a connection between tempo and music “quality” (Liu et al.), it can lead to a more determinant factor for how tempo would affect someone’s judgment of musical quality as they age. One study in Volume 80 of *Cognition* proposes that the function of interpreting music by tempo is a trait that is developed very early on in age, earlier than other musical qualities. This study showed young children music and asked them to state whether they thought it was more “happy” or “sad”. Two factors, tempo and mode (the “key” that the music is played in, e.g., major, minor, etc). As the study states,

“5-year-old children exclusively used tempo information, whereas 6–8-year-olds, like adults, utilized both tempo and mode.” (Bella et al.). Tempo is shown to be one of the earliest cues used in musical interpretation, indicating that age and tempo go hand in hand in the aspect of pure interpretation capabilities. The reaction to temporal stimuli, not just the performance of temporal activities, is affected by age, demonstrating a connection, although not fully defined, between internal and external tempo functions.

Foundational conclusions like these were quickly expanded into the perspective of preference declaration. This is the area where research has varying efficacy. In a study published in *The Journal of Research in Music Education*, people of ages ranging from third grade to college seniors were exposed to 24 instrumental music samples, each with varying tempo, and their preferences were recorded and compared with their age. The results showed that all members of this entire age range showed a “strong and statistically significant preference for music of fastest tempi” (LeBlanc et al.). But, the level of this preference shifted as the subjects aged, showing small discrepancies in general enjoyment within the demographic, with children in the youngest areas of the age range preferring the fastest tempi. In another study, age preferences of adults were compared when asked to listen to 51 samples, and they concluded that “age trends in musical preferences could be partially explained by differences in the ways individuals process the intrinsic attributes of the music with age” (Bonneville-Roussy et al.). These studies demonstrate a prominent link between the age of the listener and their interpretation of music at varying tempos, more specifically, how they prefer music based on these interpretations. Additionally, internal and external tempi seem to both converge on the downward trend of tempo with age. However, a gap also emerges. Many of these studies are not recent, and those that are tend to focus on a specific age range (such as the previously mentioned study, which only compared participants from the range of third graders to college students). Notably, LeBlanc, Bonneville-Roussy, and Bella’s studies do differ in their age demographics, which creates an early consideration that can conflate conclusions. Specifically, the tempo preferences of teenage (14-18) participants have yet to be directly compared to those of a significantly older demographic (such as 30-60) without eliminating other factors. The lack of tempo as a sole variable leads to a glaring subjectivity, as there are too many independent variables that are not able to directly isolate tempo as a sole determinant. A method in which pairs of similar samples are compared across a wide age range has yet to be implemented, and the closest comparison of a wide age range draws its conclusions from internal tempo functions (LeBlanc et al.). As a result, further research must be conducted to compare age ranges that are further apart, as well as a comparison of samples that are tonally and instrumentally similar, with tempo being the only independent variable (pairs of music with their tempo altered). A necessary elimination of other musical factors, and a need for a wider age group participating in a study exclusively responding to external tempi raised the question: “What is the extent to which age differences (discrepancies between age ranges of 14-18, 19-30, 30-39, 40-49, and 50+) of individuals affect preferences for songs after various tempo alterations, irrespective of genre?”. From the conclusions drawn from the previous studies, both when analyzing internal and

external tempo, there appears to be a large preference for music of faster tempo for those in the younger demographics (14-18, 19-30), and an inverse relationship with tempo speed and age. Therefore, it is hypothesized that those in the younger age ranges will demonstrate large preference for faster tempi, and those of older demographics (40-49, 50+) will demonstrate large preference for the slower tempi.

Method

This study encompassed a single multimedia survey that consisted of 18 multiple-choice questions and 3 open-ended short-answer questions. The majority of survey data was collated quantitatively. The questionnaire was structured by first optionally asking for the applicant's name, and then mandatorily inquiring about their age, which took up two of the three short-answer questions. The 18 multiple-choice questions consisted of question pairs that aligned with 9 song clips, with three time-stretched versions each. One version was slowed by 17.5%, one was left at its normal tempo, and one was sped up by 17.5%. They were altered to this extent to ensure that participants were capable of hearing that each variation differed in tempo. In a study conducted by Irene Gratton and a team of scientists, they found that musically uninclined individuals, when inclined to match a set tempo value with a rhythmic activity (like tapping their feet or hands), generally approximated a tempo value to about $\pm 12\%$ (Gratton et al.). To ensure a noticeable tempo change, the alteration must fall outside of this range. Although originally set to 20, it was reduced to 17.5% (which was the split difference between a slight change of 15% and an extreme change to 20%) to ensure that it did not affect the quality of the transient audio in the samples. The pitch and key of each clip were kept the same, with the only post-processing effects applied to the clips being multiband compression on the altered samples to keep sound quality similar. These songs were chosen from 9 major genres that encompassed a majority of popular genres, while also reaching across a wide range to best minimize bias in the form of genre preference. These genres were Pop, Rock, Synth Rock, Pop Country, Hip-Hop, Latin Pop, Jazz, Classical, and Big Band Jazz. Each genre had a single corresponding song selection. Each song was procured along the general guidelines listed in Section 107 of the Copyright Act that defines Fair Use (U.S Copyright Office). Each clip is 30 seconds or less in length and was acquired from trustworthy sources (they were converted to audio format from YouTube.com, each song taken from the artist's officially licensed release). The intended use, being that of research and scholarship, qualifies for Fair Use.

For Pop, the song chosen was Welcome To New York (Taylor's Version) by Taylor Swift. For Rock, the song chosen was Little Miss Strange by The Jimi Hendrix Experience. For Synth Rock, the song chosen was Radio Ga Ga (Remastered 2011) by Queen. For Pop Country, the song chosen was Chain Smokin' by Morgan Wallen. For Hip-Hop, the song chosen was Headlines by Drake. For Latin pop, the song chosen was DtMF by Bad Bunny. For jazz, the song chosen was Take Five, composed by Dave Brubeck and performed by The Dave Brubeck Quartet. For Classical, the song chosen was Symphony No. 5, composed by Beethoven, and performed by the Boston Philharmonic Orchestra. For Big Band Jazz, the song chosen was

Caravan, composed by Duke Ellington and Juan Tizol and arranged by John Wasson. These songs were chosen as they are popular representatives of their respective genres, which best eliminates subjective bias as to whether these songs actually do belong in their respective genre assignments.

The multiple-choice questions consisted of the same two questions repeated for each song triplet. After being provided the audio files for a song and its two altered counterparts, participants are then asked to rate each song on a Likert scale out of 7, 1 being that they do not enjoy it, and 7 being that they enjoy it to a high degree. Afterwards, they were asked to select which of the three versions of the song they preferred. Then, the audio files of the next song and its counterparts are provided, and the same questions are repeated for each song. Afterwards, one final open-ended question asks them to state which song and what version of the song was their favorite out of the ones they listened to.

The use of a survey was necessary to collate data without outside bias, which poses a possibility with an interview-style process. Making the survey online and able to be taken at the participants' convenience allows them to take it when their attention is relatively isolated, and without a prompter or surveyor on-site. The reason samples with altered tempo but not altered pitch were used is so that the factor of pitch could be minimized as much as possible, as a variable that could affect the outcome. In the aforementioned study in *The Journal of Research in Music Education*, it is important to note that whilst a similar, survey-style rating method was used, each sample was a completely different sample with varying style and key. Sample individuality presents two events that affected both the former and current method. Because they were all different samples, the study I used as a model had a much better coverage of genre, as the more genres included, the more negligible the factor of genre preference would be. However, with their use of completely different samples every time, elements like harmony, timbre, and style all affect a listener's experience, especially pitch. In a study conducted by Josh H McDermott, published in the NIH, it is stated how "Western listeners consistently prefer some intervals to others, and this depends little on musical training" (McDermott et al.). Listeners will prefer certain intervals and tonal arrangements, which means having completely different samples risks too many factors that cannot be controlled, such as the key and the intervals of the notes in each song, creating differences in preference and "sonic pleasantness". The study covering Western listeners is also important, as all of the participants are from America, which would pertain to this demographic. Such a consideration is important to consider, as it poses an issue that the method this study attempted to circumvent. By using altered samples, these samples had the same melodic content, simply with their tempo altered. In order to respect participants' time, 9 songs were chosen as any more would lead participants to opt out of completing the survey. More general genres needed to be selected in order to still encompass the broad taste of a large body of modern listeners. The large majority of data is meant to be quantitative, as it provides the best gauge of data comparison. It aided in establishing the most concise correlational relationship between age and their preferences, as the general trend in each song could be combined to draw a broader conclusion about each age's preferences (e.g., 70% of

participants under the age of 20 rated the sped-up versions of all songs higher on average than any other age group).

Results

The survey was taken over the course of 2 months and garnered 23 responses. This sample size is less than desired, though it provides a moderately adequate basis, especially because a majority of responses fell in the 14-18 year range. This range was extremely important as it was a large gap in the current data. The only all-age tempo studies that included adolescents were the previously identified studies, where tempo was not the only factor that was altered. The number of responses was of a smaller volume. From further inquiry of some of the participants, the reason why there were fewer responses was attributed to the length of the survey. The survey was designed to take 10 minutes at the most; however, with the time it took to load each individual music clip, this process often became arduous. With no financial incentive, this quickly deterred many participants. A majority of the responses are from Northern Burlington County Regional High School Students; some were collected from others outside of that circle, from students sharing the survey with family members. Two participants did not fit into the age ranges specified, as one was 9 years old and one was 10. Seeing as how the main focus of the study, while each age group was weighed equally, was to compare the general adolescent age group to the older individuals, their responses were considered extraneous (n=21).

The Figures below encompass the important data comparisons between three factors: Preference (both rating and type), Age Group (broken into specified age ranges), and Song Genre (broken up by individual song, as each represents a genre). Figure 1 takes the mean rating of all songs by version (that way, for each participant, there is a singular data point per rating by version, which is a consolidation), and finds the average rating of each version by age group. Figure 1 represents a conclusive comparison of each age range's general rating of all song versions, as this would allow direct analysis of the three versions by age. This was the most crucial comparison, as this analysis would correlate age range and preference through ratings. Figure 2 takes the results from the "Which version did you prefer?" question, and counts the number of participants by age group that had a majority of their responses as one option (either slowed, sped up, or unaltered), which was defined as more than 50% of songs had one choice selected by the participant. Figure 2 analyzes the "strength" of preferences. Any participant with a majority rating in one song version demonstrates a strong preference for one version, and if a larger number of participants appear to be in one area, it provides further evidence of a correlation. Figure 3 consolidates all the preference choices across all age ranges and separates them by genre, comparing the percentage of all participants who preferred each choice by song. Figure 3 provides an analysis of preference ratings by song, which are representative of genre. If there is a large difference in ratings by song, then it indicates that perception of tempo alterations is also genre dependent. Whilst having multiple genres helps to eliminate that factor, if there is a large enough discrepancy, it indicates that more genres need to be included for a more comprehensive result. Figure 4 consolidates all preference choices by all participants and

separates them by the percentage of total choices that were each of the three options (slowed, sped up, and unaltered). Figure 4 is a general comparison of overall ratings to ascertain if one version has a large preference, which can indicate the consensus of all song ratings. A Kruskal-Wallis H Test was used to determine the general statistical significance of the differences in the data collected in Figure 1, considering the unequal demographics and the nature of the data provided by the Likert scale rating. If a significant difference was detected, this would support a present discrepancy between age groups. Firstly, a null and alternative hypothesis were determined, with $H(0)$ being no significant difference in the distribution between each age group, and $H(a)$ being some existing difference in distribution. From there, each response was collated for each participant and split by age range, and was done three times, one for each song version. A p-value was then calculated from the resulting data. This data is represented in Figure 5.

Figures

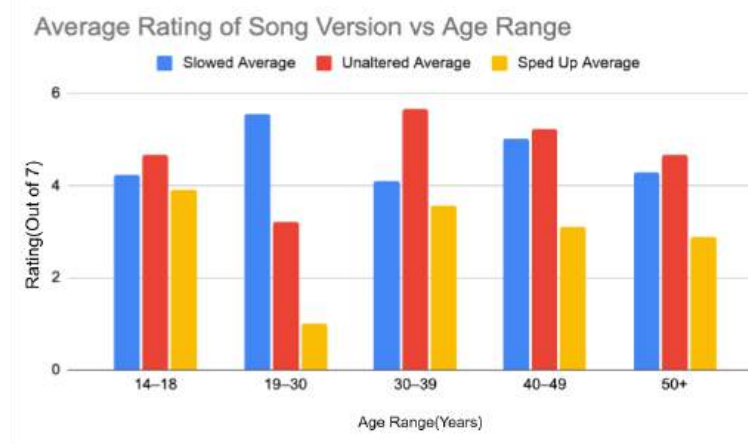


Figure 1

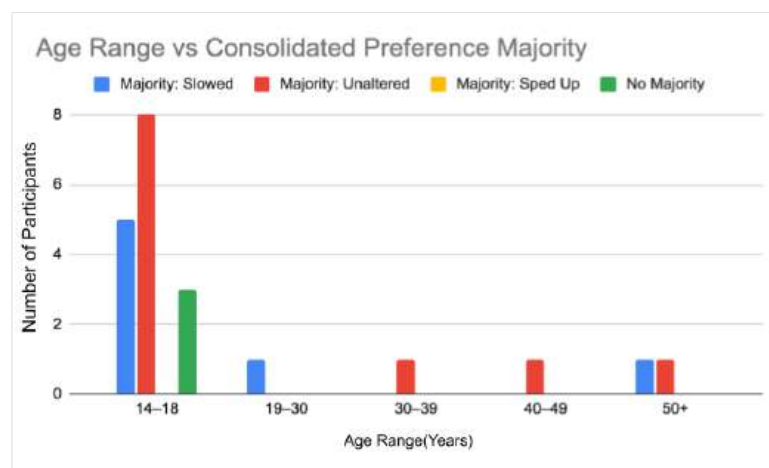


Figure 2

Slowed %, Unaltered % and Sped Up % Across All Age Ranges

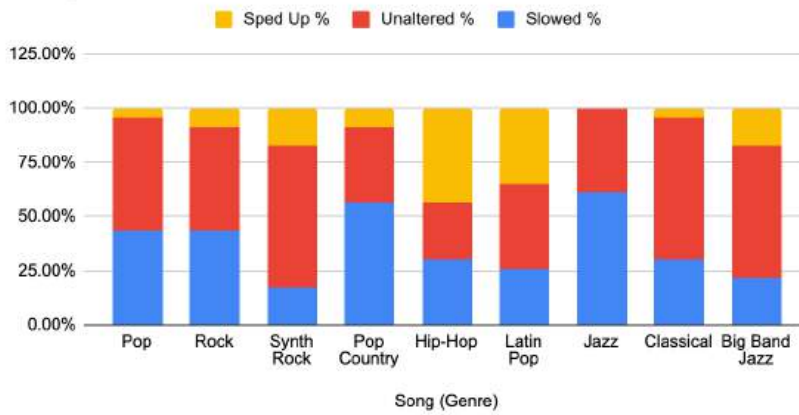


Figure 3

Percentage of All Response Preferences(consolidated)

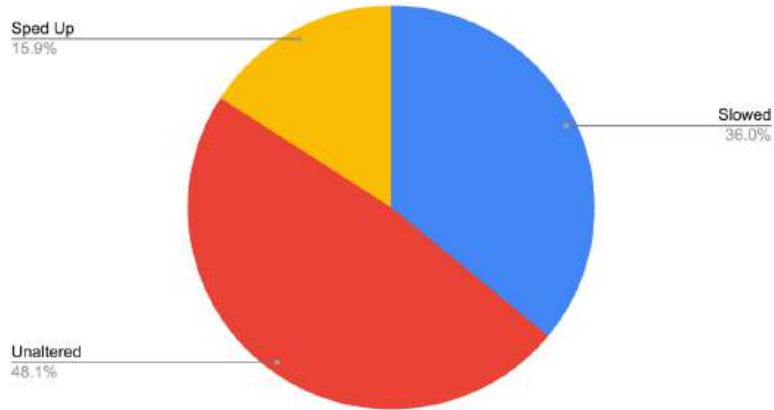


Figure 4

Version	H-Statistic	P-Value
Slowed	1.139	0.888
Unaltered	0.443	0.978
Sped Up	1.341	0.854

Figure 5

Analysis & discussion

The results both partially align with the aforesaid conclusions drawn from other studies, while also presenting new information that challenges important methodological assumptions in previous studies. Firstly, much like other studies have seen, in Figure 1, it can be seen that the preference for sped-up music clips decreases as age increases. The average rating of sped-up song versions decreased by 1.03 points from the youngest to the oldest age range. There is a decrease in approval of sped-up music, while there is a slight increase in approval (0.06 points) from 14-18 to 50+ in slowed music. This trend aligns with the aforementioned studies conducted by Vanneste and LeBlanc, respectively. They both agreed that tempo preferences “slow down” as age increases, which is seen here as well. In both Vanneste's and this current study, there is a higher likelihood for those in younger and adolescent ages to prefer faster or sped up music, and that likelihood promptly decreases as age increases. However, this trend requires a nuanced analysis. Where this study and many others differ is the overall degree of preference for clips that were sped up compared to unaltered ones. Vanneste's study in particular concluded that those in the adolescent age range preferred music of faster tempi; Across all age ranges, including the same age ranges highlighted in Vanneste's study, there was a majority preference for unaltered clips, as seen in Figures 1 and 2.

The results drawn from the p-values in Figure 5 all indicate an absence of comparative statistical significance between each age group, meaning that while there were differences, they are still minimal, and far less than previous studies suggest. Where our studies differed is that Vanneste and many similar studies used different song clips for each tempo class, whereas I altered the tempi only. This calls into question whether tempo as a sole correlational factor in their studies was enough to encompass their results. In mine, where tempo was the only changing factor across all three clip versions, many preferred an unchanged version. This study suggests the possibility that the high degree of age-differentiated tempo preferences in other studies could have been due to other elements like melody, mode, or chordal arrangement. These results argue that prior results are at least marginally impacted by a lack of isolated variables. This potentially impacts the conclusions of several previous studies, while also suggesting that more research is needed to conclude the relative importance of other musical elements and tempo to a listener's preference.

This contradiction must be considered cautiously, as the fact that the samples were altered can be a contribution to the participants' tendency to prefer the median tempo value. The samples were originally produced in their unaltered tempo, so their percussion, vocals (if applicable), and instrumentation are all tailored to that tempo. The drawback of using the same sample for all three tempo variants is that, while it isolates tempo as the only independent variable, the fact that none of the other elements change can lead to a slightly unnatural quality of the altered samples (Rashotte et al.). Such leads many to default to the original sample for a lack of said unnatural factor. The statistical significance of the unaltered samples may have been disproportionately inflated if this effect were extremely prominent. Various responses in the free-response question highlighted this, as one response stated how “Hearing the versions with

faster and slower tempi, if anything, enhanced the enjoyment and entertainment value of the original.” The consolidation of all preference ratings across all songs and participants in Figure 4 corroborates this finding, as almost 50% of all preference ratings were unaltered, indicating a large preference for music of unaltered tempo in general.

Additionally, while melodic variation was reduced to an absolute minimum, there were multiple songs in the data set to account for genre (although, seeing as how many more songs would have resulted in the survey being too arduous for the participants, genres were limited, leading to a presence of a margin of error in that aspect). To get a more comprehensive response from each participant, multiple songs and versions were used, but this ultimately led to variation by song. This is what characterized this study, as it is the least affected by melodic differences in other songs within their designated tempo ranges, but, as much of the data is drawn from averaging ratings across all 9 songs, genre variations have to be addressed. As seen in Figure 3, there are various discrepancies between the three version preferences and what song is analyzed. For instance, the Synth Rock representative song has the highest percentage of unaltered preferences among all the songs, at 65%, whereas the Hip-Hop representative has the lowest, at 26%. The variation in the altered versions varies similarly. This indicates that in order to have a more comprehensive result, more songs need to be used, as 9 songs are a small pool of genres. Given the variation in preference across the three versions by genre, the overall conclusion could be affected by the number of songs participants rated.

Limitations

In assessing the results of this study, it is important to recognize several factors that limit the generalizability of the conclusions drawn. Firstly, it is important to note the limitations pertaining to the sample size. The sample size was not only limited to the Northern Burlington County Regional High School students and individuals associated with it, but also small, being only 21 individuals who were considered when computing the data. This impact significantly affects the reliability when analysing age groups other than the adolescent age group of 14-18. Many age groups had a minimal amount of participant responses, usually 1 or 2, meaning that the responses that contributed to these age demographics have a limited amount of efficacy as to the implicit broadness of the respondents. Such could explain the high p-values, possibly resulting from the insufficient number of respondents in older demographics. The limited sample size in the 19-30 and 50+ demographics prevents a completely definitive conclusion in those ranges. The statistical significance of these age groups is affected drastically, as the participants in the older age ranges have a possibility of being outliers(those who generally prefer faster music or slower music disproportionately to their age range), which would affect the overall mean of the ratings for their age group, thereby suggesting a possible Type-II error. This greatly limits the generalizability, as this data can only be applied to those in the Northern Burlington area, and whilst the 14-18 age group had a significant amount of responses, the others did not, meaning the conclusions drawn from the older individuals have limited effectiveness.

The contrast between the possible correlative and causational effects of the age of the respondent and their perception and overall preference of the tempo of the samples is also important to consider. Previous studies have established a substantial correlation between the age of the listener and their perception of tempo, while also providing some evidence as to a possible neurological explanation. This evidence, however, is minimal. For example, an article published by Johnathan Peele establishes a deterioration of the auditory cortex, the area of the brain most commonly associated with musical stimuli, with age. He states how “Age-related hearing loss [is] reflected in the structure of human auditory cortex,” (Peele and Wingfield). However, studies like these do not provide a significant amount of information about tonal or metronomic effects in the brain. This limits the causational implications of the study and reduces the conclusion of the study to establishing a purely correlational relationship between age and tempo preference.

While the independent variable was only the tempo of the musical stimulus, other factors can affect the listener’s preference for the song at varying tempo values. For example, the genres they prefer, the music they listen to, and their aforementioned familiarity with the song can significantly affect their responses. Their familiarity can especially affect their responses, as all of the songs that were chosen are popular songs, increasing the chances that the participants had heard these songs before in a previous instance outside of the study. This concept is called the “Imprinting Effect”, and is described by Jiayu Fu as the concept that “...most songs and poems identified as ‘favorites’ were learned during early adolescence” (Fu et al.). If one has heard a song before, especially in early adolescence, they have more of an affinity for it. When listening to an altered version, this affinity can cause them to dislike the alterations and prefer the original, which could explain their preference for the unaltered version more. Because of the lack of personal information disclosed, there was no way to track whether they had heard the song before. This presents an unresolved, indiscernible distinction between whether participants are choosing the tempo they prefer or the tempo that is the most familiar to them. In future studies, original samples should be used to ensure that the Imprinting Effect is mitigated. Review over the genre classification of the aforementioned samples is imperative, as this was avoided by using popular music in this study, but presents itself as a considerable factor if using original samples.

Conclusion

This study yields three conclusions. Firstly, there is a modicum of evidence that supports a downward trend in tempo preference with increasing age. This directional decrease creates a small practical observation that, if a larger study increases statistical significance, would mean that older individuals appear to be more likely to prefer music of a slower tempo than younger individuals. By contradiction, it can also be conjectured that younger individuals would be more likely to prefer music of a faster tempo alternative than older individuals. Secondly, all individuals across all age ranges tended to prefer music of unaltered tempo to the highest degree when compared to its alternates with a slightly stronger significance, meaning the data suggests that individuals are more likely to prefer tempo towards the mean or “normal” tempo range,

regardless of age. Finally, it can be concluded that isolating tempo as a sole determinant factor seems to drastically reduce the intensity of preferential differences in participants, decreasing the amplitude of how much each tempo range is preferred across age groups. This isolation also contributes to a large decrease in overall significance in said discrepancies, meaning that while a trend is established, it is relatively minimal. The first conclusion aligns with previous works and deepens the understanding of how age increases correlate with a preference for slower music. Studies surveying the most preferred genres among elderly individuals, like the one conducted through the National Institute of Health by J. L. Jonas, indicates a preference for country music, traditional jazz, and art music (Jonas). All of these are associated with slower tempo ranges and further contribute to the idea that older individuals may prefer slower tempi. Whilst the difference is not significant, there was slight evidence to support previous studies. Commercially, these results can be used to aid demographic information for music artists, such as marketing slower or faster music to a specific age group. Scientifically, this correlation can be further studied as to how age can affect neural processing areas, and whether any physical changes to these areas directly contribute to the differences in preference seen. Moreover, in areas like music therapy, the forms of music utilized can be further adjusted considering the tempo preferences of the demographic being administered (Ma).

The second and third conclusions offer a new factor of consideration when analyzing the extent to which tempo alone can impact a listener's experience. Additionally, the nature of the conclusions from the study leads to a larger efficacy of these conclusions specifically. The general preference for unaltered tempo can clue into the human tendency towards the mean, with the general ideal or “walking” tempo being 120 beats per minute (Etani et al.). In contrast, the consideration that isolating tempo led to a lesser intensity of tempo preferences leads to the suggestion that tempo alone has a mixed effect when compared to other factors like mode, melody, and familiarity. Prior research has yet to consider such a difference, and if this conclusion holds strong in a broader study, it would lead to a need to reevaluate this factor in other studies of the sort. In all, the extent to which discrepancies in age contribute to differences in preference for songs at altered tempo is somewhat present, but to a lesser and more complex extent than previously thought.

Due to the drastic difference in efficacy of the conclusion compared to other studies, this study serves as a model of a more precise methodology that can be repeated to draw more specified conclusions in a larger sample size. This novel method demonstrates a shift in responses when a wider age range is taken and when tempo is independently isolated. From this, psychologists in the future should focus on creating even more specific conditions where tempo is the only isolated factor, as the results from this study support a slightly different outcome than previous studies, which did not account for this. The high p-values acquired from this study's data, and the existing trends in mean ratings, suggest this discrepancy in tempo preference is present, but minor. The efficacy of this claim can be tested with larger sample sizes. Such places this study as a viable pilot methodology in which future research can be based, highlighting the need for more precise investigation into the impact of tempo on musical interpretation.

Appendix A: Full Survey Instrument

Instruction: In this form, you will be asked to listen to state your name(optional) and age(required). Then, you will be asked to listen to 9 music clips(headphones optional but preferred), each with three versions(slowed, unaltered tempo, and sped up), and you will be asked to rate them on a scale of 1 to 7(1 being strongly dislike, and 7 being strongly like). At the end of each of the three clips, you will be asked to select which version you prefer out of the three.

1. Please select either option to indicate your consent(or lack thereof) of your child to participate in this survey:
2. Type your name below as an electronic signature:
3. Please select the box below to indicate your parent has written their signature in the question above or you are 18 years or older:
4. Please state your full name(optional):
5. Please state your age:
6. Listen to these three clips and rate each according to your liking:
7. Out of the prior clips, which do you most prefer?
(QUESTIONS 6-7 ARE REPEATED 9 TIMES IN TOTAL)
8. Which song and what version did you like the most?

APPENDIX B: Audio Stimuli List & Estimated Tempo Values by Version

Song	Original Tempo (Beats per minute)	Slowed(-17.5%)	Sped up(+17.5%)
Welcome To New York	117	97	138
Little Miss Strange	124	102	146
Radio Ga Ga	92	76	131
Chain Smokin'	150	124	176
Headlines	88	73	103
DtMF	100	83	118
Take Five	174	144	205
5th Symphony	108	89	127
Caravan	132	108	155

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Eliminating the “Improper Women”: Effects of the Indian Feminist Movement on the Devadasi System By Bonnie Shao

Abstract

In the early twentieth century, the Indian feminist movement increasingly targeted devadasis—female Hindu temple dancers—as prostitutes and sought to reform them by banning their traditional practice. They did so under the guise of ending the exploitation of devadasis by men. However, this paper argues that the feminists’ activism was mainly a reaction to the widespread standard that only chaste women deserved respect. In abolishing the devadasi practice, Indian feminists attempted to highlight their own purity and obtain legitimacy for the other causes that they advocated within patriarchal society.

Carrying icons of the goddesses Yellamma and Matangi, modern devadasi women of India’s Karnataka state climb into a flatbed truck, which transports them to a nearby river for a worship ritual.²⁰ In the sparkling cold water of the river, the devadasis unwrap the icons’ saris and wash their bodies. When finished, they adorn the icons with new saris.²¹ These devadasis have devoted themselves to serving Yellamma and Matangi, their chosen deities.

The term “devadasi” refers to a class of women who, since the third century C.E.,²² have dedicated themselves to Hindu temple gods through ceremonial rituals of “marriage.”²³ After dedication, devadasis spend their lives performing dance and rituals in the deity’s honor.²⁴ Historically, they also maintained sexual relationships with wealthy patrons.²⁵ By the mid-twentieth century, however, the devadasi system had been legally banned. The movement to supposedly reform the devadasis was greatly furthered by upper-caste Hindu feminists in the late nineteenth and early twentieth centuries who claimed that the system encouraged exploitative prostitution.²⁶ In contrast to this negative stereotype, however, many devadasis enjoyed exemption from the constraints of widowhood, greater flexibility in their relationships, and more financial autonomy than most other Indian women.²⁷ One might thus wonder why numerous prominent feminists fought for devadasi abolition, depriving many women of their legal right to these distinctive freedoms. In undermining devadasis’ rights, however, feminists sought another form of women’s emancipation: the esteem of patriarchal society. Reacting to the widespread

²⁰ Lucinda Ramberg, *Given to the Goddess: South Indian Devadasis and the Sexuality of Religion* (Duke University Press, 2014), 1934.

²¹ Ramberg, *Given to the Goddess*, 1935.

²² R. Kalaivani, “Devadasi System in India and Its Legal Initiatives – An Analysis,” *IOSR Journal Of Humanities And Social Science* 20, no. 2 (2015): 51.

²³ N. Anitha, “Role of Dr. Muthulakshmi Reddy in Abolition of Devadasi System,” in *The Opportunities of Uncertainties: Flexibility and Adaptation Needed in Current Climate*, ed. Shahana A. M, et al. (Lulu Publication, 2021), I: 284.

²⁴ Kalaivani, “Devadasi System in India and Its Legal Initiatives,” 50.

²⁵ Amrit Srinivasan, “Reform and Revival: The Devadasi and Her Dance,” *Economic and Political Weekly* 20, no. 44 (1985): 1869.

²⁶ Teresa Hubel, “The High Cost of Dancing: When the Indian Women’s Movement Went after the Devadasis,” in *Intercultural Communication and Creative Practice: Music, Dance, and Women’s Cultural Identity*, ed. Laura Lengel (Praeger Publishers, 2004), 128-9.

²⁷ Srinivasan, “Reform and Revival,” 1869-72.

belief among Indian men that women must maintain sexual purity to be valued, feminists manipulated the devadasis' image to highlight their own chastity. In doing so, they attempted to obtain respect from male-dominated society and legitimize their activism.

Throughout the late nineteenth and early twentieth centuries, the Indian feminist movement advocated for women's wellbeing. It sought to better women's condition within patriarchal systems, such that they would receive more favorable treatment from men.²⁸ To do so, major women's groups like the Women's Indian Association (WIA) focused on reforming issues like the unfavorable treatment of widows,²⁹ child marriage, and large age gaps between husbands and wives.³⁰ The feminist movement did not advocate the interests of all Indian women, however. In fact, in order to secure recognition within Indian society, the women's movement frequently disregarded the opinions and objectives of lower-class women,³¹ such as devadasis, and suppressed their ancient traditions.

Yet, the call for devadasi reform arose not from the feminists, but from male Hindu professionals in the late-nineteenth-century Madras Presidency.³² Encouraged by Christian standards of female passivity and chastity perpetuated by British colonial elites,³³ such professionals criticized the devadasis' matriarchal traditions and sexual freedom; they accused devadasis of intentionally spreading venereal disease to Indian youth.³⁴ Many denounced the system with journalism and presented memoranda advocating the official ban of the practice to officials. By the early twentieth century, numerous feminists had also become driving forces of the devadasi reform movement, with many successfully pushing for anti-devadasi legislation. They heavily scrutinized devadasis' dances for their portrayals of sexual themes and claimed that the system exploited women for prostitution.³⁵ One such feminist was Muthulakshmi Reddy, the first Indian woman legislator, a central figure of the women's movement, and a prominent critic of temple dedication.³⁶ Reddy championed legislation against the devadasi system, even setting forth a bill in 1927 to eradicate the dedication of women in Madras Presidency.³⁷ This bill ultimately passed in 1947 as the Madras Devadasis (Prevention of Dedication) Act.³⁸ This act

²⁸ Mytheli Sreenivas, "Creating Conjugal Subjects: Devadasis and the Politics of Marriage in Colonial Madras Presidency," *Feminist Studies* 37, no. 1 (2011): 68; Subhadra Mitra Channa, *Gender in South Asia: Social Imagination and Constructed Realities* (Cambridge University Press, 2013), 88.

²⁹ Aparna Basu and Bharati Roy, *Women's Struggle: A History of the All India Women's Conference, 1927-1990* (Manohar, 1990), 91-92.

³⁰ Channa, *Gender in South Asia*, 87.

³¹ Sreenivas, "Creating Conjugal Subjects," 71.

³² Srinivasan, "Reform and Revival," 1873.

³³ Srinivasan, "Reform and Revival," 1873; Rajalakshmi Nadadur Kannan, "Colonial Material Collections and Representations of Devadasi Bodies in the Public Sphere in the Early 20th-Century South India," *Anthropos* 114, no. 2 (2019): 534.

³⁴ Srinivasan, "Reform and Revival," 1873; Kannan, "Colonial Material," 540.

³⁵ A. Asaf Ali, *The resurgence of Indian women* (Radiant, 1991), 220; Kannan, "Colonial Material," 537.

³⁶ Basu and Roy, *Women's Struggle*, 184; Hubel, "The High Cost of Dancing," 127; Janaki Nair, "The Devadasi, Dharma and the State," *Economic and Political Weekly* 29, no. 50 (1994): 3158.

³⁷ Hubel, "The High Cost of Dancing," 127-8; Nair, "The Devadasi, Dharma and the State," 3158.

³⁸ Kunal M. Parker, "'A Corporation of Superior Prostitutes': Anglo-Indian Legal Conceptions of Temple Dancing Girls, 1800-1914," *Modern Asian Studies* 32, no. 3 (1998): 629.

abolished the practice of dedicating girls to temples.³⁹ Additionally, it outlawed women's dancing in temples, processions, and religious festivals.⁴⁰ It forbade even consensual dedication, dictating that any woman over the age of sixteen who willingly devoted herself to a deity should be arrested.⁴¹

In legally abolishing the devadasi tradition, the Madras Devadasis Act rendered many of the unique interpersonal and financial freedoms that devadasis had possessed unlawful. After her marriage rituals, a devadasi was traditionally regarded not as an immoral prostitute, but as *nitya sumangali*—an auspicious woman free from widowhood.⁴² Hindu widows were commonly forbidden from remarrying or enjoying social and economic freedom; a devadasi's marriage to her deity exempted her forever from this degradation. Further, devadasi marriages defied the rigid monogamous structure of human unions. Despite being “married,” a devadasi was also permitted to engage in sexual relations with a human patron, who supported her financially in return.⁴³ Though reformers perceived this system as degrading prostitution, it often provided devadasis with more rights than many traditional Hindu wives. Under this system, devadasis were not confined to the domestic realm. In fact, they were forbidden from cooking or performing domestic chores for men.⁴⁴ Devadasi culture also upheld matrilineal traditions that differed from India's largely patriarchal practices. Since devadasis frequently generated much of their households' income, they were often prioritized over male relatives in matters of inheritance.⁴⁵ They were also permitted to own land;⁴⁶ as T. Duraikkannu, secretary of the Madras Devadasi Association—a coalition of devadasis assembled to defend their traditions—pointed out, “women of this community alone” were allowed to do so.⁴⁷

Of course, devadasis' autonomy was also limited by patriarchal institutions. Despite their unique rights, they still answered to patrons and temple officials.⁴⁸ Moreover, as historian Teresa Hubel pointed out, the scarcity of records detailing devadasis' own views of their tradition impedes historians' ability to accurately uncover its technicalities.⁴⁹ It is therefore difficult to determine the full extent of the devadasis' ancestral freedom. While devadasis likely did not experience complete equality, however, they nevertheless possessed unique privileges in their

³⁹ Government of India, *The Tamil Nadu Devadasis (Prevention of Dedication) Act*, Act 31, January 17, 1948, 1010.

⁴⁰ Government of India, *The Tamil Nadu Devadasis (Prevention of Dedication) Act*, 1011.

⁴¹ Government of India, *The Tamil Nadu Devadasis (Prevention of Dedication) Act*, 1011-12.

⁴² Srinivasan, “Reform and Revival,” 1870.

⁴³ Srinivasan, “Reform and Revival,” 1869-70.

⁴⁴ Srinivasan, “Reform and Revival,” 1871. Some modern *devadasis* continue to view their tradition as a source of freedom. In a 2012 interview, Pandu, a *devadasi* in Sangli, commented, “I have no regrets...I can freely roam anywhere and do whatever I want. Nobody can stop me from going anywhere” (“Prostitutes of God (Documentary),” video, YouTube, posted by VICE News, August 21, 2012, 13:35-13:55).

⁴⁵ Srinivasan, “Reform and Revival,” 1872.

⁴⁶ S. Jeevanandam, “Approaches on the Devadasi System of India - A Review,” *Indian Historical Studies* 19, no. 2 (2023): 44.

⁴⁷ T. Duraikkannu, “*Cennai Uruttirakanikaiyar Cankam*,” 1927, quoted in Daves Soneji, Appendix 1 to *Unfinished Gestures: Devadāsīs, Memory, and Modernity in South India* (The University of Chicago Press, 2012).

⁴⁸ Nair, “The Devadasi, Dharma and the State,” 3161.

⁴⁹ Hubel, “The High Cost of Dancing,” 125.

relationships and financial lives that, relative to the restrictions many other Indian women experienced, represented a form of liberation. The Madras Devadasis Act of 1947 forced them to yield most of these rights.

The devadasi community reacted to this act with extreme backlash. When Muthulakshmi Reddy proposed the bill in 1927, many devadasis sprang to counter it. In 1927, twenty-five devadasis met outside Madras to “severely oppose” the resolution. They issued a statement denouncing the claim that their tradition was related to the “mean practice” of prostitution. Rather, they argued, it was an “ancient honorable religious practice,” and Reddy had set forth the bill without conducting “proper investigation” into the system.⁵⁰ Therefore, they urged the government not to approve it.⁵¹ Similarly, T. Duraikkannu wrote in 1927, “It is unbelievable that prostitution can be eliminated by putting an end to the [devadasi] practice.” On the other hand, she claimed, “[i]t is certain that the bill...will put an end to the social practices and hereditary rights of our community.”⁵² Many devadasis throughout India attempted to preserve their ancient customs, art, and freedom. Yet, their activism did not reverse the prevailing belief that, as stated by the Madras Act, the devadasi practice, “however ancient and pure in its origin, leads many...women...to a life of prostitution.”⁵³

This image of devadasis contrasted that of a woman deemed respectable by patriarchal Indian society: one who denied her sexuality and devoted herself to a monogamous marriage. Many men in the late nineteenth and early twentieth centuries believed that women should exhibit unwavering virtue.⁵⁴ As historian Subhadra Mitra Channa argued, women had to depict themselves as “worshippable” through chaste behavior in order to be esteemed by men.⁵⁵ In particular, according to historian Sujata Patel, nationalist Mahatma Gandhi framed denial of sexuality as the “most supreme attribute in the world” and maintained that women should abstain from intercourse outside of procreation.⁵⁶ Gandhi claimed in 1924 that women who practiced “self denial,” like ideal Hindu widows, garnered his “reverence.”⁵⁷ Such widows restricted their lifestyle and refused to remarry in order to exhibit devotion to their deceased husbands.⁵⁸ Their “patient suffering,” alleged Gandhi, was “impossible to rival.”⁵⁹ He further argued that chaste

⁵⁰ “The Question of *Devadāsīs*: A Meeting of the *Devadāsīs*, Cheyyur,” *Cutēcamittiraṇ*, 1927, quoted in Soneji, Appendix 1 to *Unfinished Gestures*, 229. Interestingly, Reddy herself was the daughter of a *devadasi*; this relationship may have provided her with more intimate knowledge of the system. However, a lack of sources exploring her familial relationships limits historians’ ability to understand Reddy’s mother’s experiences and how exactly they factored into Reddy’s work (Hubel, “The High Cost of Dancing,” 130).

⁵¹ Soneji, *Unfinished Gestures*, 124.

⁵² T. Duraikkannu, “Cennai Uruttirakaṇikaiyar Caṅkam,” 1927, quoted in Soneji, Appendix 1 to *Unfinished Gestures*, 230.

⁵³ Government of India, *The Tamil Nadu Devadasis Act*, 1009.

⁵⁴ Partha Chatterjee, *The Nation and Its Fragments: Colonial and Postcolonial Histories* (Princeton University Press, 1993), 131.

⁵⁵ Channa, *Gender in South Asia*, 96-7.

⁵⁶ Sujata Patel, “Construction and Reconstruction of Woman in Gandhi,” *Economic and Political Weekly* 23, no. 8 (1988): 385.

⁵⁷ Gandhi, *Collected Works of Mahatma Gandhi* (1922), 22: 524.

⁵⁸ Sarojinee Sahoo and Arun Kumar Acharya, “Caste System, Widows and their Representations in Indian Society: An Exploratory Study,” *Society and Culture Development in India* 4, no. 2 (2024): 349-350.

⁵⁹ Gandhi, *Collected Works of Mahatma Gandhi* (1922), 22: 524.

women should control decisions regarding the household by telling women, “In my home, I take a second place to my wife.”⁶⁰ He permitted women to hold authority in the domestic sphere, so long as they remained monogamous and abstained from sex. Other prominent thinkers echoed Gandhi’s ideology. Namely, writer Bhudev Mukhopadhyay suggested in the late nineteenth century that women should behave as “goddess[es]” to their husbands.⁶¹ He encouraged them to act as moral guides for men by suppressing their sexuality;⁶² in doing so, they would earn men’s respect.

With societal norms and notable men like Gandhi and Mukhopadhyay emphasizing that only chaste women were worthy of authority, feminists sought to portray themselves as pure in order to gain respect for themselves within patriarchal society. *Stri-Dharma*, a feminist journal published by the Women’s Indian Association from 1918-1936, printed an article in 1933 called “Ideals of Indian Womanhood.” This article argued that women should aspire to be like archetypal mothers; namely, they should “treat marriage as “[inviolable]” and esteem the “sacredness of religious celibacy.”⁶³ These descriptions echoed those idealized by men like Gandhi and Mukhopadhyay, who emphasized that moral women should restrict their sexuality and devote themselves to monogamous marriages. The article also noted that a perfect woman should “[look] on all men as her children.”⁶⁴ It then laid out expectations for how children—namely, male children—should treat their mothers, asserting that there were “no circumstances sufficient...to justify criticism of a mother by her child.”⁶⁵ It therefore established that by asserting her purity, a woman would earn the unconditional respect of men in the domestic sphere, just as a model mother commands the unqualified devotion of her children. By publishing this article, the WIA promoted the belief that women should depict themselves as chaste in order to obtain men’s high regard. The fact that one of India’s leading feminist organizations upheld this philosophy suggests that it was prominent within the movement.

In seeking recognition for their purity within the domestic sphere, feminists also attempted to legitimize the other women’s rights causes that they championed in public settings. Historian Mrinalini Sinha argued that “the construct of the modern Indian woman...created the climate...for women’s reforms and for women’s entry...to the male-dominated public sphere.”⁶⁶ By adhering to an idealized version of Indian womanhood, feminists sought to render their public activism—which covered controversial issues like widow abuse and child marriage—more palatable to patriarchal society.

⁶⁰ Gandhi, *Collected Works of Mahatma Gandhi* (1919), 15: 325.

⁶¹ Bhudev Mukhopadhyay, “Lajjasilata,” in *Bhudev-racanasamhar*, ed. Pramathanath Bisi (Mitra and Ghosh, 1969), 447, quoted in Partha Chatterjee, “The Nationalist Resolution of the Women’s Question,” in *Recasting Women: Essays in Indian Colonial History*, ed. Sangari Kumkum and Vaid Sudesh (Rutgers University Press, 1999), 243.

⁶² Partha Chatterjee, “The Nationalist Resolution of the Women’s Question,” in *Recasting Women: Essays in Indian Colonial History*, ed. Sangari Kumkum and Vaid Sudesh (Rutgers University Press, 1999), 243-5.

⁶³ Swami Jagadiswarananda, “Ideals of Indian Womanhood,” *Stri-Dharma* 11 (August 1933): 485.

⁶⁴ Swami Jagadiswarananda, “Ideals of Indian Womanhood,” 484.

⁶⁵ Swami Jagadiswarananda, “Ideals of Indian Womanhood,” 482.

⁶⁶ Mrinalini Sinha, “Gender in the Critiques of Colonialism and Nationalism: Locating the ‘Indian Woman,’” in *Feminism and History*, ed. Joan Scott (Oxford University Press, 1996), 483.

This mindset led feminists to turn against the devadasis; numerous feminists depicted devadasis as prostitutes who had declined from a celibate past. One such activist was Annie Besant, an Irish woman who helped found the Women's Indian Association in 1917.⁶⁷ Besant portrayed contemporary devadasis as deviating from the chastity of their predecessors. In 1919, she alleged that historically, devadasis had been "pure virgin devotees" who "dressed in the simplest sanyasi garbs and [sang] pious hymns."⁶⁸ Feminist leader Muthulakshmi Reddy corroborated these oversimplifications in a 1927 essay.⁶⁹ She argued that in ancient times, devadasis had been "pure virgins spending their time in religious study, meditation and devotional service"; however, contemporary devadasis "promote[d an] evil and immoral trade."⁷⁰ Reddy's accusations further reflected the widespread generalization that devadasis, though once completely pure, had fallen into dishonorable prostitution.

Feminists condemned devadasis' supposed sexual impurity to contrast themselves with the devadasis and highlight their own chastity. Historian Dवेश Soneji noted that "[d]evadasi reform...worked to consolidate a new nationalist modernity for middle-class women by setting up the devadasi as the marked other."⁷¹ Likewise, gender studies scholar Rajalakshmi Nadadur Kannan argued that devadasis were "used to demonstrate what impropriety and deviance looked like...their presence was necessary in order to define the 'new woman.'"⁷² The devadasis served as a model of extreme impurity; women who sought to portray themselves as "proper" constructed personas of morality by contrasting themselves with the devadasis. In order to establish this contrast, feminists advocated against the devadasi system. Kannan stated that "[h]aving juxtaposed devadasis and the new middle-class women as polar opposites, the anti-devadasi movement aimed at purging devadasi of her impropriety."⁷³ As "examples of deviants," devadasis functioned as social rejects "to be looked at and criticized but not to be included in mainstream society."⁷⁴ Through their efforts to ban the devadasi tradition and convert devadasis to monogamous marriages, feminists sought to exaggerate the differences between themselves and the devadasis. They stressed that their own beliefs did not align with the ostensibly corrupt ones of devadasis by denouncing the devadasis' immorality. In portraying devadasis as impure, they legitimized themselves as the chaste reformers that much of male society deemed worthy of recognition.

Though devadasis originally possessed more freedom than most Indian women, the feminists of the late nineteenth and early twentieth centuries stripped away their rights in order to obtain their own version of women's liberation: respect from male-dominated society for themselves and their activism. Reacting to late-nineteenth-century Indian society's equating a

⁶⁷ Hubel, "The High Cost of Dancing," 127.

⁶⁸ Muthulakshmi Reddi, *Why Should the Devadasi Institution in the Hindu Temples Be Abolished?* (Central Co-operative Printing Works, 1927), 5.

⁶⁹ Soneji, *Unfinished Gestures*, 122.

⁷⁰ Reddi, *Why Should the Devadasi Institution in the Hindu Temples Be Abolished?* 5.

⁷¹ Soneji, *Unfinished Gestures*, 20.

⁷² Kannan, "Colonial Material," 541.

⁷³ Kannan, "Colonial Material," 542.

⁷⁴ Kannan, "Colonial Material," 541.

woman's respectability with her sexual purity, the feminists attempted to frame themselves as chaste by depicting devadasis as immoral prostitutes. In doing so, they sought to legitimize their other women's rights reforms in the eyes of men. Though the devadasis and feminists could have united in order to obtain the devadasis' rights for all Indian women, the desire to appeal to patriarchal standards ultimately drove them apart. Today, expression of sexuality still negatively correlates with women's social standing in India. In fact, as noted by Subhadra Mitra Channa, women in the public eye, such as government officials, are usually addressed with familial titles like "mother." Such titles, which were also used by publications like the WIA's *Stri Dharma* to refer to chaste women, emphasize that many Indians still expect their female leaders to be paragons of purity.⁷⁵

Meanwhile, most devadasi women live on the outskirts of society, with the Madras Devadasis Act and more recent anti-devadasi legislation having rendered them unable to practice their ancestral traditions freely.⁷⁶ While some rural villagers still covertly dedicate their daughters to deities, the custom is largely viewed as a "disease."⁷⁷ As wealthy patrons have sought to distance themselves from the disgraced devadasis,⁷⁸ the devadasi system has increasingly become a means for poor parents to rid themselves of their daughters.⁷⁹ In doing so, parents frequently condemn their daughters to lives of violent exploitation. Speaking anonymously in 2008, a devadasi in Karnataka explained why her parents had dedicated her: "My father was desperate for money."⁸⁰ Ironically, in advocating against abuse of women, the devadasi reform movement contributed to the stigmas that now prevent devadasis from securing the social and financial stability they previously enjoyed.⁸¹ Though the Indian feminist movement earned numerous women more recognition, its concern with male expectations also widened the social gap between different groups of women. Many Indian female leaders today therefore stand on the shoulders of groups like the devadasis, which numerous feminists oppressed in order to increase their own power and legitimacy.

⁷⁵ Channa, *Gender in South Asia*, 96.

⁷⁶ Soneji, *Unfinished Gestures*, 21; Government of India Ministry of Home Affairs, *Abolition of Devadasi System*, Kumar Alok. 24013/16/2015-SC/ST-W, New Delhi: Ministry of Home Affairs, 2015.

⁷⁷ Kirsti Evans, "Contemporary Devadasis: Empowered Auspicious Women or Exploited Prostitutes?", *Bulletin of the John Rylands Library* 80, no. 3 (1998): 24.

⁷⁸ Nair, "The Devadasi, Dharma and the State," 3164; Nash Colundalur, "'Devadasis are a cursed community,'" *The Guardian*, January 21, 2011.

⁷⁹ Colundalur, "'Devadasis are a cursed community.'"

⁸⁰ William Dalrymple, "Serving the Goddess," *The New Yorker*, August 4, 2008.

⁸¹ Colundalur, "'Devadasis are a cursed community.'"

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"Trust the Country, Trust the Party": How Chinese Feminist Discourse Reproduces the Boundaries It Contests By Ada Hu

Abstract

This paper examines how Chinese feminist discourse on social media navigates and reproduces nationalist boundaries, using the 2022 Xuzhou Chained Woman Incident as a case study. Drawing on Antonio Gramsci's theory of cultural hegemony and Michel Foucault's analysis of disciplinary power, it argues that feminist expression in China is simultaneously constrained and enabled by the Party-state's ideological frameworks. Through Critical Discourse Analysis of Weibo posts, it demonstrates that even when feminist actors explicitly contest state authority by invoking Party ideology, appropriating revolutionary language, or appealing to socialist principles, their discourse often reproduces the very nationalist boundaries it seeks to challenge. This paradox is traced to the historical subordination of Chinese feminism to nation-building projects, which has produced an internalised horizon of political thought, in which feminist justice remains unimaginable outside of state institutions. The findings reveal that hegemony operates most powerfully through the shaping of 'common sense,' exposing the structural limits of gender critique in authoritarian contexts.

Key Words

Chinese feminism, nationalism, hegemony, Foucault, Weibo, digital activism, authoritarian governance, Critical Discourse Analysis, Xuzhou Incident, counter-conduct

Introduction

On January 27, 2022, a video circulated on Chinese social media showing a woman chained by her neck to the wall of a dilapidated shed in Feng County, Xuzhou. The woman, later identified as Xiao Huamei, had been trafficked, imprisoned, and subjected to inhumane treatment for over two decades, forced to bear eight children to her captor. As public outrage erupted across Weibo, feminist activists like Wu Yi traveled to Xuzhou to investigate, demanding accountability and justice. Yet, when local police detained Wu Yi for her activism, her response revealed a striking paradox: "不怕的, 要相信国家相信党" (Don't be afraid, we must trust the country and trust the Party).

In this paper, I answer the research question: how do Chinese feminist discourses on social media navigate and reproduce nationalist boundaries, and what does this reveal about the limits of gender critique in authoritarian contexts?

This question is important for understanding the structural constraints facing feminist movements in authoritarian contexts. While scholarship has documented the rise of digital feminist activism in China and the state's systematic repression of independent feminist voices (Fincher 2018), less attention has been given to how feminist actors themselves negotiate the boundaries imposed by nationalist discourse, and how, even in moments of explicit contestation, they often inadvertently reproduce those very boundaries – as demonstrated by feminist activist

Wu Yi's invocation of Party ideology to legitimise her own investigation of state negligence. Within feminist scholarship, less attention has also been given to the historical roots of Chinese feminism and why it is historically shaped by state-aligned interests. Understanding this dynamic illuminates both the challenges facing Chinese feminism and broader questions about the possibilities and limits of gender critique under authoritarian governance.

This paper argues that Chinese feminist discourses on social media are simultaneously constrained and enabled by nationalist boundaries. Feminist expression is permitted when it reinforces state narratives of progress and stability, but censored, co-opted, or delegitimised when it challenges patriarchal-authoritarian authority. Even when feminists emphasise the Party's own ideological commitments to gender equality, or use revolutionary language against the state, they often find themselves reinforcing the very frameworks they set out to challenge.

Nationalist boundaries in this paper refer to the discursive and ideological frameworks through which the Party-state delimits legitimate political expression by tying it to national authenticity, sovereignty, and developmental goals. These boundaries operate by constructing what is 'Chinese' versus 'foreign' (境外势力) (Fincher, 2018), what serves national interests versus what threatens social stability, and what constitutes authentic feminism rooted in China's revolutionary history versus 'inauthentic' feminism imported from the West (Karl, Liu, and Ko 2013). Often internalised by Chinese feminist actors, these boundaries produce instinctive assumptions about which kinds of advocacy are 'acceptable', and which aren't.

As this paper will demonstrate, these boundaries operate most powerfully not when they are externally enforced but when they become the 'common sense' (Gramsci 1971) through which subjects understand their own interests and articulate their demands.

This paper grounds its analysis in the 2022 Xuzhou Chained Woman Incident, in which a woman later identified as Xiao Huamei was found chained by her neck in a dilapidated shed in Jiangsu Province, having been trafficked, imprisoned, and forced to bear eight children over two decades. The incident provoked an unprecedented eruption of feminist discourse on Weibo, while simultaneously triggering aggressive state efforts to contain and co-opt public outrage, making it a uniquely revealing site for examining how nationalist boundaries are both contested and reproduced in moments of hegemonic crisis.

This paper proceeds in three sections. First, the literature review situates Chinese feminism within its historical and theoretical context, tracing the genealogy of Chinese feminism's nation-building origins from early twentieth-century reformism, through Maoist state feminism, to contemporary digital activism, while drawing on Gramsci's theory of cultural hegemony and Foucault's analysis of disciplinary power to theorise how authoritarian states maintain control over feminist discourse. Second, the methodology section explains the data collection and analysis process, focusing on Weibo posts from the Xuzhou incident and employing Critical Discourse Analysis to examine how power is enacted and contested through language. Finally, the data analysis examines how the state establishes hegemony through official narratives, how feminists deploy tactical counter-conducts under the conditions of digital

surveillance, and how feminist resistance paradoxically reproduces nationalist boundaries even in moments of explicit contestation.

Literature Review and Theory

Feminist discourse in China presents a unique paradox. While feminist activism is highly visible on digital platforms such as Weibo, where campaigns like #MeToo have sparked substantial public debate, the same discourses are constantly subject to systematic repression: censored, denounced as foreign infiltration, or absorbed into nationalist frameworks that subordinate women's emancipation to the advancement of state interests (Fincher 2018).

To examine this tension, this paper draws on a theoretical and historical framework. Antonio Gramsci's concept of cultural hegemony explains how the Party-state secures legitimacy by shaping the cultural consensus (Gramsci 1971), while Michel Foucault's *Discipline and Punish* explains the way it sustains control over discourse through surveillance and the production of self-censoring subjects (Foucault 1977). Historical scholarship on Chinese feminism highlights both reformist and radical genealogies, including the nation-building visions of Jin Tianhe and Liang Qichao alongside the anti-nationalist critiques of He-Yin Zhen (Karl, Liu, and Ko 2013). Contemporary studies, such as Leta Hong Fincher's *Betraying Big Brother* and Weibo feminism, reveal how digital activism both confronts and reproduces authoritarian boundaries (Fincher 2018). Drawing on postcolonial feminist perspectives (Ling 2002), this paper emphasises how the colonial origins of Chinese feminist discourse continue to shape and constrain its expression today.

State Authority, Cultural Hegemony, and Counterpublics

Public discourse in the People's Republic of China is largely monopolised by the Party-state. The Cyberspace Administration of China (CAC), established in 2014, functions as the primary regulatory body overseeing online content, with the authority to suspend platforms, remove content, and prosecute users for speech deemed threatening to 'social stability' or 'national sovereignty' (Roberts 2018). On Weibo specifically, real-name registration requirements – mandatory since 2012 – mean that users cannot post anonymously, creating an environment of accountability that disciplines speech before censorship even becomes necessary. Keyword filtering operates in real time, with politically sensitive terms automatically blocked or shadowbanned, while human moderators employed by platforms work alongside algorithmic systems to remove flagged content within hours of posting (King, Pan, and Roberts 2013). The Party's doctrine of *yulun daoxiang* (guidance of public opinion) further shapes this environment. State media actively floods platforms with official narratives, crowding out alternative voices and manufacturing the appearance of consensus (King, Pan, and Roberts 2013). This control is reinforced ideologically through Mao-era slogans such as "women hold up half the sky," through which the CCP projects itself as the historic defender of women's liberation while simultaneously delimiting the space for independent feminist critique.

Here, Gramsci's theory of cultural hegemony becomes analytically useful. Gramsci argues that the ruling power secures legitimacy by shaping cultural norms and 'common sense,' making its authority appear natural and inevitable (Gramsci 1971). In the Chinese context, the Party-state cultivates subjects who internalise its narratives, who genuinely believe that 'social stability' serves their interests, and who experience independent feminist critique as destabilising rather than liberating. The CAC's content removals and Weibo's keyword filters are thus only indicative of a deeper disciplinary structure.

Within a hegemonic structure, subordinated groups, such as Chinese feminist actors, face structural limits in articulating alternative visions of society (Gramsci 1971). However, Gramsci theorised that these groups can generate counter-hegemonic practices: forms of discourse and action that contest dominant norms and negotiate the conditions of legitimacy (Gramsci 1971). Social media, in this sense, becomes a site of contested hegemony. Feminist actors on Weibo navigate a digital environment where nationalist narratives structure language and reception – yet their interventions can still produce new forms of awareness, solidarity, and resistance.

Feminist Discourse in China: Discipline, Surveillance, and Subjectivity

In *Discipline and Punish*, Michel Foucault charts the emergence of disciplinary regimes that function covertly through surveillance, normalisation, and the creation of "docile bodies" (Foucault 1977). His framework is especially valuable for examining authoritarian contexts such as the PRC, where control is established through censorship, propaganda, and the orchestration of consensus, cultivating subjects who appear to consent to state legitimacy.

In China, this process is embodied in *yulun daoxiang* (guidance of public opinion), through which the Party claims to reflect popular will while in fact manufacturing it (King, Pan, and Roberts 2013). Foucault describes this as "regimes of truth," historically established systems that define what is considered commonsensical (Foucault 1977). Feminist voices that challenge the official narratives are not simply censored but also actively delegitimised by labels such as 'rumours', 'disorder,' or 'Western infiltration.' The result is a disciplinary structure in which feminist activism is permitted only when it serves state or nationalist interests, consequently producing what Foucault calls docile bodies: self-regulating subjects who either unconsciously internalise state discourse or consciously align with it as a survival tactic.

However, against this disciplinary pressure, feminist resistance persists. Foucault states that power produces its own vulnerabilities, as the very necessity of continually manufacturing public opinion (*yulun*) reveals its inherent fragility. Feminists exploit this instability through what Foucault terms counter-conducts – acts such as testimonies, digital petitions, and coded hashtags which challenge the state's monopoly on public consensus.

As common sense and nationalist myth must be continually reproduced, they are perpetually open to contestation. By exposing the constructed nature of 'public opinion,' feminist discourses reveal authoritarian legitimacy as contingent, fragile, and dependent on constant policing. The struggle over feminism in China is therefore not only a struggle for gender equality, but also a struggle over who gets to define 'truth' itself.

Chinese Feminism and the Question of Its Origins

For Chinese intellectuals and activists throughout the first half of the twentieth century, 'modernity' was a project that moved China from semi-colonisation to an industrialised nation arriving on the global stage (Spence 1999).

The earliest recognisable articulations of Chinese feminism emerged at the turn of the twentieth century, during the tumultuous final decades of the Qing dynasty and the first stirrings of the Republican era (Spence 1999). This was a period marked by profound political crises: foreign imperialism, unequal treaties, the Boxer Rebellion (1899–1901), and the looming collapse of dynastic rule. As a result, Chinese intellectuals turned urgently to Western political theory and models of modernisation, hoping to rescue a 'weak nation' from decline (Spence 1999). It was within this rise of reformist and revolutionary thought that feminism was articulated as both a necessity and a problem.

Jin Tianhe's *The Women's Bell* (1903) and Liang Qichao's essays on women's education are emblematic of this moment (Wang 1999). Both men drew explicitly from Western liberal thinkers such as John Stuart Mill and Herbert Spencer, admiring aspects of Western society and associating women's education with national strength. Women's emancipation, in their framing, was less about autonomy than about instrumental utility: educated women could become disciplined wives, nurturing mothers, and the producers of healthier children. Such reasoning reveals how early Chinese feminism was subordinated to nationalist modernisation, in which liberation was imagined only within the boundaries of nation-building.

Against this reformist model, He-Yin Zhen articulated a radically different vision (Karl, Liu, and Ko 2013). Writing in anarchist and revolutionary journals in Japan between 1907–1911, she theorised women's oppression as inseparable from political economy. Her categories of *nannü* (gender hierarchy) and *shengji* (livelihood) exposed the systemic subordination of women through both reproductive labour in the household and exploitative wage labour in the workplace. Unlike her male contemporaries, He-Yin rejected Western liberal frameworks and nationalist appropriation, instead developing a critique rooted in indigenous thought and an uncompromising call for social revolution (Zarrow 2012). Although marginalised for decades, her belated recovery demonstrates that Chinese feminism has never been reducible to foreign importation or nationalist utility, even if such narratives have dominated its reception.

Drawing on postcolonial theory, Lily Ling argues that Western imperial powers historically justified dominance over Asia by feminising the East, framing it as weak, irrational, and in need of civilisational guidance (Ling 2002). China's 'Century of Humiliation' (1839–1949), defined by unequal treaties, colonial concessions, and military defeat, exemplifies this symbolic emasculation. In response, the Party-state constructed a national identity defined by strength, order, and uncompromising sovereignty – what Ling terms the "hypermasculine Chinese Self" – projecting an image of a nation that cannot be pitied, penetrated, or fooled by Western fantasy. Within this framework, any sign of internal disorder (trafficking, gendered violence, feminist dissent) becomes a threat to the national image, facing immediate suppression.

It is precisely this enduring tension that my research tackles. While scholarship has traced both the nationalist framing of Chinese feminism and the rise of digital feminist activism, less attention has been given to how feminist voices themselves negotiate the limits of legitimacy imposed by the state. My contribution is therefore to show that the state's regulation of feminist discourse is structured by historical anxieties over its authenticity and origins, and in doing so, to foreground the strategies and voices of feminists who contest these boundaries in the digital sphere.

Methodology

Data collection took place on Weibo (Sina Weibo), China's most-used microblogging platform and primary site for real-time news aggregation and mass social commentary. With approximately 600 million active monthly users (Statista, 2025), the Chinese government and Party organisations maintain control over *yulun* (public opinion) through official accounts that disseminate Party ideology (Schlaeger & Jiang, 2014). This control is reinforced by the platform's use of algorithmic filtering, keyword blocking, and account suspensions, aligning it closely with Party-state priorities (King, Pan, & Roberts, 2017). Such mechanisms necessitate the use of coded language and tactical adaptations (counter-conducts) by feminist actors, making Weibo uniquely suited to analysing the digital structure of disciplinary power in authoritarian contexts.

A purposive sampling strategy was adopted. Incidents were defined as high-profile events generating significant feminist debate and nationalist counter-discourse on Weibo and X between 2020 and 2025. Cases were identified through triangulation of sources, including Weibo trending archives, viral hashtags, Chinese media reports, and citizen-archived screenshots. Posts from within a two-week window of each incident were sampled based on relevance, visibility (measured by engagement metrics), and discursive diversity. Only posts containing substantive discursive content were included for meaningful analysis, such as argumentation, satire, or testimony. Where posts had been deleted or censored, verified citizen-archived screenshots and reposts were used. Data integrity was further ensured by cross-referencing posts across multiple archival sources, minimising selective bias.

Data analysis was conducted through thematic coding informed by Critical Discourse Analysis (CDA) – Norman Fairclough's (1992, 1995) framework of discourse as both a form of social practice and a medium through which power is enacted and contested. Coding focused on identifying connections between discursive patterns and underlying theoretical constructs. Three central analytical themes were traced:

1. **Regime of truth and hegemony** – how certain ideas become normalised and enforced within dominant discourse;
2. **Anxiety of origins** – how tropes of foreign infiltration and inauthenticity are mobilised to delegitimise feminist voices;

3. **Docile bodies and counter-conduct** – how feminists employ coded speech, satire, and creative tactics to negotiate, resist, or reproduce disciplinary regimes

Throughout the analysis, attention was paid to both the content and form of posts, including coded language, hashtags, and user interactions, as indicators of broader negotiations between power and resistance. Ethical considerations were strictly observed, including the anonymisation of all user information and the exclusive usage of publicly available content.

Limitations

This paper's findings must be read with several limitations in mind. The focus on Weibo excludes discourse on other prominent platforms (such as WeChat, Douyin, and RED), where feminist discussion may take markedly different forms. Likewise, the Xuzhou Incident, while significant, is a single case study, and its particular intensity, coinciding with the Beijing Winter Olympics and producing an unprecedented erosion of state credibility, may have generated forms of feminist discourse more confrontational than what would typically emerge. The reliance on publicly archived posts also means that the most suppressed voices remain structurally absent from the analysis, potentially introducing a selection bias. Finally, this paper analyses feminist discourse at a textual level, without access to the intentions or material circumstances of the users behind the posts.

Case Study

The case study chosen is the Xuzhou Chained Woman Incident. The incident provided a rare moment where state apparatuses and censorship publicly failed, exposing the Party-state's systemic complicity in gendered violence and the fragility of its legitimacy. As the crisis escalated, the state intensified its efforts to reinstate hegemony by repeatedly blocking related hashtags from Weibo's trending lists, suspending the accounts of outspoken activists and citizen journalists, and intensifying platform-wide content filtering during the politically sensitive period of the Beijing Winter Olympics. This allows for analysis of how feminist actors navigate and reproduce nationalist boundaries, at times when state hegemony is challenged.

The incident came to public attention in January 2022, when a video circulated on Douyin showing a middle-aged woman chained by her neck to the wall of a dilapidated hut in Feng County, Xuzhou, Jiangsu Province (AP News 2022). The woman, later identified by authorities as Xiao Huamei ("Little Plum Blossom"), was the mother of eight children. Her captor, Dong Zhimin, initially gained online attention from a blogger who sought to praise his perseverance in raising a large family despite poverty. Investigations later revealed that Dong had kept Xiao, who had been diagnosed with schizophrenia, chained in inhumane conditions without heat, electricity, or water since the late 1990s (The China Project 2023). Subsequent reports showed that Xiao was a victim of multiple instances of human trafficking, abducted and sold several times before being ultimately 'purchased' by the Dong family (CBS News 2023; Human Rights Watch 2022).

Local Feng County authorities initially dismissed allegations of trafficking, claiming Xiao was legally married and chained due to mental instability. Facing overwhelming public outrage, higher-level Xuzhou and Jiangsu provincial authorities then issued four contradictory statements over the following weeks in an attempt to control the narrative (AP News 2022). These inconsistencies fuelled widespread suspicion toward state institutions, with many believing the government was attempting to conceal their systemic complicity in ‘bride’ trafficking. The erosion of public trust prompted unprecedented grassroots mobilisation, as activists and former journalists travelled to the village in attempts to uncover the truth. Despite heavy censorship during the politically sensitive period of the Beijing Winter Olympics, persistent online pressure ultimately forced authorities to acknowledge that Xiao Huamei was indeed a victim of trafficking (AP News 2022).

In April 2023, Dong Zhimin was sentenced to nine years in prison for abuse and unlawful detention. Five accomplices received sentences ranging from eight to thirteen years for abduction and human trafficking (CBS News 2023; The China Project 2023). The verdict remains widely criticised by Chinese netizens as too lenient, particularly for omitting charges of purchasing a trafficked person (due to the statute of limitations) and marital rape (The China Project 2023). The incident became a defining moment for Chinese feminism, exposing the systemic nature of gendered violence, the lack of legal protection for rural women, and the state’s continued control of public discourse to preserve the appearance of social harmony.

Analysis

Official Narratives and the Establishment of Hegemony

The state’s initial response to the Xuzhou Incident is articulated across four successive statements issued by Feng County and Xuzhou authorities. These statements reflect how the Party-state maintains control over public opinion under crisis, with each text representing an iterative and increasingly compromised effort to achieve discursive hegemony. Through lexical and discursive choices such as reframing and selective omission, the state attempts to reconstitute a coherent ‘truth’ that restores legitimacy and moral authority, reflecting both Gramsci’s concept of crisis of legitimacy (Gramsci, 1977) and Foucault’s notion of regime of truth (Foucault, 1971).

Statement 1 (January 28, 2022)



Figure 1: Statement 1 released by Feng Xian authorities,
<https://mp.weixin.qq.com/s/FxePNeEAXLXuAY39-2VQXg>.

Released by Feng County authorities on January 28, the first official statement sought to immediately contain public outcry by establishing a proactive and assured narrative. Phrases such as “迅速成立联合调查组进行全面调查” (immediately established a joint investigation team to conduct a comprehensive investigation) and “经初步调查核实” (upon preliminary verification) assert bureaucratic authority through an impersonal and procedural tone, positioning the Party-state as a rational arbiter of truth. It directly negated public suspicion by claiming that Deng and Xiao were legally married and that no signs of trafficking were found (“不存在拐卖行为”), while recentering the narrative around the woman’s mental illness, describing her as “经常无故殴打孩子和老人” (often beats children and the elderly for no reason).

This lexical shift pathologises her behaviour and reframes her as the source of disorder – the “Unwanted Girl” (Ling, 2002) who fails to fulfil her sanctioned role within the traditional family and state – simultaneously turning a systemic issue of gendered violence into an isolated and depoliticised medical case. The invocation of moral paternalism at the end – “对其家庭开展进一步救助” (provide further assistance to their family) and “确保过上温暖的春节” (to ensure they experience a warm Spring Festival) – restores the state’s image as a benevolent caretaker and distances it from the complicit structure that enabled the incident in the first place. This immediate, assertive stance reflects the State’s need to project a hypermasculine, orderly Chinese Self (Ling, 2002): the “cruel Asian man” who resists being fooled by Western fantasies (Ling, 2002), suppressing any sign of internal chaos that could confirm the national weakness that the hypermasculine state identity was built to resist.

Statement 4



Figure 2: Statement 4 released by the Provincial Investigation Team,
<https://news.sina.cn/kx/2022-02-10/detail-ikyakumy5270393.d.html>.

Two days later, the Provincial Investigation Team released its fourth and final statement – an announcement that signalled the state’s success in resolving a mounting crisis of legitimacy

(Gramsci 1977) and in re-establishing a definitive regime of truth (Foucault 1971). This high-level intervention became necessary precisely because the state's earlier attempts at controlling public opinion had collapsed. The first three official statements contradicted one another, shifting from "legal marriage" (S1) to the claim that the woman was a "wandering beggar taken in" (S2), and then to a "lost person with a specific origin" (S3). These inconsistencies exposed the fragility of state authority and the limits of its discursive control, compelling the Party-state to reassert its legitimacy. The statement achieves this by asserting factual certainty, using DNA verification ("It was determined that Yang Mouxia was Xiao Huamei") to refute public suspicion with state-sanctioned, scientific truth. The subsequent naming of specific perpetrators (Dong Moumin, Sang Mouniu, Shi Mouzhong) limits criminal responsibility to particular individuals and negligent grassroots officials, thereby shielding the central Party apparatus from systemic critique. The final paragraph then pivots entirely to demonstrating the state's paternalistic care. By enumerating welfare measures such as subsistence allowance protection ("低保保障"), student aid ("学生资助"), and support from the All-China Women's Federation, public emotion is shifted from anger over systemic injustice to gratitude for social stability. In doing so, it re-establishes the 'common sense' that the Party is the indispensable guardian of social stability, regaining public consent and reinstating its cultural hegemony (Gramsci 1977).

Docile Bodies and Counter-Conducts

The pervasiveness of digital surveillance is evidenced by the near-impossibility of locating uncensored discourse on the incident, even within encrypted hashtags such as #奉县八孩母亲 (#Feng County Mother of Eight Children) and #铁链女 (#Chained Woman). The surviving counter-conducts are thus characterised by tactical evasion, where users employ satire, irony, and dark humour to adhere to the form of compliant language.

"9年, 她被关了几年?" (Nine years, how many years was she locked up for?)

"小花梅被囚禁20年, 董只判了9年 *kissy face emoji*...何况小花梅这20年过得日子还不如下大牢 *kissy face emoji*." (Xiao Huamei was imprisoned for 20 years, while Dong was only sentenced to 9 years... Moreover, Xiao Huamei's life during those 20 years was worse than that of a prisoner.)

This is demonstrated by the top comments under a post by the Beijing Youth Daily. By almost playfully juxtaposing the victim's twenty-year captivity against the perpetrator's nine-year sentence, they maintain the syntactic neutrality needed to evade algorithmic censorship, while delegitimising the state's discursive authority and narrative of judicial justice. Yet, the reliance on irony and encryption also illustrates the structural limits of feminist resistance under authoritarian digital governance. Such counter-conducts remain fragmented, tactical, and restrained. They cannot coalesce into a unified political force without triggering immediate repression.

The Reproduction of Nationalist Boundaries Through Feminist Resistance

However, even when feminist actors directly contest nationalist boundaries, their discourse often inadvertently reproduces them. The case of feminist activist Wu Yi (pseudonym) exemplifies this paradox. In early February 2022, dissatisfied with the authorities' negligence, Wu Yi travelled to the hospital where Xiao Huamei was transferred to ensure her safety and investigate the incident (Wu Yi, 2022). When a concerned commenter asked whether she feared detention by police, Wu Yi responded:

"不怕的, 要相信国家相信党" (Don't be afraid, we must trust the country and trust the Party).

The "trust" Wu Yi talks of here is not just trust in the fairness of the state, but in the commitments to gender equality and women's liberation, upon which the Party ideology was built. In doing so, she invokes the moral authority of the Party-state to position herself not as challenging the system but as defending its principles against local corruption, thereby legitimising her activism.

After gaining widespread attention on Weibo, commentary bloggers began accusing Wu Yi and her feminist counterparts of being influenced by "境外势力" (foreign forces), a commonly used trope of inauthenticity that has delegitimised Chinese feminism since its inception. In a discussion about this under Wu Yi's page, a commenter exemplifies the reproduction of nationalist boundaries, even within sympathetic feminist discourse, writing:

"首先。警惕境外势力是对的。我们要不信谣不传谣, 相信我们的国家和党我们实实在在感受到现在的生活是美好的...只要公开真相, 让触犯法律的人得到应有的惩罚...我们就满意了" (First of all, being vigilant against foreign forces is correct. We must not believe or spread rumors, trust our country and Party— we genuinely feel our lives are good now... As long as the truth is made public and lawbreakers receive due punishment... we'll be satisfied).

Unlike Wu Yi, the commenter is not performing nationalist loyalty tactically to evade censorship; they trust the Party, genuinely feels gratitude for China's development under reform-era policies, and genuinely believes that vigilance against foreign interference is necessary to protect Chinese sovereignty. It is difficult for her to articulate feminist demands outside the frame of Party legitimacy, because both her historical knowledge (the Party ended feudal practices such as foot-binding and child marriage) and lived experience (standards of living have improved) confirm the narrative that the Party is fundamentally committed to women's welfare, even if local implementation sometimes fails. This reveals that nationalist boundaries operate most powerfully not when they are externally imposed, but when they become the internalised horizon of political thought, where individuals cannot possibly conceive of feminist justice except as something that the state must deliver (Foucault, 1975)

In a different approach, multiple commenters utilised official Party ideology to legitimise their critiques.

"不愿闭嘴就是境外势力不爱国??? 改革必然痛苦, 希望所有人都有改革的勇气与毅力" (Not willing to shut up makes you foreign forces and unpatriotic??? Reform is inevitably painful; I hope everyone has the courage and perseverance for reform).

The diction used here, “reform” (改革) and “courage and perseverance for reform” (改革的勇气与毅力), echo slogans and motifs vocalised by party leaders during Mao’s transformation of feudal society, Deng Xiaoping’s “Reform and Opening Up” (改革开放) that initiated China’s economic modernisation, and Xi Jinping’s rhetoric about persevering through difficulties and deepening reforms (King, Pan, and Roberts 2013). By inverting this language, the commenter asserts revolutionary legitimacy, casting feminist critics as the true ‘revolutionaries’ and their silencers as illegitimate ‘counterrevolutionaries’ clinging onto feudal remnants. Yet the very sophistication of this strategy reveals its limitations. The commentator can only legitimise feminist critique by appropriating the Party’s monopoly on revolutionary discourse, thereby reinforcing that very monopoly. Likewise, the premise that ‘foreign forces’ constitute a legitimate threat to Chinese sovereignty is not rejected. The boundary remains intact; political legitimacy still must align with nationalist ideals, and authenticity is still defined as opposition to foreignness. Such examples epitomise Gramsci’s hegemony: when the ruling ideology becomes ‘common sense,’ even oppositional discourse must work within its terms to be comprehensible (Gramsci 1971).

On February 11, Wu Yi was detained by Xuzhou police, along with another activist, Quan Mei. Following her release, she posted a lengthy reflection that marks a dramatic rupture in her earlier expressions of trust in state authority:

"沛县抓我的这段经历是我最严重的精神创伤...我一直信任的, 国家一直教导我的, 全部变成了谎言, 成了刺进我身体的剑...我一直不断问我自己, 这就是警察吗? 这真的是公安局吗? " (My arrest experience in Peixian was my most severe psychological trauma... Everything I trusted, everything the nation taught me, all became lies—swords piercing my body... I kept asking myself: Are these really police? Is this really a public security bureau? Wu Yi's anguish stems not merely from physical detention but from the shattering of her faith in the very institutions she had invoked to legitimize her activism. Her repeated questioning—"Are these really police?"—reveals her desperate need to separate the "authentic" Party-state (which should protect women and uphold socialist values) from the "corrupt" local apparatus (which detained her unjustly). Yet no one would affirm this distinction for her.

"能疗愈我的, 只有伤害了我的人受到最重的惩罚! 只有他们受到司法的雷霆之怒, 才能平息我的怒火与创伤" (Only the heaviest punishment for those who harmed me can heal me! Only judicial thunder fury can calm my rage and trauma).

When the state uses violence against those who have operated within its own discursive rules, invoking Party ideology to legitimise their activism, the fragility of hegemonic consent is revealed (Gramsci, 1971). The regime of truth that once produced and sustained common sense for Wu Yi, has now been unveiled as fabricated and contingent. Her repeated questioning – "Are these really police?" – reveals her desperate need to discursively separate the ‘authentic’ Party-state from the ‘corrupt’ local authorities. Even after experiencing state violence firsthand, Wu Yi cannot imagine a form of justice outside of state institutions, using the metaphor of a “judicial thunder fury” to invoke the image of a righteous, protective, punitive state – the very form of power that was wielded against her. The Party-state has thus successfully positioned

itself as the sole imaginable source of legitimacy and protection, and the nationalist boundary is once again reproduced.

Taken together, these responses reveal the limits and generative potential of feminist discourse in authoritarian contexts: from strategic nationalism (invoking Party ideology to legitimise critique), to subversive nationalism (turning Party rhetoric against the state). Yet even at the most transgressive end of this spectrum, the nationalist boundary continues to structure what can be imagined and articulated.

Conclusion

This paper set out to examine how Chinese feminist discourses on social media navigate and reproduce nationalist boundaries, using the 2022 Xuzhou Chained Woman Incident as a case study. Through Critical Discourse Analysis of Weibo posts and official state statements, it has demonstrated that the relationship between feminist activism and nationalist discourse in China is considerably more complex than simply one of repression and resistance. Feminist expression is permitted when it reinforces state narratives of stability, censored when it challenges patriarchal-authoritarian authority, and most critically, often reproduces the very nationalist frameworks it seeks to contest. Wu Yi's invocation of Party authority to legitimise her activism, the commenter's genuine belief in the necessity of vigilance against foreign forces, and Wu Yi's post-detention appeal to "judicial thunder fury" all reveal the same underlying dynamic: the Party-state has so thoroughly monopolised the terms of political legitimacy that feminist actors find themselves appealing to state authority for redress even when contesting it.

This finding suggests that in authoritarian contexts, the most powerful form of ideological control is the shaping of political imagination and the production of subjects who cannot conceive of justice except as something the state must deliver. Fundamentally, the historical roots of Chinese feminism, its subordination to nation-building from the first articulations of Jin Tianhe and Liang Qichao, reveal that this dynamic is a structural feature of Chinese feminist discourse that long predates social media.

This paper does not conclude that feminist resistance in China is futile. The Xuzhou Incident forced the state into four contradictory statements and produced forms of solidarity that outlasted the incident itself. The question is thus not whether Chinese feminism can resist, but whether it can imagine a vision of justice that does not depend on the state for its fulfilment. For now, that remains an open question.

Importantly, the most pressing gap this paper identifies is the absence of class and rural/urban dimensions from existing scholarship on Chinese digital feminism. The Xuzhou Incident is, at its core, a story about a rural woman whose vulnerability was produced by rural poverty and weak legal infrastructure. Yet the feminist discourse it generated was overwhelmingly urban, digitally literate, and middle-class. Future research should examine how rural women themselves narrate and respond to such incidents, and what this reveals about the class politics that urban-centred analysis cannot capture.

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From Rocket Science to Front Lines: Educated Unemployment and War Mobilization in Post-Sanctions Russia By Kangdi Zhao

Abstract

Western sanctions and Russia's wartime economic reorientation have placed heavy pressure on parts of the country's educated workforce since the 2022 full-scale invasion of Ukraine. Official unemployment numbers can make the economy look surprisingly resilient, but these figures hide a distorted labor market shaped by mobilization, emigration, sanctions, and the expansion of war-related employment.

This paper investigates how economic hardship and underemployment among educated Russians may contribute to military recruitment through dynamics resembling a "poverty draft." Drawing on anonymous interviews with Russian citizens, including recent graduates, managers, public-sector workers, and underemployed professionals, the paper argues that financial incentives and the collapse of credible civilian prospects often matter more than ideological commitment. The central system is the political economy through which weak civilian opportunity, state incentives, and war needs reinforce one another.

The findings suggest a vicious cycle: war and sanctions weaken parts of the civilian economy that might otherwise absorb educated labor, while military recruitment packages redirect some of that labor into the war effort. The paper concludes by considering policy interventions that could reduce the conversion of economic desperation into military manpower, including support for small business, stronger graduate-to-work pathways, and more careful sanctions design.

Introduction

Russia's full-scale invasion of Ukraine in 2022 generated not only a major interstate war but also a profound restructuring of Russia's domestic political economy. Western sanctions, corporate withdrawals, and the redirection of state resources toward war production have altered the country's labor market, investment patterns, and career structures. More than 1,000 firms have curtailed operations in Russia since the invasion, and the cumulative losses associated with foreign exits exceeded \$107 billion by March 2024, underscoring both the scale of economic rupture and the loss of foreign expertise from key sectors of the Russian economy (Yale School of Management; Reuters, "Foreign Firms' Losses").

At the same time, official labor-market statistics have created an appearance of stability that can be misleading if read in isolation. BOFIT reported that Russia's unemployment rate fell from 4.3% in January 2022 to 2.9% in January 2024 and 2.4% by June 2024, but interpreted this decline as evidence of an exceptionally tight labor market driven by mobilization, emigration, demographic pressure, and shortages of skilled workers rather than by the healthy expansion of civilian opportunity (Bank of Finland Institute for Emerging Economies, "Unemployment"). In other words, low unemployment in wartime Russia does not necessarily signal a flourishing labor market; it may also reflect labor scarcity and the militarization of economic demand (Gorodnichenko, Korhonen, and Ribakova).

These pressures have also been intensified by outward migration. Policy analyses and reporting describe the post-2022 outflow as the largest brain drain from Russia since the collapse of the Soviet Union, with hundreds of thousands leaving the country after the invasion and subsequent mobilization, including large numbers of young, educated, and highly mobile professionals. Reuters reported that the Russian government estimated in late 2022 that around 100,000 IT specialists were working abroad, while PONARS Eurasia characterized the exodus as a major loss of qualified labor and human capital (Kamalov et al.; Reuters, "IT Worker Flight").

For those who remained, however, civilian opportunities have often narrowed rather than expanded. Russian small and medium-sized enterprises have faced inflation, weaker consumer demand, severed supply chains, and higher borrowing costs under sanctions, while new wartime fiscal pressures have placed further strain on local businesses (Barinova et al.; Associated Press). As a result, professional and entrepreneurial pathways that might otherwise absorb educated youth have become more fragile and uncertain.

This study examines one troubling consequence of these dynamics: the movement of underemployed or economically insecure educated Russians into military service. We approach this issue sociologically, asking how structural economic forces can redirect human capital from classrooms, offices, and small businesses into the armed forces. In this context, the notion of a "poverty draft" is increasingly relevant: recent research argues that economic incentives and uneven regional poverty are central to understanding patterns of military recruitment in Russia, especially among more vulnerable populations (Yusupova).

Military pay has become a key part of this picture. Reuters reported in July 2024 that the minimum monthly payment for a private serving in the "special military operation" had been set at 204,000 rubles, pushing minimum first-year compensation above 3.25 million rubles, more than three times the average wage in Russia. By April 2026, Reuters further documented how universities were marketing drone-force roles to students with annual compensation packages ranging from roughly 5.5 to 7 million rubles, alongside tuition relief, accommodation, and other benefits (Reuters, "Moscow Hikes Pay"; Reuters, "Woos Students"). These figures help explain why military service can appear economically rational even to individuals trained for civilian careers.

The central argument of this paper is therefore not simply that Russia is recruiting more soldiers, but that wartime economic restructuring is helping produce those soldiers by weakening the civilian sectors that might otherwise retain educated labor. Our interviews suggest that many participants are motivated less by ideology than by constrained opportunity, low civilian pay, and diminishing prospects for professional advancement. This dynamic matters not only for military recruitment but also for long-term social development. As broader research on armed conflict shows, war damages human development through disruption, diversion, devaluation, and longer-term harms to livelihoods, migration patterns, and social well-being (Vesco et al.).

Against this backdrop, our research focuses on three interrelated questions:

How have recent economic conditions in Russia - especially post-sanctions - affected employment opportunities for educated young adults?

What motivates educated individuals to consider military service?

What are the broader social implications of this educated-to-military pipeline, and what alternatives could reduce it?

We hypothesize that economic drivers will far outrank ideology in the decision to enlist among our educated respondents. Those with higher education but underemployment or debt will be the most likely to view military contracts as a lifeline. Conversely, individuals who perceive stable civilian options - even if modest - are expected to avoid military service, citing risks and family well-being. Additionally, we anticipate that our findings will highlight a critical mismatch: even as Russia's war economy hungry for manpower offers quick income, it simultaneously undermines the very economic sectors that could employ educated youth in peace.

By combining qualitative interviews with supporting quantitative and policy data, this paper aims to illuminate an under-discussed feedback loop between sanctions, socioeconomic decline, and war mobilization. Ultimately, this is a study in the sociology of conflict - how large-scale political-economic forces translate into individual choices that collectively fuel war. In shedding light on this issue, we seek to inform both academic understanding and policy debates on curbing the cycle of war and suffering.

This paper also draws on analytical frameworks I developed while taking Harvard Extension School's GOV-T 1203, Capitalism and Democracy in Central and Eastern Europe. That course sharpened the comparative lens used here by foregrounding the relationship between communist legacies, market transition, institutional redesign, and democratic backsliding. Although contemporary Russia differs from the East-Central European cases often used to theorize postcommunist transformation, the course's conceptual vocabulary, especially historical legacies, regime diversity, and path contingency, helps explain why economic dislocation in Russia has not merely produced hardship, but has been politically channeled into a militarized labor market (Ekiert and Hanson; Kitschelt; Johnson).

Literature Review

1. Sanctions, war, and Russia's economic strains.

The Western sanctions regime imposed after Russia's 2022 full-scale invasion of Ukraine was intended to constrain Russia's access to foreign financing, advanced technology, and inputs relevant to military production. In important respects, those restrictions have mattered: analysts note that financial sanctions cut Russia off from much foreign financing, while export controls and technology bans restricted access to many Western goods and components. At the same time, however, Russia adapted more effectively than many early forecasts anticipated by redirecting trade, deepening links with non-Western partners, and sustaining growth through large-scale state spending, especially on the war economy (Bank of Finland Bulletin, "Russia's Economy"; Carnegie Endowment; Reuters, "Foreign Firms' Losses").

This apparent macroeconomic resilience should not be confused with balanced development. Reuters noted in February 2024 that Russia's rebound relied heavily on state-funded arms and ammunition production, while BOFIT and the Bank of Finland have stressed that record-low unemployment reflects an exceptionally tight and distorted labor market rather than an abundance of healthy civilian opportunity. Inflationary pressure has remained high, consumer purchasing power has been eroded, and fiscal priorities have increasingly favored defense over other public needs (Reuters, "GDP Boost"; Bank of Finland Institute for Emerging Economies, "Unemployment"; Bank of Finland Bulletin, "Russia Further Increases").

2. The plight of educated youth and "brain waste."

The war has intensified structural problems that predated 2022. BOFIT has argued that Russia's longer-term growth prospects were already weak before the full-scale invasion because of poor productivity growth, unfavorable demographics, and underinvestment, and that the war has worsened these constraints rather than resolving them. In this context, many educated young Russians face a labor market in which formal employment exists, but stable, well-paid, and professionally meaningful employment is much harder to secure (Bank of Finland Bulletin, "War Has Degraded"; Bruegel, "Why Russia's Economic Model").

Evidence from the public sector illustrates the problem clearly. Reporting from Tatarstan shows that teachers remain poorly paid relative to workload and qualifications: RFE/RL reported an average basic salary of about 53,000 rubles per month, with many actual starting salaries lower still. Such cases support the broader claim that higher education does not reliably translate into economic security in contemporary Russia. Our interviews point in the same direction, with respondents describing professional frustration, low wages, and the need to leave socially valuable work for more commercially viable sectors. In this sense, the issue is not simply "brain drain" abroad, but also "brain waste" at home, as skills are diverted away from the fields for which people were trained (RFE/RL, "No Time to Breathe"; Kamalov et al.).

At the same time, small-business and entrepreneurial alternatives have become harder to sustain. Research on Russian SMEs under sanctions points to falling consumer demand, higher borrowing costs, disrupted supply chains, and heightened bankruptcy risk, while more recent reporting shows that wartime tax changes have added further pressure to already fragile small firms. These conditions narrow legitimate civilian opportunity structures for educated youth and make underemployment, career downgrading, and precarious side work more common outcomes than upward mobility (Barinova et al.; Associated Press).

3. Military service as economic opportunity: toward a Russian "poverty draft."

This environment helps explain why military service can come to resemble what observers in other contexts have called a "poverty draft," in which economic insecurity makes enlistment materially attractive even without strong ideological commitment. Research on military participation has long linked service patterns to socioeconomic disadvantage, while recent work on Russia emphasizes how enlistment intersects with regional inequality, rurality,

and constrained mobility. In the Russian case, the term should be used carefully, but it captures an important structural logic: when civilian opportunities narrow, military service becomes easier to frame as an economically rational decision (Bareis and Mezuk; Yusupova).

Recent recruitment policies have made that logic unusually visible. Reuters reported in July 2024 that Moscow offered a 1.9 million-ruble upfront payment to new contract soldiers, lifting first-year compensation to about 5.2 million rubles, or more than five times the average annual wage. Around the same time, the minimum monthly payment for a private serving in Ukraine was reported at 204,000 rubles, pushing minimum first-year compensation above 3.25 million rubles. By April 2026, Reuters further reported that Russian universities and regional authorities were marketing drone-force roles to students with annual packages of roughly 5.5 to 7 million rubles, plus tuition relief, accommodation, and other benefits (Reuters, "Moscow Hikes Pay"; Reuters, "Putin Doubles"; Reuters, "Woos Students").

These pay differentials matter sociologically because they convert war into a labor-market solution. Reuters reported that officials said about 490,000 contracts were signed in 2023 and roughly 190,000 volunteers had joined in 2024 by late July, while analysts warned that the state's recruitment drive was contributing to wage distortions and intensifying competition for labor across the wider economy. Notably, Reuters also observed that Moscow, where much of Russia's educated workforce is concentrated, initially lagged behind other regions in volunteer rates, which supports the argument that stronger civilian opportunities reduce the appeal of war employment. The problem is therefore not simply patriotism or coercion, but a political economy in which economic insecurity is increasingly translated into military manpower (Reuters, "Moscow Hikes Pay"; Reuters, "Putin Doubles"; Fenton and Kolyandr).

Social consequences and human-capital loss

The funneling of educated youth into the armed forces carries social consequences well beyond recruitment statistics. War removes young adults from the life stage in which they would normally be building careers, families, businesses, and civic networks, and the effects extend from mortality and injury to interrupted education, lost productivity, family strain, and weakened communities. More broadly, recent review research on armed conflict emphasizes that war damages human development through disruption, diversion, devaluation, displacement, and long-term harm to social well-being. In the Russian case, these losses are compounded by outward migration among educated professionals, including the large exodus of IT workers and other skilled employees after the invasion and mobilization measures (Vesco et al.; Reuters, "IT Worker Flight"; Kamalov et al.).

The war's human cost also directly affects the reproduction of skilled labor. In December 2023, Reuters reported that U.S. intelligence assessed Russian forces had suffered around 315,000 casualties since the start of the invasion. Whether one uses that figure or more recent estimates, the key sociological point is the same: the state is consuming a large share of the very cohorts on which future civilian development depends. Our interview material reflects this tension repeatedly, especially in accounts of young parents, engineers, and teachers who

understand enlistment as financially tempting but socially destructive (Reuters, 2023b; Vesco et al., 2025).

A postcommunist comparative perspective helps clarify why this pattern should not be reduced to sanctions alone. Scholarship on postcommunist transformation has long stressed that outcomes after communism vary because reform unfolds through historically embedded institutions, uneven political coalitions, and inherited state-society relations, not merely through short-term policy shocks. This lens is useful here because it explains how weak support for small enterprise, downward mobility among the educated, and labor-market insecurity can coexist with superficially resilient macro-level indicators. Wartime pressures have not displaced these older institutional patterns so much as intensified them, making military service more legible as an economically rational strategy within an already uneven postcommunist order (Ekiert and Hanson; Kitschelt).

Methods

Approach and Sample: To explore these issues, we adopted a mixed-methods approach, centered on qualitative interviews supplemented by analysis of economic indicators. We conducted semi-structured interviews with 18 young adults (aged ~22-35) across different regions of Russia between May and August 2024. Given the sensitivity of the topic, all participants were assured anonymity and were identified only by descriptors (e.g. "a recent graduate in engineering from Siberia," "a schoolteacher from Tatarstan"). The sample was purposefully selected to include individuals with higher education or specialized training who faced economic difficulties. Among them were unemployed or underemployed professionals (e.g. an out-of-work software developer, a nurse turned taxi driver), as well as some who were employed in their field but considering a career change due to low pay (e.g. teachers, engineers). We also interviewed a few participants with stable jobs in non-professional sectors (such as retail or skilled trades) for contrast. Interviewees were recruited through personal networks, online forums, and referrals, following ethical guidelines for voluntary participation. Each interview lasted 60-90 minutes and was conducted in Russian (then translated to English for analysis). In addition, we analyzed secondary data on regional wages, employment rates, and military recruitment figures - drawing from Russian statistical releases, independent news outlets, and international reports - to contextualize and triangulate the interview findings.

Interview Protocol: Interviews were conducted using an ethics-conscious semi-structured interview guide that covered several thematic sections, being:

Personal and Economic Background: We asked participants about their education, current occupation, and daily life, probing how the local/regional economy was perceived. (e.g. "How would you describe the economic situation in your region for young professionals like yourself?").

Employment Aspirations and Barriers: Questions explored what jobs or careers respondents aspired to, what opportunities they saw around them, and whether they felt adequately supported in achieving their goals. We also discussed any experiences with entrepreneurship or further training.

Perceptions of Military Service: We inquired about attitudes toward the military draft and contract service. Participants were asked if they or people they know have considered enlistment, and what factors might influence such a decision (financial incentives, patriotism, family tradition, etc.).

Economic Pressures and Incentives: We delved into specific pressures (debt, cost of living, social expectations) that might make military service appealing or necessary. For example: "Have financial difficulties (such as loans, lack of jobs) ever made you consider options you otherwise wouldn't - like joining the army?"

Alternatives and Solutions: Finally, we invited thoughts on what could help young people avoid involuntary enlistment - be it government programs, community initiatives, or changes in policy. We also asked for any advice they would give to organizations trying to help in this area.

Throughout the interviews, we took care to maintain neutrality on political questions. Participants were not asked to take a stance for or against the war; instead, the focus was on their personal economic reality and choices. We also reminded interviewees that they could skip any question or stop at any time, to ensure comfort given the atmosphere of censorship and fear in Russia. In a few cases, respondents hesitated when touching on sensitive issues (such as overt criticism of authorities or mention of friends who protested); the interviewer allowed them to either express in their own terms or move on without pressure.

Data Analysis: The interview recordings were transcribed and coded using a thematic analysis approach. We identified key themes corresponding to our research questions: for example, "low wages in professional jobs," "perceived stability of service-sector vs. professional work," "motivations for enlistment - financial," "motivations for enlistment - duty," "role of small business," "impact of debt/loans," and "family responsibilities." We also noted the frequency of certain responses (e.g. how many out of 18 mentioned salary as a reason to consider military, how many cited patriotism, etc.) to detect patterns. Given the qualitative sample size, these counts are not statistically generalizable but help illustrate dominant versus minority viewpoints among participants. In parallel, we examined economic data: for instance, we compiled average salary figures for various professions and regions mentioned by interviewees, and cross-referenced enlistment reward figures. This quantitative context was used to either validate or contrast with interview perceptions (for example, if a respondent believed "everyone is losing jobs," we checked regional unemployment trends for alignment or discrepancy). Finally, we integrated all findings into a coherent narrative. Direct quotes from interviews (with minor linguistic polishing for clarity) are used in the results to amplify the participants' voices. All such quotes were reviewed to ensure they contain no identifying details.

Limitations

It is important to acknowledge the limitations of this study. Our interview sample, while geographically varied, is relatively small and is likely skewed toward respondents willing to discuss economic hardship and career frustration. It may therefore underrepresent young educated Russians who are economically secure, politically supportive of the war, or firmly opposed to military service on moral grounds.

Research under wartime conditions in Russia also creates serious access and reliability problems. The country's increasingly repressive information environment, including prosecutions for anti-war speech and broader crackdowns on dissent, plausibly increases self-censorship and makes some potential participants less willing to speak openly, particularly on politically sensitive issues such as enlistment, state policy, or economic decline (Human Rights Watch, "World Report 2024"; Human Rights Watch, "World Report 2025"). Qualitative work on wartime Russia likewise notes that self-censorship and selective silence remain major obstacles to gathering candid testimony (Arslanova).

The broader statistical environment is also difficult. Official Russian indicators, including labor-market and macroeconomic data, sometimes present a more optimistic picture than independent analyses suggest. BOFIT reported that unemployment fell to a post-Soviet record low of 2.4% by June 2024, yet both Reuters and SITE-linked analysis have emphasized that wartime growth and headline labor strength can mask deeper distortions, opacity, and structural weakness in the civilian economy (Bank of Finland Institute for Emerging Economies, "Unemployment"; Reuters, "GDP Boost"; Free Network).

Finally, our analysis focuses mainly on the 2022-2024 period. Because Russia's war economy, recruitment practices, and labor market continue to evolve, some conclusions may shift as conditions change. Even so, the mixed-method design used here offers a valuable window into how economic strain and military enlistment intersect in contemporary Russia.

Results

Economic Hardship Among Educated Youth

All participants described economic strain, though its intensity varied by location and profession. Dissatisfaction with wages in jobs requiring higher education was nearly universal: 15 of 18 explicitly described incomes as "low," "inadequate," or "unstable." A 29-year-old male in sales with an engineering degree remarked: "Most essential professions - doctors, teachers, engineers - are poorly paid here. The only way to have a decent income is to go into buying and selling." Several university-educated respondents had already left white-collar or public-sector roles for better-paid service or trade jobs, including retail management, delivery work, and small-scale commerce. As a former nurse who moved into cosmetic sales put it, "I make more money painting nails than I did in the hospital. What does that say about our economy?"

These accounts fit wider evidence that Russia's labor market is tight but uneven, and that low pay in socially important professions remains a serious problem. Reuters reported record-low unemployment alongside a defense-driven labor squeeze rather than balanced civilian prosperity, while RFE/RL documented low and often unsustainable teacher pay in Tatarstan, including an average basic salary of about 53,000 rubles and lower real starting salaries for many new teachers (Reuters, "Soaring Wages"; RFE/RL, "No Time to Breathe").

Underemployment was also common in our sample: half of respondents were working outside their degree field at the time of interview. Reasons included a lack of openings, extremely competitive entry routes into formal professions, and burnout from earlier low-pay

roles. A young lawyer from Siberia described a cycle of unpaid internships and short-term gigs before leaving law to work in a cousin's bakery, saying the few desirable positions "all went to people with connections." About a third of interviewees raised concerns about nepotism and blat. An IT graduate from central Russia commented, "In the provinces, unless you have a relative or patron in an organization, it's very hard to get a good job, so many give up and leave or do something else." These frustrations often fed into internal migration and professional downgrading.

Participants' views of the wider economy, however, diverged. Some interviewees, especially in parts of central Russia, described local conditions as "stable" or improving in sectors such as construction and logistics. A 30-year-old regional bank employee observed, "Unemployment has fallen here; loans and mortgages are more accessible now than last year." That perception is not entirely inconsistent with official data showing falling unemployment and sustained employment growth during the war economy, even if those indicators do not necessarily reflect good job quality (Bank of Finland Institute for Emerging Economies, "Unemployment").

Others rejected any notion of stability. One respondent from an industrial town argued that "the country is in a very dire state due to pressure and sanctions... plants are shutting down, there's no growth in production." Such contrasts suggest significant regional disparities, uneven sectoral effects, and a lag between macroeconomic indicators and lived experience. Even where jobs existed, many respondents felt that the available work did not match their education, ambitions, or expectations.

Despite these differences, almost all participants agreed that their economic prospects fell short of what they had expected from higher education. Their disillusionment carried real emotional weight, with phrases such as "no hope," "wasted potential," and "frustrating" recurring across interviews. As one philosophy graduate concluded, "We study, we try to better ourselves, and in the end we either drive taxis or we sit at home useless. It's a tragedy for our generation."

Motivation for Military Enlistment: "Duty" vs. "Deal"

Attitudes toward military service ranged from reluctant acceptance to firm refusal, but the underlying reasons for considering or rejecting enlistment were overwhelmingly economic. Of the 18 interviewees, only two said they personally felt a patriotic duty or inclination to serve, and both immediately added that family obligations prevented them from enlisting. By contrast, nine participants had either considered contract service themselves or had close friends or relatives who had done so recently, and all nine described financial incentives or lack of other work as the primary driver.

One 25-year-old unemployed man from a small town recounted being approached by a recruiter at a local employment center and offered 300,000 rubles a month to join a new battalion. This mirrors reporting by Important Stories, which found that unemployed Russians registered with labor exchanges were being offered contracts and promised around 300,000 rubles per month for service in Ukraine (Important Stories).

He ultimately refused on moral grounds but believed others present accepted: "I can't blame them - when you have loans and no work, 300k a month can save your family, even if it might cost your life." Many accounts revealed a similar calculus: the weighing of life-threatening risk against financial rescue.

Financial motivation dominated almost every interview. Respondents repeatedly cited high monthly pay, signing bonuses, debt pressure, and the appeal of long-term benefits. This logic is consistent with the structure of Russia's current recruitment system. Reuters reported in July 2024 that the minimum monthly payment for a private fighting in Ukraine was 204,000 rubles, raising minimum first-year compensation to 3.25 million rubles, while Moscow residents could receive total first-year compensation of around 5.2 million rubles after city and federal incentives. Reuters also noted that such payments were more than three times the national average wage, and more than five times the average for Muscovites in the latter case (Reuters, "Moscow Hikes Pay"; Reuters, "Putin Doubles").

Interviewees framed these offers in starkly comparative terms. As one respondent put it, "Here in the countryside, 30-40 thousand rubles is normal pay. They're offering 200 thousand plus bonuses to go to Ukraine - you do the math." Another described contract service as a high-risk but potentially transformative earnings opportunity: "You go for a year or two, make a fortune by local standards, and hope to come back." Recent Reuters reporting from April 2026 suggests that this incentive model has expanded even further into educated populations, with universities promoting drone-force roles to students using packages worth 5.5 to 7 million rubles annually plus fee relief and accommodation support (Reuters, "Woos Students").

Patriotism and duty did appear in the interviews, but usually as a secondary theme. Around a third of participants acknowledged that some people enlist out of patriotism, family tradition, or a sense of obligation, typically describing this as something true of "others" rather than themselves. One 32-year-old man with previous army experience said, "Some go out of duty - the motto 'Who if not me?' - I know guys like that. But others go because there is no work or to solve their finances." A recent humanities graduate similarly said, "I'd defend my country if it was truly in danger, but I don't see this war that way." Among our urban, university-educated respondents, unqualified ideological enthusiasm for the war was absent.

Reluctance and refusal were also clear. Family responsibility was the most common deterrent. Several interviewees said that children, spouses, or dependent parents made military service unacceptable. Fear of death or injury was stated bluntly, often in practical rather than abstract terms. One respondent even compared potential lifetime earnings with promised death compensation before deciding that "no amount of compensation is worth the lifelong trauma to my loved ones if I die or come back disabled." Such reasoning is consistent with publicly documented compensation schemes, including a 5 million-ruble payment ordered in 2022 for families of deceased National Guard members, alongside other credit and support measures tied to war service (Reuters, "Putin Orders"; Reuters, "Moscow Hikes Pay").

Distrust of state promises was another deterrent. Two participants said they had heard of soldiers or families failing to receive the pay and benefits they were promised, making the

bargain seem even riskier. That concern is supported by both Reuters and RFE/RL reporting that wounded soldiers and families of contract soldiers have, in multiple cases, struggled to obtain compensation, bonus pay, or other promised benefits (Reuters, "Some Wounded"; RFE/RL, "Russia Violates").

Overall, our interviews show that economic incentives are the dominant driver of enlistment among struggling educated youth, while fear, family ties, and distrust of the state act as counterweights. Idealism appears present but limited. As one participant summarized, "If tomorrow I got a job that pays even half of what the army pays, but I know I'll be safe and home, I'd take it in a heartbeat. People go [to war] because they don't see that option."

The Role of Small Business and Entrepreneurship

A notable and somewhat unexpected theme was the perceived importance of small-business opportunity in keeping educated youth out of the army. When asked what could help young people pursue non-military paths, more than half of respondents mentioned support for small business. They saw entrepreneurship, even on a modest scale, as a viable alternative to both dead-end jobs and enlistment if barriers were reduced.

This perception is consistent with broader evidence on Russia's SME sector. Research on Russian small and medium-sized enterprises under sanctions found that inflation, reduced consumer demand, rising borrowing costs, and broken supply chains placed severe pressure on smaller firms, while job losses in other sectors created pressure toward "forced entrepreneurship" among displaced workers (Barinova et al.). More recent Reuters reporting has shown that additional war-related tax changes have further strained small firms, with some owners warning of closures or movement into the shadow economy (Reuters, "Small Businesses Cry Foul").

A 28-year-old management graduate in our sample wanted to start an IT consulting firm but felt that the post-2022 environment had effectively shut that path down: "After 2022, investors vanished and regulations got tighter. If I had a real shot at my own business, I'd never think of joining the army for money." Others proposed grants, micro-loans, incubators, free legal support, and mentoring as practical ways to redirect talented young people away from military labor markets and toward civilian initiative.

Several participants were already engaged in small-scale entrepreneurship, including freelance design, tutoring, and handmade goods, but complained about taxes, weak legal protections, and the difficulty of scaling anything in an uncertain environment. One respondent described current policy support as mismatched: "There are programs, but they're mostly for trades or basic retraining. Nothing serious for those of us with higher education and business ideas." Another put the issue more bluntly: "There are so many talented people here, but no opportunity to show it to the world."

Asked about existing programs, some mentioned local job centers and retraining schemes. One woman in her late 20s described a labor exchange that offered a modest stipend during retraining and helped place people into manual trades. While she regarded this as a positive initiative, she also emphasized that it did little for university graduates seeking

professional or entrepreneurial pathways. Taken together, these interviews suggest that expanding credible civilian opportunity, especially through small-business support, may be one of the clearest ways to reduce the economic pull of military enlistment.

Mandatory job placement

Two older participants (early 30s) proposed mandatory work placements for graduates, organized by universities or local authorities. For example, they suggested guaranteeing every medical and teaching graduate a posting in a hospital or school for two years. They argued this would both strengthen public services and reduce graduates' drift into unrelated jobs or the military. A few interviewees cited Soviet-era graduate assignments as a model (though they preferred a voluntary, incentivized version). For them, the main "failure point" was the transition from university to the labor market. As one fresh graduate explained, "When you're handing out CVs for months and nothing happens, and then you see an ad offering 'Adventure, honor, and 200,000 rubles a month' for contract service - you start to think about it." Smoother pathways into jobs aligned with degrees might therefore reduce the attraction of enlistment.

Urban vs. rural gaps

We also found geographic differences in attitudes toward entrepreneurship. Urban interviewees (e.g. from Kazan, Voronezh, and a Moscow suburb) were more likely to have tried or considered starting a business. Those from smaller towns often saw it as unrealistic. One rural respondent asked, "In my village, even if you open a shop, who's going to buy? People have no money." This suggests that business support alone is not enough; in poorer regions it must be combined with broader economic development. Otherwise, educated rural youth face stark choices: migrate, accept low-quality work, or respond to military recruitment, which has been especially active in economically depressed areas .

Social Reflections: Personal Toll and Community Impact

Finally, the interviews highlight the human and social consequences of these economic and military dynamics. Several cross-cutting issues emerged:

Precarity and mental strain. Even participants who called their regional economy "stable" described living with constant uncertainty. Many feared sudden layoffs, business closures, or conflict with stressed managers. They spoke of peers defaulting on loans, losing property, or accumulating debt. Some said young people had stopped planning for mortgages or long-term careers and instead "live day by day." For a few, the military offered not only money but structure: "At least in the military you know what you're doing each day, you have a goal, even if it's dangerous." Others emphasized mental health problems - friends feeling depressed, alienated, or "checking out," including references to the Russian phrase "тушить себя" (to numb oneself). The war and sanctions have created a sense of drift that, sociologically, can increase the appeal of belonging to a larger institution or cause, including the army.

Family and gender dynamics. Family roles strongly shaped decisions. Women stressed the need for husbands or themselves to remain at home with children, and fathers often cited

their duty to family as a reason not to enlist. This suggests that married, educated men with children are among the most resistant to military service despite financial incentives, whereas single men with fewer dependents may find it easier to volunteer. Respondents' stories of acquaintances who enlisted typically involved young men, single or without stable careers, sometimes from broken homes or highly precarious backgrounds. Some female interviewees described war as "a man's escape": "Men have this out (the army) if they can't feed their family; we women don't have that luxury, we just endure or find another way," said one woman working two jobs. This captures how economic burdens are gendered, with men sometimes externalizing pressure through enlistment and women bearing much of the strain at home.

Information and trust. A subtler theme was the erosion of trust in information and institutions. One participant noted that since 2022, independent media and some social networks have been sharply curtailed, making it difficult to "objectively assess situations or solutions." In this environment, people rely on rumor and informal networks, which fuels anxiety: "We hear rumors - that the economy is actually worse than reported, or that more mobilizations are coming - and it makes everyone paranoid," said one respondent. Several suggested that if they trusted the government and media more, they might consider official programs (whether employment or military) with less skepticism. As it stands, high distrust likely affects would-be entrepreneurs and job seekers too: doubts about long-term stability may discourage investment in new ventures and push some towards short-term, high-pay options such as military contracts.

Community fabric. Most participants were unaware of strong community-level responses beyond state programs. A few cited informal support networks - group chats among teachers coping with low pay, or families of soldiers sharing updates. One interviewee volunteered at a local NGO providing career counseling to youth: "We try to show them paths other than the army," she said. Yet such efforts appeared limited in scale, and many respondents described a sense of isolation. "Everyone is quiet, just surviving. There's no protest, no union - nothing. Some leave, some join the army, the rest just keep their head down," one interviewee concluded.

In sum, the educated young Russians in this study are caught between aspirations built on their education and harsh economic and political realities. They value their degrees and often dream of careers aligned with their training, but conditions force painful compromises. The war appears simultaneously as a source of their difficulties and, paradoxically, as a tempting escape route. The next section situates these findings within the broader literature and considers what they imply for Russia's future and for possible policy interventions.

Discussion

Our results can be interpreted not only as evidence of immediate wartime hardship, but as part of a broader postcommunist political economy in which institutional legacies shape both labor-market outcomes and state strategies of incorporation. From this perspective, Russia's recruitment of educated but underemployed citizens is not merely a response to patriotic mobilization or temporary recession. It reflects a deeper configuration in which the state absorbs dislocated human capital into coercive and military institutions, thereby converting economic

insecurity into political utility. The central issue, therefore, is not simply poverty, but the transformation of postcommunist dislocation into state capacity for war. Framing the case in this way places the paper in dialogue with comparative scholarship on postcommunist regime diversity, unemployment politics, and path contingency, while also clarifying why formally educated citizens may experience blocked mobility, constrained entrepreneurship, and authoritarian integration at the same time (Ekiert and Hanson; Kitschelt; Baxandall; Johnson).

Economic Forces and the De Facto Draft

Our data strongly support the idea of a de facto "poverty draft": for many respondents, enlistment appears driven less by ideology than by economic necessity. What is especially striking in the Russian case is that this pressure now reaches beyond the traditionally poor or least educated and extends into parts of the educated or aspirant middle class. Military pay has been raised far above what many civilian careers can now offer. Reuters reported in July 2024 that the minimum monthly payment for a new private fighting in Ukraine was 204,000 rubles, implying a first-year minimum package of 3.25 million rubles, while Moscow raised first-year compensation for its own recruits to about 5.2 million rubles. Recent research on Russian enlistment also links recruitment patterns to poverty, rurality, and limited mobility, suggesting that financial incentives are increasingly central to understanding who serves and why (Reuters, "Moscow Hikes Pay"; Reuters, "Putin Doubles"; Yusupova).

From a labor-economics perspective, the Russian state is recruiting from a labor pool that is formally "employed" in statistical terms but often insecure, mismatched, or trapped in low-return civilian pathways. Record-low unemployment does not mean that the civilian economy is healthy. BOFIT reported that unemployment fell to 2.4% by June 2024, while Reuters and the Bank of Finland stressed that this tight labor market reflects mobilization, emigration, labor shortages, and heavy state spending on defense rather than balanced civilian prosperity (Bank of Finland Institute for Emerging Economies, "Unemployment"; Reuters, "Soaring Wages"; Bank of Finland Bulletin, "War Has Degraded"). In that setting, the war economy produces a push-pull dynamic: weak civilian prospects push people out of professional careers, while military wages and bonuses pull them toward service.

Regional patterns reinforce this interpretation. Reuters reported that Moscow initially lagged behind other regions in volunteer recruitment and had to raise compensation sharply, which suggests that where civilian opportunities are relatively better, educated people are less inclined to choose war work. By contrast, scholarship on Russia's enlistment patterns shows that peripheral, poorer, and more weakly connected populations remain especially vulnerable to military recruitment pressures (Reuters, "Moscow Hikes Pay"; Yusupova).

Whether this is temporary or transformative is a crucial question. If the war continues, Russia could see the consolidation of something like a military middle class: a stratum whose relative prosperity depends on military or adjacent state employment rather than productive civilian work. That is an inference, but it is consistent with Reuters reporting on wage-pressure effects and with Bank of Finland analysis showing that military expenditure is crowding out

other financing needs and distorting the wider economy (Reuters, "Soaring Wages"; Bank of Finland Bulletin, "Russia Further Increases").

Societal Repercussions

This economic-military dynamic is also straining Russia's social fabric. Our interviews suggest a collapse of the older promise that education would deliver stability, dignity, and upward mobility. In broader terms, this is consistent with research showing that armed conflict damages human development not only through death and destruction but also through disruption, diversion, devaluation, and long-term harm to livelihoods and social well-being. In Russia, that strain is compounded by repression and self-censorship, which narrow public agency and make it harder for citizens to articulate dissatisfaction openly (Vesco et al.; Human Rights Watch, "World Report 2025"; Arslanova).

Family and demographic pressures are part of this picture. Russia already faces population decline and low birth rates: BOFIT reported that the country's natural population growth has been negative or flat since 2000 and that births in 2023 fell to their lowest level since 2000. Reuters also noted in late 2024 that Russia's birth rate had fallen to its lowest level in a quarter century, underscoring the depth of the demographic problem. In that context, the removal of young adults from civilian life through death, injury, migration, or military service is especially damaging (Bank of Finland Institute for Emerging Economies, "Population"; Reuters, "Child-Free Propaganda").

The cost is not only demographic but social and psychological. Reuters reported in December 2023 that a declassified U.S. intelligence assessment put Russian losses at 315,000 dead and injured since the invasion began. Whatever estimate one adopts, the broader sociological point remains that a large share of the country's young adult cohort is being removed from normal life trajectories at the very stage when they would otherwise be building careers, families, and civic ties. Research on conflict and development suggests that the effects of such losses are long-lasting and extend well beyond the battlefield itself (Reuters, 2023b; Vesco et al., 2025).

Our findings also suggest a partial class inversion. In the wartime economy, some blue-collar, logistical, or defense-linked roles may now offer more reliable returns than prestigious but underpaid professions such as teaching or nursing. Reuters reported soaring wages in tight labor markets, especially in areas tied to defense production, while RFE/RL documented that teachers in Tatarstan still faced basic average pay of around 53,000 rubles a month, with many real starting salaries lower. If these conditions persist, it is plausible that some young people will view higher education as a weaker investment than practical trades, migration, or military employment (Reuters, "Soaring Wages"; RFE/RL, "No Time to Breathe").

Breaking the Cycle: Possibilities and Challenges

Breaking the feedback loop of educated underemployment, enlistment, and deeper militarization is extraordinarily difficult while the war continues. Even so, our data point to several pressure points where intervention could matter.

Strengthen support for small and medium enterprises

Participants repeatedly described entrepreneurship as one of the few imaginable civilian alternatives to dead-end work or enlistment. That view is supported by wider evidence. Research on Russian SMEs under sanctions found that inflation, falling demand, disrupted supply chains, and rising borrowing costs have all increased bankruptcy risk and narrowed room for business formation, while Associated Press reporting in 2026 showed that new wartime tax burdens were placing additional pressure on already fragile small firms. Expanding micro-grants, low-interest loans, tax relief, mentoring, and simpler registration processes would not solve everything, but it would widen the space for civilian futures (Barinova et al.; Associated Press).

Improve the education-to-employment pipeline

A repeated theme in our interviews was that the most dangerous moment comes when a graduate concludes that no meaningful civilian path is open. This is especially damaging in a country that simultaneously reports labor shortages and underuses educated labor. BOFIT's labor-market analysis points to an economy with historically low unemployment and acute shortages, while RFE/RL's reporting on teacher shortages in Tatarstan shows clear unmet social need in essential sectors. Public-sector hiring initiatives, graduate placements, and stronger university-employer partnerships could therefore operate as both labor-market and anti-recruitment measures (Bank of Finland Institute for Emerging Economies, 2024; RFE/RL, 2024).

Address wage disparities and civilian retention

The gap between military and civilian pay remains central. The state is unlikely to match military compensation across the board, but it has already shown that it can selectively shield strategically important groups. Reuters reported in 2022 that some IT specialists, bankers, and journalists were exempted from mobilization, implicitly recognizing that the state cannot afford to lose all skilled labor to the war effort or to emigration. Extending that logic to teachers, engineers, medical workers, and other critical professions through hardship allowances, housing subsidies, and accelerated pay progression would make civilian service more viable (Reuters, "Russia Excludes"; Reuters, "Moscow Hikes Pay"; Reuters, "Putin Doubles").

Refine sanctions with greater social precision

Our findings point to a real paradox: sanctions designed to constrain Russia's war-making capacity can also deepen hardship in civilian sectors and thereby enlarge the recruitment pool. That does not mean sanctions should be abandoned. It means they should be calibrated carefully. Bruegel, Carnegie, and CSIS all argue in different ways that sanctions have imposed real costs but that Russia has also adapted, meaning that policymakers should learn from where the pressure bites and where it spills over inefficiently into wider society. More targeted pressure on war-related industries and elites, alongside carve-outs or indirect support for civilian education, exchanges, and independent economic activity, may reduce unintended recruitment effects

without easing pressure on the war machine itself (Bruegel, "Sanctions Against Russia"; Carnegie Endowment; Fenton and Kolyandr).

Support psychosocial and community resilience where possible

Many interviewees described not only poverty, but also isolation, drift, and loss of direction. Community-based mentoring, retraining, legal guidance for start-ups, and peer-support networks would not end the war, but they could reduce the sense that military service is the only structured path left. The problem, of course, is that Russia's repressive environment sharply limits what independent organizations can do. Human Rights Watch and PONARS both document how repression, fear, and self-censorship constrain collective action and public expression, which means that even small civic interventions have to be understood against a hostile political background (Human Rights Watch, "World Report 2024"; Human Rights Watch, "World Report 2025"; Arslanova).

Recommendations

Based on our findings, the most credible interventions are those that expand civilian opportunity quickly enough to compete with the military labor market.

Expand SME support through micro-grants, subsidized credit, lighter compliance burdens, and practical start-up support for graduates in IT, engineering, education, and services, since smaller firms remain one of the clearest non-military absorbers of skilled labor (Barinova et al.; Associated Press).

Create stronger graduate-to-work pathways by funding placements in schools, clinics, infrastructure, and local administration, especially in underserved regions where shortages already exist (Bank of Finland Institute for Emerging Economies, 2024; RFE/RL, 2024).

Raise retention incentives in critical civilian sectors through targeted housing support, hardship allowances, and faster pay progression for professions that are strategically important but currently under-rewarded relative to military service (Reuters, "Russia Excludes"; Reuters, "Moscow Hikes Pay"; Reuters, "Putin Doubles").

Review sanctions for unintended recruitment effects while maintaining pressure on war-enabling sectors and elites, since broad economic damage that primarily weakens civilian opportunity can indirectly help military recruitment (Bruegel, "Sanctions Against Russia"; Carnegie Endowment; Fenton and Kolyandr).

Preserve community-level support and professional networks where possible, because in repressive settings even modest mentoring, retraining, and peer coordination can counter fatalism and reduce the quiet normalization of enlistment as the default option (Human Rights Watch, "World Report 2025"; Arslanova).

Conclusion

This study examined why some of Russia's educated young people are helping sustain a war that many do not appear to support ideologically. Our interviews and the wider evidence point in the same direction: financial motives and the collapse of credible civilian opportunity

often matter more than patriotic fervor. Reuters' reporting on recruitment pay, bonuses, and student-targeted drone-force campaigns, together with recent scholarship on the socioeconomic mechanisms of enlistment, suggest that for many young Russians military service increasingly functions as a labor-market decision under constraint rather than an expression of conviction (Reuters, "Moscow Hikes Pay"; Reuters, "Putin Doubles"; Reuters, "Woos Students"; Yusupova).

The broader social cost is immense. Russia is not only losing people on the battlefield; it is also hollowing out classrooms, clinics, laboratories, and small firms. Research on armed conflict shows that such losses have enduring developmental effects, while reporting on casualties, teacher shortages, and skilled-worker flight illustrates how the war is already consuming or displacing precisely the cohorts needed for recovery and long-term growth (Vesco et al.; Reuters, "IT Worker Flight"; Reuters, "U.S. Intelligence Assesses"; RFE/RL, "No Time to Breathe").

The comparative implication is also important. Postcommunist systems do not move uniformly toward stable markets or stable politics; outcomes depend on how inherited institutions, state strategies, and social coalitions shape the conversion of economic change into political order. In Russia's wartime case, those dynamics have taken a particularly destructive form: dislocated human capital is increasingly being redirected into militarized and authoritarian structures rather than into innovation, entrepreneurship, or public service. Seen through postcommunist institutional scholarship, this looks less like simple wartime improvisation than a harsh form of authoritarian adaptation (Ekiert and Hanson; Kitschelt; Bank of Finland Bulletin, "War Has Degraded").

That trajectory is not inevitable. If policymakers inside and outside Russia want to reduce the flow of educated youth into war participation, they must take civilian opportunity seriously. SME support, graduate hiring, wage reform in essential professions, and more socially precise sanctions will not end the war by themselves, but they can shrink one of its most neglected recruitment mechanisms: economic desperation among people who wanted to build civilian lives and found those lives narrowing in front of them (Barinova et al.; Associated Press; Fenton and Kolyandr).

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How School Culture and Parental Expectations Differently Affect the Academic Motivation and Mental Health of High School Students Across Gender Identities By Anshika Gupta

Abstract

This study examines how school culture and parental expectations impact the academic motivation and mental well-being of high school students, with an emphasis on the role of gender identity. In many high-pressure academic environments, students are pushed to earn perfect grades, build highly competitive resumes, and meet both parental and peer expectations of academic success. Although widespread, these pressures are experienced and managed differently depending on a student's gender, specifically affecting their academic motivation and mental health. In this paper, academic motivation is considered in terms of intrinsic drivers, such as curiosity and personal growth, as well as extrinsic drivers, such as grades and external approval. Mental health is examined in relation to these motivational factors, focusing on how academic pressure and gender-based expectations contribute to anxiety and overall emotional well-being.

According to Bronfenbrenner's Ecological Systems Theory, with a focus on the microsystem and mesosystem levels, this study examines the ways in which interactions with parents, teachers, and peers, along with the interconnections among these interactions, impact students' emotional health and motivation toward learning. For example, poor communication with parents or teachers can increase a student's sense of isolation or stress. This paper investigates two central research questions: How does school culture impact high school students' academic motivation and mental health? And how do parental expectations influence these same outcomes?

To further explore these issues, the review paper includes studies that address how gender identity moderates the effects of these interactions between family, school, and peers. Sub-questions include: In what ways do students' gender identities influence how school culture affects their motivation and mental health? And how do gendered family expectations differently shape students' academic experiences?

This paper highlights the need for more welcoming and supportive learning contexts based on understanding how gendered expectations in families, schools, and individual motivation relate to the academic motivation and mental health of students. Without this type of review, there would be little understanding of the interconnections between pressures from home and school in promoting students' well-being. Educators and parents may try to alleviate stress with classroom strategies or family dynamics without understanding how these factors relate to each other. This paper helps fill that gap by illustrating the dynamics of external and internal stress, as well as the importance of supporting students in the best way. This strengthens the ability of educators and families to develop approaches that mitigate experiences of stress based on gender and promote the mental health of all students.

Introduction

In the United States, the mental health of high school students has become a growing concern. In 2023, 40% reported persistent feelings of sadness or hopelessness, 20% seriously considered suicide, and 9% attempted suicide in the past year (Centers for Disease Control and Prevention). Academic motivation is equally important, as it is closely tied to students' learning, achievement, and overall adjustment in school (Eccles and Wigfield). This emphasizes the urgency of addressing the factors that influence students' emotional well-being, especially in environments where academic and social pressures are high.

The mental well-being of high school students matters because it not only affects their life during adolescence but also their academic success in the future. In the U.S., high school students, typically ages 14-18, covering grades 9 through 12, are at a stage that bridges adolescence and prepares them for higher education or the workforce. This age range is especially significant in development, as adolescents are forming a stronger sense of identity and purpose. Poor mental health can make it difficult for students to stay focused and manage daily challenges, while good mental health supports resilience and healthy relationships. Academic motivation is also important as it drives students to set goals and work toward their future aspirations. When motivation is low due to stress, pressure, or lack of support, students may struggle to reach their potential, and this can have lasting effects on both individual and societal outcomes in the future. This could mean reduced access to higher education, or difficulty building confidence or resilience.

Students' feelings and the conditions that influence them are both important for understanding the developmental processes of high school. During this phase of life, different pressures, identity changes, and future expectations interact and put students at particular risk for stress and shifts in motivation. This paper, through studying the factors that affect both mental health and academic motivation, points out the necessity of developing family and school settings that attend to students' emotional needs while also fostering positive involvement in learning. Knowledge of these factors is vital for helping students cope with high school and for establishing a stronger foundation for their mental health in the future.

Operationalizing Key Constructs

This paper uses specific definitions for the key concepts it examines.

School culture refers to the values, norms, and expectations that shape a student's day-to-day experience at school. It includes elements such as rules and routines, as well as less visible elements like how students are treated by teachers and peers. Here, school culture is defined as the environment that encourages certain behaviors and attitudes toward academic achievement. In competitive schools, this often means high pressure to succeed, which can affect both motivation and mental health.

Parental/family expectations are the aspirations and beliefs parents or guardians hold about their children's academic success and future opportunities. These expectations can inspire students to work hard, but they can also become a source of stress if students feel they cannot

meet them. Iqbal et al. note that parents' expectations often stem from factors such as family experiences, socioeconomic background, and cultural standards. While striving for excellence can motivate students to achieve, unreasonably high expectations can increase stress and anxiety, sometimes hindering learning rather than supporting it. Studies have found that excessive parental pressure can contribute to feelings of depression and burnout among students. This paper considers how students respond to such pressures, particularly when families have different expectations for children based on gender.

Academic motivation is the drive that inspires students to learn, achieve in school, and pursue their goals. It may come from a genuine interest in learning, external rewards, or the pressure to meet expectations. Allen et al. describes academic motivation as the extent to which students are motivated to learn and do well in school, involving both cognitive elements such as planning and goal setting and behavioral elements such as pursuing those goals through effort and performance.

Mental health refers to students' emotional well-being and their ability to handle stress, challenges, and day-to-day demands. Good mental health supports focus, motivation, and resilience; poor mental health can interfere with learning and overall functioning. This paper specifically examines how home and school environments affect students' mental health, particularly when expectations are high.

Gender identity refers to how a person understands and experiences their own gender. Girls, for example, based on gender stereotyping and socialization, may feel pressure to be perfect, boys may feel pressure to appear strong and independent, and nonbinary students may face misunderstanding or lack of support. These gendered assumptions can influence how students experience both school and home, as well as how they cope with stress.

By focusing on high school students in the United States and considering the ways school culture, parental expectations, and gender identity shape academic motivation and mental health, this study aims to better understand how these factors interact and how they can be addressed to create more supportive learning environments.

Theoretical Framework

This review paper is grounded in Bronfenbrenner's Ecological Systems Theory (EST). This theoretical grounding informs my approach to the review. EST provides a way to understand how different layers of environment shape adolescent development. Rather than seeing students as influenced only by individual traits, EST emphasizes that outcomes emerge from interactions across multiple systems (Tudge and Rosa). The purpose of this paper is to examine these interactions primarily within the microsystem, mesosystem, and macrosystem levels, focusing on how relationships between home, school, and cultural contexts influence students, rather than addressing all levels of the model in depth.

At the microsystem level, adolescents are directly influenced by settings like family, school, and peers. These immediate contexts shape daily experiences of stress and support, which in turn affect both motivation and well-being (Cherry). The mesosystem highlights how

these microsystems connect, such as when school culture interacts with family expectations. Alignment between the two can reinforce either healthy motivation or harmful pressure.

Apart from the direct contexts mentioned, the exosystem and macrosystem show how broader forces still connect back to students from a wider perspective. The political context is one of these forces. For example, national debates over gender rights, including political targeting of transgender students or efforts to roll back women's health protections, shape the broader climate in which gender norms and ideologies are formed. These cultural messages influence how communities and families think about gender roles, which can amplify or lessen the pressures students face at home and at school (Cherry). These outer systems suggest that student outcomes cannot be understood only within the school setting but must also be considered in relation to wider social and cultural contexts.

As Tudge and Rosa explain, a key feature of Bronfenbrenner's later bioecological model is the concept of proximal processes, the regular interactions between an individual and the people and objects in their environment that drive development over time (Bronfenbrenner and Morris). These ongoing interactions, such as supportive relationships with parents or teachers, are central to understanding how environments influence adolescent motivation and mental health.

By applying EST, this study situates academic motivation and mental health within a network of interrelated environments. This perspective highlights how family, school, and cultural expectations combine to shape adolescents' developmental pathways, particularly across different gender identities.

Methods

Positionality

Being at a competitive Silicon Valley high school, I've first-hand felt the academic pressure that comes with it. As someone who identifies as a girl, I have always felt like I need to work even harder. While many guys are seen as "naturally smart," girls are expected to perform well while staying calm. That double standard has made me realize how differently students are treated and how it impacts the way we see ourselves.

I have dealt with a lot of academic stress because I constantly feel the need to prove how "smart" I am to my teachers, my peers, and especially my family. Everyone in my family has a strong educational background, so there is definitely pressure to live up to those standards, which often leads to a lot of stress and anxiety about school.

That is why this topic matters so much to me. After talking to friends going through the same thing, I want to understand how school culture and parental expectations affect student motivation and mental health, especially through the lens of gender. Hopefully to help us better understand that, maybe school can become a place that supports us instead of one that constantly asks us to prove ourselves.

Search Strategy

I created a Google Sheets database to organize academic articles related to my research topic. This allowed me to track the source details, abstracts, and their relevance to my research questions. I used combinations of my operationalized key terms: school culture, parental expectations, academic motivation, mental health, and gender identity to conduct searches in Google Scholar and ERIC (via EBSCO). Related terms and synonyms were also used to broaden the search. To ensure my review reflects current social and educational dynamics, I prioritized studies published in the 2020s while including earlier foundational works when relevant. For each article, I recorded the APA citation, the study's abstract, and key connections to my research question in the database. I approved each source according to the key terms it addressed and grouped them by their relevance to different aspects of the topic. This process allowed me to identify patterns across studies and compare how school culture and parental expectations intersect with academic motivation and mental health, with attention to the moderating role of gender identity.

To execute this operation, I searched Google Scholar and ERIC, opening articles I thought might be useful in new tabs. I looked at the title and abstract to determine if the study included two of my key terms. If relevant, I entered the full APA citation and abstract into my Google Sheets tracker, along with a short comment on how it related to my research question. I excluded papers that were out of scope, such as those that focused exclusively on college students or those that did not address school culture or mental health. Once I had looked through and collected all the possible studies, I returned to the database to read the abstract to check that it met my inclusion criteria before making a final decision regarding the selection.

The literature search strategy in total gave me 26 studies, from which I applied the inclusion criteria to 11 studies and they were included in the final review. Of these studies, 7 were quantitative, and 4 were qualitative, thus providing statistical and narrative insight.

Inclusion and Exclusion Criteria

To be included in the review, studies needed to meet the following criteria:

- Directly addressed at least two of the main key terms: school culture, parental expectations, academic motivation, mental health, or gender identity.
- Focused on high school students in the United States.
- Published in English and available through the selected databases.
- Clearly connected to students' academic or mental health experiences.

For example, a paper discussing anxiety alone was not sufficient; it needed to address anxiety in the context of high school students and gender, or anxiety alongside another relevant factor such as school culture or parental expectations. Studies focusing solely on adults, elementary school children, or international populations were excluded, as were studies lacking a clear connection to the academic or mental health experiences of high school students.

Literature Review

Parental Expectations and Mental Health

Studies have repeatedly shown that parental expectations can be both a strong source of motivation and a significant source of stress for high school students. Almroth et al. found that adolescents were about 50 percent more likely to show high externalizing symptoms when parents held low academic expectations compared with students whose parents expected a university education. The study also reported that when both parents and children shared low expectations, the odds of externalizing problems more than doubled. Mismatched expectations between parent and child were also associated with elevated risk, suggesting that misalignment itself can create strain within the parent-child relationship. Within Bronfenbrenner's ecological framework, parental expectations are part of the microsystem, the immediate setting that shapes students' daily experiences. When expectations align with a student's own goals, they can support intrinsic motivation and a sense of direction. However, when expectations are unrealistic or markedly lower than the student's aspirations, they can contribute to frustration, lower self-esteem, and disengagement from school. Almroth et al. emphasize that shared high expectations are associated with more positive outcomes, highlighting the importance of supportive and open communication about academic goals.

In a U.S. based study, Lewis et al. explored how gender moderated the relationship between family and depression in adolescents. They found that girls were more likely than boys to develop depressive symptoms when they lacked closeness in relationships with their parents or when there were big family conflicts. These findings connect directly to this review's research questions by showing how parental expectations and gender identity intersect to influence students' mental health, and how the microsystem of the family can contribute to gender differences in emotional outcomes.

Cultural and societal norms in parental expectations also play a role. Iqbal et al. reported that in competitive academic environments, parents often set exceptionally high benchmarks for performance. While these expectations can encourage resilience and drive, they also heighten anxiety when students feel unable to meet them. Brooks et al. similarly found that when academic achievement is heavily emphasized, adolescents, particularly girls, were more likely to feel overwhelmed.

Other research included in this literature review show that the way expectations are communicated are also vital. Fan found that supportive and encouraging parents were associated with stronger emotional well-being, while overly critical approaches correlated with higher rates of anxiety and depression. This strengthens the idea that parental expectations influence not only academic outcomes but also the student's emotional atmosphere.

Overall, literature suggests that parental expectations can be both positive and negative. When they are supportive, flexible, reasonable, and agree with a student's own goals, they foster learning and success. When they are too strict and have unattainable expectations, students face negative mental health effects, and this is often stronger for female students. These patterns

highlight the need for families to find a balance between promoting high standards while still prioritizing emotional well-being, recognizing that both are essential to long-term success.

School Culture and Academic Motivation

School culture: the shared values, and expectations within an academic environment, plays a major role in shaping how students approach learning. In many high-pressure settings, such as competitive college-preparatory or high-achieving schools where students feel pushed to take advanced classes and maintain perfect grades, the culture boosts competition and perfectionism. The culture emphasizes competition and perfectionism to achieve high academic performance. These elements can motivate students, but they can also create a lot of stress for a student. Within Bronfenbrenner's ecological systems theory, school culture is part of the microsystem, while its interaction with family life forms part of the mesosystem. When school and family values don't align, it can lead to even more pressure for students.

Lester et al. found that schools promoting teacher–student trust and collective support contained students with higher engagement and more intrinsic motivation. In contrast, schools that placed strong emphasis on standardized testing and ranking tended to encourage extrinsic motivation, which often led to burnout over time.

The School Values study (Allen et al.) compared schools that prioritized mental health and belonging with those that mainly focused on grades and academic attainment. Students in supportive environments reported greater enjoyment of learning and motivation that lasted longer. In highly competitive schools students often pursued grades over good mental-health and showed less resilience when facing setbacks.

Gender differences also shape how students respond to school culture. Fan found that girls in competitive school environments were more likely to follow peer pressure, which increased motivation but also heightened anxiety. Boys were more likely to show disengagement or behavioral issues, depending on peer norms. Brooks et al. found similar patterns, noting that girls in high-achievement cultures often balanced motivation with anxiety, while boys' reactions varied more based on social influences.

The interaction between school culture and parental expectations can make these effects stronger. Iqbal et al. reported that when both home and school held rigid expectations, students felt caught in a cycle of proving themselves, which reduced genuine interest in learning. If at least one environment encouraged growth, effort, and emotional well-being, students were more likely to sustain motivation in a healthy way.

In general, school culture not only influences whether or not students are motivated, but also the kind of motivation they feel, and whether or not it can be maintained.

Academic Motivation and Mental Health

The connection between academic motivation and mental health is complex. The source of motivation strongly influences emotional well-being. Intrinsic motivation: driven by personal interest or desire to master a subject, is linked to better mental health outcomes. In contrast,

extrinsic motivation: driven by grades/rewards, or the fear of failure, is often connected to higher levels of stress. Within Bronfenbrenner's ecological systems theory, this relationship is shaped by the microsystem and the mesosystem.

Lester et al. found that students motivated by enjoyment and personal growth reported lower anxiety and fewer depressive symptoms than those motivated mainly by external pressures. This protective effect was especially strong in schools that encouraged learning rather than comparing performances.

High levels of extrinsic motivation, particularly in competitive environments, can worsen mental health problems. Suldo et al. studied academically gifted students and found that those motivated mainly by the need to keep high grades often experienced emotional exhaustion and lower life satisfaction. The risk was higher when students tied their self-worth directly to academic performance.

Gender differences also play a role in how motivation affects mental health. Fan reported that girls often showed high motivation regardless of stress, but when their motivation came from external pressures it was more strongly linked to burnout. Boys were more likely to disengage completely and give up if they felt they couldn't succeed, even though they were occasionally less academically motivated. However, they also tended to experience less emotional stress when extrinsically motivated.

Increasing intrinsic motivation can help protect mental health. Students in settings that encouraged curiosity, independence, and teamwork were more resilient and showed fewer signs of stress, according to Allen et al. Healthy motivation patterns were also associated with programs that incorporated mental health education into the classroom (Bastian and Rohlik).

In general, academic motivation is an important factor in students' success, but its source matters. Research shows that pressure-based, extrinsic motivation can lead to higher stress and poorer emotional outcomes, particularly for girls. Lewis et al. found that girls were more likely to show depressive symptoms when experiencing relational tension or conflict at home, suggesting that external pressures tied to expectations can increase psychological distress. In contrast, intrinsic and self-driven forms of motivation are associated with stronger academic performance and better emotional well-being. This contrast highlights the importance of understanding whether motivation stems from internal interest or external demands, which is central to the research questions in this paper.

Synthesis of Findings

The research shows that parental expectations and school culture work together to shape both the motivation and mental health of high school students in the United States. When both environments focus on achievement without balancing emotional support and one's own goals, students are more likely to develop extrinsically driven motivation, which is linked to higher stress, being overwhelmed, and burnout. When either the family or school environment prioritizes belonging and well-being, students are more likely to develop intrinsic motivation, which supports resilience and long-term engagement with learning.

Using Bronfenbrenner's ecological systems theory helps make sense of these patterns. Family and school each form a microsystem, and the way they connect, the mesosystem, can either lessen pressure students face or intensify it. If parents and teachers communicate well and share balanced goals, students can sustain motivation without sacrificing mental health.

Gender identity changes how these forces play out. Girls often maintain high motivation but face increased emotional strain when expectations are strict and performance-based. Boys' motivation tends to be more sensitive to peer norms, sometimes giving up in competitive settings that don't have opportunities for success. Nonbinary and transgender students face additional challenges, including limited representation and institutional support, which can make keeping up motivation even more difficult.

In general, the literature makes it clear that academic outcomes cannot be understood by only examining home and school together, motivational characteristics alone, or the differences in how those experiences differed by gender identity. Improving student well-being efforts should focus on all three (family, school, and individual), because focusing on just one doesn't address how pressures in one area could affect students in their daily lives.

Conclusion

This review explored how school culture and parental expectations influence the academic motivation and mental health of high school students, with gender identity shaping how these pressures are experienced. Upon synthesizing findings, when both home and school focus heavily on achievement without balancing emotional support, students are more likely to develop extrinsic motivation, working mainly for grades or approval, which leads to higher stress and burnout. Supportive environments tend to foster intrinsic motivation, which supports resilience and long-term success. Gender identity plays a key role: girls often remain highly motivated but experience more emotional strain under rigid expectations, boys' motivation is more influenced by peer norms, and nonbinary students face unique challenges due to limited representation and institutional support.

These findings align with Bronfenbrenner's Ecological Systems Theory, which emphasizes how relationships with parents, teachers, and peers (microsystem) and the connections between them (mesosystem) shape students' experiences. When expectations from home and school conflict or both focus solely on performance, stress increases. When they align around both achievement and emotional support, students show greater resilience and healthier motivation.

Limitations

This review is limited by its reliance on published literature and may not capture all perspectives, particularly those of marginalized student groups. Restricting the focus to U.S. studies and sources retrieved from the selected databases may have excluded relevant international or unpublished work. Emphasizing recent publications may have led to overlooking some of the significant older studies.

Additionally, although a formal reliability check was not completed, the review process was guided by clear inclusion criteria to create a reasonably consistent approach to identifying and analyzing articles. Research on nonbinary and transgender students remains especially limited and often relies on small sample sizes, which makes it difficult to generalize findings. Long-term studies are also scarce, leaving questions about how motivation and mental health change over time, and how combined pressures from home and school influence whether motivation becomes intrinsic or extrinsic.

Another limitation of relying solely on published literature is that it tends to favor quantitative studies over qualitative or mixed-methods research. As a result, student experiences and contextual factors, specifically related to identity, may be underrepresented. Much of the existing work focuses on measurable outcomes such as achievement or stress levels.

Future Research

In future research, adding a reliability check would strengthen accuracy in the selection process. Future studies should also use student self-reports or participatory methods that include students directly in the research process. These approaches would help capture their real experiences and emotions more accurately, which is often missing in current studies that rely heavily on quantitative data. Future research would also benefit from drawing on a broader range of cultural and educational contexts, including international studies, unpublished research, and community-based qualitative reports. Expanding the participant base to include students from different sociocultural and socioeconomic backgrounds, as well as a variety of school settings, would provide a more complete picture beyond high-achieving or well-resourced environments. Incorporating a consistency check among reviewers and combining both qualitative and quantitative approaches would create a more comprehensive understanding of how school culture and parental expectations shape motivation and mental health across gender identities.

The findings suggest that schools can integrate mental health education into academics, promote self-driven goals, and reduce emphasis on grades and test scores. Families can maintain high expectations as long as they communicate support and flexibility. By aligning academic and emotional goals at both the family and school level, consistent with Bronfenbrenner's model, students of all gender identities can be supported to stay motivated while protecting their mental health.

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The Exhaustion Divide: Gender Based Discrepancies in Sleep By Sofia Norton

Just as the workplace has its wage gap and the healthcare system has its pain gap, the bedroom too has its discrepancies: a persistent gender sleep gap that leaves women uniquely exhausted. For decades the gender sleep gap has been blamed on unequal division of labor, the societal and familial pressures that keep women up at night. This mental load keeps women's minds spinning long after the day has ended, a day lengthened by the infamous "second shift" as a disproportionate amount of household labor and caregiving is assigned to them. However, focusing solely on this perspective disregards new research that focuses on the internal reality. Women have fluctuating endocrine levels and unique sleep architecture that dictate their quality and length of rest. Gendered social expectations further influence how sleep disparities are experienced, but it is biological sex that shapes those physiological sleep processes. Ultimately, biological differences such as hormonal fluctuations, neurobiology and sex-specific physiology lay the essential foundation for the gender sleep gap, while social expectations further intensify these biological effects.

Biological differences in endocrine function and neurobiology predispose men and women to different sleep experiences. From fetal development onward, males and females begin to show significant biological differences, which later in life impact how respective genders experience sleep. On a basic biological level, women are more likely to report worse sleep and experience greater sleep-related impairment, while men tend to show less sleep efficiency. The origins of these factors can be studied through differences in hormonal fluctuations, neurobiology, and genetic predispositions. Alosta notes that women are more likely to experience sleep latency, nighttime awakenings, nightmares, suggesting biological predispositions. Reports show that "3,778 young adults (20.6 ± 0.86 years) indicate a higher prevalence of poor sleep quality in females than males (65.1% vs. 49.8%)" data that proved to have biological causes, as it accounted for variables such as depression, lifestyle and socio-demographic factors (Fatima). Mallampalli argues that sleep differences stem from "... biological and physiological differences" with "sex chromosomes and... gonadal hormones primarily contributing to these differences at the cellular, organ, and system levels". On an anatomical and physiological basis, biology predisposes women and men to varying levels of sleep quality, duration and perception.

Sleep architecture studies reveal measurable sex based differences in how sleep is regulated and experienced by the brain. Results from these studies show that "women obtained longer sleep durations and higher SE than men" (Hrozanova). Similarly, Mong notes that "women have less wakefulness after sleep onset, less light Stage 1 sleep, more slow-wave sleep (i.e., non-REMS), and more slow-wave activity (SWA; delta, $\sim 0.4\text{--}4$ Hz)" as opposed to their male counterparts. Together, these findings suggest that biological sex influences not only sleep duration, but its quality as well. Research into sleep architecture patterns continues to further insight into how biological sex influences regulation and structure of sleep. Another contributing biological factor is certain genetic and neurobiological markers that determine sleep and sleep quality between genders. Ovarian steroids, primarily progesterone and estrogen, "influence

circadian rhythms, including of sleep-wake activity, through direct or indirect effects on the master pacemaker – the suprachiasmatic nucleus” (Alzueta). This suggests that reproductive hormones directly impact neurological sleep regulation. Studies into neurobiology support this conclusion; Perger describes the brain as a “sexually dimorphic organ”, suggesting that neurology fundamentally shapes sleep differences between sexes.

Hormonal fluctuations across the lifespan are among the most significant biological contributors to gender sleep disparities. Earlier in life, these hormonal differences can be seen during puberty and with the start of the menstrual cycle, as “... fluctuating ovarian hormonal milieu [influencing] sleep throughout a woman's lifespan” (Mong). These effects can be seen in studies completed during the luteal phase, showing that during this time period “REM sleep had an earlier onset and REM sleep episodes were shorter” correlating with changes in progesterone and estradiol levels. These hormonal changes were further complicated by menstrual pain, affecting 84.1% of young women, contributing to additional sleep disturbances (Alzueta). In a menstruation study, Hrozanova observed that many women reported mood changes, stomach and back pain, headaches and tender breasts which worsened their sleep. While more subtle, studies have also found a tentative relationship between testosterone levels in men and their sleep (Mong), but more research needs to be done to get a full grasp on the correlation. The combination of endocrine variations and the pain and discomfort that accompanies menstruation provide evidence for how intrinsic biologic factors separately affect sleep between sexes.

Pregnancy, perimenopause, and menopause further demonstrate how female specific hormonal fluctuations disrupt sleep throughout adulthood. Pregnancy causes a myriad of sleep disturbances, the severity of which is dependent on each case. Some of these disturbances “... particularly during the third trimester of pregnancy, can be due to many factors such as discomfort, urinary frequency, fetal activity, RLS, or OSA” (Mallampalli). Pregnancy, a biological process that exclusively affects women, is a major event that not only affects the women's sleep while it is happening, but also after the baby is born, especially when it comes to breastfeeding. In menopause, estrogen levels generally decline in women and this leads to vasomotor symptoms such as night sweats, which is a purely biological sleep disruptor (Decker). One of the most common complaints of menopausal women is about their sleep, reported by 33-51% of women, as well as reports of increased insomnia (Mong). In a study, “... late-reproductive aged women in SWAN, found a 5% decline in sleep efficiency and a 25-minute decrease in total sleep time in the premenstrual week relative to the prior week” (Alzueta). Menopause further demonstrates how significant biological changes in women's lives, with limited equitable comparisons in men's life, greatly contributes to gender sleep discrepancies.

Biological differences also influence the prevalence and presentation of sleep disorders and conditions among men and women. Research has found that female adolescents have a high reported prevalence for sleep disorders with possible reasoning including biological evolutions across the lifespan (Alost). An example of a sleep disorder that specifically impacts women is polycystic ovary syndrome (PCOS) which is shown to have a huge impact on women's ability to fall asleep (Alzueta). Other common sleep disorders found in women are restless leg syndrome

(RLS), obstructive sleep apnea (OSA) (Mallampalli), and insomnia, with women “at a 40% greater risk for developing insomnia”, a risk factor that only increases with age (Mong). It is important to note that, historically, sleep apnea was seen as significantly more common in men due to certain structural differences in upper airway anatomy, with recent studies finding “that the average prevalence of OSAS is 27% in men and 22% in women” (Perger). This relatively small difference, combined with evidence of possible underdiagnoses, may ignore that rates of OSAS are not as low in women as previously thought. This exposes a possible link between how biological differences in symptom presentation may also play a critical role in how sleep disorders are diagnosed and treated. Additionally, women are also more likely to experience chronic pain that interferes with sleep such as migraines, rheumatological issues and musculoskeletal pain (Casale), and Mallampalli notes that fibromyalgia is a chronic pain condition that “affects 7% of women over the age of 60 years and is characterized by poor sleep efficiency”. These findings provide insight to how interconnected women's sleep conditions are with biological origins. Disorders and conditions that affect sleep are often experienced differently between men and women and can be explained by biological differences such as structural anatomy and endocrine variations.

Despite often showing higher sleep efficiency than men, women consistently report poorer subjective sleep quality. Polysomnography studies generally show that women get objectively better sleep than men. Mong notes that women experience “... less wakefulness after sleep onset, less light Stage 1 sleep, more slow-wave sleep..., and more slow-wave activity” than men. Despite this, women consistently report subjectively poorer sleep quality, suggesting that hormones and neurobiology might have a role, not just on how sleep is measured, but also in how it is experienced. Mong suggests that this paradox comes from the fact that, while women have higher sleep efficiency, they are more affected by their inherent hormonal fluctuations and shorter internal circadian rhythms. These findings may also suggest that women's bodies compensate for hormonally induced sleep issues by increasing their sleep efficiency and quality.

Along with the impact of biological factors, societal expectations and gender based labor roles also significantly impact sleep disparities. In a research study about insomnia, Zeng found that “[f]emales are more vulnerable to negative socioeconomic factors, such as lower income or education level”. While biology may provide the template for sleep differences, it is society that dictates how sleep is experienced. Due to familial responsibilities and household duties, women have less time in a day to sleep, especially accounting to “changes in sleep routines and sleep quality due to caregiving responsibilities” (Alosta). Sleep quality assessments often have strong correlation with factors such as anxiety and depression which are more common in women due to societal influences. This becomes a compounding issue as these variables increase insomnia (Mong). Due to societally and biologically induced poor sleep, “caffeine intake is more frequent in females predisposing them to sleep disturbances and short sleep duration” and women's sleep is more likely to be impacted by both their job and family, rather than men whose sole focus is work (Alosta). Additionally, a study showed that women have more nighttime awakenings which “is often marked by child-rearing, which is associated with increased psychological and

behavioral demands for women, many of which impact sleep” (Decker). Biology plays a central role in all aspects of sleep discrepancies and lays the groundwork for societal influences to further exacerbate.

Social expectations when it comes to caregiving and emotional availability significantly intensify biologically rooted sleep disparities for women. Women are often expected to manage nighttime caregiving responsibilities, attending to infants, and taking care of sick children, which greatly increases sleep disruptions and chronic sleep deprivation. Parenting responsibilities often fall disproportionately on mothers, which is a significant sleep disruptor not only due to the children themselves but trying to balance all that comes with keeping a household running. Decker recognizes that sleep is a result of many factors as “gender differences (individual) in sleep health are influenced by family roles (interpersonal) and policies such as family leave and health care (societal)”. These societal pressures do not replace any biological factors, but they amplify physiological vulnerabilities and help provide a fuller picture on gender affected sleep.

Ultimately, the gender sleep gap cannot be explained by any one factor. It is a multifactorial issue combining biology caused hormonal fluctuations and neurological differences, with society and its disproportionately assigned caregiving roles. Biology creates underlying issues for society to exacerbate; it underscores all aspects of life and provides the vulnerabilities for societal expectations to exploit. In the end, understanding these correlations is essential for addressing sleep inequalities as it helps improve sleep medicine, workplace expectations and caregiving systems in ways that better account for the distinct physiological demands placed on women throughout their lifespans.

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Playing for Passion or Pressure: Intrinsic vs. Extrinsic Motivation in Chinese Sports and Education By Cengan Zhang



Abstract

Chinese education and youth sport have long relied on external pressure, high-stakes evaluation, and visible achievement as markers of success. This paper examines whether that model aligns with how people in China themselves understand healthy and sustainable motivation. Drawing on original survey data from Suzhou ($N = 104$), we analyze perceptions of intrinsic motivation—participating for enjoyment, interest, or personal meaning—versus extrinsic motivation, such as parental pressure, prizes, rankings, and status competition. Across the survey, respondents consistently associated intrinsic motivation with stronger mental health, longer participation, lower burnout, greater happiness, and even broader academic and life benefits. At the same time, the data do not support a simplistic rejection of all external incentives: respondents also acknowledged that extrinsic motives can contribute to short-term success. The stronger pattern, however, is that passion is seen as more durable than pressure.

These perceptions matter because they challenge the logic of two influential Chinese institutions: exam-oriented education and medal-centered sport development. Existing scholarship in self-determination theory, Chinese education, and sport psychology similarly suggests that environments emphasizing autonomy, mastery, and interest tend to support persistence and well-being, while highly controlling climates increase anxiety, dropout risk, and psychological strain (Ryan and Deci; Yu et al.; O'Rourke et al.; Wu et al.). The recent popularity

of Jiangsu's grassroots football league, commonly known as "SuChao," offers a concrete social example of how accessible, low-pressure, community-centered sport can mobilize enthusiasm at scale ("China Grassroots Football League Kicks Off Cultural Tourism Boom").

Using sport as a lens into wider educational culture, this paper argues that China does not need to choose between excellence and enjoyment. Rather, the evidence suggests that more intrinsically supportive environments may produce both healthier students and more sustainable achievement. The paper concludes with recommendations for schools, families, coaches, and policymakers seeking to rebalance motivation toward passion over pressure.

Keywords: intrinsic motivation, extrinsic motivation, China, youth sport, exam-oriented education, SuChao, self-determination theory

Introduction

The contrast between intrinsic and extrinsic motivation has become increasingly important in contemporary debates about education, sport, and youth development. Self-determination theory defines intrinsic motivation as engagement driven by inherent satisfaction, while extrinsic motivation refers to activity pursued for separable outcomes such as rewards, approval, or avoidance of punishment (Ryan and Deci). That distinction matters in China because many of the country's most visible success stories have emerged from systems built on disciplined effort, competition, and measurable outcomes. In both schooling and sport, achievement has often been pursued through external pressure: exam scores, rankings, admissions, medals, and family expectations.

That model has produced real gains. China's education system has been associated with high test performance, and its elite sports system has delivered major international success. Yet a growing literature argues that these outcomes have often been accompanied by hidden developmental costs. Scholars using self-determination theory to analyze Chinese education argue that high-stakes structures can frustrate students' needs for autonomy, competence, and relatedness even while producing strong surface-level achievement (Yu et al.). Related research in Chinese classrooms shows that autonomy-supportive teaching improves students' interest, perceived competence, and self-endorsed engagement, including in settings often assumed to be too collectivist or authority-centered for such dynamics to matter (Zhou, Ma, and Deci). In other words, the issue is not whether Chinese students and athletes can perform under pressure. It is whether pressure alone is the best way to cultivate long-term excellence.

Sport offers a useful window into this broader question. Youth sport is highly visible, emotionally charged, and socially embedded. It reveals what a society rewards, what it fears, and how adults shape children's aspirations. International sport psychology research has repeatedly found that mastery-oriented climates support persistence and enjoyment, whereas heavily controlling or ego-oriented climates are associated with anxiety, amotivation, and dropout (García Calvo et al.; O'Rourke et al.; Wu et al.). In China, this distinction is especially meaningful because organized sport has historically been split between a medal-driven elite pathway and a weaker tradition of broad, joy-centered community participation.

Recent reforms also suggest that official institutions recognize the problem. The 2021 “double reduction” policy, for example, explicitly sought to reduce excessive homework and off-campus tutoring burdens in compulsory education (“China Reiterates Implementation of ‘Double Reduction’ Policy”). Recent developments in Jiangsu point in a similar direction from below. The explosion of interest in the grassroots football league known as “SuChao” showed that a low-threshold, community-centered, amateur competition could attract massive audiences and activate local pride, tourism, and participation without relying on elite professional status (“China Grassroots Football League Kicks Off Cultural Tourism Boom”). That example does not prove that all competitive structures are harmful. It does, however, suggest that when sport becomes accessible, humorous, communal, and emotionally inviting, people respond with enthusiasm rather than obligation.

This paper uses original survey data from Suzhou to examine how respondents perceive intrinsic and extrinsic motivation in sports and education. Three questions guide the analysis. First, how do respondents evaluate the effects of intrinsic and extrinsic motivation on success, mental health, and sustained participation? Second, how do they interpret the current social culture surrounding youth sport, especially parental pressure, competition, and status-seeking? Third, what reforms do they appear to support, particularly in light of the SuChao phenomenon? The central argument is that the survey reveals not simply a preference for “fun,” but a broader critique of a pressure-heavy developmental culture. Respondents broadly recognize that external incentives may help produce short-term achievement, yet they associate intrinsic motivation with healthier and more sustainable outcomes.

Literature Review

Intrinsic and Extrinsic Motivation

Self-determination theory provides the core conceptual framework for distinguishing intrinsic and extrinsic motivation. Ryan and Deci argue that intrinsic motivation supports high-quality learning, creativity, and persistence because it arises from interest and enjoyment rather than external control. Deci and Ryan further contend that contexts supporting autonomy, competence, and relatedness help individuals internalize goals in healthier ways, whereas highly controlling environments undermine volition and long-term well-being. This does not mean all external motives are equally harmful. The theory allows for a spectrum, including forms of extrinsic motivation that become more self-endorsed through internalization. Still, the general pattern across the literature is that when rewards, surveillance, and pressure dominate, intrinsic interest tends to weaken.

Chinese Education and the Problem of Pressure

This framework has been especially influential in scholarship on Chinese education. Yu et al. argue that self-determination theory offers a useful way to understand the paradox of Chinese schooling: exceptionally strong academic outcomes can coexist with reduced autonomy, constrained creativity, and high psychological strain. Their analysis of high-stakes testing

suggests that exam-oriented structures may secure performance while undermining deeper learning and broader development. Zhou, Ma, and Deci, working with rural Chinese children, similarly found that autonomous motivation predicted interest, competence, and perceived choice, while controlled motivation was associated with weaker classroom adjustment. Their study is important because it challenges the claim that autonomy is merely a Western cultural value. Even within more authority-centered Chinese educational contexts, autonomy support mattered.

At the same time, the literature is more nuanced than a simple intrinsic-good/extrinsic-bad dichotomy. A large longitudinal study of Chinese students found that intrinsic motivation positively predicted academic performance, but that extrinsic motivation could also help when intrinsic motivation was low; its effects were most problematic when layered onto already high intrinsic engagement (Liu et al.). This nuance matters for the present paper. The question is not whether external incentives should disappear entirely. It is whether current Chinese institutions lean too heavily on pressure, rankings, and instrumental reward, thereby crowding out more sustainable forms of engagement.

Chinese Sports, Anxiety, and Dropout

Research in sport psychology points in a similar direction. García Calvo et al. found that adolescent athletes were more likely to persist in sport when self-determined forms of motivation were stronger and basic psychological needs were better satisfied. O'Rourke et al. showed that high parental pressure, especially when combined with low-mastery or high-ego motivational climates, was associated with elevated trait anxiety in young athletes. In the Chinese context, Wu et al. found that task-oriented climates were positively associated with basic psychological needs and healthier motivation among winter sports athletes, whereas ego-oriented climates were linked to amotivation, psychological distress, and burnout. Together, these studies suggest that the emotional climate surrounding participation matters as much as the activity itself.

This literature is particularly relevant for China because sport has often been valued instrumentally. Youth sport may be encouraged for school admissions, status competition, or elite development rather than for broad participation and enjoyment. In such settings, children can easily learn that sport is another arena in which they are judged, compared, and pushed. The likely result is not only stress but also a narrower base of long-term engagement. A society may produce a small number of highly successful performers while discouraging a much larger number of ordinary participants.

SuChao and the Social Meaning of Grassroots Sport

The popularity of SuChao provides an especially useful contemporary example because it appears to reverse that logic. Reporting from People's Daily described the league as an amateur tournament featuring players from many walks of life, from students to delivery workers and office employees, and emphasized its "carnival" atmosphere of local pride, humor, and accessible participation ("China Grassroots Football League Kicks Off Cultural Tourism Boom"). The same reporting linked the league not only to football enthusiasm but also to cultural

tourism, local consumption, and renewed city identity. In policy terms, this matters because the Chinese state has increasingly stressed mass participation and public fitness alongside elite sporting ambitions ("Outline Released to Build China into Sports Leader").

SuChao therefore matters sociologically as well as athletically. It suggests that enjoyment, identity, and belonging can be socially productive. The league's success indicates that low-pressure sports do not have to be trivial or unserious. They can generate attention, discipline, local commitment, and economic spillovers. This makes SuChao a particularly useful case for thinking about motivation in a wider Chinese context: it shows that when people are invited into sport through community and pleasure rather than only competition and prestige, participation can deepen rather than decline.

Methodology

This study draws on an original survey titled "Sports Motivation and Its Impact on Students and Society." The survey included thirty-five closed-ended questions and one open-ended question. Most of the items asked respondents to compare the effects of intrinsic motivation—participation for enjoyment, interest, or personal meaning—with extrinsic motivation, including rewards, competition, or parental pressure. The final section focused specifically on the Jiangsu grassroots football league "SuChao," asking respondents how they interpreted its local and social significance.

The survey was distributed online in Suzhou through social media and school-community networks. To keep the chronology consistent with the SuChao questions, the survey is treated here as a 2025 instrument rather than a 2024 one. A total of 104 valid responses were collected. Participants included students, educators, and community members familiar with local sports culture. Because sampling was voluntary and non-random, the dataset should not be treated as nationally representative. It is best understood as a localized but revealing snapshot of how one urban Chinese community thinks about motivation, youth sport, and reform.

Each closed-ended question offered a yes/no response. The analysis therefore focuses on descriptive percentages rather than inferential statistics. To make the findings clearer and more readable, the results are grouped into four thematic areas: success and persistence, mental health and burnout, the perceived culture of sport, and views on reform and SuChao. The open-ended responses were reviewed qualitatively for recurring themes, particularly around community, social connection, and the educational value of sport, though the present paper prioritizes the quantitative findings because they provide the clearest overview of the pattern.

Results

The survey results reveal a strong and consistent pattern: respondents broadly favored intrinsic motivation as the healthier and more sustainable basis for student sport. They did not entirely reject extrinsic motivation, but they repeatedly associated external pressure with stress, shorter participation, and a less desirable sporting culture.

Table 1 summarizes the most important findings related to motivation, mental health, persistence, and reform.

Table 1. Selected Survey Findings on Motivation and Student Outcomes

Survey Item	% Yes
Athletes are more successful when they participate out of interest	89.4%
Participating in sports for fun benefits students' mental health	96.2%
Extrinsic motivation has a more negative effect on mental health than intrinsic motivation	63.5%
Intrinsic motivation enables longer-term participation in sport	97.1%
Students participating only for external rewards are less likely to continue	69.2%
Students who enjoy sports are happier than those driven mainly by pressure	80.8%
Schools should offer more interest-based sport choices	90.4%
Reducing pressure would increase student participation in sports	87.5%

Note. Percentages indicate “Yes” responses.

Several points stand out immediately. First, respondents associated intrinsic motivation not only with enjoyment but also with success. Nearly nine in ten agreed that athletes are more successful when they participate out of interest (89.4%), and 93.3% believed that students who are passionate about sports are more likely to succeed at higher levels of competition. This is a crucial result because it challenges the assumption that “fun” and “serious achievement” are opposed. For these respondents, passion is not a soft alternative to excellence; it is one of the main conditions that makes excellence sustainable.

Second, the mental-health pattern was even stronger. Fully 96.2% agreed that participating in sports for fun benefits students’ mental health, while 63.5% said extrinsic motivation has a more negative impact on mental health than intrinsic motivation. A majority also associated externally driven participation with greater stress and anxiety. These responses suggest that respondents do not simply see intrinsic motivation as more pleasant. They see it as psychologically safer.

Third, intrinsic motivation was strongly linked to persistence. Almost all respondents (97.1%) agreed that students motivated by enjoyment or personal interest are more likely to remain involved in sport over time. Conversely, 69.2% believed that students who participate only for external rewards are less likely to continue. This aligns closely with the wider sport literature on dropout and supports the view that external incentives may be able to mobilize effort in the short term while weakening long-term commitment.

Fourth, respondents extended the value of intrinsic motivation beyond sport itself. More than 90% believed that interest-driven sports projects can improve academic performance, and 93.3% believed that students who pursue sport from genuine interest are more likely to succeed in other areas of life. These responses suggest that respondents understand sport as part of a wider developmental ecology. Enjoyment, in their view, is not a distraction from success; it is one pathway toward it.

The survey also revealed an important nuance: respondents did not deny that extrinsic motivation can produce results. In fact, 82.7% agreed that athletes can be successful when driven

by competition or rewards. This is one of the paper's most important findings because it prevents an overly simplistic interpretation of the data. Respondents are not arguing that external incentives never work. Rather, they appear to believe that such incentives are more fragile, more stressful, and less likely to sustain engagement over time.

Perceived Culture of Sport

The survey results were also critical of current social attitudes surrounding youth sport. A majority of respondents believed that parents often push children into sport for reasons other than enjoyment, such as competition, admissions, or status. Likewise, many respondents believed that Suzhou's sports culture emphasizes competition more than cultivating a genuine love of sport, and 70.2% agreed that sports are often used to improve social status or impress others.

These results matter because they suggest that the respondents' preference for intrinsic motivation is also a critique of the surrounding culture. The issue is not only individual psychology. It is also a social environment in which children may learn that sport matters mainly as performance, display, and comparison. In this kind of environment, even potentially joyful activities can become extensions of academic pressure and family reputation. The survey therefore points to a broader sociological pattern: when institutions and families frame sport instrumentally, participation risks becoming another form of competition rather than a source of flourishing.

Reform and the Meaning of SuChao

Respondents were highly receptive to reforms that would make sport more interest-driven and less pressured. More than four in five agreed that sports would be more attractive in Suzhou if programs emphasized fun and interest rather than rewards and competition. More than 90% believed schools should provide greater opportunities for students to choose sports based on interest instead of performance requirements. An equally strong majority believed that lowering pressure would increase student participation.

Table 2 presents the survey findings related specifically to SuChao.

Table 2. Survey Responses on the Social Significance of SuChao

Survey Item	% Yes
SuChao can promote football culture and improve football ability in Jiangsu	94.2%
SuChao's popularity has made people recognize football's importance	86.5%
Other provinces could organize similar grassroots competitions	89.4%
SuChao has improved the status of football in Chinese sport	90.4%

Note. Percentages indicate "Yes" responses.

The SuChao responses were overwhelmingly positive. Most respondents believed that the league promotes football culture and skill development in Jiangsu, raises awareness of football's importance, and improves the status of football in Chinese sport. They also believed that other

provinces could create similar competitions. These responses imply that respondents see SuChao not merely as entertainment but as a workable model of reform.

One result complicates this picture in a useful way. A majority still believed that many people prefer to watch more professional competitions than SuChao. This suggests that respondents do not imagine grassroots sport as a replacement for elite sport. Instead, they seem to see it as a necessary complement: a broader, more participatory foundation that can coexist with higher-level professional competition. That distinction is important. The survey is not anti-competition. It is anti-monopoly by competition.

Discussion

The findings point to a consistent social logic: respondents appear to believe that pressure can produce effort, but passion produces durability. This distinction helps explain why the survey does not reject extrinsic motivation wholesale, yet still shows such strong support for intrinsic motivation across well-being, persistence, and reform. In the respondents' view, the problem is not that rewards, rankings, or competition exist. The problem is that they have become too dominant in key institutions of youth development.

This interpretation fits the broader scholarship. The survey's strong support for intrinsic motivation aligns with self-determination theory's claim that interest, autonomy, and felt competence support higher-quality engagement over time (Ryan and Deci; Deci and Ryan). It also echoes Chinese educational research showing that autonomy support matters in Chinese classrooms, despite assumptions that obedience or external discipline are more culturally appropriate forms of motivation (Yu et al.; Zhou, Ma, and Deci). In sport, the same basic pattern appears: controlling or ego-centered climates tend to intensify anxiety and burnout, while mastery-oriented climates better sustain participation (García Calvo et al.; O'Rourke et al.; Wu et al.).

At the same time, the survey's nuance is analytically important. Because respondents also acknowledged the effectiveness of extrinsic incentives in producing success, their answers do not support a romantic anti-structure argument. They are closer to a balance argument. This is similar to the finding by Liu et al. that extrinsic motivation in China can sometimes coexist with good academic outcomes, particularly when intrinsic motivation is weak. The problem arises when systems over-invest in external pressure and under-invest in meaning, enjoyment, and student ownership. Under those conditions, success may remain possible, but its costs increase and its durability weakens.

The responses about parental pressure, status competition, and Suzhou's local sports culture also suggest that motivation in China should be treated as a social phenomenon, not only an individual trait. Children do not encounter sport in a vacuum. They encounter it through schools, family expectations, admissions systems, local norms, and status displays. That broader culture shapes which motives are seen as legitimate. If a child learns that sport matters only when it helps one get ahead, then even leisure becomes instrumental. The survey indicates that many respondents recognize this dynamic and are dissatisfied with it.

SuChao is especially revealing here because it offers a counter-model. The league's appeal appears to come not from technical superiority or national prestige, but from community identity, accessibility, humor, and participation. That is precisely why it matters for this paper. SuChao suggests that an intrinsically appealing environment can still generate commitment, loyalty, and even economic impact. It is therefore a powerful rejoinder to the assumption that lowering pressure necessarily lowers seriousness. In this case, lowering pressure seems to have broadened engagement.

There is also a wider implication for Chinese education. The survey repeatedly linked enjoyable sport to better academic and life outcomes. While such perceptions do not prove causation, they do reveal a wider public intuition that well-being and achievement need not be opposed. This matters in a social environment often shaped by the language of sacrifice, competition, and "involution." The respondents appear to be articulating a different developmental ideal: not the rejection of effort, but the integration of effort with interest.

The study has limitations. The sample is small, localized, and non-random. The responses are perceptual rather than behavioral, so they capture attitudes more than verified outcomes. Some participants may also have answered in ways that reflect ideals rather than daily practice. Even so, the consistency of the pattern is analytically useful. The survey does not claim to provide a definitive account of all Chinese students or parents. It does, however, reveal a coherent set of beliefs that resonates strongly with both existing scholarship and visible social developments such as SuChao.

Recommendations

The survey findings support several practical recommendations for educators, schools, coaches, parents, and policymakers.

First, schools should expand student choice in physical education and extracurricular sport. If students are allowed to choose among a wider range of activities, they are more likely to develop personal ownership and sustained interest. This recommendation is directly supported by the survey and is also consistent with autonomy-supportive educational research in China (Zhou, Ma, and Deci).

Second, schools and sports programs should shift from purely performance-centered evaluation to a mixed model that values participation, improvement, teamwork, and enjoyment alongside competitive results. This does not require abandoning standards. It requires redefining success more broadly so that more students can experience sport as meaningful rather than merely judgmental.

Third, teacher and coach training should explicitly address motivational climate. Research shows that mastery-oriented environments reduce some of the harms associated with pressure-heavy settings, while ego-centered climates heighten anxiety and amotivation (O'Rourke et al.; Wu et al.). Coaches and PE teachers should therefore be trained not only in technique and discipline, but also in how to create psychologically supportive environments.

Fourth, parent education matters. Because respondents frequently identified parents as a source of instrumental pressure, schools should communicate more clearly that sport's value lies

not only in admissions advantages or trophies but also in long-term health, resilience, identity, and social connection. This is especially important in contexts where children experience sport as another domain of resume-building rather than enjoyment.

Fifth, cities and provinces should continue building low-threshold, community-centered sports events. SuChao's popularity suggests that grassroots sport can serve educational, cultural, and economic goals at once ("China Grassroots Football League Kicks Off Cultural Tourism Boom"). Expanding similar leagues would not weaken elite development; it would likely strengthen the broader ecosystem from which stronger athletes, fans, and communities emerge.

Conclusion

This paper set out to examine how people in one Chinese urban context understand intrinsic and extrinsic motivation in sports and education. The findings are clear. Respondents overwhelmingly associated intrinsic motivation with stronger mental health, greater happiness, lower burnout, longer participation, and broader developmental success. They did not deny that extrinsic motivation can produce achievement, but they consistently treated it as less stable and more psychologically costly.

The sociological importance of this result lies in its challenge to a deeply familiar Chinese developmental script. For decades, schooling and sport have often been organized around externally visible performance, and this arrangement has indeed produced impressive outcomes. Yet the survey suggests that many people increasingly understand the limits of pressure-heavy systems. They appear to want a more balanced model—one in which effort remains important, but enjoyment, autonomy, and meaningful participation are no longer treated as optional luxuries.

SuChao makes that possibility easier to imagine. Its success shows that broad participation, local pride, and genuine enjoyment can create seriousness without coercion. In that sense, the league is more than a football story. It is a glimpse of what a less instrumental culture of development might look like.

The broader lesson is straightforward: if China wants not only high-performing students and athletes but also healthier, more resilient, and more deeply engaged young people, then motivation must be reconsidered. Lighting a fire within may not eliminate the role of structure, discipline, or competition. But it may produce a stronger and more lasting flame than pressure alone.

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Feeling Invisible: AAPI Identity, Media Influence, and the Psychological Impact of Curriculum Absence By Evelyn Leung

Abstract

This paper discusses how insufficient representation of Asian American and Pacific Islander (AAPI) in formal education and the increasing prominence of media may determine both individual identity construction and the general opinion of a wider population. In spite of growing focus on diversity in education, AAPI histories are still underrepresented in the U.S. curriculum, which frequently leads to a lack of knowledge and inadequate cultural awareness. A mixed-methods design was employed, utilizing quantitative survey data and qualitative responses to measure both the familiarity of the participants with major AAPI events, their main sources of information, and the psychological outcomes of the gaps in the curriculum. The results show that there is usually little or no coverage of AAPI content in formal schooling, and that more than 80% of the participants have insufficient exposure. As a result, people use media, especially digital and social media, as the main sources of learning. While the media increases accessibility, it also contributes to uneven awareness and reinforces selective narratives. Moreover, around 70% of participants mentioned that their sense of identity and belonging were negatively affected, which emphasized the psychological impact of underrepresentation. The paper also reveals that there is a disconnection between perception and historical reality because familiarity is organized around recent and highly visible events in addition to underlying historical experiences. The results emphasize the importance of more inclusive curricula and better media depiction.

Introduction

The discourse of cultural identity has taken center stage in recent years, especially in learning areas, such as representation and inclusion. In spite of this change, Asian American and Pacific Islander (AAPI) history is significantly underrepresented in U.S. educational programs. Their stories tend to be overlooked, given a brief mention, or not mentioned at all, which illustrates a larger trend of structural invisibility underpinning mass culture and identity formation.

A deficiency in organized AAPI education also leads to misconceptions, disjointed knowledge, and the reproduction of stereotypes. Although media can be utilized to address these gaps, it can also be used to misrepresent cultural knowledge by simplifying or choosing what to convey (Sleeter, 2011). Thus, the absence of systematic and correct educational frameworks will lead to the persistence of these gaps and the further evolution of incomplete thoughts about the experience of AAPIs.

This paper analyses the role of curriculum absence in AAPI identity, the ways in which media addresses or misinterprets this absence, and the disconnect between the public and history. The position of this paper is that curriculum disjuncture and media impact work together to create both the world-view and self-construction, frequently boosting invisibility and

constraining full comprehension of cultural identity. It is a belief that media and curriculum gaps can reciprocally influence the exterior perception and inward identity, which are likely to reinforce invisibility and suppress in-depth understanding of AAPI communities.

Literature Review

AAPI in Education.

Historically, the underrepresentation of Asian American and Pacific Islander (AAPI) histories in the U.S. curriculum is a structural issue that has persisted. Conventional textbooks and classroom histories give greater focus to Eurocentric views and commonly relegate AAPI experience to peripheral or back-story status (Sleeter, 2011). Such restricted integration is part of a larger trend of cultural invisibility, which is limiting the exposure of students to different historical stories.

The marginalization of AAPI histories in the curriculum is symptomatic of more systemic forms of marginalization. An (2016) identifies that Asian Americans are often not encompassed in the mainstream U.S. history curriculum or only within specific contexts, like immigration history or participation in war. These subjects, traditionally covered, have usually not included enough depth or continuity, creating fragmented knowledge of AAPI history. This is indicative of a wider trend of tokenization, by which AAPI content is presented as a few lines or individualized lessons instead of part of the national narrative.

Ethnic Studies is a set of corrective scholarship, as determined by scholars. According to the arguments by Selena Leiter (2011), inclusive curricula based on marginalized communities promote academic interaction, critical analysis, and identity formation. With a focus on agency, diversity, and intersectionality, Ethnic Studies disrupts the mainstream discourse and defies stereotypes, including the myth of the model minority. Nevertheless, access to these curricula has been unequal, and most schools still use standardized material that creates gaps in representation (Lee, 2009). Consequently, students resort to other sources, especially the media, to acquire information about AAPI identities and histories.

Media Representation and Culture Learning.

Where formal education is not thoroughly developed, the media serves more as a cultural educator, influencing the way people perceive AAPI identity. Media channels like television, film, and most notably social media afford easier avenues of cultural education, whereby AAPI voices get a chance to narrate stories and create awareness (Sleeter, 2011). Besides, social media platforms such as Instagram, TikTok, and YouTube have increased the possibilities of representation because they allow individuals to create and share content. Such platforms have the power to elevate neglected histories and offer more alternative views than mainstream media (Lee, 2009). However, their influence is shaped by algorithmic systems that prioritize engagement over accuracy, often promoting simplified or sensationalized content.

This process is explainable by using Cultivation Theory, which postulates that the attitude to social reality is formed by repeated exposure to media (Gerbner et al., 1986). By being repeatedly exposed to a few or limited representations, the stereotyped representations can be made normalized. In a similar manner, the Social Identity Theory emphasizes the importance of representation in the development of the self-concept (Tajfel and Turner, 1979). When media representations lack depth or are stereotypical, people might find it difficult to create a whole and valid sense of identity (Lee, 2009). The media, therefore, plays a dual role, where on the one hand, it contributes to the dissemination of cultural knowledge, and on the other, it runs a risk of distortion and simplification, especially in cases where there are no proper educational backgrounds.

Stereotypes and Perception Formation

The media is a major contributing factor to reinventing AAPI stereotypes, especially the model minority myth and the perpetual foreigner stereotype. The model minority is a narrative that creates the image of uniformly successful and independent AAPI people, which conceals internal diversity and structural inequalities (Lee, 2009). Although this stereotype appears to be a good thing, it fosters unrealistic expectations and systemic silencers.

On the other hand, the perpetual foreigner stereotype portrays AASI people as outsiders regardless of nationality or generational identity. This perception strengthens the exclusion of nationality and is more frequently reinforced by media images, which underline the cultural difference (Tuan, 1998). Such stereotypes influence more than just interpersonal perceptions but also wider societal views. The result of this is that without a thorough education, people tend to follow media-based narratives, and simplified representations may prevail. This leads to a lack of empathy, misunderstanding of AAPI experiences, and the continuation of prejudiced attitudes.

Mental Effect and Development of Identity.

The cumulative impact of curriculum absence and media representation is witnessed in identity development and psychological welfare. An important notion is identity validation, or how well people feel their experiences are represented in the narratives of a society. In cases where AAPI histories are not included in an educational process, the individuals can feel invisible and detached, which can complicate the formation of a coherent identity. Also, inclusive curricula have been found to enhance self-esteem, belonging, and cultural awareness. Without them, people are prone to seeking validation in the media (Sleeter, 2011). Although media may offer other platforms within which identity is explored, it may also strengthen stereotypes or exert pressure to conform to dominant discourse.

A habitual exposure to limited representations can result in internalization of societal narratives, impacting self-perception and desired aspirations. This may dilute cultural identity and lessen a sense of belonging. On the other hand, precise and varied representation, either educational or through media, can promote cultural pride, self-awareness, and a sense of belonging. As such, the literature tends to indicate that media influence and curriculum absence

are two forces that are profoundly intertwined and construct perception and identity. These gaps should be addressed to facilitate more precise knowledge and facilitate healthier identity development.

Methodology

Research Design

The research design of this study applies a mixed-methods cross-sectional research design to test the association between curriculum absence, media, and AAPI identity formation. The mixed-methods approach combines quantitative survey data with qualitative analysis, allowing for both statistical analysis and deeper interpretation of participants' experiences. To obtain perceptions and attitudes at one given time, a cross-sectional design was chosen to obtain a snapshot of current awareness, educational exposure, and identity-related outcomes.

Participants

The participants were recruited by distributing a structured survey online. The sample consists of people of various levels of education, such as middle school, high school, undergraduate, graduate, and adult learners. The respondents also depict a diversity of ethnicities, such as East Asian, Southeast Asian, South Asian, Pacific Islanders, and mixed-heritage identities. Also, the participants are geographically dispersed in various regions in the United States. This diversity makes the study more effective at getting a range of views on AAPI representation and identity formation.

Instrument

A survey in the form of a structured questionnaire was used to collect data and evaluate four main areas. First, knowledge of AAPI history was assessed using knowledge of major historical events. Second, respondents indicated their major knowledge sources, such as formal education, media, and community-based learning. Third, the survey assessed views on curriculum representation, including aspects of the frequency and intensity of AAPI coverage in schools. Lastly, the psychological influence and identity formation were evaluated by using Likert-scale items and open-ended questions about the feeling of belonging, representation, and connection to their culture.

Data Analysis

Descriptive statistics with frequencies and percentages were used for the quantitative data and provided insights into patterns in awareness, sources of knowledge, and perceptions of representation. Thematic analysis was utilized to analyze qualitative responses and identify common themes, including identity, invisibility, and representation (Tuan, 1998). This integrated method allows one to learn about the quantifiable tendencies and the subjective experiences of the situation.

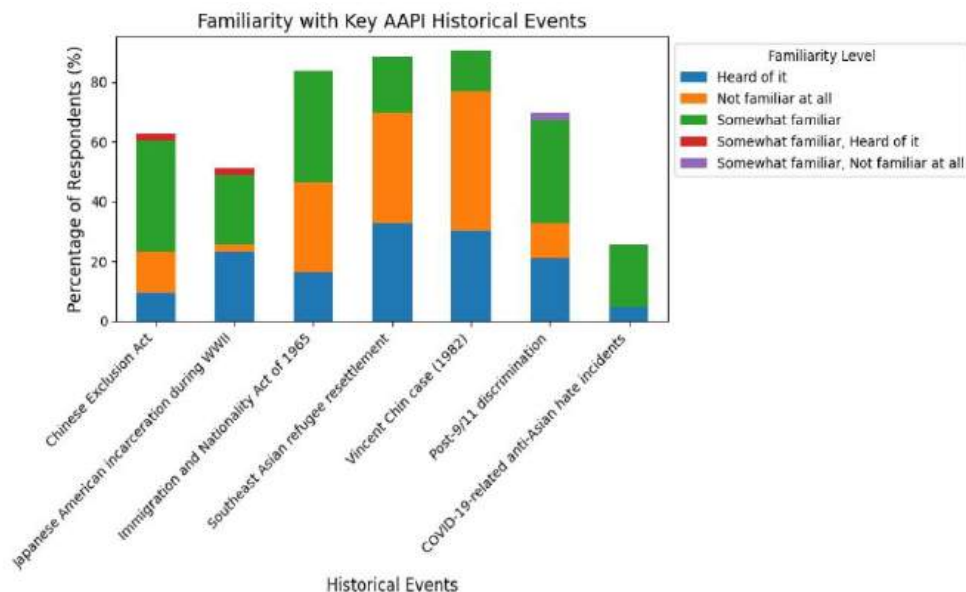
Table 1*Shows the Survey Structure Overview*

Section	Focus	Type of Data
Awareness of AAPI History	Familiarity with key events	Quantitative
Sources of Knowledge	School vs media learning	Quantitative
Curriculum Representation	Frequency and depth of inclusion	Quantitative
Psychological Impact	Identity, belonging, perception	Mixed (Quantitative + Qualitative)

Findings / Results

Awareness of AAPI History

Survey findings (N = 43) demonstrate the lack of equal familiarity with events of AAPI history, with a noticeable trend of preference towards the recent and the highly debated problems. Respondents proved to be more conversant with modern issues, especially anti-Asian hate and post-9/11 discrimination, based on their knowledge. Conversely, previous historical events, including the Chinese Exclusion Act, the Vincent Chin case, and Southeast Asian refugee resettlement, were much less familiar. This gap implies that awareness, to a large extent, is influenced by the visibility and recency of the media but not the systematic exposure to education.

Figure 2*Bar chart illustrating levels of familiarity across historical events.*

Sources of Knowledge

The evidence shows a heavy dependence on informal sources of learning, specifically media sources. Although formal education was mentioned as the source of knowledge, it had a

minor percentage (around 9%). Conversely, online platforms and social media were more often mentioned by participants as their main sources of learning. This tendency reveals an enormous departure of institutional learning into media-based learning. The comparatively smaller contribution of schools to the growth of the AAPI awareness of history supports the suggestion that formal education is not the main stimulus that helps the individual to seek the less structured and possibly uneven sources.

Curriculum Representation

Feedback on curriculum representation proves that AAPI content is scarcely incorporated in educational institutions. Specifically:

- 46.5% indicated that the representation of Asians is infrequent.
- 37.2% reported occasional inclusion
- Only 7.0% said they were frequently represented.
- 9.3% reported no exposure at all

These results suggest that more than 8 out of 10 participants have lower or uneven exposure, which holds the reasoning that AAPI histories are not systematically taught in the curriculum. Rather, there is sporadic and frequently shallow representation, which indicates tokenization trends in previous studies. This trend supports the necessity of a more systematic and global inclusion of AAPI histories into the system of education.

Table 2

The frequency of AAPI representation in education.

Representation Frequency	Number of Respondents	Percentage (%)
Frequently	3	7.0%
Occasionally	16	37.2%
Rarely	20	46.5%
Never	4	9.3%

Psychological Impact

The statistics show that there is a serious psychological effect connected with the lack of representation of Asian-Americans in education. Most respondents admitted being negatively affected in terms of identity and cultural attachment:

- Only 55.8% of the respondents experienced somewhat negative effects.
- 14.0% indicated strongly negative outcomes.
- 23.3% reported neutral responses

- 7.0% were unsure

These findings suggest that around 70% of the respondents have undergone some sort of adverse effect, such as disconnection, underrepresentation, or invisibility.

Figure 3

Psychological Impact of Curriculum Absence on Identity and Belonging

These findings are further supported by the qualitative responses. Respondents said they wanted to be more integrated by organizing cultural events, reforming the curriculum, and conducting sensitization campaigns. An example of this was when one respondent proposed the inclusion of Asian American narratives and books; another focused on increasing school-driven cultural holiday events. These reactions reveal a need for increased institutional visibility and representation.

Discussion

Curriculum Absence as Structural Invisibility

There is solid empirical backing to the argument that curriculum absence is a kind of structural invisibility, which is supported by findings. Considering that more than 83% of surveyed individuals have had infrequent or minimal exposure to AAPI content, it becomes clear that this background is not being introduced in the education systems in a structured way. Such underrepresentation makes the scarcity of awareness seem commonplace and nostalgizes the marginalized role of the AAPI narratives in the institutional body of knowledge.

Media: A Double-Edged Sword.

The information proves that the media has succeeded in becoming a prominent contributor to the learning of the culture, bridging the gap in formal education. Nevertheless, the percentage of school-based learning (around 9%) is relatively low, indicating a major transition towards institutional sources of knowledge. Whereas media makes content more accessible, its reliance on operations of the algorithm raises concerns of biased exposure and its preference for simplified narratives, which is also consistent with the Cultivation Theory.

Gap between Perception and Reality.

The disparity between high awareness of recent events and low familiarity with foundational history illustrates a clear perception–reality gap. This trend shows that the media visibility of the AAPI history is what shapes the understanding of the subject among participants, rather than thorough education. Consequently, knowledge is biased and fragmented, further supporting the hegemony of media-based narratives.

Implications

The results of this research have profound educational, social, and psychological implications. Educationally, the findings highlight the necessity to integrate Ethnic Studies and inclusive curricula that take the AAPI past into meaningful consideration. Such inclusion, as noted in previous studies, not only boosts academic engagement but also fosters critical thinking and cultural awareness (Sleeter, 2011). The elimination of gaps in the curriculum can lead to the minimization of the use of informal and possibly untrustworthy sources of knowledge, creating a more precise and thorough interpretation of history.

Socially, making AAPIs more visible and more accurate may help reduce stereotypes and enhance cross-cultural awareness. Through challenging the dominant narratives like the model minority myth and the perpetual foreigner stereotype, educational and media organizations can contribute to the restructuring of public perception and promote empathy and inclusivity. From a psychological standpoint, better representation can enhance feelings of belonging, esteem, and culture in people (Sleeter, 2011). The more people learn of their past and their experiences in educational and cultural accounts, the more likely they will grow to have a positive and unified identity about themselves. Therefore, curriculum inclusivity and responsible media reporting are two of the necessary measures that contribute to the improvement of an individual and the overall society.

Conclusion

This paper sheds light on the interwoven nature of curriculum gaps and mediated media spaces in forming the external representations of AAPI communities as well as the internal mechanisms of identity formation. The results indicate that the relative absence of AAPI histories on the formal educational curriculum is part of a larger tendency of structural invisibility, in which the latter experiences take a back seat or are entirely excluded. Because of this, people tend to resort to media as a substitute source of cultural knowledge, thus forming a dynamic where informal learning environments become central to the formation of understanding.

Even though the media provide greater visibility and access, it does not entirely balance the lack of thorough education. Rather, it tends to recreate reduced or selective accounts that perpetuate stereotypes and add to uneven awareness. The findings of this research indicate that there is a significant divergence between perception and historical reality, with the respondents showing more knowledge of well-reported events or recent ones than about the foundations of AAPI history.

Ultimately, the issue of invisibility is not incidental but structurally embedded within both educational systems and media representations. This can be tackled in two ways, where one is the incorporation of inclusive, evidence-based curricula, and the second is encouraging more objective and varied media representations. With these gaps closed, one may have a better-informed population, the persistent stereotypes will be challenged, and more robust cultural identities will be built. By doing so, education and media can stop perpetuating invisibility to engage in representation, recognition, and belongingness actively.

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Ain't Worried: Selective Calm as a Systems Outcome in China's AI Moment By Liren Cao

Abstract

In the middle of a weak labor market and a loud AI boom, I expected to find sharper personal fear. Instead, my 2025 survey of 200 Beijing residents revealed a more complicated pattern: respondents often believed artificial intelligence would change work in a major way, but most did not think their own jobs were in immediate danger. Only 21.5% worried that AI would replace their own jobs, even though 58% worried about AI replacing other people's jobs. This paper uses that paradox as a systems problem rather than as a narrow AI opinion problem. The object of analysis is not simply artificial intelligence; it is the system of institutions, incentives, narratives, and habits that teaches people where to place anxiety and where to feel safe. Drawing on Harvard Extension's course *The Future of Humanity: A System Approach*, I treat AI anxiety as an interconnected problem involving labor markets, education, media narratives, governance, technological sovereignty, family strategy, and everyday platform life (Harvard Extension School). I also use Heidegger and Marcuse to interpret why optimization can feel normal and why technological administration reaches not only production but culture, attention, and consciousness (Heidegger; Marcuse). The paper adds a brief analysis of China's current AI situation in 2025-2026: mass adoption, the AI Plus policy push, DeepSeek's rise, pressure toward domestic chips, and persistent youth labor stress. These developments matter not because the paper is about AI as a technical object, but because they intensify the feedback loops that produce selective calm. Respondents worried most about misinformation, privacy, monopoly power, and social disorder, while job displacement felt less urgent. Yet they also predicted automation across clerical, legal, programming, customer-service, and media work, and supported AI in hiring, lending, and policing. I therefore argue that China's apparent lack of personal AI anxiety should not be read as simple stability. It may reveal a false sense of security produced by the system itself - a system that makes adaptation sound like protection before the real disruption has fully arrived.

Introduction

I began this project with what felt like a very reasonable expectation. China entered the current AI boom while already facing a weak youth labor market. The official youth unemployment rate reached 21.3% in June 2023 before authorities suspended publication of the series, then reappeared under a revised method excluding students; by August 2025, the 16-24 rate under the new measure had climbed to 18.9%, and in March 2026 it stood at 16.9% after rising from February (Chen and Zhang; "China's Youth Jobless Rate Rises"). At the same time, AI was not arriving as a distant laboratory idea. By December 2025, China had 602 million generative AI users and a 42.8% adoption rate, while the State Council's AI Plus initiative called for deep integration of AI into industry, consumption, public services, governance, and global cooperation ("China's Internet User Base"; "China Issues Guideline"). In a context like this, I

expected to find sharper personal fear, especially among adults who think not only about themselves but about children, family strategies, and the future value of education.

That is not what I found. In my survey of Beijing residents, people were very willing to say that AI would transform work, education, and public life. They were also very willing to support regulation, labeling, safety review, and retraining funds. But they were much less willing to say that their own jobs were in danger. Only 21.5% worried that AI would replace their own jobs, while 58% worried about AI replacing other people's jobs. That difference is too large for me to treat as noise. It points to a patterned way of seeing risk: broad systemic recognition paired with personal distance.

At first glance, that attitude can look almost healthy. A society that faces technological change without panic may be more capable of adapting. The same survey showed that respondents strongly supported reducing rote memorization and increasing critical thinking and creativity in education. That sounds forward-looking. It also sounds sensible. If AI handles repetition more easily than humans do, then schools should not keep preparing students as if repetition were the safest route to the future. In that sense, part of the calm I found is understandable. Respondents could see reasons not to panic because they believed adaptation was possible, education could change, and institutions could intervene.

But the more closely I read the data, the less comforting that calm became. The respondents were not just describing hope. They were also revealing a false sense of security. They predicted automation across a surprisingly broad set of occupations. They accepted AI in gatekeeping roles such as hiring, lending, and police risk assessment. They often spoke in the language of inevitability, adjustment, and management rather than in the language of collective bargaining, structural protection, or serious labor disruption. In other words, many people could see the wave coming, but they still assumed it would break somewhere else. That is the paradox at the center of this paper.

To make sense of that paradox, I drew on systems material that had already shaped Eagle's work. The most useful concepts are feedback loops, leverage points, unintended consequences, path dependence, competing values, and the difference between surface symptoms and deeper system structure. *The Future of Humanity: A System Approach* frames major contemporary problems not as isolated variables but as interacting systems in which one intervention often shifts pressure elsewhere (Harvard Extension School). That framework matters because AI anxiety is not simply a feeling about robots or software. It is shaped by labor markets, schooling, media narratives, state legitimacy, family strategy, technological sovereignty, and everyday digital habits all at once. If I isolate only one variable, I lose the pattern.

The second body of material is philosophical. Heidegger helps me think about the deeper normalization of optimization - the way technology teaches people to encounter the world as ordered, available, calculable, and ready for use (Heidegger). Marcuse helps me think about how that technological rationality moves beyond production into entertainment, attention, and consciousness, so that people can be managed not only through factories or offices but through culture and everyday life (Marcuse). Marx remains in the background because the labor question

never disappears, but Heidegger and Marcuse let me explain why the labor question can become emotionally displaced even when people are not actually ignorant about it (Marx; Marcuse; Heidegger).

My central argument is therefore twofold. First, the apparent absence of strong personal AI anxiety in this sample is not simple ignorance. It is a systems effect produced by interacting forces: a difficult but familiar labor market, confidence in state regulation, heavy exposure to algorithmic systems, a reformist narrative around education, the national push to make AI a productive infrastructure, and a technological culture that normalizes optimization. Second, that same system's effect may be dangerous. When I compare what respondents say they fear with the systems they are willing to authorize and the occupations they themselves mark as vulnerable, I see not reassurance but selective attention. People are worried, but in the wrong places or at least in an incomplete pattern. Because of that, I think they should be more worried than many of them currently are - not more panicked, but more systemically alert.

This paper proceeds in six parts. I first review the literature on AI optimism, labor anxiety, educational pressure in China, public responses to algorithmic governance, and China's current AI policy and deployment environment. I then set out the theoretical framework I use, combining the systems approach with Heidegger and Marcuse. After that, I explain the survey design and sample, present the main findings, and develop a discussion of what the contradictions in the data mean. I close by arguing that the right response is not panic but foresight: broader public debate, more serious educational reform, stronger labor protections, and a deeper willingness to ask what kinds of human judgment should not be casually reorganized into a machine-administered process.

Literature Review

Any serious interpretation of low personal AI anxiety in China has to begin from the fact that existing literature does not point in only one direction. The most obvious expectation is the economic one: if labor markets are weak and automation is accelerating, people should become more anxious about work. But that expectation competes with at least three other patterns already visible in the literature. First, public optimism about AI can coexist with nervousness and distrust. Second, educational pressure in China often converts structural anxiety into adaptation strategy. Third, acceptance of algorithmic systems can be surprisingly high even where fairness concerns remain unresolved. My survey sits at the intersection of all three.

The first important body of work concerns global public opinion on AI and the broader acceleration of AI capability. The 2026 Stanford AI Index describes a central tension: AI capability and adoption are moving quickly, while public trust and expectations remain uneven (AI Index Steering Committee et al.). The report argues that the U.S.-China model performance gap has effectively closed, while also noting that China leads in publication volume, citations, patent output, and industrial robot installations even as the United States continues to lead in private investment and production of top-tier models (AI Index Steering Committee et al.). That mixed picture matters because it prevents me from treating optimism as the opposite of worry. A

public can believe its country is advancing in AI, believe AI is useful, and still be anxious about misinformation, privacy, labor, or political control. The question is not whether optimism exists. The question is where concern is directed inside a fast-moving technological system.

The labor literature complicates things further. Popular commentary on automation often assumes that if machines become more capable, workers will become more fearful. Yet survey evidence does not always work that way. The original draft of my paper already noted the 'third-person effect,' the pattern in which people recognize risk to others more readily than risk to themselves. That pattern has analogues beyond China. Public opinion data in the United States, for example, show that many people distrust algorithmic judgment in principle while still assuming their own situation may be protected or different (Smith). The Stanford AI Index also suggests that beliefs about whether AI will improve the overall job market do not neatly match beliefs about one's own job, and that attitudes vary sharply across countries (AI Index Steering Committee et al.). This matters for my paper because the core puzzle is not whether respondents saw any labor risk at all. They did. The puzzle is why they saw so much labor risk in general while locating so little of it personally.

Chinese labor conditions make that puzzle sharper rather than weaker. Reuters reporting on 'rotten-tail kids' describes a generation of graduates pushed into weak matches, low-paying work, and delayed adulthood in a sluggish economy (Woo and Wang). The youth labor question is therefore not abstract. It is part of a wider climate of stalled expectations, credential inflation, and institutional strain. Under those conditions, one might expect AI to intensify insecurity. But another possibility is that a population already accustomed to economic difficulty begins to interpret technological disruption not as a new emergency but as one more condition to be managed. That second possibility is closer to what I found. The respondents were not blind to disruption. Many seemed already trained to metabolize disruption as the normal background of social life.

A second relevant literature concerns educational anxiety in China. This matters because my survey repeatedly suggested that respondents convert labor anxiety into educational discourse. Instead of speaking only in the language of unemployment, they speak about reskilling, creativity, adaptability, and future-proofing children. Research in BMC Public Health finds that parents' educational anxiety in China is shaped by income, educational attainment, mortgage pressure, and educational expenditure, showing how closely family strategy and structural insecurity are bound together (Zhang et al.). Other work points to a wider culture of parental educational anxiety driven by competition, status pressure, and the fear of falling behind (Zhang et al.; "Excessive Anxiety of Chinese Parents about Education"). When I read those studies beside my own data, I see a useful clue: in China, social anxiety often gets routed through schooling rather than expressed directly as labor fear. Education becomes the site where uncertainty is made actionable.

That helps explain why the creativity question in my survey is so important. Respondents overwhelmingly said rote memorization had become risky and that schools should emphasize creativity and critical thinking instead. This could be read as evidence of sophistication, and I do

think there is real sophistication there. But the educational turn can also do ideological work. If anxiety is translated into an educational challenge, then structural labor insecurity becomes a problem that families and schools are supposed to solve by upgrading themselves. The problem is individualized and future-oriented. The focus moves from protecting workers in the present to preparing students for a better competition tomorrow. That is a powerful narrative because it sounds hopeful. It is also convenient because it can hide how little protection exists when the future arrives unevenly.

A third body of literature concerns public responses to algorithmic decision-making. Here the comparative contrast is especially useful. Pew's work on algorithmic decision-making found that Americans were broadly skeptical of AI systems evaluating job interviews or personal finance, with many seeing such systems as unfair or unacceptable (Smith). By contrast, the Beijing respondents in my survey were quite open to AI in recruitment screening and at least moderately open to it in lending and policing. This is not a small difference. It suggests that acceptance of algorithmic authority is not simply a function of technical capability. It also depends on political culture, institutional trust, everyday habituation, and local moral priorities.

Chinese views on AI regulation also matter here. A Jamestown survey of elite Chinese students found a pattern that is especially relevant to my paper: relatively high optimism about AI's social value paired with substantial support for regulation and caution (Corvino and Li). That pairing resembles what I found in a broader urban sample. It also fits China's regulatory landscape. China has not yet adopted a single comprehensive AI law, but it has built a patchwork of rules for generative AI, recommendation algorithms, deep synthesis, labeling, cybersecurity, and data governance (CMS). People can therefore welcome AI and still expect the state to manage it. In fact, those two things may reinforce one another: if one believes regulation will handle the main risks, then personal fear becomes less necessary. Regulation operates psychologically as reassurance before it operates materially as protection.

China's current AI situation makes this loop stronger. The State Council's AI Plus initiative presents AI not as a single industry but as an infrastructure to be integrated into science and technology, industrial development, consumption upgrades, people's well-being, governance, and global cooperation ("China Issues Guideline"). CNNIC-related reporting described 602 million generative AI users by December 2025, rapid adoption across consumer and industrial sectors, and more than 6,000 AI enterprises in China ("China's Internet User Base"). DeepSeek's 2025-2026 rise, its low-cost model reputation, its reported fundraising talks, and its collaboration with Huawei chips also show how AI has become tied to technological sovereignty and national industrial strategy (Baptista et al.; "DeepSeek Could Be Valued"). For this paper, these facts matter less as a technology ranking and more as system context. They make it easier for AI to appear inevitable, nationally important, institutionally managed, and personally usable at the same time.

Yet that loop may be misleading. The problem is not only whether more AI education or more AI industry creates enough good jobs. The problem is also whether the response narrows the imagination of what education, work, and human judgment are for. If humanities and

interpretive disciplines are cut in the name of future competitiveness, then a society may become better at adjusting to technological systems while becoming worse at criticizing the values those systems carry. That is one reason I insist on keeping this paper humanities-oriented even though the topic looks partly sociological and partly technological. The empirical results become much more legible once I ask not only what respondents believe about AI, but what kind of system is teaching them which dangers should feel urgent.

This is where the philosophical literature enters the review more directly. Marx remains foundational because he ties technological development to production, class structure, and the organization of labor (Marx). But Marx alone is not enough for my purposes. Heidegger shifts attention from production to disclosure. He asks how modern technology reveals the world, reducing beings to resources that can be ordered and optimized (Heidegger). Marcuse, in turn, shows how technological rationality becomes totalizing in advanced industrial society, extending into culture, entertainment, and consciousness rather than remaining limited to the factory or workshop (Marcuse). The lecture materials Eagle worked from were especially useful here because they emphasized Marcuse's move beyond a narrow productive-forces account: technology no longer acts only inside production but also shapes the wider culture in which domination becomes livable and even pleasurable (Technology and the Critique of Capitalism Beyond Marx).

I am not using these thinkers to decorate empirical results after the fact. I am using them because the pattern in my survey already demands a theoretical account richer than simple opinion polling. A public that predicts widespread automation, supports AI in bureaucratic gatekeeping, worries intensely about misinformation, and still feels relatively secure personally is not just expressing random preferences. That public is living inside a particular configuration of technology, governance, labor, and culture. The literature therefore pushes me toward a hybrid framework. From empirical studies, I take the insight that optimism, anxiety, and regulation can coexist. From educational literature, I take the insight that insecurity in China is often processed through family strategy and schooling. From philosophical literature, I take the insight that technology shapes not just tasks but the horizon through which people recognize some dangers and not others. That combination sets up the theoretical framework I develop next.

Theoretical and Analytical Framework

The most important conceptual move in this paper is to refuse the idea that AI anxiety is an isolated attitude. The Future of Humanity: A System Approach repeatedly emphasizes that contemporary crises are interconnected, nonlinear, and resistant to simple one-cause explanations. The course language of complexity, systems mapping, foresight, design, and coalition-building pushed me to ask a different question from the one I started with. At first I was asking, 'Why are people not more scared of AI?' After working through the system's material, I began asking, 'What combination of institutions, narratives, habits, and incentives makes this specific pattern of concern feel normal?' That is a better system question because it asks how anxiety is produced, redirected, and stabilized.

A system lens changes the paper in at least four ways. First, it prevents me from reading each survey answer in isolation. Low personal job anxiety means something different once it is placed beside high concern about misinformation, high support for regulation, strong faith in creativity-focused education, and broad acceptance of AI in recruitment screening. Second, it forces me to think in terms of feedback loops. Labor pressure feeds educational pressure; educational pressure feeds self-upgrading narratives; self-upgrading narratives make labor insecurity feel individually manageable; that sense of manageability then lowers pressure for collective labor protection. Third, it makes me look for leverage points, meaning places where a small change in public conversation, school design, regulation, or labor policy could shift the whole pattern. Fourth, it makes selective calm itself analytically significant. What people are not worried about can be a product of the system just as much as what they are worried about.

Heidegger helps me give that systems account an ontological depth. In 'The Question Concerning Technology,' modern technology is not simply a collection of tools. It is a mode of revealing that presents the world as calculable, available, and orderable. Humans are drawn into that order too. Once the world is encountered in that way, optimization no longer feels like one value among others. It begins to feel like common sense (Heidegger). This is extremely relevant to my survey. Respondents often treated AI in hiring, lending, or public administration not as a philosophical rupture but as a practical upgrade. I do not think that reaction can be explained by convenience alone. It also reflects a horizon in which sorting, filtering, predicting, and optimizing already feel natural.

Still, Heidegger is not fully enough for me. His account is powerful, but it can remain too abstract if I want to explain why people accept technological rationality not only cognitively but socially and emotionally. That is where Marcuse becomes necessary. In *One-Dimensional Man*, Marcuse argues that advanced industrial society stabilizes itself not only through productive organization but through the management of needs, desires, entertainment, and consciousness. Technological rationality becomes social rationality. People adapt to systems that diminish them because those systems also organize comfort, aspiration, and everyday legitimacy (Marcuse). When I read my respondents' calm through Marcuse, I see not simply misperception but successful adaptation to a world in which permanent adjustment has become virtuous.

This is why the false sense of security I describe should not be reduced to a personal error. It is not that respondents are foolish and I am wise. The more disturbing possibility is that the system itself makes their selective calm seem rational. If one lives inside a society where technological modernization is publicly celebrated, where education is constantly reconfigured as competitive adaptation, where algorithmic systems are ordinary, and where regulation is expected to contain major risks, then personal alarm begins to feel excessive or unproductive. Worry becomes displaced onto safer objects - misinformation, privacy leakage, monopoly power, deepfakes, and so on. Those concerns are real. But their relative urgency may also help conceal labor vulnerability.

Marx remains relevant here because the labor question never disappears. Technology still matters politically through production, class position, and the distribution of gains and losses

(Marx). But one of the insights I take from the Marcuse materials Eagle studied is precisely that technology in late modern life cannot be understood as a mere subset of production anymore. It spills into the cultural and perceptual environment itself. That broader field is what allows labor disruption to become emotionally muted. A person can be objectively exposed to automation and still feel subjectively secure if the system has already taught them to imagine that the main answer is personal adaptation within a generally beneficent technological order.

Using these thinkers together lets me define the core claim of the paper more precisely. I am not arguing that respondents 'should be worried' in a simplistic emotional sense. Panic is not the goal. What I am arguing is that the distribution of concern in the survey does not match the distribution of vulnerability implied by respondents' own answers. They predict broad automation. They support AI in administrative domains that can displace or devalue human judgment. They describe adaptation in generic terms. Yet they still rank their own job risk low. The systems framework helps me see this as a mismatch between expressed personal security and the institutional world respondents themselves are authorizing. Heidegger explains why optimization feels normal inside that world. Marcuse explains why people can live inside it with disciplined calm. The systems approach explains why those philosophical forces connect to labor, schooling, governance, media, and industrial policy rather than remaining abstract ideas.

Methods

I administered an online survey in Chinese in June 2025 under the title "Beijing Residents' Views on AI Ethics." Before entering the main questionnaire, respondents had to confirm that they lived in Beijing, were at least 15 years old, and agreed to anonymous participation. A total of 206 people initially responded, but six failed the screening criteria, leaving a final sample of 200 Beijing residents. Participation was voluntary and unpaid. The survey link circulated online through social media and community networks, which means the sample is best understood as an urban convenience sample rather than a probability sample. I do not claim national representativeness. I do claim that the responses offer a valuable window into one influential urban demographic.

The sample skewed toward educated, middle-aged respondents. Adults aged 35-44 made up 35.5% of the sample and those aged 45-54 made up 36%. Younger adults were less represented. Women constituted 60% of respondents, men 38.5%, and a small remainder preferred not to say. The educational profile was high: 54.5% held bachelor's degrees and 26.5% held graduate degrees. Office workers formed the largest occupational group, followed by service workers, retirees, self-employed respondents, and those in the government or public sector. This matters because a highly educated urban sample may feel both more familiar with AI and more insulated from some forms of automation, at least in the short term.

The questionnaire combined closed-ended and open-ended items. One section asked directly whether respondents worried that AI would replace their own jobs and whether they worried it would replace other people's jobs. Another asked which occupations they thought were vulnerable in the next five years, including clerical work, basic programming, legal

drafting, customer service, tutoring, media production, manufacturing quality control, transportation, and finance. A separate battery measured concern on a 0-5 scale across issues such as job replacement, misinformation, privacy leakage, inequality, bias, safety accidents, monopolies, and academic integrity.

I also included normative and governance questions. Respondents were asked whether AI should be used in domains such as hiring, lending, judicial sentencing, police risk assessment, social media recommendation, public transport scheduling, exam grading, classroom work, and wartime targeting. They were asked whether important AI systems should be explainable, whether AI-generated content should be labeled, whether high-risk systems should require permits or real-name verification, whether the government should establish national AI safety standards, whether workers displaced by AI should receive retraining support, and whether legal liability for AI-caused harm should be clearly defined. In addition, I included moral-value questions on truthfulness and restriction, along with an autonomous-vehicle dilemma that tested whether respondents wanted stable rules or context-sensitive exceptions.

Finally, I included two open-ended prompts. The first asked what respondents would plan or think if AI replaced them or their family or friends at work. The second asked how large they thought AI's future impact would be and whether they had thought about it before. I translated the open-ended responses into English for analysis and read them thematically, looking especially for recurring patterns of adaptation, resignation, confidence, helplessness, and selective concern.

My analytic approach was interpretive but grounded. I first summarized the quantitative frequencies and distributions. Then I compared answers across different sections to see where they aligned and where they did not. I was especially interested in inconsistencies that were too patterned to dismiss: predicting automation but feeling personally safe, supporting AI in gatekeeping domains while ranking labor risk low, or wanting both strict restrictions and truth-telling from AI systems. Those contradictions are not a flaw in the dataset. They are one of its most important findings. Because my goal is not only descriptive but interpretive, I read the numbers through the systems framework and through the philosophical contrast between Heidegger and Marcuse outlined above.

Findings

Self-other displacement and the structure of selective calm

The most striking result in the entire survey is the gap between personal and general labor anxiety. Only 21.5% of respondents said they worried AI would replace their own jobs. By contrast, 58% said they worried about AI replacing other people's jobs. I do not think this is a small discrepancy. It tells me that the respondents were not denying labor disruption in general. They were distancing themselves from it. Put more bluntly, they could imagine a society transformed by AI while still imagining that their own position, somehow, would remain manageable.

This pattern matters even more once I compare it with the occupations respondents themselves identified as vulnerable. Large majorities predicted that AI would replace or heavily disrupt data entry, basic programming and software testing, customer service, cashier work, routine legal drafting, medical image analysis and report writing, manufacturing quality control, and parts of accounting and finance. In other words, respondents were not underestimating AI's range. They did not think automation was confined to a few obsolete clerical roles. They anticipated risk in both blue-collar and white-collar domains, including occupations that many educated urban respondents might previously have assumed were protected by skill or status.

That is why I think the calm in the survey cannot be read as simple lack of awareness. The people answering my questionnaire were often very aware of AI's capabilities. The harder question is why that awareness did not translate into stronger personal concern. One answer is occupational self-exemption: people assume their own tasks are less routine than the tasks of others. Another answer is temporal distancing: respondents may believe disruption is real but not immediate. A third answer, which I think is the deepest, is systemic normalization. If a society constantly presents technological change as something to adapt to rather than contest, then a person can learn to describe serious structural risk in a calm register without ever fully processing its personal implications.

Jobs at risk, but adaptation imagined in general terms

The open-ended responses reinforce that interpretation. When asked what they would do if AI replaced them or their family and friends, many respondents described adaptation in generic terms: "keep up with the times," "improve oneself," "learn AI," "use AI as a tool," or "find work AI cannot replace." Those are not foolish answers. In fact, they often sounded pragmatic and disciplined. But they were usually vague. Very few respondents described concrete pathways of transition, institutional supports beyond general retraining, or sector-specific strategies. The dominant tone was composed adjustment rather than detailed planning.

Another common theme was emotional or creative refuge. Some respondents argued that AI could solve problems but not replace emotional exchange, empathy, or authentically human connection. Others imagined that jobs involving creativity, interpersonal care, or uniquely human expression would remain safer. Again, this may be partly true. But it also reveals a pattern: respondents recognize that routine work is exposed, then leap quickly to a category of supposedly human work without always confronting how broad the exposed zone already is. The movement from risk recognition to reassuring exceptions happens very fast.

There was also a striking undercurrent of fatalism. Many respondents described AI as an "inevitable trend," used phrases similar to "there is no way around it," or implied that survival would belong to those who adjusted fastest. These answers are important because they show that the absence of panic does not necessarily indicate confidence. Sometimes it indicates resignation. People can sound calm because they feel they have no meaningful alternative except adaptation. That is not the same thing as feeling safe.

Education as both insight and reassurance

One of the most coherent findings in the survey concerned education. Eighty-three percent of respondents agreed that heavy reliance on rote memorization had become risky in the age of AI, and 87% said schools should reduce memorization and increase training in critical thinking and creativity. I think this is one of the genuinely strong insights in the dataset. Respondents understood that if AI can already outperform humans at many repetitive cognitive tasks, then an education system built around repetition becomes strategically weak.

At the same time, the educational finding also helps explain the false sense of security I am tracing. The idea that schools will become more creative and students more adaptable offers a story in which society is already pivoting in the right direction. It softens the labor problem by translating it into a curriculum problem. That story is not entirely false. Educational change does matter. But it can become too reassuring if people assume that a rhetorical shift toward creativity is enough to protect them from structural disruption.

This is especially important in the Chinese context, where official and semi-official discussion often links AI pressure to educational modernization. Reporting on China's "AI+" strategy and on universities restructuring toward AI-oriented training gives the impression that the country is actively future-proofing itself (Colville, "China's AI Job Mirage"; Colville, "Shrinking Humanities for AI"). In one sense, that is real. In another sense, it may deepen the problem by making people think the system is already adjusting at the same pace as the technology. My survey suggests that respondents have absorbed the educational part of the story very strongly. What is less clear is whether they have matched that hope to a realistic sense of how uneven labor transition could be.

The hierarchy of concern: disorder before displacement

When I asked respondents to rate how concerned they were about different AI-related issues, job replacement ranked surprisingly low. The highest concern by far was misinformation and deepfakes, with 54% choosing the maximum concern score. Privacy leakage came next, followed by safety accidents and monopoly power. Job replacement had the lowest mean concern score of the entire set. That hierarchy is, to me, one of the most revealing patterns in the paper.

It shows that respondents were not living in a state of generalized trust. They were worried. But they were worried first about disorder: false content, data abuse, opacity, concentration of power, and physical risk. Those are real issues and not minor ones. In fact, many of them are urgent. But the ranking still matters because it displaces labor disruption downward. If job replacement is both broadly predicted and emotionally deprioritized, then selective calm becomes visible not as absence of concern, but as reallocation of concern.

I do not think this reallocation is accidental. Concerns about deepfakes and privacy have a vividness that job displacement often lacks. A fake video, a fraud, a data leak, or a manipulative algorithm can be imagined immediately. Labor disruption is more diffuse, slower, and easier to imagine as something that will happen to another sector, another age group, or

another class of worker. That makes it psychologically easier to rank lower even when it could become socially devastating. The survey results fit that pattern very well.

Authorization of AI in gatekeeping roles

The strongest evidence that respondents may be underestimating their own exposure comes from the questions about where AI should be used. Seventy-eight percent supported AI in recruitment screening. Seventy-seven percent supported AI in personalized advertising. More than half supported AI in bank loan approval and police risk assessment. There were limits, of course: majorities rejected AI in judicial sentencing and wartime targeting, and many were cautious about AI in medicine or in grading. But the broader pattern is clear. Respondents were broadly comfortable with AI where it appeared to offer administrative efficiency or large-scale sorting.

That matters because gatekeeping systems do not simply sit above labor markets. They shape them. If AI filters resumes, recommends who gets interviews, helps determine credit access, and classifies risk, then AI is not just replacing jobs. It is reorganizing the pathways into jobs, capital, and social legitimacy. I think this is where the "what people do versus what they say" contrast becomes most visible. Respondents say they are not very worried about their own job displacement. Yet they are prepared to authorize systems that may change the very conditions under which employment and opportunity are allocated. That is not a minor contradiction. It is the heart of the false security problem.

To put it another way: many respondents seemed comfortable with AI so long as it was imagined as a manager rather than a conqueror. But management itself can be transformative. Administrative AI may not look dramatic in the same way as a humanoid robot taking over a workplace. Still, it can quietly reorder who gets seen, who gets sorted, who gets opportunity, and whose skills are judged legible. The low level of personal labor anxiety therefore sits uneasily beside the public willingness to hand over exactly those gatekeeping functions that structure labor futures.

Governance optimism and trust in management

Governance responses were extraordinarily strong. Eighty-three percent supported explainability in important AI decisions. Ninety-five and a half percent supported independent safety review before deployment. Ninety-seven percent supported labeling AI-generated content. Ninety-four and a half percent supported permits or real-name requirements for high-risk systems. Ninety-seven and a half percent supported national AI safety standards. Eighty-eight percent supported government retraining funds for displaced workers. And 96.5% wanted clear legal liability for AI-caused harm.

I read this as evidence of two things at once. First, respondents are not anti-regulation. They do not think the market will solve AI's problems by itself. Second, strong expectations of regulation may operate as emotional reassurance. If one assumes the government will create standards, labeling regimes, review procedures, and retraining support, then personal fear

becomes easier to contain. In that sense, governance optimism may reduce private urgency before the actual effectiveness of those protections has been tested.

That is one reason I resist simply calling the respondents complacent. Complacency implies laziness or indifference. What I found is more complex. Many respondents were intensely regulation-minded. They wanted the system managed well. The problem is that management expectations can create confidence that outruns actual preparedness. A society can trust regulation long before it has built institutions strong enough to handle the scale of disruption ahead.

Moral contradictions and the difficulty of wanting everything at once

The survey also surfaced several contradictions that tell me something important about the moral texture of public AI attitudes. Respondents wanted AI to follow restrictions, but many also wanted it to tell the truth even when rules blocked full disclosure. They wanted predictable autonomous vehicle rules, but many changed their position when the endangered pedestrian was described as a child. They supported AI in some sensitive domains while reporting limited concern about bias and discrimination. These contradictions are not random. They reflect the difficulty of wanting safety, efficiency, truth, fairness, and moral nuance all at the same time.

I do not treat these contradictions as evidence that respondents were irrational. On the contrary, they show that people sense the moral insufficiency of purely technical rules. But they also reveal an underdeveloped public conversation. Many respondents wanted AI to produce outcomes that human institutions themselves often fail to reconcile. They wanted systems that are rule-bound but compassionate, restricted but honest, efficient but fair, powerful but fully accountable. That is understandable. Yet the wish for all of those things at once may also make it easier to underestimate how disruptive AI could become when trade-offs turn out to be much harsher in practice.

Open-ended views of AI's future impact

In the final open-ended question, most respondents said AI's impact would be large or even enormous. Some compared it to earlier technological revolutions. Some said it would touch not only jobs but human relations, moral life, and the structure of society. Others were more restrained and described AI as "just a tool," comparable to computers or smartphones. What matters to me is that the broad expectation of major social impact was already there. The respondents did not need to be convinced that AI matters.

This is exactly why their low personal labor anxiety is so revealing. It is not the product of thinking AI is trivial. Rather, many respondents thought AI was extremely important while still picturing themselves at some manageable distance from the danger. That combination - large acknowledged impact, wide predicted automation, strong support for regulation, broad authorization of AI gatekeeping, and low personal labor fear - is the empirical core of the paper. It is why I argue that the issue is not lack of awareness in general, but selective awareness shaped by the system itself.

Discussion

Why people seem calm, and why that calm is not fully safe

Part of the calm in my data is reasonable. I want to state that clearly because otherwise the argument becomes too easy. Respondents had good reasons not to interpret AI as a pure disaster. Many had already used AI tools. They could see efficiency gains in certain domains. They did not rank every change as necessarily dehumanizing. They understood that education systems built around memorization are becoming less defensible. They also trusted that regulation, if done properly, could reduce some risks. A population that sees these things is not stupid. In some ways it is more serious than a population that responds only with panic.

But that is only half the story. The same reasons for calm can become mechanisms of false security. Educational reform sounds reassuring, but a narrative of creativity does not guarantee labor absorption. Trust in regulation sounds stabilizing, but high expectations of governance can outrun actual institutional capacity. Familiarity with AI sounds empowering, but everyday use can also normalize systems before their consequences are visible. The AI Plus policy environment can make technological integration appear already planned and managed, even though the distributional effects are still uncertain. Pragmatic adaptation sounds mature, but adaptation without concrete pathways can become a polite word for individualized burden. This is why I keep insisting that the respondents seem unworried for reasons that make sense and for reasons that should make us less calm.

Comparing what respondents say with what they authorize makes this especially visible. They say their own jobs are not in immediate danger. Yet they identify wide swaths of clerical, routine cognitive, and administrative work as vulnerable. They say creativity matters more now. Yet they also authorize AI in hiring, lending, and police assessment, domains that can reshape access to work and status before creativity has much chance to rescue anyone. They say regulation is necessary. Yet those high governance expectations may already be doing emotional work by lowering perceived personal risk. Put differently, the respondents are often calm because they think the system is adjusting - but the same survey suggests the system they are authorizing could become part of the disruption itself.

The educational question is central here. A push for more creativity is not wrong. I think it is one of the best instincts in the survey. The problem is that creativity can become a comforting slogan when it is detached from institutional detail. Who gets access to this new kind of education? How quickly can schools and universities actually transform? What happens to workers who are not children anymore and cannot simply restart their formation? What happens if creativity is invoked while humanities and interpretive disciplines are cut in favor of narrow technical alignment with state or market priorities? The language of future creativity can easily reassure those who are already comparatively advantaged, while offering much less to those already exposed.

This is also why the systems framework matters so much. If I interpret the survey psychologically, I get optimism bias and stop there. That is not wrong, but it is incomplete. If I

interpret it sociologically, I get demographic skew and institutional trust. That is useful, but still incomplete. If I interpret it philosophically, I get Heidegger's enframing and Marcuse's managed consciousness. That is powerful, but also incomplete alone. Only when I place these together do I see the full pattern: a technological culture that normalizes optimization, a governance culture that promises management, a labor market that has already trained people to adapt, an educational discourse that converts insecurity into competitive self-improvement, and an industrial-policy environment that frames AI as national destiny. Inside that configuration, selective calm is not an anomaly. It is almost what one should expect.

Why they should be more worried

When I say respondents should be more worried, I do not mean they should become apocalyptic or irrational. I mean that their current distribution of concern does not adequately match the institutional world they are entering. If one predicts that AI will reshape programming, legal drafting, clerical work, tutoring, customer service, media production, and financial processing, then low personal labor anxiety becomes hard to defend, especially in a weak labor market. If one supports AI in recruitment and lending, then one is not only anticipating labor disruption but helping normalize systems that mediate entry into economic life itself. If one assumes education will solve the problem, then one still needs to ask whether education can transform fast enough, widely enough, and humanely enough to compensate.

The deepest issue is that labor disruption rarely announces itself all at once. People often look for dramatic replacement and miss subtler forms of devaluation: harder entry into jobs, downgraded tasks, increased competition, algorithmic filtering, wage pressure, status loss, or the slow conversion of professional judgment into machine-supervised routine. My survey makes me think many respondents are still imagining danger in overly theatrical terms. They are looking for the moment a machine takes "your job," not for the many quieter ways AI can hollow out a labor market before that moment becomes obvious. Because of that, the strongest anxieties cluster around misinformation and privacy, which feel more visible and immediate, while labor vulnerability remains underfelt even when it is conceptually acknowledged.

Heidegger and Marcuse help me say why this underfeeling is not accidental. Heidegger suggests that technological ordering makes calculability appear normal, so administrative AI can look like common sense rather than dispossession. Marcuse suggests that advanced technological society trains people to adapt to systems that manage them, even to experience that adaptation as realism or maturity. Those frameworks do not prove that every respondent is wrong about their own future. But they do explain why a broad social pattern of underworry could emerge even among people who are intelligent, informed, and not naive.

Implications for public conversation

The practical implication is not that China needs a louder panic about AI. Panic is often politically manipulable and analytically shallow. What is needed instead is a different public conversation. First, labor disruption should be discussed more concretely and less abstractly.

People need to see how AI can reshape hiring, task composition, credential value, and wage dynamics even before straightforward substitution becomes visible. Second, educational reform must go beyond slogans about creativity. It needs real curriculum change, wider access, and protection against a future in which the humanities are dismissed as expendable just when society most needs ethical and interpretive judgment.

Third, governance should not be treated as reassurance by itself. My survey shows that respondents strongly support regulation, which is important. But support for regulation is not the same thing as evidence that regulation will be adequate. Public discourse should therefore be less triumphal about management and more honest about uncertainty, trade-offs, and enforcement limits. Finally, AI ethics in China should not be narrowed to misinformation, privacy, and order alone. Those matter. But labor, dignity, and human judgment matter too, and they should not stay secondary simply because they are less visually dramatic.

In one sense, the people I surveyed were calm because they believed things were going more or less in the right direction: schools would adapt, technology would be managed, and society would absorb the change. In another sense, that very belief may be the danger. If the public waits until labor disruption is undeniable before taking it seriously, then much of the damage will already have been organized into the normal functioning of everyday institutions. That is why I think the survey is not just interesting. It is a warning.

China's Current AI Situation as a System

The current Chinese AI situation should be read as a system rather than as a scoreboard. A scoreboard asks whether China is ahead or behind the United States, whether DeepSeek is better than one American model, or whether Huawei chips can replace Nvidia chips. Those questions matter, but they are not the main question for this paper. The system's question is different: how do adoption, industrial policy, labor pressure, education, regulation, media narrative, and platform life interact to make AI feel both disruptive and manageable? Once I ask that question, China's 2025-2026 AI situation becomes directly relevant to the survey. It helps explain why respondents can recognize major change while still feeling personally calm.

The first subsystem is mass adoption. By the end of 2025, China had 602 million generative AI users and a 42.8% adoption rate, according to reporting based on the China Internet Network Information Center ("China's Internet User Base"). That scale means AI is not only a policy discourse or a corporate technology. It is becoming part of ordinary digital life. This matters because everyday use changes the emotional texture of risk. A tool that appears in study, customer service, office work, shopping, entertainment, and platform recommendation does not feel like an alien invader. It feels like the next layer of infrastructure. The more ordinary the tool becomes, the easier it is to imagine that ordinary adaptation will be enough.

The second subsystem is state-directed integration. The AI Plus guideline is explicitly cross-sectoral: it pushes AI into science and technology, industrial development, consumption, public services, governance, and international cooperation ("China Issues Guideline"). From a systems perspective, this is important because it makes AI a general organizing layer, not a

narrow product category. When a technology is framed as a national modernization infrastructure, personal anxiety can become politically and emotionally awkward. The more AI is narrated as the route to productivity, sovereignty, and future growth, the more individual fear may be displaced into a more acceptable language of adaptation, retraining, and responsible management.

The third subsystem is technological sovereignty. Stanford's 2026 AI Index argues that the U.S.-China model performance gap has effectively closed, while Reuters reporting on DeepSeek's V4 and Huawei collaboration shows Chinese developers trying to reduce dependence on foreign chips and hardware ecosystems (AI Index Steering Committee et al.; Baptista et al.). DeepSeek's possible fundraising at a valuation of up to \$50 billion, led by China's national AI fund according to Reuters sources, also shows how frontier AI is becoming integrated with state-backed capital and strategic competition ("DeepSeek Could Be Valued"). Again, the point is not to celebrate or dismiss this. The point is that AI becomes tied to national resilience. Once a technology is placed inside the story of sovereignty, questioning its social costs can easily sound like a secondary concern.

The fourth subsystem is labor stress. The youth unemployment rate for 16-24 year-olds excluding students rose to 16.9% in March 2026, while the 25-29 rate rose to 7.7% ("China's Youth Jobless Rate Rises"). That does not prove AI is causing unemployment. It does show that AI is arriving in a labor market already under pressure. Systems thinking matters here because it causes overlap. Weak demand, real estate stress, credential inflation, demographic shifts, platform work, and AI adoption may interact without one simple cause dominating. In that kind of environment, people may treat AI disruption as one more pressure in a crowded field rather than as a distinct emergency. This can make calm appear rational even when cumulative risk is rising.

The fifth subsystem is regulation. China's current AI governance environment is not a single comprehensive law, but a layered framework involving generative AI services, recommendation algorithms, deep synthesis, labeling, cybersecurity, and data governance (CMS). This patchwork creates a particular kind of public expectation: AI is risky, but risk should be managed through standards, labeling, review, and state supervision. That expectation matches my survey, where respondents strongly supported labeling, safety review, explainability, liability, and retraining support. The danger is that regulation can become a psychological substitute for preparedness. People may feel safer because the system has a management vocabulary, even before they know whether the management tools are adequate.

The sixth subsystem is education. In China, labor risk often becomes a question of schooling: what should children learn, what should universities emphasize, and how can families remain competitive? This is why the survey's education results matter so much. When 83% of respondents agree that rote memorization has become risky and 87% want more creativity and critical thinking, they are not just commenting on pedagogy. They are identifying a leverage point. But leverage points can be misused. If creativity becomes only a slogan attached to more

competition, or if AI literacy replaces rather than complements ethical, historical, and interpretive education, then reform may reinforce the same system it claims to fix.

Seen this way, China's current AI situation supports the core argument of the paper. It is not a background detail. It is the environment in which selective calm becomes plausible. Mass adoption makes AI familiar. State strategy makes it inevitable. Sovereignty competition makes it nationally important. Labor stress makes it one pressure among many. Regulation makes it feel manageable. Education converts it into a project of self-upgrading. These subsystems form a loop of reassurance. The loop does not mean people are foolish. It means the system gives them reasons to underfeel certain dangers while correctly feeling others.

Implications for Education, Labor, and Public Discourse

If my interpretation is right, then the main implication is not that China needs stronger anti-AI rhetoric. The deeper need is for a more honest language about transition. Public discussion currently has a tendency to oscillate between triumphalism and generic caution. AI is either celebrated as national modernization or criticized in broad ethical terms that remain detached from everyday institutional life. My survey suggests that what is missing is a middle register: concrete public discussion of how AI changes pathways into jobs, alters the value of credentials, shifts the internal structure of work, intensifies competition, and changes the rules of administrative visibility long before direct substitution becomes obvious.

Education is the first major implication. Respondents were right to reject rote learning as a sufficient strategy in an AI-saturated world. But a serious educational response cannot stop at saying 'be more creative.' Creativity without institutional support easily turns into another moral demand placed on individuals. Schools and universities would need to redesign curricula, assessment, teacher training, and student support in ways that genuinely cultivate judgment, interpretation, collaboration, and the ability to work across technical and humanistic domains. If universities simply add more AI literacy while stripping away humanities and social critique, then they may produce graduates who are more fluent in the system and less able to question what the system is doing.

Labor policy is the second major implication. The survey's strong support for retraining funds shows that respondents already expect the state to play a transitional role. I think that expectation is reasonable, but retraining alone is too narrow. A labor-centered AI response should also consider placement assistance, protections against opaque algorithmic hiring filters, public scrutiny of AI-mediated recruitment, and more transparent debate about which sectors are likely to absorb displaced labor and which are not. If the public conversation stays at the level of "workers should reskill," then responsibility gets pushed downward onto individuals even when structural adjustment is the real issue.

There is also a major implication for the design of AI governance. Many respondents strongly endorsed explainability, labeling, review, standards, and liability. That is an important base of public legitimacy for regulation. But regulation should not be framed only as reassurance. It should also be framed as a contestation over what kinds of automation are

acceptable, where human judgment must remain meaningful, and how institutional accountability should work when machine-assisted systems reshape opportunity. In other words, regulation should not be only about making AI safer. It should also be about deciding where AI should not quietly become the default manager of human worth.

Public discourse around misinformation and privacy needs to widen too. Those concerns are real, and the respondents were right to care about them. But when misinformation becomes the most vivid AI problem, it can crowd out slower-moving forms of institutional change. A society can become very alert to fake videos and still be underprepared for job-market restructuring, educational narrowing, or algorithmic gatekeeping. I think one of the paper's main practical lessons is that visibility is not the same as severity. The dangers that appear most dramatic in the media are not always the ones that most deeply reorganize ordinary life.

Finally, there is an implication for the humanities themselves. One reason I wanted this paper to remain humanities-oriented is that technological disruption is not just a technical or economic story. It is also about what kinds of people institutions are trying to make. When respondents say schools should produce creativity instead of memorization, they are already asking a humanistic question, even if they do not name it that way. They are asking what human capacities matter when machine capacities expand. That question cannot be answered by engineering alone. It requires philosophy, history, ethics, political theory, and serious reflection on what forms of life should remain irreducible to speed, prediction, and optimization. If AI policy sidelines those forms of inquiry, then society may become better at deploying systems while becoming worse at recognizing when those systems are reshaping human value in quietly harmful ways.

For me, then, the biggest implication is quite simple. The conversation about AI in China should become less enchanted by adaptation and more interested in distribution: distribution of vulnerability, institutional protection, educational opportunity, and the right to resist being measured only through optimized systems. The system-level question is not whether AI is good or bad. It is who benefits from its integration, who bears the transition costs, who gets to contest the metrics, and which human capacities remain protected from reduction into speed, prediction, and administrative efficiency. My respondents already sensed many of the right pressures. What they lacked was not intelligence. It was a framework that connected those pressures strongly enough to their own lives. That is precisely the connection public discourse needs to make more forcefully.

Limitations

This study has clear limitations. The sample was limited to Beijing residents with internet access and skewed toward educated urban adults, many of whom likely occupy positions that feel more buffered from immediate disruption than other sectors of Chinese society. I therefore do not treat these findings as representative of all workers or all regions in China. It is very possible that younger graduates outside top cities, migrant workers, or those in more precarious service positions would show stronger labor anxiety.

The survey also measures perception rather than actual labor outcomes. Respondents may be wrong in both directions: underestimating some risks while exaggerating others. The point of the paper is not to prove that every person who feels secure is objectively mistaken. It is to show that the pattern of concern in the sample does not line up neatly with the vulnerabilities implied by respondents' own judgments about AI's reach.

Finally, the open-ended responses add interpretive richness, but they do not replace long interviews, participant observation, or longitudinal tracking. I can identify themes such as adaptation, fatalism, and selective calm, but I cannot fully reconstruct the life histories and institutional experiences that produced them. I therefore present this paper as an interpretive intervention grounded in original survey evidence, not as the final word on Chinese AI attitudes.

Conclusion

I started this project expecting to find sharper personal fear in the middle of a weak labor market and an aggressive AI boom. Instead, I found something more complicated and, in a way, more unsettling. My respondents recognized AI's broad power. They predicted automation across many forms of work. They supported educational reform, strong regulation, and multiple governance safeguards. They were not absent-minded about the technology. And yet most still did not think their own jobs were in immediate danger.

I argue that this is not best understood as simple ignorance. It is better understood as a system's outcome. A difficult labor market has already trained people to adapt. Educational pressure translates structural uncertainty into the language of creativity and future preparation. Everyday exposure to algorithmic systems makes administrative AI feel normal. Confidence in regulation reduces the felt need for private alarm. Media emphasis on misinformation, privacy, and social disorder makes those dangers feel more immediate than labor reorganization. Current industrial policy and technological sovereignty narratives make AI feel like a national infrastructure, not simply a labor threat. Put together, these forces do not eliminate anxiety. They redirect it.

That is why I describe the survey as revealing a false sense of security. The respondents are not unworried because everything is going well. They are unworried, in part, because the system gives them plausible reasons to feel that adaptation is already underway and that management will handle what remains. But when I compare what they say with what they authorize, the reassurance becomes unstable. They predict wide exposure. They accept AI in gatekeeping systems. They offer mostly generic adaptation strategies. They trust that regulation and educational change will catch up. That is not nothing. Still, it is not enough.

Heidegger helps me understand why optimization has become such a normal way of encountering human and institutional life. Marcuse helps me understand why that normalization can be socially reproduced through administration, culture, and everyday habits rather than through coercion alone. The systems framework from Harvard Extension helps me hold those insights together with labor markets, schooling, governance, media discourse, industrial policy, and technological sovereignty. The result is a picture of selective calm, not empty calm.

I do not end by calling for panic. I end by calling for foresight. If AI is going to reorganize opportunity, work, and judgment, then the response has to be broader than technical enthusiasm or hopeful slogans about creativity. It requires real labor preparation, deeper educational reform, stronger public debate, and more willingness to say that some risks are being undervalued precisely because the system has made them easy to underfeel. In the end, the people I surveyed were not saying that AI is harmless. They were saying something more subtle and maybe more dangerous: AI is powerful, AI will change things, but I will probably be fine. History is not always kind to that sentence.

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Water Quality Impacts on Reproductive, Birth, and Health Outcomes: What's in Your Water? By Hyeon-Joo Cha

Abstract

Seven legacy contaminants—arsenic, disinfection by-products, fracking related substances, lead, nitrates, per- and polyfluorinated alkyl substances (PFAS), and uranium—remain in community water systems and present various kinds of health risks. Contaminants have been known to be linked to certain health complications such as cancer, cardiovascular disease, and early reproductive and developmental abnormalities. To what extent does exposure to contaminants in Community Water Systems possibly precipitate disease and health outcomes, and especially influence rates of premature birth and post-birth developmental complications in populations? This research assessed data from the CDC's National Environmental Public Health Tracking Network, utilizing the Drinking Water Dashboard to analyze different health indicators and measures at the county-level across New Jersey, New York, and Pennsylvania. Results show that detectable levels of nitrates and disinfection by-products are associated with infant mortality rates and cancer prevalence. Exposure to PFAS was also found to be linked to low birth rates and very preterm births. However, some contaminants such as arsenic and uranium revealed unexpected outcomes in relation to certain health outcomes, indicating possible external factors that may be influencing these results.

Introduction

There are seven legacy contaminants in drinking water that are impacting the health of communities in the US: arsenic, disinfection by-products, fracking related substances, lead, nitrate, per- and polyfluorinated alkyl substances (PFAS), and uranium. Health impacts include increased risks of cancer, cardiovascular disease, kidney dysfunction, and respiratory complications. Some studies have found that these legacy contaminants contribute to risk of stillbirth, developmental toxicity, reduction in fetal growth, and chromosomal damage that can cause epigenetic modifications which in turn influence immune and metabolic function. Closer evaluation of the impacts of these legacy contaminants should be conducted to understand health risks, particularly for those related to infant and childhood development since cumulative exposure can increase long-term health risks. Increased risks may be associated by geography according to the community water sources being accessed by residents.

Literature Review

Drinking Water

The EPA defines public water systems (PWS) as systems that provide water for human consumption to at least 15 service connections or serves an average of at least 25 people for at least 60 days a year. PWSs can be public or private and can be categorized as community water systems (CWS) or noncommunity water systems. CWSs supply water to the same population

throughout the year while noncommunity water systems can either supply it for at least 25 people over a minimum of six months per year or supply it to locations where people typically do not remain. Public water systems provide water to 90% of Americans (EPA, 2025), with private wells supplying to the rest of the population. Although Community Water Systems account for one-third of Public Water Systems (~49,600 of 150,000), they serve about 95% of the US population (Levin et al., 4).

The Safe Drinking Water Act (SDWA) was established to protect the quality of drinking water in the US. Under the SDWA, the US Environmental Protection Agency (EPA) sets legal standards for contaminants in public drinking water known as the Maximum Contaminant Level (MCL). MCLG stands for Maximum Contaminant Level Goal, a level at which water contaminant levels are low and there are little to no risks to our health. This goal is utilized by the EPA to set a margin of safety. Using the MCLG, the EPA then determines an enforceable MCL based on health data, technological feasibility, and cost (Levin et al., 4; EPA “National Primary Drinking Water Regulations”)

The Contaminant Candidate List (CCL) is a roster of drinking water contaminants that are not subject to any proposed or promulgated national primary drinking water regulations, but are known or anticipated to occur in PWSs. Under the SDWA, the EPA must publish a new list of CCL every 5 years, considering both occurrence data and potential health effects of unregulated contaminants. Publishing CCL does not impose any requirements on PWSs, if the EPA decides to regulate a contaminant on the list in the future, the EPA will begin a separate process during which they engage the public for input and comments (EPA “Basic Information on the CCL and Regulatory Determination,”).

Once a contaminant appears on the CCL, the EPA enters the next stage known as Regulatory Determinations. This is a formal decision made by the EPA on whether the agency should initiate a process to develop a national primary drinking water regulation for a specific contaminant. US law requires the EPA to make regulatory determinations for at least five contaminants from the most recent CCL within five years of completion of the previous round of regulatory determinations (EPA, N.D.). Importantly, the EPA is not limited to making regulatory determinations for certain amounts of contaminants.

When a water system fails to meet an EPA-mandated drinking water standard, a drinking violation can be issued. Allaire et al. (2079) found that there have been irregular increases in the number of violations over the history of the SDWA because of several factors: regulatory changes, enforcement capacity, raw water quality, and treatment capabilities. They underscore how increases in the number of violations are partly impacted by the implementation of new regulations—with increases from 22 in 1974 to 91 at present—as well as the time needed for PWSs to learn and adapt to the new regulations. Of the 95,754 violations between 1982-2015 reviewed, Allaire et al found that the most common violations involved total coliform bacteria (36.7%) and other contaminants, which predominantly included disinfection-by-products (DBPs) as well as Radionuclides (35.7%). Violations were especially concentrated in rural and low-income areas, with “hot spots” in the Southwest and Midwest, including Idaho, Oklahoma,

and Texas. It is notable that violations increased after new EPA rules (such as the Stage 1 DBP Rule) suggesting that smaller systems struggled to meet stricter standards implemented by the EPA. Rural systems had higher violation rates due to limited funding and technical resources (essential for efficient regulatory determinations), while private or wholesale systems were more likely to stay in compliance. Overall, the study concludes that while quality violations are generally infrequent, some do reflect a risk to human health. Identifying hot spots and factors associated with violations can allow public policies to target underperforming water systems. Ultimately, reducing water quality violations can lead to improved health outcomes and less disparity in water service.

Health Impacts of Legacy Contaminants

Levin et al. (7-12) underscore the health impacts of seven legacy contaminants that ought to be monitored and present their corresponding risk profiles and health effects: arsenic, disinfection by-products, fracking related substances, lead, nitrate, per- and polyfluorinated alkyl substances (PFAS), and uranium. These are the most commonly-occurring chemical contaminants that affect many Americans in the US. Each of these substances have a prominent history of occurrence in US water systems and continue to pose a health threat due to geological, environmental, and toxic conditions. Arsenic and uranium are naturally occurring substances in the ground, but human exposure to these elements have intensified because of mining activities.

Nitrate levels are linked to uses of inorganic fertilizer and animal manure in agricultural areas, while DBPs form when disinfectants react with various compounds including organic matter, bromides, and nitrogenous compounds. PFAS are a concern across the US and globally due to its far-reaching toxicity and bioaccumulation potential. They persist in the environment for decades and can accumulate in the body, increasing health risks such as cancer. Lead, on the other hand, is often found in old plumbing and industrial sites. Due to its level of corrosion associated primarily in plumbing pipes, lead has been progressively restricted from plumbing use in the US. Lead can lead to serious health risks that can cause issues like impairment of neurological development in children. All together, these contaminants illustrate how chemicals can linger in water for large amounts of time and pose long-term public health challenges.

Arsenic

Rahman et al. (189-197) overviews the health effects of arsenic (As) on various human populations across the world. Main exposure to arsenic to humans are commonly from ingestion of As through contaminated drinking water and food. As has been found to have a statistically significant association with cancers including cancers of the skin, lung, bladder, and kidney. Other potential health impacts include, but are not limited to: respiratory complications, pulmonary hindrance, increased risks of cardiovascular disease and hypertension, gastrointestinal ailments, hematological abnormalities, hepatic damages, kidney dysfunctions, dermatological conditions and abnormalities, neuropathic blockages, stillbirth and developmental toxicity, immunological responses, and mellitus.

Although Rahman et al. (197) posit that further evaluation is needed to confirm with greater certainty the positive correlations between exposure to arsenic and specific health conditions, the International Agency for Research on Cancer (IARC) has already classified arsenic as a Class I human carcinogen. Studies have also confirmed the relationship (Smith et al., 265; Speer et al., 19-20). As research has progressed and more technically-advanced equipment has been developed, stronger evidence for organ-specific carcinogenicity in skin, lung, bladder, and kidney, mainly associated with the ingestion and inhalation of inorganic arsenic have emerged (Speer et al., 1-20). Although the relationship between arsenic exposure and lung and bladder cancers—specifically involving smelt workers, residents who reside near mining sites, miners, and those exposed to arsenic contaminated areas—had been found to be inconclusive in early studies (Smith et al., 260), more recent studies have confirmed that lung cancer incidence was statistically higher among workers in the copper mining and smelting industries because of the ingestion of contaminated water (Speer et al., 4).

Disinfection By-Products

Disinfection by-products (DBPs) form when chlorine and other disinfectants react with organic matter in drinking water to create haloacetic acids (HAAs) and trihalomethanes (THMs), resulting in widespread human exposure through ingestion, inhalation, and dermal absorption. According to Villanueva et al. (107), long-term exposure to trihalomethanes (THMs) has shown a consistent epidemiological association with elevated bladder cancer risk, while evidence for other cancer sites remains mixed. Beyond carcinogenicity, DBPs have also been linked to reproductive impacts such as altered menstrual cycle parameters, reduction in fetal growth, sperm quality changes, and respiratory effects—particularly among individuals highly exposed to swimming pool environments, such as lifeguard and professional swimmers. More recently, Shi et al. (8-9) demonstrated a clear positive dose-response relationship: higher concentrations of total trihalomethanes (TTHMs) in drinking water, longer exposure duration, and greater cumulative intake were each associated with progressively increased cancer risk. Their analysis also showed that every 10 µg/L increase in TTHM corresponds to a measurable rise in overall cancer incidence, with bladder cancer exhibiting a strong linear association (RR= 1.08 per µg/L). Notably, cancer risk increased substantially above 40 µg/L for men and 55 µg/L for women, indicating that chronic DBP exposure is a meaningful contributor to cancer.

Lead

Lead is one of the seven legacy contaminants of concern due to its well-documented toxicity across multiple organ systems. Lead adversely affects the reproductive, renal, hepatic, immune, and nervous systems, and that the renal cortex is the primary target organ for lead-induced carcinogenesis Silbergeld (317). Silbergeld highlights that lead's carcinogenic potential likely roots from indirect and permissive processes rather than direct DNA mutation: lead can impair DNA repair fidelity, induce chromosomal damage, promote oxidative stress, inhibit key zinc-dependant transcription factors, and disrupt gene regulation. It is important to

note, however, while animal studies demonstrate clear renal and possible brain carcinogenicity, gaps still remain; such as the absence of studies testing carcinogenicity in the developing nervous systems of rats. Human evidence further supports concern for cancer risk. In the nationally representative NHANES III cohort, Schober et al. (1540) found that even moderate blood lead levels (5-9 µg/dL) were associated with a 44% increased risk of cancer mortality, and levels less than or equal to 10 µg/dL were associated with a 69% increase, demonstrating a strong dose-response pattern. Together, these epidemiological findings reinforce that lead acts as a facilitator of carcinogenesis.

Nitrate

Nitrate is a widely occurring environmental contaminant, particularly in agricultural regions where fertilizer runoff infiltrates groundwater and drinking water systems. The most common form of consumption of nitrogen fertilizers is through ingestion of contaminated water. High doses of nitrates have been associated with the competitive inhibition of iodine, which can induce hypertrophic changes in the thyroid (Ward, 2). Epidemiological studies have linked chronic exposure to elevated nitrate levels in drinking water with increased risks of stomach, bladder, ovarian, kidney, and colorectal cancers, but several studies have found statistically significant associations for only brain and colon cancers and no statistically significant associations with other cancers including breast, pancreatic, kidney, bladder, and esophageal (Essien et al., 4-5).

Uranium

Other naturally occurring elements in groundwater can also contribute to long-term health risks. Uranium is a heavy metal present in soils and groundwater, and human exposure primarily occurs through ingestion of contaminated drinking water. Once absorbed, uranium exerts both chemical toxicity and radiological effects, with the kidneys serving as the principal target organ. High or prolonged exposure damages the renal proximal tubules and can lead to acute nephrotoxicity, tubular dysfunction, and renal failure (Bjørklund, 2). Beyond these impacts, ingested uranium can also cause gastrointestinal symptoms, hematological changes, hepatotoxicity, and accumulation in bone, where approximately two-thirds of the body burden resides for up to 1.5 years, potentially altering bone turnover (Kurtio et al., 72; Bjørklund, 2). While the basis for uranium's potential carcinogenicity is established, recent epidemiological evidence indicates that associations between naturally occurring radionuclides in drinking water—particularly uranium, radium, and radon—and cancer risk remain inconsistent and methodologically limited. Nonetheless, due to uranium's ability to deposit in high-dose tissues and emit alpha radiation, its presence in drinking water continues to be considered a long-term carcinogenic risk.

PFAS

Per- and polyfluoroalkyl substances (PFAS) are persistent “forever chemicals” that accumulate in water systems and the body, where they have been linked to a broad spectrum of adverse health effects. As summarized in Li et al. (426), PFAS exposure disrupts endocrine pathways, alters lipid metabolism, induces oxidative stress, and can cause epigenetic modifications that influence immune, reproductive, and metabolic function. Li et al. (428-429) found positive associations between PFAS in drinking water and multiple cancers: PFBS (Perfluorohexane Sulfonic Acid) was associated with a 33% increase in oral (pharyngeal) cancer, PFNA (Perfluorononanoic Acid) and PFHpA (Perfluoroheptonic Acid) were linked to endocrine cancers, particularly thyroid cancer, PFBA (Perfluorobutane Sulfonic Acid) and PFHxS (Perfluorohexane Sulfonic Acid) were associated with digestive system cancers (including colon, rectum, and liver), and PFOA (Perfluorooctanoic Acid) exposure predicted higher lung cancer incidence. The National Cancer Institute’s Division of Cancer Epidemiology and genetics (DCEG) further highlights that PFAS such as PFOA and PFOS are classified as carcinogenic or possibly carcinogenic, noting that PFAS can promote tumor formation through hormone disruption, chronic inflammation, and DNA methylation changes. Recent reviews also show that PFAS accumulate in blood and tissues and persist long enough to exert chronic biological effects, even at low environmental doses. Sinclair et al. (11) note that while many laboratory toxicity thresholds are higher than typical environmental concentrations, sub lethal and chronic molecular effects remain poorly characterized and could meaningfully contribute to long-term disease burden, indicating that risk at real-world exposure levels may be underestimated. Altogether, these findings reinforce that PFAS contamination in drinking water presents a measurable cancer hazard.

Research Question

To what extent does exposure to contaminants in Community Water Systems possibly precipitate disease and health outcomes, and especially influence rates of pre-mature birth and post-birth developmental complications in populations?

Data & Methods

Data for this study were collected from the National Environmental Public Health Tracking Network, maintained by the Centers for Disease Control and Prevention (CDC). This network integrates health, exposure, and environmental hazard data from national, state, and local sources to monitor the relationship between environmental contaminants and health outcomes. Specifically, the Drinking Water Dashboard was utilized to obtain data regarding various contaminants—such as arsenic, disinfection by-products, nitrates, and uranium—as well as childhood lead poisoning, total cancer incidence, and reproductive and birth outcomes. This dataset includes information on the people served by CWS and certain measures based on health indicators.

To investigate the public health impact of drinking water quality, data was organized by the type of health indicator. For each contaminant and health indicator, CWS were divided into

1) two groups based on whether they exceeded or fell below federally mandated regulatory thresholds or 2) two groups based on whether they exhibited any contamination or some contamination regardless of exceedance of thresholds. A two-tailed two-sample t-test was employed to determine if statistically significant differences existed between these groups. Prior to performing the t-tests, the equality of variance had to be evaluated using Levene's test to determine whether to use the t-test of equal or unequal variance. Depending on the resulting p-value of Levene's test (threshold of $p=0.05$), either a t-test of equal variance or unequal variance was selected. Health outcomes were considered to have statistically significant differences if the t-test yielded p-values of less than 0.05, allowing for the rejection of the null hypothesis with at least a 95% confidence. However, the t-test with a p-value of greater than 0.05 was considered to exhibit no statistically significant difference with its indicators, meaning that there were no statistically significant differences in health outcomes for the two groups tested.

Results

Table 1. T-test Results for Disinfection By-Products in CWS

	Counties that had no detectable levels of DBP	Counties that had detectable levels of DBP
Percent very low birth weight of live singleton births (<2,5000 g) over a 5-year period: $p = 0.2183$	1.096	.9621805
Percent of babies who were born very preterm (< 22 weeks gestation) over a 5-year period: $p = 0.4285$	1.176	1.079429
Annual infant (< 1 year of age) mortality rate per 1,000 live births over 5 years: $p = 0.0001$	3.831667***	5.548819***
Average age-adjusted prevalence of cancer (2020-2021) $p = 0.0435$	5.491667**	6.125347**

Note: * = $p < 0.1$; ** = $p < 0.05$; *** = $p < 0.01$

Table 2. T-test Results for Nitrates in CWS - Lower Thresholds to Categorize Counties

	Counties that had no detectable levels of Nitrates	Counties that had detectable levels of Nitrates
Percent very low birth weight of live singleton births (<2,5000 g) over a 5-year period: p = 0.3564	1.055	0.9630303
Percent of babies who were born very preterm (< 22 weeks gestation) over a 5-year period: p = 0.5826	1.141667	1.080216
Annual infant (< 1 year of age) mortality rate per 1,000 live births over 5 years: p = 0.0003	4.072857***	5.549021***
Average age-adjusted prevalence of cancer (2020-2021) p = 0.0603	5.6*	6.124476*

Note: * = $p < 0.1$; ** = $p < 0.05$; *** = $p < 0.01$

Table 3. T-test Results for Nitrates in CWS - Higher Thresholds to Categorize Counties

	Counties that had less than the maximum contaminant level for Ar	Counties that had more than the maximum contaminant level for Ar
Percent very low birth weight of live singleton births (<2,5000 g) over a 5-year period: p = 0.8576	0.9652679	0.9746154
Percent of babies who were born very preterm (< 22 weeks gestation) over a 5-year period: p = 0.8665	1.081008	1.090769
Annual infant (< 1 year of age) mortality rate per 1,000 live births over 5 years: p = 0.0001	5.628065***	4.774615***
Average age-adjusted prevalence of cancer (2020-2021) p = 0.0005	6.147984***	5.871154***

Note: * = $p < 0.1$; ** = $p < 0.05$; *** = $p < 0.01$

Table 4. T-tests for Arsenic in CWS

	Counties that had less than the maximum contaminant level for Ar	Counties that had more than the maximum contaminant level for Ar
Percent very low birth weight of live singleton births (<2,5000 g) over a 5-year period: p = 0.8576	0.9652679	0.9746154
Percent of babies who were born very preterm (< 22 weeks gestation) over a 5-year period: p = 0.8665	1.081008	1.090769
Annual infant (< 1 year of age) mortality rate per 1,000 live births over 5 years: p = 0.0001	5.628065***	4.774615***
Average age-adjusted prevalence of cancer (2020-2021) p = 0.0005	6.147984***	5.871154***

Note: * = p < 0.1; ** = p < 0.05; *** = p < 0.01

Table 5. T-tests for Uranium in CWS

	Counties that had less than the maximum contaminant level for Uranium	Counties that had more than the maximum contaminant level for Uranium
Percent very low birth weight of live singleton births (<2,5000 g) over a 5-year period: p = 0.9295	0.9678761	0.9632
Percent of babies who were born very preterm (< 22 weeks gestation) over a 5-year period: p = 0.8051	1.08025	1.0948
Annual infant (< 1 year of age) mortality rate per 1,000 live births over 5 years: p = 0.0000	5.67536***	4.504***
Average age-adjusted prevalence of cancer (2020-2021) p = 0.0000	6.1552***	5.824***

Note: * = p < 0.1; ** = p < 0.05; *** = p < 0.01

Table 6. T-tests for PFAS in CWS

	Counties with no PFAS reported	Counties with PFAS reported
Percent very low birth weight of live singleton births (<2,5000 g) over a 5-year period: p = 0.0537	0.9491228*	1.052083*
Percent of babies who were born very preterm (< 22 weeks gestation): p = 0.0484	1.063306**	1.180833**
Annual infant (< 1 year of age) mortality rate per 1,000 live births over 5 years: p = 0.1044	5.5432 ???	5.1648 ???
Average age-adjusted prevalence of cancer (2020-2021) p = 0.0087	6.1264***	5.968***

Note: * = $p < 0.1$; ** = $p < 0.05$; *** = $p < 0.01$

Discussion

As expected, counties that had detectable levels of DBP had a statistically significant higher average annual infant mortality rate per 1,000 live births over 5 years (5.548819) than counties that had no detectable levels of DBP (3.831667) at a p-value of 0.0001 or at a 99% certainty level. Similarly, counties that had detectable levels of DBPs had a statistically significant higher prevalence of cancer (6.125347) than counties that had no detectable levels of DBPs (5.491667) at a p-value of 0.0435 or at a 95% certainty level (see Table 1). Also as expected, counties that had detectable levels of nitrates had a statistically significant higher average annual infant mortality rate per 1,000 live births over 5 years (5.549021) than counties that had no detectable levels of nitrates (4.072857) at a p-value of 0.0003 or at a 99% certainty level. Similarly, counties that had detectable levels of nitrates had a statistically significant higher average prevalence of cancer (6.124476) than counties that had no detectable levels of nitrates (5.6) at a p-value of 0.0603 or at a 95% certainty level (see Table 3).

However, when this research divided counties according to the higher threshold amount detected for arsenic (i.e., counties with less than the maximum contaminant level vs counties with more), opposite results were obtained. This difference reveals how the criteria we use to categorize groups can possibly impact results.

Counter to what was expected, counties that had more than the maximum contaminant level reported for arsenic had a statistically significant lower average annual infant mortality rate per 1,000 live births over 5 years (4.774615) than counties that had less than the maximum contaminant level reported for Ar (5.628065) at a p-value of 0.0001 or at a 99% certainty level (see Table 4). Similarly, counties that had more than the maximum contaminant level for arsenic had a statistically significant lower average prevalence of cancer (5.871154) than counties that had less than the maximum contaminant level for Ar (6.147984) at a p-value of 0.0005 or at a 99% certainty level.

A similar pattern was observed for uranium. Counties that had more than the maximum contaminant level for uranium also had a statistically significant lower average annual infant mortality rate per 1,000 live births over 5 years (4.504) than counties that had less than the maximum contaminant level for uranium (5.67536) at a p-value of 0.0000 or at a 99.99% certainty level. Similarly, counties that had less than the maximum contaminant level for uranium had a lower average prevalence of cancer (5.824) than counties that had more than the maximum contaminant level for uranium (6.1552) at a p-value of 0.0000 or at a 99.99% certainty level (see Table 5).

PFAS Contamination produced mixed results. As expected, counties that had PFAS reported at least once had a statistically significant higher percentage of low birthweight over a 5-year Period (1.052083) than counties that had no PFAS reported at least once (0.9491228) at a p-value of 0.0537. Likewise, counties that had PFAS reported at least once had a higher percentage of babies who were born very preterm (1.180833) than counties that had NO PFAS reported at least once (1.063306) at a p-value of 0.0484 or at a 95% certainty level (see Table 6). However, counter to what was expected, counties that had PFAS reported at least once had a lower average prevalence of cancer (5.968) than counties that had NO PFAS reported at least once (6.1264) at a p-value of 0.0087 or at a 99% certainty level.

Conclusion

Our data consistently finds that water contamination leads to statistically significant differences in the annual infant mortality rate and age-adjusted prevalence of cancer in NJ, NY, and PA counties. However, the different rates for counties by specific water contaminants led to expected and unexpected results. Higher rates of infant mortality and cancer were associated with detectable contamination by disinfection by-products and nitrates in community water systems as expected, but lower rates were associated with contamination by arsenic, uranium, and PFAS.

This research also finds that there are no statistically significant associations between 1) the rate of very low birth weight or 2) the rate of babies born very preterm and water contamination. The only exception was for PFAS contamination, which was found to be statistically associated with higher rates of very low birth weight and higher rates of babies born very preterm, at p-values of 0.0537 and 0.0484 respectively.

These results point to the likelihood that counties with detectable contamination by disinfection by-products and nitrates in community water systems are different from counties with detectable contamination by arsenic, uranium, and PFAS. What could be occurring is that counties with arsenic, uranium, and PFAS contamination have better access to healthcare, which in turn helps mitigate health concerns. This would be a potentially fruitful line of research to pursue in the future.

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The Hidden Cost of Fashion: How Globalisation Shapes Sweatshops Worldwide

By Natasha Bhatia

Abstract

Fast fashion has become a major feature of the global economy, driven by low prices, rapid trend cycles, and globalised supply chains. In many countries, clothing collections now change weekly rather than seasonally, encouraging consumers to treat garments as disposable rather than durable goods. Companies such as Shein, Zara, and H&M encourage overproduction and overconsumption, transforming how clothing is produced, consumed, and discarded. From raw material extraction to garment disposal, fast fashion contributes to environmental degradation through excessive water use, chemical pollution, high carbon emissions, and textile waste, while also relying on the exploitation of low-wage labour. This raises the question: how has globalisation shaped the development of sweatshops and environmental harm across both the Global South and the Global North, and why does exploitation in wealthier countries receive far less attention? This article argues that globalisation has created an interconnected production system that shifts environmental damage and labour exploitation across borders, concentrating visible harm in the Global South while concealing similar patterns within the Global North. By comparing labour and environmental impacts across both regions, this article demonstrates that fast fashion's harms are structural and global rather than isolated to developing economies.

Outline

- Globalisation as the link between the Global South and Global North
- Sweatshops in the Global South
- Sweatshops in the Global North
- Comparing environmental and labour impacts across North and South
- Conclusion

Introduction

Consider a T-shirt. Before reaching the consumer, it would have travelled across many countries. Cotton grown in India, spun into yarn in Indonesia, woven and manufactured in China, and finally sold and worn in the United States. This complex worldwide journey is made possible by globalisation, which has also fuelled the rise of fast fashion. By enabling low-cost production, rapid manufacturing, and worldwide distribution, fast fashion has changed the clothing industry into one characterised by speed, low cost, and disposability. When you walk past a store, the clothing on display is likely to be replaced within a week by a new trend produced on the other side of the world, all made possible by globalization. This has allowed firms to design, manufacture, and export goods at an unprecedented speed. This rapid expansion of fast fashion has created serious consequences for the environment, workers, and long-term economic sustainability, suggesting that globalisation has played a central role in developing an industry that does more harm than good. Fast fashion is highly demanding and polluting. The industry

contributes around 10% of global greenhouse gas emissions and is one of the largest consumers of water, requiring thousands of litres for just one garment (UNEP; Niinimäki et al.). Toxic dyes and chemical waste from textile processing contaminate water systems, especially in countries where environmental protections are weak (Sallai and Pepper). While existing literature extensively documents the environmental intensity of fast fashion production, much of this research focuses primarily on pollution in garment-producing countries in the Global South. There is comparatively less analysis examining how globalised consumption patterns in the Global North sustain and normalise these environmental pressures across supply chains.

These clothes are produced in sweatshops where workers face long hours, harsh conditions, and are often paid below-average wages. Bangladesh has over 4,000 documented garment factories where poor working conditions are frequently reported and widely shown in the media, reflecting patterns seen across much of the Global South. Whilst health and safety laws exist, they are rarely followed (Hossain). However, countries in the Global North, such as the UK, the US, and those within the EU, also contain thousands of garment workshops that have similarly harsh working conditions and low wages. This highlights how sweatshop labour is not confined to one region but is instead a structural issue within the global fashion industry. Multinational fashion brands often prioritise low production costs and fast turnaround times, which puts pressure on suppliers to reduce expenses, frequently at the expense of workers' rights and safety. In countries like Bangladesh, weak enforcement of labour laws allows unsafe practices to persist, while in the Global North, subcontracting and informal workshops enable companies to avoid accountability. This suggests that sweatshops are sustained not only by unfair local regulatory failures but also by global consumer demand for cheap clothing and corporate profit-maximisation. These cases receive far less attention, partly because they challenge the assumption that labour exploitation is confined to developing countries. Although environmental damage in the Global North is less visible and less frequently reported, it remains widespread. Pollution arises from the improper disposal of dyes, chemicals, and textile waste, as well as high carbon emissions caused by energy-intensive machinery, heating, and frequent transportation within supply chains.

With rapid globalisation, advancements in the fashion industry caused a major rise in the number of sweatshops present. Social media and existing literature often talk about sweatshops in the Global South, e.g. Bangladesh, China, etc. This makes people think workers are only exploited in those places, although that is not the case. Sweatshops are also commonly found in the Global North, yet nobody pays nearly as much attention to them, even though they are just as harmful for workers and the environment. This article explores the key similarities and differences between sweatshops in the Global South and those in the Global North, and examines why exploitation in wealthier countries receives far less attention in public discourse. Ultimately, ignoring sweatshops in the Global North allows exploitation to persist unchecked and reinforces a false narrative that injustice only exists elsewhere. This paper argues that the hidden costs of sweatshops persists all around the world, not just in the global south.

Globalisation as the link between the Global South and Global North

Globalisation has increased connectedness between countries in the Global South and Global North, especially in the fast fashion industry. It enables materials, production, and clothing to move between countries quickly, creating global supply chains where different parts of a garment are made in different places (Li). Brands in the Global North outsource production to countries where labour and materials are cheaper, while selling the finished products in their own markets. This system has been adopted due to the need to keep costs low and produce clothes quickly to meet constantly changing trends. For example, over 80% of the global garment workforce is based in the Global South, with major production hubs in Bangladesh, India, and Vietnam, where garment workers often earn less than \$100 per month (ILO). Whilst consumers in the Global North benefit from this by giving them affordable clothes and new styles all the time, it also creates risks for workers in both regions (Kühl et al.). Many employees receive low wages, work long hours, and experience unsafe working conditions. The environment suffers because of water and energy use, chemical pollution, and textile waste. An example is the 2013 Rana Plaza factory collapse in Bangladesh, which killed over 1,100 garment workers producing clothes for large companies, including Primark (BBC News). This incident demonstrates the extreme consequences of weak regulation and corporate pressure within the fast fashion supply chain. In spite of visual cracks in the building, workers were forced to enter to meet production targets for Global North brands, showing how profit and speed were prioritised over human safety. Primark was also linked to a sweatshop scandal in 2008, where workers in India were found to be paid extremely low wages under poor conditions (BBC Panorama). This interconnected system has helped fast fashion grow rapidly, it has also contributed to the rise of sweatshops in both the Global South and the Global North.

Globalisation has also increased the dependence of countries on each other in the fashion industry as many countries in the Global South provide raw materials, labour, and manufacturing expertise, while countries in the Global North focus on design, branding, and marketing. This creates a system where the economies of both regions are affected by the other. For example, delays or price changes in raw materials can disrupt production worldwide, showing how closely connected the two regions have become. Globalisation has also encouraged the growth of large multinational corporations that control multiple stages of production and distribution, giving them the ability to control terms with smaller suppliers. This abuse of power generally controls what gets produced, and what labour and environmental standards are followed, in spite of regulations being put in place in most countries. Additionally, global competitiveness encourages brands to constantly lower costs and innovate, which can further amplify the level of pressure on workers and resources. These links show that globalisation doesn't just make trade easier, as it is commonly said. It has increased interdependence between the Global South and Global North, where changes in one region can have a knock-on effect on the other.

Sweatshops in the Global South

Sweatshops in the Global South are a feature of the fast fashion industry. These factories operate in countries such as Bangladesh, India, and China. These factories are a part of globalised supply chains, as they produce clothing for multinational firms that are based mainly in the Global North (Hayat). Most factories in the Global South operate under high pressure to keep costs low and meet strict deadlines set by global brands. Consequently, workers generally tend to experience extremely long working hours, low wages below minimum wage requirements, and dangerous, unsafe conditions, in spite of labour laws existing in many of these countries. Health and safety regulations are frequently overlooked, which has led to many accidents and ongoing risks for workers. Environmental damage is also common, as the production of textiles in these regions involves the release of harmful chemical waste and dyes into local water systems (Ekenga). Sweatshops in the Global South receive significant media attention, which has created a public perception of labour exploitation as only a problem in developing countries.

Sweatshops in the Global South are also affected by global economic dependence on the fast fashion industry. In many countries, garment production accounts for a large percentage of employment and export income, which makes governments hesitant to impose strict regulations on factories, as doing so could risk capital flight and job losses (Cambridge Core). This structural dependence limits the power of states to prioritise labour rights over economic growth. This dependence allows multinational brands to move production to other countries if costs rise, giving them strong bargaining power over suppliers. NGOs such as the Clean Clothes Campaign and Human Rights Watch highlight that this power imbalance enables brands to demand low prices and short turnaround times while avoiding responsibility for labour conditions further down the supply chain (Clean Clothes Campaign). As a result, factory owners may cut corners on wages, safe working conditions, and environmental standards to remain competitive.

Workers often have limited alternatives for employment, which reduces their ability to escape unfair conditions or organise for more rights. The International Labour Organization (ILO) reports that weak enforcement of labour laws and high levels of informal employment in countries that produce garments further undermine workers' bargaining power and access to protections (ILO). In addition, some production takes place in unregistered factories that operate outside official oversight, increasing the risk of exploitation and unsafe conditions (Cambridge Core). These conditions show how sweatshops in the Global South are not just caused by local factory practices, but due to global economic pressures that prioritise competitiveness, flexibility, and profit over worker welfare.

Sweatshops in the Global North

Sweatshops are not only found in the Global South, but they also exist in the Global North, although they are far less visible and rarely discussed in the media. A well-known example is the Leicester garment industry in the UK, where investigations found workers being paid as little as £3–£4 an hour, well below the legal minimum wage, which is £12.21 an hour (Gov.UK). Unsafe working conditions were also found, particularly during the COVID-19

pandemic. Reports by NGOs such as Labour Behind the Label indicate that informal production, migrant labour, and integration into fast fashion supply chains are structural features of the garment industry rather than isolated abuses. Similarly, the UK “Operation Tacit” investigation shows the limitations of enforcement in supposedly well-regulated markets, as the absence of prosecutions despite widespread inspections suggests that labour exploitation can remain hidden and unchallenged. Together, these findings show that regulatory frameworks alone are insufficient to prevent abuse, particularly when economic incentives prioritise low costs and rapid production (Business & Human Rights Resource Centre). Historical cases in the United States also show a similar incident where armed police raids in El Monte, California, uncovered an unauthorised garment sweatshop, where 72 Thai workers were held in near-slavery conditions, forced to work long hours for inadequate pay under surveillance and threats (Bales). Despite these incidents, there is relatively little academic research or media coverage focusing on sweatshops in the Global North compared to the extensive attention given to those in the Global South.

The lack of attention given to sweatshops in the Global North makes it hard to understand how common labour exploitation really is within the fast fashion industry (Rights Lab). One reason for this is the widespread assumption that strong labour laws and higher wages in wealthier countries prevent sweatshop conditions from existing, which reduces both media interest and public concern (Taplin). When cases such as Leicester are treated as isolated scandals rather than part of a wider issue, the problem appears less serious than it actually is (Business & Human Rights Resource Centre). In addition, exploitation in the Global North is often hidden within informal supply chains, subcontracting, and the use of migrant labour, making it harder to identify and report (Bales). This limited coverage also leads to less pressure on governments and brands to take accountability or introduce stronger enforcement (ILO). As a result, poor working conditions can continue unnoticed, even in countries that are assumed to have strong labour laws. This shows that sweatshops are not just a problem of poorer countries, but a hidden issue within wealthier economies as well (Taplin).

Comparing environmental and labour impacts across North and South

When comparing the environmental and labour impacts of fast fashion across the Global North and Global South, clear similarities and differences can be seen. In the Global South, labour exploitation is often more visible and severe, with low wages, long working hours, unsafe factories, and weak enforcement of labour laws being common (ILO). These conditions are reinforced by economic dependence on garment exports, which limits governments’ willingness to impose stricter regulations for fear of losing investment. Environmental damage is also concentrated in these regions, as textile dyeing, water pollution, and waste disposal are frequently outsourced to countries with fewer environmental protections, resulting in long-term harm to local ecosystems and public health (UNEP). This demonstrates how the Global South bears a disproportionate share of the social and environmental costs of fast fashion production.

In contrast, the Global North is often perceived as having stronger labour and environmental regulations, creating an assumption that exploitation is largely absent. However, evidence suggests that abuse still exists, though in more concealed forms. Sweatshops in the Global North, such as those documented in Leicester, show that low pay, informal contracts, and poor working conditions can persist despite legal protections, largely due to weak enforcement, subcontracting, and the use of vulnerable migrant labour (Business & Human Rights Resource Centre). Environmentally, while direct pollution from production is less visible in the Global North, high levels of consumption fuel overproduction, textile waste, and carbon emissions across global supply chains. This indicates that environmental harm is not reduced but displaced, reinforcing the idea that fast fashion creates interconnected impacts across both regions rather than isolated problems limited to the Global South (Niinimäki et al.).

Conclusion

Fast fashion is often presented as a measure of globalization. However, as this article has shown, its global success is built upon a system that shifts social and environmental costs across borders. Globalisation has connected the Global South and Global North through complex supply chains that allow brands to minimise costs, accelerate production, and maximise profit. In doing so, it has enabled the expansion of sweatshops, environmental degradation, and structural inequalities that are embedded within the industry itself. While existing literature heavily documents labour abuse and environmental pollution in the Global South, there is significantly less research and public attention focused on sweatshops and environmental harm within the Global North. This gap is dangerous because it reinforces the assumption that exploitation is confined to poorer countries, allowing similar practices in wealthier economies to remain hidden and normalised. When cases in places such as Leicester or El Monte are treated as isolated scandals rather than structural issues, the broader pattern of exploitation goes unnoticed. The lack of sustained academic and media attention reduces pressure on governments and corporations to improve enforcement, meaning that informal labour, underpayment, unsafe working conditions, and migrant worker exploitation can continue with limited accountability.

Furthermore, the Global North contributes to environmental harm not only through hidden production sites, but through high levels of consumption, overproduction, and textile waste. The demand for constantly changing trends drives excessive manufacturing, carbon-intensive transportation, and the disposal of millions of garments each year. Even when pollution is geographically displaced to production countries, it is economically driven by consumption patterns in wealthier markets. Ignoring this connection separates responsibility from impact and prevents a full understanding of how globalised supply chains operate. By comparing labour and environmental impacts across both regions, this article addresses the limited focus within existing literature and highlights the interconnected nature of fast fashion's harms. Recognising exploitation in the Global North does not minimise the severe conditions faced in the Global South; instead, it reveals that these issues are systemic rather than regional. Ultimately, meaningful change requires acknowledging that responsibility lies across the entire

supply chain. Without confronting the hidden costs within wealthier economies as well as developing ones, fast fashion's model of speed, low cost, and disposability will continue to reproduce inequality and environmental damage on a global scale.

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Humanized vs. Dehumanized Opponents in First Person Shooter Games: How Character Design Impacts Adolescent Aggression By Sophia Hijjawi

Abstract

Violent video games are games where the player intentionally inflicts harm upon other players or characters. Most often, these games are first-person shooters, where gameplay is from a first-person perspective of the character inflicting harm. The relationship between first-person shooters and real-world violence is a highly contested topic. Investigating the general relationship between violent gaming and real-world violent behavior was out of this project's scope in terms of complexity and resources. Instead, it focused on how the character design of in-game targets influences aggressive reactions towards those characters. In this study, aggression was defined as the decision to react violently to a character and the speed of those violent reactions. Reaction time and mistake rates were used as measures of aggression because, given the implicit study design, faster or mistaken reactions to hit a target reflect automatic hostile responses. Character designs differ by their humanity ("most human" vs. "least human"). This within-subject experiment asks participants to play a simple shoot/no shoot game to determine if there are significantly differing reactions to more or less human character designs. This study found some evidence that participants expressed more automatic hostile reactions towards less human character designs. Due to limitations of sample size and time in this study, further research is needed to verify these findings.

Background

In the United States, the prevalence of adolescent violence—school shootings in particular—has raised concerns about the impact of violent gaming on aggressive behavior (Anderson and Dill 772, 789; "Mass Shooting Factsheet"; Matthews; Molina). This question has only increased in importance as years pass and violent video games continue to gain popularity. While Markey et al.'s investigation into the relationship between violent video game releases and violent crime rates in the U.S. found no relationship, Anderson and Dill's study displayed a positive correlation between violent and delinquent behavior as a result of playing violent games (Markey et al. 14-16, Anderson and Dill 788-789). There is persuasive evidence that the impact of violent video game exposure on behavior and cognition is negligible, and has an exaggerated reported effect due to publication bias and interference from other variables (Borrego-Ruiz and Borrego 1, 17-18; Willoughby 61). The conflict of current evidence in the face of rapidly proliferating adolescent violent video game exposure makes clear the need for deeper investigations of the true relationship between gaming and violence (Borrego-Ruiz and Borrego 3).

This field is difficult to investigate for technical and ethical reasons. Aggression is hard to measure or define because of social desirability bias and overlap with other angry, irritable, or alert behavior (APA Dictionary of Psychology, Evans et al. 278). In a lab setting, another major challenge is eliminating confounding variables such as competitiveness, difficulty, and pace of

action, which can strongly affect research findings (Willoughby 56, 58-60; Dowsett and Jackson 22-23). Ethically, the potential of causing a violent change to a participant's personality makes it problematic to investigate the effect of long-term gaming on aggressive personality or serious violent crime like assault or school shootings (Griffiths 204).

While the topic has proven to be a challenge to study, there are a few popular theories about the impact of gaming on personality and about aggression as a whole directing current research. One of these theories is the social learning theory, a prominent psychological framework, which predicts that exposing an individual to violent behavior, even through a screen, encourages that behavior and desensitizes them to it (Griffiths 204, Borrego-Ruiz and Borrego 4-6). This concern is emphasized for adolescents, who may learn inappropriate behavior from violent games in a vital period of life when moral and ethical values are developing (Borrego-Ruiz and Borrego 7). Contributing to social learning theory are theories about moral disengagement factors, which are hypothesized to be the underlying cause of real-life aggression increasing as a consequence of violent gaming. Moral disengagement factors are aspects of violent games that decrease irritation, guilt, or disgust in players. They increase the allure and social acceptability of violent games by justifying the player's violent acts. Some of these strategies include euphemistic labeling of violence; attributing blame to victims, adversaries, or circumstances; and dehumanizing victims of violence, reducing guilt by stripping targets of humanity (Hartmann et al. 310-311, 324).

While prior research has broadly explored aggression and gaming, few studies isolate singular variables like character dehumanization, the principal focus of this project. This study focuses on the impact of NPC design—humanized or dehumanized—on adolescent aggression. This study reduces the impact of social desirability bias on findings by using an implicit measure of aggression, a strategy employed in studies investigating racially-motivated violence and instantaneous perceptions of danger in a crowd (Alt et al. 154, Correll et al. 1324-1328, Payne et al.). Implicit, or automatic, reactions are being used in this study in order to investigate unconscious and immediate hostile biases and associations. Correll et al.'s study, which dug into the issue of racially-motivated police brutality using an implicit design, lent this study a vital framework for game design and methodology (1315-1317). Its simple "shoot/no-shoot" game design fit the purposes of this study very well; while it restricted the experiment from perfectly simulating a modern first-person shooter game, it allowed for the sole investigation of the effect of NPC design on aggressive behavior.

This study investigated subconscious differences in adolescent perceptions of less or more human NPC designs through participants' reaction time when shooting at said NPCs. Understanding how adolescents express violence toward different character designs will contribute to future gaming and behavioral research.

Research Question and Hypotheses

To what extent do adolescents aged 14-19 implicitly react to humanized vs. dehumanized targets of violence in first-person shooter video games with measurably different aggression?

Before conducting the experiment, it was hypothesized that: A) participants will react more quickly to shoot armed dehumanized character designs than armed humanized character designs, B) participants will react more quickly to spare unarmed humanized character designs than unarmed dehumanized designs, C) participants will more often incorrectly shoot unarmed dehumanized character designs than unarmed humanized character designs, and D) participants will more often incorrectly spare armed humanized designs than armed dehumanized designs.

Methodology

1. Participants

Ages of participants ranged between 14 and 19 years old. They were recruited from a high school in Salt Lake City, Utah, using flyers and email announcements. Participant gender varied, though a majority were female (16 woman/girl, 9 man/boy, 2 other). 22 had played a first-person shooter before. Some (n=9) had played one to two of the games from which stimuli were sourced, and one participant had played 6 of them.

2. Video Game Design

2.1 Instructions

To support participant wellbeing, the contact information of a mental health professional, who was employed at the school and therefore easily accessible, was given to them before commencing the experiment.

Before beginning the game, participants received verbal instructions on how to access the game and a general overview of what the game would have them do. In the game, participants were given more instructions, including specific instructions on how to play the game and an explanation of the point system.

2.2 Gameplay

The video game was based off of the study design originated by Correll et al., which was a shoot/no-shoot simulator (1314). The game was designed in PsychoPy and run through the Pavlovia website.

First, participants completed a practice block to ensure they understood the instructions. The practice block was one eighth of the real trial (12 stimuli compared to 96). The practice stimuli were completely different and unrelated to the ones being used in the trial of interest.

The characters that appeared on screen had humanized designs or dehumanized designs. There were 8 different character designs; each character appeared 12 times, 6 times armed and 6 times unarmed. They appeared at random places on the screen with a 1 second pause between. Participants assessed whether the figure on the screen was holding a firearm or a can, clicking their laptop's "k" key to shoot them and the "s" key to not

shoot. Participants were given 850ms to respond, the game moving on automatically if they didn't answer quickly enough. There was a 1500ms pause in between each stimulus.

2.3 Point System

Hits were +10 points. Correct Rejections were +5 points. False Alarms were -20 points. Misses were -40 points. Timeouts were -10 points. This point system, reflecting that from Correll et al.'s study, was chosen to force participants to respond quickly while reducing error and nonresponse (1314, 1317).

Effortful gameplay was encouraged by incentivizing scores above 650 with double the candy prize. Participants were not allowed to view their scores after the experiment concluded to avoid comparison, and scores were not displayed on screen at any point during the experiment. The researcher checked scores after the experiment concluded and privately rewarded those who reached 650 points the extra prize. In this study, no participants had enough points to win the double prize.

2.4 Opponent modification/manipulation

As previously stated, this game includes 8 character designs, each with 2 variations: armed and unarmed. Character designs were chosen from popular existing first-person shooter games with NPCs or opponents. Images were pulled from the Wikipedia pages for the selected games, which can be found in the "Works Cited" section. The labeling of a character as "humanized" or "dehumanized" was decided by a survey (n=8). Survey takers were asked to rank each character from 1-8, most to least human, without repeating any number. Characters were ranked by the sum of the character's ranking from each survey participant (see table 1). The top four most human designs (lowest cumulative ranking) were labelled as "humanized characters", and the four least human designs (highest cumulative ranking) were labelled as "dehumanized." This measure of humanity is purely exploratory, and further investigation is necessary to deem its validity.

Character	Final score (sum of all responses)	Character	Final score (sum of all responses)
 Battlefield 4 Brodie Paragon 4 Soldier	8	 Ultimate Marquell	42
 The Last of Us Ellie	17	 Ultimate Y1	46
 Borderlands 2 Combs	25	 Rainbow Six Siege X Breacher	55
 Metro Exodus Humanimal	34	 Halo: Master Chief Collection Stern Grunt	61

Table 1: Character source images sorted from most to least human. Image sources can be found in “Works Cited”

3. Procedure

Participants completed the experiment in an empty classroom when they had free time during school hours (e.g. lunchtime, office hours, etc.). After the experiment, all participants were debriefed and received their compensation (a candy bar) before returning to school.

4. Risks and Mitigation

4.1 Psychological Risks

One risk was triggering PTSD or invoking traumatic memories in people with a negative history with guns and gun violence. People with a history with gun violence or violent trauma were barred from participating in the study. All participants acknowledged potentially upsetting content in consent forms.

A second risk was causing a negative or violent personality change in participants. While prolonged exposure to violent video games is hypothesized to cultivate aggressive behavior, the relationship was found to be insignificant when other social factors were taken into account (Borrego-Ruiz and Borrego 12). Studies have shown that a single gaming session can temporarily prime aggressive thinking, but have no effect on long-term aggressive behavior or mindset (Anderson and Dill 788, Dowsett and Jackson 25-26).

As previously stated, another risk was harming participant self-esteem with score comparison, countered by barring participants from knowing their scores.

All participants provided informed consent (or obtained guardian informed consent and provided informed assent), were debriefed, and were aware of their right to back out of the experiment at any time. Data is anonymous, and all school ethical guidelines were adhered to.

4.2 Data Protection

Signed paper consent forms were kept secure in a lockbox in a locked closet in a secure classroom. Participant IDs were written on the consent forms, and no other personally identifying information exists beyond the IDs on the consent forms.

All data is stored offline on a password-protected laptop. The only personally-identifying information that exists is the consent forms, which will be shredded after seven years, when the anonymized data will also be deleted.

4.3 Compensation

As compensation for participating in this experiment, individuals received a king-sized candy bar; the options were widely varied to accommodate dietary restrictions. Participants who scored above 650 points were rewarded with two candy bars instead of one, although no participant in this study reached that threshold.

Results and Discussion

This study compared the speed and accuracy of reactions between humanized and dehumanized characters (see table 2). Matched-pair *t*-tests for differences were conducted for statistical analysis, comparing reaction time and mistake rate between the humanized and dehumanized conditions. *P*-values under the predetermined alpha level of .05 were determined to be statistically significant.

Across all sections of analysis, negative mean differences indicate more hostile reactions toward dehumanized characters.

One concern when analyzing the results was that the reactions of participants with a high Timeout rate were skewed due to random chance. To counter this, separate analyses were conducted for the entire group ($n=27$) and the smaller group ($n=20$) with Timeout rates above 25% .

	P-value (entire group, df=26)	Mean Difference (entire group, df=26)	Humanized mean (entire group, df=26)	Dehumanized mean (entire group, df=26)	T-value (entire group, df=26)	P-value (small group, df=19)	Mean Difference (small group, df=19)	Humanized mean (small group, df=19)	Dehumanized mean (small group, df=19)	T-value (small group, df=19)
Hits (ms)	0.053	-0.01	0.664	0.653	-1.679	0.171	-0.0074	0.652	0.644	-0.973
Correct Rejections (ms)	0.401	-0.0019	0.671	0.673	-0.254	0.493	-0.00017	0.6645	0.6646	-0.017
Misses (proportion of Misses to all armed reactions)	0.021	-0.036	0.13	0.093	-2.147	0.202	-0.014	0.114	0.099	-0.853
False Alarms (proportion of False Alarms to all armed reactions)	0.511	0.00055	0.125	0.125	0.029	0.299	-0.011	0.122	0.133	-0.535

Table 2: *P*-values, means, and *t*-values of *t*-tests comparing humanized/dehumanized mean Hit and Correct Rejection reaction times and the humanized/dehumanized mean Miss and False Alarm rates

When all participants were included, a statistically significant difference was observed between the Miss rates for humanized characters and the Miss rates for dehumanized characters (see table 2, fig. 1). The results found for the difference between Hit reaction times for humanized characters and Hit reaction times for dehumanized characters approached statistical significance (see table 2, fig. 2). No significant results were found in mean Correct Rejection reaction time differences of False Alarm rate differences (see table 2, fig. 3, fig. 4).

When participants with high Timeout rates (exceeding 25% of all trials) were excluded from analysis, no statistically significant results were found in any group (see table 2, fig. 1-4)

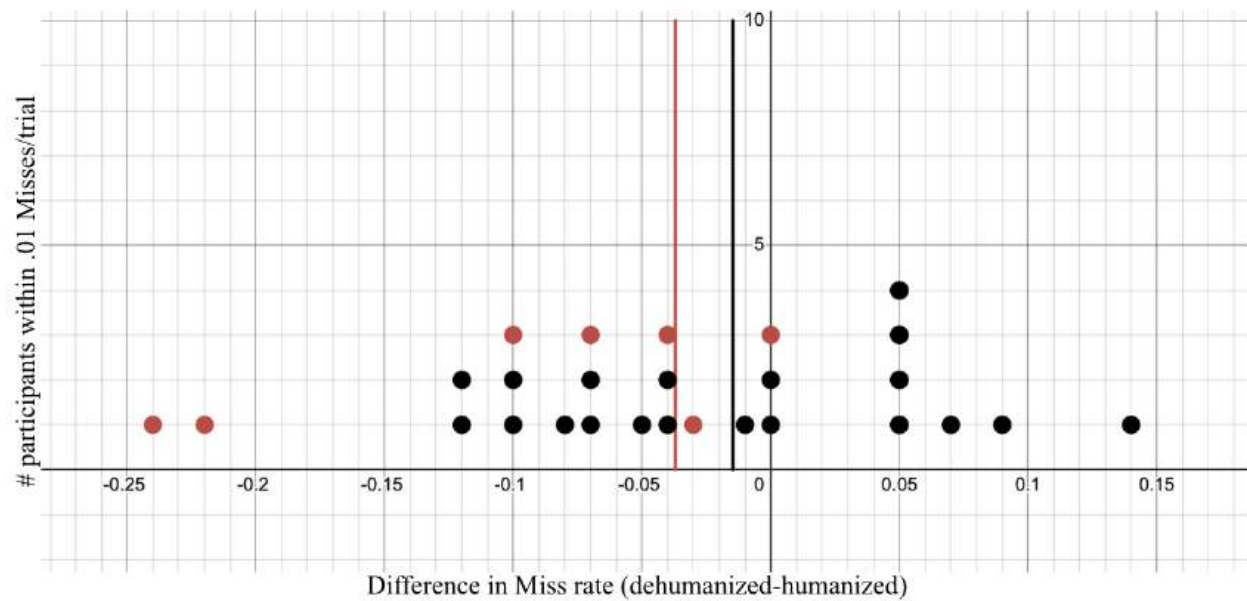


Figure 1: Dotplot depicting differences in mean Miss rate. Red dots indicate data excluded for low Timeout rate analysis. The black line indicates the whole population mean, and the red indicates the reduced population.

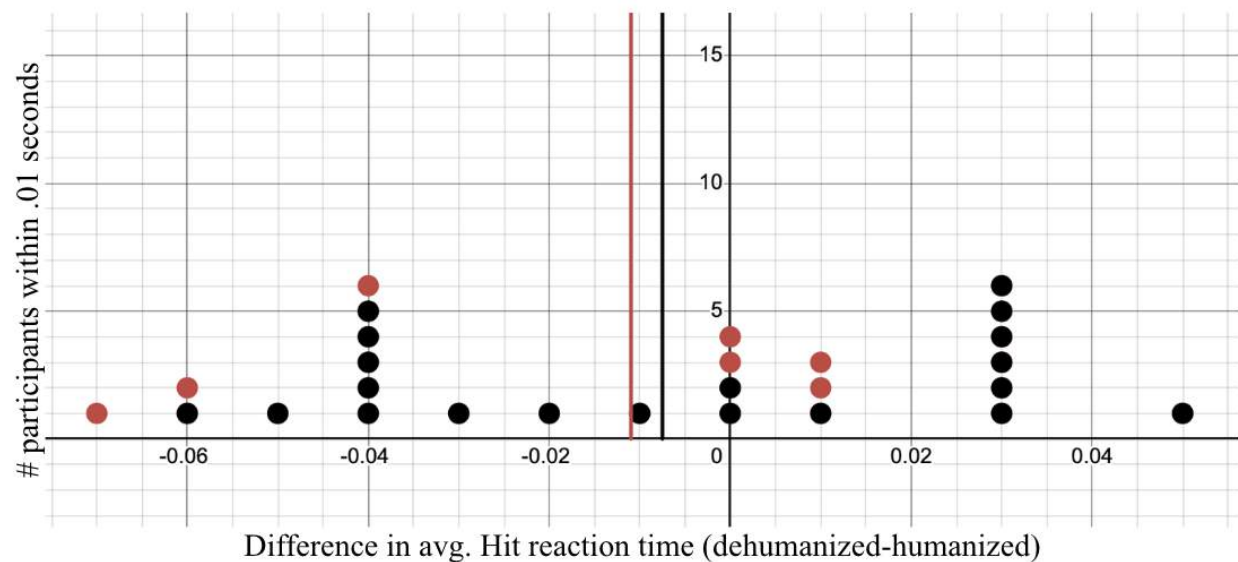


Figure 2: Dotplot depicting differences in mean Hit reaction times. Red dots indicate data excluded for low Timeout rate analysis. The black line indicates the whole population mean, and the red indicates the reduced population.

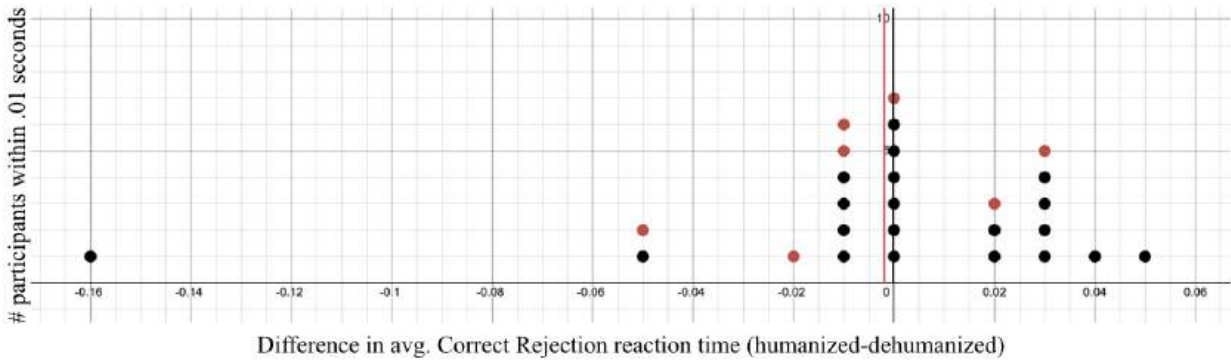


Figure 3: Dotplot depicting differences in mean Correct Rejection reaction times. Red dots indicate data excluded for low Timeout rate analysis. The black line indicates the whole population mean, and the red indicates the reduced population.

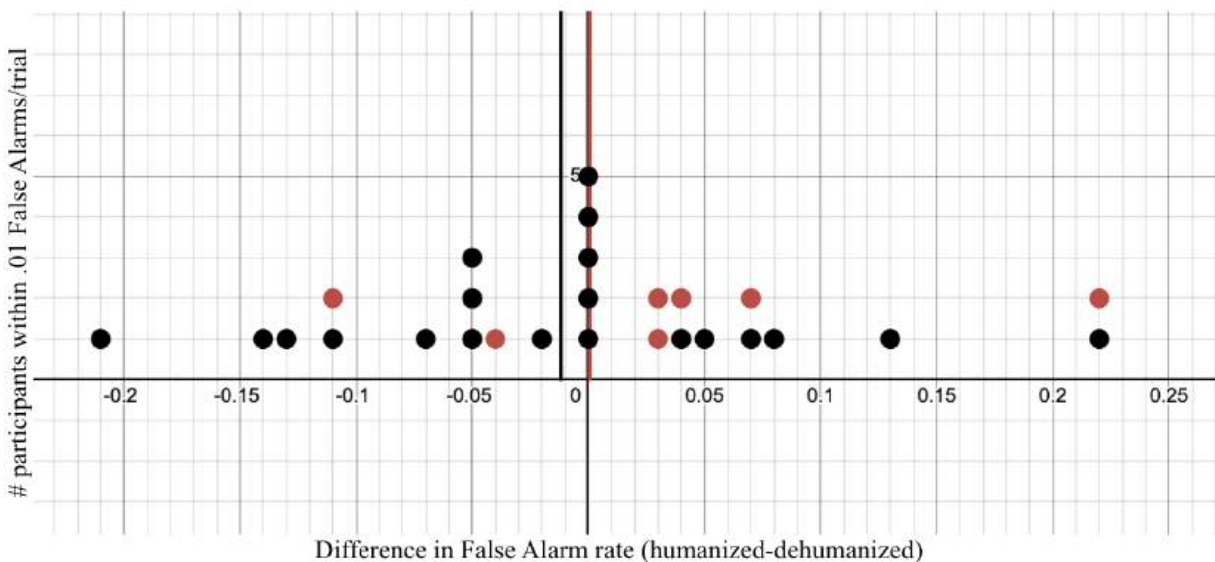


Figure 4: Dotplot depicting differences in mean False Alarm rate. Red dots indicate data excluded for low Timeout rate analysis. The black line indicates the whole population mean, and the red indicates the reduced population.

The weak correlation between number of timeouts and mistake rate indicates that a repeat of this study is warranted. The presence of obvious outliers from the high timeout rate group greatly reduces the validity of the strong Miss rate difference found in the large group analysis (see fig. 1, table 3, fig. 6). Further studies should be conducted to see if a stronger relationship between character design and Hit reaction time emerges across a broader field of participants. Yet, given the absence of visible outliers for high Timeout rate participants in the Hit data, it is reasonable to assume that the near-significant difference between humanized and dehumanized Hit reaction times is valid (see fig. 2). Additionally, the correlation between Timeout rate and the difference in mean Hit reaction times was extremely weak, further reinforcing this result as credible (see table 3, fig 5).

Regardless of this encouraging result, further investigation is warranted across all sections, and results are encouraged to be interpreted with caution.

	Correlational Coefficient (all data)	Correlational Coefficient (excluding outliers, if applicable)
Timeouts to difference in mean Hit reaction time	0.000087	N/A
Timeouts to mistake rate	0.008	0.144

Table 3: Correlational Coefficients of the relationships between Timeouts and Hit reaction time differences and Timeouts and mistake rate.

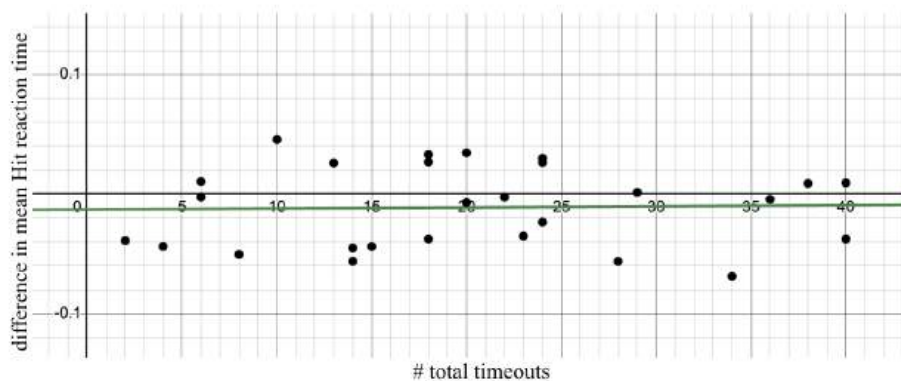


Figure 5: Dotplot depicting relationship between total Timeouts and difference in mean Hit reaction time.

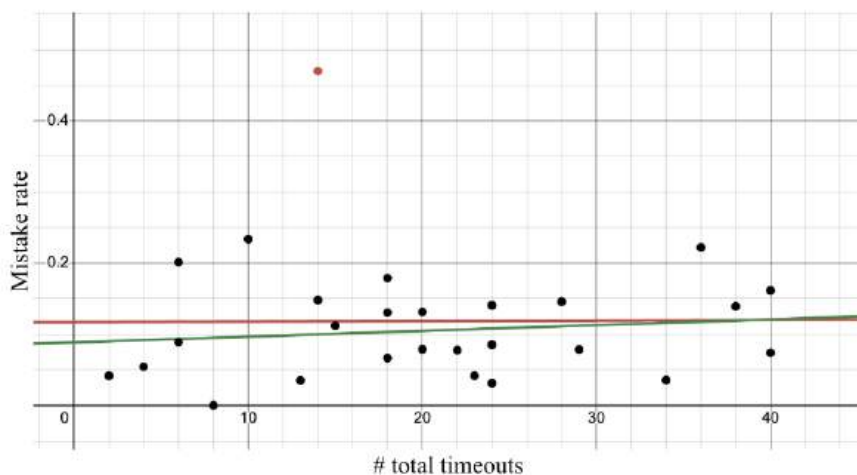


Figure 6: Dotplot depicting relationship between total Timeouts and mistake rate. Red dots indicate outlier data. Redline LSRL including outlier data, green is LSRL without

In conclusion, evidence suggests that dehumanized character designs encouraged faster Hits than humanized designs within this sample. Other results were inconclusive and require

deeper investigation. Potentially, this evidence was a result of reduced empathy for dehumanized characters, and a consequent increase in automatic reactions, manifesting as faster Hit times. This is a purely hypothetical claim that needs further scientific backing.

Limitations of this study include small sample sizes, lack of a quantitative measure of humanity, and limited statistical analysis. Conducting this experiment at a larger high school or university, improving the metric used to measure humanity of characters, and conducting a deeper, multi-level statistical analysis could produce completely novel results. The candy reward may have also had an impact on participant's gameplay, potentially introducing competition as a conflicting variable. Additionally, no deeper statistical analysis was done controlling for demographic information like gender or game experience.

Future research may explore the possibility of a relationship between bias and reflexes; if given a wider time frame, would stronger results emerge? It is important to note the subjectivity of humanity; future research should find a better way to quantify humanity, greatly expand survey size when ranking character designs, and increase the variety of targets incorporated into gameplay. Future research should also control for gaming experience as a potential confounding variable, as a larger sample size and broader statistical expertise will allow.

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From Bully to Offender: A Mixed-Methods Study of How Bullying Can Escalate, the Role of Opportunity and Money, and How Media Distorts Our Understanding

By Abhishek Ra

Abstract

Bullying is frequently dismissed as a routine feature of childhood. However, converging evidence from developmental criminology and public health identifies bullying perpetration as an early, identifiable risk marker for later criminal offending. This paper advances a conditional escalation thesis: bullying is not a deterministic cause of crime, but rather a developmental warning sign whose progression toward wider transgression depends on identifiable and interruptible mediating conditions. These include peer reinforcement of aggression, availability of exploitable opportunities, moral disengagement from the consequences of harm, poor emotional regulation, and repeated failures of institutional response. A mixed-methods framework integrates quantitative evidence—including meta-analyses reporting that school bullies carry significantly elevated odds of later offending (OR = 2.50, 95% CI: 2.03–3.08; Ttofi et al., 2011) and weapon carrying (OR = 3.25, 95% CI: 2.72–3.89 for perpetrators; van Geel et al., 2014)—with qualitative analysis of publicly available transcripts from judicial and sentencing professionals. An additional analytical layer examines how crime media distorts public understanding through forensic expectation inflation, cultivation-driven fear amplification, and copycat scripting dynamics. The integrated framework describes a conditional pathway: bullying → reinforced dominance habits → wider and bolder transgressions → (for some) financially motivated manipulation and crime, with media narratives and institutional responses either accelerating or interrupting this cascade. Implications for early intervention, diversion policy, and responsible media framing are discussed.

Keywords bullying, criminal offending, developmental risk marker, escalation, weapon carrying, moral disengagement, media distortion, CSI effect, restorative justice, labelling theory

Introduction

Bullying is generally defined by three co-occurring features: (a) intentional aggression—the harm is deliberate, not accidental; (b) repetition—it occurs more than once or carries a credible threat of recurrence; and (c) a power imbalance—the perpetrator holds some advantage, whether physical, social, or numerical, over the target (Shetgiri, 2013; Wolke & Lereya, 2015). This tripartite definition, widely adopted in public health and paediatric literature, distinguishes bullying from ordinary peer conflict. Bullying is not incidental unkindness; it is a patterned, deliberate strategy of dominance.

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Longitudinal evidence consistently demonstrates that children who bully others are more likely to engage in criminal behaviour later in life. Ttofi, Farrington, Lösel, and Loeber (2011) conducted a systematic review and meta-analysis of 18 prospective longitudinal studies and reported that school bullies had substantially elevated odds of later offending (OR = 2.50, 95% CI: 2.03–3.08). Crucially, this association attenuated but did not disappear after controlling for major overlapping childhood risk factors (adjusted OR = 1.82, 95% CI: 1.55–2.14). In a subsequent update encompassing additional longitudinal datasets, Farrington, Lösel, Ttofi, and Theodorakis (2012) confirmed the adjusted association at a OR = 1.89 (95% CI: 1.60–2.23). The partial attenuation after adjustment is not a weakness—it is informative. It tells us that bullying shares a common risk architecture with other childhood adversities, while retaining independent predictive value.

This paper avoids two tempting but misleading narratives. The first is *moral essentialism*—the idea that bullies are simply bad people—which collapses a complex developmental trajectory into a character judgement and forecloses useful intervention. The second is *narrative determinism*—the notion that every bully inevitably becomes a criminal—which contradicts available data and generates stigmatising public beliefs. The productive question is conditional: *what specific combinations of individual traits, social rewards, available opportunities, and system responses increase the likelihood that bullying grows into repeated transgression?*

To answer this question, the paper employs a mixed-methods framework. Quantitative evidence draws on prospective meta-analyses linking bullying to later offending and to bridge behaviours—physical fighting and weapon carrying—that occupy intermediate positions between school-level aggression and criminal justice involvement. Qualitative evidence comes from publicly available transcripts of justice professionals who articulate how the system attempts, and sometimes fails, to interrupt escalation. An additional analytical layer examines how crime media distorts public understanding through three specific mechanisms: forensic expectation inflation (the “CSI effect”), cultivation-driven fear amplification, and copycat scripting. The integrated framework describes a conditional cascade: bullying → reinforced dominance habits → wider and bolder transgressions → (for some) financially motivated manipulation and crime, with media narratives and institutional responses either pushing this cascade forward or holding it back.

MATERIALS AND METHODS

This paper uses a mixed-methods framework—combining quantitative (numbers-based) and qualitative (narrative-based) evidence—to address a question requiring both forms of understanding.

Quantitative Evidence

The quantitative strand draws on two categories of source material. First, prospective meta-analytic estimates from studies that followed the same cohorts of children over multiple

years and pooled results across independent samples. The primary benchmarks are Ttofi et al. (2011), who synthesised 18 longitudinal studies on bullying perpetration and later offending, and Farrington et al. (2012), who updated these estimates with additional datasets. Both report odds ratios (OR) and adjusted odds ratios (aOR) with 95% confidence intervals. Second, the paper draws on van Geel, Vedder, and Tanilon's (2014) meta-analysis of bullying and weapon carrying, which pooled data from 15 studies involving over 236,000 participants for the perpetrator subgroup. Weapon carrying and physical fighting are treated as “bridge behaviours”—intermediate escalation steps that sit between school aggression and formal criminal justice contact. Trend data from Perlus, Brooks-Russell, Wang, and Iannotti (2014), analysing four waves of the WHO Health Behaviour in School-Aged Children (HBSC) survey (combined $N > 50,000$), provide temporal context on how bullying, fighting, and weapon carrying have changed over a 12-year period.

Qualitative Evidence

Qualitative evidence comes from publicly available transcripts and podcast episodes produced by two institutional sources. The first is the Federal Judicial Center's *Off Paper* podcast, which features federal probation officers, judges, and educators discussing sentencing logic, reentry, and systemic challenges (Federal Judicial Center, 2018–present). The second is the Queensland Sentencing Advisory Council's *Sentencing Matters* podcast, which emphasises public education about how sentencing actually works, with recurring attention to youth justice and early intervention (Queensland Sentencing Advisory Council, n.d.). Both sources are legally available and institutionally produced. Short excerpts are used (rather than extended quotations) to remain within standard academic fair-use practice and to keep the analysis centred on evidence rather than quotation volume.

Media and Narrative Analysis

This section synthesises research on three specific mechanisms through which media can distort public understanding of crime. First, forensic expectation inflation—sometimes called the “CSI effect”—is the phenomenon whereby crime dramas lead audiences to expect forensic evidence to be universally available, accurate, and rapid (Shelton et al., 2006, 2011). Second, cultivation-style fear amplification refers to how sustained exposure to crime-heavy media shifts viewers' mental model of social reality toward overestimation of crime prevalence and danger (Gerbner & Gross, 1976; Dolliver et al., 2018). Third, copycat dynamics refers to the possibility that media depictions of crime provide mental scripts—methods, targets, sequences—to vulnerable individuals (Shah & Kaushik, 2025). This paper does not claim that fiction causes crime. It argues that media ecosystems can shift expectations, available mental models, and policy preferences in ways relevant to courts, policing, and public policy.

Ethical Stance

Throughout, the paper maintains careful distinctions between correlation and causation; between risk markers and deterministic causes; and between recommending individual-level intervention and making moral condemnations of people. This ethical stance is consistent with developmental science, which recognises that people can change and that outcomes are probabilistic, and with how sentencing institutions describe their own obligations.

RESULTS

Three central findings emerge from this integrated analysis.

Finding 1: Bullying Predicts Later Offending, With Important Nuance

Prospective longitudinal studies consistently support the association between bullying perpetration and later criminal offending. Ttofi et al. (2011) reported that the probability of offending up to 11 years later was substantially higher for school bullies than for non-involved students (OR = 2.50, 95% CI: 2.03–3.08). After adjusting for major childhood risk factors—including family problems, peer influences, self-regulation difficulties, and neighbourhood context—the association attenuated to aOR = 1.82 (95% CI: 1.55 -- 2.14). Effect sizes were smaller when the follow-up period was longer and larger when bullying was assessed in older children. In an updated analysis, Farrington et al. (2012) confirmed the adjusted figure at aOR = 1.89 (95% CI: 1.60–2.23). The partial attenuation means bullying is best understood as an early risk marker embedded in a wider network of risks, rather than a standalone cause of crime.

Complementary evidence from the Cambridge Study in Delinquent Development, which followed 411 South London males from ages 8–10 to ages 48–50, found that bullying at age 14 predicted violent convictions between ages 15 and 20, drug use at ages 27–32, and an unsuccessful life at age 48, even after controlling for explanatory and behavioural childhood risk factors (Farrington & Ttofi, 2011). Wolke and Lereya (2015), in a comprehensive review, reported that bullies showed higher rates of antisocial behaviour, serious criminal charges, and illegal drug use relative to non-involved peers, and that a dose-response relationship was evident: more frequent, severe, or chronic bullying was associated with progressively worse outcomes.

Finding 2: Bullying Is Linked to Specific Bridge Behaviours

Bullying is associated with two specific behaviours that occupy intermediate positions between school aggression and official criminal justice contact: weapon carrying and physical fighting. Van Geel et al. (2014) conducted a meta-analysis encompassing studies with over 250,000 participants and reported that bullying perpetrators had significantly elevated odds of weapon carrying (OR = 3.25, 95% CI: 2.72–3.89, $k = 15$ studies, $n = 236,145$). Bully-victims—those who both perpetrate and experience bullying—showed the highest odds (OR = 4.95, 95% CI: 3.77–6.50, $k = 8$ studies). In the United States specifically, the bully-victim subgroup showed odds as high as OR = 7.84 (95% CI: 6.02–10.21).

Trend data from the HBSC surveys (Perlus et al., 2014) showed that between 1998 and 2010, bullying perpetration declined from approximately 16.5% to 7.5%, and physical fighting

decreased from 23.4% to approximately 18.4%. However, weapon carrying showed no overall decline and actually increased among certain demographic subgroups. These bridge behaviours matter both substantively (as escalation indicators) and practically: fights generate injury documentation, and weapons trigger legal thresholds, both increasing the probability that a young person enters official systems where labelling effects can accelerate criminal identity consolidation.

Jochman, Cheadle, and Goosby (2017), using HBSC 2005–2006 data (N = 8,066), demonstrated that risk behaviours including substance use, social withdrawal, and fighting mediated the bullying–health link, with substantially more mediation for perpetrators than victims. This confirms that bullying rarely exists in isolation; it clusters with other risk behaviours, suggesting a common underlying risk architecture.

Finding 3: Justice Professionals Consistently Emphasise Early Individualised Intervention

Practitioners—judges, probation officers, sentencing educators—repeatedly articulate two principles in the publicly available transcripts analysed for this paper. First, responses to youth offending must be individualised and proportionate—calibrated to interrupt harm without generating new harm through overly punitive responses. Second, early intervention is not merely ethically preferable; it is practically necessary to prevent minor youth-system contact from hardening into adult-system entrenchment.

This professional logic receives strong empirical support from labelling theory research. Petrosino, Turpin-Petrosino, and Guckenburg (2010) conducted a Campbell Collaboration systematic review of 29 randomised controlled experiments involving 7,304 juveniles and concluded that formal juvenile justice processing did not reduce delinquency and in fact appeared to increase it. Bernburg, Krohn, and Rivera (2006), using panel data from approximately 870 urban adolescents, found that justice system intervention predicted subsequent serious delinquency even after controlling for prior offending, and that this effect operated partly through increased association with deviant peer groups. More recently, Cauffman et al. (2021) followed 1,216 adolescent boys for five years after first arrest and found that formally processed youth had worse outcomes that amplified over time, consistent with labelling theory predictions.

Table 1
Longitudinal Quantitative Evidence Linking Bullying to Later Offending

Source	Outcome Measured	Key Finding	Interpretation
Ttofi et al. (2011); 18 longitudinal studies	Criminal offending up to 11 years later	OR = 2.50, 95% CI: 2.03–3.08 (unadjusted); aOR = 1.82, 95% CI: 1.55–2.14 (adjusted)	Bullying is a significant independent risk marker. Partial attenuation after adjustment indicates shared risk architecture with other childhood adversities.

van Geel et al. (2014); 15 studies, n = 236,145	Weapon carrying	Bullies: OR = 3.25, 95% CI: 2.72–3.89; Bully-victims: OR = 4.95, 95% CI: 3.77–6.50	Weapon carrying is a bridge behaviour between school aggression and justice system contact, driven by threat perception and dominance motivation.
Jochman et al. (2017); N = 8,066	Clustered risk behaviours	Risk behaviours (substance use, fighting, withdrawal) mediate bullying–health link; stronger mediation for perpetrators	Bullying rarely exists alone. It clusters with other externalising problems, suggesting common underlying causes.
Farrington & Ttofi (2011); Cambridge Study, N = 411, follow-up to age 48–50	Violent convictions, drug use, life outcomes	Bullying at age 14 predicted violent convictions at 15–20, drug use at 27–32, unsuccessful life at 48	Effects persist across decades, even after controlling for childhood risk factors. Dose-response relationship confirmed.

Table 2

Evidence-Based Conceptual Map: How Bullying Escalates—An Interruptible Chain

Developmental Stage	Observable Behaviours	Testable Mechanisms	Evidence-Based Intervention Points
Early childhood	Testing limits, forced dominance, targeting specific peers	Peer attention reinforces aggression; aggression modelled in environment; boundaries absent or inconsistent	Positive classroom management; caregiver boundary-setting; immediate repair after harm. Anti-bullying programmes reduce perpetration by ~20% (Gaffney et al., 2021; OR = 1.31).
Early adolescence (11–14)	Repeated bullying, public status performances, group-level aggression	Moral disengagement ($r = .31$ with bullying perpetration; Killer et al., 2019); poor emotional regulation ($r = .33$ with aggression; Smith et al., 2025); deviant peer affiliation	School-wide anti-bullying programmes; consistent adult responses; targeted social-emotional learning interventions.
Mid-adolescence (14–16)	Physical fighting, weapon carrying, intimidation, association with delinquent peers	Opportunity structures (unguarded targets, weak supervision); expanded behavioural repertoire; retaliatory attitudes fuelling violence cycles (Cunningham et al., 2012)	Diversion programmes; structured prosocial alternatives; restorative justice (17% recidivism reduction; Fulham et al., 2025). Avoid formal processing (Petrosino et al., 2010).
Late adolescence / early adulthood	Wider offending range; instrumental use of intimidation for financial gain	Deliberate cost-benefit reasoning applied to existing dominance toolkit; prior justice contact cements criminal identity (Bernburg et al., 2006)	Minimally sufficient sentencing; avoid fast-tracking minor offences; individualised risk-responsive programming. RJ conferences cost-effective at 3.7–8.1:1 benefit ratio (Sherman et al., 2015).

Discussion

The pathway from bullying to crime becomes clearer when framed as a **script-and-opportunity model** rather than a purely personality-based account. In early life, bullying can function as a successful strategy from the perpetrator's perspective: it delivers social status, compliance from others, and sometimes material gains at low immediate cost. Repetition, then, is not a reflection of fixed character but of a learning loop reinforced by social rewards. When adults respond inconsistently—or intervene only after serious harm has occurred—this loop can consolidate into a full behavioural repertoire in which intimidation, manipulation, and escalation become default tools for navigating social environments.

A key point for courts, schools, and clinicians is that escalation is *probable, not guaranteed*. The meta-analytic findings confirm elevated risk while simultaneously showing that this risk diminishes after adjusting for overlapping factors. This partial attenuation demonstrates that bullying and later crime share a common risk architecture—encompassing family stress, peer context, poor self-regulation, and neighbourhood opportunity structures—rather than constituting a single origin point. This is precisely why early, multi-level intervention matters: when risks accumulate, even small positive trajectory changes can compound over time.

Bridge Behaviours and Investigative Visibility

Physical fighting and weapon carrying deserve particular attention within a coherent escalation framework. They occupy an intermediate position between school-level bullying and criminal justice contact. The van Geel et al. (2014) meta-analysis provides robust evidence of this linkage: perpetrators carried weapons at more than triple the rate of non-involved peers, and bully-victims—the subgroup at greatest overall risk for adverse outcomes (Wolke & Lereya, 2015)—showed odds approaching five times the baseline rate. In practical criminological terms, these behaviours also create investigative visibility: fights produce injury documentation; weapons trigger legal thresholds; both increase the probability that a young person enters official systems.

Once a young person enters the justice system, labelling effects can accelerate identity consolidation. The Campbell Collaboration review by Petrosino et al. (2010), synthesising 29 randomised experiments with 7,304 juveniles, concluded that formal system processing does not reduce delinquency and, if anything, increases it. Bernburg et al. (2006) provided a mechanism: official labelling predicted subsequent serious delinquency partly through increased embeddedness in deviant social networks, with delinquent peer associations accounting for nearly half the direct effect. Cauffman et al. (2021) confirmed that these labelling effects amplified over a five-year follow-up. The implication is clear: the very systems designed to correct early offending can inadvertently entrench it.

Moral Disengagement and Emotional Dysregulation as Escalation Mechanisms

Two psychological mechanisms merit special attention as they bridge the bullying literature and the broader offending literature. Bandura's (1996, 1999) theory of moral

disengagement describes how individuals neutralise self-censure through cognitive restructuring (moral justification, euphemistic labelling, advantageous comparison), displacement or diffusion of responsibility, minimisation of consequences, and dehumanisation or blaming of victims. Gini, Pozzoli, and Hymel (2014), in a meta-analysis of 27 samples encompassing 17,776 young people, found that moral disengagement correlated with aggressive behaviour at $r = .28$ (95% CI: .23–.32). Killer et al. (2019), with a larger pool of 47 samples and 43,809 participants, reported a specific correlation between moral disengagement and bullying perpetration of $r = .31$ (95% CI: .27–.34). Longitudinal evidence from Bjärehed et al. (2021) confirmed that changes in moral disengagement predicted changes in bullying within the same individuals over time, strengthening the case for a causal contribution.

Emotional dysregulation provides the second mechanism. Smith et al. (2025), in the most comprehensive meta-analysis to date (137 articles, 171 studies, $N = 252,605$), reported that maladaptive emotion regulation strategies were associated with aggression at $r = .33$. The association was strongest for reactive aggression—precisely the impulsive, emotionally driven form of violence most relevant to the early stages of escalation. Garofalo, Neumann, and Velotti (2021) demonstrated that emotion dysregulation mediated the psychopathy–aggression link in both community and offender samples, with effects on reactive aggression more than double those on proactive aggression. These findings align with the escalation model: early-stage bullying may be partly reactive (driven by poor emotional control), while later-stage offending increasingly incorporates proactive, instrumental elements.

The Instrumentalisation of Dominance

Financial motives enter the escalation pathway most defensibly as what this paper terms the *instrumentalisation of dominance*: a bully’s existing skills—strategic intimidation, social engineering, coercive control, reputational leverage—gradually shifting toward financially motivated behaviour. This is not a separate category of crime; it is an extension of the same dominance toolkit. The translation is conditional and mechanism-based: a dominance script becomes financial when (i) the target is financially exploitable, (ii) guardianship is weak, and (iii) the offender perceives a low risk of detection. While precise global statistics on the proportion of crimes with financial motives are not uniformly tracked across jurisdictions, the UNODC Global Study on Homicide (2019) classifies robbery-related homicide as a documented subcategory of criminal homicide, confirming that economic instrumentalisation is central to serious violence for a definable subset of offenders.

Cycles of Aggression: Empirical Evidence on Retaliation and Perpetuation

A critical feature of the bullying–offending pathway is how cycles of aggression become self-perpetuating when uninterrupted. Widom’s (1989) landmark prospective study followed 908 abused and neglected children for approximately 20 years and found that childhood maltreatment increased the risk of juvenile arrest by 53%, adult arrest by 38%, and violent crime arrest by 38%. Importantly, the majority of maltreated children did not become criminals—the cycle is

probabilistic, not deterministic, paralleling the conditional logic of the bullying–offending association. Cunningham et al. (2012), studying 129 assault-injured adolescents, demonstrated that retaliatory attitudes at baseline predicted greater aggression and more frequent fighting over an 18-month follow-up, suggesting that retaliation fuels ongoing violence cycles.

Restorative justice approaches offer one empirically supported mechanism for interrupting these cycles. Latimer, Dowden, and Muise (2005), in a meta-analysis of 22 studies, found that restorative justice produced small but significant reductions in recidivism ($d = 0.07$, 95% CI: 0.02–0.12), with 72% of effect sizes favouring the restorative approach. Sherman et al. (2015), focusing exclusively on 10 randomised controlled trials involving 1,880 offenders, found that restorative justice conferences were particularly effective for violent crime and were cost-effective at a 3.7–8.1 benefit-to-cost ratio. Fulham et al. (2025), in the most recent meta-analysis, reported a 17% reduction in general recidivism across 27 studies. These findings directly support the practitioner emphasis on early, non-punitive intervention—and suggest that cycles of aggression can be interrupted without relying exclusively on formal justice processing.

Media Distortion: Forensic Expectations, Cultivated Fear, and Copycat Scripts

The role of forensic science in this paper is not to explain forensic methods themselves, but to show how forensic expectations, shaped by the media, interact with the justice system. Shelton, Kim, and Barak (2006) surveyed 1,027 prospective jurors in Washtenaw County, Michigan and found that 46% expected to see some form of scientific evidence in every criminal case: 22% expected DNA evidence, 36% expected fingerprint evidence, and 32% expected ballistic evidence. Only 14% would convict in a rape case based solely on victim testimony. Shelton et al. (2011), combining data from two empirical studies ($N \approx 2,266$), concluded that these elevated expectations reflected a broad “technology effect” rather than a CSI-specific phenomenon. Whether the effect is narrow or broad, the practical consequence is the same: public confusion between narrative closure—everything resolved neatly at an episode’s end—and evidentiary reality.

Beyond forensic expectations, media narratives distort base rates and calibrate fear. Gerbner and Gross (1976) proposed cultivation theory, which posits that sustained, heavy television exposure shifts viewers’ perceptions of social reality toward overestimation of violence, danger, and mistrust—what they termed the “mean world syndrome.” Dolliver, Kenney, Reid, and Prohaska (2018), using a nationally representative U.S. sample ($N = 1,311$), confirmed that media consumption had significant direct and indirect effects (mediated through fear of crime) on support for controversial criminal justice policies including three-strikes laws, the death penalty, and stand-your-ground legislation. For a global audience, this matters: fear-driven policy tends to push for harsh, visible action—including aggressive school discipline and heavy policing—even when prevention-oriented approaches would be more effective and cost-efficient.

Copycat dynamics require careful handling to avoid moral panic. Shah and Kaushik (2025), synthesising research through the lens of Akers’ social learning theory, concluded that

approximately 25% of offenders in studied samples claimed media influenced their actions, and that approximately 40% of serious and violent juvenile offenders reported copycat crime histories (citing Surette, 2002). The media, the authors argue, functions more as a catalyst or rudder—shaping how crimes are committed—than as a direct motivational cause. The responsible claim is not that media causes crime, but that for a vulnerable subset, depictions can provide scripts that lower inhibitions and suggest methods.

Cross-National Evidence: The Indian Context

The bullying problem is not limited to Western schools. Rana, Gupta, Malhi, Grover, and Kaur (2020) studied 667 adolescents in Chandigarh, North India, and found a 25.6% bullying involvement rate (16% victimised, 5.2% perpetrating, 4.3% bully-victims). Males were 2.5 times more likely to be involved (OR = 2.5, 95% CI: 1.5–4.2). Critically, 44% of students reported that adults in school never intervened when bullying occurred. This finding is directly relevant to the escalation thesis: perceived impunity and weak supervision are not merely moral failures—they constitute behavioural reinforcement. When a bully observes that no one intervenes, that absence of response is itself a signal that the behaviour is tolerable, or even acceptable.

Limitations

Several limitations warrant transparency. First, the quantitative evidence base relies predominantly on Western samples, particularly from Europe and the United States. While the Rana et al. (2020) study provides Indian data, the meta-analytic effect sizes reported by Ttofi et al. (2011) and van Geel et al. (2014) may not generalise uniformly to all cultural contexts. Second, the qualitative component uses publicly available institutional transcripts rather than original interview data. While this approach avoids ethical complications of primary data collection, it limits the depth of practitioner insights to what was already made public. Third, the media analysis is synthetic—it integrates findings from separate research traditions (the CSI effect literature, cultivation theory, and copycat research) into a coherent framework, but does not present original empirical data on media effects. Fourth, while the paper argues that bullying is a developmental risk marker, the cross-sectional design of several cited studies limits causal inference. The prospective longitudinal designs of Ttofi et al. (2011) and Farrington et al. (2012) provide the strongest temporal evidence, but unmeasured confounders remain a possibility in all observational research. Fifth, effect sizes for the bullying–offending link, while statistically significant, are modest after adjustment (aOR = 1.82), which is appropriate for a risk marker embedded in a complex causal network but means bullying alone has limited predictive power for any individual case.

Conclusion

Bullying is not merely a school problem. It is a visible, measurable, early-appearing signal within a broader developmental trajectory that, under specific and identifiable conditions, can extend toward wider transgression and criminal offending. The evidence reviewed in this

paper supports a conditional model: bullying perpetration significantly elevates the odds of later offending (OR = 2.50, aOR = 1.82), is linked to intermediate bridge behaviours including weapon carrying (OR = 3.25 for perpetrators) and physical fighting, and shares mechanistic underpinnings—including moral disengagement, emotional dysregulation, and opportunity structures—with more serious forms of antisocial behaviour.

Justice professionals consistently confirm that early, individualised, and minimally sufficient intervention can prevent minor youth-system contact from cascading into adult-system entrenchment—a position now supported by robust experimental evidence on the iatrogenic effects of formal processing (Petrosino et al., 2010) and the comparative effectiveness of restorative approaches (Sherman et al., 2015; Fulham et al., 2025). Media ecosystems add a further layer of complexity by inflating forensic expectations, amplifying fear of crime, and providing behavioural scripts to vulnerable individuals.

The integrated framework advanced here—bullying → reinforced dominance habits → wider and bolder transgressions → (for some) financially motivated manipulation and crime—is explicitly conditional and mechanism-based. Its value lies not in predicting which individual bully will become an offender, but in identifying the constellation of factors that can be interrupted. Every stage in the escalation chain presents an opportunity for intervention. The question is not whether bullying leads to crime—for most individuals, it does not. The question is whether we will recognise and act on the warning signs while the window for interruption remains open.

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Appendices

Appendix A: Key Terms Used in This Paper

Term	Operational Definition	Measurement
Bullying perpetration	Repeated, intentional aggression characterised by a power imbalance between perpetrator and target (Shetgiri, 2013).	Self-report survey items (e.g., HBSC, Olweus BVQ); school disciplinary records; structured interviews.
Bridge behaviours	Behaviours that occupy intermediate positions between school-level aggression and formal criminal justice contact, specifically weapon carrying and physical fighting.	Self-report items on weapon carrying and fighting frequency; injury reports; legal statute triggers.
Moral disengagement	Cognitive mechanisms through which individuals neutralise self-censure for harmful conduct (Bandura, 1996).	Bandura 32-item Moral Disengagement Scale ($\alpha = .82$); adapted versions for specific age groups.
Institutional miscalibration	System responses that inadvertently worsen outcomes—whether too harsh, too inconsistent, or entirely absent.	Absence of adult intervention (e.g., Rana et al., 2020: 44% report no adult response); recidivism rates post-formal processing.
Labelling effects	The process through which official justice system contact cements a criminal identity and increases subsequent offending (Becker, 1963; Bernburg et al., 2006).	Comparison of formally vs. informally processed youth on recidivism (Petrosino et al., 2010: 29 RCTs, $N = 7,304$).
Narrative distortion	Public beliefs shaped by media representations rather than empirical base rates of crime.	CSI effect surveys (Shelton et al., 2006: $N = 1,027$); cultivation-fear SEM (Dolliver et al., 2018: $N = 1,311$).

Appendix B: Coding Scheme for Media/Narrative Analysis

Code Category	What Is Analysed	Criminological Relevance
Forensic omniscience	Depictions of forensic evidence as universally available, definitive, and rapid.	Inflates juror expectations (46% expected scientific evidence in every case; Shelton et al., 2006), distorting evidence evaluation.
Genius-offender trope	Framing of criminals as uniquely brilliant and uncatchable.	Glamorises crime; obscures situational and developmental explanations for offending.
Base-rate neglect	Presenting rare crimes as if they occur constantly.	Amplifies fear of crime and increases support for punitive policies (Dolliver et al., 2018); drives fear-based rather than evidence-based responses.
Script clarity	Step-by-step depictions of how specific crimes are committed.	Raises concerns about copycat behaviour among vulnerable individuals (~25% of offenders report media influence; Shah & Kaushik, 2025).

Appendix C: Consolidated Effect Size Reference Table

Study	Measure	Effect Size	Sample
Ttofi et al. (2011)	Bullying → offending (unadj.)	OR = 2.50, CI: 2.03–3.08	k = 18 studies
Ttofi et al. (2011)	Bullying → offending (adj.)	aOR = 1.82, CI: 1.55–2.14	k = 18 studies
van Geel et al. (2014)	Bullies → weapon carrying	OR = 3.25, CI: 2.72–3.89	k = 15, n = 236,145
van Geel et al. (2014)	Bully-victims → weapons	OR = 4.95, CI: 3.77–6.50	k = 8, n = 199,563
Gini et al. (2014)	Moral disengagement → aggression	r = .28, CI: .23–.32	k = 27, N = 17,776
Killer et al. (2019)	MD → bullying perpetration	r = .31, CI: .27–.34	k = 47, N = 43,809
Smith et al. (2025)	Maladaptive ER → aggression	r = .33	k = 171, N = 252,605
Widom (1989)	Child abuse → juvenile arrest	53% increased risk	N = 1,575
Gaffney et al. (2021)	Anti-bullying → reduced perp.	OR = 1.31, CI: 1.24–1.38	k = 100 evaluations
Petrosino et al. (2010)	Formal processing → delinquency	Increases delinquency	29 RCTs, N = 7,304
Fulham et al. (2025)	RJ → recidivism reduction	17% reduction	k = 27 studies

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Cardiac Arrest Prediction Using AI: A Literature Review

By Devan Sharma

Introduction & Rationale

Cardiac arrest remains one of the leading causes of death worldwide because it often happens without warning. Recognizing the signs of cardiac arrest early can make the difference between life and death (S et al., 2025). In the United States, more than 350 thousand out of hospital cardiac arrests occur per year. Additionally, survival rates remain below 10% in many regions of the country. However, artificial intelligence is improving the detection and management of cardiac arrest by making faster, more accurate identification of high-risk patients than previously used clinical methods. My interest in this topic stems from the growing use of AI in healthcare and its potential to be a faster and more accurate alternative to human decision-making in high-risk situations. Furthermore, a couple of years ago, my uncle tragically passed away from cardiac arrest. This event made me think about how cardiac arrest truly works and drove me to fill the existing gaps in the research for better prevention. Over the past years, researchers have looked at numerous predictive models that identify cardiac arrest risk using factors such as characteristics, ECGs, and EMS data like heart rate or time to CPR (Hannan et al., 2024; Kawai et al., 2022; Nikolaj Blomberg et al., 2023). By analyzing this, AI can not only predict cardiac arrest but also inform clinical decision-making, ultimately improving patient outcomes and saving lives. Many of these studies showed improved predictive accuracy compared to traditional accuracy. However, other research raises concerns about generalizability, inconsistent validation methods, and limited testing for EMS personnel in the real world (Amacher et al., 2024). **Although AI-based models show strong potential for predicting cardiac arrest using patient and EMS data, existing research has struggled to develop models with real-world applicability--highlighting the need for stronger external validation, exploration of more predictors, and more focus on implementation in pre-hospital emergency settings.**

Foundational Findings across the Literature

Most of the articles' researchers agree that machine learning models can improve early detection of cardiac arrest risk. This is important because it allows for timely intervention and can save lives before cardiac arrest occurs. A number of papers show that AI algorithms beat older scoring systems because they are able to pick up patterns that humans might miss (Hannan et al., 2024; Kawai et al., 2022). Many of these studies use a retrospective cohort design and measure model performance using statistical tools such as the receiver operating characteristic curve (AUROC), sensitivity, specificity, precision, and recall. These tools show how accurately the model identifies high-risk patients and minimizes the amount of false positives or negatives. One study developed a deep learning algorithm trained on over 47,000 ECGs from two hospitals to predict cardiac arrest within 24 hours. The algorithm resulted in impressive AUROC values of 0.913 for internal validation and 0.948 for external validation, showing high predictive accuracy

in both cohorts (Kwon et al., 2020). A similar article applied an AI ECG model to survivors of sudden cardiac arrest with hypertrophic cardiomyopathy. This is a common genetic heart condition that causes the left ventricle muscle to thicken and stiffen, leading to restricted blood flow, shortness of breath, chest pain, or fainting. The study found that AI detected risk even when data collection was incomplete. Their AUC values of 0.913 and 0.948 proved that the AI model could reliably identify patients at risk, highlighting its potential as an important tool in healthcare (Swain et al., 2025). Even though things like the study populations, healthcare environments, and types of cardiac data all varied, the high predictive performance of these models highlights the potential importance of machine learning in risk assessment in the future. Additionally, the use of multiple performance metrics makes it a more complete evaluation of each model's real-world use, rather than relying on one measure of accuracy. These findings indicate that AI provides grander advantages in identifying high-risk patients than traditional methods.

Beyond ECG-based prediction, AI has been applied to stroke detection and neurological outcome prediction, which are closely related to cardiac arrest. A study conducted a systematic review and meta-analysis of AI detection of ischemic stroke in MRI scans. This study reviewed a total of 33 studies and found that 15 of them were low bias. The results show that AI could detect stroke lesions with 93% sensitivity and specificity, and they also resulted in a positive likelihood ratio of 12.6 (Bojsen et al., 2024). Clearly, AI models have the potential to accurately identify high-risk patients not just in cardiac arrest, but also in different sections of health care, such as stroke. Back to cardiac, Amacher and his study explored AI's role in predicting outcomes after cardiac arrest had occurred using ChatGPT-4, an everyday AI chatbot. In this study, 713 adult cardiac arrest patients compared AI predictions to three validated clinical scores for mortality prediction and neurological outcome. ChatGPT-4 got an AUROC of 0.85 for mortality prediction and 0.84 for neurological outcome. However, the model was found to have been producing hallucinations, with 14 % of outputs being illogical. This highlights the importance of human monitoring, as AI is still prone to mistakes (Amacher et al., 2024). These studies show how AI can extend beyond cardiac arrest prediction to areas such as stroke and neurological assessment and increase the accuracy and guidance of decision-making in those fields as well.

In prehospital settings, AI and machine learning models have improved the prediction of OHCA (out-of-hospital cardiac arrest). A study by Wang and other coresearchers developed a multicenter cohort prediction model for EMS-witnessed traumatic cardiac arrest. Using multivariable logistic regression, the model effectively separated survivors from non-survivors by using several variables such as age, injury mechanism, and resuscitation efforts, which were strongly associated with survival. This helped the team prove their hypothesis: specific patient, injury, and prehospital factors can be used to accurately predict survival outcomes in EMS-witnessed traumatic cardiac arrest (Wang et al., 2023). Other researchers created a prediction model for 252,771 adult non-traumatic EMS patients in Taipei City, showing an AUROC of 0.94. Key predictors included abnormal heart rate, breathing rate, low blood pressure, low oxygen saturation, male sex, and poor consciousness (Hsieh et al., 2020).

Furthermore, a third research team examined how prehospital time factors interact to affect outcomes, building a neural network model with 24 features to predict neurological outcomes one month after OHCA in 8,274 patients in Japan. The model achieved an AUROC of 0.94 and revealed nonlinear interactions between transport time and defibrillation or drug administration (Kawai et al., 2022). These findings suggest that AI can optimize EMS protocols by identifying which prehospital interventions have the greatest impact, allowing paramedics to prioritize actions that improve survival and neurological recovery.

Gaps in the Research and Future Research Needs

Although AI models have shown promising results in hospital settings, there are still major gaps in research, especially in EMS pre-hospital environments (Nikolaj Blomberg et al., 2023). Many prediction models are trained on limited or non-diverse datasets, which may reduce their accuracy when they are applied to different populations or regions (Amacher et al., 2024). In addition, a majority of the existing research relies on retrospective hospital data rather than real-time EMS data collected in the field (Amacher et al., 2024; Kawai et al., 2022). This is not ideal because they are limited by their reliance on pre-existing data, making them open to selection bias and missing information. There are also challenges related to the small number of cost-effective and small/convenient devices that can gather reliable and strong data from patients on scene. There are also limited models that are specifically developed and validated for EMS use compared to in-hospital prediction (Nikolaj Blomberg et al., 2023). Another limitation is that emergency environments are highly unpredictable, which makes it difficult for models trained in controlled clinical settings to perform well in the EMS settings. In many EMS situations, the personnel have to rely on incomplete or constantly changing patient information. This serves as a disadvantage because it has the potential to reduce the reliability of AI prediction. Additional studies that collect real-time EMS data and test models across multiple emergency systems are necessary to improve reliability and real-world implementation. These gaps show that while predictive performance may appear strong in controlled settings, more EMS-focused research and larger datasets are needed to move towards real-world application.

Future Work and Conclusions

Overall, current research shows that AI models can improve cardiac arrest prediction, but there are still clear issues with out-of-hospital applications. While many studies show strong accuracy, conflicts appear when models are tested outside their original datasets, and gaps remain in real-time pre-hospital use and dataset diversity. This creates concerns about how reliable the models are in real-world emergencies, particularly in pre-hospital settings where data can be limited or inconsistent. In addition, there is also an area of concern with what characteristics to feed the models in order to produce the best outcome of risk prediction. Future work should focus on validating models across different EMS systems, improving dataset diversity and reliability, and pointing out the most important indicators so that prediction tools can be used more effectively in real-time emergency response. Based on the articles read and

reviewed, a key research direction is understanding how prediction models are created and which predictors work the best in clinical applications. This direction directly builds from the patterns identified in my research, where many studies report strong predictive performance but also acknowledge limitations in consistency concerning variables and real-world implementation. Looking at how models perform across different emergency systems and patient populations could help determine whether these tools are reliable enough to assist dispatchers and emergency medical teams during critical situations. Moving forward, continued research on external validation, real-time EMS data interrogation, and improved demographic selection will help guide the development of more practical and dependable cardiac arrest prediction. Ultimately, making these areas of research stronger could potentially make AI prediction models a valuable tool for emergency medical response teams and potentially improve survival outcomes for patients who experience cardiac arrest.

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Academic Accommodations: A Flawed System By Amber Olson

When Section 504 was passed in 1973, it was a landmark victory for students with disabilities; but today, 504 plans have transformed into a “must have” for those looking to gain a competitive edge. Ultimately, both 504 plans and IEPs were introduced into the American education system to further equity in schools for students with physical and invisible disabilities. Their original intent was to serve those in need and to equalize academic opportunities. Eventually, the equality and success of this system began to falter due to a variety of factors. There was an increase in the number of students with “invisible” disabilities receiving accommodations, which correlated with heightened over-diagnosis of “invisible” disabilities in kids. Similarly, kids from affluent backgrounds receiving accommodations began to trend upward. Additionally, it became unclear if students relying on accommodations would actually be prepared for future workplace success. Although 504’s and accommodations support many students, they should be strengthened and redefined by clear standards that guide their implementation; such standards could prevent overuse, ensure that accommodations promote equity, and better prepare students for future academic and professional demands.

Background

For decades, Americans with disabilities faced systemic exclusion, lack of legal protection from discrimination, and minimized educational opportunities. However by 1973, Section 504 was passed into law, prohibiting discrimination on the basis of disability in programs receiving federal funding and allowing opportunities for disabled Americans through developments in equity. In 1975, IDEA (Individuals with Disabilities Education Act) was introduced to promote equity within the educational system, by requiring both primary and secondary schools to assess and create educational plans for students with clearly identified disabilities. Schools could work with teachers to grasp the students’ needs in order to develop an IEP (Individualized Education Plan). Clint Pinion, a teacher/dean explains how these IEPs work to, “ensure the student learns in an atmosphere of inclusion,” (Pinion 1). IEPs are implemented in severe cases, (often known as interventions), and bridge larger educational gaps. Students struggling with reading and math can be provided with better tools and separate curriculums in order to catch up to their peers. By contrast, 504 plans are given to students that have smaller educational gaps (known as accommodations), allowing them access to tools that enable them to work with the same curriculum as their peers. These include separate testing rooms with minimized distractions, or extended time on tests/assignments. The goal of these resources is promoting equity in education spaces, allowing disabled students to be on the same plane as their peers. Notably in 1990, the Americans with Disabilities Act was created to prohibit discrimination against individuals with disabilities in all areas of public life. The ADA offers “reasonable accommodations” in schools and work places; however, the problem is that there are no clear definitions of what “reasonable accommodations” actually are. Emily Krebs (director of Fordham University’s Disability Studies program) regarding guidelines from the U.S. Office of

Disability Rights, “[it] doesn’t offer concrete definitions for how such ‘reasonability’ can be determined for other cases,” (Krebs). Despite controversy, the ADA, IDEA, and Section 504 have served many Americans and have worked to reduce stigma and protect rights for disabled individuals. Nevertheless, with the increasing prevalence of accommodations in recent years, questions surrounding their success in promoting equity in education have arisen. One example of this was the 2017 College Board policy change; nearly all students with formalized IEPs or 504 plans received automatic approval for accommodations on all college board exams. Consequently, there was a significant increase in students requesting and receiving accommodations, especially in affluent areas. Generally, the laws established were intended to serve disabled Americans and provide them with the same educational opportunities as their peers; but some remaining gaps lead to misuse.

Increase in Over Diagnosis May Correlate with Increase in Accommodations

Since the 1970s, when accommodations were introduced into public schools, the number of students receiving these services have increased, particularly within the past few years. There has been a simultaneous increase in mental disorders in kids/adolescents, particularly Attention Deficit/Hyperactivity Disorder (ADHD). A study performed by Merten, a pediatric psychiatry specialist, examined these statistics and found that the past 50 years have seen a worldwide increase in diagnosed mental disorders in children/adolescents observed by special education programs. Merten also found there was a stark increase in psychostimulant prescription rates for ADHD. She notes potential concerns when she says, “Research shows that children who do not fulfill ADHD criteria are treated with psychostimulants,” (Merten). Additionally, Merten describes a study showing a strong case for overdiagnosis of ADHD, “In total, 16.7% of psychotherapists diagnosed ADHD although diagnostic criteria was not met...Only 7% gave no diagnosis, although the case vignette fulfilled diagnostic criteria for ADHD. Therefore, there were significantly more false-positive than false-negative diagnoses,” (Merten). Merten’s statistics illustrate the link between increasing numbers of accommodations and the increasing number of children with mental disorder diagnoses. The fact that many of these diagnoses may be false—or a product of overdiagnosis—underscores that not all students receiving accommodations truly qualify. Although this may seem minor, both Krebs and Pinion reiterate the difficulties disabled students face obtaining accommodations; therefore, an influx of students with questionable diagnoses can reduce available resources for truly disabled students.

Inequalities in Accommodations

Once accommodations are granted, disparities appear among those receiving these resources. Lovett and Liu (a PhD student and professor) found that 504 plans disproportionately benefit white and affluent students. Students in the wealthiest districts are nearly twice as likely to receive accommodations, compared to students in poorer districts (Liu). After the College Board change, there was a significant rise in 504 plan enrollment, especially in affluent schools. Liu highlights results of this policy: “...increases in 504 plan enrollment among White students

that were, on average, about 11.1% greater than those among Black students, 26.9% greater than those among Hispanic students, and 29.4% greater than those among Asian students” (Liu). The current system provides no guidelines to ensure that those receiving accommodations are coming from varying socioeconomic or racial backgrounds. Similarly Lovett explains, “The most common accommodation—extended time on tests—has repeatedly been found to raise the performance of students regardless of disability status”. Because of performance gains from these resources, socially advantaged families likely recognized 504 plans as a way to raise academic achievement. Both Lovett and Liu examine strong statistical evidence that accommodations are not granted equitably, and therefore might not be serving those in need. Rather, serving those taking advantage of the system. It is evident that barriers faced by disabled individuals can end up exacerbated based on socioeconomic status, and race.

Diagnostic Inequalities

America’s healthcare system is unaffordable for many, especially those without insurance (including 12.5% of college adults). Krebs highlights advantage for students who come from socioeconomic privilege, who are far more likely to receive diagnoses prior to college. In an extreme case, “Some upper class students even claim to have duped the system and received false diagnoses for LDs,” (Krebs). Furthermore, wealthier parents can afford independent disability evaluations from private practice diagnosticians who have financial incentives to provide a diagnosis. Krebs also found in primary and secondary schools, disabled people of color are much less likely to be identified as disabled than their white peers—therefore, less likely to be placed in special education programs. White students are represented disproportionately in disability categories acknowledged in general educational settings (like ADHD), or other learning disabilities. Conversely, racial minority students are often put into categories like “emotional disturbance” and “intellectual disability” which Liu explains, “carry stronger behavioral stigma and are more closely tied to exclusionary placements,” (Liu). These clear diagnostic disparities highlight barriers to students of color getting accommodations, as compared to their white counterparts. For example, *The Wall Street Journal* described some wealthy schools with implausible disability identification rates. One third of students in one school had 504 plans; the journal notes, “Such rates of disability are improbable on their face—indeed, taken literally, they would suggest a public health crisis,” (Lovett). Ongoing societal repercussions of racial and socioeconomic distinctions, along with classroom inequities, should be eliminated for the success of all students.

Accommodations May Not be Helpful

When students obtain accommodations, research shows that they can be of little benefit. Lovett highlights a trial sampling of students aged 11-15 with ADHD. Some students were placed in an “accommodation group,” receiving a copy of class notes and 50% additional time to complete assignments. The other students, the “intervention group,” were given more resources including training on organizing their own assignments, instruction on note-taking, and

computerized interventions for attention skills. Afterwards, they compared students' performance across groups and learned the "intervention group" greatly outperformed those in the "accommodation group." They showed more engagement in class, and completed assignments successfully, while the "accommodations group" had reduced motivation in some cases. Although this study was focused on ADHD, it can't be assumed that providing accommodations for all students will always guarantee success in the classroom. Additionally, similar findings can be observed when accommodations are given to students with emotional disorders, specifically anxiety. Clinical psychologists have found that providing accommodations for anxiety tends to make the anxiety worse over the long-term, and may help students avoid situations they need to learn to cope with, (Lovett). The goal should be teaching coping skills and providing interventions to help students overcome challenges, rather than promoting normalcy with avoidance. When schools have limited resources, kids can end up with services that provide little benefit, and are actually shielded from scenarios they should be learning to overcome.

Accommodations in the Workforce

The initial benefits of accommodations may lapse once the workforce is entered. Typically, accommodations for students do not give an accurate picture of what employers would provide. In Rebekah Wanic's opinion piece, she explains how deadlines are non-negotiable, and students who have become reliant on accommodations like extra time may struggle to adapt to pressures of completing tasks efficiently and timely. Also, accommodations cater to individual needs, and therefore students may not be versed in working collaboratively under standard conditions (Wanic). While universities can adjust standards to ensure their students are accommodated, employers might not be able to do the same, and students reliant on these forms of support may face challenges when entering the workforce without the same level of support. Thus, students may face frustration, reduced job satisfaction, and overall trouble succeeding. One study reviewed how students dependent on accommodations may not be prepared for a career in healthcare since their settings are extremely fast-paced or unpredictable. Victor (an associate professor) outlines, "...patient safety and privacy [could] be compromised when using assistive technology or a note-taker or scribe. Lawsuits highlight the tension between the legal requirements for providing accommodations and the functional demands or technical standards of healthcare jobs..." (Victor). It is indisputable that current accommodations may not support the students' long-term growth in professional settings. Students may lack the skills required for unpredictable environments, suggesting reevaluation of accommodations might be necessary for the success of students.

Conclusion

Years ago, the introduction of accommodations was meant to serve as a mark of progress in American education. However, the overwhelming statistics proving overdiagnosis of learning disorders, coupled with the rapid pursuit in seeking accommodations is evident. While some statistics do show that students—those greatly in need—can benefit from these services;

unfortunately, overdiagnosis and demand for resources can dilute the support available. While they can be helpful for students, accommodations often exacerbate existing disparities, driving socioeconomic and racial divides deeper. Students may not be set up for success as they become reliant on services that aren't realistic in a working environment. Ideally, an emphasis on refining accommodations is essential through setting clearer standards, frequent reevaluations, and access to affordable diagnostic services to ensure accommodations are distributed equitably. Schools should also place greater emphasis on interventions to build specific skills so students can be prepared for long-term success. By reforming the current system, accommodations can continue serving their original purpose to promote educational equity for all students.

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Congressional Stock Trading and Market Performance: An Analysis of Government Officials' Investment Returns By Lucas Ahern and Eliot Meese

Abstract

Government officials play a significant role in shaping economic policy and regulating financial markets. Concerns have been raised regarding whether access to private information provides advantages in stock trading. This paper evaluates whether the investment performance and trading activity of U.S. government officials differ significantly from standard market benchmarks. Using congressional trading disclosures, portfolio tracking databases, and historical S&P 500 data, this study analyzed quarterly portfolio returns, high-volume trading activity, and lagged correlations between government trades and future market performance. Government officials achieved average quarterly returns of 4.58% compared to 3.21% for the S&P 500, with statistical testing showing the excess returns were in fact statistically significant ($p = 0.0047$). In addition, elevated trading activity was positively correlated with future S&P 500 returns over longer time horizons, with the strongest relationship occurring at the 30-day time interval ($r = 0.143$, $p = 0.009$). While short-term correlations remained insignificant, the emergence of significant positive correlations over longer intervals suggests that periods of elevated government trading activity are followed by favorable market conditions. This paper evaluates whether these observed patterns reflect conventional investment skills or broader concerns regarding transparency, market fairness, and the effectiveness of current financial disclosure regulations.

Keywords

Congressional Stock Trading, Insider Trading Regulations, Government Transparency, Market Anomalies, STOCK Act

Introduction

Government officials play a significant role in shaping economic policy, influencing public companies, and regulating financial markets. Because of their access to classified information and their direct ability to impact legislation, questions have been raised about whether their personal involvement in the stock market creates potential conflicts of interest (Pei 1). In the United States, members of Congress are legally allowed to buy and sell stocks. However, concerns regarding whether the current safeguards in place, such as the STOCK Act (Chu 1), are sufficient enough to prevent the use of private information wrongly.

The STOCK Act was enacted to increase transparency and aimed to prohibit insider trading by government officials, requiring full disclosure of financial transactions. Despite these regulations, there have been challenges, such as enforcing the act and delaying disclosure reports, that have raised concerns about the effectiveness of the law. Critics argue that even when disclosures are made, they often are disclosed after the trades have already been executed (Caputo et al. 3). This detail limits the law's usefulness in preventing unethical use of private

information. Moreover, the complex nature of financial markets and the various ways that policy decisions can influence stock prices make it difficult to determine whether trades are based on private information or standard investment strategies. Recently, ongoing public skepticism about the fairness of allowing government officials to actively trade in financial markets has sparked controversy.

Prior research has suggested that a large number of government officials have historically achieved investment returns that outperform the broader market (Caporal 8). This research has led to speculation that access to private information may provide an advantage in making financial decisions. While some studies argue that these returns can be explained by other factors, such as wealth or financial education (Tanwar 1-7), other studies emphasize patterns of trades occurring before major announcements or market-moving events that can be exploited (Wei et al. 2-3). These findings raise important details about market fairness, public trust in government institutions, and transparency.

However, other analyses emphasize the timing of trades rather than the overall returns. In particular, some investigations have identified instances where stock transactions occur before major policy announcements (Rasheed 1). While these patterns do not definitely prove misconduct, they raise important details regarding the informational benefits government officials receive that impact the financial markets. These concerns are especially significant because they directly relate to issues of market fairness, transparency, and public trust. If investors believe that certain individuals have unequal access to information, confidence in the integrity of financial markets may be undermined.

Despite ongoing discussion and debate, there remains an undecided consensus on whether the investment performance of government officials is statistically abnormal when compared to the standard benchmarks of the market. Additionally, the extent to which their portfolio returns is still not fully agreed upon, whether it is due to advanced conventional investment strategies or potential private information advantages. Addressing these concerns is important for evaluating the effectiveness of current regulations and for implementing effective future policies.

In this study, we analyzed the stock portfolios of government officials to assess whether their investment returns are statistically significant in differing from market benchmarks. We hypothesized that government officials achieve returns that were statistically significant due to the returns being higher than expected compared to standard indices, which would suggest that there is a reason for accelerated returns for government officials, and the possibility of the misuse of private information.

Materials and Methods

Portfolio Return Analysis

Quarterly portfolio return data from U.S. government officials were collected using publicly available congressional trading databases and portfolio trackers, including Unusual

Whales (Evans 2) and Capitol Trades (Capitol Trades 1-8). Three government officials with sufficient publicly disclosed portfolio data across consecutive quarters were selected for analysis: Tim Moore (Quiver Quantitative 1), John Fetterman (InsiderCat 1), and April McClain Delaney (Quiver Quantitative 1). Quarterly percentage returns for each official were recorded across eight quarters spanning Q2 2024 to Q1 2026. Mean quarterly returns were calculated by averaging the percentage gains of the selected officials for each quarter.

Performance of the broader market was measured using the Vanguard 500 Index Fund (VFIAX), which tracks the S&P 500 (Yahoo Finance 1). Quarterly percentage returns for VFIAX across the same time period were collected and used as the benchmark comparison. A one-sample significance test comparing the mean difference between government officials' returns and the S&P 500 benchmark was conducted using an alpha of 0.05. A 95% confidence interval for the mean difference in quarterly returns was also calculated to evaluate the magnitude of excess returns.

Trading Volume and Marking Timing Analysis

To analyze whether government trading activity corresponded with significant market movement, publicly disclosed congressional trades exceeding \$250,000 were collected from Capitol Trades (Capitol Trades 1). The number of qualifying trades occurring on each date was recorded over an approximately three-year period. Market performance during the same interval was measured using historical VFIAX data representing the S&P 500 (Nasdaq 1-50). Trading activity and market performance were overlaid graphically to visually evaluate whether periods of elevated government trading volume occurred near significant market movements.

Lagged Correlation Test

A lagged correlation test was performed to evaluate whether elevated government trading activity was associated with future S&P 500 performance. Daily counts of congressional trades exceeding \$250,000 were compared against forward returns of the S&P 500 at multiple time intervals, including 1-day, 2-day, 3-day, 5-day, 10-day, 20-day, 30-day, and 60-day periods. Correlation coefficients (r) were calculated for each lag interval to measure the strength and direction of the relationships. Two-tailed significance tests were then performed to determine whether observed correlations differed significantly from zero. Statistical significance was defined as $p < 0.05$.

Data Visualization

All statistical analyses, confidence intervals, and graphical visualizations were generated using spreadsheet-based calculations and statistical software tools, including Excel and Google Docs. Figures were created to compare quarterly returns, visualize trading volume, and display lagged correlation coefficients across multiple future time horizons.

Results

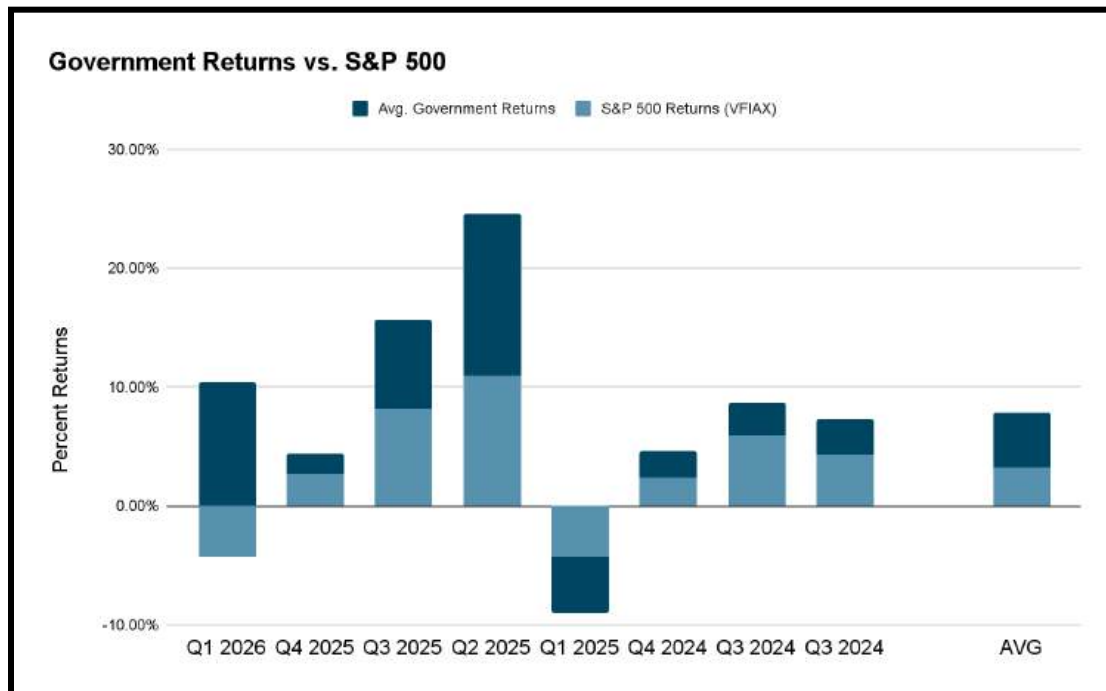


Figure 1. Government Returns vs. S&P 500.

The light blue bars represent the S&P 500's percentage gains or losses each quarter through the mutual fund, VFIAX (Yahoo Finance 1). The darker blue bars represent the return of many government officials' investments per quarter (Evans 2). These returns were calculated from averaging the percentage gain of two ETFs, one for republicans and one for democrats. The ETFs track holdings that members of each party buy in disclosed files. Keep in mind that the top of the dark blue bar is not the government officials' returns; The length between the top and bottom points of the dark blue bar represents the accurate returns. An average of each group's returns was calculated and displayed as a separate bar.

Four of the eight quarters display the government officials outperforming the S&P 500. One quarter shows the government officials being in line with the S&P 500. Finally, three quarters show the government officials underperforming the S&P 500. Looking at the average returns for the whole eight quarters, the government officials had a 35.1% higher return than the S&P 500, with a 4.58% average gain compared to a 3.21%. These results show that government officials achieved an average excess return of 1.37 percentage points per quarter relative to the S&P 500.

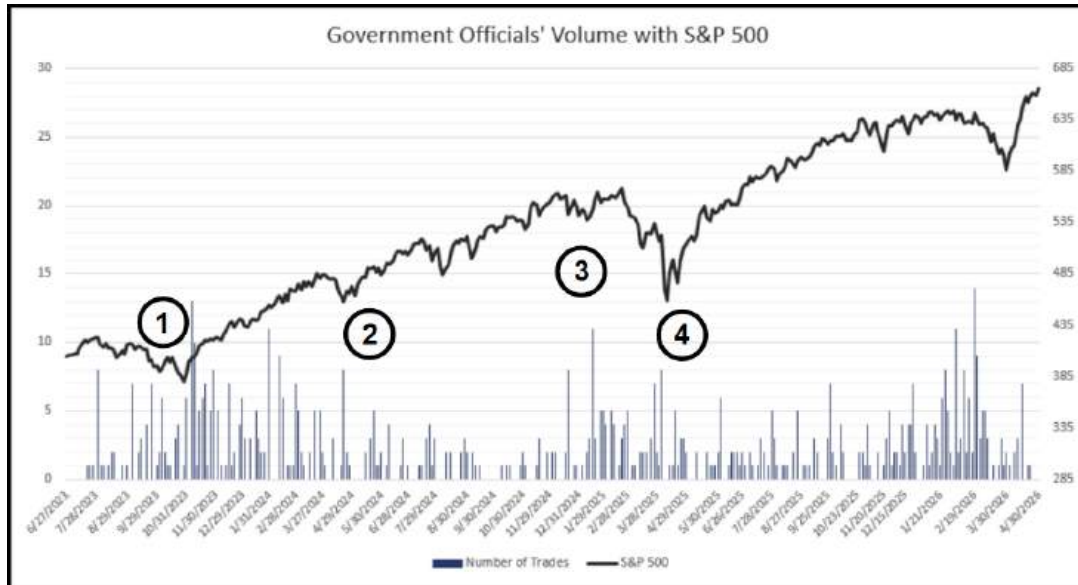


Figure 2: Government Officials' Volume with S&P 500.

The black line represents the performance of the S&P 500 over approximately the past three years, using the Vanguard 500 Index Fund (VFIAX) (Nasdaq 1-50). The blue bars represent the number of reported trades exceeding \$250,000 conducted by government officials on each date (Capitol Trades 1-8). These variables are overlaid to evaluate whether periods of increased trading activity correspond with substantial market movements.

Numbered regions (1–4) highlight intervals in which trading activity appears to cluster around notable market inflection points. In several cases, increased trading volume occurs near local market lows or during periods of heightened volatility, followed by upward price movements. These patterns suggest that trades may be occurring at times that precede favorable market conditions.

Across the highlighted intervals, elevated trading activity is frequently observed at exact downturns in the market, with all instances followed by substantial increases in the S&P 500. Given that consistently timing market bottoms is widely considered impossible and implausible for investors (Wing 5-6), the observed clustering of large trades near these periods provides concern about whether such outcomes can be explained by chance alone.

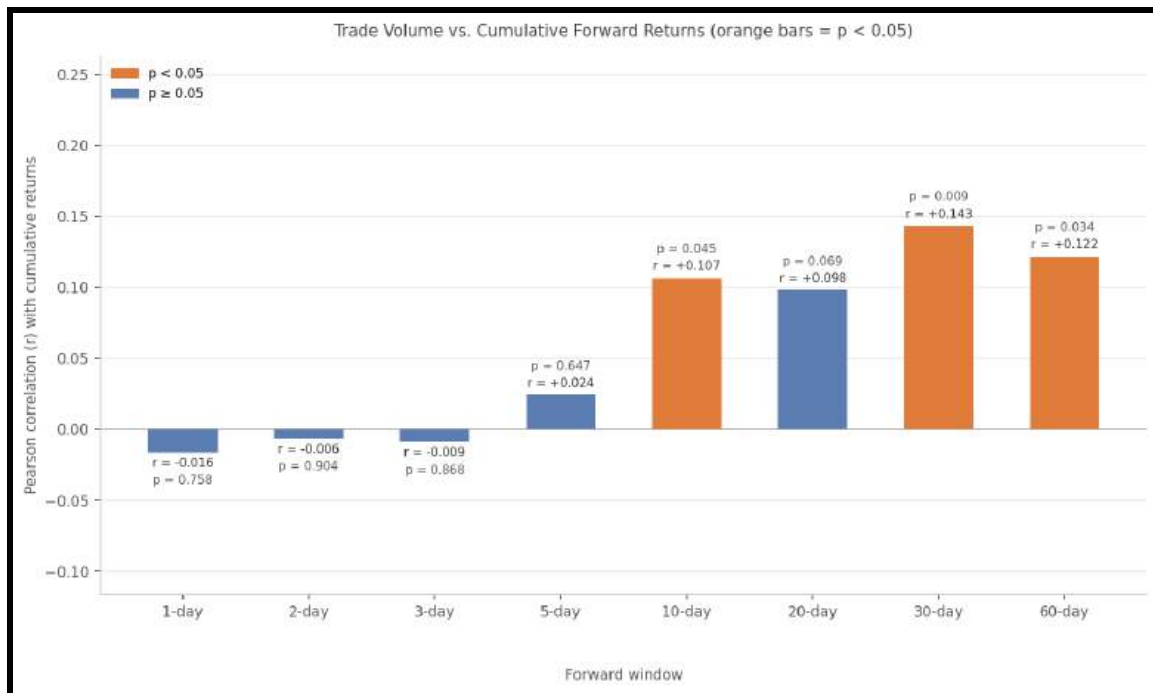


Figure 3: Government Officials’ Trade Volume vs. Cumulative Forward Returns.

A lagged correlation analysis was conducted to evaluate the relationship between government trading activity (Capitol Trades 1-8) and the corresponding performance of the S&P 500 (Nasdaq 1-50) across multiple time horizons. Pearson correlation tests, along with significance tests to calculate two-tailed p-value, were used to assess statistical significance. The orange bars represent time intervals where the results were, in fact, statistically significant and where the Pearson correlation was the highest. The specific time intervals display the amount of time after government officials’ trades where returns were elevated.

At shorter intervals, no statistically significant relationship was observed. The 1-day ($r = -0.016$, $p = 0.758$), 2-day ($r = -0.006$, $p = 0.904$), 3-day ($r = -0.009$, $p = 0.868$), and 5-day ($r = +0.024$, $p = 0.647$) correlations were all close to zero and not statistically significant. These results suggest that trading activity does not have an immediate or short-term relationship with market performance. However, the relationship becomes more pronounced over longer time horizons. At the 10-day interval, the correlation becomes positive and reaches statistical significance ($r = +0.107$, $p = 0.045$), indicating that higher levels of trading activity are moderately associated with positive market returns over the following two weeks. This relationship strengthens further at the 30-day interval ($r = +0.143$, $p = 0.009$), representing the strongest observed correlation in the dataset. At 60 days, the relationship remains statistically significant ($r = +0.122$, $p = 0.034$). The 20-day interval ($r = +0.098$, $p = 0.069$) approaches significance but does not meet the conventional threshold.

The absence of significant correlations at short time horizons suggests that government officials are not simply reacting to publicly available information that is quickly incorporated into market prices. Instead, the emergence of statistically significant positive correlations over

longer horizons, such at the 30-day interval, indicates that periods of elevated trading activity are associated with following positive market performance. The decline in correlation at the 60-day horizon further suggests that this relationship is not solely driven by broader market trends, which would likely produce more persistent effects. Overall, the findings are consistent with the possibility that elevated government trading activity contains forward-looking information related to market movements. However, as this analysis is observational, it does not establish causation, and alternative explanations such as general market conditions or behavioral timing strategies cannot be excluded.

Taken together, these findings indicate that elevated trading activity by government officials is statistically associated with positive cumulative S&P 500 returns over the following 10 to 60 days, a pattern consistent with the idea that congressional trading contains forward-looking information about market movements, although this analysis cannot distinguish informational advantage from generalized market-timing skill or correlation with unobserved factors.

	Q1 2026 Gain	Q4 2025 Gain	Q3 2025 Gain	Q2 2025 Gain	Q1 2025 Gain	Q4 2024 Gain	Q3 2024 Gain	Q2 2024 Gain
Tim Moore	9.5%	20.5%	31.5%	15.0%	6.5%	4.0%	3.2%	2.8%
John Fetterman	29.6%	22.7%	15.8%	28.1%	5.9%	20.0%	2.2%	26.9%
April Delaney	5.1%	8.2%	4.9%	4.0%	3.5%	2.8%	2.4%	2.0%
Avg. Gains	14.7%	17.1%	17.4%	15.7%	5.3%	8.9%	2.6%	10.6%
S&P 500 (VFIAX)	(4.3%)	2.64%	8.11%	10.93%	(4.3%)	2.4%	5.9%	4.3%

Table 1: U.S. Government Officials' Investment Returns Compared to the S&P 500.

The table represents the performance of investment portfolios of three current U.S. Senators going back two years, every quarter. The data was based on the senators Tim Moore (Capitol Trades 1), John Fetterman (InsiderCat 1), and April Delaney (Capitol Trades 1). The portfolio's percent gains for each quarter were calculated into an average. The percentage gain, or loss in parentheses, for each quarter of the S&P 500, through the mutual fund, VFIAX, was also tracked in comparison (Yahoo Finance 1).

When testing if the average return of the government officials from each quarter is greater than the S&P 500's gains not due to chance, the p-value that is outputted is 0.0047. Since the p-value is less than alpha ($0.0047 < 0.05$), there is convincing evidence to reject the hypothesis that these three government officials as a group generate equal gains as the S&P 500. Moreover, when conducting a 95% confidence interval for the mean difference between the three government officials and the S&P 500's quarterly returns, the interval is outputted as (2.77, 13.89). The entire confidence interval is greater than 0 and goes all the way up to 13.89. This calculation suggests that not only are the government officials outperforming the S&P 500, but they are also greatly outperforming the index quarterly.

Conclusion

This study evaluated whether the investment performance and trading activity of U.S. government officials differed significantly from standard market benchmarks. By analyzing quarterly portfolio returns, high-volume congressional trading activity, and lagged correlations with future S&P 500 performance, several statistically significant patterns were identified. Government officials achieved average quarterly returns that exceeded the S&P 500 benchmark, and statistical testing demonstrated that these excess returns were unlikely to have occurred by chance alone. In addition, periods of elevated congressional trading activity were associated with positive future market performance over longer time periods.

Although these findings do not provide insight to insider trading or the causation of the misuse of private information, this study's findings raise important concerns regarding transparency, market fairness, and the effectiveness of current disclosure regulations, such as the STOCK Act. The absence of significant short-term correlations, combined with the emergence of statistically significant longer-term relationships, suggests that government trading activity may contain forward-looking information about future market conditions. However, alternative explanations cannot be excluded.

These results contribute to ongoing debates regarding whether government officials should be permitted to actively participate in financial markets while simultaneously influencing economic policy and legislation. Due to public trust in financial institutions and government transparency being essential to market integrity, further research examining larger data sets, individual trading, and attempting to infer causation may help better determine the extent to which informational advantages contribute to these observed patterns. Future policy discussions may also consider whether additional disclosure requirements or trading restrictions are necessary to maintain confidence in the fairness of U.S. financial markets.

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The Interactions Between Gut-Microbiome Metabolites and T-cells in the Pathogenesis of Pediatric Celiac Disease By Reya Pandya

Abstract

Coeliac (Celiac) disease is an autoimmune disease that occurs in genetically predisposed individuals who develop an immune reaction to gluten when consumed. Individuals with Celiac disease often have intestinal dysbiosis, an unhealthy balance of gut bacteria in which opportunistic pathogenic bacteria become damaging to the gut lumen due to traits that allow them to adhere more strongly to the gut lining, trigger inflammation, and produce toxins. Recently, studies have demonstrated that the gut microbiome, which is the community where microorganisms such as bacteria, viruses, and fungi reside, is associated with the metabolic processes that can appear before the clinical signs of celiac disease in some children. These data demonstrate that gut microbiome products and immune cell interactions may, in part, cause the disease state, rather than being an effect of intestinal damage. Additionally, scientists measured metabolites in children with celiac disease through stool and blood samples. These data revealed that children with celiac disease have distinct gut bacterial and metabolite profiles, specifically in terms of chemicals such as short-chain fatty acids produced by gut bacteria. Although the studies demonstrated some variability in findings, researchers agree that these metabolite changes are important for understanding and predicting celiac disease. However, the specific mechanisms linking metabolite changes, gut microbiota adjustments, and T-cell responses in the early stages of Celiac Disease remain selectively understood, specifically in pediatric populations. This review article aims to address specific T-cells and their importance to celiac disease, how their relationship with metabolites shapes the disease as a whole, and how certain metabolites can be used as a biomarker/predictor in children.

Keywords Celiac disease, pediatric patients, gut microbiome, metabolites, T-cells

Introduction

What Is Celiac Disease and How Does It Damage the Intestine

Celiac disease (CD) is a chronic, autoimmune disorder that is characterized by an immune response to gluten proteins. Under normal conditions, the small intestine mucosa inhibits tall, finger-like villi that provide a large surface area for efficient nutrient absorption (Figure 1a). But in CD, the body mistakenly reacts to gluten, causing damage and inflammation to the intestine. Gluten proteins can be found in sources like barley, wheat, and rye. Gluten consists of two primary types of proteins: gliadins, which are monomeric proteins, and glutenins, which are aggregated proteins. Gliadins are smaller, single-chained proteins that contribute to the consistency and thickness of the dough, whereas glutenins are larger, multi-chained proteins that form strong networks together that are important for elasticity. Together, they make up gluten's structural properties, but also make it highly resistant to being broken down and digested in the

gastrointestinal tract. (Ahmadzadeh and Rezaei-Tavirani, 2025). These proteins are notably high in proline and glutamine and poorly digested in the gastrointestinal tract. This condition affects the mucosa of the small intestine, leading to malabsorption, maldigestion, and villous atrophy (Figure 1b). These markers are especially seen in pediatric populations. Diarrhea, loss of appetite, bloating, and failure to thrive are the most common symptoms seen in children (Salarian et al., 2023). CD begins when gluten is broken down into small pieces and changed by the enzyme tissue transglutaminase (tTG). After the gluten is changed, it is then presented to the immune cells. In individuals with the correct genetic background, it causes CD4+ T cells to activate, releasing inflammatory signals. Additionally, CD8+ T-cells kill intestinal epithelial cells. This combination of immune responses destroys the intestinal lining, resulting in impaired nutrient absorption and defines the pathology for CD (Salarian et al., 2023).

Genetic Risk Factors and Epidemiology

Genetic susceptibility plays a key role in celiac disease, but the genetic component alone is not enough to cause the disease. An important feature of CD development is the genetic susceptibility given by certain human leukocyte antigen (HLA) class II molecules. These genes are HLA-DQ2 and HLA-DQ8, which account for half of the risk for CD by inheritance. More than 90% of people with CD carry the HLA-DQ2.5 gene, and the rest carry HLA-DQ8 (Taylor et al., 2025). Although 20-40% of the general population holds the HLA-DQ2 and HLA-DQ8 predisposing alleles, only 1-3% of those people develop CD. This indicates that additional environmental and immunological factors are necessary for the disease onset, including environmental influences such as early-life infections, antibiotic use, infant feeding methods, and shifts in the gut microbiome, alongside immunological factors such as a weakened immune tolerance and increased activation of gluten-specific T-cells that cause intestinal inflammation. (Kelley, Girdhar, et al., 2024). The frequency of CD in pediatric populations varies globally but is generally similar to that of adults, with a prevalence of 1% in many Western countries. First-degree relatives of CD patients have a higher risk, as studies show that up to 10% of siblings develop biopsy-confirmed CD, often while being asymptomatic (Sahin and Mermer, 2022). Additionally, twin studies show high heritability, with the occurrence of development of CD in monozygotic twins is 80%, compared to a lower occurrence in dizygotic twins with 20%. CD is diagnosed in females 2 to 3 times more than in males (Ahmadzadeh and Rezaei-Taviran, 2025). While specific HLA genes increase the risk of celiac disease, most genetically predisposed individuals don't develop it, underlining the importance of other factors such as the composition of the gut microbiome, infections, environmental exposures in early life, immune system regulation, and microbial metabolites.

Gut Microbiome and Disease Development

Despite the large role of gluten exposure and genetics, not everyone with the HLA risk gene develops CD, showing that other factors play a role. Within these factors, the gut microbiome is a crucial environmental influencer. The gut microbiota comprises trillions of

microorganisms such as *Bacteroides*, *Lactobacillus*, *Bifidobacterium*, and *Clostridium*, which contribute to nutrient metabolism, mucosal immune development, and protective barrier function. An altered microbiota composition known as dysbiosis has been found in pediatric CD patients, compared to those who do not have the disease. This includes a reduction in beneficial gut bacterial species, such as *Bifidobacterium* and *Lactobacillus*, and an increase in potentially harmful bacteria, including *Escherichia coli*, *Clostridium difficile*, *Staphylococcus aureus*, *Bacteroides fragilis*, and *Enterococcus faecalis* (Said et al., 2025). Longitudinal studies have shown that individuals with genetically predisposed CD often show clear microbial and metabolic profiles years before disease onset, suggesting that early microbiome changes can influence the risk of losing gluten tolerance and starting an autoimmune response (Kelley, Dogru, et al., 2025). Furthermore, certain gut microbes, more specifically bile acid-transforming bacteria such as the *Cloustridium* species and other Firmicutes capable of the process of 7 α -dehydroxylation (Ridlon et al., 2016), can produce metabolites such as secondary bile acids like taurodeoxycholic acid, which control immune signaling and can intensify mucosal inflammation when present with abnormally elevated levels (Kelley, Girdhar, et al., 2024). Emerging evidence suggests that the connection between metabolites and T-cell function is crucial to disease initiation. Metabolites generated by microbial metabolism can influence T-cell differentiation and cytokine production, changing the balance of how immune cells act towards a pro-inflammatory response that is part of CD pathogenesis (Yang and Cong, 2021), and therefore, variations in the gut microbiome and metabolites can act as environmental triggers that impact immune responses and contribute to CD onset.

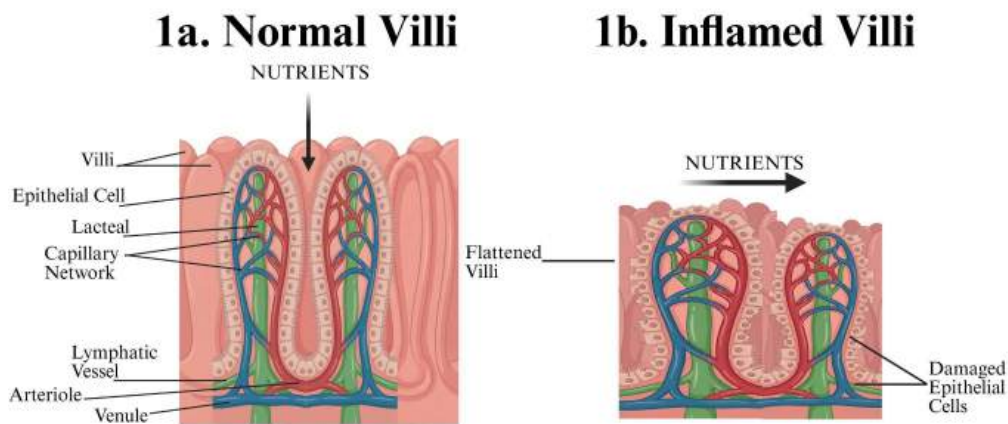


Figure 1: Schematic comparison of normal and inflamed intestinal villi

1a. The mucosa of the normal small intestine showing tall, finger-like villi. The large surface area allows for optimal nutrient absorption.

1b. In Celiac disease, allergens like gluten trigger an immune-mediated inflammatory response in genetically susceptible individuals. The mucosa contains villous atrophy, which is the flattening or blunting of the villi. The absorptive surface area is reduced, leading to impaired absorption of nutrients.

Section 1: T-Cell Immunology and Immune Activation in Celiac Disease

In individuals with CD, gluten peptides activate the adaptive immune system, taking place in the small intestine. T-cells are central to this process, as they act as the main coordinators of the immune response that leads to tissue damage and intestinal inflammation. Beyond just reacting to pathogens, specific types of T-cells in CD can recognize gluten fragments as harmful, triggering inflammatory signals that destroy intestinal structure and function. It is essential to understand how T-cells develop, recognize antigens, and become activated, as it explains why gluten triggers inflammation in genetically susceptible individuals. The sections below outline the importance of T-cells in the pathogenesis of celiac disease, from recognizing antigens to causing intestinal damage.

1.1 T-cells

T-cells are a type of white blood cell that plays a fundamental role in the adaptive immune system, which is the branch of immunity that learns how to recognize and remember foreign molecules, such as antigens. Antigens protect the body from infections that have reached the inside of cells. T-cells are produced in bone marrow, mature in the thymus, and circulate throughout the body, ready to respond to their matching antigen when encountered (Patt et al., 2023). After maturing in the thymus, they develop receptors that allow them to recognize foreign antigen fragments, which are bound to major histocompatibility complex (MHC) molecules on cells. In humans, MHC class II molecules are known as human leukocyte antigens (HLAs) and are crucial for presenting peptides to CD4⁺ helper T-cells. In celiac disease, the immune response is triggered by gluten (du Pré and Sollid, 2015, Figure 2). There are several different subtypes of T-cells, but there are a few that are important to CD. CD4⁺ helper T-cells have the ability to coordinate and increase the power of immune responses by secreting cytokines, which have an influence on the behavior of other immune cells, such as cytotoxic T lymphocytes and B cells. CD8⁺ cytotoxic T-cells can directly kill infected cells, which is one of the factors that contribute to tissue cell damage in CD. Regulatory T-cells help suppress incorrect immune reactions, though they are not active in CD. If regulatory T-cells are inactive, they fail to suppress the immune response to gluten. This leads to chronic intestinal inflammation and mucosal damage. In children with CD, gluten-specific CD4⁺ T-cells are the main drivers of pathogenic inflammation (van Bergen et al., 2015).

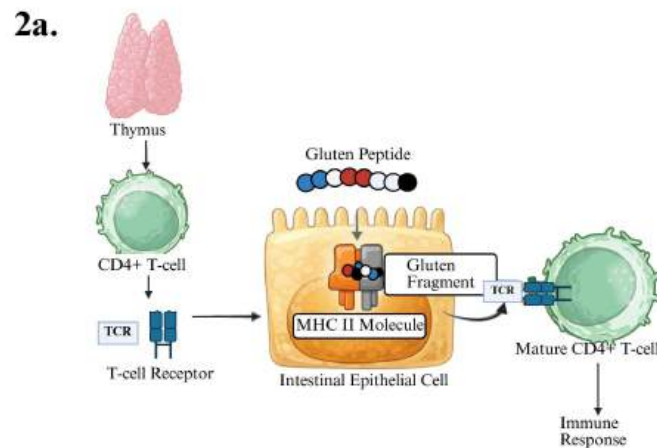


Figure 2: Gluten Peptides and CD4+ T-cell Activation in Celiac Disease

2a. CD4+ T-cells mature in the thymus and develop T-cell receptors (TCRs) that have the ability to recognize specific antigen fragments, like gluten. In the small intestine, gluten peptides are presented to HLA Class II (MHC II) molecules on the intestinal cells. When a T-cell receptor recognizes the gluten fragment and HLA complex, the CD4+ T-cell becomes activated, this triggers an immune response that causes the tissue damage and inflammation seen in CD.

1.2 How a Naive Helper T-cell Recognizes Its Antigen

Before a naive helper T-cell can see an antigen, the antigen has to be processed and brought onto MHC class II molecules by an antigen-presenting cell (APC). A naive Helper T-cell is a mature T lymphocyte that has left the thymus, but has not yet encountered its specific antigen. Every CD4+ T-cell has a unique T-cell receptor (TCR), which is created by a genetic recombination process known as V(D)J recombination. Each TCR binds to a specific peptide when it is displayed by the HLA class II molecule on an antigen-presenting cell (APC) (Risnes et al., 2018). To start the antigen-presenting process, APCs such as B cells and dendritic cells engulf proteins from the environment, which includes the gluten fragments from digested food, and break it down into short peptides. Once broken down, these peptides are brought onto the MHC class II molecules and more specifically, the HLA-DQ2.5 and HLA-DQ8 molecules. This process happens inside the APC and is then transported to the surface of the cell. Naive CD4+ T-cells constantly circulate through lymphoid tissues, such as Peyer's patches and mesenteric lymph nodes, where the TCRs scan the newly created peptide-HLA complexes on APCs. Only T-cells with TCRs that match the specific peptide-HLA combination (DQ2/DQ8) will bind strongly enough to receive activation signals and go through proliferation. These T-cells increase in numbers rapidly and therefore create a highly specific immune response (Patt et al., 2023) Through this mechanism, specificity is ensured. Out of hundreds of millions of TCRs, only those

that are able to bind to the specific gluten peptide presented by the disease associated with the DQ2/DQ8 HLA molecule will become activated. Most T-cells that never find their matching peptide-HLA combination remain inactive or eventually die, as previously activated cells may become memory cells. Memory cells are long-lived immune cells that keep information on previously encountered antigens, allowing the immune system to respond more quickly if the antigen is reencountered. In CD, gluten-specific memory T cells reside in the intestine and rapidly trigger inflammatory responses when gluten is consumed again, contributing to the already persistent intestinal damage. (Risnes et al., 2018) In pediatric celiac disease, this mechanism leads to the activation of gluten-specific CD4⁺ T-cells because certain gluten fragments, particularly after being modified by the enzyme tissue transglutaminase (tTG), bind more strongly to the disease-causing HLA molecules and stimulate an adaptive immune response (Tjon et al., 2010). Together, this mechanism highlights how central HLA-restricted immune activation is to the development of CD.

1.3 DQ2/DQ8 Genes Produce Negative T-cell Response to Gluten

As mentioned earlier, most individuals with CD carry at least one of the HLA class II risk alleles, HLA-DQ2 or HLA-DQ8. The reason that these specific genes are associated with disease risk is that only these specific HLA variants have the ability to present the remodeled gluten peptides to CD4⁺ T-cells. Both HLA-DQ2 and HLA-DQ8 alleles tend to bind peptides with negatively charged amino acids that fit into specific amino acid side chains that insert into binding pockets of another molecule, or HLA binding grooves, needed to stabilize complex formation. So, gluten peptides by themselves have poor binding abilities to HLA class II molecules because they require the negatively charged amino acids that fit into the HLA binding grooves. The enzyme transglutaminase (tTG) modifies gluten peptides through deamidation, a process in which glutamine residues are converted into glutamic acid, which is a charged amino acid. Gluten is now a high-affinity binding peptide for HLA-DQ2 and HLA-DQ8 (Tjon et al., 2010). Since gluten consists of a high content of glutamine (Q) and proline (P), the sequences of QXXP, QXP, and QP, given that X represents any amino acid, are regularly found in gluten peptides. As presumed, only the sequence QXP, where glutamine is converted, results in the particular introduction of negative charges present in the gluten peptides. Glutamine itself is not a negatively charged acid, but after deamidation, it becomes glutamic acid, which has a negative charge after it is converted (Vader et al., 2002). Studies show that the length and stability of the peptide binding differ between HLA alleles. For instance, HLA-DQ2.5 forms a more stable peptide-HLA combination than HLA-DQ2.2, which explains why HLA-DQ2.5 carries a higher disease risk (Michael Bodd et al., 2012). Children who lack the disease-carrying HLA molecules rarely develop CD because their APCs are unable to present gluten peptides in the way that causes activation of harmful gluten-specific T-cells (Patt et al., 2023). In other words, this emphasizes that genetic susceptibility is a key factor of disease risk, as the absence of the specific HLA molecules prevents the initiation of the harmful immune response to gluten.

1.4 Roles of Gluten-Specific CD4+ T-cells in Intestinal Inflammation

Once the naive helper T-cells are activated by peptide-HLA combinations, gluten-specific CD4+ T-cells in the intestinal mucosa initiate multiple processes that contribute directly to disease pathogenesis. These T-cells secrete pro-inflammatory cytokines such as interferon- γ (IFN- γ) and interleukin-21 (IL-21), which promote a Th1-type immune response (Bodd et al., 2010). IFN- γ mostly enhances inflammation, recruits more immune cells, and changes the regulatory systems that normally limit too much immune activation. Also, gluten-specific helper T-cells interact with B cells, which provide signals that support the production of autoantibodies against gluten and tTG, which is a distinguishing marker for CD (M. Bodd et al., 2010, du Pré and Sollid, 2015). Even though CD4+ T-cells cannot directly kill intestinal epithelial cells, their cytokines that they secrete amplify innate immune signals, one being IL-15. This activates cytotoxic intraepithelial lymphocytes, or CD8+ cells, which damage the intestinal epithelium and cause the villous atrophy and flattening of the villi in the small intestine as seen in a biopsy for CD. (Voisine et Abadie, 2021). Both mechanisms together provide insight into how gluten-specific adaptive immune responses in genetically susceptible children are the drivers of the tissue injury and chronic inflammation characteristics of pediatric celiac disease. (van Bergen et al., 2015)

Section 2: Gut Microbiome Metabolites in Pediatric Celiac Disease

While the exposure of gluten and genetic susceptibility are critical to the development of celiac disease, they do not explain why the immune system becomes dysregulated in a certain group of individuals. Evidence shows that the gut microbiome in early life is an important environmental link between diet, metabolism, and the immune activation present in celiac disease. Microbial communities that inhabit the intestine create a diversity of metabolites that directly interact with epithelial and immune cells, such as T-cells. Changes to gut microbiome composition and the production of metabolites can influence whether gluten ingestion results in pathogenic immune activation or immune tolerance (Rahman et al., 2023). It is important to understand how gut-derived metabolites influence immune pathway signaling, as it provides comprehension for disease origination, progression, and even early biomarkers for celiac disease.

2.1 Metabolites and the Gut Microbiome

In the background of the GI tract, metabolites can arise from host or microbial metabolism, and from dietary compounds that are modified by host or microbial enzymes (Liu et al., 2022). Most of the metabolic activity in the human gut is driven by the trillions of microorganisms that live in the intestinal tract, known as the gut microbiota. These microbes contain genes that encode enzymes capable of breaking down and digesting substances that humans cannot, such as complex carbohydrates and fiber. After these substrates undergo fermentation, they produce microbial metabolites that must interact with the host cells, which also include immune cells in the gut lining. The microbes contain multiple genes that encode enzymes that help with digesting compounds humans are unable to break down, such as complex

carbohydrates and fiber. (Rahman et al., 2023). Metabolites derived from the gut microbiota can be organized into major classes: short-chain fatty acids (SCFAs) like acetate, propionate, and butyrate, bile acids and their derivatives that are microbially modified, amino acid-derived compounds like tryptophan metabolites, and any other small molecules such as polyamines and branched-chain fatty acids. All of these compounds and more have a job of acting as chemical messengers that have an influence on immune responses, gut physiology, and gut barrier integrity. (Liu et al., 2022)

2.2 Metabolites Linked to Celiac Disease in Children

In pediatric celiac disease, multiple metabolic patterns and specific metabolites have been connected with the disease progression, sometimes even years before clinical onset or diagnosis (Girdhar et al., 2023). Furthermore, research centering on metabolic analysis in CD has found significant patterns and connections that are vital to CD development. Firstly, longitudinal studies in pediatric patients at genetic risk for CD have discovered specific metabolic changes such as shifts in fatty acid pathways, glycolipid metabolism, and an imbalance between free radicals and antioxidants in the body, leading to cellular and tissue damage, known as oxidative stress (Girdhar et al., 2023). Secondly, taurodeoxycholic acid (TDCA) is a secondary bile acid produced by gut microbes, and is notably elevated in children who developed CD. In a cohort study, it was found that TDCA exposure caused changes in intestinal structure and immune cell diversity similar to early celiac pathogenesis, showing a direct link between this metabolite and CD mechanisms (Kelley, Girdhar, et al., 2024). Thirdly, SCFAs such as butyrate, acetate, and propionate are the most abundant types of metabolites found in the gut. They play important roles in maintaining gut barrier integrity and regulating inflammation (Zafeiropoulou et al., 2020). Pediatric CD studies have noted changes in SCFA profiles and the bacteria that produce them, causing dysregulated immune responses and long-term inflammation, which can both be seen in CD pathogenesis. (Zafeiropoulou et al., 2020) Lastly, metabolomic studies comparing controls with CD patients have found differences in amino acid metabolites such as tryptophan and glutamine in the mucosa of the small intestine, urine, and blood plasma. The differences in amino acid metabolites are a direct reflection of improper immune activation, altered energy metabolism, and the stress on the intestines found in CD (Upadhyay et al., 2020).

2.3 Influence of Early-Life Gut Microbiome Composition

The composition of the gut microbiome in early life plays a huge role in the amount and diversity of metabolites found in the intestinal environment as the child matures. From birth, the infant gut microbiota is rapidly assembled through contact with microbes from the mother and the surrounding environment. This is influenced by environmental factors, mode of delivery, breastfeeding, introduction of solid foods, and antibiotics. This sequential change in the structure, function, and compounds within the microbiome continues till about age 2-3, when the microbiome starts resembling an adult-like microbial community (Sarkar et al., 2021). As these microbial communities continue to develop and diversify, their metabolic activities change, most

significantly leading to an increased amount of SCFAs such as acetate, propionate, butyrate, and secondary bile acids, among other metabolites. SCFAs are produced through the fermentation of undigested carbohydrates and dietary fibers and play a role in influencing immune and epithelial cell functioning, whereas secondary bile acids are changed through the transformation of host-derived primary bile acids. The amount and type of microbiota correlate with the abundance of types of bacterial taxa. For instance, bacteria like *Bifidobacterium* and *Bacteroides* are linked to metabolic profiles in infancy, and changes in diet, such as the introduction of solid foods, and can alter metabolic end products by fermentation of new substrates (Aatsinki et al., 2025). Additionally, longitudinal infant microbiome studies have shown that when the diversity of the microbiota increases with age and diet changes, the metabolic community also progresses, showing that microbiome composition changes metabolic functioning, rather than being completely determined by age (Wu et al., 2024). However, in consequence, changes in the structure of the early-life gut microbiome through mode of delivery, environmental exposures, or feeding choices can cause differences in metabolite activity, with negative impacts on immune cell maturation, epithelial barrier functioning, and even disruption of metabolic signaling paths that can influence disease risk, like CD, later in childhood (Roager et al., 2023).

2.4 Metabolites as Biomarkers

Metabolites can be measured in samples of the blood, urine, or stool after birth and throughout childhood, often before villous atrophy or gluten-specific antibodies are detectable. Several metabolites mainly residing in the gut have surfaced as favorable biomarkers for predicting CD onset in children, often detected even months or years before diagnosis. Longitudinal studies of genetically at-risk children with positive HLA-DQ2 and HLA-DQ8 genes have found elevated levels of TDCA in plasma samples collected before the appearance of celiac autoantibodies. TDCA production comes from microbial modification of host bile acids and induces inflammation in the intestines and activation of immune responses, showing that it directly contributes to early CD pathogenesis (Girdhar et al., 2023). Additionally, SCFAs are different in fecal samples from children who later develop CD, with a decrease in butyrate-producing pathways seen before diagnosis. Since SCFAs play crucial roles in protecting intestinal barrier integrity and regulating immune responses, an observable early decrease in SCFAs can reflect a gut community dismissive to T-cell activation (Zafeiropoulou et al., 2020). Amino acid metabolites, specifically the ones involved in tryptophan metabolism, can influence immune signaling pathways and can manipulate the commands inside T-cells, which determine what type of specialized T-cell it becomes (Upadhyay et al., 2020). Overall, these findings hint that metabolite analysis, along with environmental changes, could contribute to the immunopathogenesis of pediatric celiac disease.

Section 3: Metabolite-T-cell Interactions and Biomarker Potentials

Although interactions between the gut microbiome and immune system are involved in pediatric autoimmune diseases, the relationship between T-cell activation and gut-derived

metabolites in CD is shaped uniquely by the presence of an environmental antigen, gluten, and the specific disease-carrying alleles. In genetically susceptible pediatric patients, gut-derived metabolites can influence whether the immune tolerance to gluten stays or is lost, even before clinical signs appear. CD is specifically characterized by the early metabolite signaling pathways that cause the expansion of pathogenic CD4⁺ T-cells, while blocking off regulatory systems. These interconnections occur in the mucosa of the intestinal tract, where metabolites directly impact T-cell differentiation, cytokine production, and epithelial barrier integrity. Understanding how metabolites affect T-cell behavior shows an important mechanism link between immune system dysregulation, the gut microbiome, and the chronic inflammation that defines celiac disease.

3.1 Metabolite- T-cell Interaction

The metabolite-T-cell interactions that are most connected to predicting the onset of pediatric CD are the interactions that have an influence on T-cell activation in the gut and immune tolerance. Together, metabolites formed from early-life microbe activity form the intestinal community in ways that either boost or prevent gluten-specific CD4⁺ T-cell responses to occur in genetically susceptible individuals. As seen before, interactions containing SCFAs and amino-acid metabolites are important, as they regulate T-cell and cytokine production at mucosal sites. Butyrate promotes regulatory T-cell development and overpowers unnecessary Th1-type immune responses, supporting oral tolerance of antigens ingested. In the later development of pediatric celiac disease, a reduced abundance of SCFA-producing bacteria has been seen before diagnosis, which shows a loss of functioning metabolic signals that normally limit gluten-reactive T-cell growth. At the same time, metabolites acquired from tryptophan metabolism can influence the behavior of T-cells by causing an activation of host cell receptors involved in immune regulation (Zafeiropoulou et al., 2020). The inflammatory presence in the gut is caused by the disruption of these metabolites and host cell receptors (Zafeiropoulou et al., 2020). In children with HLA-DQ2/DQ8 genes, the metabolite-driven effects on T-cells are more consequential, as even just the presence of gluten peptides causes a tiny shift in immune regulation, and can push the balance towards pathogenic CD4⁺ T-cell activation (Figure 2a, Risnes et al., 2018). So, the most predictive metabolite-T-cell interactions are the ones that show early breaks in immune tolerance systems, rather than just the presence of intestinal inflammation alone. Overall, these findings emphasize that shifts in microbiome-derived metabolites are not just consequences of the disease, but instead are contributors to the loss of immune tolerance and initiation of harmful T-cell responses in CD.

3.2 Reduced SCFAs on Regulatory T-cell function

SCFAs, which are produced from gut microbes in dietary fiber, are responsible for regulating the immune system and promoting regulatory T-cell (Treg) development in the intestine. In general, SCFAs enhance the immune system and create a stronger defense against pathogens. More specifically, SCFAs allow for the expansion of Foxp3⁺ Tregs, which is the

primary subset of Tregs that has a responsibility of maintaining immune tolerance to harmless antigens it comes across in the gut, and is done through many molecular mechanisms. One important pathway is epigenetic regulation. Histone deacetylases (HDACs) are inhibited in T-cells by butyrate and propionate. This increases the epigenetic accessibility of Foxp3 and other genes that are important for Treg functioning and differentiation. The histone pathway also assists in the production of anti-inflammatory cytokines, like IL-10, which represses excessive T-cell responses (Kim, 2021). In other studies, supplementation of SCFAs increases Foxp3 and Treg abundance in the gut, whereas a reduction in SCFAs leads to a lower quantity of Tregs and an increased amount of intestinal inflammation, supporting that there is a direct link between SCFAs and their effect on Treg functioning and differentiation in immune tissues (Zaiss et al., 2019). To add on, studies using human blood cells show that butyrate and propionate elevate the activity of Treg-associated genes and increase their suppressive functions, highlighting that SCFAs not only support the quantity of Tregs, but also their functioning ability to limit T-cell division and cytokine production (Hu et al., 2022). A reduction of SCFA levels can be seen in early gut dysbiosis or low-fiber diets in susceptible children, and therefore, the influence of immune cells such as Tregs weakens. Mainly, this causes fewer regulatory signals in the gut, reduced IL-10 secretions, and allows for inflammation-driven T-cells to remain active. In the context of CD, weaker Treg functioning causes the immune system to have an intolerance to gluten peptides, which increases the likelihood that gluten-specific CD4⁺ T-cells expand into other inflammatory types and contribute to the mucosal wall damage in the intestine. So, SCFA reduction compromises the development and functioning of Tregs, changing the mucosal membrane immunity towards inflammation, and directly suppresses the regulatory mechanism on pathogenic T-cell immune responses. In genetically susceptible pediatric patients, early gut dysbiosis and a reduced fiber intake can cause a decrease in SCFA production, causing regulatory immune mechanisms to weaken in the intestine. In the context of CD, the reduction in SCFA-mediated regulatory T-cell function affects the oral tolerance of gluten, allowing gluten-specific CD4⁺ T-cells to grow in abundance and acquire inflammatory causing mechanisms. CD uniquely combines an impaired regulatory mechanism with a specific environmental trigger and HLA gene restrictions on T-cell activation (Girdhar et al., 2023). Consequently, a reduced abundance of SCFAs directly feeds into constant intestinal inflammation, epithelial cell damage, and the villous atrophy distinguishing factors of celiac disease. These findings underline how microbiome-derived metabolites like SCFAs directly regulate the T-cell balance in the intestines, specifically through their effects on Treg function and development. Since variations in SCFA levels can be seen in early stages of gut dysbiosis, these metabolites can be used as important indicators of immune system dysregulation and possible biomarkers for predicting the development of CD in genetically susceptible pediatric populations.

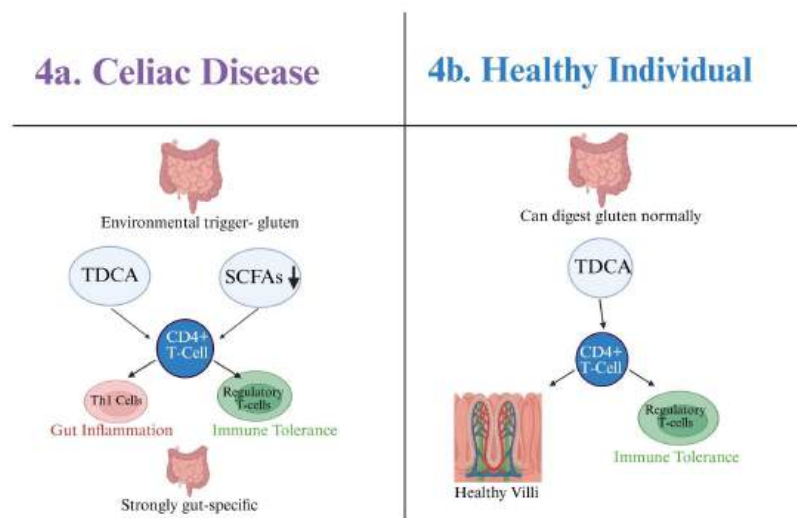


Figure 4: Comparison of metabolite-T-cell interactions in celiac disease in contrast to a healthy individual

4a. Taurodeoxycholic acid (TDCA) and a reduced amount of short-chain fatty acids (SCFAs) impacts CD4+ T-cell development in the gut. CD4+ T-cells develop into regulatory T-cells (green) which support immune tolerance or cause a Th1 response, causing inflammation (red). Gluten is the main environmental trigger, and its effects are strongly tied to the gut, highlighting gut specialization.

4b. Taurodeoxycholic acid (TDCA) and sufficient levels of short-chain fatty acids (SCFAs) support a balanced level of CD4+ T-cell development in the gut. T-cells differentiate into regulatory T-cells (green), which promotes immune tolerance. Gluten is digested effectively without triggering an inflammatory immune response, preserving healthy intestinal structures, such as the villi, and maintaining a gut-specific immune regulation.

Discussion/Results

From a clinical and patient-focused perspective, these findings highlight the importance of looking beyond just a genetically based understanding of CD, toward a more diverse model that incorporates the gut microbiome and metabolic activity. For clinicians, this supports the potential for earlier conformation of susceptible children through approaches such as stool or blood metabolite profiling, even before the hallmark features or intestinal damage appears (Girdhar et al., 2023). For patients and their families, specifically with known genetic susceptibility, this paper emphasizes the role of early life factors such as proper diet, fiber intake, antibiotic exposure, and gut health, in impacting immune responses (Rahman et al., 2023). Although the current treatment is a gluten-free diet, these findings might suggest that future treatment strategies may focus on incorporating gut microbiome focused interventions such as dietary modulation, probiotics, prebiotics, or even metabolite-centered therapies focused on rebuilding immune tolerance (Girdhar et al., 2023). On a wider scale, the observation that

metabolic and microbial changes are seen before disease onset has important public health implications, as it creates opportunities for preventative strategies for at-risk pediatric populations (Kelley et al., 2025). Future research may focus on confirming reliable metabolite biomarkers, supporting the mechanisms present between microbial metabolites and T-cell dysfunction, and using long-term studies that integrate the gut microbiome, metabolite, genetic, and immune cell data to identify the specific early-life periods when an intervention may be the most successful in preventing the disease (Liu et al., 2022).

Conclusion

Pediatric CD emerges from a combination of factors, including environmental exposures, genetic susceptibility, immune system dysregulation, and the T-cell-metabolite interactions that link these mechanisms together. Though most children carry the genetic risk alleles HLA-DQ2 and HLA-DQ8, only a small percentage go on to develop the disease, showing the importance of non-genetic risk factors, such as metabolites shaping the occurrence of immune responses. Studies on at-risk patients for developing CD show a variation in the gut microbiome and its metabolic pathways that emerge months to years before clinical onset, showing specific pathways in children who go on to develop this disease compared to children who do not. For instance, children who continue towards CD development show early differences in the abundance of microbial taxa and metabolite pathways that are connected to immune modulation and inflammation, and can occur as early as age one, years before a diagnosis (Kelley, Dogru, et al., 2025). Specific metabolite differences impact the behavior of T-cells in ways that are crucial to CD pathogenesis. Metabolites such as taurodeoxycholic acid (TDCA) have been shown to make inflammatory changes in the intestine and also enhance the activation of CD4+ T-cells, while reducing regulatory T-cells in models, showing how metabolites can have an impact on the balance between inflammation and immune tolerance (Girdhar et al., 2023). Additionally, other metabolic changes, like changes in secondary bile acid pathways and amino acid metabolism, have a tendency to correlate with immune system dysregulation, showing that metabolic markers are a presentation of emerging pathogenic T-cell responses even before antigen markers and inflammation are present (Leonard et al., 2021). Most importantly, metabolic patterns do not just show genetic risk; they also show a predictive value because they show the interactions between the gut microbiome, diet, and the growing immune system. By knowing what metabolite patterns appear before clinical onset of the disease, and linking those patterns to T-cell differentiation, cytokine production, and regulatory functioning, researchers are finding out what biomarkers can be used to predict disease pathogenesis in genetically susceptible children (Kelley, Dogru, et al., 2025). These biomarkers have the potential to lead to earlier diagnosis, better monitoring of symptoms, and the development of future risk-reducing or even preventative interventions for genetically susceptible patients. All together, the evidence shows a system where T-cell-metabolite interactions bond genetic susceptibility and environmental exposures, changing the understanding of pediatric CD pathogenesis and opening discoveries for preventative and predictive measures for at-risk children.

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Purkinje Cell Dysfunction in Autism Spectrum Disorders By Aastha Pendyala

Dr. Catherine Berry
Genetic Beyond Mendel

Autism Spectrum Disorders (ASDs) are complex developmental disorders in the neurological system of the body. Difficulties in social interaction, communication, and habitual behaviors are used to characterize ASDs. Approximately 1 in 36 adolescents are affected by Autism Spectrum Disorders in the United States, with a higher rate of diagnosis in males over females. Multiple environmental, genetic, and epigenetic factors can be used to describe the etiology of ASDs. The cerebellum is highlighted as harboring the most imperative role in the development of Autism Spectrum Disorders and the expression of neurodevelopmental mechanisms. Large neurons in the cerebellar cortex, known as Purkinje cells (PCs) are credited with playing a significant role in the pathophysiology of ASDs. Functions of this specialized cell include regulation of cerebellar signals to the rest of the body, thus controlling motor and non-motor coordination. PCs moderate cellular processes important to neural functions such as calcium signaling, synaptic transmission, and intracellular protein trafficking. In addition to this the release of GABA, is released by PCs, interconnecting motor and cognitive networks (Zhu *et al.*, 2018). The aim of this study is to analyze Purkinje cell dysfunction in Autism Spectrum Disorders through the exploration of discrepancies in cellular processes, external environmental causes, and genetic and molecular mechanisms.

The importance of Purkinje cells in relation to brain function can be derived from their structure, location and function. PCs are particularly large and interconnected to one another through the intricate branching of these neurons, serving as the output cells of the cerebellum. Input is received from parallel fibers and climbing fibers, sending output to the cerebellar nuclei. The vast extension and growth of dendrites, signal receiving parts of the neuron, allows Purkinje cells to integrate large masses of signaling inputs as a part of dendritic arborization processes (Tam *et al.*, 2011). These dendrites allow for up to 200,000 synapses per cell from parallel fibers, essentially processing signals from the human body system through the cerebellum. These synaptic inputs are used in order to form specific motor movements. This input of synapses are sent to cerebellar nuclei, as stated before, by GABA, a neurotransmitter and relayed to other parts of the nervous system (Zhu *et al.*, 2018). This pathway in the brain connects the entire nervous system together. To add to this, cognitive processes such as timing and learning are inhabited by PCs in the cerebellum, making this part of the brain a central part in regulating complex functions. The Purkinje cells' ability to process large amounts of information involve the ability to control high-order brain functions such as emotional, learning, procedural, and sensory regulation and integration. Any dysfunction or loss of Purkinje cells can disrupt motor or cognitive abilities due to the preciseness of neural processes moderation, correlating this to a plethora of neurodegenerative disorders.

Human postmortem studies have displayed evidence of PC abnormalities, specifically a deficiency, in Autism Spectrum Disorders. In posterior and lateral regions, a reduction of

Purkinje cells can be observed in consistent neuropathological results. Often showing patterns in regional reduction, studies suggest that loss of PC is not random, but rather targeting disruption in the developmental sectors of the brain. As Purkinje cells are the main pathway of connectivity to the prefrontal cortex and limbic system, high-order brain functions, a decrease in PC leads to a reduced function of emotional regulation, motor skills, and social behavior, typically used to characterize Autism Spectrum Disorders. To support this further, functional imaging technology, picturing evidence that modified cerebellum patterns and presence of PC in individuals with Autism Spectrum Disorders influences reaction to tasks involving motor skills and various cognitive abilities. In the findings of imaging technology, genes associated with ASD expression in Purkinje cells include TSC1, TSC2, SHANK3, MECP2, FMR1, and PTEN. Mutations within these genes are germline, insinuating that Autism Spectrum Disorders can be inherited, in either autosomal dominant or recessive. Some mutations such as MECP2 are X-linked, ensuing gender specific phenotypic expression. Through analysis of these studies and technology, Purkinje cell degeneration shows patterns of behavioral traits most similar to traits described in ASD. Anomalies in the cerebellar cortex, in relation to the consistent or inconsistent presence of Purkinje cells, highlights the importance of understanding ASDs and their neurobiology.

Complex interconnectedness of genetic and environmental factors take hand in causing Purkinje cell dysfunction in ASDs, disrupting cerebellar development. Mutations have been found in several genes, such as the deletion of TSC1, developing cell death, autistic-like behavioral patterns, and abnormal firing patterns in neuron signaling, insinuating that mutations affecting Purkinje cells are directly linked to Autism Spectrum Disorders (Nishiyama et al., 2024). Synaptic development, regulation of cell growth, and intracellular signaling, processes critical for Purkinje cell integrity, are all affected by these mutations. Apart from genetics, environmental factors contribute to Purkinje cell dysfunction. Exposure to infections, hypoxic events, or teratogens such as a valproic acid during pregnancy has shown to decrease the amount of PC and damage cerebellar function. For example, VPA exposure to rodents resulted in PC loss and rodents exhibited abnormalities characterized by autistic behaviors. In addition to this, ASD etiology can be further studied through epigenetic regulation. MECP2, a gene that encodes proteins in PC moderate chromatin structure and DNA methylation, therefore controlling gene expression without DNA sequence alteration (Li et al., 2020). Failure to regulate epigenetic modifiers can disrupt critical neurodevelopmental phases in Purkinje cells, causing long lasting effects on cerebellar function and structure. Signaling to the prefrontal cortex and thalamus is affected due to the loss of or loss of function of Purkinje cells, disrupting a pathway of circuits influencing social behaviors, attention, and language. Summatively, genetic, environmental, and epigenetic dysfunction in Purkinje cells affect the cerebellum resulting in behaviors used to describe Autism Spectrum Disorders.

This study has reviewed the role of the Purkinje cells as a central point in the cerebellum. PC function plays a fundamental role in the development and expression of Autism Spectrum Disorders. This paper has analyzed function, structure, and evidence of dysfunction of PC in Autism Spectrum Disorders, as well as possible causes and its resulting behavior. The cerebral

cortex has been the main focus of study in Autism research for years. The evidence of changes in the Purkinje cell has become a central figure in new studies due to its key role in genetic, environmental, and epigenetic factors. Disruptions in processes such as synaptic development, gene expression, and calcium signaling in Purkinje cells lead to broader neural pathway communications. Exploring these abnormalities through the main focal point of PC has brought in a new perspective into scientific research of ASD. Through the research of pathway connection disruptions and mutative causes in PC, future solutions and more new study strategies may be unlocked. Purkinje cell research has proven promising, however overdue. Cerebellar abnormalities have been observed for years but research in Purkinje cells has started fairly recently, leaving many burning questions. Due to this, much remains unknown. For instance, scientists ask: Can we use brain imaging technology to detect Purkinje loss in early stages? What are Purkinje cells most vulnerable in Autism Spectrum Disorders? Future studies in this direction could open the door to effective treatments such as medications or neurostimulation therapies. As the ever-evolving understanding of Autism Spectrum Disorders continue, regions of study in the brain must evolve with it. Recognizing the Purkinje cells' key role in ASD will change the way we diagnose, study and treat this disorder.

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Which Machine Learning Classifier Model Most Accurately Detects Exoplanets and How Does Fast Fourier Transform Enhance This Process? By Carolina Camara

Abstract

Exoplanets are planets outside of our solar system. The discovery and classification of exoplanets have become central to modern astrophysics, particularly in the search for habitable worlds beyond our solar system. This study investigates the application of five machine learning (ML) classifier models to enhance the detection of exoplanet candidates using NASA's Kepler data. In this study, a range of architectures including classical models and deep learning approaches are implemented and evaluated for accuracy and reliability. Through confusion matrices and comparative analysis, the most accurate classifier is identified. Within this context, the study also examines how Fast Fourier Transform (FFT) extracts frequency-domain features from light curves, improving model sensitivity to exoplanet transit signals.

Introduction

An exoplanet – also called an extrasolar planet - is a planet outside our solar system orbiting a star other than the Sun. Exoplanets exist outside our solar system in the Milky Way Galaxy and based on statistical analyses from NASA missions it is now widely accepted that most stars in our galaxy host at least one exoplanet [1]. To date, researchers have discovered more than 6,000 exoplanets outside of our solar system [2] with the help of NASA's space missions such as Kepler (NASA's first space telescope launched in 2009), K2 (Kepler's second mission), and TESS (NASA's Transiting Exoplanet Survey Satellite, launched in 2018 as the successor to Kepler) [3]. In recent years, rapid advances have been made in the detection and classification of exoplanets with machine learning (ML) classifier models.

Detecting exoplanets plays a critical role in the search for habitable worlds and expanding our knowledge of their properties - such as composition and formation - has the potential to significantly advance the field of space exploration. Machine learning can help speed up this process. ML is a subset of artificial intelligence and has significantly accelerated candidate identification by automating the analysis of large astronomical datasets. Each classifier model presents its own challenges - such as sensitivity to noise, class imbalance, overfitting, interpretability – and this research implements a range of complex machine learning architectures to evaluate their comparative performance.

This research aims to identify which machine learning classifier model demonstrates the highest accuracy in detecting true exoplanet signals. Furthermore, this study explores how Fast Fourier Transform (FFT) is used in the analysis of light curves to enhance machine learning classifier models.

Detecting Exoplanets

Exoplanets are detected by using various methods: transit method, radial velocity method, gravitational microlensing, direct imaging and astrometry. In our research we look at the

transit method. An exoplanet transit is when a planet outside our solar system passes in front of its star from our point of view on Earth. When this happens, the exoplanet blocks a tiny fraction of the star's light. For a brief period of time, that star's light gets dimmer, making a transit light curve - a dip in light - when viewed on a graph which can be seen in Fig. 1 [4]. Transit Photometry is the actual measurement of the star's brightness over time, typically using light curves. Fig. 2 shows another example of a light curve, that of HAT-P-32b, a hot Jupiter discovered by the HATNet survey [5]. NASA's Kepler and TESS generate enormous amounts of light curve data and machine learning models help sift through this data to spot the subtle dips caused by exoplanets transiting their stars.

Because space telescopes such as Kepler and TESS can detect the small, periodic dip in the star's brightness in the time domain, we see a series of dips spaced by the orbital period, like inverted bell curves. By measuring these dips, astronomers can make inferences about properties of the exoplanet such as the planet's size (the deeper the dip, the larger the planet), the orbital period (how often transits repeat tells how long the planet takes to orbit its star), the orbital distance (how far the planet is from the star), and in certain cases even the exoplanet's atmosphere. ML classifier models can distinguish between true exoplanet signals and false positives such as binary stars, stellar activity and instrumental noise. This study discusses the accuracy and reliability of the five most commonly used ML classifier models and the use of Fast Fourier Transform in these models.

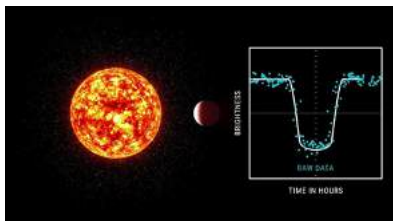


Figure 1. Example of an exoplanet transit. [4]

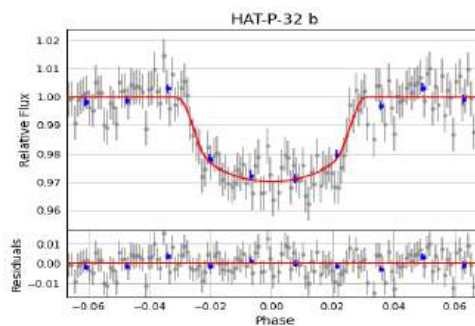


Figure 2. Light curve data for HAT-P-32 b. [5]

Machine Learning Classifier Models

Fig. 3 illustrates the five machine learning classifier models used in this study. All five models were trained and tested to determine which one resulted in the highest accuracy rate when detecting exoplanets.

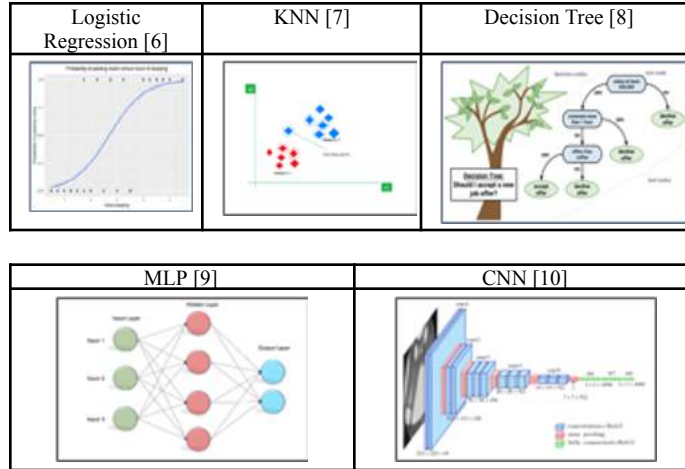


Figure 3. Visualization of classifier models implemented in this study.

The workings of each model are summarized below.

Logistic Regression model - predicts a binary (0 or 1) based on a probability of 0-1 from the sigmoid activation function.

K-Nearest Neighbors (KNN) model - classifies a data point based on the similarities. It's best for smaller datasets as it can be slow with large data sets.

Decision Tree model - categorizes data used in decisions to split the data into branches.

Multi-Layer Perceptron (MLP) model - a type of neural network that is made up of layers of connected neurons.

Convolutional Neural Network (CNN) model - a special type of deep learning neural network that is designed to handle image or grid-like data and works by learning patterns and features from the images.

Methods

Data Collection

The Kepler data used in this research study was sourced from Kaggle. In the Kaggle dataset, NASA's raw data is reshaped into a big matrix of flux values for each star or candidate. The Kaggle dataset consists of 5,087 rows (individual stars/observations) and 3,198 columns where column 1 is the label (exoplanet vs non-exoplanet). Columns 2 through 3,198 (3,197 flux measurements) are the flux/timeseries features (i.e. brightness measurements over time). According to this dataset 37 are labelled as exoplanets / positive class and 5,050 are labelled as non-exoplanets / negative class [11]. Flux is defined as how bright a star appears at one point in time as it is perceived by another object [12]. In this case, flux means how bright a star appears from the perspective of a satellite analyzing it. When the flux decreases dramatically, this is a strong suggestion that the star is harboring an exoplanet, as the drop and apparent brightness can be explained by an exoplanet crossing in front of it.

Data Preprocessing

Further preprocessing of Kepler data from the Kaggle dataset is crucial for machine learning classifier models. The preprocessing pipeline in this study were as follows.

1. Exploratory Data Analysis (EDA). Visualized distributions of key features like orbital period, stellar mass and flux. Checked Correlations to identify redundancy and detected outliers using boxplots or z-score methods.
2. Handle Missing Values. Dropped columns with excessive missing data and imputed missing values using mean/median for numerical features and mode (or a placeholder) for categorical features.
3. Feature Engineering. Created new features from existing ones using ratios and log transformations and encoded categorical variables.
4. Feature Scaling. Standardized or normalized features to ensure consistent input ranges to improve model performance.
5. Class Balancing. Applied Fourier-based data augmentation and SMOTE to address class imbalances and enhance model generalization.
6. Train-Test Split: Categorized the dataset into 80% training and 20% testing subsets.

Python Tools and Libraries

Python's modular design and extensive ecosystem make it well-suited for scientific computing in each stage of the full machine learning pipeline. In order to implement and evaluate the machine learning classifiers, this study utilized the suite of Python tools and libraries as displayed in Table 1 below, which support efficient data analysis and model development.

TABLE 1. PYTHON CLASSIFICATION WORKFLOW

Purpose	Library
Data Manipulation	pandas, numpy
Preprocessing	scikit-learn
Classical ML models	scikit-learn
Deep learning models	tensorflow
Visualization	matplotlib, seaborn

Data Augmentation

Since the dataset only contains 37 confirmed exoplanets, the classes are extremely imbalanced making reliable exoplanet detection a challenging task. This results in ML modelling becoming difficult unless class balancing techniques such as data augmentation are applied. Because ML models tend to bias toward the majority class (i.e. non-exoplanets), they predict the majority class as an answer most of the time because they don't see enough variation in what an exoplanet transit looks like. Data augmentation artificially increases the diversity of the positive class so the models generalize better. This study used Fourier-based data augmentation and

SMOTE (Synthetic Minority Oversampling Technique) to make the classifier models less imbalanced. By utilising Fourier-based augmentation techniques, the study generated many variations that still looked like real transits. This reduced overfitting and also enriched the negative class (false positives) with more realistic noise patterns. SMOTE was used to generate synthetic samples for the minority class by interpolating between existing minority samples and their nearest neighbours to better define classification regions [13].

Results and Discussion

Data Analysis

The confusion matrix for each classifier model summarizes its classification performance on the training and testing data, showing the number of true positives, true negatives, false positives, and false negatives. In order to compare the different models, the study evaluated the confusion matrices for each model. Based on each classifier model's initial performance the study concluded that due to the large amount of non-exoplanet data compared to exoplanet data the models did not perform very well. In order to improve model performance, the study used data augmentation which aided in balancing the data. After re-running the code post-data augmentation there was a large improvement in each classifier's accuracy. Fig. 4 presents the post-data augmentation confusion matrices for training and testing data for each classifier. The final accuracy results are shown below in Table 2.

TABLE 2. COMPARATIVE DATA ANALYSIS

Model	Post-Data Augmentation Accuracy
Logistic Regression	99.5%
KNN	92.8%
Decision Tree	99.0%
MLP	99.1%
CNN	99.86% (testing) 100% (training)

For training data, the logistic regression model post-data augmentation showed 3519 true positives, 3548 true negatives, 0 false negatives and 3 false positives. The decision tree model post-data augmentation had no false positives or negatives, 3493 true positives and 3577 true negatives. The KNN model post-data augmentation showed 3493 true positives, 3216 true negatives, 0 false negatives, and 361 false positives. MLP post data-augmentation showed 3511 true positives, 3506 true negatives, 53 false positives and 0 false negatives. Finally, CNN post data-augmentation showed 3511 true positives, 3559 true negatives, 0 false negatives and 0 false

positives with a total accuracy of 100%. For testing data, the logistic regression model post-data augmentation showed 1535 true positives, 2049 true negatives, 1 false negative, and 15 false positives. The decision tree model post-data augmentation showed 1555 true positives, 2009 true negatives, 7 false negatives, and 29 false positives. The KNN model post-data augmentation showed 1562 true positives, 1778 true negatives, 0 false negatives, and 260 false positives. MLP post data-augmentation showed 1542 true positives, 2024 true negatives, 2 false negatives and 32 false positives. Finally, CNN post data-augmentation showed 1543 true positives, 2052 true negatives, 1 false negative and 4 false positives with an accuracy rate of 99.86%. The study concluded that the best model with the highest accuracy rates was CNN as it had the highest accuracy rates both in training and testing data, as illustrated in Fig. 4.

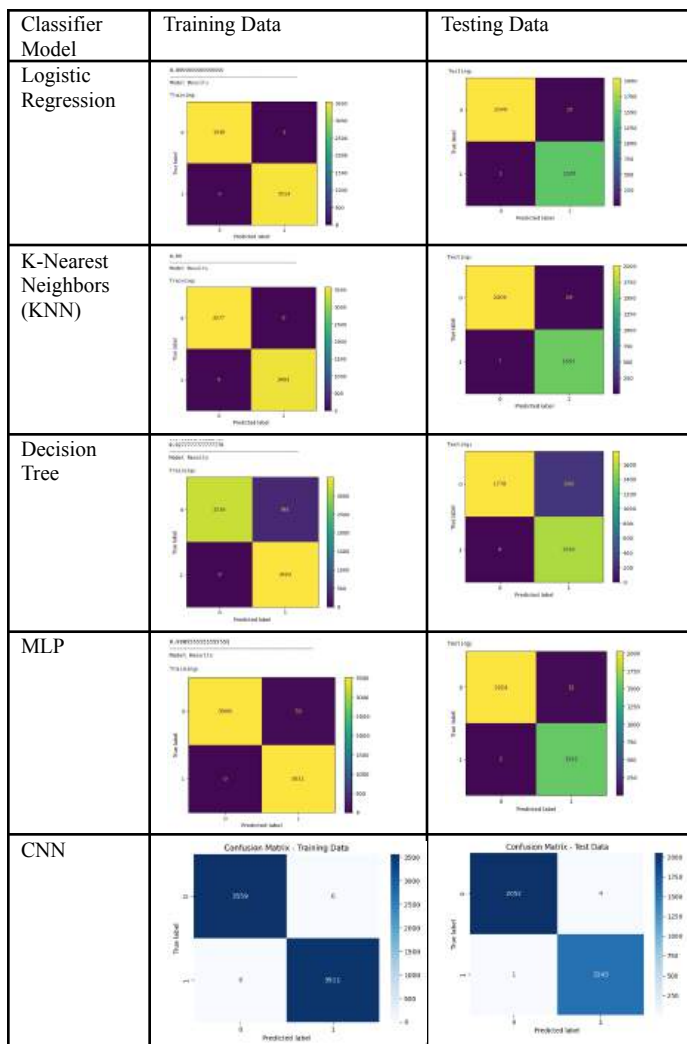


Figure 4. Confusion matrices post-data augmentation for Logistic Regression, KNN, Decision Tree, MLP and CNN models for training and testing datasets.

Extended Evaluation of MLP vs CNN compared with Classical Models

This study implemented both deep learning and classical architectures to evaluate each model's performance in detecting exoplanets. MLP and CNN are both deep learning neural network models, whereas Logistic Regression, KNN and Decision Tree models are classical models. MLP and CNN are better suited for capturing transit shapes and subtle flux variations in light curve data. Classical models still perform well, especially with strong feature engineering, but may miss deeper temporal and nonlinear patterns. They can be faster and easier to interpret but may struggle with complex patterns. In this study, the MLP, supported by scikit-learn and imported as `MLPClassifier`, was trained using SMOTE-augmented data (`smote_X_train`, `smote_y_train`) and evaluated via accuracy metrics and confusion matrices. To explore more flexible architectures, a Sequential model was built using tensorflow, consisting of a hidden Dense layer with 10 units and ReLU activation, followed by an output Dense layer with a sigmoid function. While MLPs treat input features independently, CNNs are designed to detect patterns regardless of their position in the input. This is particularly useful for light curve data, where transit signals may occur at varying time offsets. To accommodate CNN input requirements, the augmented data was reshaped into a 1D format suitable for convolutional layers. A Sequential CNN model was then constructed to learn spatially invariant features, improving robustness to shifts in flux measurements.

This comparison highlights a key trade-off between simplicity and spatial awareness: MLPs offer straightforward implementation and fast training, while CNNs provide greater flexibility in capturing localized patterns critical to exoplanet detection.

Application of Fast Fourier Transform (FFT)

The Fast Fourier Transform (FFT) is a mathematical transform widely used to analyze time-series data, such as stellar light curves [14], [15]. By converting brightness measurements from the time domain into the frequency domain, FFT enables researchers to identify periodic signals that may indicate the presence of an exoplanet because random noise is spread across frequencies, making periodic components easier to detect. These periodic dips in brightness, caused by a planet transiting its host star, manifest as distinct peaks at specific frequencies within the transformed spectrum. Beyond detecting transit signals, FFT is instrumental in separating planetary signatures from stellar variability and instrumental noise. This filtering capability enhances the clarity of the data and improves the reliability of subsequent classification models. As such, FFT serves as a foundational tool in the preprocessing pipeline for exoplanet detection, allowing machine learning algorithms to focus on the most informative features of the signal. It is important to note that while FFT is not itself a data augmentation method, it can be used within an augmentation strategy.

For a planet with an orbital period P the main frequency is written as $f = 1/P$. You also see harmonics at multiples of that frequency ($2f$, $3f$, $4f$...). Harmonics appear because the transit shape is not a perfect sine wave but instead more like a flat line with sharp dips. Sharp, non-sinusoidal signals naturally produce harmonics in Fourier space as these harmonics reinforce the periodicity and make the planet's signal stand out from random noise. In summary,

in the frequency domain the repeating dips from a planet's orbit turn into a series of regularly spaced peaks (harmonics), which highlight the planet's presence and make faint signals easier to detect.

How Complex Numbers Are Used in FFT

Complex numbers are extended from the real number line into the two-dimensional domain to handle the square root of a negative number, which in the real domain does not exist. A complex number is expressed as:

$$a + bi$$

where a is the real part, b is the imaginary part, and i is the imaginary unit $\sqrt{-1}$. Fast Fourier Transform expresses a time-domain signal, like a light curve, as a sum of sinusoidal components, each with a specific frequency, amplitude, and phase. These sinusoidal components are represented using complex numbers. FFT analyzes the light curve to find repeating patterns and identifies frequency peaks that suggest a consistent orbital period. These peaks can be matched to known transit durations as well as used to flag new exoplanet candidates. Each frequency component is represented as a complex number ($a + bi$) which allows precise modelling of both amplitude and phase. The FFT algorithm uses Euler's formula to efficiently compute the components:

$$e^{ix} = \cos(x) + i\sin(x)$$

Future Work

The development of ultra-stable optical systems and high-contrast imaging should help accelerate the discovery and characterization of exoplanets. Future studies could explore the integration of time-series feature extraction techniques such as wavelet transforms and more advanced Fourier-based methods into deep learning pipelines. This could potentially enhance the sensitivity of classifiers to subtle transit signals embedded in noisy data. Further research could also investigate the use of transfer learning in order for models trained on Kepler data to be applied to newer missions like TESS or PLATO as this would enable cross-mission generalization.

Novel Contribution of This Study

Machine learning represents a recent advancement in computational science, and this study contributes by demonstrating how Python-based tools and libraries facilitate the implementation and analysis of diverse classifier model architectures. By integrating complex-number operations such as Fast Fourier Transform (FFT), this study underscores the importance of an interdisciplinary approach and highlights the convergence of physics, mathematics, and astrophysics in the pursuit of exoplanet detection.

Conclusion

This study evaluated the performance of five machine learning classifier models in detecting exoplanets using Kepler mission data - Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Multi-Layer Perceptron (MLP), and Convolutional Neural Network (CNN). Due to the extreme class imbalance in the dataset, data augmentation techniques were applied to improve model sensitivity. Among the models tested, the CNN model achieved the highest accuracy, with a training accuracy of 100% and a testing accuracy of 99.86%. Unlike the other four classifiers, the CNN model was able to analyze the full dataset holistically and able to identify complex patterns across multiple features. Its ability to extract spatial and temporal relationships enabled superior classification of exoplanet candidates. It furthermore demonstrated the potential of deep learning in light curve analysis. This study specifically addressed the challenges posed by extreme class imbalance and noisy observational signals. While traditional preprocessing and resampling techniques such as SMOTE are essential for balancing the dataset, feature extraction remains equally critical for improving model performance. The Fast Fourier Transform (FFT) offers a powerful method using complex numbers for transforming time-domain light curve data into the frequency domain. In doing so, it reveals periodic patterns indicative of planetary transits. By isolating dominant frequencies and reducing noise, FFT enables classifiers to focus on the rhythmic signatures of exoplanetary motion rather than stochastic variability. Although FFT does not directly balance the dataset, it enhances the signal-to-noise ratio and improves the sensitivity of models to rare transit events. Incorporating FFT into the preprocessing pipeline thus represents a meaningful step toward more accurate and robust exoplanet detection, especially when combined with data augmentation and advanced classification techniques. Future research should explore hybrid models that integrate frequency-domain features with deep learning architectures to further refine detection capabilities.

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Targeting Immune Evasion in Hodgkin Lymphoma: A Review of Modern Immunotherapy

By Bhavin Patlola

Introduction to Immune Evasion in Cancer

Cancer is a type of disease in which cells divide and multiply uncontrollably. Many types of cancer also avoid immune system detection through various immune evasion tactics. The immune system is one of the main body systems that is targeted in cancer. It is the system that is responsible for fighting viruses, bacteria, and diseases. Under normal circumstances, macrophages and dendritic cells detect the foreign invader in the body. After which, the antigen is presented which activates T-cells, which are part of the adaptive immune response. These active T-cells become cytotoxic and begin to kill the cancerous cells. Cancer usually avoids the immune system by overexpressing PD-L1, which blocks T-cells activation by binding to PD-1 receptors, reducing MHC I and II expression, which hides it from the cytotoxic T-cells, creating an environment where regulatory T-cells and suppressor cells weaken the immune system, and by releasing cytokines, which suppress immune activation (“Immunotherapy in Advanced Hodgkin Lymphoma”). Immunotherapy is a cancer treatment in which the patient’s immune system is modulated, which helps it recognize cancer cells and kill them (American Cancer Society). At first glance, this treatment is attractive since it uses the patient’s immune system to target and destroy cancer cells. However there are several restrictions for this type of treatment.⁸² This review examines how Hodgkin’s evades the immune system and how modern immunotherapies target these mechanisms, and these limitations.

Overview of Hodgkin’s Lymphoma

Hodgkin’s Lymphoma, named after Dr. Thomas Hodgkin, who discovered the disease in 1832, is a subtype of Lymphoma, which is a type of blood cancer, that affects the white blood cells in the lymphatic system and causes abnormally large cells to attract normal cells to the lymph nodes, causing discomfort and swelling. The primary cause of this type of cancer is mutations, which are errors that happen during the DNA replication process, that cause immune dysfunction and resistance to cell death. The mutations take place on the B cells, a type of lymphocyte, that develop throughout a patient’s lifetime. These mutations cause abnormally large cells called Reed-Sternberg cells to form. These larger-than-normal cells release substances such as cytokines and chemokines that draw normal immune cells into the lymph nodes, causing swollen lymph nodes. Over time, these cancerous cells spread throughout the lymphatic system to organs such as the thymus and spleen. The disease is most common in males between the ages of 15 and 30 and those males that are 55 and older, though a rare subtype called nodular sclerosis is common in females. The disease is common in less-developed countries such as African countries and various countries throughout Asia, due to the lack of sufficient healthcare and the failure to diagnose the cancer in its earliest stages. Despite this, Hodgkin’s is associated with a high remission rate. Hodgkin’s is known to be highly curable if it is diagnosed in its earliest

⁸² “Immunotherapy in Advanced Hodgkin Lymphoma,” *Cancer Today*; “Immunotherapy for Hodgkin Lymphoma,” *American Cancer Society*.

stages and if treatment is started soon after diagnosis. Survival rates were seen to be very high if the latter is true and if the patient is young. However, patients who do not start therapy within 5 years of the cancer showing up can die, meaning that Hodgkin's affects normal body functions like all other cancers but is more slow-acting than most. Hodgkin's also has many different possible therapies such as chemotherapy, immunotherapy, radiation therapy, and targeted therapy(American Cancer Society). Immunotherapy is a type of therapy in which the immune system is modulated to help it fight the cancer on its own and is used as an alternative if the patient does not respond to initial treatment, usually chemotherapy or radiation therapy. Drugs such as nivolumab and pembrolizumab assist the body's immune system in recognizing cancerous cells("Immunotherapy for Hodgkin's Lymphoma"). Immunotherapy is also one of the safer ways to treat cancer as it has fewer side effects because it uses the body's own immune system, making it a common treatment for younger patients. It is used as a first-line treatment in tandem with chemotherapy in the later stages of Hodgkin's. This combination has been proven to be highly effective, with many patients achieving remission after two years of this treatment("Immunotherapy in Advanced Hodgkin Lymphoma").⁸³

Mechanisms of Immune Evasion in Hodgkin Lymphoma

Hodgkin Lymphoma has many nuanced ways of evading one's immune system such as overexpressing the protein PD-L1, manipulating the tumor microenvironment, lessening the presentation of antigens, and the secretion of cytokines. Cancerous Hodgkin Lymphoma(HL) cells often overexpress a protein called PD-L1, which binds to PD-1 receptors on T-cells, causing a failure to activate T-cells, which allows cancerous cells to evade immune detection. HL cells also create an immunosuppressive microenvironment, that is filled with regulatory T-cells and myeloid-derived suppressor cells, which further suppress the immune response. Reed-Sternberg cells, which are the abnormal cells in Hodgkin's, often lose expression of MHC class I and II molecules, which affects their ability to present antigens and be recognized by cytotoxic T-cells. HL cells also secrete cytokines like IL-10 and TGF- β that suppress immune cell activation and encourage immune tolerance(Frontiers Editorial Office).

Modern Immunotherapy Approaches for Hodgkin Lymphoma

Common immunotherapies used to treat Hodgkin's are checkpoint inhibitors and monoclonal antibodies. Checkpoint inhibitors such as, Nivolumab and Pembrolizumab block PD-1 receptors on T-cells, which prevent PD-L1 from binding to the cells, causing the T-cells to reactivate. This type of therapy has shown high response rates with a significant portion of Hodgkin patients achieving remission("The immunotherapy real estate of Hodgkin lymphoma"). The immunotherapy of monoclonal antibodies utilizes Brentuximab vedotin, which targets CD30 on HL cells, which, after binding, deliver a cytotoxic agent, called MMAE, that disrupts microtubules and induces apoptosis(Present Status and Future Strategies). When this

⁸³ "Immunotherapy for Hodgkin's lymphoma," *Blood*, American Society of Hematology; *Frontiers*, Editorial: Innovative immunotherapy strategies for enhanced treatment of Hodgkin and non-Hodgkin lymphomas.

immunotherapy is paired with chemotherapy, it has been shown to produce high response rates and remission in most patients.

Limitations and Challenges of Current Immunotherapies

These treatments are very effective when started at the correct time, with very high response and remission rates. However there are several limitations to these therapies some of which include, patients not responding properly, immune-related adverse events developing, and the high costs and limited access to these treatments in some regions. A patient's response rate depends on the stage the cancer is in and how developed it is. For example a patient with cancer in very early stages might respond very well because the cancer is not developed enough to combat the drugs. However a patient with cancer in the later stages might not respond well because the cancer is well-equipped to resist and evade the drugs. Immune-related adverse events might also occur in some patients because every patient has a unique response to the drugs, though they are rare when they occur they can be life-threatening as some therapies cause cytokine release syndrome, which can cause a fatal immune reaction. These therapies are also very limited because of their advanced nature and highly personalized treatment, making it very expensive. Under-developed countries may not have access to these treatments which limits it to a small number of countries and some treatments can cost as much as \$104,244(“Cost-Effectiveness of Immune Checkpoint Inhibitors”). While expensive these treatments have been proven to be effective as the FDA has approved multiple immunotherapies for HL such as Nivolumab, which is used as part of first-line treatment and also has a high response rate, Pembrolizumab, which is used for refractory or relapsed HL, and Brentuximab Vedotin, which is used as part of initial treatment in combination with chemotherapy or alone; it is mainly a maintenance therapy(American Cancer Society).⁸⁴

Current Clinical Trials and Emerging Therapies

In the ongoing battle against Hodgkin lymphoma, innovative immunotherapy trials are exploring new ways to harness the immune system for more effective treatment. Some examples of current ongoing clinical trials regarding Hodgkin Lymphoma in 2025 are Prologolimab Monotherapy or in Combination With Bendamustine for r/r Classical Hodgkin Lymphoma (Prologo-HL) and BrECADD Therapy in Stage 2B-IV Hodgkin Lymphoma (BrECADD). Prologo-HL is an ongoing clinical trial that as of August 11 2025 is in the recruiting phase. The main purpose of this trial is to evaluate the safety and efficacy of Prologolimab, which is a PD-1 inhibitor, in patients with relapsed or refractory classical HL. It is currently in Phase II where the researchers are testing how effective Prologolimab is for treating HL and determining the side effects of this treatment. The inclusion criteria for this trial goes as follows: patients must be 18 or older, they must have a confirmed diagnosis of classical HL and it must be measurable per imaging standards, they must show relapse or refractory disease after prior treatments, they must

⁸⁴ “Immunotherapy in Hodgkin Lymphoma: Present Status and Future Strategies”; “Cost-effectiveness of immune checkpoint inhibitors in NSCLC according to PD-L1 expression,” *PMC*.

be in acceptable physical condition, and they must provide informed consent. The exclusion criteria goes as follows: the participant has active autoimmune diseases that require treatment, they are currently receiving other investigational drugs, they have uncontrolled infections or serious medical conditions, they are pregnant or breastfeeding, or they have received Prologolimab in the past. This therapy targets T lymphocytes, which play a key role in recognizing and attacking cancer cells. In HL, cancerous cells often evade immune detection by overexpressing PD-L1, a protein that binds to the PD-1 receptor on T cells. This interaction suppresses T cell activity, allowing uncontrolled cancer growth. Prologolimab blocks the suppressive signal on T cells so that they can attack the cancer cells. This trial represents a shift towards more effective treatment options for classical HL than past trials(ClinicalTrials.gov NCT05757466). BrECADD is an ongoing clinical trial that as of August 11 2025 is in the recruiting phase of Phase II. The purpose of this trial is to evaluate the effectiveness of the BrECADD regimen in patients with advanced-stage HL and to explore the use of circulating tumor DNA (ctDNA) as a biomarker for minimal residual disease (MRD) and treatment response. The inclusion criteria for this trial goes as follows: patients must be diagnosed with Stage 2B to IV HL, they must be eligible to receive chemotherapy, they must be able to provide informed consent, and they must be willing to comply with study procedures. The exclusion criteria for this trial goes as follows: participants must not have a presence of significant comorbidities that could interfere with treatment, must not have uncontrolled infections or other serious medical conditions, they must not be pregnant or breastfeeding, and they must not have any prior exposure to similar chemotherapy regimens in the past. This therapy targets Reed - Sternberg cells, which are the malignant cells found in classical HL. These cells evade immune detection by overexpressing CD30, creating an immunosuppressive microenvironment, and by releasing cytokines and chemokines. This therapy uses Brentuximab Vedotin, which binds to CD30 on HRS cells and delivers a cytotoxic agent directly to the cell, which bypasses the immune evasion and directly kills the HRS cells. This trial addresses earlier limitations by reducing toxicity, targeting only Reed-Sternberg cells, circulating tumor DNA, and by using patient-centered toxicity monitoring(ClinicalTrials.gov NCT07002216).⁸⁵

Conclusion

In conclusion, Hodgkin's Lymphoma is a type of blood cancer, which affects B-lymphocytes and causes mutations on these cells, which cause abnormally large cells called Reed-Sternberg cells to form. In general Hodgkin's is a relatively easy disease to slow down with remission rates from immunotherapy being very high. This is so because the Reed-Sternberg cells have distinct features that help in diagnosis. The main immunotherapies of HL are PD-1 inhibitors and Monoclonal Antibodies. In general, Monoclonal Antibodies have shown strong effectiveness, especially when paired with chemotherapies, as when combined with chemotherapy remission rates are significantly higher than the remission rates of PD-1

⁸⁵ Study Details | NCT05757466 | Prologolimab Monotherapy or in Combination With Bendamustine for r/r Classical Hodgkin Lymphoma | ClinicalTrials.gov; Study Details | NCT07002216 | BrECADD Therapy in Stage 2 B-IV Hodgkin Lymphoma | ClinicalTrials.gov.

inhibitors(Present Status and Future Strategies). Main obstacles people are trying to overcome are immune-related adverse events after the treatment. These events can be minor or severe varying by patient, with some potentially being fatal. Despite this, immunotherapy has improved significantly over the years, making it one of the most effective treatments for cancer.⁸⁶

⁸⁶ Ibid

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Immunotherapy in Advanced Hodgkin Lymphoma | Cancer Today
Immunotherapy for Hodgkin Lymphoma | Hodgkin Antibody Treatment | American Cancer Society
Immunotherapy for Hodgkin's lymphoma
The immunotherapy real estate of Hodgkin lymphoma | Blood | American Society of Hematology
Frontiers | Editorial: Innovative immunotherapy strategies for enhanced treatment of Hodgkin and non-Hodgkin lymphomas
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Cost-effectiveness of immune checkpoint inhibitors in NSCLC according to PD-L1 expression - PMC

The Effects of Indian Classical Music on Mood By Ira Shellikeri & Sarah L. Hennessy

Abstract

Previous research has demonstrated that music, specifically cultural music, can have therapeutic benefits for individuals, such as improved mood. However, there are relatively few studies examining the impact of Indian classical music on mood and how familiarity influences this effect. To address this gap in the literature, we investigated whether listening to Indian classical music affects a person's mood, and how this effect is influenced by familiarity with the genre. To begin, we created and distributed a survey that played either Indian classical music or white noise, with participants randomized to one of the two conditions, stratified by familiarity with Indian classical music. Then, we measured positive affect with the PANAS Questionnaire and conducted a two-way ANOVA on the positive mood scores. We found there was a main effect of condition, such that music improved positive mood scores more than white noise. There was no main effect of familiarity. That is, familiarity with the music had no effect on positive mood scores. By extension, there was no interaction effect, which means the positive effect of music was consistent across music and white noise. The main effect of the condition aligns with previous studies demonstrating that listening to music improves mood. However, familiarity did not enhance this effect, contrary to our hypotheses and previous literature. Additional studies need to assess the effect of familiarity in a larger, more representative sample. Overall, these findings suggest Indian classical music may hold promise as a therapeutic tool to improve mood, and should be further studied to understand who may benefit most.

The Effects of Indian Classical Music on Mood: The Role of Familiarity

Recent studies have highlighted the potential of music as a medical intervention. For example, in one study, music therapy reduced self-reported anxiety and improved depression scores in breast cancer patients (Xu et al. 1). In another, exposing individuals with schizophrenia to music appeared to reduce symptom severity (Tseng et al. 4). Scholars posit that these therapeutic benefits may stem from music's ability to act as an "auditory anchor", or a stable and predictable sensory baseline that modulates emotional processing in the limbic system, grounding affective responses and supporting neurological resilience to psychological stress (Rus et al.). Further, music appears to help patients redirect their focus from painful events to a more pleasant stimulus, and perceptions of music styles, sounds, and rhythms vary cross-culturally (Reynaud et al. 2). Together, these findings establish music's potential as a therapeutic tool.

However, the effectiveness of music as a therapeutic tool may depend on characteristics of both the listener and the music itself, such as the listener's familiarity and cultural resonance. Music helps patients redirect their focus from painful events to a more pleasant stimulus, and perceptions of music styles, sounds, and rhythms vary cross-culturally (Reynaud et al.). Therefore, familiarity with cultural music can further enhance positive mood change in a therapeutic setting (Hämäläinen et al.). Familiarity with music, rather than objective characteristics such as tempo or energy, can also drive relaxation responses and enhance positive

mood change in a therapeutic setting such that familiar energetic music may be perceived as more relaxing than unfamiliar calm music (Hämäläinen et al.; Scarratt et al.).

Nevertheless, there are many areas for improvement in using music therapy. This approach does not account for music enculturation, the process of understanding the structure of cultural music through repeated exposure. Current studies demonstrate music enculturation affects emotional responses to music, suggesting that the cultural resonance of the music being used may increase the efficacy of music therapy (Athanasopoulos et al.). Assuming the universal efficacy of music therapy in promoting positive emotional responses without considering important listener and music characteristics may obscure the broader picture. For example, Western music emphasizes the importance of major and minor keys, with major keys typically being used to signal happiness while minor keys are typically utilized to signal sadness or negative feelings. However, researchers have found that among some remote Asian and African tribes, individuals were not able to distinguish between major and minor compositions (Egermann et al.). Therefore, Asian and African people may not have the same emotional experience with Western music as Western listeners. It stands to reason that music enculturation, a learned ability, differentially affects listening experiences and benefits during music therapy. To effectively maximize the benefits of music therapy, further research is needed to explore the link between familiarity and mood in music therapy when using cultural music.

One type of music that has not been well studied for its potential therapeutic benefits is Indian classical music. Indian classical music originated in Southeast Asia almost 6,000 years ago. While divided into two types of music (Carnatic and Hindustani), both are influenced by cyclical patterns of nature, like times of day and seasons. Musicians in this art form express themselves through singing or various instruments, such as the sitar, sarod, and harmonium. Both the vocal and instrumental traditions are grounded in improvisation, making each performance creative and unique (“An Introduction To”). Previous literature on Indian classical music has established that listening has health-related benefits. For example, listening to Indian classical music helped reduce anxiety, stress, and depression among adults (Pareek and Shekhawat). Furthermore, healthy elderly men who listened to Raag Todi experienced a significant decrease in systolic pulse rate, breathing, and respiratory rate (Chatterjee and Mukherjee). However, most literature on Indian classical music does not explore the effect of familiarity with the music on positive mood change in therapeutic settings.

To address this gap in the literature, this study aimed to investigate how familiarity and exposure to Indian classical music affect mood. By assessing the difference in mood between our control group (white noise) and our experimental group (Indian classical music), we sought to establish a causal relationship between listening to Indian classical music and an increased positive mood. We hypothesized that the group listening to Indian classical music would have higher positive mood scores after exposure than the white noise group. Additionally, we hypothesized that greater familiarity with Indian classical music would lead to a stronger positive effect on mood.

Methods

Research Participants

To satisfy our inclusion criteria, participants were required to live in the United States and be between the ages of 18-50 years old. Upon selection, participants were sorted into two groups: familiar and unfamiliar with Indian classical music. Familiarity was defined as responding to the question “*How familiar are you with Indian classical music?*” with “unfamiliar” (never heard of Indian classical music before), or “maybe familiar” (heard of Indian classical music before but never listened). Consequently, unfamiliar participants were defined as responding to the question with “unfamiliar” (never heard of Indian classical music before), “maybe familiar” (heard of Indian classical music before but never listened), or slightly familiar (listened once or twice). Participants were recruited primarily from online Facebook communities centralized in the United States. Forty four people (23 males and 21 females) were recruited, and the average age was 40.36 years old ($SD = 7.17$).

Recruitment Procedures

We used Facebook to recruit participants because our target demographic uses it as a primary social media platform, and features such as Facebook Groups allowed us to target specific communities. Participants familiar with Indian classical music were recruited through Indian classical music forums, specifically the “Lovers of Hindustani Classical Music” forum. Participants unfamiliar with Indian classical music were recruited through general United States forums, specifically the “United States of America (USA)” forum, as American participants were assumed to have limited exposure to Indian classical music.

To recruit participants, we posted a flyer on each online forum advertising the opportunity. Participation was completely voluntary, and no compensation was provided. The flyer included a QR code and a link to the survey on Qualtrics. Upon entering the survey, participants completed a prescreening survey. If participants did not meet the inclusion criteria, we informed them of their ineligibility and thanked them for their time. If they were eligible, we informed them of the survey’s expectations and requested that they sign an informed consent form, reminding them that participation was voluntary and that withdrawing from the study was an option at any time. Finally, the participant took the main survey.

Study Design

Upon selection, participants were sorted into two groups: familiar and unfamiliar with Indian classical music. The question was “How familiar are you with Indian classical music?”, and answers were measured on a five point Likert scale, the scale items ranging from one (not at all familiar) to five (extremely familiar). Familiarity was defined as responding to the question “*How familiar are you with Indian classical music?*” with “unfamiliar” (never heard of Indian classical music before), or “maybe familiar” (heard of Indian classical music before but never listened). Consequently, unfamiliar participants were defined as responding to the question with

“unfamiliar” (never heard of Indian classical music before), “maybe familiar” (heard of Indian classical music before but never listened), or slightly familiar (listened once or twice).

This study was a 2x2 between-subjects design. The first independent variable was familiarity with Indian classical music. The question was “How familiar are you with Indian classical music?”, and answers were measured on a five point Likert scale, the scale items ranging from one (not at all familiar) to five (extremely familiar). The second independent variable was Indian classical music exposure, which was manipulated. Participants were either exposed to a 5-minute white noise clip or a five minute recording of Raag Bhairavi, which has been shown to have a calming effect in previous studies (Jain and Maheshwari). Finally, the dependent variable was positive mood. Participants rated their emotional state after listening to the piece of Indian classical music using the positive portion of the Positive and Negative Affect Schedule (Watson et al.).

Stimuli

The Indian classical music clip was of Raag Bhairavi, taken from a YouTube video titled “Raga Bhairavi | राग भैरवी | Pt. Hariprasad Chaurasia | Flute Music | Indian Classical Instrumental”, and the white noise clip is taken from a YouTube video titled “10 min white noise for stress” (Hämäläinen et al.; Scarratt et al.).

Procedures

Following recruitment, participants were asked about their age and gender. Both groups, participants familiar with Indian classical music (“familiar participants”) and unfamiliar with Indian classical music (“unfamiliar participants”), were randomly assigned to a condition, stratified such that there were an equal number of familiar and unfamiliar participants in each condition. There were two conditions: Indian classical music and White Noise (see **Stimuli**). More detailed participant demographics by music condition are outlined in Table 1. After reporting age and gender, participants listened to their assigned listening condition for five minutes. After listening, participants completed the Positive and Negative Affect Schedule (PANAS) questionnaire. Since this study measured positive affect, the questionnaire was shortened to ten items measuring only positive emotions on a Likert scale.

Analysis

The results of this survey were analyzed by creating a sum score for the positive affect scores for each participant after listening. We compared the means between conditions and familiarity groups using ANOVA. Then, we identified whether there were significant main effects of familiarity with Indian classical music, condition, and a significant interaction effect of familiarity by condition on positive affect. Statistical significance was determined by comparing the *p*-value to an alpha value of 0.05, which represents the threshold for significance (ability to reject the null hypothesis). The null hypotheses, that 1) Indian classical music has no positive effect on mood , 2) familiarity with Indian classical music does not have a positive effect on

mood, and 3) familiarity with Indian classical music does not affect the effect of condition on mood, are rejected if the p-value is less than the alpha value. Statistical analyses were performed using JASP.

Results

Participant Characteristics

We determined that the majority of our sample (68%) was familiar with Indian classical music. About twenty four people (55%) received Indian classical music, while twenty people (45%) received white noise, with the small discrepancy arising from stratified random assignment, which guaranteed the proportional representation of both familiar and unfamiliar subjects within both conditions. The table below contains more comprehensive information regarding participant distribution.

Table 1
Participant Demographics by Music Condition

Variable	Music	White Noise
<i>N</i>	24	20
Female	13	8
Male	11	12
Age (<i>M, SD</i>)	(40.08, 7.77)	(40.70, 6.55)
Age Range	18 - 50	30 - 50

Table 2
Participant Distribution by Familiarity and Condition

	Music	Noise	Total
Familiar	18 (41%)	12 (27%)	30 (68%)
Unfamiliar	6 (14%)	8 (18%)	14 (32%)
Total	24 (55%)	20 (45%)	44 (100%)

Effect of Condition and Familiarity on Positive Affect

Scores on the PANAS questionnaire ranged from 10 to 50. Among all the participants, the overall mean was 23.8 and the standard deviation was 10.8. The means and standard deviations for each condition are displayed in the table below.

Table 3
Descriptives Statistics for Each Condition - Positive Mood

Condition	Familiarity Binary	<i>N</i>	<i>M</i>	<i>SD</i>	<i>SE</i>	<i>CV</i>
Music	Familiar	18	30.33	9.267	2.184	0.306
	Unfamiliar	6	29.83	6.014	2.455	0.202
Noise	Familiar	12	16.00	8.213	2.371	0.513
	Unfamiliar	8	16.25	8.120	2.871	0.500

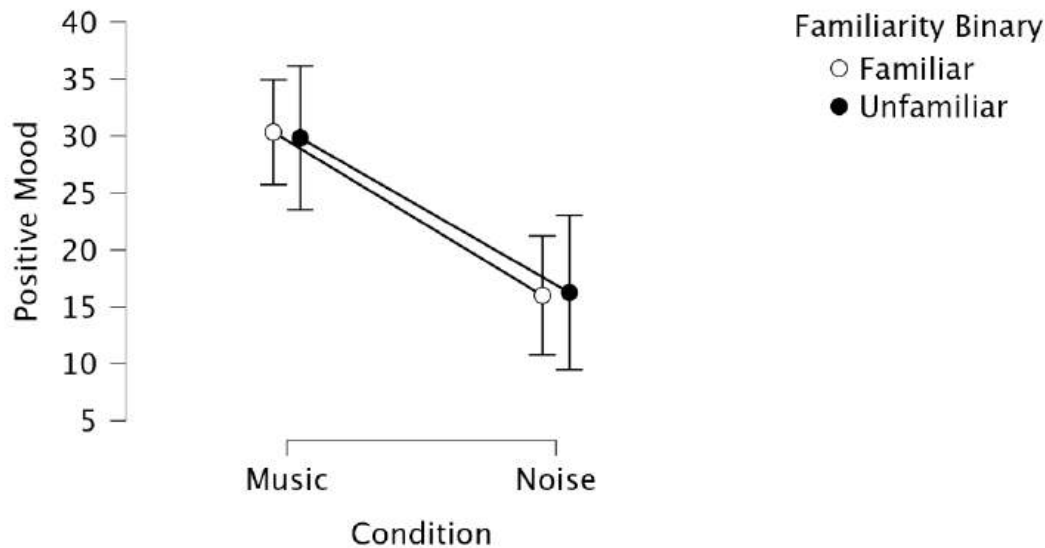
We conducted a two way analysis of variance on the positive mood scores to assess the main effects of each factor as well as the interaction effect. There were two levels of condition (white noise, Indian classical music) and two levels of familiarity (familiar, unfamiliar).

The main effect of condition was statistically significant ($F(1, 40) = 25.455, p < .001$), indicating that the mean positive mood score was significantly greater following Indian classical music ($M = 30.21, SD = 8.45$) than for white noise ($M = 16.10, SD = 7.96$). The main effect of familiarity was not statistically significant ($F(1, 40) = 0.002, p > 0.05$), indicating that the mean change score was not significantly different in the familiar group ($M = 24.60, SD = 11.27$) than in the unfamiliar group ($M = 22.07, SD = 9.90$). The interaction of condition and familiarity was non-significant ($F(1, 40) = 0.018, p > 0.05$).

Table 4
ANOVA - Positive Mood

Cases	Sum of Squares	<i>df</i>	Mean Square	<i>F</i>	<i>p</i>
Familiarity Binary	0.145	1	0.145	0.002	0.964
Condition	1810.081	1	1810.081	25.455	<0.001
Familiarity Binary Condition	1.306	1	1.306	0.018	0.893
Residuals	2844.333	40	71.108		

Figure 1
Positive mood scores by Condition and Familiarity



Discussion

In this study, we conducted a two-way analysis of variance to test the hypotheses that Indian classical music has a positive effect on mood, familiarity with Indian classical music has a positive effect on mood, and familiarity with Indian classical music increases the positive effect of condition on mood. We found that participants who were exposed to 5 minutes of Indian Classical Music reported an improvement in mood, and that familiarity with the music did not increase mood or enhance the effect of the condition on mood.

Regarding conditions, we found that listening to Indian Classical Music significantly enhanced mood compared to white noise, supporting our hypothesis that exposure to Indian classical music improves positive affect scores. This is in line with previous studies. For example, listening to Indian classical music helps reduce anxiety, stress, and depression among adults (Pareek and Shekhawat) and improves physiological markers of stress (Chatterjee and Mukherjee).

These findings are important because they indicate that Indian classical music can be used in music therapy by anyone, regardless of their familiarity with the music. This adds to music selection beyond typical Western music choices, which may be useful for patients who have not responded to other treatments in music therapy. This research builds on previous research suggesting that Indian classical music has a positive effect on mood.

Regarding familiarity, we discovered no significant main effect of familiarity, which means that familiarity with Indian classical music did not influence positive mood scores regardless of condition. This does not support our hypothesis that familiarity with Indian classical music increases positive mood scores. This does not align with previous studies, which

have found that familiarity with cultural music, such as Yoik music, can enhance positive mood in therapeutic settings (Hämäläinen et al.). There are several possible explanations for this finding. First, it is possible that familiarity does not have a positive effect on mood in Indian Classical Music. Second, this result may be explained by the Hawthorne Effect, which is when individuals modify their behavior because they feel they are being observed (Landsberger) or due to demand characteristics, wherein participants receive cues that reveal the experiment's purpose and alter their behavior (Orne). Third, uneven group sizes between the familiar ($n = 30$) and unfamiliar ($n = 14$) participants may have reduced statistical power, making it difficult to detect a true effect of familiarity if it existed.

Regarding the interaction effect, we found that the difference in positive mood scores between the Indian classical music and white noise condition was not affected by participants' familiarity with the music. This does not support our hypothesis that familiarity with Indian classical music affects the effect of condition on mood. This is also not consistent with previous literature, which demonstrated that familiarity enhances positive therapeutic outcomes in the case of Yoik music (Hämäläinen et al.). This null effect can be explained in two ways. 1) It is possible that familiarity with Indian classical music does not affect the condition of mood. 2) The Hawthorne Effect or demand characteristics may have impacted the participant's responses, resulting in a null effect. 3) Uneven group sizes between the familiar ($n = 30$) and unfamiliar ($n = 14$) groups may have reduced statistical power, meaning the study may not have had enough sensitivity to detect a true difference in mood between the two groups if it existed.

Limitations

There are several limitations to this study that must be considered when interpreting these results. First, our sample suffered from sampling error. The group sizes for familiarity were uneven (thirty familiar participants, fourteen unfamiliar participants). This may have reduced our statistical power to detect differences between the groups. Secondly, this study had a small sample size. There were only 44 participants, or approximately 8 participants per group. Having more participants would have increased the study's accuracy. The small sample size might mean we did not find an effect that truly exists. The findings of this study are not generalizable beyond our specific subsample. Our sample only represents the United States; therefore, the results are not generalizable globally. Additionally, our sample was limited in terms of age (30 - 50). These findings may not apply to adolescents, young adults, or older adults. Last, our stimulus selection was not comprehensive. Because this study tests how Raag Bhairavi affects mood, these findings are not generalizable for all raags. Similarly, the white noise control may not have been an ideal comparison. Some participants may have been calmed or irritated by the white noise clip, affecting their positive mood scores. Finally, future research designs should better control for music background when recruiting subjects.

Conclusion

In this study, we found that listening to Indian classical music increased positive mood scores, familiarity with Indian classical music did not influence positive mood scores, and familiarity did not influence the effect of condition on positive mood scores by conducting a two-way ANOVA on positive mood scores. These findings are important because they indicate that Indian classical music may be generally helpful, regardless of familiarity with the music, and may expand the utility of current music therapy. In the future, studies must be conducted with a larger and more representative sample to ensure the results are generalizable to the overall population. Additionally, more research must be conducted to assess the interaction effect and the familiarity condition to ensure results are not influenced by the Hawthorne Effect or demand characteristics.

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The Effects of Commuter Rail Transit on Density: Evidence from American Cities

By Yuncong Evan Liu

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Abstract

The benefits of public transit infrastructure projects are unclear, which often delays expansions in the United States. This project investigates the impacts of passenger rail on urban and regional economic outcomes. I hypothesize that the areas in close proximity to rail transit will see job and worker density increases. Quantitative analysis employs two types of data. First, I use publicly available data provided by the Bureau of Transportation to obtain geographic locations. Second, I use publicly available data from the National Historical Geographic Information System from 2002 to 2019 to obtain the geographic locations and socioeconomic outcomes of census tracts. Out of the five major American cities that received transit expansion post-2002, I find that worker density in treated census tracts increased in Salt Lake City and Minneapolis, and decreased in Orlando, Portland, and Albuquerque. Similar trends are found across other outcomes, with results varying drastically across cities. These results suggest that the success of public transit expansions is heavily influenced by the mechanisms of land use around stations and transit accessibility options. My findings provide new insights for policymakers considering rail transit.

I. INTRODUCTION

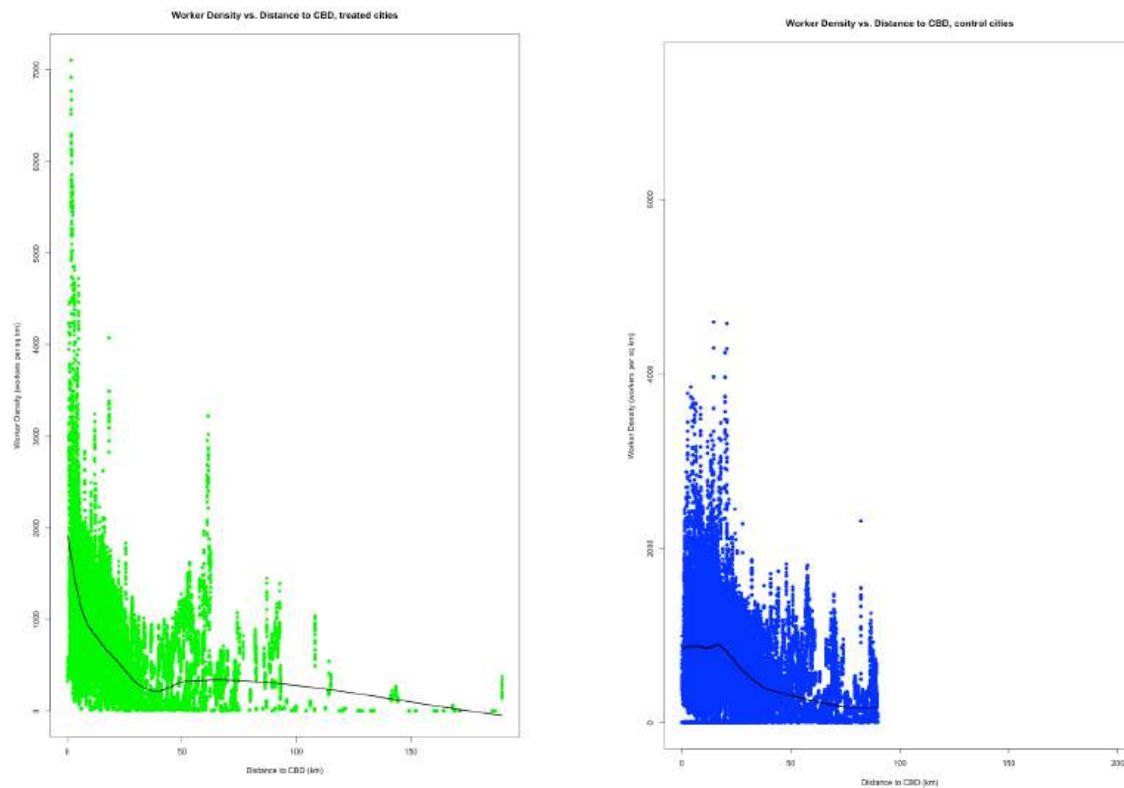
Public transportation is a topic of heated discussion in the US, with a wide variety of opinions. Transit advocates claim that bus and train lines help population growth, centralize cities, and serve as faster and more convenient means of transportation, especially during rush hour. On the other hand, car-centric infrastructure has made it incredibly difficult for expansions to occur. Even for cities with established transit networks, very few gains have been made since the spike in car ownership. Chicago's most recent major expansion began service in 1993, and New York City has not opened a new subway line since World War II. Because challenging geography makes laying down new tracks difficult and expensive, the majority of large cities in the US have not substantially changed their transit networks for decades.

While new subways and other forms of transportation are rare, environmentalism and increasing traffic problems in cities have resulted in several new expansions of a different kind: commuter rail. Typically operating from densely populated business districts to residential suburbs, commuter rail can reach higher average speeds by making fewer stops. Beyond geography, commuter rail provides more capacity per train, although frequencies are significantly lower.

Commuter rail appears to have three main advantages. First, these networks specialize in serving suburban areas dominated by car infrastructure. Being able to reach these residential districts allows commuter rail to replace driving for those who live close to a station. A larger group of people may choose to drive to a station, then ride the train for the remainder of their commute. The ability to serve sparsely populated suburbia is crucial as cities become increasingly more decentralized. However, commuter rail may not attract high ridership on the weekends, as the majority of its passengers are 9 to 5 workers. Additionally, they may not fully replace cars in the areas they serve. Second, commuter rail can be built on existing freight railroads, eliminating the need to lay new tracks. This is a significant advantage because it lowers the initial capital investment required to open a new service, making the expansion much more likely to succeed amongst policymakers. Of course, this also means that commuter rail is largely limited to existing freight tracks. For this reason, its ability to serve substantial portions of a city may be limited. Third, commuter rail can still be useful to riders even with low frequency. Unlike subways and so-called “rapid transit” systems, commuter rail specifically targets commuters whose movements around the city tend to stay the same. As a result, as long as train timetables are designed to be appealing to normal workers, the service may still prove to be popular. This operating strategy culminates in a set of relatively frequent trains running from the suburbs to downtown in the morning, almost no service midday, and another set of trains running from downtown back to the suburbs in the evening. By cutting trains that do not serve commuters primarily (i.e., those that serve offpeak hours or operate in the opposite direction of rush hour travel), agencies may cut down on operating costs, thereby making commuter rail an attractive option for cities seeking to save money while also providing public transit. At the same time, low frequencies make non-work-related trips almost impossible, with most services studied in this paper ceasing to run on weekends. These advantages have resulted in several commuter rail expansions in the past two decades across the United States. However, the comparatively low cost of these projects also raises questions about the economic benefits they actually bring to cities.

There are five major US cities that received new commuter rail after 2002: Salt Lake City, Albuquerque, Minneapolis, Orlando, and Portland. In contrast, there are also several major US cities that did not participate in opening any commuter rail line at all: Detroit, Columbus, Phoenix, San Antonio, and Las Vegas. Figure 1.1 plots the worker density for each tract in treatment cities from 2002 to 2019 against distance from the Central Business District. In essence, the graph visualizes how “centralized” the cities are. Figure 1.2 presents the same statistic for the control group.

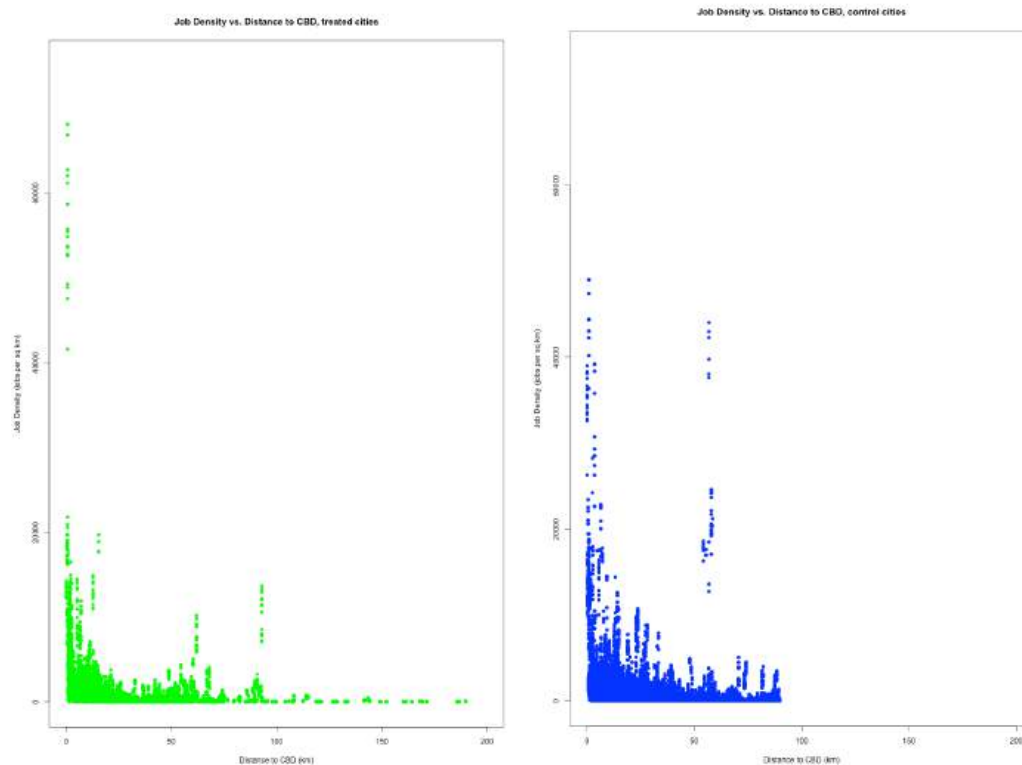
Figure 1.1, Figure 1.2



A cursory look at the data suggests that cities with commuter rail tend to be denser and centralized. This is visually verified in two main ways. First, the treatment group has tracts that are much higher in density close to the central business district. Second, the density of the treatment cities seems to decrease faster with distance from the city center than the control.

Figures 1.3 and 1.4 show the relationship between job density and distance from the CBD. Again, the treated cities have much higher peak densities, with a few tracts containing over sixty thousand jobs per square kilometer. Also, jobs decrease more quickly as distance increases in the cities with commuter rail, similar to the trend observed with workers.

Figure 1.3, 1.4



This implies that commuter rail centralizes both the workers and jobs of a city about the central business district, but it could also just be that denser cities are more likely to get commuter rail. The causal effects are unclear.

In this paper, I examine the impact of commuter rail transit expansion on the worker and job density in surrounding areas. I prefer to focus on the two aforementioned types of density instead of regular population density because commuter rail is most likely to affect the choices of those in the workforce. Residents who do not regularly commute, such as students or seniors, are unlikely to respond substantially to rail expansions because they rarely use the service.

I hypothesize that census tracts in close proximity to the new train stations will experience an increase in both worker and job density, particularly within a 2 km radius, due to improved accessibility. To test this hypothesis, I combine tract-level outcome data with transit data that represents the treatment. I examine the five major US cities that received commuter rail expansions post-2002. I identify two levels of treatment based on a tract's distance to the nearest train station. Then, I employ standard differences-in-differences estimations as well as event studies to infer the causal relationship. I find that, contrary to my hypotheses, both worker and job density decreased for the treated tracts. However, I also find a high degree of heterogeneity amongst cities, where a select few actually saw increases in density.

This paper provides new insights into more recent transit developments and their impacts. Additionally, my research focuses on commuter rail, a type of transit that seems to be popular with cities but has not been extensively studied in recent decades. This paper asks: How does the introduction of commuter rail affect job and worker density in surrounding areas? The remainder

of the work proceeds as follows: Section II discusses existing literature on the topic, Section III reviews the data and methodology that the research employs, Section IV describes the results, Section V presents a discussion on the underlying mechanisms that could explain the findings, and Section VI concludes.

II. LITERATURE REVIEW

The layout of a city plays a key role in its overall wealth, population, quality of life, and growth. City design heavily influences where residents are able to live and work, given their socioeconomic standing. At the same time, different modes of transportation are thought to facilitate varying types of activity and development. The monocentric city model simplifies the situation by assuming that all jobs are contained within a single point in space known as the Central Business District (CBD). Furthermore, every worker is thought to be identical, with each one needing to commute to the CBD to earn their wage. Therefore, the monocentric city model relates the population density at a point to the distance from the CBD. The price of housing falls as a function of distance, meaning residents would derive more utility from the same cost when living further from the CBD. This comes at the cost of a longer commute, which can be mitigated by various modes of transportation often installed by publicly funded infrastructure projects. This model shows intuitively that highways decentralize cities, as the cost of commuting is decreased with faster transportation (Arnott & McMillen, 2000). Similarly, transit lines are expected to cause the same, although perhaps to a lesser extent. The monocentric city model is not immediately consistent with my hypothesis, as it suggests a negative change in density. While the city itself may become less dense as a result of commuter rail, I argue that density would increase along the transit line. This is a reasonable claim because workers will be incentivized to live close to stations for more convenient access. The monocentric city model only states that residents would live further from the CBD, which is not at odds with the idea that they would live closer to train stations. For example, it could be that someone chooses to move further away from the CBD if it means living closer to a commuter rail station, which might give them convenient access to a potentially more efficient means of transportation.

One mode of transportation is the highway. Throughout the late twentieth century, the United States' rapid expansion of highways across the country is commonly believed to have caused rapid suburbanization, where residents moved out of traditional urban settings in favor of comparatively large single-family homes. In the context of the monocentric city model, this means that the construction of highways substantially lowered the cost of commuting per unit of distance, so much so that it even outweighed the initial burden of owning a car. The literature supports this conclusion. Baum-Snow (2007) confirms that highways did indeed contribute to decreasing population density in American cities, and that such cities would have been more populous had the Interstate system not been built. Again, the monocentric city model verifies that these results are robust. Extending this to commuter rail, Baum-Snow demonstrates that rail transit may achieve similar effects. This would depend on how workers choose to access public transportation, which, in most cases, would increase density surrounding stations.

Another mode of transportation provided to commuters is public transit. Often thought of as the alternative to highways, public transit is useful in dense urban areas where car traffic results in significant slowdowns. As a result, it is commonly believed that transit makes cities denser. However, a large-scale analysis of subway systems across the world found that, much like highways, subways also cause cities to decentralize (Florida, 2016). This is robust in the monocentric cities model, because fast transit allows commuters to avoid traffic congestion downtown, lowering the cost of their travel. Additionally, the analysis concluded that while transit rarely resulted in significant urban growth, it was possible that it rearranged the economy within a city. For example, residents could agglomerate near stations for easier transit access. This could result in an increase in density along the transit line. Tsivanidis 2018 verifies this narrative through an examination of Colombia's bus rapid transit, finding that downtown areas saw an increase in employment density, while outer districts saw a rise in residential density. Given that the bus line was a faster alternative to both low and high-skilled workers, the finding is again consistent with the monocentric city model because decreasing the cost of commuting allows for residents to spend more on housing. This study indicates that workers will indeed respond to transit and will shape their housing choices around it, as seen in the reorganization of Bogota into "residential" districts in the suburbs and "employment" districts downtown. Overall, the new route did result in a significant welfare increase across all income groups, both supporting and challenging the trends found by Tsivanidis 2018. Although it was not studied, it is likely to assume that Bogota's overall worker density decreased as more people elected to move away from the CBD. At the same time, residents clearly agglomerated around the bus stations, providing justification for my own hypothesis as well.

When it comes to rail transit, however, Baum-Snow and Kahn 2005 reach less favorable conclusions. Analyzing evidence from sixteen American cities, they found that, while commute times did slightly decrease, transit use did not substantially rise following a passenger rail expansion between 1970 and 2000. On the other hand, Gao et. al. 2024 studies the implementation of commuter rail lines in Japan. Using methods that split effects into "population catchment zones" similar to the two types of treatments defined in this paper, they find a positive and statistically significant correlation between commuter rail and population density within 750 m of the station. This directly supports the hypothesis that housing will develop around new transit stops as residents try to more easily access transit by walking or biking. However, this work is a purely cross-sectional analysis, which means that it, by definition, cannot infer a causal effect between the treatment and outcome. For example, it could be that density did not change, only that higher density areas were more likely to get new stations. Gao et. al. also motivates this research paper to investigate if commuter rail is actually the reason for changes in density.

In conclusion, the existing literature reports mixed results on the impact of trains on urban planning and outcomes. Even so, more rail networks have been built in the United States since then, the effects of which have yet to be thoroughly established.

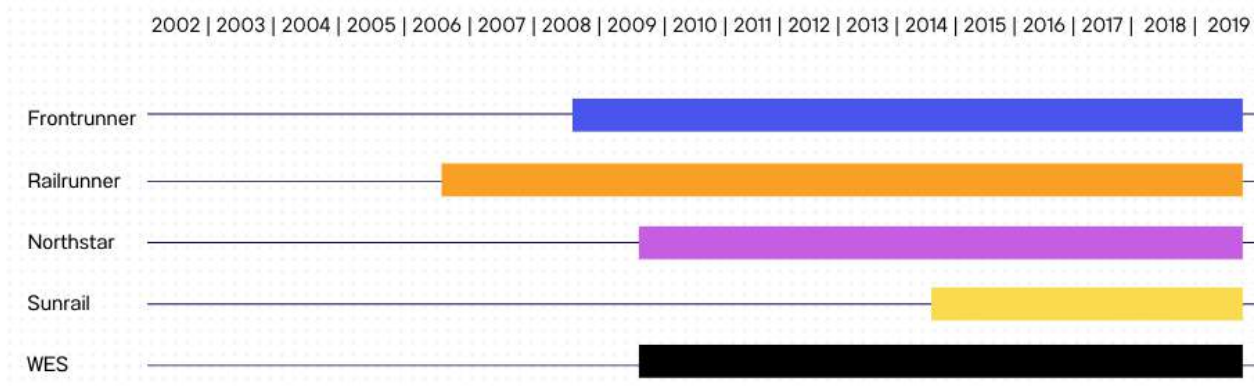
III. METHODOLOGY

A. Data Collection

My research utilizes two primary data outlets, followed by a synthesis that assembles the final panel data. First, I obtained transit data from the Bureau of Transportation Statistics (BTS), which gives the names and geographic locations of all commuter rail stations in the country in the year 2025. Since the outcome data covers the time period from 2002 to 2019, I investigate cities that opened commuter rail after 2002. Because of this constraint, I study only the following five services.

Frontrunner, which opened in 2008, serves Salt Lake City, Utah. This line covers 130 km with 17 stations. Railrunner, which opened in 2006, serves Albuquerque, New Mexico. This line covers 160 km with 11 stations. Northstar, which opened in 2009, serves Minneapolis, Minnesota. This network is 64 km long with 7 stations. SunRail, which opened in 2014, serves Orlando, Florida. It covers 98 km and has 17 stations. Westside Express Service (WES), which opened in 2009, serves Portland, Oregon. This network is just 24 miles long with 5 stations. Between their opening dates and 2025, these services have undergone negligible changes to their networks, which allows me to exploit present-day data from the BTS. Figure 3.1 displays the timing of each treatment in a Gantt-style timeline.

Figure 3.1



Second, tract-level outcome data originates from the United States Census Bureau's LEHD Origin-Destination Employment Statistics (LODES) dataset. The census block-level data has been aggregated by the Urban Data Catalog at the tract level. From the summary files, I use the Residence Area Characteristics (RAC) and Workplace Area Characteristics (WAC) datasets. RAC provides data for the number of workers living in a given tract for each year from 2002 to 2019, as well as more granular breakdowns by characteristics such as income and age. WAC provides data for the number of jobs contained in a given tract for each year from 2002 to 2019, with more detailed groups similar to ones seen in RAC also provided. From these data, the area of each tract is calculated using the "tigris" package in R. Because census tracts typically do not undergo large changes, tract area data is standardized as 2000 boundaries for all years. From this,

I obtain the worker and job density for each tract. Furthermore, the centroid of each unit is also recorded, which is used to calculate distances. When interpreting and discussing the results of this analysis, it is assumed that all residents and jobs live and occur at the tract centroid.

Most tracts I study are those in close enough proximity to the 5 cities of interest. I use a simple monocentric city model where tracts exist in one dimension: distance to the Central Business District (CBD). In other words, the units that are chosen are located within a certain radius of the CBD. I determine the radii for each city based on the size and location of the native commuter rail. It is worth noting that all commuter rails studied consist of a single line, but the location of the line relative to the CBD varies. For example, some routes run approximately through the city center on a north-south or east-west trajectory, while others do not directly serve the CBD. Nevertheless, the circle drawn around each CBD always encompasses the commuter rail line with room to spare. The radii are assigned as follows: 85 km for Salt Lake City's Frontrunner, 190 km for Albuquerque's Railrunner, 75 km for Minneapolis' Northstar, 105 km for Orlando's SunRail, and 40 km for Portland's WES.

For the purpose of comparison, I also collect outcome data on large American cities with no commuter rail. These cities may have once had expansive streetcar networks, but the unifying feature is that none of them have ever had a commuter line served by heavy rail. The 5 control group cities are Columbus, OH; Detroit, MI; Phoenix, AZ; San Antonio, TX; and Las Vegas, NV. The control group was chosen to be as geographically diverse as possible while also consisting of relatively large and well-known cities.

B. Data Summary

Next, I synthesize the outcome and transit data to build a complete dataset. The first way a tract can be considered "treated" is if its centroid is within 2 km of a new train station. This definition is identical to the one used in Baum-Snow and Kahn (2016). Furthermore, this definition is reasonable because 2 km is approximately the maximum distance someone would be willing to walk.

Among the treatment and control groups, there are 98,004 and 31,815 tract-year observations, for a total of 129,819 units. Using the treatment method described above, 1,075 tract-year observations are within a 2 km radius of a commuter rail station. The city-level breakdowns are as follows: in Salt Lake City, there are 330 tract-years, out of 5,112; in Albuquerque, there are 325 tract-year,s out of 3,888; in Minneapolis, there are 160 tract-year, out of 12,015; in Orlando, there are 150, out of 5,994; in Portland, there are 110, out of 4806.

A secondary form of treatment that I consider occurs when a tract centroid is between 2 km and 5 km away from a new train station. At this distance, most residents would not be able to directly walk to the transit stop, but the tract outcomes may still be affected by other dynamics. For example, some people may choose to bike to the station. Because there are no tracts in the dataset that are exactly on the 2 km circumference, the two levels of treatment are mutually exclusive.

For this type of treatment, 3,530 tract-year observations are between 2 and 5 km away from a station. The city level breakdowns are as follows: in Salt Lake City, there are tract-years, out of 5,112; in Albuquerque, there are 420 tract-years, out of 3,888; in Minneapolis, there are 940 tract-years, out of 12,015; in Orlando, there are 730, out of 5,994; in Portland, there are 260, out of 4806.

In terms of outcomes, the median tract in a treatment city (across all years) has 610.3045 workers per square kilometer. The lower and upper quartiles are 259.8311 and 1010.0956, respectively. The median control group tract has a lower worker density of 603.7265, with the lower and upper quartiles being 262.4539 and 965.9360. However, the mean density for the treatment cities is 715.2487, which is noticeably higher than the 657.5665 workers per tract seen in the control cities. Given these summary statistics, both distributions are slightly skewed right.

C. Empirical Specification

The tracts that get the treatment are not randomly assigned. For example, it could be that train stations are constructed in the highest-density areas. Alternatively, the existing freight railroads on which new commuter rails run on may pass through less developed areas. These possibilities may represent results as too positive or too negative. To account for this endogeneity, I employ differences-in-differences. I first estimate the effect of commuter rail transit on various economic outcomes using a standard differences-in-differences model.

$$(1) Y_{it} = \alpha_i T_{it} + \gamma_i + \delta_t + \epsilon_{it},$$

where Y_{it} denotes the log outcome of tract i in year t , including worker and job density, as well as those statistics broken down into different age and income groups. T_{it} is an indicator variable that represents a specific treatment status for a given tract i in year t , which can be either the closest 2 km circle or the second ring that covers 2 to 5 km away from the new station. Then, γ_i are tract fixed effects, δ_t are time fixed effects, and ϵ_{it} is the error term. I cluster standard errors at the tract level to account for correlation across tracts over time. Standard errors must be clustered at the treatment level because there is assumed to be correlation within “clusters.” As Abadie et. al. (2023) emphasizes, the sampling variation of the treatment effect estimator comes from the number of clusters receiving treatment, not the number of individual observations. Using non-clustered standard errors will treat all units as independent, which is intuitively incorrect. For example, two randomly chosen tracts from just Minneapolis will be more similar than two randomly chosen tracts from the entire United States. Without clustering, the analysis will overestimate the t-statistic and errors will be biased downward.

I consider two types of treatment because I anticipate multiple mechanisms at play. The 2 km circle takes into account the effect of new residents moving in to get within walking distance of the station. From 2 to 5 km, we can consider biking to be a more effective mode of transportation. The “ring” treatment has been shown to be effective in adjacent economic literature.

I also employ another specification to measure the effects of both treatments at once.

$$(2) Y_{it} = \alpha_1 C_{it} + \alpha_2 R_{it} + \gamma_i + \delta_t + \epsilon_{it},$$

where Y_{it} is still the independent variable, C_{it} is an indicator variable that equals 1 when the tract is considered treated under the 2 km circle definition, R_{it} is an indicator variable that equals 1 when the tract is considered treated under the 2 km to 5 km ring definition, γ_i are tract fixed effects, δ_t are time fixed effects, and ϵ_{it} is the error term. Equation (2) is expected to be superior to equation (1) because the latter considers the alternative treatment type to experience no effect from the treatment. The results table for equation (1) will show that this is not true for both the circle and the ring.

Differences-in-differences infers causation under the parallel trends assumption. To check if treatment tracts systematically differ from control tracts within the 5 commuter rail cities, I estimate an event study framework that takes the following expression:

$$(3) Y_{it} = \sum_k \alpha_k \cdot \mathbf{1}\{t - T_i = k\} + \alpha_i + \gamma_t + \epsilon_{it},$$

where Y_{it} is again the measured outcome of tract i in year t , T_i represents the year tract i was treated, $t - T_i$ converts standard time into event time, γ_i are tract fixed effects, δ_t are time fixed effects, and ϵ_{it} is the error term. This estimation allows me to visually observe the validity of the parallel trends assumption.

IV. RESULTS

This section discusses my empirical findings. I systematically show differences-in-differences results of the impact of commuter rail stations on various census tract outcomes. I estimate two regressions, one that examines a 2-kilometer radius around train stations, and another that includes a 2 to 5-kilometer ring around the treatment. The dataset does not include the control cities, only the 5 cities that received commuter rail expansion

A. Worker density across all Cities

I first estimated the impact of commuter rail stations on the log of worker density.

Table 1

Effect of Commuter Rail Transit on Worker Density, given by Equation (1)		
	D < 2 km only	2 km < D < 5 km only
D < 2 km	-0.0183*** (0.0070)	-
2 km < D < 5 km	-	-0.0632*** (0.0041)

Fixed Effects	Yes	Yes
Observations	31,815	31,815
R ²	0.99147	0.99154
Within R ²	0.00023	0.00798

I estimate a significant negative causal effect between worker density and commuter rail under the parallel trends assumption. As shown in Column 1 of Table 1, the presence of a commuter train station within 2 km of a tract centroid lowers the log worker density by roughly 1.83 percent. Tracts between 2 and 5 km away from a new train station experience an even bigger decrease with a 0.0632 decrease in log density. For small changes, logarithms can be interpreted as percent changes. Therefore, tracts in the ring saw a 6.32% decrease. In the rest of this paper, results will often be presented as a percent change as well.

Both results are statistically significant with $p < 0.001$. Both estimations see an R^2 of over 0.99, reflecting that most variation in worker density occurs across units. The within R^2 is 0.00023 and 0.00798, respectively, indicating that both types of tracts see only a small variation in worker density over time. This result is expected due to the binary explanatory variables used in the estimation and the fact that most units do not change status during the time period. Nevertheless, because both results are statistically significant, I infer a causal effect of commuter rail stations on density.

Using only one treatment at a time as the regressor introduces bias. For example, when the estimation uses the second ring treatment, all tracts within 2 km of the train station are considered to be “untreated,” even though there is a clear effect, as measured by column 1 of Table 1. Therefore, I now turn to equation (2), which estimates the coefficients of both treatments simultaneously.

Table 2

The Effects of Commuter Rail Transit on Worker Density, given by Equation (2)

(2)	
D < 2 km	-0.0296*** (0.0070)
2 km < D < 5 km	-0.0649*** (0.0041)
Fixed Effects	Yes
Observations	31,815
R ²	0.99154
Within R ²	0.00857

Table 2 shows the results from equation (2). In this case, tracts within 2 km of the train station see a decrease in worker density of 2.96%, which is larger than the one reported by the single variable estimation. This effect is qualitatively robust. When the regressor is the circle treatment, the “control” group also includes those treated by the ring. This means that the control will have a lower average worker density than if it just included the tracts greater than 5 km away from the treatment. For tracts between 2km and 5 km away, I observe a coefficient of -0.0649, which is again greater in magnitude than the one given by equation (1).

Mathematically, the more negative coefficients are also robust. If $\square^I_C$ represents a coefficient measured in equation (1), $\square^2_C$ is the coefficient in equation (2) of the circle treatment, and $\square^2_R$ is the coefficient in equation (2) of the ring treatment, then the omitted variable relationship gives

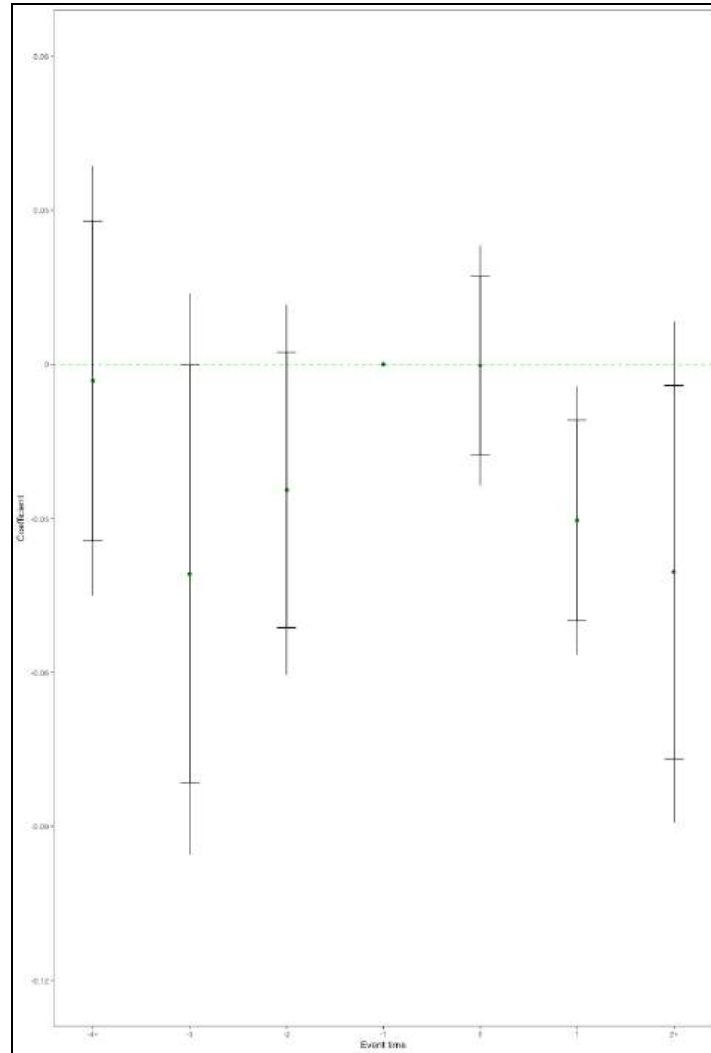
$$\square^I_C = \square^2_C + \square^2_R \text{Cov}(C_{it}, R_{it}) / \text{Var}(C_{it}).$$

Since both treatments are mutually exclusive, the covariance term is negative. The variance term is, by definition, always positive. It has already been shown that both $\square^2_C$ and $\square^2_R$ are negative, so the entire term $\square^2_R \text{Cov}(C_{it}, R_{it}) / \text{Var}(C_{it})$ must be positive. As a result, $\square^I_C$ is greater (less negative) than $\square^2_C$.

Next, I check the parallel trends assumption required to derive causality from differences-in-differences. Figure 4.1 displays the estimated coefficients of worker density relative to event time for within 2 km tracts.

Figure 4.1

Event Study: Aggregate Worker Density, 2 km



The coefficients during the pre-period are statistically indistinguishable from zero, showing that the treatment group was not systemically different from the control. 1 year after the treatment is introduced, the coefficient begins to decrease, consistent with the results suggested by the estimation.

Figure 4.2

Event Study: Aggregate Worker Density, 2 to 5 km

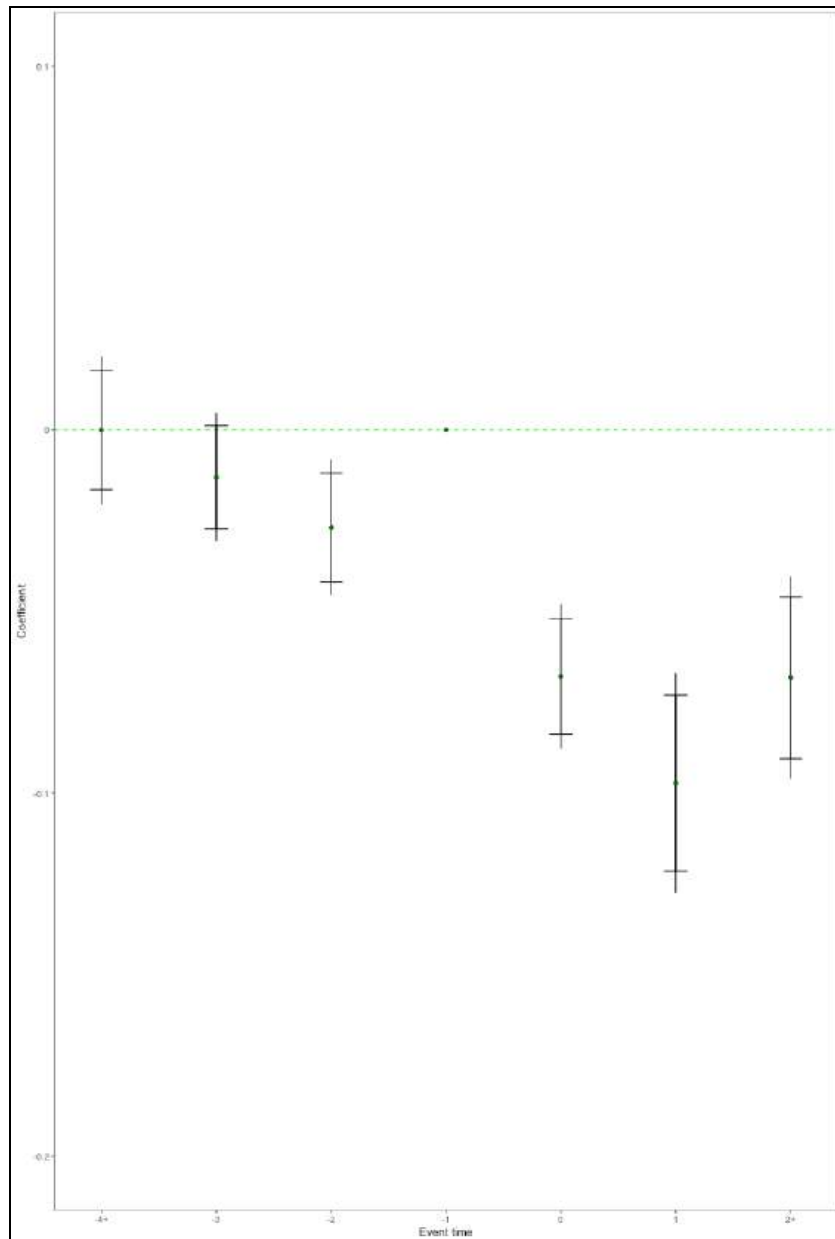


Figure 4.2 plots a similar event study, this time where the treatment group consists of tracts between 2 and 5 km away from a transit stop. Here, the coefficients show a downward slope prior to the treatment. However, this trend is smaller in magnitude than post-treatment effects, being only about one-third in size. Even so, I interpret results from this treatment group with more caution. While my findings from Table 1 Column 2 still remain valid, the event study suggests that the regression has overestimated the magnitude of the impact of the treatment.

Having established the causal effect of the policy on worker density, I now turn to a very similar metric: job density. Analyzing job density is useful because firms and companies respond differently to infrastructure changes compared to citizens. On one hand, firms are much wealthier

and may be able to afford relocations where an individual worker may not be able to buy another house; on the other hand, relocations are costly and affect all workers, meaning firms must be careful when making such decisions.

B. Job Density across all cities

Table 3

Effect of Commuter Rail Transit on Job Density, given by Equation (1)

	D < 2 km only	2 km < D < 5 km only
D < 2 km	-0.0635*** (0.0152)	-
2 km < D < 5 km	-	-0.0356*** (0.0088)
Fixed Effects	Yes	Yes
Observations	31,791	31,791
R ²	0.97468	0.97468
Within R ²	0.00058	0.00054

Table 3 shows a strong negative correlation between commuter rail and log job density. Tracts within 2 km experience a change of -0.0635, whereas those further than 2 but within 5 km see a -0.0356 effect. Again, both coefficients are statistically significant. Similar to the worker density results, the R² squared is large while within the R² is small, showing that the independent variable explains mostly within-unit variation.

As was done with worker density, I again regress job density on both treatment types to obtain a more complete understanding of the effects. Table 4 reports the results.

Table 4

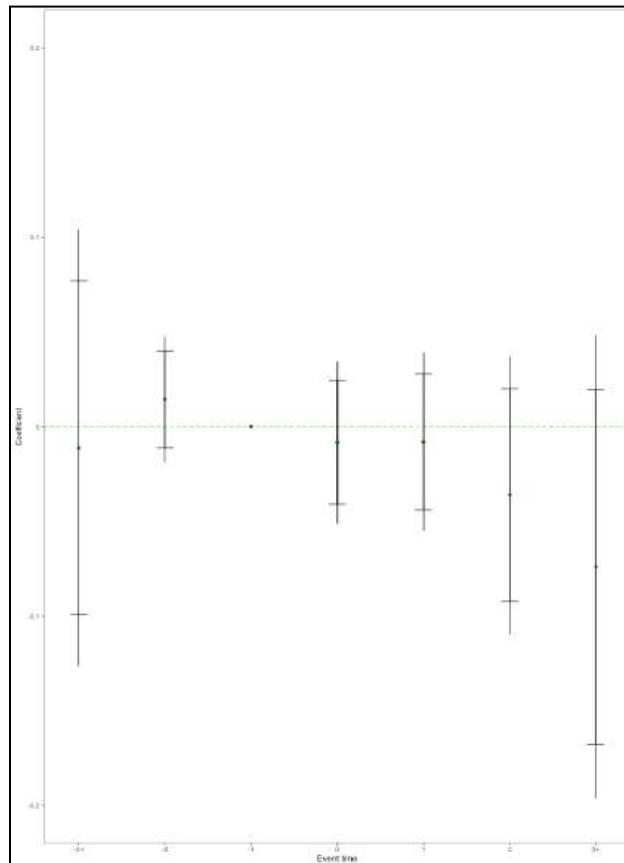
Effect of Commuter Rail Transit on Job Density, given by Equation (2)

	(2)
D < 2 km	-0.0704*** (0.0152)
2 km < D < 5 km	-0.0398*** (0.0089)
Fixed Effects	Yes
Observations	31,791
R ²	0.97470
Within R ²	0.00125

Again, as expected, both coefficients are more negative than the ones from equation (1). Closer tracts experience a 7.04% decrease, while the units slightly further only see a 3.98% decrease. Nevertheless, both effects are statistically significant.

Figure 4.3

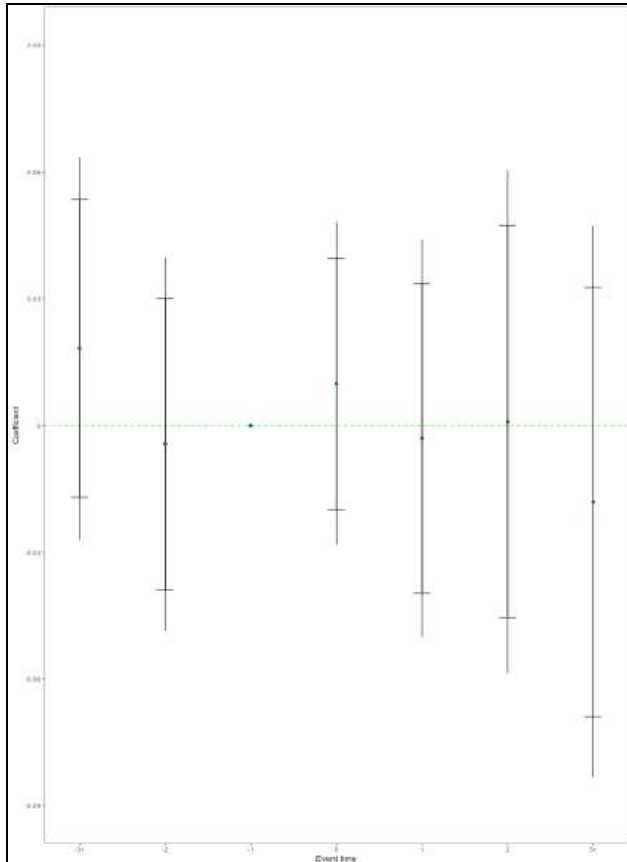
Event Study: Aggregate Job Density, 2 km



Again, I test for parallel trends to verify the underlying assumptions. Figure 4.3 plots the coefficients against event time. Before treatment, there is no difference between the tracts that got new stations and the ones that didn't. After treatment, the confidence intervals still cross zero, but there is a slight downward trend in the coefficients. The event study suggests that the effect of train stations on the job density of tracts within 2 km is present but less than the effect experienced by worker density.

Figure 4.4

Event Study: Aggregate Job Density, 2 to 5 km



I repeat the process with Figure 4.4, which shows coefficients for tracts between 2 and 5 km. Similarly, I observe no pre-trends for this treatment group, although the effect on job density is even less obvious. Furthermore, the data appears to be quite noisy, with the confidence intervals behind visibly larger. This shows that firms further away experience effects that are more varied and also smaller in magnitude.

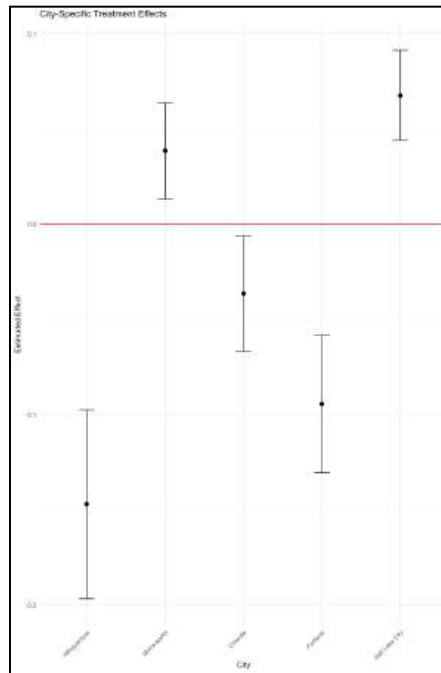
C. Heterogeneous worker density effects across cities

The aggregate effects may cover up variation across cities. As mentioned earlier, commuter rail transit expansions differ heavily from city to city. Moreover, cities differ in many other factors, such as road infrastructure, political attitudes, and income composition, which all may affect transit use. To examine more specific results, I estimate city-level effects.

Figure 4.5 plots specific effects on worker density for each of the five cities studied. The results vary substantially across cities: Salt Lake City and Minneapolis saw worker density increase following the introduction of commuter rail, while Albuquerque, Orlando, and Portland experienced sharp decreases.

Figure 4.5

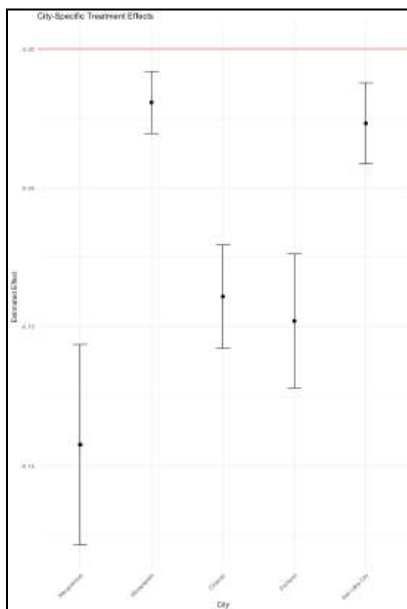
City-Specific Treatment Effects: Worker Density, 2 km



All results are statistically significant, although to different degrees: Salt Lake City, Minneapolis, and Portland with $p < 0.001$, Minneapolis with $p < 0.01$, and Orlando with $p < 0.05$.

Figure 4.6

City-Specific Treatment Effects: Worker Density, 2 to 5 km

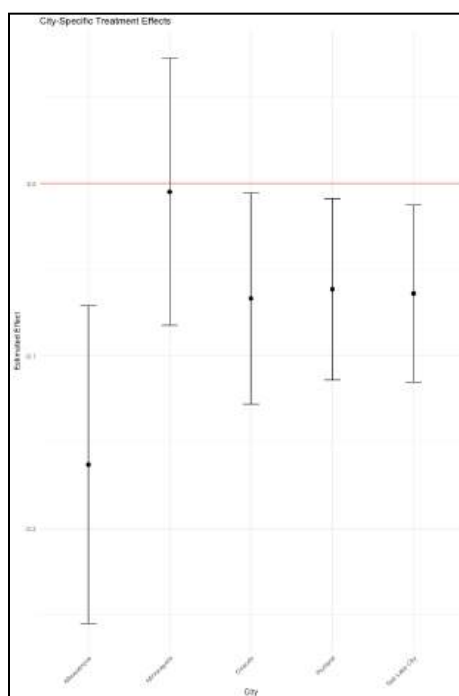


Next, I investigate the city-level effects for tracts between 2 and 5 km of the new commuter rail station. As seen in Figure C, all five cities show decreases in worker density. Albuquerque still experiences the most negative change out of the dataset. Interestingly, Salt Lake City and Minneapolis, the only two cities with positive effects for within-2 km tracts still have the most positive coefficients. This result implies that the mechanism behind the greater decrease for tracts further away is roughly independent of the mechanism that caused the initial increase for the within-2 km units. Despite the heterogeneity observed for the closer tracts, results from the second ring shows that the negative effect of commuter rail applies relatively uniformly to all cities.

D. Heterogeneous job density effects across cities

Figure 4.7

City-Specific Treatment Effects: Job Density, 2 km

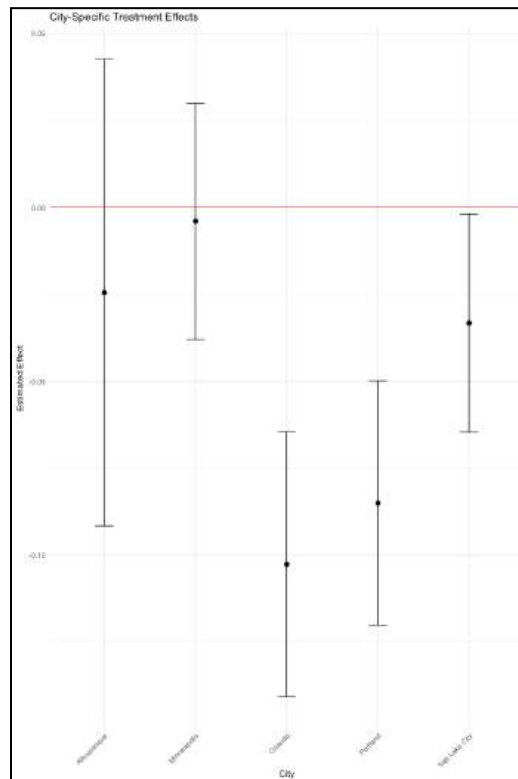


In this section, I examine the city-level effects for job density. Similarly, I start with tracts within a 2 km radius of a train station. For this cohort of units, Figure 4.7 plots the results.

All cities except Minneapolis saw a clear decrease in job density. The effect for Minneapolis is indistinguishable from zero. However, all coefficients are statistically significant (Salt Lake City, Orlando, Portland, with $p < 0.05$; Albuquerque with $p < 0.001$) with the exception of Minneapolis. Figure 4.8 plots the results for tracts in the second ring.

Figure 4.8

City-Specific Treatment Effects: Job Density, 2 to 5 km



With increased distance from the train stations, the treatment now has virtually no effect for both Albuquerque and Minneapolis. The previously observed negative correlation still holds for the remaining three cities, with all results besides Minneapolis being statistically significant.

The effects of job density are much less clear than those of worker density. Additionally, the effects are quite noisy, which makes them harder to interpret. The trend that employment opportunities are less responsive to changes in transit than individual workers still holds.

V. DISCUSSION

Overall, the analysis finds that commuter rail has a strong negative effect on both worker and job density. However, some cities do experience statistically significant increases in certain metrics, and no noticeable change at all in other metrics. This indicates that nuanced mechanisms are at play. This section first discusses possible explanations for the negative effects that the

aggregate dataset shows, then takes into account the heterogeneity of treatments in the different cities to speculate on causes for the variation in city-level effects.

To start, the results are counterintuitive. It has been shown previously that public transit decentralizes cities by providing a fast mode of transportation to and from the central business district (Florida, 2016). The intuition is that commuters will choose to use transit extensively, which allows them to live further away. Therefore, although overall city density may decrease, people should tend to agglomerate close to transit stops where they can access the service as easily as possible. Evidently, this is not what is seen in most cities.

A. Mechanism 1: Noise Disamenities

The most intuitive explanation for the observed effects is that transit, especially commuter rail, introduce an additional disamenity to the surrounding communities that may not be appreciated by residents. Train horns, bells, and railroad crossings are incredibly loud, capable of being heard even from several kilometers away. As a result, home builders may be disincentivized from developing property within the area. Furthermore, houses that have already been built may not as easily sell given the added noise pollution.

Relative to other mechanisms discussed in this paper, this is the one that best explains why density decreases even for tracts that are further than 2 km away. However, it is also potentially the smallest in magnitude, because commuter rail services examined by this work are infrequent, with only a few round trips per day. Additionally, most operations are concentrated during the weekdays and rush hour times, where residents are least likely to be home. Therefore, while I consider this to be a reasonable explanation, it is perhaps not the most important one.

A. Mechanism 2: Land Use Constraints through Parking Infrastructure

The most compelling mechanism that could explain the overall decrease in the implementation of “park and ride.” As aforementioned, commuter rail serves daily 9-to-5 workers who live in the suburbs but work downtown. Because it may be hard to access transit due to the sprawling nature of suburban residential districts, agencies seek ways to allow commuter rail to be frequently useful also to those who do not live close enough to walk to a station. Park and ride (PAR) is a system in which riders travel to a transit stop by car and park their car before transferring onto a form of public transportation. PAR necessitates that there be parking structures next to stations that are large enough to match the capacity of commuter rail. If the construction of a station is frequently associated with an additional parking lot, then the treatment would effectively decrease the amount of land in the tract that can be used for commercial purposes.

Image 5.1



I verify these initial intuitions with qualitative evidence in the form of images extracted from Google Maps. Image 5.1 shows a top-down view of Elk River station on the Northstar line serving Minneapolis. The station is surrounded by some residential buildings to the east as well as a few commercial buildings to the west. However, the most noticeable detail is the 8-acre parking lot situated right next to Elk River station. This piece of infrastructure reinforces the idea that commuter rail is meant to be an extension of car use, not a replacement. Riders are not expected to live next to the station itself; in fact, they are clearly discouraged to, because the land has been used to support PAR.

Image 5.2



Image 5.2 offers even further insights into why PAR might decrease density around newly introduced train stations. SunRail's DeLand station in Orlando, Florida, features a massive parking lot meant for PAR. However, the land immediately surrounding the station is sparsely populated, with the station itself being on the edge of DeLand. Interestingly, the land around the station is noticeably underdeveloped compared to the rest of the city, indicating that SunRail riders are not meant to live near its stations. The lack of economic activity around stations could be explained by state or agency ownership of the land. If a transit authority buys the land next to a station without adjusting policy to encourage growth in the region, then the area will stay underdeveloped, whereas other parts of a city may continue to grow. This mechanism could explain the observed decrease in density associated with a new computer rail station.

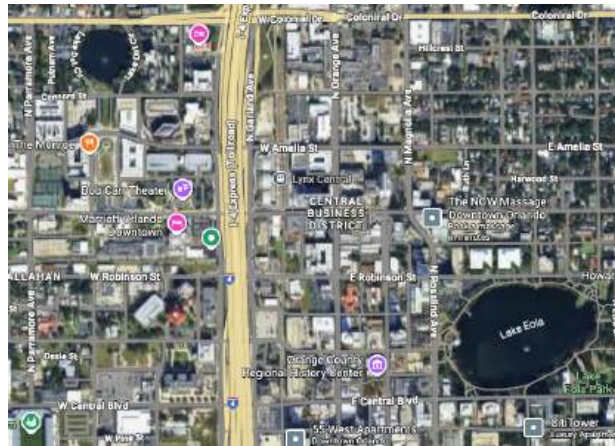
A more unique piece of non-parking lot roadway that is expected to contribute to the negative density change is seen in Image 5.3. A visual inspection reveals a “roundabout” shaped road that is likely designed to drop off passengers. Although this is not specifically consistent with the PAR model, it still supports the idea that riders are not meant to walk or bike to their local station.

Even if parking lots and roadways do not contribute much area, it is still possible that transit agencies will acquire land beyond such infrastructure for legal or planning purposes. In any case, the land will inevitably be underdeveloped, being made unavailable for housing or businesses.

Image 5.3



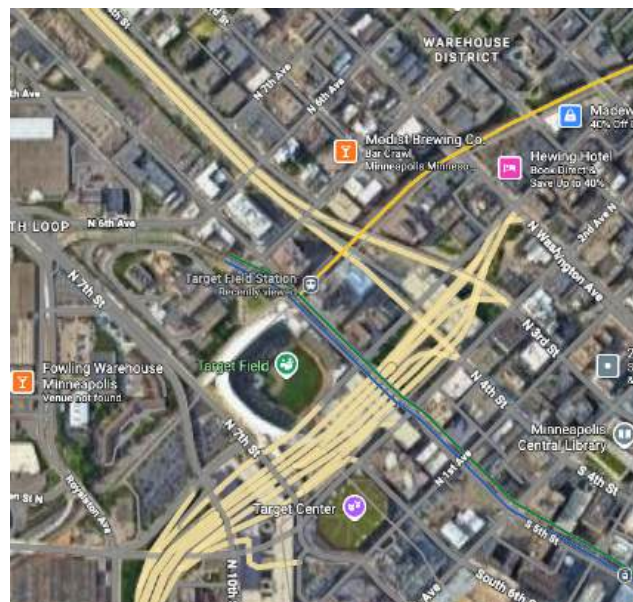
Image 5.4



B. Mechanism 3: Downtown Station Dynamics

The trend of parking lots being built in tandem with train stations has one major exception: the downtown terminus. SunRail's Lynx Central Station directly serves Orlando's Central Business District and does not feature the commonly seen parking lot. This is likely because most commuters disembark at this station, meaning they would not need a place to park their car. Instead, the Lynx Central is surrounded by infrastructure that is associated with transit-oriented development, such as a bus connection terminal. Additionally, the dense nature of central business districts means that agencies may not have the ability to create a parking lot even if they wanted to.

Image 5.5



The absence of parking lots in central stations continues with Target Field Station in Minneapolis. Northstar's southern terminus is situated beneath a bridge and has a quick connection to the Metro Transit Blue line and Green line trams, similar to Lynx Central's bus connections. Again, transit-oriented development seems to thrive downtown where connections to other modes of transit are necessary.

In these cases, I anticipate the initial hypothesis of increased density around stations to hold true. However, because PAR stations vastly outnumber stations that serve the CBD, the aggregate effect would still be negative. I verify this claim quantitatively. 50 out of 57 stations, or 88%, studied had an easily identifiable parking lot visible in a 512 by 512 pixel satellite image. The main takeaway of this statistic is that PAR is an extremely prevalent service model for suburban stations. If each network has 1 downtown terminus that does not have a parking lot, then 50 out of 52 suburban stations, or 96%, have a parking lot.

Meanwhile, it is important to recognize that the results of decreased density through PAR do not necessarily imply that commuter rail is a poor policy choice. It could be that density and development increase in neighborhoods that offer the best roads or most direct access to the stations through driving. These "feeder" neighborhoods could see growth, which is a possibility that should not be neglected. Additionally, we must consider the potential density increases in downtown stations. If a city becomes more dense at its CBD (or the commuter rail's urban terminus station), then that could align with a transit agency's economic goals. In any case, evaluating whether or not changes in density are productive is a process that should be done on a case-by-case basis, taking into account city-specific problems and solutions. Nevertheless, this discussion shows that extensive use of park and ride as a method to increase transit access is a reasonable cause for the observed decrease in both worker and job density.

C. Mechanism 4: Ridership and Service Quality to Explain Heterogeneity

Next, I address the high level of heterogeneity seen at the city level. Indeed, there is a substantial variation in the degree of treatment assigned to different cities. Interestingly, all commuter rail systems studied in this paper, except WES, use the exact same combination of rolling stock, with Bombardier Bi-level coaches and the MPI MPXpress locomotive, meaning that there is no service that is considerably faster than the rest. For this reason, I determine the most important factor to distinguish the treatments to be the service frequency. Using this metric, Frontrunner and Northstar stand out as the outliers, with 25 and 2 round-trips per weekday, respectively. These two services are also the ones that led to a positive coefficient for worker density within 2 km of the station. All other services are roughly similar, with 10 to 20 round-trips per weekday.

I propose that there are two main mechanisms at play: first, the area lost from the acquisition of land around the station, along with the parking lot, and second, the gain in workers or jobs as a result of individuals in companies seeking to take advantage of transit. Applying the observed results, the former clearly outweighs the latter in most scenarios. However, more people may move into a tract if the service is good enough, which could explain the positive

coefficient for Frontrunner in Salt Lake City. Frontrunner's impressive ridership numbers, the highest of any network studied in this paper, defend the quality of transportation it provides. On the other hand, if the service is underfunded (like Northstar, with only 2 round-trips per weekday), the agency may also acquire less land around its stations, meaning more land can be used for housing development. Northstar attracted under 130 thousand riders in FY 2024, which is by far the lowest of any of the other 5 routes. Even prior to the pandemic, the service was not very popular in relation to similar commuter rails in the country. This might have decreased land acquisition around stations for parking lots, as demand was perceived as low. In the three other cities that observed a decrease in density, it may be that authorities obtained too much land relative to how popular the commuter rail actually turned out to be. This would lead to an increase in the effect of the first mechanism and a decrease in the effect of the second mechanism.

Overall, these patterns suggest that the effects of commuter rail are linked closely to the systems of operation that accompany its implementation. Specifically, when transit agencies purchase land without developing the housing market nearby, the park-and-ride model may not actually increase the density of the surrounding area. In fact, density may actually decrease as individuals choose to live further away from the transit stop because they will still be able to access it by car. While my interpretations of the results are largely speculative, they provide a coherent explanation for the observed effects and can be anecdotally verified. Conclusively, the impacts of commuter rail are highly contingent on the overall vision of how the service might be made more accessible to its riders.

VI. CONCLUSION

This paper examines the quantitative impacts of commuter rail expansion on density. In general, the analysis showed that worker and job density tend to decrease in the presence of transit, although some cities showed an increase. These findings contribute to the literature by providing insight into the implications of recent expansions, which have not been thoroughly investigated. These results suggest that policymakers should be mindful of nuanced ways in which transit can affect surrounding areas, as the specific responses also depend upon how riders access the stations. Even so, results from Salt Lake City and Minneapolis show that even lackluster levels of service quality still have the potential to achieve the intended agglomeration.

There are several limitations. First, the sample size does not encompass the full array of commuter rail networks in the US. Specifically, transit systems that were opened prior to 2002 are not included in the analysis due to limitations in data availability. At the same time, the group of cities that had transit prior to 2002 has substantially more successful systems than the cities studied in this paper. As a result, the full effects of transit may not be fully captured. Second, service quality is not included in the empirical analysis, even though it may still play a role in influencing residents' housing choices. This means that the impact of train speed, frequency, and station amenities has yet to be investigated. Future studies should quantify how service quality dimensions, such as frequency and multimodal connectivity, influence ridership and surrounding

density. Future work should apply a quantitative lens to analyze the impacts of more successful transit systems, which may require studying recent expansions from other countries. In conclusion, the findings presented in this paper provide a foundation for understanding the nuanced implications of commuter rail on city dynamics.

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The Transformation of Language's Metaphorical System in Algorithmic Media

By Yaxuan He

In today's algorithmically curated media, the metaphoric system of language has been reshaped by algorithms. This research paper discusses the metaphorical change of language based on criticism of the entertainmentization of media as seen in Neil Postman's *Amusing Ourselves to Death: Public Discourse in the Age of Show Business*, analyzing the causes and effects of language's change through its structure and frequency of use.

Neil Postman argues in *Amusing Ourselves to Death: Public Discourse in the Age of Show Business* that every medium has a non-neutral metaphor. From the typographic era to televisual time, the metaphor embedded in different forms of communication has significantly reshaped the public discourse from explaining truths and the reasons behind them to entertaining the audience for more viewers, using his words, "a show business" (Postman). Many television programs became fast-paced, less serious, shifting from one subject to another quickly in order to continuously seize the audience's attention, forming a culture full of disinformation deprived of a "coherent understanding of the world." In the early 2000s, as the new medium of digital internet prevailed, algorithms, which are automated procedures that select, prioritize, and present content based on users' preferences and behaviors, emerged as a response to the entertainment-oriented public demand. When implemented, they do not merely adapt to this logic but further intensify the entertainmentization of media. The algorithmic mechanism is highly personalized so that every user can receive the optimized content designed for them. It constantly exposes people to new information they are interested in, changing the pace and pattern with which people engage with society. Such a transformation sets the condition for change in language, the medium through which this engagement is articulated.

Algorithms' prioritization of immediacy and emotional expression over reasoning reshapes the linguistic system. Postman states that when the medium becomes entertainment, people prioritize entertainment instead of what is being informed (Postman). Extending this trend, in today's digital media world, algorithms further fragment logic and reasoning. Algorithms provide users with unceasing entertaining information, which triggers the human brain to secrete more dopamine. Under this biological stimulation that is less intense but increasingly frequent, humans get accustomed to the dopamine-based incentive mechanism that brings them chronic pleasure. The threshold of people's happiness increases, creating a desire for rapid, frequent, and emotionally charged responses. This, in turn, pressures the development of a mode of social interaction where both expression and response become highly emotionalized. Ultimately, the primary emotion outlet—language—adapts in response. People's expression on social media gets shorter, simpler, often only portraying their emotion at that instant. Algorithms also prioritize easily shareable and repeatable language. As a result, the usage of "hahaha," "LMAO," "LOL," and "lowkey" rises; emojis and meme culture also exemplify emotions conveyed through symbols rather than contexts of what incites these feelings. The frequent use of symbolic language with minimal cognitive content changes the function of language from a

medium for exchanging thought to a vehicle for expressing emotions with no substantive meaning, fundamentally different from the typographic culture that encourages making complicated comments. People pay less attention to the metaphors behind complex language structure because comprehending them is time consuming and interrupts their fast-paced experience online. Hence, human expression risks being reduced to an emotional mechanism caused by algorithms instead of real-world actions stemming from individual choice. Language becomes flattened, losing its capacity to convey nuanced meanings. This shift in metaphorical structure of language is a resulting consequence of algorithmic integration.

K-Pop provides a clear case study of how algorithmic systems operate within digital entertainment, reshaping language's metaphorical structure. Intense visual stimulation, high emotional appeal, and fast-paced rhythms are the most prominent features of the K-Pop industry. These characteristics continuously capture the audience's attention, prolonging customers' engagement and consumption. Algorithms, whose main goal is to maximize users' interest and engagement, are therefore particularly well suited for this industry, further amplifying K-Pop's stimulation on human emotions. In this process, social media has become a place for emotional resonance rather than for rational discourse. Fans' strong admiration for K-Pop idols reinforces a frequent use of emotionally charged content using words such as "he looks insane," "he's killing it," and "I can't handle this." Phrases that once signified essential changes in one's life now appear ubiquitously on social media to express their algorithmically stimulated emotions, causing language's connotations to become increasingly superficial. For example, algorithms actively shape the shift in contemporary Chinese K-Pop fans' metaphoric conception of "happiness." In Korean, 행복 (haengbog), which means happiness, is a common expression that people use daily. It shows a person is in a good mood and often implying an easy and daily connotation. In comparison, although "xingfu" in Chinese also translates as happiness, it carries a heavier semantic weight that is typically associated with meanings more significant than everyday pleasure, often referring to emotions formed through long term affection. However, as K-Pop fans' experience of "happiness" is shaped more by algorithmic engagement than by lived experience, the emotion becomes trivial, turning into an automatic affective response largely triggered by algorithmically curated stimulation. People start to ignore the meanings behind the expression of "happiness." Their disregard allows algorithms to shift the frequency of using "xingfu" and its linguistic environment unnoticeably. The repeated presentation of the Korean rendering of happiness on K-Pop fans' social media normalizes the use of "xingfu" on a daily basis. Fans become accustomed to encountering this concept and its flattened meaning not only in Korean but also through its recurring translation in Chinese. "Xingfu," therefore, is used far more frequently in everyday Chinese than before under algorithmic calculations. Moreover, as "happiness" also no longer indicates an experiential state derived from concrete events but instead reflects a mechanically induced response shaped by algorithms, "xingfu's" metaphorical structure is flattened, causing "xingfu" to lose the former semantic gravity and function primarily as an ordinary expression of emotion. Thus, the algorithmic system contributes to the scale of language's metaphorical shift, altering the metaphors of language by prioritizing emotional

immediacy over experiential meanings. In the early twentieth century, Postman demonstrated that the medium of television turns the Age of Exposition into the Age of Show Business. It creates a new public discourse that lacks coherent language and is characterized by passive exposure to information without corresponding action. People adapt to the new discourse with delight and think it's just fine (Postman). In today's algorithmic era, algorithms push the fragmentation and reshape the public discourse a step further: people are not only "amused to indifference" like what Postman says, but they no longer possess the ability to express themselves without using "indifferent language," deprived of the capacity of making complex arguments and delivering thoughts precisely. Algorithms inaugurate an era where not only the way people receive information becomes entertainment-driven and less rational, but the outlet for expressing thoughts is also simplified and transformed through rapid information exchanges on social media.

The simplification of metaphorical structure in the linguistic system results in individuals' self-alienation and the gradual disappearance of public discourse. Without a linguistic system with diverse expressions, people may become detached from the capability to understand and document their emotions using language, therefore distancing individuals from a perception of "self." The capacity for people to articulate thoughts also becomes increasingly shallow and unfamiliar, making it more difficult to locate meanings in life. These changes would challenge individuals to find an in-depth self-narrative in society and isolate them from the feeling of being "present," contributing to problems in public mental health. Meanwhile, public discourse may diminish as fewer people are able to use the right word to express thoughts. As expressing emotions becomes the major outlet for language, social conflicts tend to take on more emotional forms, where rational deliberation is often secondary. Differences in opinions could also be increasingly influenced by algorithmic calculations. Over time, people may experience a reduction in the autonomy to decide their thoughts and actions, immersed in shallow, algorithmically curated pleasure, with algorithms mediating the dynamics of public disputes.

In conclusion, Postman's argument of how the medium influences social behaviors is still applicable in today's algorithmic era, explaining algorithms' profound impact on reshaping language's metaphorical structure. Given humans' predisposition toward entertainment, algorithms facilitate the transformation of metaphors in the linguistic system. By prioritizing the transmission of instantaneous emotional expressions over nuanced arguments, algorithms reshape language's metaphors. The decline of linguistic metaphor challenges humans' self-identification and negatively affects the maintenance of a rational, coherent public discourse. This shift in human expression is gradual, guided by the calculation of algorithms, as humans have participated less in the process of procuring information. As humans' role gradually diminishes in this process, people may not notice this linguistic change, potentially allowing the change in language's metaphorical system to continue evolving unchecked. If this trend persists, the weakening of language's capacity to sustain more complex and reflective expressions may become a growing concern.

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Happiness In a Box:

A Quantitative Correlational Approach: How Blind Boxes Impact Hedonic and Eudaimonic Happiness in Singapore By Brooklynn Chow

Abstract

Blind boxes, defined as “packaged goods that contain random products unknown to consumers until purchase” (Huang & Suo, 2025), have gained massive international popularity recently. Because this activity is fairly recent, very limited research has gone into examining psychological impact, specifically well-being. In the existing literature, research has shown that purchasing blind boxes leads to emotional consumption that drives consumer demand (Liu, 2022). That being said, little research has looked at blind boxes' impact on happiness. This study aims to address this gap by measuring hedonic, short-term pleasure, and eudaimonic, long-term, value-driven fulfillment, happiness from blind box consumption.

Using a quantitative method with a survey that involves likert scales that measure participants' self-reported levels of hedonic and eudaimonic happiness, I used Confidence Intervals for my analysis to grasp ideas and patterns of where people stand with their happiness levels from blind box consumption using a small Singapore sample. Findings suggest that blind boxes are more highly associated with hedonic happiness, yet there is evidence that blind boxes also create some association with eudaimonia as well. This research can be highly useful as considerations for businesses who sell blind boxes, as well as those who consume themselves. If blind boxes show that they are more associated with a certain type of happiness, it can help businesses understand better what drives consumers to buy their products. It also allows the consumers to recognize that blind boxes might not bring them happiness for a longer period of time than they might think. However, this study's sample size is very limited and only in Singapore, thus more data needs to be collected in order to make a broader statement.

Key words: Blind boxes, hedonia, eudaimonia, happiness, well-being.

Introduction

Recently, the terms “Labubus” and “Pop-Mart” may have come up in daily conversation. Social media, trends, and successful marketing campaigns have led up to these strange collectibles taking their place on the world stage. But why? Why are people spending so much money on these mass-produced, random figurines, toys, and collectibles? These toys that are so popular right now are referred to as blind boxes. The “blind box” concept originated from Japan, known as “lucky bags” or “gacha eggs.” (Xia et al., 2025). Blind boxes are “packaged goods that contain random products unknown to consumers until purchase” (Huang & Suo, 2025). Usually, blind box contents “are part of a collection, such as dolls, hand pieces, limited edition toys, or peripheral products, and each collection has multiple different styles, while some styles may be limited or rare, adding to the fun and challenge of collecting” (Chen, 2024). Furthermore, in recent years, blind box popularity has skyrocketed. In 2021, “blind boxes overtook assembled toys to become the largest popular toy segment in China,” having a market value of around 1.4

billion USD (Lim, 2025). Because of this, recent literature has looked into intentions for blind box purchases, as well as the psychology behind these products.

Amongst the ongoing blind box craze, not much attention has specifically paid attention to the impact of these small toys on consumers' happiness. However, purchasing blind boxes can lead to emotional consumption that drives consumer demand (Liu, 2022). In an effort to explore more deeply into the details on consumer happiness driven from blind box consumption, this study aims to investigate this rather recent, and open-fielded phenomenon through a quantitative correlational approach.

Literature Review

In the overall literature surrounding blind boxes, most of the research done on consumer demand and popularity is fairly recent and roughly around the last 4-5 years. Many studies have looked into possible reasons and push factors for the spike in popularity of purchases in blind box toys. Literature has shown that in a blind box, because the actual product inside the box is unknown to the consumer, the main factor of the appeal to blind boxes is the randomness. The randomness and uncertainty draws people in. When the consumer gets a toy or figure they desired by chance, it holds much more value than purchasing a desired toy or figure they knew they were guaranteed to get (Xia et al., 2025).

While Xia and associates found that the randomness of blind boxes is the main push factor for purchases, other research found that various social factors explain blind box appeal. For example, the development of subcultures, influence from social media, and fear of missing out on trends are all additional reasons for the popularity of blind box consumption in recent times (Wei & Yu, 2025). Wei and Yu found that people bond when sharing similar activities such as blind box consumption which can lead to smaller, niche social groups called subcultures. Within these subcultures, they can “foster social connections, as collectors form communities where they share experiences and exchange items, fulfilling their need for relatedness. In the blind box community, these social interactions enhance a sense of belonging and connection among consumers in order to maintain a better social relationship” (Wei & Yu, 2025).

Adding onto this, researcher Gunarian and associates found social factors as an incentive to purchase in Jakarta.. They found that social factors such as social media and community through shared interests were main reasons for purchase intentions. Their findings showed “that social value, influenced by peer pressure and social media-exposure, significantly enhances purchase intentions” (Gunarian et al., 2025). However, not only do they corroborate with Wei and Yu, they also found that purchase intentions can vary in different places, so for Jakarta's case, according to Gunarian's research, social factors are the purchase intentions there.

Something to note is that Gunarian's source is one of the few papers surrounding blind box consumption in a South East Asian context. There is little to no research regarding blind box consumption in Singapore which starts to reveal a gap in the overall literature surrounding blind boxes.

Consumer Happiness

Amongst the literature surrounding consumer happiness, most shared a commonality as they talked about two kinds of happiness: Hedonia and Eudaimonia. Lots of research has found ways to measure hedonic and eudaimonic happiness or well-being. In this paper, the terms “well-being” and “happiness” are used interchangeably.

Hedonia associates with pleasure and short, high bursts of pleasure, while eudaimonia associates with deeper understandings of moral values and long-term happiness. Both are needed for a higher overall well-being. (McMahan & Estes, 2011).

Similarly, Delle and associates conducted research that corroborated with McMahan and Estes findings. This team of researchers studied hedonia and eudaimonia through quantitative and qualitative methodology using several hundreds of people from different countries. They further substantiated that both hedonic and eudaimonic happiness leads to an overall higher well-being (Delle et al., 2010).

Further research has found that different purchase types correlate with different kinds of happiness. Researchers Van Boven and Gilovich ran a study that researched if material purchases or experiential purchases made people more happy. They found that materialistic purchases like tangible objects and goods correlate more with hedonia, while experiential purchases like vacations correlate more with eudaimonia. Van Boven and Gilovich also identified that eudaimonic happiness is more meaningful and longer-lasting than hedonic happiness since hedonia is short bursts of pleasure.

Additionally, research has found that our activities also affect hedonic and eudaimonic well-being. Researchers Huta and Ryan found that activities that are more “carefree” tend to resonate more highly with hedonia, while more meaningful activities resonated more highly with eudaimonia (Huta & Ryan, 2010).

Yet, no research has specifically looked at blind box activity and its effects on hedonic and eudaimonic happiness which highlights the gap in the literature. Additionally, there is little research around blind box consumption in South East Asia, specifically Singapore. This is important because with blind boxes’ East Asian origin, South East Asia has strong exposure to East Asian products such as blind boxes. Therefore, there is value in studying this specific region. Since Huta and Ryan found that our activities can impact hedonia and eudaimonia, blind box activity must have some sort of impact on the two kinds of happiness as well. This has led me to investigate the impact of blind box consumption on hedonic and eudaimonic happiness in Singapore.

Therefore, this research will aim to answer this question: Through a quantitative correlational approach, how does blind box consumption impact hedonic and eudaimonic well-being in Singapore? This research could help consumers raise awareness on how purchasing these products might affect them both short and long-term, leading to a better understanding of what to keep in mind when purchasing.

Hypothesis

I hypothesized that blind box consumption would impact both hedonic and eudaimonic happiness in Singapore; however, I thought that blind box consumption would have a greater effect on hedonia over eudaemonia, since blind boxes are more materialistic purchases, and prior research has shown that materialistic purchases correlate more with hedonic happiness (Van Boven & Gilovich, 2003). On top of this, purchasing blind boxes “can also be considered a form of leisure activity” (Wei & Yu, 2025). Hence, blind box consumption activities seem to lean more towards carefreeness than meaning which would suggest higher correlation with hedonia according to Huta and Ryan’s study.

Method

This study replicates parts of the quantitative correlational study laid out by Huta and Ryan (2010). Huta and Ryan’s 2010 study: “Pursuing Pleasure or Virtue: The Differential and Overlapping Well-Being Benefits of Hedonic and Eudaimonic Motives” aimed to research the impacts of an individual’s activities on their hedonic and eudaimonic happiness. Because my study aims to measure the same thing, except substituting blind box consumption as the activity, parts of Huta and Ryan’s 2010 study are replicated in my methodology. This study is framed around my research question: *Through a quantitative correlational approach, how does blind box consumption impact hedonic and eudaimonic well-being in Singapore?* This study received ethical approval on November 26, 2025 by passing an Institutional Review Board. This document ensured informed participant discretion and anonymity; the documents also ensured that the participant knew they had full control over whether they wanted to participate in my research or not.

To measure hedonic and eudaimonic well-being from blind box consumption, I used a survey that contained three 7-point likert scales to collect data. My survey was split into five sections. The first section included an informed consent form (see Appendix A) . The second section included two qualifying questions that filtered out responses. Then, the third, fourth, and fifth sections were for each of the three quantitative likert scales. The following quantitative scales were the Hedonic and Eudaimonic Motives for Activities (Huta & Ryan, 2010), Diener and Emmon’s 1984 Positive and Negative Affect scale, and the Hedonic and Eudaimonic Meaningfulness scale developed by Huta and Ryan in their own 2010 study. All were developed as 7-point likert scales, with 1 indicating zero agreement or resonance with the given statement, and 7 meaning very high agreement or resonance with the given statement. The Hedonic and Eudaimonic Motives for Activities, or the HEMA, was used to measure hedonic and eudaimonic motives for blind box consumption and activity. Diener and Emmon’s 1984 Positive and Negative Affect scale measured the typical self-reported well-being of oneself to see if they exhibit positive or negative emotional responses from blind box consumption. Thirdly, Huta and Ryan’s Meaningfulness scale measured hedonic and eudaimonic well-being from blind box consumption. These scales are replicated from Huta and Ryan’s study that measured the impact of one’s activities on hedonic and eudaimonic well-being, validating my replication of parts of their study. Since my study uses these scales to measure the hedonic and eudaimonic well-being

from the specific activity blind box consumption, some aspects of Huta and Ryan's study were irrelevant to what my study aims to focus on. For example, Huta and Ryan also measure the presence of meaning in life using a life satisfaction scale from Steger et al. (2006). However, since my study focuses on well-being from just blind box consumption and activity, measuring a broader life satisfaction is not something to be measured.

The participants in this study were selected randomly from the popular "313 Somerset Mall" in Singapore on Orchard Road. This location was selected due to its popularity and generous availability of authorized retailers and resellers that sell blind box products. Since this study aims to gather an overall understanding of blind box activity using a Singapore sample, there was no specific population I was looking out for— just those who have experience with blind box consumption.

Participation was voluntary and those who participated were aware that they had complete control whether or not they wanted to partake in the study in order to adhere to the Institutional Review Board. An informed consent statement is embedded in the survey used for the study. This consent statement clearly stated that participation is voluntary and the participant can stop at any point during the process. Also, this study does not require any form of personal information, thus ensuring that personal information will not be collected and hence not shared. Once agreed to, the participant is able to move on to the next section of the survey:

Qualifications to participate. For the first qualification, participants had to be 18, so they would not need parental consent before participating, as this would be difficult to obtain from randomly selected participants, especially if they were not currently accompanied by a parent or guardian. For the second qualification, participants must have purchased a blind box at least once before, ensuring that the individual has valid opinions and experience with blind boxes. These qualifications were determined by the participant responding a "Yes/No" to the following statements in the survey: (1) "I have purchased a blind box at least once in my life," and (2) "I am 18 or older." If one or both qualifications received a "No," the survey would lead to a submission page that informed the individual that they did not pass the qualification to fully participate in the survey, and they would not get to fill out the actual scales. If the participant passes both qualifications, they will be led to the actual scales that measure well-being from blind box activity.

Finally, to analyze my data, I found 95% Confidence Intervals (95% CI) through T-Tests for each item on the survey. Across the three scales, there were 40 items, so I ran 95% Confidence Interval T-Tests for all with the help of the TI-Nspire CX Student Software.

Pilot Study

Prior to the final round of data collection, I ran a pilot study primarily to run through the data collection process and to see if there were any issues with the survey. My cohort changed to convenience sampling, as I had classmates fill out the survey. As this convenience sample consisted of high school peers, the '18 years or older' qualification was waived. In my pilot, I gathered nine responses, one of which the individual indicated they had not purchased a blind

box before, so their answer was excluded from the data analysis. During the time I ran my pilot, my data analysis method was not fully solidified yet. That said, from calculating the average scores (see Appendix B) of each likert scale, I gathered an idea of where my pilot cohort stood. The results from the pilot showed a higher resonance with hedonic motivation, positive emotions, and hedonic meaning. So from my pilot study, my results showed agreement with my hypothesis. However, because my data analysis method was not identified yet, these conclusions might not have been fully accurate. This pilot was essential in helping me gain an understanding of what the data collection process would look like and ensured that participants did not have any trouble with the survey.

In both my pilot and my real study, since I collected data on people's feelings and emotions, I assumed that the participants were truthful in their answers. Additionally, I am assuming that their answers reflect the feelings and emotions they feel from blind boxes only, not outside factors too. I made assumptions to keep in mind when looking at my data.

Findings

In total, I got 17 responses to my survey. However, four responses were not included in the data analysis as they did not meet the qualifications to participate. All four responses that were excluded from the survey were under 18 years of age, and thus did not qualify to participate. Hence, I got 13 responses that were eligible for data analysis.

Hedonic and Eudaimonic Motives for Activities Scale

For the Hedonic and Eudaimonic Motives for Activities (HEMA) scale, the scale is a mix of both hedonic motive items and eudaimonic motive items. In the survey, these items are mixed.

Figure 1. 95% CI from Hedonic Motive Items from HEMA Scale Table

Hedonic Motive	Sample Size (n)	Mean (M)	Margin of Error	95% CI Lower	95% CI Upper
			(ME)		
Seeking to take it easy	13	5.69	0.67	5.02	6.36
Seeking enjoyment	13	6.92	0.17	6.76	7.09
Seeking relaxation	13	5.23	0.82	4.41	6.05
Seeking fun	13	7.00	0.00	7.00	7.00
Seeking pleasure	13	6.77	0.27	6.50	7.03

As seen in Figure 1, hedonic motives show consistently relatively high mean values, ranging from 5.23 to 7.00, with relatively narrow confidence intervals, suggesting strong agreement among my sample. In particular, the item "seeking fun" yielded a mean of 7.00 with a margin of error of 0, showing no variability and unanimous maximum endorsement across all

respondents. Other hedonic items, such as “seeking enjoyment” ($M = 6.92$, 95% CI [6.76, 7.09]) and “seeking pleasure” ($M = 6.77$, 95% CI [6.50, 7.03]) further reinforce the idea that blind box activity and consumption has high correlation with hedonia.

Figure 2. 95% CI from Eudaimonic Motive Items from HEMA Scale Table

Eudaimonic Motive	Sample Size (n)	Margin of Error		95% CI Lower	95% CI Upper
		Mean (M)	(ME)		
Seeking to pursue excellence or a personal ideal	13	3.31	0.52	2.79	3.82
Seeking to use the best in yourself	13	3.23	0.61	2.62	3.84
Seeking to develop a skill, learn, or gain insight	13	2.38	0.68	1.71	3.06
Seeking to do what you believe in	13	3.54	0.68	2.86	4.22

In contrast, Figure 2 shows that the eudaimonic motives from the HEMA scale demonstrate lower mean values, ranging from 2.38 to 3.54, along with wider confidence intervals, indicating greater variability in responses. For example, “seeking to develop a skill, learn, or gain insight” had the lowest mean ($M = 2.38$, 95% CI [1.71, 3.06]), suggesting that in my cohort, they generally do not associate blind box consumption with personal growth or learning. Similarly, items related to purpose and self improvement, such as “seeking to pursue excellence or a personal ideal” ($M = 3.31$) and “seeking to use the best in yourself” ($M = 3.23$), were comparatively low.

Diener and Emmon’s Positive and Negative Affect Scale (1984)

For Diener and Emmon’s Positive and Negative Affect scale (PANAS), Figures 3 and 4 show that there is a notable difference between the positive and negative feelings that come with buying a blind box. The order of the positive and negative items were randomized on the survey.

Figure 3. 95% CI from Positive Affect Items from Diener and Emmons 1984 PANAS Table

Positive Affect	Sample Size (n)	Mean (M)	Margin of Error		95% CI Lower	95% CI Upper
			(ME)			
Delighted	13	6.85	0.23		6.62	7.07
Contented	13	5.46	0.63		4.83	6.10
Glad	13	4.92	0.63		4.3	5.55

Joy	13	6.92	0.17	6.76	7.09
Happy	13	6.85	0.23	6.62	7.07
Enjoyment/Fun	13	6.92	0.17	6.76	7.09
Pleased	13	6.00	0.49	5.51	6.49

Shown in Figure 3, Positive Affect items consistently exhibit elevated mean values, ranging from 4.92 to 6.92, accompanied by relatively narrow confidence intervals. The emotions "joy" (M = 6.92, 95% CI [6.76, 7.09]), "enjoyment/fun" (M = 6.92, 95% CI [6.76, 7.09]), and "delighted" (M = 6.85, 95% CI [6.62, 7.07]) are especially high, showing that participants often have very strong positive emotional responses. The small margins of error suggest that all of the respondents agree with each other. That said, items "contented" (M = 5.46, 95% CI [4.83, 6.10]) and "glad" (M = 4.92, 95% CI [4.30, 5.55]) were on the lower side relative to the other Positive Affect items.

Figure 4. 95% CI from Negative Affect Items from Diener and Emmons 1984 PANAS Table

Negative Affect	Sample Size (n)	Mean (M)	Margin of Error (ME)	95% CI Lower	95% CI Upper
Angry	13	1.62	0.58	1.03	2.2
Annoyed	13	2.15	0.42	1.74	2.57
Fear/Anxiety	13	1.08	0.17	0.91	1.24
Gloomy	13	1.00	0.00	1.00	1.00
Frustrated	13	2.15	0.34	1.82	2.49
Sad	13	1.31	0.29	1.02	1.6
Depressed	13	1.00	0.00	1.00	1.00

On the other hand, Figure 4 shows that the Negative Affect items have much lower mean values, between 1.00 and 2.15. Both items "gloomy" and "depressed," have a mean of 1.00 with no variation, which means that participants consistently reported very little negative emotional states. Other negative emotions, including "fear/anxiety" (M = 1.08, 95% CI [0.91, 1.24]) and "sad" (M = 1.31, 95% CI [1.02, 1.60]), persist at low levels with relatively narrow confidence intervals, indicating that negative affect is both constrained and uniform among participants.

Huta and Ryan Hedonic and Eudaimonic Meaningfulness Scale (2010)

The Meaningfulness Scale results presented in Figures 5 and 6 reveal a distinction between hedonic and eudaimonic meaningfulness associated with blind box consumption.

Figure 5. 95% CI from Hedonic Items from Huta and Ryan's 2010 Meaningfulness Scale

Hedonic Meaningfulness	Sample Size (n)	Mean (M)	Margin of Error (ME)	95% CI Lower	95% CI Upper
Detached from my troubles	13	4.38	0.91	3.48	5.29
Easygoing	13	5.08	0.84	4.24	5.91
Carefree	13	5.00	0.7	4.30	5.70
Free of Concerns	13	4.23	0.79	3.44	5.02

In Figure 5, hedonic meaningfulness items demonstrate moderate to moderately high mean values, ranging from 4.23 to 5.08, with relatively consistent confidence intervals. Items such as “easygoing” (M = 5.08, 95% CI [4.24, 5.91]) and “carefree” (M = 5.00, 95% CI [4.30, 5.70]) suggest that participants experience a sense of relaxation, ease, and freedom from concerns. That said, margins of error are wider than those observed in the hedonic motive data, showing that hedonia in this collective sample, is more strongly identified with motivation for blind box activity, while comparably less strongly identified with actual meaning.

Figure 6. 95% CI from Eudaimonic Items from Huta and Ryan's 2010 Meaningfulness Scale

Eudaimonic Meaningfulness	Sample Size (n)	Mean (M)	Margin of Error (ME)	95% CI Lower	95% CI Upper
Valuable	13	4.62	0.76	3.85	5.38
I could see how they all added up	13	3.54	1.04	2.5	4.57
Precious	13	5.31	0.8	4.51	6.1
Playing an important role in some broader picture	13	2.46	0.53	1.93	2.99
Dear to Me	13	4.69	0.87	3.82	5.56
Meaningful	13	4.15	0.69	3.46	4.84
Something I could Treasure	13	4.38	0.87	3.51	5.26
They contributed to various aspects of	13	2.69	0.52	2.18	3.21

myself					
Making a lot of sense to me	13	2.38	0.8	1.58	3.19
They contributed to my community and broader world	13	1.62	0.39	1.22	2.01

Lastly, Figure 6 presents Eudaimonic Meaningfulness items that show more variability and the widest range of mean values, from 2.46 to 5.31 in this study. Certain items, such as “precious” (M = 5.31, 95% CI [4.51, 6.10]) and “valuable” (M = 4.62, 95% CI [3.85, 5.38]), suggest that participants attribute personal significance to their experiences. However, other items related to broader purpose and contribution, including “playing an important role in some broader picture” (M = 2.46, 95% CI [1.93, 2.99]) and “they contributed to various aspects of myself” (M = 2.69, 95% CI [2.18, 3.21]), are notably lower. This indicates that while blind box consumption may hold some personal value, it is less strongly associated with deeper meaning, self-development, or contribution to a larger context.

New Understanding

In my study, my findings showed that blind box consumption correlated with relatively high levels of hedonic motivation throughout all items on the HEMA scale. Meanwhile, blind box consumption did not have relatively high ratings for eudaimonic motivation compared to that of hedonic motivation. The eudaimonic items showed a mean range from [2.38, 3.54]. Contrasted with the hedonic motivation items with lower margin of error, the results show that we can be more confident when making the statement that blind box consumption is more influenced by hedonic motivation.

For Diener and Emmon’s Positive and Negative Affect Scale, the results showed blind box consumption having high levels of positive affects and low levels of negative affects, reinforcing that blind box consumption is an activity that correlates with positive feelings and emotions. This is important because it shows that blind box consumption has impacts positively on consumers happiness.

While I found relatively high agreement with blind box consumption being more hedonically motivated, Huta and Ryan’s Hedonic and Eudaimonic Meaningfulness scale showed a more nuanced pattern. While blind box consumption does have a more consistent high correlation with hedonic happiness than eudaimonic happiness, participants still show that blind box consumption also impacts eudaimonic happiness. For the ten Eudaimonic Meaningfulness items, about half showed relatively low mean scores ranging from [1.62, 3.54]. That said, the other half of the eudaimonic items showed moderate to high means. For example, the eudaimonic meaningfulness item “Precious” had the highest mean of 5.31, which actually was the highest average score from the entire Meaningfulness scale. Similarly, a few of the other eudaimonic items on the Meaningfulness scale showed relatively high mean scores and wide

confidence intervals such as “valuable” ($M = 4.62$, 95% CI [3.85, 5.38]). This shows that certain eudaimonic items showed mixed feelings about how blind consumption impacted their eudaimonic well-being. Therefore, we cannot be too confident about making a generalization in my study that blind box consumption has little or high impact on eudaimonic happiness because participants are across the board in terms of their answers. That said, these results do suggest that blind box consumption can have some impact on eudaimonic happiness, suggesting that purchasing blind boxes might not be a purely material purchase.

Discussion

My findings can be applied to the existing scholarly conversation surrounding consumer happiness and how different kinds of purchase impact different kinds of happiness. In Van Boven and Gilovich’s “To Do or to Have? That Is the Question,” they found that materialistic purchases are more highly correlated with hedonic happiness, while experiential purchases are more highly correlated with eudaimonic happiness. Since blind boxes are considered materialistic purchases, Van Boven and Gilovich’s findings suggest that blind boxes should also highly correlate with hedonic well-being. However, my findings found not only does it correlate with hedonia, it also has an impact on eudaimonia. Taken with Van Boven and Gilovich’s findings, this potentially suggests that blind boxes might also create a sort of experience for the consumer which would explain the eudaimonic impact despite blind boxes being more material in nature. A possible explanation from previous research on blind boxes could answer this, as research found that blind box activities can create connections between collectors and other fellow consumers (Wei & Yu, 2025; Gunarian et al., 2024). These kinds of experiences and the development of relationships from a shared activity could be what ties eudaimonia and blind boxes together.

Also, McMahan and Estes explained that both hedonia and eudaimonia are necessary for a more fulfilling and higher rate of well-being in an individual. Therefore, since my findings suggest that blind box activity shows correlation with both hedonic and eudaimonic happiness, put together with Huta and Ryan’s findings, blind boxes are purchases that could contribute to a higher rate of well-being.

With my findings, I have addressed the gap in the literature surrounding blind box consumption and its effects on consumer happiness in Singapore. Since there is no particular research in the field that has touched on this yet, my research can serve as a basis for further investigation on a larger scale that would produce more accurate results. My findings suggest that blind box consumption impacts not just hedonic happiness, but also touches on eudaimonic happiness. This is important to consider because it might suggest ties between blind boxes having experiential aspects to the purchase.

Limitations

My study includes a few limitations that should be properly addressed. For example, one major limitation of my study is that since I was collecting data through surveying people in public areas, I was the one choosing potential participants. While I tried my best to select

randomly, unconscious biases still could have occurred when choosing participants to ask. Another limitation and something to consider from my study is that since my method included the decision to leave out a cohort of participants that are under the age of 18, there is a large population of blind box consumers that was not included in my study. Therefore, if further research is done and minors are included in the study, my findings from this research will no longer be applicable. Additionally, there are also limitations with the method. For example in the Huta and Ryan Meaningfulness scale, I chose to leave out scales that evaluated fuller life satisfaction from activities as this was not relevant to the more instant gratification I was trying to measure from blind box consumption. Therefore, the full methodology of Huta and Ryan's study was not replicated, which could leave potential discrepancies that would make my findings different to theirs. Another limitation of my method is the imbalance in the Huta and Ryan Meaningfulness Scale, where more eudaimonic items than hedonic items may have led to uneven measurement. This could make comparisons between hedonic and eudaimonic outcomes less reliable, as one construct is more extensively represented than the other. Since this was replicated from Huta and Ryan's study, this also applies as a limitation to their method and the method that I chose.

Implications

On the other hand, my research and findings provide several implications for blind box consumers and companies. Firstly, since my findings show that blind box consumption is more hedonically motivated, it can serve as something to keep in mind when consumers think about purchasing blind boxes. Since hedonia is more associated with short-term bursts of pleasure, it is helpful to keep this in mind in the case of impulse buying which can lead to overconsumption. Therefore, it is valuable to the consumer that they keep in mind their motives behind purchasing. Is this purchase something that is acted on impulse or something more well thought out? Additionally, these implications can be applied to companies as well. There could be measures that limit how many blind boxes a customer can get with the knowledge that consumption is mainly hedonically driven.

Conclusion and Future Direction

My implications around blind box regulation have actually been discussed in Singapore recently. On February 12, 2026, Home Affairs Minister K. Shanmugam came public about the Singapore government drafting blind box regulations due to suspected association with gambling behavior. These regulations would potentially include additional regulatory measures "such as age-based restrictions and probability disclosure, [but] will be considered subsequently, as more time is needed to study them" (Sun, 2026). This shows the relevance and emergence of blind boxes, not only in Singapore, but globally too. The fact that Singapore has begun to flag blind boxes as potential risk for gambling tendencies shows that more research is needed in the field to make further conclusions. Therefore, I suggest that future research should not only look into

blind box consumption and their effect on happiness on a larger, more global scale, but also whether blind boxes have other risks associated with its practices.

To further explore this, I came in contact with an expert advisor during my research process. Together, we discussed how other communities could be affected by this potential risk. We discussed that if more research supports the development of unhealthy gambling behavior due to blind box activity, it would be important to look into how children and teens' exposure to blind boxes can affect them. With this, it would be important for parents to understand potential harm or risks associated with blind box practices, so that they are aware of how this could affect their children. This would support Singapore's drafting of blind box regulations— especially if they are considering placing age restrictions on these products.

All in all, my research on blind boxes barely scratches the surface. Blind boxes are products that have been implemented all over the world, making their appearance on the global stage. Many companies have implemented blind box products as successful marketing tools to appeal to its customers like the popular chocolate drink brand Milo who put randomized plush figurines in some of their packaged products to drive consumer demand (Li, 2026). With its recent global popularity spike, blind boxes are an area that the research community should continue to look deeper into to explore the psychological effects it has on consumers.

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Is Cryptocurrency the Future of the World's Reserve? By Hayoon Jung

Abstract

Recent policy decisions by the United States, such as withdrawal from key international agreements, signal a shift in its approach to international welfare. This has prompted some scholars to question the long-term stability of the U.S. dollar as the world's primary reserve currency. Should it continue to occupy this dominant role, given the lack of predictability? Although alternative fiat currencies such as the euro, British pound, and Japanese yen are often proposed as alternatives, their dependence on national governments exposes them to similar political and economic vulnerabilities. Consequently, attention has increasingly turned toward non-fiat alternatives, particularly private digital currencies such as cryptocurrencies, which operate outside of centralized state control.

This paper evaluates the viability of cryptocurrency as a potential replacement for the U.S. dollar in the global reserve system. Through a review of existing economic and digital security literature, it investigates characteristics required of a reserve currency, including stability, security, and scalability. Results indicate that cryptocurrencies fall short on several of these dimensions. Their limited supply mechanisms can prevent liquidity, and price volatility undermines their function as stable stores of value, also rendering cryptocurrency less trustworthy than fiat currency. In addition, security risks further weaken their reliability. These limitations suggest that cryptocurrencies are not currently viable substitutes for the U.S. dollar as the world's reserve currency.

Recent decisions by the current U.S. administration, such as withdrawing from the Iran Deal, Paris Climate Deal, and the World Health Organization, signals a shift in how the United States will position itself on the global stage. Doubtful of the benevolence of the new U.S. policies, some experts are asking: Should the U.S. dollar still serve as the world's reserve currency? While it is tempting to replace the U.S. dollar with the euro, British pound, or Japanese yen, other fiat currencies will likely encounter the same issue if policies undergo a similar shift. This is why non-fiat currencies, such as private digital currencies (e.g. cryptocurrencies) that are not linked to a government entity, have emerged as an option. The purpose of this paper is to evaluate whether cryptocurrency is a good candidate for replacing the U.S. dollar as the world's reserve currency. Review of economic analysis and digital security literature reveals that cryptocurrency may not be the holy grail of currency that some believe it to be. Generally speaking, cryptocurrency supply is limited, value is too volatile, and they carry immense security risks. These issues make them a poor choice to replace the U.S. dollar.

Qualities of World Reserve Currency

The U.S. dollar has been the world reserve currency since the Bretton Woods agreement of 1944 when 44 allied countries agreed to link their currencies to the dollar, which was backed by gold. Even after the dollar was de-linked from gold in 1971, it remained as the world reserve currency "due to the size and stability of the U.S. economy, its deep financial markets, and its

central role in global trade, especially commodity trading” (Emmert). As the world’s reserve currency, the U.S. dollar is the foreign currency held by governments or central banks as part of their foreign exchange reserves. This means it is used generally for the following:

- a) International payments as a medium of exchange
- b) Interventions in the national and global markets
- c) International investments and as a portfolio asset to hold excess central bank capital or otherwise function as a store of value

While there is some disagreement about what exactly the next world reserve currency should be, scholars agree that the currency should be:

- a) Stable and able to function as a holding asset
- b) Liquid and widely circulating; that is, having an accessible market to move quickly and easily for international transactions
- c) Able to expand and contract during times of crisis to meet global demand
- d) Unaffiliated with government entity to avoid manipulation for policy goals

Generally, privately owned digital currencies fail to meet these criteria, save for the fourth, which has its own pitfalls and will be discussed in conjunction with the others below.

Security Risks

If we follow the assertion that the ideal world’s reserve currency should be one that is de-linked from a government entity, cryptocurrencies become an interesting option because they are often decentralized. Cryptocurrency exists at the intersection of economics, computer science, and cryptography, meaning it is produced and behaves vastly differently from fiat currency. In fact, it is not a currency in the traditional sense; it is a digital token created by cryptographic algorithms and then moved across cyberspace using peer-to-peer networking protocols. Because it is secured by cryptography, it is vulnerable to digital attacks.

Take, for example, Bitcoin, the first and most widely used decentralized cryptocurrency. To use bitcoins, an individual needs to install a wallet on their desktop or mobile device, and this wallet stores a set of public and private keys exclusive to the owner which are used to prove ownership and authorize transactions. While this can prevent counterfeit, it also means that Bitcoin is particularly vulnerable to cyber theft. If the wallet is compromised due to a cyber attack, stolen device, or other online threats, “the client will suffer immediate and irrevocable monetary losses” (Conti et al.). To lessen the risk, some individuals choose to keep their bitcoins with online exchanges. However, online exchanges are frequently targeted by cybercriminals and the losses are usually unrecoverable. For instance, Mt. Gox, an online exchange that handled about 70% of all bitcoin transactions during its prime, was subject to distributed denial of service attacks (DDoS) and lost hundreds of thousands of bitcoins (the equivalent of 450 million USD),

forcing the exchange to suspend trading and file for bankruptcy in 2014 (Feder et al.). Similarly, the Ethereum network was subject to a security breach due to weaknesses in its base code. In an attempt to retrieve stolen assets, and prevent hackers from using stolen cryptocurrency, they created a fork of the currency (a snippet of code to distinguish it from a new currency), a decision which led to considerable user backlash (Cryptopedia Staff). These are only two of the numerous attacks on exchanges, not to mention the amount of cryptocurrency loss due to mismanagement and fraud.

Due to its nature, cryptocurrency is even more vulnerable to cyberattacks than fiat currency, undermining user confidence in it as an asset and medium of exchange. Secondly, it is apparent that securing cryptocurrency requires cutting edge technology and interdisciplinary understanding of computer science and cryptography; given the digital gap between nations, some countries will be unable to secure cryptocurrency if it were used as a world reserve currency. This would limit its market and usage for international transactions. Finally, even though it is not linked to a government, cryptocurrency is still managed by some entity that can have its own goals and suffer from mismanagement, leading to manipulation or loss of currency.

Price Volatility

Cryptocurrency is also not stable, further undermining public confidence in its ability to store value. Clark and Mihailov (2019) conducted a quantitative study of ten private cryptocurrencies from their creation until the end of 2018. Results are shown in Figure 1 below.

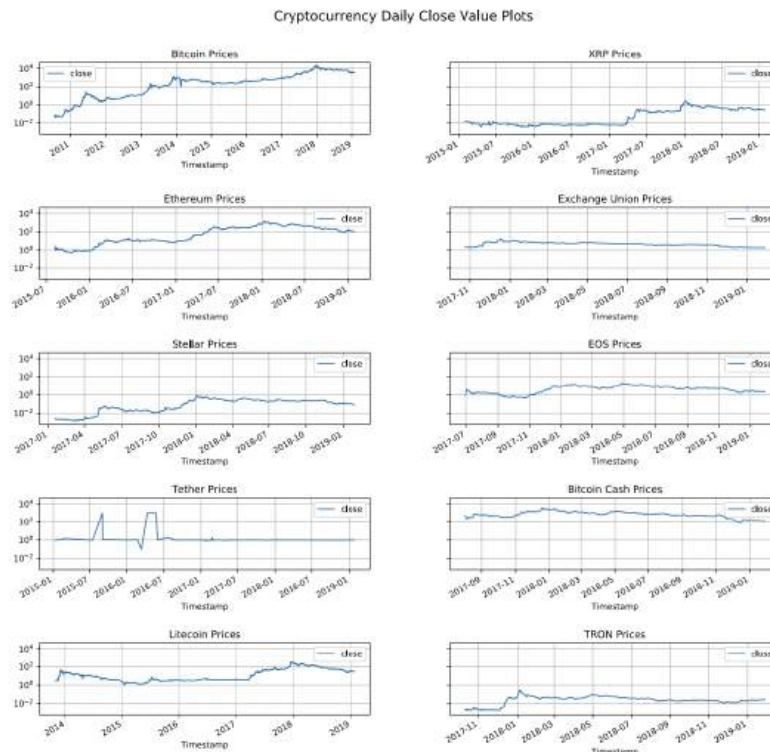


Figure 1: Daily USD Prices (log scale) of Top 10 Cryptocurrencies (Clark and Mihailov)

The cryptocurrencies have all had price dynamics that can be described as volatile. For example, the USD price of Bitcoin has varied from 10^{-1} to 10^{-4} since its creation, and Tether has had two sudden spikes in price. Although Exchange Union, Bitcoin Cash, and TRON seem to be the most stable, it is important to note that they have only existed since 2017, so their long-term behavior is unknown.

For comparison, Figure 2 below shows the USD prices for traditional reserve currencies such as gold. The USD price range enjoys less volatility, with the price of gold hovering around 10^3 over roughly the same eight year period. The basket of major world currencies and IMF's Special Drawing Rights (SDR) are even more stable.

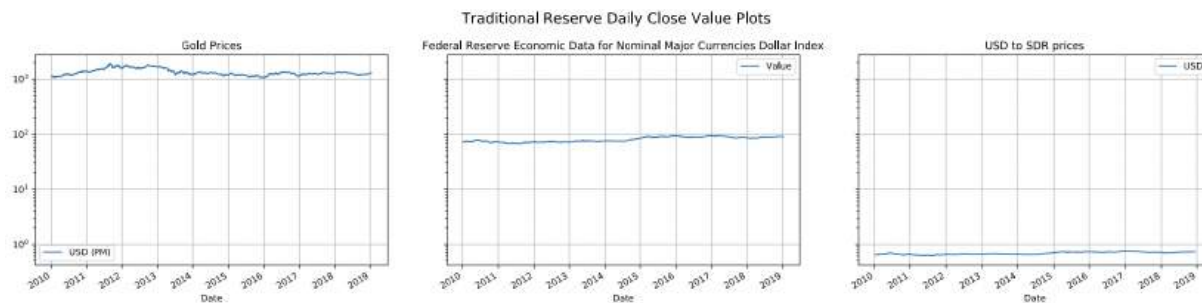


Figure 2: USD Prices (log scale) of Traditional Reserve Currencies (Clark and Mihailov)

Therefore, even though the information we have about cryptocurrencies is limited due to how recently some were created, we cannot say their market behavior is predictable. This volatility is another reason why replacing the U.S. dollar with a private digital currency would be a major confidence risk, making it unsuitable as a world reserve currency.

Limited Supply

Third, cryptocurrency is unable to expand and contract because many have a capped supply. For example, the maximum amount of bitcoins is capped at 21 million, a number which was arbitrarily chosen by its creator. As of 2025, about 20 million bitcoins are already in circulation, so when the final 1 million are mined, no new bitcoins will be created because the hard-coded rules prevent this from happening (Kraken). On one hand, the hard limit makes it less susceptible to inflation than fiat currency, which can be printed at the discretion of central banks. However, the limited supply also makes it a poor currency for managing the money supply.

Keynesian economists argue that increasing the money supply could improve economic growth by reducing interest rates and driving investment (Cesaratto and Pariboni). For example, Bajrami and colleagues in an analysis of transitional economies in Western Balkan countries, finds that increasing the money supply “improves liquidity, facilitating investment, and promoting consumption.” Thus, the inability to change the amount of available cryptocurrency means it is unable to be used to regulate economies. This is a critical failure as the world’s reserve currency is often used for major economic interventions.

Though it could be theoretically possible for Bitcoin's maximum supply limit to change if the creator and the majority of users came to a consensus, this would undermine its original design and its integrity, making it more of a risk to investors. It is also possible to consider a cryptocurrency that does not have a hard cap such as Ethereum; however, the lack of hard cap also makes it more susceptible to price volatility, which would also undermine confidence and stability.

Conclusion

For the reasons described above, it is not feasible for privately owned digital currencies, specifically cryptocurrencies, to become the world's reserve. Despite being delinked from a government entity, cryptocurrency still suffers from high volatility, cyber attacks, and mismanagement, all of which undermine its ability to be used as a store of value and public trust. Additionally, the hard-limit on many cryptocurrencies means it is not scalable; as such it is unfit for large scale international economic interventions or other such emergencies that require manipulating the money supply. This is a critical failure because this is a major function of a reserve currency. Even if cryptocurrency is not a viable option, however, the reasons for de-linking the world reserve currency from political entities, especially the U.S. government, are valid. Future research should consider other alternatives more closely. For example, the Better Than Stable Coin (BTSC), is owned and operated by a nonprofit organization and pegged to a basket of most freely convertible currencies. There is also potential for central banks to produce their own digital currencies and allow households and firms to open accounts with them, but central banks would have to undergo extensive development for this to happen. Future scholarship focusing on the potential performance and pitfalls of such alternatives would pave the way for a more sustainable financial future on a global scale.

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The Science of Wormholes: Exotic Matter and the Plausibility of Human or Object Space Travel By Noelle Villani

Abstract

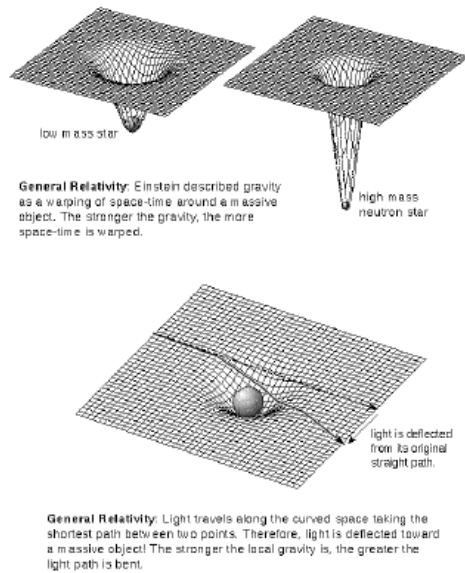
Space travel is often imagined in science fiction through dramatic scenes involving hyperspace, faster-than-light travel, and shortcuts across the universe. Movies such as *Star Wars* show spaceships moving through hyperspace to cross huge distances in a short amount of time. While this idea may seem fictional, wormholes offer a theoretical way to imagine how shortcuts through spacetime could be possible. What if we actually could become a Jedi and move that fast through space? This could all be theoretically possible through wormholes. Wormholes, also known as Einstein-Rosen bridges, are hypothetical structures that may connect two distant points in space and time. In order for humans to travel through traversable wormholes, it is predicted that exotic matter is required. This paper explores the theoretical structure, types, and possible uses of wormholes, as well as the conditions needed to make them stable and traversable. It also discusses the role of exotic matter and the challenges involved in using wormholes for human space travel. By examining these theories and diagrams, this paper explains how wormholes could expand our understanding of spacetime and possibly help humans explore distant regions of the universe.

The Plausibility of Human Space Travel via Wormholes

In general relativity, is exotic matter required to sustain traversable wormholes, and how does this compare with wormhole solutions in modified theories of gravity?

Wormholes, also known as Einstein-Rosen bridges, are theoretical structures with physical features in spacetime, predicted by Einstein's Theory of General Relativity. They connect two far points in space, which acts like a shortcut to get from one point in space to another. This unique structure could allow for space travel faster than the speed of light, however they are very unstable and are thought to require exotic matter to stay open. Wormholes currently aren't well studied because they haven't been confirmed observationally, but could help explain the fundamental nature of the universe as well as help to bridge the gap between general relativity and quantum mechanics. Einstein claimed gravity was not a force at all but rather the result of space-time curved by the presence of mass.

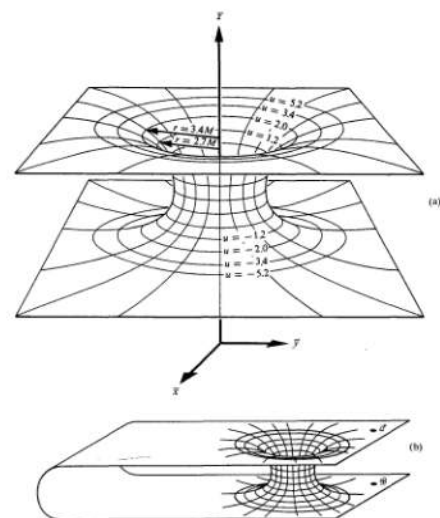
General Relativity explains how gravity warps spacetime based on an object's mass. The more mass an object has, the greater of an impact it has on the space around it. Einstein's Field Equation ($G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$) solutions have predicted the existence of black holes, which have now been confirmed observationally. A black hole is a singularity in spacetime where gravity is so strong that not even information can escape. Black holes and wormholes are both solutions to Einstein's Field Equation,



where a wormhole can connect two black holes, or a black hole and a white hole. In this paper, we will discuss the theoretical existence of wormholes and how they affect spacetime around them. We will expand on the types of wormholes, and if exotic matter is truly needed for traversable vs. non-traversable wormholes. We will also discuss the two solutions to wormholes including a black hole vs. white hole. We want to prove the plausibility of human space travel through great distances, which are possibly faster than the speed of light, via the use of traversable wormholes. Prior to Einstein's Theory of General Relativity, the accepted model for gravitational interactions was the Newtonian Model, which describes gravity as an invisible force pulling two objects together.

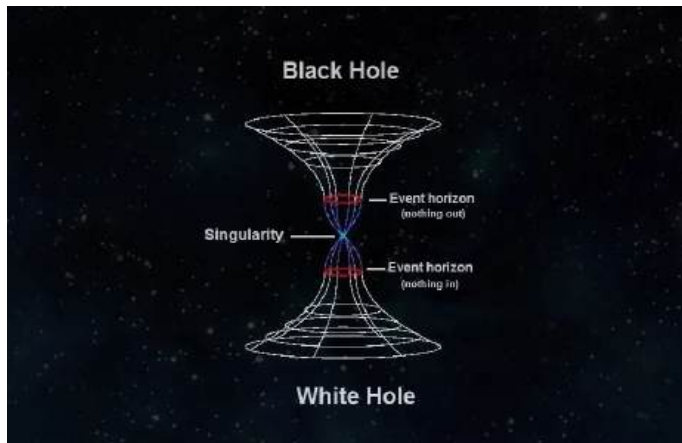
While this model works for general astrophysical cases, for more extreme physics General Relativity is needed to explain gravitational behavior. General Relativity predicts how gravity warps spacetime based on an object's mass (DOE 2021). Einstein's Theory of General Relativity predicts the existence of Black Holes and Wormholes, and is the foundation for understanding how wormholes interact with spacetime. Solutions to Gravitational Field Equations predict that wormholes exist. According to Stephen Hawking, macroscopic wormholes may provide the mechanism for black holes to evaporate and disappear completely, while microscopic wormholes seem to have an important effect on physical constants, particularly the cosmological constant. General relativity relates an object's mass to its effect on spacetime; as shown in Figure 1. A small object with a large mass has a greater effect on the spacetime around it in comparison to a larger object with less mass. For example, a high mass neutron star would have a greater impact on its space than a star with less mass. Therefore, the radius of the celestial body doesn't directly correlate to how massive the object is. The first wormhole solution to Einstein's Field Equation is the Schwarzschild wormhole which is interchangeable with the second solution, the Einstein-Rosen Bridge.

Einstein-Rosen Bridges are a solution for a specific type of wormhole that is considered non-traversable. The Einstein-Rosen Bridge theory states that two points in space can be connected by a shorter path of travel through a bridge, or tunnel, from one point to the other, also known as a wormhole. As shown in Figure 2, there is a point F (north of the bridge) and point X and Y (south of



the bridge). This diagram shows that the shortest distance between the two points is the bridge, or throat, in between the two points. In this theory, the traveler would travel faster than the speed of light to get from one point to another by going through the bridge, whereas if you went the long route without using the bridge, it would take an exceedingly longer time.

The basic foundation of an Einstein-Rosen Wormhole includes a black hole and a white

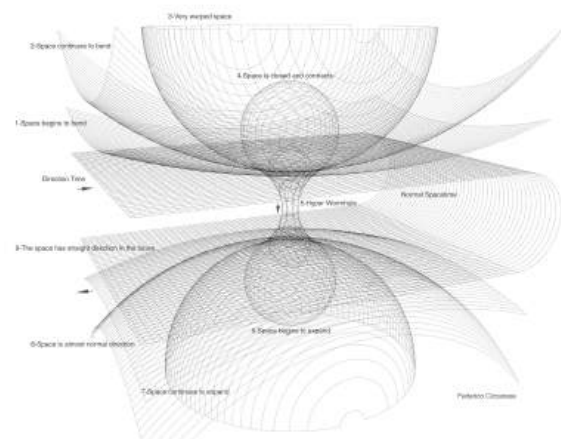


hole merging together, shown in Figure 3.

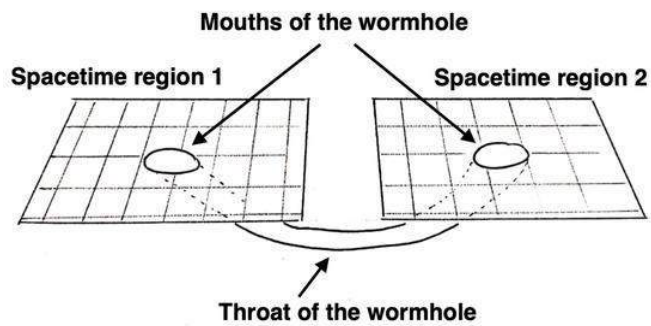
Another proposed solution is if two black holes merge together. Finally, the last proposed solution explores the idea that the wormhole connects a black hole to an alternate or parallel universe. The existence of wormholes is dependent on black holes which are also predicted by Einstein's Theory of General Relativity. Black holes have been observed, but it's unclear what lies on the other side. A wormhole is generally thought to be connected by two black holes, but there's

the theoretical case of a wormhole connecting a black hole to a white hole. To get through a

black hole one must travel through a white hole which could be connected to another black hole in another point of space, or another white hole. There is no observational confirmation of how wormholes are created. They are theoretical, which makes it difficult for scientists to fully understand the mechanics. As seen in Figure 4, the main theory is that wormholes may connect a black hole and white hole. This could hypothetically be because while black holes go backwards slower, white holes go forward and faster. Black holes also tend to pull objects into themselves, whereas white holes like to push things out. (We predict) that since black holes are black, and have an orange and yellow frame around itself, white holes are most likely white, but with a more purple and blue frame to it. This is because one possibility sucks in light, and the other spits out all the light.



Another key concept for wormholes is the Cosmic String and String Theory. String theory is that the distance from both spacetimes (or points in space) is so far away, but can still be faster than the speed of light to get to via a wormhole.



In the image, the throat of the wormhole can be as long as possible. One spacetime can even be in a completely different galaxy, and it would take barely a millisecond to get to. Cosmic string theory also helps keep wormholes traversable (able to go through). Cosmic strings “tug” on the

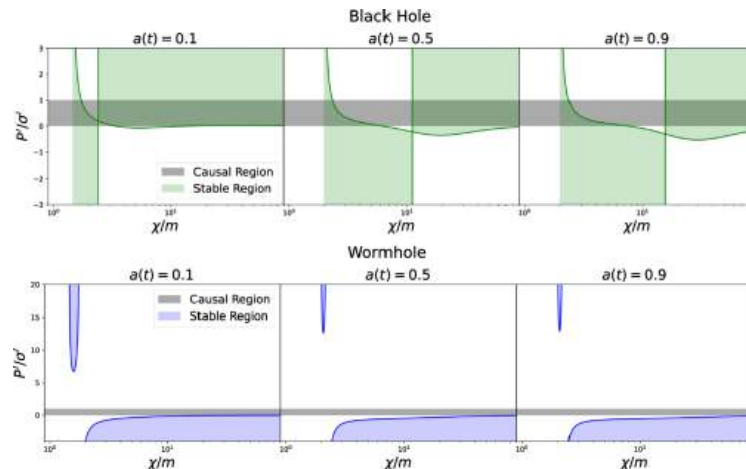
mouths of a wormhole by threading it. This, along with exotic matter, helps keep wormholes stable and open.

Exotic matter is a theoretical form of negative matter that is thought to keep wormholes open and help turn non-traversable wormholes traversable (Garattini 2021). It is a form of matter that is far from the “normal” matter we have seen here on earth, and has not been observed yet, hence why it is a theoretical area of wormholes that we study. In this definition, the term “exotic” means that the matter is made of different, and very rare, forms of elements. Other than normal matter, exotic matter is used like a popper, and normal matter is a puller. For instance, if we put exotic matter into a collapsing wormhole, it would push open the walls or throat) of the wormhole out, and since gravity seems to squeeze or push object together, the exotic matter would keep the wormhole open in response to that, whereas normal matter would just cause it to collapse faster.

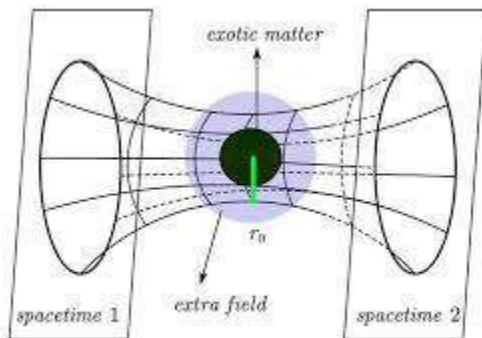
Wormholes are extremely unstable, which connects to the idea of **traversability**, or whether something could safely pass through them. Some wormholes, such as Einstein-Rosen bridges, are considered non-traversable because they would collapse too quickly for matter or people to cross. In contrast, Morris-Thorne wormholes are theoretical models designed to remain open long enough for travel. Garattini explains that “an Einstein-Rosen (ER) bridge is a non-traversable theoretical construct tied to the geometry of black holes, whereas a Morris-Thorne (MT) wormhole is a theoretical model specifically engineered to be traversable by matter” (Garattini, 2021, p. 2). In other words, a traversable wormhole would need to stay open at both ends without collapsing. A non-traversable wormhole, however, would close too quickly, likely destroying anything inside. This is why exotic matter is so important: its unusual negative properties could theoretically hold the wormhole’s throat open and make safe passage possible.

However, exotic matter may not be required in every possible wormhole theory. In standard general relativity, traversable wormholes usually need matter that violates certain energy conditions, which is why exotic matter is often treated as necessary (Visser et al., 2003). However, (Visser et al. 2003) also argue that some wormhole models may require only very small amounts of this unusual matter, meaning exotic matter might still be needed, but not in large quantities. Another possible scenario is modified gravity, where the rules of gravity are changed or expanded. In these theories, the wormhole could be supported by extra curvature

effects in spacetime instead of by exotic matter itself (Lobo & Oliveira, 2013). Therefore, exotic matter is probably required for traversable wormholes in standard general relativity, but some alternative theories suggest that wormholes might stay open through modified gravity, special spacetime geometry, or very small amounts of exotic matter rather than large quantities of it.



In general relativity, traversable wormholes are usually predicted to require exotic matter, but modified theories of gravity suggest this may not always be the case. Einstein's theory allows for the mathematical possibility of wormholes, including Einstein-Rosen bridges, also known as



Schwarzschild wormholes. These structures could connect two regions of spacetime, but they are considered non-traversable because they would collapse too quickly for matter, light, or spacecraft to pass through safely. In contrast, models such as the Morris-Thorne wormhole suggest that traversable wormholes could theoretically exist if their throats were kept open.

The biggest challenge is that most traversable wormhole models require matter with unusual properties, especially matter that violates the null energy condition. This is often called the exotic matter condition. Since exotic matter has not been observed, wormholes remain purely theoretical and are currently supported more by mathematical models than by physical evidence. However, wormhole research is still important because it helps scientists test the limits of general relativity and explore deeper connections between gravity, black holes, quantum theory, and the structure of spacetime. Although no wormhole has been observed yet, studying them helps us better understand how spacetime might behave and whether future discoveries could one day make travel across vast regions of the universe possible.

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Analyzing How Noticing Hostile Architecture Affects Midtown Manhattan Residents' Empathy Toward Homeless Individuals By Alexa Zohar

Abstract

This study investigates the relationship between exposure to hostile architecture and empathy toward individuals experiencing homelessness among residents of Midtown Manhattan. Hostile architecture, also referred to as defensive or exclusionary urban design, includes features such as segmented benches, anti-homeless spikes, barriers, and other structures intentionally designed to discourage certain uses of public space. While existing research has focused primarily on the identification, ethical implications, and social consequences of hostile architecture, limited research has examined how these designs influence public perception and empathetic attitudes. To address this gap, the study used a mixed-methods experimental survey design combining quantitative analysis with qualitative interpretation of participant responses. Participants were divided into two survey conditions in which they viewed either hostile or non-hostile urban environments before responding to Likert-scale empathy questions and short-answer prompts. Quantitative analysis included descriptive statistics and independent sample t-tests to compare empathy levels between groups, while qualitative responses were analyzed to identify recurring patterns in interpretation and emotional response. Findings revealed that participants exposed to non-hostile environments maintained higher levels of empathy, emotional concern, and support for homelessness-related assistance policies. In contrast, participants exposed to hostile architectural features demonstrated lower empathetic responses and frequently justified exclusionary designs as practical, functional, or necessary for maintaining order. The findings suggest that hostile architecture may normalize exclusion and reduce emotional concern toward marginalized populations. These findings indicate that hostile architecture functions not only as a tool of urban regulation, but also as a social signal that shapes public perception and reinforces attitudes surrounding belonging and exclusion in urban spaces.

Introduction/ Literature Review

Public spaces are often perceived as open, accessible environments designed for all members of society, yet in many cities, subtle design choices quietly determine who is able to occupy them. A person sitting on a park bench, resting near a storefront, or seeking shelter in a train station may encounter physical barriers that appear ordinary, but are intentionally designed to prevent usage. These features, ranging from segmented seating to obstructive surfaces, reflect a broader pattern in urban design where space is not only organized, but controlled. As cities continue to prioritize order, safety, and aesthetics, questions emerge about who benefits from these decisions and who is consistently excluded.

Within this context, researchers have examined hostile architecture, also referred to as defensive or exclusionary urban design, to understand how urban spaces are purposefully designed to control public behavior. Current research establishes that hostile architecture is not

accidental or solely aesthetic, but is an intentional strategy designed to regulate who occupies public spaces and how those spaces are used.⁸⁷ This group of scholarly studies assess how urban design discourages certain behaviors. Scholars analyze hostile architecture through observational studies, ethical frameworks, and urban theory, revealing how design functions as a subtle, yet powerful form of social control. For the purpose of this study, hostile architecture is defined as intentionally designed urban features that restrict or discourage specific uses of public space.

Defining and Identifying Hostile Architecture

Scholars first examine hostile architecture by documenting how it appears and operates within everyday urban spaces. In “Discipline and Order: Hostile Architecture in Copenhagen,” Egill Karlsson conducts field observations to identify exclusionary design features throughout Copenhagen. These include sloped surfaces, anti-skateboard studs, and metal barriers, particularly concentrated in high traffic areas, where there is greater access to public space and a higher concentration of unhoused individuals.⁸⁸ Rather than simply listing these observations, Karlsson situated them within broader discussions of urban discipline. He argues that hostile architecture regulates behavior by enforcing societal norms of belonging, revealing that these features are intentionally embedded in architectural planning practices that value order and aesthetics over inclusivity.

While Karlsson illustrates hostile architecture through field observations, Jessica Annan’s study of hostile architecture in Calgary, Canada, similarly investigates exclusionary design by combining in-person observations with visual documentation and urban theory. Annan identifies common exclusionary designs in transit spaces, such as segmented seating and discouraging fixtures, but she enhances this analysis by examining the use of public art as a tool of exclusion. Specifically, she analyzes strategically placed statuettes welded over heating vents, which were previously accessible to unhoused individuals seeking warmth.⁸⁹ Through this approach, Annan reveals how aesthetic justification can cover up exclusionary intent, demonstrating that hostile architecture can be both visually appealing and socially harmful. Together, Karlsson and Annan demonstrate that while hostile architecture may take different forms across the world, its underlying purpose of controlling access to public space through design, remains the same. These findings establish how hostile architecture is physically implemented, which provides a foundation for understanding its broader social and ethical consequences.

Ethical and Health Impacts on Vulnerable Populations

Building on these observational studies, scholars examine the physical, ethical, and health related consequences of hostile architecture, emphasizing that these designs are not accidental. This research mainly highlights how such urban designs unfairly target and exclude vulnerable populations, particularly individuals experiencing homelessness. Brendan McCreath argues that hostile architectural designs worsen existing vulnerabilities among people experiencing

⁸⁷ "Hostile Urban Architecture: A Critical Discussion of the Seemingly Offensive Art of Keeping People Away."

⁸⁸Karlsson, Egill. *Discipline and Order: Hostile Architecture in Copenhagen*.

⁸⁹Annan, Jessica. *An Uncomfortable City: A Community-Based Investigation of Hostile Architecture*.

homelessness by restricting access to public spaces, and causing both physical harm and ethical injustice. This is evident in features such as armrest-divided benches, locked trashcans, sealed ventilation grates, and obstructive surfaces, which are deliberately created to limit access to rest, warmth, and basic resources. These designs force unhoused individuals into more dangerous and isolated areas of the city, resulting in greater harm than they initially face.⁹⁰ This shift from documenting design features to analyzing their consequences deepens the scholarly conversation by emphasizing that hostile architecture does not just merely exclude those in poverty but also actively harms them.

This argument is reinforced by Andreas Albertsen in the *British Journal of Political Science* article “Anti-Homeless Hostile Design as Wrongful Discrimination,” which states that these architectural choices function as signals that communicate exclusion and determine who is unwelcome in public spaces. By preventing access to basic survival needs, hostile architecture transforms public areas into spaces that are only functionally accessible to socially “acceptable” populations, depicting that hostile architecture operates as a discriminatory practice embedded within urban design. The article further describes that “Notable examples are...adding spikes to surfaces to prevent rest; sprinkling pavements to soak those who might want to sleep there; and closing off ventilation grates to remove the option of obtaining heat.”⁹¹ These specific features illustrate how hostile architecture directly targets basic survival needs, deliberately creating conditions that exclude homeless individuals from public spaces. Together, these articles demonstrate that hostile architecture is a deliberate design strategy that targets basic needs and forces people to leave public spaces due to their lack of alternatives. While these studies emphasize the harm caused by hostile architecture, they also raise questions about how such exclusion is justified within urban design.

Aesthetic Justifications and Design Rationalizations

Beyond ethics and health, other scholars have been focused on the aesthetic rationalizations used to justify hostile design. Karl de Fine Licht argues that exclusionary features are often framed as harmless or necessary design choices, which hides the ethical responsibility of designers and planners. Features such as benches designed to prevent sleeping are not simple functional design elements, but intentional choices that shape how cities look and function.⁹² In “Hostile Architecture’ and Its Confederates: A Conceptual Framework for How We Should Perceive Our Cities and the Objects in Them,” de Fine Licht asserts that aesthetic rationalization allows cities to justify exclusion while avoiding accountability. In this way, hostile architecture regulates behavior and reinforces a social order that defines “proper” uses of

⁹⁰ McCreath, Brendan. *Hostility in the City: The Implications of Hostile Architecture on Health and Ethics*.

⁹¹ Albertsen, Andreas. "Anti-homeless Hostile Design as Wrongful Discrimination." *British Journal of Political Science*.

⁹² Licht, Karl de Fine. "'Hostile Architecture' and Its Confederates: A Conceptual Framework for How We Should Perceive Our Cities and the Objects in Them."

public space while excluding others. These justifications highlight how design can normalize exclusion and connect to broader systems of urban control and governance.

Hostile Architecture as Urban Control and Governance

Additional research situated hostile architecture within the larger context of urban control and regulation. Setha Low argues that modern public spaces are increasingly shaped by privatization and securitization, limiting access through design rather than law.⁹³ James Petty expands this argument by describing hostile architecture as a form of soft policing, in which behavior is regulated without direct enforcement.⁹⁴ Although architects and urban planners often defend these designs as necessary for maintaining order and aesthetics, many scholars raise serious ethical concerns about their role in social exclusion. This intersection of exclusion and urban aesthetics highlights how deeply embedded these features are in the planning and appearance of modern cities. Rather than just being physical structures, hostile design functions as a tool that determines who is visibly welcome in public spaces. Such intentional shaping of environments prompts important ethical questions about the use of urban design as a means of social control.

Critics argue that hostile architecture is particularly unethical because it excludes vulnerable groups, especially people experiencing homelessness, and disregard their basic needs. This pattern reflects a broader system of social exclusion rooted in design. Urban theorist Don Mitchell argues that these practices transform public spaces into areas that only serve the socially acceptable.⁹⁵ He explains that when governments and developers prioritize aesthetic order and property value over accessibility, they reinforce inequality and deny vulnerable groups their rights to public spaces. In doing so, modern cities place order, property value and visual cleanliness above the rights of marginalized populations. David Harvey argues that urban design has become a tool of neoliberal control, where the physical layout of the city determines who belongs and who doesn't.⁹⁶ Hostile architecture, in this sense, is just one visible aspect of a larger effort to manage social order through exclusionary planning. These perspectives reveal that hostile architecture is not solely about homelessness, but is also about maintaining a controlled, desired image of the city.

Gap of the Research

While existing research provides insight into the identification, purpose, and consequences of hostile architecture, there remains a significant gap in understanding how these designs are perceived by the general public. Much of the current literature focuses on the intentions behind hostile architecture and its impacts on vulnerable populations, particularly individuals experiencing homelessness. However, there is limited research examining whether

⁹³ Low, Setha. "Behind the Gates: Life, Security, and the Pursuit of Happiness in Fortress America."

⁹⁴ Petty, James. "The London Spikes Controversy: Homelessness, Urban Securitisation and the Question of 'Hostile Architecture,'"

⁹⁵ Mitchell, Don. *The Right to the City Social Justice and the Fight for Public Space*

⁹⁶ Harvey, David. *Rebel Cities: From the Right to the City to the Urban Revolution*

everyday individuals recognize these exclusionary features or how such awareness may influence their attitudes toward those affected. This gap is critical because hostile architecture does not only shape physical environments, but also communicates societal values regarding belonging and exclusion.

This study seeks to address this gap by investigating the relationship between awareness of hostile architecture and empathy toward individuals experiencing homelessness. The guiding research question for this study is: “To what extent does noticing hostile architecture influence empathy toward individuals experiencing homelessness among residents of Midtown Manhattan?” This research question creates several subquestions. These subquestions include: “Are individuals able to recognize and identify features of hostile architecture in urban spaces,” “Does awareness of hostile architecture correlate with higher or lower levels of empathy toward individuals experiencing homelessness,” “How do individuals interpret the purpose and justification of hostile architectural features,” and “Do perceptions of hostile architecture vary based on prior beliefs about homelessness or urban design.” My hypothesis in response to these questions and prior research is that exposure to hostile architectural features will increase empathy. By addressing these questions, this study expands existing research by connecting physical urban design with public perception, offering a better understanding of how hostile architecture functions not only as a structural tool of control, but also as a social signal that may shape attitudes and empathy.

Methodology and Research Design

Design of Study

This study uses a mixed methods experimental survey design, allowing for the analysis of both numerical trends in empathy and patterns in how participants interpret what they see. Rather than relying solely on opinion-based questions, participants were shown real images of public spaces and asked to respond to them. Quantitative research allows researchers to systematically measure relationships between variables using statistical analysis, thereby increasing validity and reliability.⁹⁷ A between-subjects design was chosen to prevent participants from being influenced by exposure to both hostile and non-hostile images. This approach reduces response bias and demand characteristics by limiting participants' ability to infer the purpose of the study and adjust their response accordingly.⁹⁸ To maintain immediate, unbiased reactions, each participant was exposed to only one condition. This meant that some participants viewed images containing hostile architecture, while others viewed comparable spaces without those features. This design allowed for direct comparison of group means to determine whether visual exposure influenced empathy responses. Unlike prior studies that primarily rely on observational or theoretical analysis of hostile architecture, this study directly measures how exposure influences individual empathy levels, addressing a gap in existing research.

⁹⁷ Slater, Paul, and Felicity Hasson. “Quantitative Research Designs, Hierarchy of Evidence and Validity.”

⁹⁸ Charness, Gary, et al. “Experimental methods: Between-subject and within-subject design.”

Participants

The target population consisted of Midtown Manhattan residents age 18 and older. This demographic was selected because it represents an accessible and relevant population for the research setting, aligning with convenience sampling methods that prioritize accessibility and proximity when studying location based occurrences.⁹⁹ Additionally, focusing on one location increased the likelihood that participants were familiar with similar environments, making their responses more realistic. Furthermore, the 18+ age requirement ensured that participants had reached adulthood and possessed sufficient cognitive maturity to evaluate the affective stimuli used in the study.¹⁰⁰ Participants were recruited through convenience sampling via posters containing QR codes displayed at Berkeley College and Baruch College in Midtown Manhattan (Appendix B). These locations were chosen due to their high concentration of public spaces where hostile architecture is commonly present. While convenience sampling may limit broader generalizability, it strengthens relevance by focusing on a specific population directly connected to the research context.¹⁰¹

The study aimed to collect approximately 200 total responses, with 100 participants per survey condition. Sample size distribution was designed to support independent sample t-tests, which are more reliable with balanced group sizes. Prior power analysis research indicates that dividing participants into approximately equal groups increases statistical power and improves the likelihood of detecting true differences between experimental conditions in quantitative studies.¹⁰² The surveys remained open for two weeks to maximize participation. Research on online survey methodologies suggest that defined, yet flexible response periods improve participation rates without significantly increasing survey fatigue or reducing response quality.¹⁰³ Participants were required to confirm Midtown Manhattan residency before starting the survey, and if participants indicated that they were not a Midtown Manhattan resident they were excluded from the final analysis.

Data Collection Methods

Visual Documentation

Photographs of hostile architecture were collected through direct field observation in Midtown Manhattan to ensure that participants were evaluating real architectural conditions. Research on visual survey methods suggests that incorporating images into surveys can help respondents better interpret and evaluate spatial environments.¹⁰⁴ Images included segmented

⁹⁹ Etikan, Ilker, et al. "Comparison of Convenience Sampling and Purposive Sampling"

¹⁰⁰ Cohen, Alexandra Ochoa, et al. "When Is an Adolescent an Adult? Assessing Cognitive Control in Emotional and Nonemotional Contexts."

¹⁰¹ Jager, Justin, et al. "More than Just Convenient: The Scientific Merits of Homogeneous Convenience Samples"

¹⁰² Buckley, Jeffery. "Conducting Power Analyses to Determine Sample Sizes in Quantitative Research: A Primer for Technology Education Researchers Using Common Statistical Tests"

¹⁰³ Dillman, Don. "Internet, Phone, Mail, and Mixed-Mode Surveys: The Tailored Design Method."

¹⁰⁴ Shr, Yau-Huo (Jimmy), et al. How Do Visual Representations Influence Survey Responses? Evidence from a Choice Experiment on Landscape Attributes of Green Infrastructure

benches, anti-homeless spikes, and ghost amenities, along with corresponding non-hostile spaces. These images were then paired to ensure that the only major difference between conditions was the presence or absence of hostile design (Appendix C). The use of visual analysis is supported by Rose, who emphasizes that systematic visual methodologies allow researchers to interpret how spatial design communicates meaning and influences perception.¹⁰⁵ By categorizing images prior to survey implementation, it ensured consistency across survey conditions.

Survey Instrument Design

Two Google form surveys were created (Appendix A). Both versions included identical Likert-scale empathy questions; however, one survey presented images of hostile architecture, while the other contained corresponding non-hostile environments. Participants were given one of the two surveys, each divided into three sections. The first section measured baseline empathy to account for pre-existing attitudes toward individuals experiencing homelessness. The second section presented seven images, where participants evaluated each image using Likert-scale questions related to recognition, fairness, and emotional response, along with short-answer explanations describing how they interpreted each design. The final section measured post-exposure empathy, allowing for direct comparison after viewing the images. Empathy was measured using self-report Likert scales, which is a widely accepted method for assessing attitudes.¹⁰⁶ The survey questions were designed to measure key components of empathy, including emotional response, perceived fairness, willingness to help, and personal responsibility. This structure is supported by Davis's model of empathy, which validates the use of self-report instruments to assess both cognitive and emotional aspects of empathetic responses.¹⁰⁷ Responses were recorded on a five-point scale ranging from strongly disagree to strongly agree. When multiple Likert items are combined using a scale, researchers support analyzing them through means and standard deviations.¹⁰⁸ This is because the combined Likert responses tend to allow for a more advanced statistical analysis. A pilot study was conducted prior to full distribution to improve clarity and reduce potential error. Pilot testing strengthens reliability by identifying unclear wording and procedural flaws.¹⁰⁹

Procedure

The study was conducted in four phases. First, hostile architectural features were photographed in Midtown Manhattan and categorized by design type and location (Appendix C). Second, two versions of the survey were developed and pilot tested to ensure clarity. Third, recruitment posters containing QR codes were placed at Berkeley College and Baruch College (Appendix B). Participants completed the survey digitally and anonymously over the two week

¹⁰⁵ Rose, Gillian. Visual methodologies: An introduction to researching with visual materials

¹⁰⁶ Sullivan, Gail M., and Anthony R. Artino, Jr. "Analyzing and interpreting data from likert-type scales"

¹⁰⁷ Davis, Mark H. "Measuring individual differences in empathy: Evidence for a multidimensional approach"

¹⁰⁸ Norman, Geoffrey. "Likert scales, levels of measurement and the "laws" of statistics"

¹⁰⁹ The Importance of Pilot Studies

period. The survey was conducted anonymously to reduce social desirability bias and interviewer effects, both of which can influence participants to provide responses aligned with perceived social expectations rather than their true attitudes. Research in survey methodology indicates that anonymity increases honesty and response validity in attitude studies involving sensitive topics such as empathy and social judgement.¹¹⁰ At the beginning of the survey, participants reviewed an IRB-approved consent form outlining the purpose of the study and their right to withdraw at any time (Appendix D). Residency in Midtown Manhattan was confirmed before proceeding (Appendix D). After data collection, responses were exported into a spreadsheet for statistical analysis. Spreadsheets are a common method for organizing research data, and are widely used as a tool for data analytics because they provide an easy way for calculations and basic statistical processing.¹¹¹

Statistical Method

Responses were coded numerically on a 1 to 5 scale, where 1 represented low agreement and 5 represented high agreement. Instead of combining all responses into a single score immediately, averages were first calculated for individual questions, allowing for clearer comparison across different aspects of empathy. Descriptive statistics, including means, standard

Mean (Average Empathy Score) $\bar{x} = \frac{\sum x_i}{n}$ Formula Key x_i = individual Likert-scale responses n = total number of responses - Used to calculate each participant's overall empathy score	Frequency Distribution (Response Patterns) $f(x)$ = number of times a value x occurs Formula Key $f(x)$ = frequency of a specific response x = a Likert-scale value (1–5)
Standard Deviation (Variability of Responses) Formula Key s = standard deviation (how spread out the empathy scores are) x_i = each participant's empathy score $\bar{x}$ = average (mean) empathy score n = total number of participants $\sum$ = adds all the values together $n - 1$ = adjustment used because the data comes from a sample $s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$	

To determine whether exposure to hostile architecture influenced empathy levels, independent sample t-tests were conducted to compare the average empathy scores between participants exposed to hostile architecture and those who were not. An independent sample t-test is commonly used to evaluate whether the difference between two group averages is large enough to be significant rather than due to chance, making it essential for determining whether the independent variable had a measurable effect.¹¹² The formula used to calculate the t-tests is

¹¹⁰ Yan, Ting. "Sensitive Questions in Surveys." *Psychological Bulletin*, Sept. 2007.

¹¹¹ Rebman, Carl M., Jr., et al. "An Industry Survey of Analytics Spreadsheet Tools Adoption: Microsoft Excel vs Google Sheets"

¹¹² Palmer, Alfonso, et al. "Procedures for Comparison of Two Means in Independent Groups with R"

shown below:

Independent Samples t-Test (Comparing Group Means)
$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Formula Key

- t = t-value (used to determine if the difference is statistically significant)
- $\bar{x}_1$ = mean empathy score of the **hostile architecture group**
- $\bar{x}_2$ = mean empathy score of the **non-hostile group**
- s_1^2 = variance of the hostile group
- s_2^2 = variance of the non-hostile group
- n_1 = number of participants in the hostile group
- n_2 = number of participants in the non-hostile group

Additionally, awareness responses were analyzed using comparative statistical methods to examine whether participants who explicitly identified hostile architectural features demonstrated different empathy scores than those who did not. This type of comparative analysis is commonly used in survey research to identify relationships between perception and attitudes. In addition to quantitative analysis, qualitative responses from short-answer questions were examined to identify recurring themes in how participants interpreted the images. This provided deeper insight into how participants understood and reacted to hostile architecture.

Ethical Considerations

This study received IRB approval prior to implementation. Participation was voluntary and anonymous. No personally identifiable information was collected. Participants are informed of their right to withdraw at any time with no consequence. This study posed minimal risk, as it involved only viewing public urban images and responding to attitudinal questions.

Limitations

Several limitations may affect the findings. First, convenience sampling may introduce selection bias, as participants were recruited from college campuses. Additionally, the study focused exclusively on Midtown Manhattan residents, which limits geographic generalization. Finally, empathy was assessed through self-reported scales, which may be affected by social desirability bias.

Results and Analysis

A total of approximately 100 participants completed the surveys, with a relatively equal distribution between the hostile architecture and non-hostile architecture survey conditions. This allows for direct comparison across multiple measures of empathy. Results are presented through

an analysis of baseline empathy levels, participants interpretation of architectural images, post-exposure empathy levels, and qualitative response patterns used to assess how exposure to hostile architecture influences empathy toward individuals experiencing homelessness.

Baseline Empathy

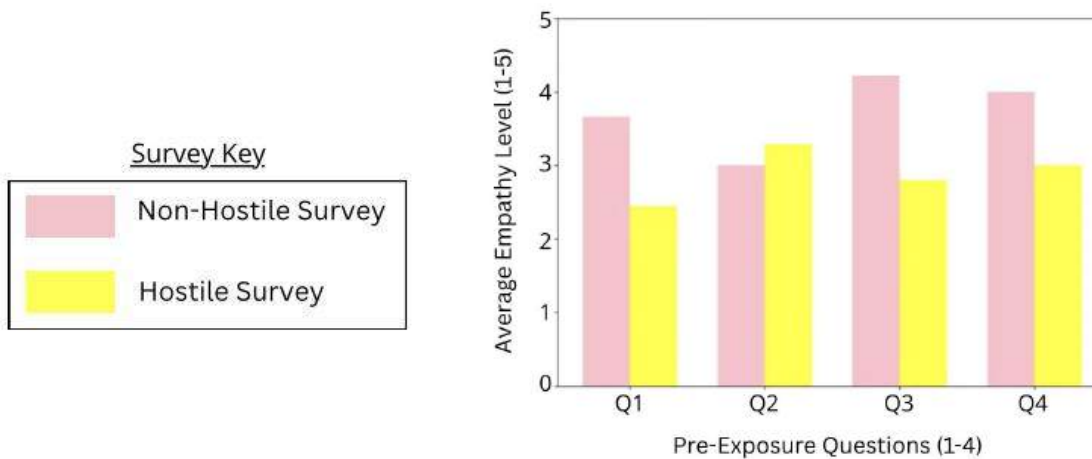
Pre-exposure responses revealed clear differences in baseline empathy levels between participants in the two conditions. Participants completing the non-hostile architectural survey demonstrated higher initial empathy, with responses concentrated in the mid to high range across all four questions. For example, 66.7% of participants selected a value of 3, and 33.3% selected 5 when asked about their willingness to help individuals experiencing homelessness. Similarly, when asked about their likelihood of supporting community programs designed to assist individuals without stable housing, 66.7% selected 5 and 33.3% selected 4. These results suggest that participants completing the non-hostile survey began the study with a relatively strong baseline empathy prior to image exposure.

In contrast, participants in the hostile architecture condition demonstrated a generally lower distribution of baseline empathy responses. For instance, more than half of participants selected values between 1 and 3 when asked about their willingness to help individuals experiencing homelessness. This lower mean score indicates that participants in the hostile condition entered the study with a comparatively lower baseline empathy.

This difference suggests that the two groups were not identical prior to exposure, introducing a potential uncontrollable variable that must be considered when interpreting post-exposure results. However, this variation also reflects the diversity of real-world attitudes, indicating that the sample captures a broad range of perspectives.

To visually represent these baseline differences, Figure 1 displays the average pre-exposure empathy scores across all four questions for both survey types. Across most measures, participants in the non-hostile condition demonstrated higher average empathy levels than those in the hostile condition, with the largest disparity appearing in question 3, which measures support for community programs. This pattern reinforces the presence of baseline differences between groups, suggesting that participants exposed to different survey types did not begin the study with identical levels of empathy. An exception to this overall pattern appears in question 2, when participants are asked about their understanding of the challenges faced by individuals experiencing homelessness. In this case, the hostile group reported a slightly higher average response to the non-hostile condition. This indicates that feeling less empathy does not necessarily mean participants were less aware of the challenges faced, showing a gap between understanding something and emotionally responding to it.

Figure 1. *Average pre-exposure empathy scores across survey types (non-hostile vs. hostile architecture)*



Question Key

Q1: Imagine a hypothetical situation in which you walk past a person experiencing homelessness. How strong is your impulse to offer help?
Q2: When you think about the daily challenges faced by people without stable housing, how confident are you that you understand the challenges they face?
Q3: If you heard about a community program designed to support people without homes, how likely would you be to support such efforts?
Q4: Imagine you come across a homeless individual who seems to need help on the street. How willing would you be to offer direct assistance? (money, food, etc.)

Image Analysis

Analysis of responses to individual images provides insight into how participants interpreted the architectural environments they were exposed to. In the non-hostile condition, participants generally perceived the environments as neutral or inclusive. When asked whether designs discouraged use, responses clustered between values of 1 and 3, indicating limited recognition of exclusionary intent. Emotional responses were also neutral or slightly positive, with participants frequently selecting options such as “no emotion” or “somewhat positive.” Open-ended responses reinforced this interpretation, with participants describing the spaces as “normal,” “functional,” or “accessible.” These findings suggest that non-hostile environments are viewed as standard features of urban design and do not significantly alter emotional perception or attitudes towards marginalized groups.

In contrast, responses to hostile architectural design images revealed a more complex pattern. While some participants identified exclusionary features such as spikes, dividers, and barriers, this recognition did not consistently lead to increased empathy. Instead, many participants rated these spaces as fair, with responses often falling within the upper range of 4 to 5 when evaluating fairness for community members. Additionally, emotional responses were predominantly neutral, with a large proportion of participants selecting “no emotion” across multiple images. This indicates that even when exclusionary design elements are recognized, they do not necessarily provoke strong emotional reactions.

Further analysis of open-ended responses revealed consistent patterns in how participants interpreted these features. Many participants engaged in rationalization, reframing exclusionary designs as reasonable or necessary. For example, one participant explained that bench dividers “force a more equitable sharing of space,” suggesting that design features intended to prevent rest were instead perceived as promoting fairness. Similarly, spikes and barriers were frequently interpreted through a lens of functionality, with participants describing them as necessary to maintain order, prevent damage, or regulate space usage. These interpretations indicate that participants often reinterpret exclusionary features in ways that align with perceived social norms, reframing them as reasonable design choices rather than harmful practices.

Together, these patterns highlight a critical disconnect between awareness and emotional response. Although some participants were able to identify hostile design elements, this awareness did not lead to increased emotional concern or charitable attitudes. Instead, recognition often led to justification, suggesting that awareness alone is insufficient to generate empathy. This pattern is consistent with prior research by de Fine Licht, which suggests that aesthetic and functional justifications can obscure the ethical implications of exclusionary design. The findings reinforce the idea that individuals may recognize hostile architecture without critically evaluating its social impact.

Post-Exposure Empathy

Following exposure to the images, patterns emerged in participants' empathy levels, indicating that visual stimuli influenced emotional responses. In the non-hostile condition, empathy levels remain high and, in some cases, increased slightly. Post-exposure responses were concentrated at the higher end of the Likert scale. For example, when asked about their likelihood of supporting policies aimed at assisting individuals experiencing homelessness, 66.7% of participants selected the highest value of five. Responses related to sympathy and emotional engagement were similarly distributed across moderate to high values, with no responses in the lowest categories. These findings suggest that exposure to non-hostile environments does not diminish empathy and may reinforce pre-existing attitudes.

In contrast, participants in the hostile architecture condition demonstrated a noticeable shift toward lower empathy levels following image exposure. Responses to post-exposure questions were more concentrated in the lower and middle ranges of the Likert scale. For instance, when asked about their emotional response to individuals experiencing homelessness, over half of participants selected values of 1 or 2, indicating lack of emotional investment. A similar pattern appeared when participants were asked how emotionally moved they felt when considering the daily challenges faced by individuals without stable housing, with average responses remaining at a similarly low level. Together, these results suggest that exposure to hostile architectural features may reduce empathetic responses. To further examine whether these differences were meaningful, independent sample t-tests were conducted to compare mean empathy scores between the hostile and non-hostile conditions. The results suggested a difference in average empathy levels between groups; however, due to the relatively small

sample size and variation in baseline empathy, these findings cannot be considered statistically significant.

Figure 2 illustrates these post-exposure differences, showing that participants in the non-hostile condition maintained higher empathy levels across all measures, while those in the hostile condition demonstrated lower average responses. The most prominent difference appears in measures of policy support, suggesting that exposure to hostile architecture may reduce willingness to act.

Figure 2. Average post-exposure empathy scores across survey types (non-hostile vs. hostile architecture)

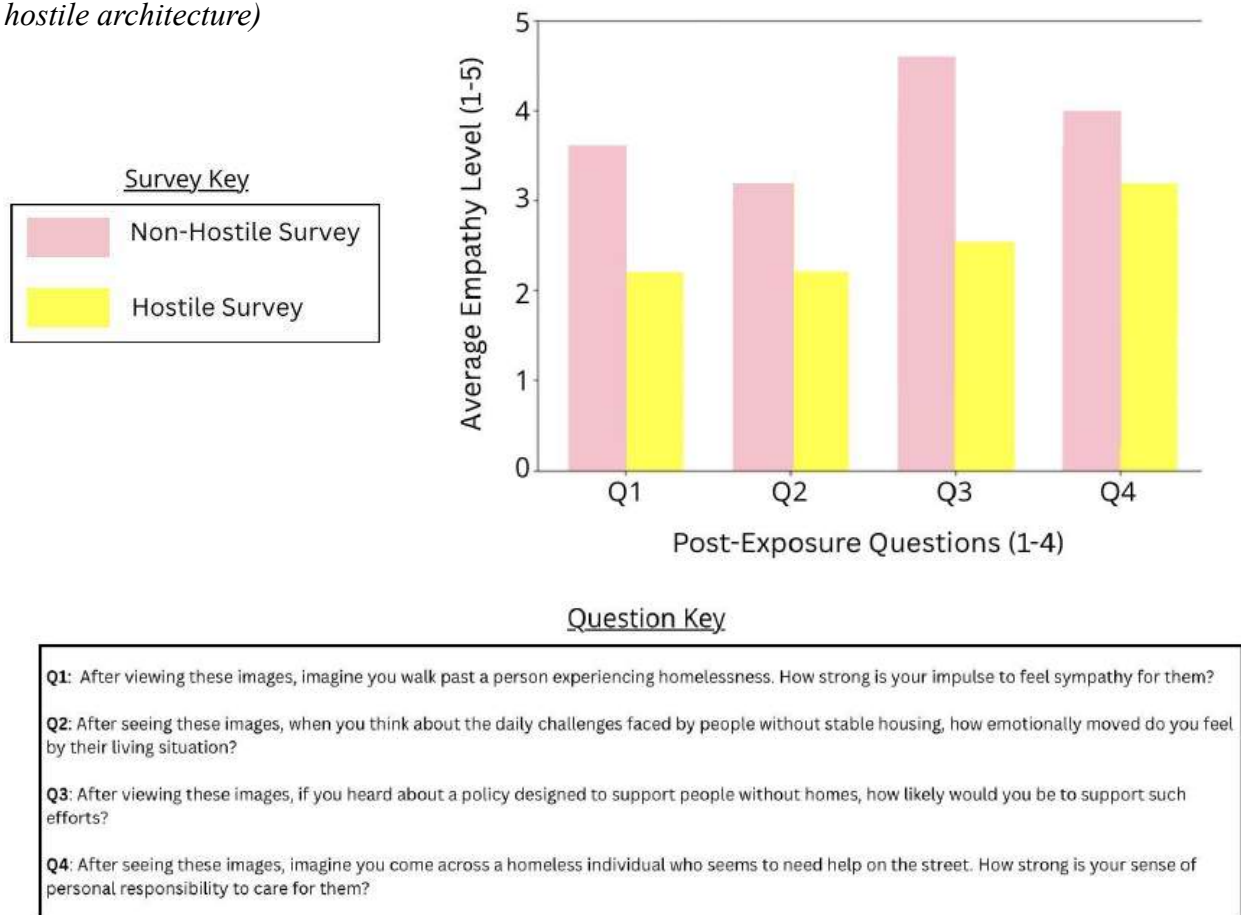
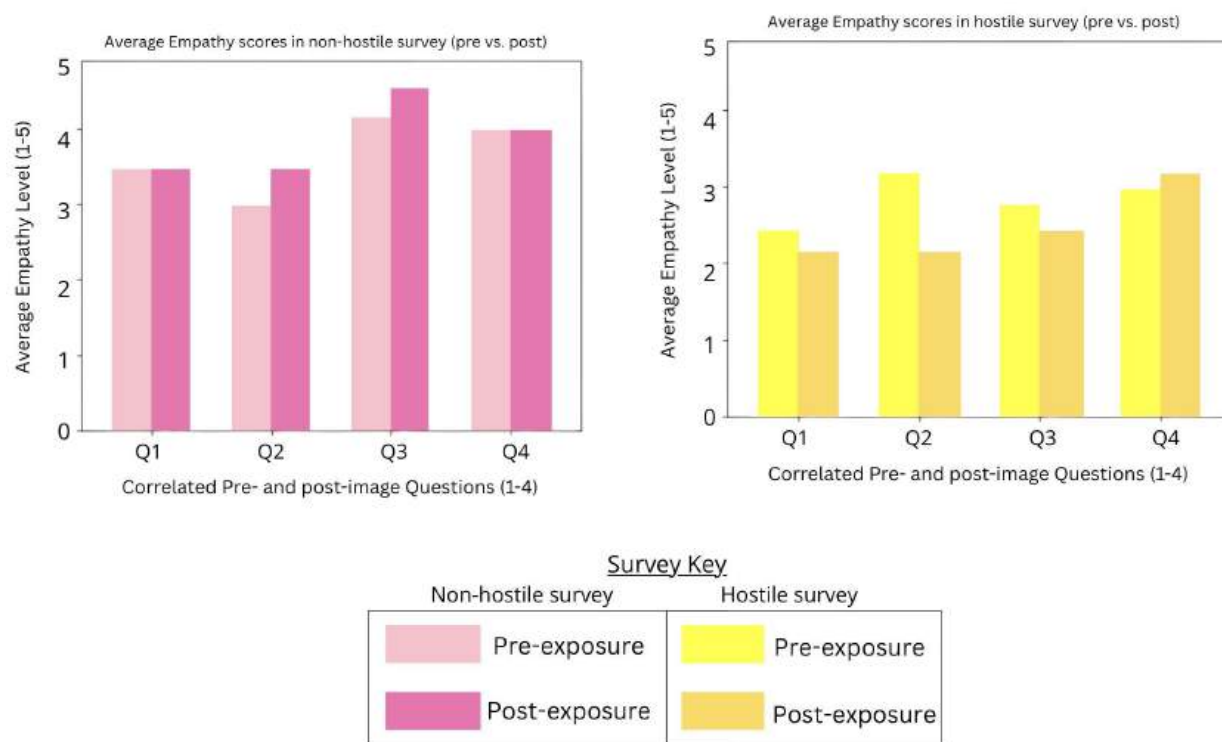


Figure 3 illustrates the change in empathy from pre to post-exposure across both conditions. While the non-hostile group shows relatively stable or slightly increased empathy levels across all measures, the hostile condition demonstrates decline for the majority of measures. This pattern suggests that exposure to hostile architectural features may actively suppress empathetic responses.

Figure 3. Average empathy scores (pre-exposure and post-exposure) across survey types



Pre-Exposure Question Key

- Q1:** Imagine a hypothetical situation in which you walk past a person experiencing homelessness. How strong is your impulse to offer help?
- Q2:** When you think about the daily challenges faced by people without stable housing, how confident are you that you understand the challenges they face?
- Q3:** If you heard about a community program designed to support people without homes, how likely would you be to support such efforts?
- Q4:** Imagine you come across a homeless individual who seems to need help on the street. How willing would you be to offer direct assistance? (money, food, etc.)

Post-Exposure Question Key

- Q1:** After viewing these images, imagine you walk past a person experiencing homelessness. How strong is your impulse to feel sympathy for them?
- Q2:** After seeing these images, when you think about the daily challenges faced by people without stable housing, how emotionally moved do you feel by their living situation?
- Q3:** After viewing these images, if you heard about a policy designed to support people without homes, how likely would you be to support such efforts?
- Q4:** After seeing these images, imagine you come across a homeless individual who seems to need help on the street. How strong is your sense of personal responsibility to care for them?

Overall

Across both quantitative and qualitative findings, the data reveals consistent patterns in how participants respond to different architectural environments. Participants in the non-hostile

condition maintained higher levels of empathy across measures such as emotional response, willingness to help, and policy support. In contrast, participants exposed to hostile architectural features demonstrated lower emotional engagement and weaker support for assistance based policies.

Importantly, these findings do not support the original hypothesis, which predicted that exposure to hostile architecture would increase empathy. Instead, the results indicate the opposite trend, suggesting that recognizing exclusionary design does not necessarily lead to greater concern for affected individuals. These differences are not solely explained by baseline variation. The qualitative data suggests that hostile architecture operates not only as a physical barrier, but also as a social cue that shapes interpretation. Even when participants recognized exclusionary features, this awareness often didn't translate into empathy. Instead, many participants justified these designs as practical or necessary, reflecting a disconnect between awareness and emotional response.

This pattern aligns with the concept of soft policing, where behavior and perception are shaped indirectly through environmental design. Rather than prompting reflection, these spaces encouraged interpretations that reinforced existing beliefs about order and fairness, ultimately normalizing exclusion. Although the relatively small sample size limits generalizability, the consistency of trends across multiple measures strengthens the reliability of the results. Overall, the data directly addresses the research question by demonstrating that exposure to hostile architecture does not increase empathy and may instead contribute to its reduction.

Implications

The findings of this study have important implications for urban planners, policymakers, and organizations involved in public space design. While hostile architecture has traditionally been justified as a functional response to issues such as safety, maintenance, or public order, these results suggest that such designs may also shape public attitudes towards marginalized populations in unintended ways. Because exposure to hostile architectural features is associated with reduced empathy and greater justification of exclusion, designers and policymakers should consider not only the physical outcomes of these structures, but also their broader social and psychological effects, as these impacts can shape public attitudes and influence support for exclusionary practices.

For urban planners, this research highlights the need to reevaluate the use of exclusionary design elements in public spaces. If these features contribute to the normalization of social exclusion, their implementation may reinforce negative perceptions of vulnerable populations, especially the homeless demographic. As a result, planning strategies could shift toward more inclusive design approaches that prioritize accessibility and shared use, rather than prevention.

In addition, these findings are relevant for advocacy organizations and local governments seeking to address homelessness and social inequality. Public attitudes play a huge role in shaping policy support, and the normalization of hostile architecture may reduce public

willingness to engage with or support solutions aimed at assisting marginalized groups. By recognizing how environmental design influences perception, the advocacy groups may be better equipped to challenge exclusionary practices and promote alternative approaches that encourage empathy and social responsibility.

Finally, this study opens avenues for further research on the relationship between environmental design and social cognition. Future studies could examine how long-term exposure to hostile architecture shapes attitudes over time, and how different demographic groups respond to these environments.

Limitations

Several limitations must be considered when interpreting the results of this study. First, the sample was relatively specific, as participants were primarily recruited from college campuses in Midtown Manhattan. This means the findings largely reflect younger, urban individuals, limiting generalizability to other demographics. Additionally, the study did not reach its intended sample size, which reduces statistical strength and makes it harder to detect small differences between conditions. A larger sample would improve the reliability of statistical comparisons and strengthen overall conclusions.

Another important limitation is the difference in baseline empathy levels between the two groups. Participants in the non-hostile condition began with higher empathy than those in the hostile condition, making it more difficult to isolate the true effect of image exposure from participants' pre-existing attitudes.

Finally, the study relied on self-reported Likert scale data. Although anonymity was maintained to reduce social desirability effects, participants may still have responded in ways they perceived socially acceptable, rather than their true attitudes. In addition, because participants responded to static images rather than interacting with physical environments, their reactions may not fully reflect real-world experiences, where direct exposure could produce stronger emotional responses.

Future Research

Future research should build on these findings by examining how hostile architecture influences empathy in real-world settings over longer periods of time. Because this study relied on short-term exposure to images, future studies could use field-based or longitudinal studies to assess how repeated, real-world exposure to hostile architecture shapes attitudes over time. For example, researchers could track participants' interactions with urban environments over several weeks or months to determine whether prolonged exposure produces more lasting changes in empathy and perception.

In addition, future research should expand the scope of analysis by investigating how different forms and intensities of hostile architecture influence perception. Rather than treating hostile design as a uniform concept, studies could isolate specific features such as spikes or

barriers to determine which designs produce stronger psychological responses than others. This would allow researchers to identify which designs have the greatest social impact.

Further research should also incorporate more diverse populations to examine how individual differences shape responses to hostile architecture. Variables such as age, socioeconomic status, prior beliefs about homelessness, and familiarity with urban environments may influence how individuals interpret these spaces. By integrating these factors into experimental or mixed-methods designs, future studies could better account for variation in perception and provide a better understanding of how environmental design interacts with social perception.

Finally, future research could explore whether these changes in perception translate into actionable behavior. While this study focuses on self-reported empathy and attitudes, future studies could investigate whether exposure to hostile architecture influences actions, such as supporting public policies or donating to homelessness initiatives. Examining behavioral outcomes would allow researchers to determine whether shifts in perception lead to changes in decision making.

Conclusion

Overall, this study demonstrates that hostile architecture extends beyond its role as a physical tool of urban control and may also influence public attitudes towards individuals experiencing homelessness. These findings indicate that exposure to hostile design does not increase empathy, and may instead contribute to reduced emotional responses by normalizing exclusionary practices.

Importantly, the study reveals that awareness of hostile architectural features does not necessarily lead to increased concern. Instead, participants often rationalize these features, framing them as functional or fair. This suggests that hostile architecture shapes not only behavior but also perception, reinforcing social norms that justify exclusion.

These findings support and expand upon existing literature by providing empirical evidence that urban design influences psychological and social processes. As cities continue to prioritize order, aesthetics, and regulation, it is essential to consider the broader implications of these design choices. Hostile architecture does not simply determine who can occupy public space; it also shapes how society perceives and responds to those who are excluded from it.

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Bio-Inspired Aerodynamics in Modern Aircraft Design By Q. Khanh Hoang

Abstract

This review examines selected examples of bio-inspired aerodynamics and considers how far they can realistically inform aircraft design. The paper focuses on birds, insects, and marine animals because these groups show well-studied mechanisms for lift generation, drag reduction, vortex control, maneuverability, and lightweight structures. The strongest engineering links are found in hummingbird and insect-inspired flapping systems for micro air vehicles, shark skin-inspired surface treatments, and blended-wing or morphing concepts influenced by streamlined animal forms. At the same time, many biological mechanisms cannot be copied directly because they depend on scale, flexible tissues, sensory feedback, and complex control. The review therefore treats animal flight and swimming not as ready-made blueprints, but as sources of design principles that require careful testing before use in aircraft engineering.

Keywords

Physics; Biomimicry; Aerodynamics; Aircraft Design; Bio-inspired Engineering; Aerospace Engineering; Aerodynamic Efficiency; Drag Reduction; Vortex Control; Blended Wing Body.

Introduction

Engineers have often looked to animals for ideas about movement, stability, and flow control. In aerodynamics, birds, insects, and marine animals are useful not because aircraft can simply copy them, but because their bodies and movements show different ways of handling airflow or fluid flow. Some of these mechanisms, such as wing flexibility, vortex control, streamlining, and drag reduction, relate directly to problems faced in aircraft (1-3).

Bio-inspired aerodynamics brings together biology, fluid mechanics, materials science, and aerospace engineering. Recent tools such as computational fluid dynamics (CFD), high-speed imaging, additive manufacturing, and smart materials have made it easier to study animal movement and test selected features in engineered systems (2, 4, 5). These tools have supported research on morphing wings, flapping-wing micro air vehicles (MAVs), adaptive surfaces, and drag-reducing coatings, although most examples remain experimental rather than routine aircraft technologies (1, 2, 4, 5).

This review focuses on three animal groups. Birds are most relevant to gliding, hovering, streamlining, and wing morphing (6). Insects are especially useful for understanding low-Reynolds-number flight and unsteady lift production. Marine animals are included because their movement through water provides fluid-dynamic lessons for drag reduction, vortex control, and body-surface design, even though direct transfer from water to air requires caution (1, 3, 7, 8).

Observation of birds influenced early aviation, including the work of Leonardo da Vinci, George Cayley, Otto Lilienthal, and the Wright brothers (9). Modern work is different: it uses

animal systems as test cases for aerodynamic principles rather than copying their shapes directly. This distinction is important because a feature that works well in a bird or insect may fail when scaled to an aircraft.

This review asks which animal-inspired aerodynamic ideas have the clearest relevance to aircraft, UAV, or MAV design, and which ideas remain more speculative. The discussion is organized around birds, insects, and marine animals, with attention to lift generation, drag reduction, structural efficiency, maneuverability, and adaptive control.

The literature was synthesized using a narrative review approach. Studies were grouped first by animal model and then by engineering mechanism. This made it possible to compare whether each biological example supports a practical design principle, such as drag reduction or low-speed maneuverability, or whether the link to aircraft design is still indirect.

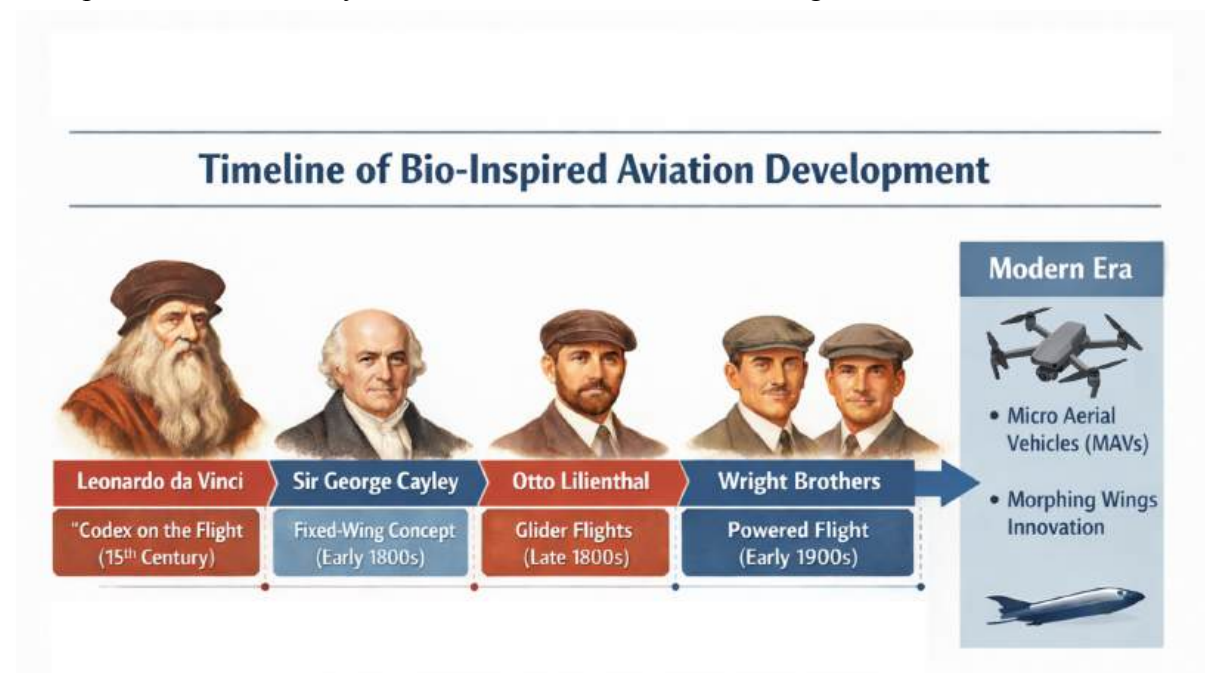


Figure 1. Historical development of bio-inspired aviation, from early observations of bird flight by Leonardo da Vinci to modern technologies such as micro aerial vehicles and morphing wing systems.

Method

This paper was prepared as a structured narrative review. Sources were identified mainly through Google Scholar, ResearchGate, journal websites, and academic books available to the author. Search terms included combinations of 'bio-inspired aerodynamics', 'animal-inspired flight', 'bird-inspired aircraft', 'flapping-wing micro air vehicle', 'morphing wing', 'dragonfly wing structure', 'shark skin drag reduction', and 'blended wing body'. The search was not intended to be a systematic review or meta-analysis; instead, it was used to identify representative studies that connect biological mechanisms with aircraft or UAV design.

Sources were included if they met at least two of the following conditions: relevance to biological aerodynamic or fluid-dynamic mechanisms or discussed relevance to aircraft.

Peer-reviewed papers were prioritized. Books, NASA reports, and preprints were used when they provided historical context, engineering background, or recent design examples. Sources were excluded if they were mainly descriptive, lacked technical detail, or made broad biomimicry claims without a clear aerodynamic link. 55 articles were identified, 20 included after relevance screening.

The selected studies were read qualitatively and organized into themes: drag reduction, lift generation, vortex control, maneuverability, wing flexibility, and structural efficiency. Because the evidence was diverse and not directly comparable across animals, the review emphasizes patterns and limitations rather than numerical ranking. Claims about full-scale aircraft applications are treated more cautiously than claims about MAVs, drones, or surface-level design features.

Case Studies of Animal-Inspired Aerodynamics

Birds

Birds are a natural starting point for bio-inspired aerodynamics because their wings, bodies, and feathers combine lift production with control. However, bird-inspired design is strongest when it focuses on specific mechanisms, such as wing morphing or streamlining, rather than trying to imitate a whole bird. Studies of avian biomechanics and flight evolution show that wing shape, body posture, and feather control all affect aerodynamic performance (3, 6).

Hummingbirds are useful models for low-speed maneuverability and hovering. Unlike most birds, they generate lift during both the upstroke and downstroke, allowing controlled hovering in small spaces (10, 11). Their figure-eight wing motion, shown in Figure 2, helps maintain lift throughout the wingbeat cycle. Recent work also shows that wing inertia contributes to fast turning and body control, not just aerodynamic force production (12). This makes hummingbirds more relevant to flapping-wing MAVs than to large aircraft, because MAVs operate in similar low-Reynolds-number conditions and require compact maneuverability.

The peregrine falcon represents a different design problem: high-speed stability. During dives, its streamlined body and swept wings help reduce drag and control flow separation (13, 14). CFD studies suggest that its body shape and wing posture may be useful for thinking about high-speed UAVs, fuselage shaping, and adaptive control surfaces. Still, this example should be interpreted carefully because a falcon controls its body and feathers continuously, while aircraft structures are much more constrained. The kingfisher is another useful case, especially for drag reduction at the air-water interface. Its head and body shape have been discussed in relation to low-noise, streamlined design, although the strongest evidence comes from water-entry studies

rather than aircraft testing (15, 16).

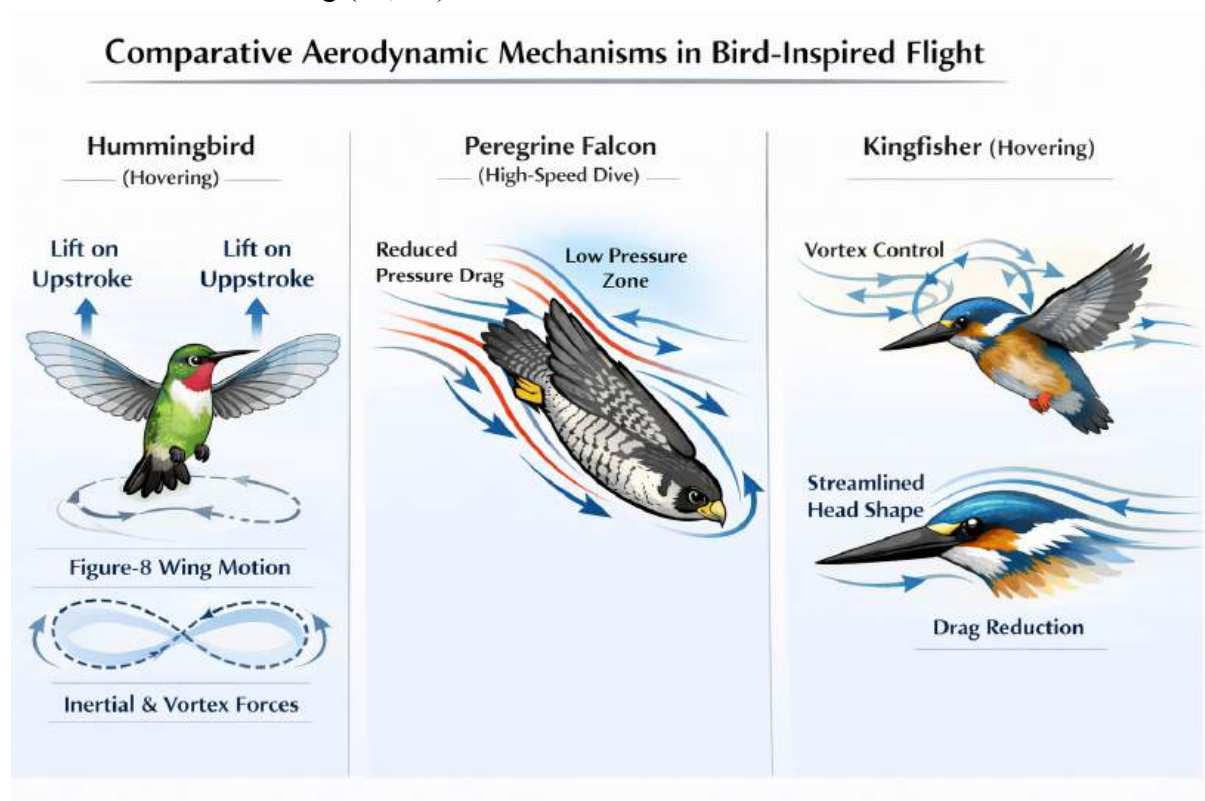


Figure 2. The hummingbird exhibits figure-eight wing motion, generating lift during both the upstroke and downstroke, supported by vortex and inertial forces that enable hovering. The peregrine falcon's streamlined body reduces drag during high-speed dives. The kingfisher demonstrates vortex control and a streamlined head for low-speed stability and drag reduction. These flight strategies provide models for MAV and UAV design.

Insects

Dragonflies provide one of the strongest insect examples because their flight combines unsteady aerodynamics with a lightweight wing structure. Their relevance is mainly to MAVs and small UAVs, where low-Reynolds-number effects are important. Dragonflies can hover, accelerate, and turn rapidly because their forewings and hindwings interact with vortices in ways that fixed wings do not reproduce easily.

A key aerodynamic feature is the leading-edge vortex (LEV). During flapping, attached LEVs can delay separation and help maintain lift at low Reynolds numbers. Figure 3A summarizes how counter-stroking forewings and hindwings create vortex interactions that can increase lift. Wing torsion and deformation also matter because the flexible wing changes shape passively during motion. These features are attractive for MAV design, but they also create engineering difficulties because artificial wings must combine flexibility, strength, repeatability, and controllability.

The dragonfly wing is also important structurally. As shown in Figure 3B, primary and secondary veins form polygonal membrane cells and distribute loads through Y-, T-, and cross-shaped joints. Zheng et al. showed that this network can be interpreted through

force-balance and truss-like principles (7). This is a more direct engineering contribution than simply saying dragonfly wings are lightweight: the vein layout provides a model for designing stiff, low-mass cellular structures.

Recent computational studies have taken this further by using reciprocal mapping and machine-learning methods to generate dragonfly-inspired structural networks (7). These methods do not copy the insect wing exactly; instead, they extract structural rules and adapt them to engineering geometry. This is a useful example of mature biomimicry because it translates a biological pattern into a design method rather than a visual imitation.

Overall, dragonflies offer two separate lessons. Their flapping aerodynamics are most relevant to MAV control and lift at small scales, while their vein networks may inform lightweight aerospace structures. The first lesson is difficult to scale up, but the second may be more broadly useful because structural optimization can be applied independently of flapping flight.

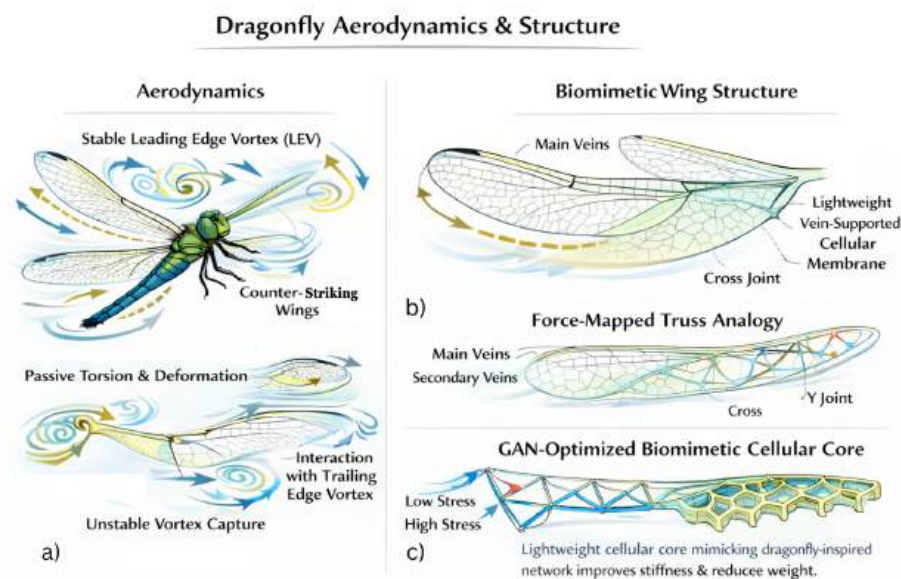


Figure 3. (A) Aerodynamic mechanisms of dragonfly flight, including the generation of stable leading-edge vortices (LEVs) and vortex interactions produced by counter-stroking forewings and hindwings. (B) Structural organization of dragonfly wings showing the hierarchical vein network composed of primary and secondary veins, forming polygonal membrane cells with specialized Y-, T-, and cross-shaped joints that distribute aerodynamic loads while maintaining a lightweight structure. (C) Force-mapped truss analogies and machine-learning-generated cellular cores, which replicate the force-balanced architecture of natural wings and provide improved stiffness-to-weight performance for aerospace structures and unmanned aerial vehicles.

Marine Animals

Marine animals move in water, not air, so their relevance to aircraft must be treated as an analogy rather than a direct imitation. Still, fluid-dynamic principles such as drag reduction,

boundary-layer control, vortex management, and streamlining apply across fluids, even though density, viscosity, and Reynolds number differ. Figure 4 summarizes three examples: shark skin, dolphin-like body shaping, and manta ray-inspired gliding forms.

Shark skin is probably the clearest marine example. Its surface denticles can affect the boundary layer and reduce drag under certain flow conditions. Domel et al. reported aerodynamic improvements from shark skin-inspired designs at low angles of attack (1). This does not mean that shark skin coatings will automatically improve all aircraft surfaces, but it does show that biological microstructures can guide surface engineering when tested under defined flow conditions.

Dolphins and porpoises are often cited for streamlined body shapes. Their forms may be relevant to fuselage, nacelle, and UAV body design, but the comparison should not be overstated because aircraft operate under different structural and operational constraints (17). In this case, the lesson is mainly about pressure-drag reduction through smooth body integration rather than copying a dolphin shape.

Manta rays are relevant to gliding and flexible body motion. Their broad, smooth forms have been compared with blended-wing-body or flexible-wing concepts (18, 19). The connection is useful but indirect: most manta ray studies are hydrodynamic or robotic, while aircraft applications require different materials, loads, and control systems.

Other marine animals, including tuna and swordfish, show how vortices can support efficient propulsion. For aircraft design, these examples are most useful as reminders that vortex control is not always harmful; under the right conditions, vortices can be managed to improve stability, lift, or propulsion. However, these ideas need careful experimental validation before being treated as aircraft design solutions.

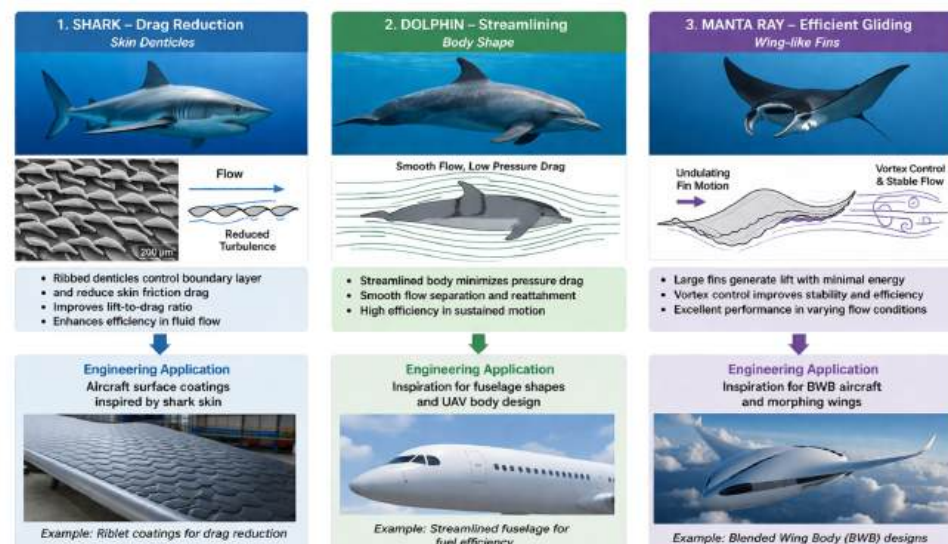


Figure 4. Marine animal-inspired aerodynamic mechanisms and their engineering applications. Shark skin denticles reduce drag through boundary layer control, dolphin body shapes minimize pressure drag through streamlining, and manta rays demonstrate efficient gliding and vortex control. These biological principles have inspired aircraft surface coatings, fuselage design, and blended wing body configurations.

Discussion

The clearest conclusion from the reviewed literature is that bio-inspired aerodynamics is strongest when it extracts a specific mechanism from nature and tests it under engineering conditions. The evidence is strongest for small-scale systems such as MAVs, where insect and hummingbird flight operate in similar aerodynamic regimes, and for surface or structural ideas such as shark skin-inspired coatings and dragonfly-like cellular networks. The evidence is weaker when biological shapes are used only as broad inspiration for full-size aircraft without direct testing.

Bird, insect, and marine examples therefore contribute different kinds of design information. Birds are most useful for adaptive wing posture, gliding, streamlining, and maneuvering. Insects are more relevant to unsteady lift, flapping control, and lightweight wing structures. Marine animals contribute ideas about drag reduction and integrated body shaping, but their water-based motion makes the aircraft connection less direct.

Current Applications in Aircraft Design and Future Directions

The reviewed studies suggest that animal-inspired ideas are already influencing aircraft research, but at different levels of maturity. Blended-wing-body (BWB) concepts, for example, are not simply animal copies; they are engineering designs that share some principles with streamlined natural bodies, such as integrated lifting surfaces and reduced wetted area. NASA and Boeing's X-48 program and related BWB studies show that this configuration can reduce drag, fuel use, and noise compared with conventional tube-and-wing designs (17, 19). This is one of the more practical large-aircraft directions discussed in the literature.

Traditional aircraft rely mostly on fixed wings, rigid control surfaces, and steady-flow assumptions. These approaches remain highly successful, but they are less suited to very small aircraft, gusty environments, or missions requiring rapid low-speed maneuvering. This is where bird- and insect-inspired systems are most relevant. For example, the RAVEN robot reported by Shin et al. uses avian-inspired multifunctional legs for ground-to-air transition, showing how biological movement can inspire specific aircraft capabilities rather than complete aircraft replacement (20).

Insect and marine examples add two different application routes. Insect-inspired systems point toward flapping wings, flexible membranes, and adaptive control for small UAVs. Marine-inspired systems point more toward surface design, streamlining, and vortex control. These are useful directions, but many remain at the prototype or computational stage. The most convincing studies are those that compare a bio-inspired design against a conventional baseline under the same test conditions.

A realistic future direction is not a single aircraft that copies many animals at once, but a selective combination of tested principles. A drone might use insect-inspired wing flexibility, a shark skin-inspired coating might be applied to a limited aerodynamic surface, or a large aircraft might use BWB geometry while incorporating local adaptive surfaces. Progress will depend less

on dramatic biomimicry claims and more on measurable improvements in drag, lift, weight, noise, control, or energy use.

Category	Traditional Aircraft	Blended Wing Body (BWB)	Bio-inspired / Morphing Aircraft
Design Structure	Tube-and-wing, separate fuselage and wings	Integrated wing + fuselage (single lifting body) (19, 21)	Flexible, articulated, shape-changing wings (22)
Aerodynamic Principle	Steady-flow aerodynamics	Reduced drag via lift distribution across the body (19)	Unsteady aerodynamics (vortex, flapping lift) (22)
Efficiency	Moderate fuel efficiency	High fuel efficiency (reduced drag, lower weight) (19, 21)	High efficiency at small scale (low Reynolds number) (22)
Control Mechanism	Fixed control surfaces (ailerons, rudder, elevator)	Advanced control with distributed surfaces (21)	Dynamic wing morphing, flapping, twisting (22)
Maneuverability	Limited, especially at low speeds	Stable but less agile than small UAVs	Highly agile and adaptable (22)
Applications	Commercial aviation, military jets	Future commercial transport, cargo aircraft (19, 21)	UAVs, drones, surveillance, research (22)
Advantages	Proven, reliable, scalable	Fuel savings, lower emissions, noise reduction (19, 21)	Adaptability, efficiency in complex environments (22)
Limitations	Drag, lower efficiency at some conditions	Complex design, airport compatibility issues (21)	Structural complexity, limited payload/scale (22)
Future Potential	Incremental improvements	Next-gen sustainable aviation (19)	Integration with AI + smart materials (22)

Table 1. Comparison of traditional aircraft, blended wing body (BWB), and bio-inspired morphing aircraft in terms of design structure, aerodynamic principles, efficiency, control mechanisms, maneuverability, applications, advantages, limitations, and future potential.

Challenges and Limitations

Animal-inspired aerodynamics still faces major obstacles. The first is biological complexity. Natural wings are flexible, anisotropic, self-repairing to some extent, and controlled by living sensory systems. Engineering materials can reproduce some of these traits, but usually with tradeoffs in weight, durability, cost, or reliability.

Control is another serious issue. Animals coordinate muscles, nerves, and sensory feedback continuously during movement. A MAV or aircraft must replace that with sensors, actuators, software, and power systems. This increases mechanical complexity and can reduce reliability, especially in small vehicles with limited payload and battery capacity.

Scaling is perhaps the most important limitation. Mechanisms that improve performance in a dragonfly or hummingbird do not necessarily work for a large aircraft. Reynolds number, structural load, energy supply, and safety requirements all change with size. Therefore, laboratory success at a small scale should be viewed as a starting point, not proof of practical aircraft performance.

Finally, there is still a gap between biological description and engineering design. Biologists may describe how an animal moves, while engineers need quantitative data, repeatable tests, and manufacturable designs. Better collaboration between biology, fluid mechanics, materials science, and aerospace engineering is needed to turn biological observations into usable technologies.

Conclusion

This review shows that animal-inspired aerodynamics is most useful when it is treated as a source of testable design principles rather than a collection of natural shapes to copy. Birds provide examples of wing morphing, gliding, and high-speed streamlining; insects are especially relevant to low-Reynolds-number flight and lightweight structures; and marine animals offer ideas for drag reduction and fluid-dynamic shaping. The strongest current links are found in MAVs, UAVs, surface treatments, and structural optimization, while direct application to large aircraft remains more limited.

Future progress will depend on careful validation. Bio-inspired designs should be compared with conventional designs using measurable outcomes such as drag, lift, stability, weight, energy use, and durability. Advances in CFD, smart materials, additive manufacturing, and machine learning will help, but they should support experimental testing rather than replace it. Overall, bio-inspired aerodynamics remains a useful research direction when its claims are specific, evidence-based, and matched to the correct engineering scale.

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Effect of GLP-1RAs for Teenage Obesity on Anxiety By Rhea Srinivas

Abstract

This paper explores the relationship between anxiety and obesity, specifically examining the role that GLP-1, a class of medications utilized to treat diabetes and obesity, has on anxiety in adolescents. This paper argues that GLP-1s offer an effective path forward for addressing obesity in adolescents; however, the drugs should be used with caution, given limited research on their potential long-term effects on younger, adolescent populations, making this an imperfect solution. With obesity and anxiety being big problems, and anxiety being further exacerbated by obesity, both issues and their relationship are worth studying. GLP-1 use does not provide a solution for the obesogenic factors. However, because the problems are so significant and the long-term consequences of obesity are well established, it is imperative to conduct further research. Adolescents on these drugs should be given regular psychological evaluations to ensure effective and safe weight loss, but external factors that lead to anxiety and obesity should be addressed, and more research should be done on the long-term side effects of taking GLP-1s.

Introduction

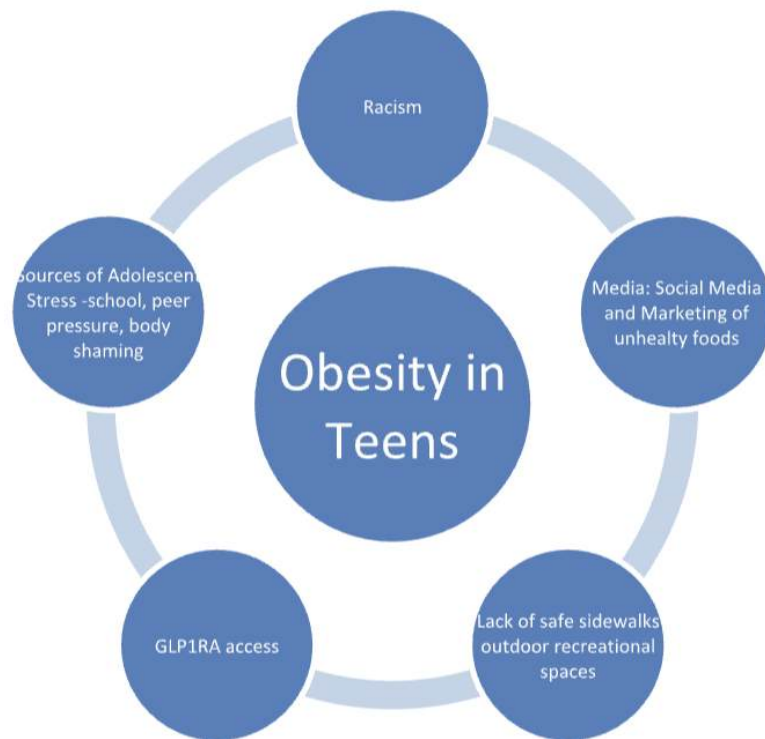
Adolescent obesity rates and anxiety rates are rising, making both of them important matters of concern (Anderson et al. 1–2; Sanyaolu et al. 1). Roughly one in five children and adolescents is living with obesity ("About Obesity"), and its prevalence has been steadily climbing over the recent decades (Noiman et al. 1, 4–6). Concurrently, there is a substantial increase in anxiety symptoms among teens, who are undergoing rapid growth and developmental changes surrounded by environments with social media and academic pressures (Anderson et al. 1–4). They are at an age where they are especially vulnerable to societal standards on body image and self-esteem issues (Voelker et al. 150). Anxiety disorders in adolescence matter not only because of their immediate distress but also due to the potential for problems worsening during life. Studies indicate that adolescent anxiety can be chronic, impair performance at school, and affect social relationships (Stromájer et al. 1548).

Early anxiety is a risk factor for adult depression, and is associated with a higher risk of substance use, including drugs and alcohol use, as a way of managing emotions (Siegel and Dickstein 1). With obese people being at an elevated risk for anxiety, taking a GLP-1RA may seem like a solution to their obesity-related mental health issues as well as their weight.

Literature Review

Glucagon-like peptides (GLP-1s) are a hormone naturally produced by the small intestine that regulates blood sugar levels ("GLP-1 Agonists"). Insulin is naturally released from the pancreas to lower the amount of glucose in the blood, while glucagon is a natural hormone released to raise the blood sugar ("GLP-1 Agonists"). Glucagon-like peptides control blood sugar levels by triggering the release of insulin and blocking the release of glucagon ("GLP-1 Agonists"). Type 2 diabetes involves impaired insulin secretion and insulin resistance (DeFronzo

773), and GLP-1 receptor agonists (GLP-1 RAs) are one class of medication used to lower blood glucose by stimulating insulin release (Hinnen 202). The GLP-1 RAs mimic the naturally occurring GLP-1 hormone and activate GLP-1 receptors in the brain's hypothalamus, with their effects including improving insulin sensitivity, slowing gastric emptying, and increasing feelings of satiety (Zheng et al.). A hormone is defined as a chemical messenger that the body creates in glands and other tissues to release into the bloodstream, where it travels to different parts of the body to alter biological processes ("Hormones").



Certain GLP-1 RAs (such as Wegovy and Saxenda) have recently been approved by the FDA for obese people who are not necessarily diabetic due to their weight loss effects ("FDA Approves First Treatment"; "FDA Approves New Drug Treatment"; "FDA Approves Weight Management Drug"). Some potential benefits of taking GLP-1RAs include a decrease in blood pressure, lowering the risk of heart disease, reducing hunger, and slowing gastric emptying (Zheng et al.). Some common side effects of taking GLP-1RAs for weight loss include nausea, vomiting, diarrhea, and having a loss of appetite ("GLP-1 Agonists").

The WHO defines obesity as having a body mass index (BMI) of 30 or above ("Obesity and Overweight"), which fails to account for the difference between fat, muscle, and bone mass (National Academies of Sciences, Engineering, and Medicine), but should be considered along with other factors such as medical history, lab tests, and physical health readings for a more comprehensive understanding of a person's health (Rubino et al. 230). Romero-Corral et al.'s study on the accuracy of BMI for diagnosing obesity, compared to body fat percent–defined obesity (BF% > 25% in men, > 35% in women), showed that even though there was a good correlation between the two, BMI failed to recognize over 50% of the individuals with excess fat, showing its limitations (Romero-Corral et al. 963). For the purposes of this paper, despite

this definition's limitations, obesity will be defined as a BMI of 30 or above ("Obesity and Overweight").

Obesity predisposes individuals to serious long-term risks, including high blood pressure, heart disease, stroke, fatty liver disease, and some cancers (colon, rectum, breast, uterus, gallbladder) ("Health Risks"). For these reasons, it is important to control weight, with the most common recommended ways being to consume foods rich in nutrients (vitamins, minerals, fiber) as opposed to foods dense in calories, salt, sugar, and fat, and getting more physical activity ("Weight Control").

Obesity and Anxiety

The existing research shows that obesity is associated with elevated rates of anxiety (Gariépy et al.). Discrimination and stigma were identified as significant contributors to anxiety symptoms among people with higher BMI (Emmer et al.). Physical mechanisms also play a role, as chronic stress and obesity both involve dysregulation of the HPA (hypothalamic-pituitary-adrenal) axis (Bose et al. 340), which leads to altered cortisol patterns that heighten the risk of anxiety (Ariño-Braña et al. 593).

Another study confirms that obesity and anxiety have a positive correlation, which is independent of other socioeconomic factors (Lindberg et al.). Their data suggest that higher BMI is predictive of increased anxiety rather than the reverse (Casanova et al.), which reinforces theories that psychological and biological stressors linked to obesity drive psychological distress (Tomiya). Less social support reinforces this effect, emphasizing the importance of close relationships.

Weight stigma is an important connector in this relationship. A study of psychosocial distress in adults with overweight and obesity found that experiences of weight stigma were associated with elevated depression, anxiety, and loneliness, and that these associations weakened when social support and family function were accounted for, suggesting that strong social support systems may buffer the psychological harms of weight stigma (Timkova et al.). The internalization of stigma has been consistently shown to contribute to emotional eating and uncontrolled food intake, which establishes a feedback loop perpetuating obesity and anxiety.

GLP-1 Use in Adolescents

GLP-1RAs have recently gained attention for pediatric weight management due to their strong weight-reducing effects. The existing research shows that GLP-1RAs are effective in weight loss for adolescents, but do pose extra risks due to their age. A meta-analysis demonstrated that GLP-1RAs effectively reduce BMI z-score and waist circumference among adolescents with obesity taking them (Romariz et al. 505). Gastrointestinal adverse events were more common in patients taking GLP-1RAs (Romariz et al. 506). Another meta-analysis published in the *Journal of Clinical Research in Pediatric Endocrinology* reached a conclusion that GLP-1RAs significantly reduced body weight in adolescents with overweight/obesity and/or

type 2 diabetes (Dai et al. 327). These two findings both show a pattern of adolescent weight reduction as a result of taking the GLP-1RAs.

Recent literature does warn of the psychological and developmental risks of GLP-1RA use in adolescents. A 2023 paper pointed out concerns that the dramatic suppression of appetite and restriction of calories during adolescence can disrupt healthy physical growth (Cooper et al.) Adolescents and young adults using prescription weight-loss medications, most of which are GLP-1 receptor agonists, show significantly higher rates of binge eating, purging, and overall eating disorder psychopathology compared to non-users (Ganson et al.). Krug et al. emphasize that adolescents prescribed GLP-1RAs require careful psychiatric monitoring and longitudinal evaluation, given that adolescence is a peak risk period for eating disorders and body image disturbances and the long-term developmental and psychosocial effects of these medications remain largely unknown ("Adolescents and Young Adults").

Structural and Behavioral Factors Affecting Treatment Outcomes

The existing research shows that individuals can achieve successful weight loss with consistent diet and exercise habits, but this isn't always possible for people, as there are structural and social factors that hinder progress, such as low socioeconomic status. A 2023 study found that barriers to obesity treatment are often significantly affected by socioeconomic status, racial inequalities, and disparities in healthcare access (Bell Washington et al. 432). Patients from lower socioeconomic backgrounds often have reduced access to GLP-1RAs due to insurance restrictions and cost (Eberly et al.), and experiences of unequal treatment in healthcare settings have been linked to elevated anxiety symptoms more broadly (Williams et al.).

Behavioral determinants of adherence to treatment are also important. A 2023 systematic review on motivators and barriers to weight loss in people with obesity found that effective weight-loss strategies include consistent dietary patterns and regular physical activity, but maintaining these behaviors is impeded by barriers such as time constraints, limited access to safe exercise spaces, emotional states like stress, and insufficient social support (Binsaeed et al.). Among adolescents, poor emotion regulation — a form of self-regulation — predicts greater engagement in emotional eating, including eating in response to anxious feelings (Shriver et al.). These behaviors increase caloric intake beyond the levels supported by the satiety induced by the GLP-1s.

A systematic review notes that while demographic factors such as age and gender are not strong predictors of maintenance of weight loss treatment (Varkevisser et al. 195), social and environmental support systems play a key role (Karfopoulou et al. 511). Family-based multicomponent behavioral interventions - combining counseling on diet and physical activity with parent skills training - are recommended as the first-line treatment for adolescent obesity (Bean et al.), and umbrella reviews of the broader pediatric literature find them effective for child weight management (Chai et al.). This shows that social support enhances metabolic and emotional outcomes for teens undergoing GLP-1RA treatment.

Major Sources of Anxiety among Teenagers

Adolescents experience anxiety due to a combination of interpersonal, structural, and societal factors, including pre-existing mental health conditions and socioeconomic status (Anderson et al.). A narrative review of the current literature identifies six contributing domains: biological factors, digital technology and social media, nuclear family dynamics, academic pressure, extracurricular activities, and political and environmental uncertainty (Anderson et al.). Academic pressure was a consistently identified driver of anxiety, especially in competitive academic environments where high performance is linked to future college admission and perceived economic security (Anderson et al.). Students report higher stress related to academic performance and time spent studying, and these pressures are consistently associated with increased anxiety (Anderson et al.).

Socioeconomic conditions, which include family income and education, influence both exposure to stress and access to resources such as higher-quality education and mental health services (Kirkbride et al. 59–60). Studies have linked lower socioeconomic status and financial strain with higher levels of adolescent anxiety (Kirkbride et al. 60; McLaughlin et al., "Socioeconomic Status" 1745; McLaughlin et al., "Childhood Socio-Economic Status" 1092). Insecurity of basic needs, including unstable housing and limited access to nutritious food, is associated with an increased risk of developing anxiety (Keen et al. 818; Fang et al.). Family relationships are another factor, as harsh parenting, inconsistent discipline, and inter-parental conflict are associated with a higher risk of internalizing problems including anxiety (Yap and Jorm 424; Yang et al.; Pinquart 613; Gorostiaga et al.), and exposure to community violence similarly heightens this risk (Fowler et al. 227), whereas parental warmth shows a protective association with lower internalizing symptoms (Yap and Jorm 424; Pinquart 613) and can buffer against the effects of academic pressure (Quach et al. 106). Adolescents with chronic health conditions (Pinquart 141) or obesity may also face judgement from their family or criticism of their body, which can cause body-related anxiety (Pont et al. e20173034). Technology and social media are also contributors to contemporary adolescent anxiety, as frequent use of social media platforms increases exposure to cyberbullying, social comparison, and unrealistic body goals (Popat and Tarrant 332), which are associated with higher anxiety symptoms (Keles et al. 88).

Adolescents manage anxiety through many strategies, ranging from clinical treatment to sometimes harmful coping behaviors (Leonard et al.). These choices are affected by access, stigma, and social contexts (Aguirre Velasco et al.). Evidence-based treatment options for anxiety include cognitive behavioral therapy and selective serotonin reuptake inhibitors, which, in combination, effectively reduce anxiety symptoms (Walkup et al.). Because untreated adolescent anxiety predicts later substance use disorders, effective treatment may reduce the risk of harmful coping behaviors such as substance abuse (Wolitzky-Taylor et al.). School-based counseling and mental health services are additional options for adolescents seeking support. By providing care directly in schools, these services help reduce common barriers to mental health care, such as lack of insurance, shortage of providers, stigma, and transportation challenges (Richter et al. 2).

Less formal coping, such as skipping classes, is a common strategy adolescents report to manage their stress and anxiety, with anxiety often manifesting as persistent avoidance behaviors that interfere with school engagement (Wang and Qin). Peer relationships can aid in emotional validation and provide some help (Carapeto et al.), but avoiding school is linked to worse academic outcomes and, gradually, can reinforce internalizing symptoms over time (Rogers et al.). For adolescents with obesity, avoiding physical education, medical appointments, or social events due to weight-related anxiety and internalized stigma may further worsen their mental and physical health (Pont et al.; Braddock et al.). Use of addictive substances such as alcohol, cannabis, nicotine, and misuse of prescription drugs is a prominent but concerning coping mechanism for adolescents with anxiety (Scott and Hulvershorn). Over time, this self-medication pattern increases the risk of dependence and contributes to a mutually reinforcing cycle in which anxiety symptoms worsen (Anker and Kushner).

GLP-1's Physiological Function

GLP-1 is a hormone naturally produced by intestinal enteroendocrine L-cells after the intake of nutrients, and GLP-1RAs are drugs created to mimic natural GLP-1 to activate the GLP-1 receptors (Paternoster and Falasca). These receptors are in the pancreas, brain, and peripheral tissues, which allow the GLP-1RAs to influence the secretion of insulin, glucagon suppression, and appetite regulation (Paternoster and Falasca). Within the CNS, GLP-1 receptors are present in the hypothalamus and the brainstem. Activation of these receptors promotes satiety and may modulate reward-related eating behavior (Baggio and Drucker 4223–24), and clinical trials in adolescents with obesity have shown that GLP-1RAs such as semaglutide produce significant reductions in body weight and BMI compared to placebo (Weghuber et al. 2245–46).

GLP-1RAs suppress the secretion of glucagon from pancreatic alpha cells and increase the secretion of insulin from pancreatic beta cells, which improves glycemic control and lowers the risk of hypoglycemia (Paternoster and Falasca). GLP-1 signaling also slows gastric emptying, leading to a slower delivery of nutrients to the small intestine, though this effect is subject to tachyphylaxis and is not the primary driver of long-term weight loss; rather, GLP-1RAs reduce food intake largely through direct action on hypothalamic neurons, particularly POMC/CART neurons in the arcuate nucleus, which produces the feelings of satiety, smaller meal sizes, and reduced snacking that drive the weight loss effects seen in patients taking them (Baggio and Drucker 4224–25).

RCTs of GLP-1 RAs in adolescents with obesity show significant reductions in BMI z-score, BMI as a percentage of the 95th percentile, and body weight compared to placebo with lifestyle intervention alone (Weghuber et al.; Kelly et al.; Liu et al.). These weight changes are accompanied by improvements in cardiometabolic markers including HbA1c and lipid levels, suggesting that GLP-1 RAs can improve both the severity of obesity and its downstream medical consequences (Weghuber et al.).

Despite their benefits, GLP-1 RAs do pose side effects that should be considered. The delayed gastric emptying and reduced GI motility caused by these medications can lead to

nausea, vomiting, diarrhea, abdominal pain, and constipation, particularly during initiation and dose escalation (Schalliol et al.).

Weight Stigma: Connecting Obesity and Anxiety

People with obesity often experience depression and anxiety at higher rates than the general population, with stigma, not body weight itself, emerging as a key driver of this distress (Westbury et al. 16). Stereotypes that frame people with obesity as lazy, undisciplined, or lacking willpower can make everyday social encounters a source of stress and self-consciousness (Westbury et al. 14; Bannuru et al.). These experiences are associated with negative self-perceptions, social withdrawal, and depressive symptoms (Segal and Gunturu). Obesity stigma is commonly defined as prejudiced, stereotyped, and discriminatory views or actions toward people with obesity (Westbury et al. 11). It includes both explicit discrimination and implicit, interpersonal bias, such as assumptions that people with obesity lack self-control (Bannuru et al.). Such stigma contributes to depression, anxiety, and body-image dissatisfaction among those affected (Westbury et al. 16). In healthcare and employment, weight-based bias leads to unequal treatment that can reinforce internalized stigma (Westbury et al. 17–18). The research consistently indicates that internalized stigma, rather than obesity itself, accounts for much of the psychological distress people with obesity experience (Westbury et al. 16).

Traditional public health messages often emphasize diet and exercise as an effective approach to treating obesity, but this framing overlooks the challenges faced due to structural and emotional issues (Hampl et al.). The AAP acknowledges that obesity is not just a matter of personal choice, but a chronic disease that is influenced by genetics, environment, and social determinants of health (Hampl et al.). For patients who are already struggling with weight stigma and internalized weight bias, being told to lose weight may not come across the right way, as weight-related discussions can elicit strong emotional responses including sadness or anger, which has the potential to negatively impact mental health (Hampl et al.). The AAP does warn that stigmatizing language can reinforce bias, whether or not intended, contributing to avoidance of health care and disrupted clinician-patient relationships (Hampl et al.).

In addition to this, there are also practical barriers that can deter attempts made to implement these standard recommendations of a healthier diet and more exercise (Hampl et al.). Many people do not have access to affordable, healthy food or safe spaces for physical activity, which is especially common in underresourced communities (Hampl et al.). Other people are faced with the issue of limitations in transportation, cost, and availability that make accessing healthy foods and obesity treatment impractical (Hampl et al.). These challenges are faced across most racial and economic groups, but disproportionately affect marginalized groups who experience higher rates of obesity and depression (Hampl et al.).

The AAP recognizes the effects of environmental and social factors promoting high-energy intake and sedentary behavior by describing these influences as an "obesogenic environment" (Hampl et al.). This makes it so that peoples' decisions on food and exercise are not just on an individual level, but largely shaped by factors such as marketing, transportation,

and resource availability (Hampl et al.). The popularity of advertising fast food and branded foods, especially to children, influences their dietary choices and behaviors, as younger children are more susceptible to advertising and more likely to interpret it as factual (Hampl et al.).

The combination of systemic barriers to weight loss and the stigma surrounding people with obesity often reinforces psychological distress by overwhelming individuals with blame for their situation while failing to acknowledge that the problem is exacerbated by structural and societal limits (Westbury et al. 11–15).

Structural Roots of Obesity

Individually focused approaches to adolescent obesity are practically inadequate and risk widening health inequities because they neglect the structural drivers of obesity and do not provide a durable public health solution without broader policy change (Jensen et al.; Krug et al.).

The rapid expansion of GLP-1 receptor agonists for adolescent obesity raises concerns that pharmacological treatment may obscure the structural drivers of the obesity epidemic. Weight stigma is rooted in neoliberal ideologies of personal responsibility, yet obesity disproportionately affects adolescents from low-income and racial/ethnic minority communities - populations shaped by obesogenic food environments, food insecurity, and chronic stress from racism (Jensen et al.; Krug et al.). Without attention to these upstream factors, treating adolescent obesity primarily at the individual level risks deepening rather than narrowing health inequities. GLP-1RAs can reduce weight and obesity-related health issues in adolescents, but their benefit depends on sustained use, close monitoring, and attention to mental health, especially around eating disorders and concerns of body image (Krug et al.). When treatment is framed as an individual choice without consideration of structural determinants, adolescents might feel that they bear all responsibility for their condition, despite the effects of access, affordability, and marketing of processed foods being out of their control (Jensen et al.; Krug et al.; Harris et al.).

Evidence from adult trials and pediatric work shows that weight regain is common after discontinuing GLP-1 treatment, suggesting that medication alone does not provide an adequate solution to the environmental and social conditions that promote weight gain. Reviews of adult pharmacotherapy trials consistently find that cessation of GLP-1 receptor agonists leads to weight regain even when lifestyle intervention is continued (Bin Ahmed). In the STEP 1 extension trial, adults who discontinued semaglutide regained roughly two-thirds of their prior weight loss within one year off-treatment (Wilding et al.). A similar pattern has been observed in adolescents: in the STEP TEENS trial, BMI began to rebound within seven weeks of stopping semaglutide (Weghuber et al.), and earlier work with liraglutide in adolescents showed weight returning nearly to baseline within 26 weeks of discontinuation (Kelly et al.). Starting GLP-1RAs in childhood carries the potential for lifelong reliance on the medication, which is concerning given that pediatric dosage forms can be more expensive than adult formulations (Cooper et al.).

For these reasons, obesity should be addressed at a structural, not individual level, since upstream social and structural determinants of health are more influential on population obesity outcomes than individual-level factors (Williams et al. 617). One way to prevent GLP-1 treatment from becoming a lifelong financial burden is to expand insurance coverage, including in public programs like Medicare and Medicaid, since current coverage gaps disproportionately affect lower-income populations who already face higher rates of obesity (Pearson et al. 22). Despite alleviating part of the cost for families, this is insufficient without addressing broader structural drivers like food environments, neighborhood conditions, and access to healthful foods (Williams et al. 620–621), and may even be impractical given that public systems face significant budgetary constraints - the Congressional Budget Office estimated that Medicare coverage of GLP-1s for obesity would add approximately \$35 billion in federal spending over eight years (Pearson et al. 19).

Interventions increasing walkability, access to parks, and opportunities for physical activity have been linked to healthier weight outcomes in many studies, though associations vary by geographic and socioeconomic context (Sridhar and Gumpeny). Components of a supportive built environment include sidewalks, green spaces, bike lanes, and access to public transportation (Sridhar and Gumpeny). However, implementing these interventions requires substantial investment of time, money, and political will, and isolated changes - such as installing sidewalks in low-density neighborhoods without accompanying density or destinations - may not be cost-effective on their own (National Academies of Sciences, Engineering, and Medicine 11, 13). Statutory regulation could reduce children's exposure to unhealthy food marketing (Smith Taillie et al. 787) because marketing ultra-processed products normalizes diets that promote weight gain (Harris and Smith Taillie 213).

Conclusion

GLP-1RAs do offer effective treatment of obesity for adolescents (Kelly et al. 2117; Cooper et al.), but the current evidence and safety concerns suggest using them with caution rather than being universally accessible (Cooper et al.). Long-term effects in youth still remain unknown, so use of these drugs requires monitoring and attention to mental health (Cooper et al.; Zaitoon et al.), and they should be used alongside broader structural efforts to reduce obesity and stigma rather than as a singular, standalone solution (Heitmann). Future research should study the long-term effects of taking GLP-1RAs for children and adolescents, to see their impacts on metabolic health, puberty, and mental health, rather than just on short-term weight loss.

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